Enterprise Firewall Administrator Study Guide

FortiOS 7.6

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1/30/2026

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Change Log

This table includes updates to the *Enterprise Firewall 7.6 Administrator Study Guide* dated 5/12/2025 to the updated document version dated 1/30/2026.

Change	Location
Typo fixed	Lesson 1 slide 14
Quotes updated for CLI commands	Lessons 3,4,6,8, and 10
Update images to version 7.6.3 and remove the step to create Zone2	Solutions slides 6, 33, and 38

In this lesson, you will learn about the network security architecture in an enterprise network.

Objectives

- Understand the enterprise firewall solution
- Design secure networks using Fortinet solutions

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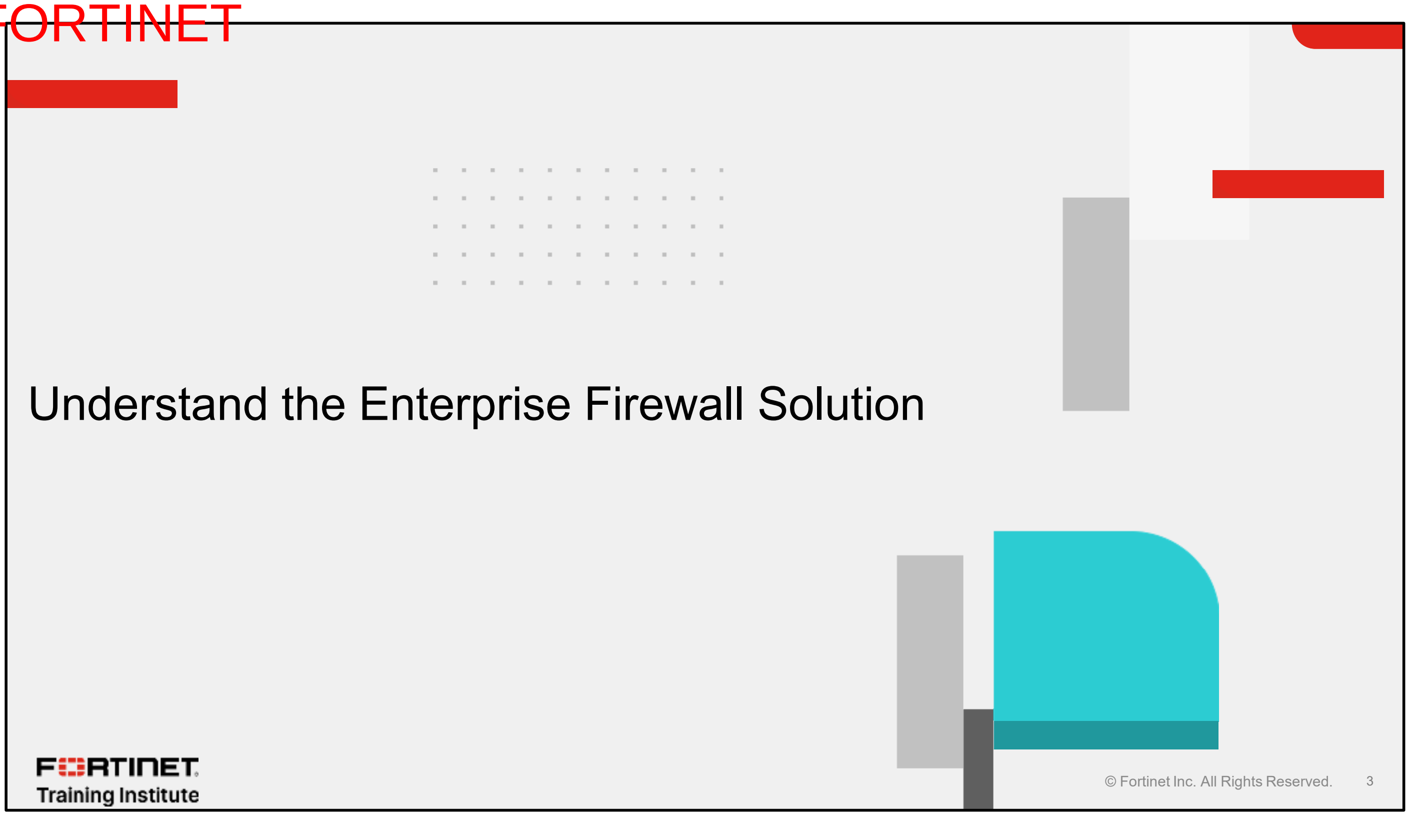
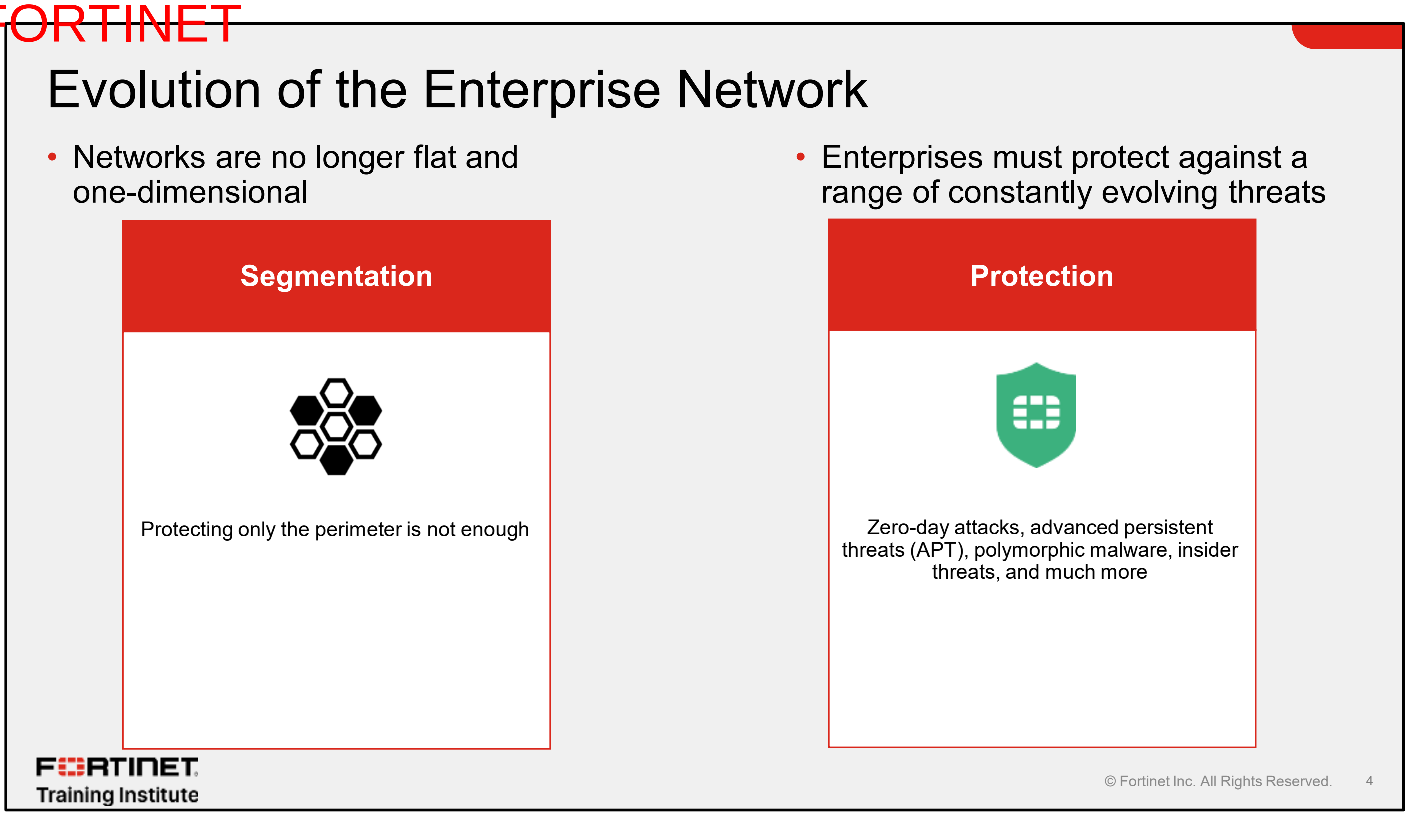
After completing this lesson, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in Fortinet network security architecture, you will be able to understand the enterprise firewall solution and the network security reference architecture, and the Fortinet products it comprises.

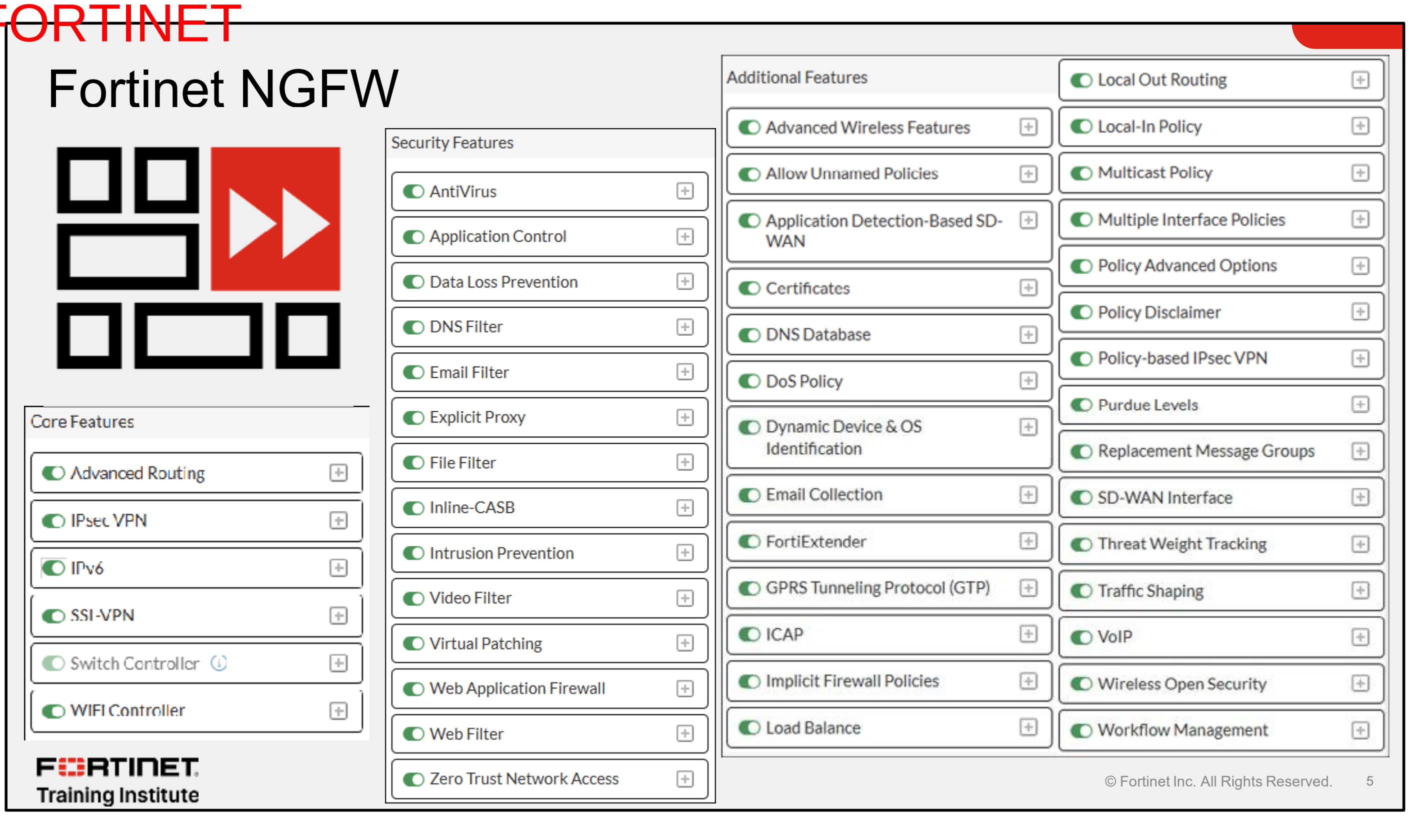


In this section, you will learn about the Fortinet enterprise firewall solution at a high level.



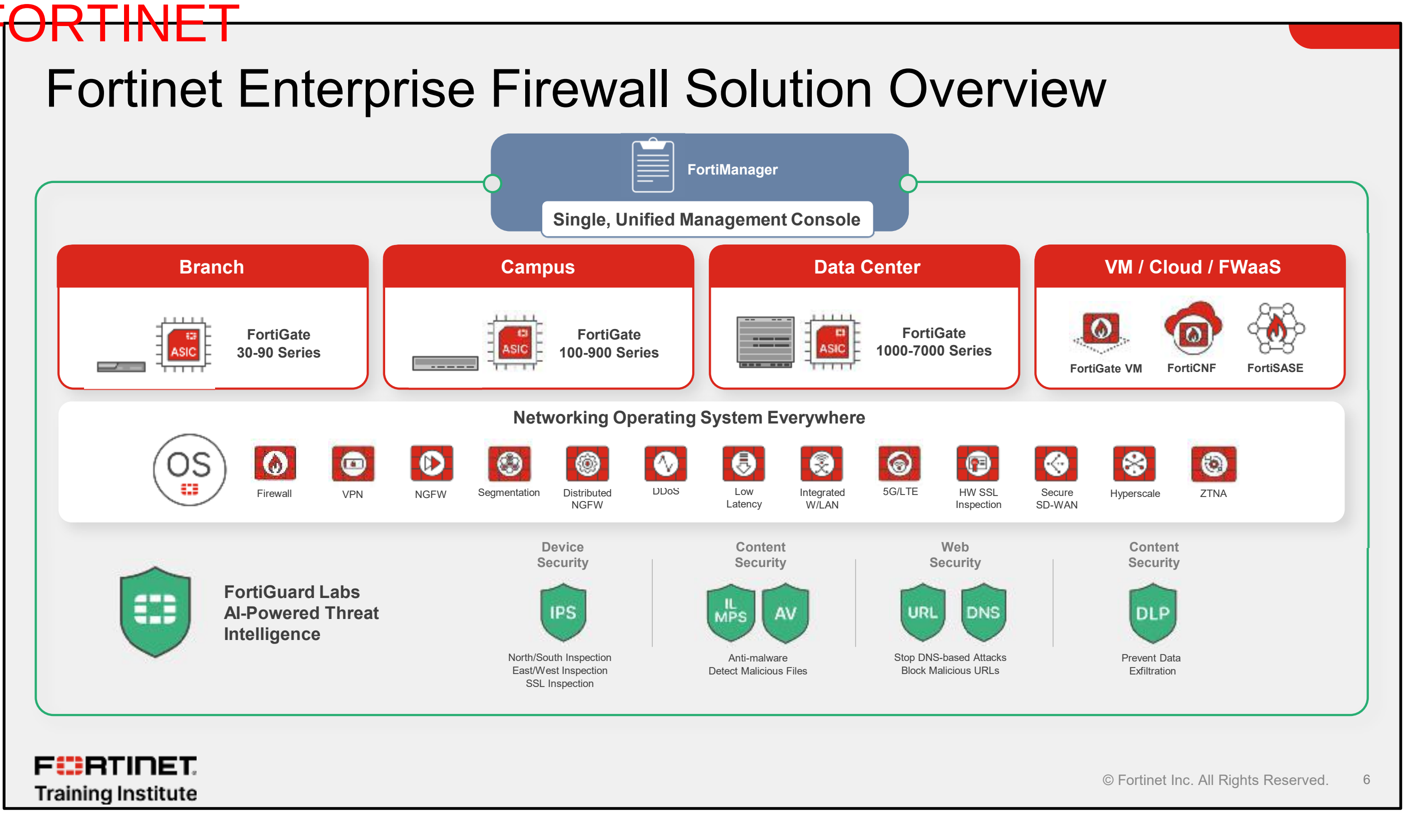
The traditional way of protecting a network by securing the perimeter has become a thing of the past. Network and security administrators today must protect against a wide range of threats, such as zero-day attacks, APTs, polymorphic malware, and more. They must also protect the network from any potential insider threats.

Malware can easily bypass any entry-point firewall and get inside a network. This could happen through an infected USB stick, or an employee's compromised personal device being connected to the corporate network. Additionally, we can no longer be sure that everything and everyone inside the network can be trusted. Attacks can come from anywhere, take many different forms, and affect all parts of a network. To secure an entire network against such wide array of potential attacks with multiple potential targets, you must apply the zero-trust approach to network security.



In response, a next-generation firewall (NGFW) not only incorporates traditional firewall capabilities (such as access control lists (ACL) and filtering based on ports and IP addresses) but also includes core security features (also known as unified threat management (UTM) profiles) and additional features to analyze the application layer, as shown on this slide.

This comprehensive approach allows NGFWs to provide more in-depth visibility and control over network traffic, enhancing protection against sophisticated cyberthreats by examining the actual content of data packets.



The Fortinet enterprise firewall solution encompasses a wide range of features and concepts.

In this diagram, you can see the enterprise infrastructure with the single, unified management console—FortiManager can manipulate and control all devices in the network.

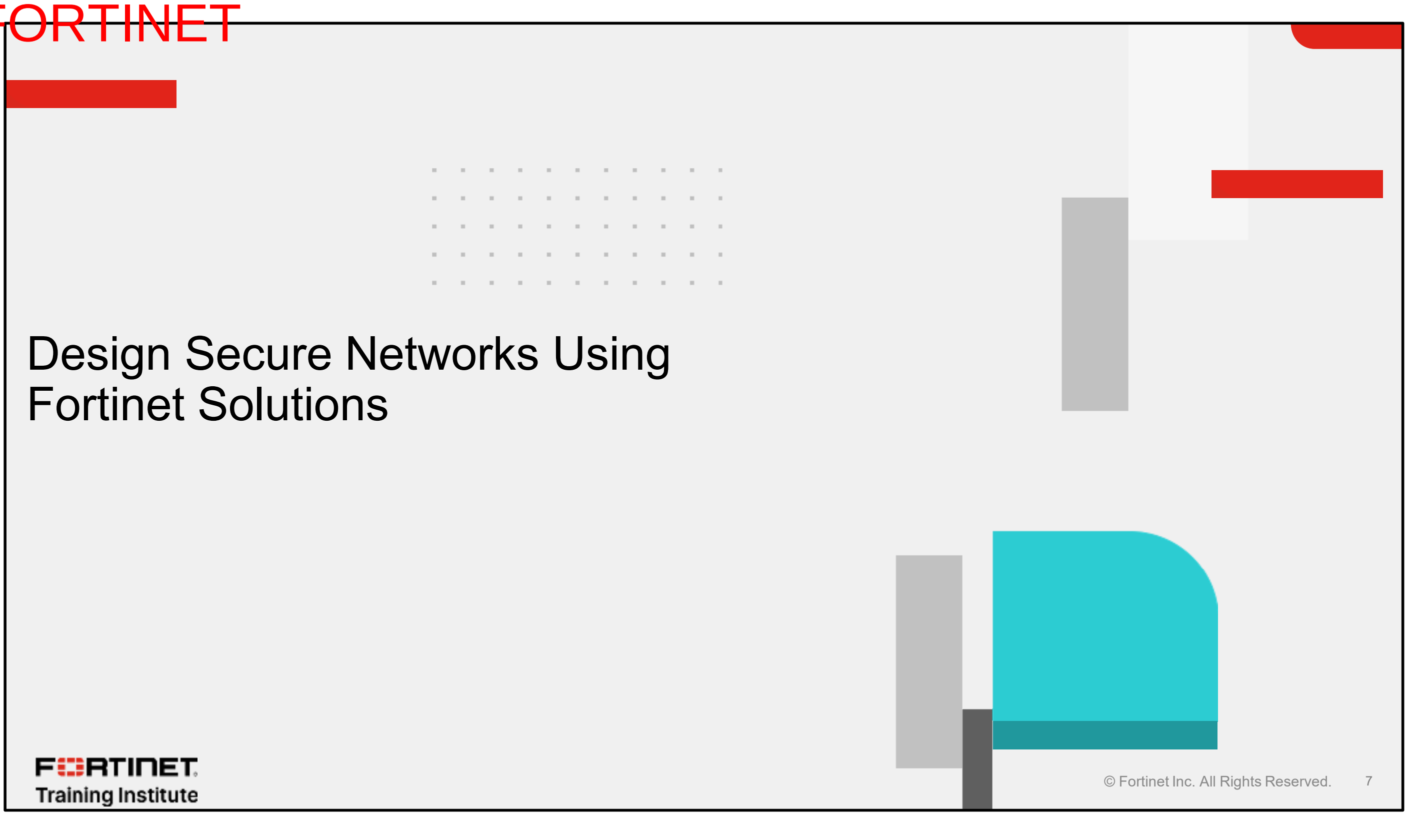
Fortinet offers a broad spectrum of devices, from the compact FortiGate 30-90 series dedicated to branch sites—sometimes known as Fortinet distributed enterprise firewall (DEFW) —to the FortiGate 100-900 series for campuses, also known as Fortinet internal segmentation firewalls (ISFW), and the Fortinet data center firewalls (DCFW) with the FortiGate 1000-7000 series. Additionally, virtual devices, like FortiGate VMs, are provided for private and public environments, as well as the FortiGate Cloud Native Firewall (FortiCNF).

The devices deliver a variety of solutions such as firewalls, segmentation, VPN, NGFWs, distributed NGFWs, integrated LANs, integrated WANs, 5G, LTE, hardware SSL inspection, secure SD-WAN, and zero trust network access (ZTNA). They also address several issues such as DDoS attacks and low latency.

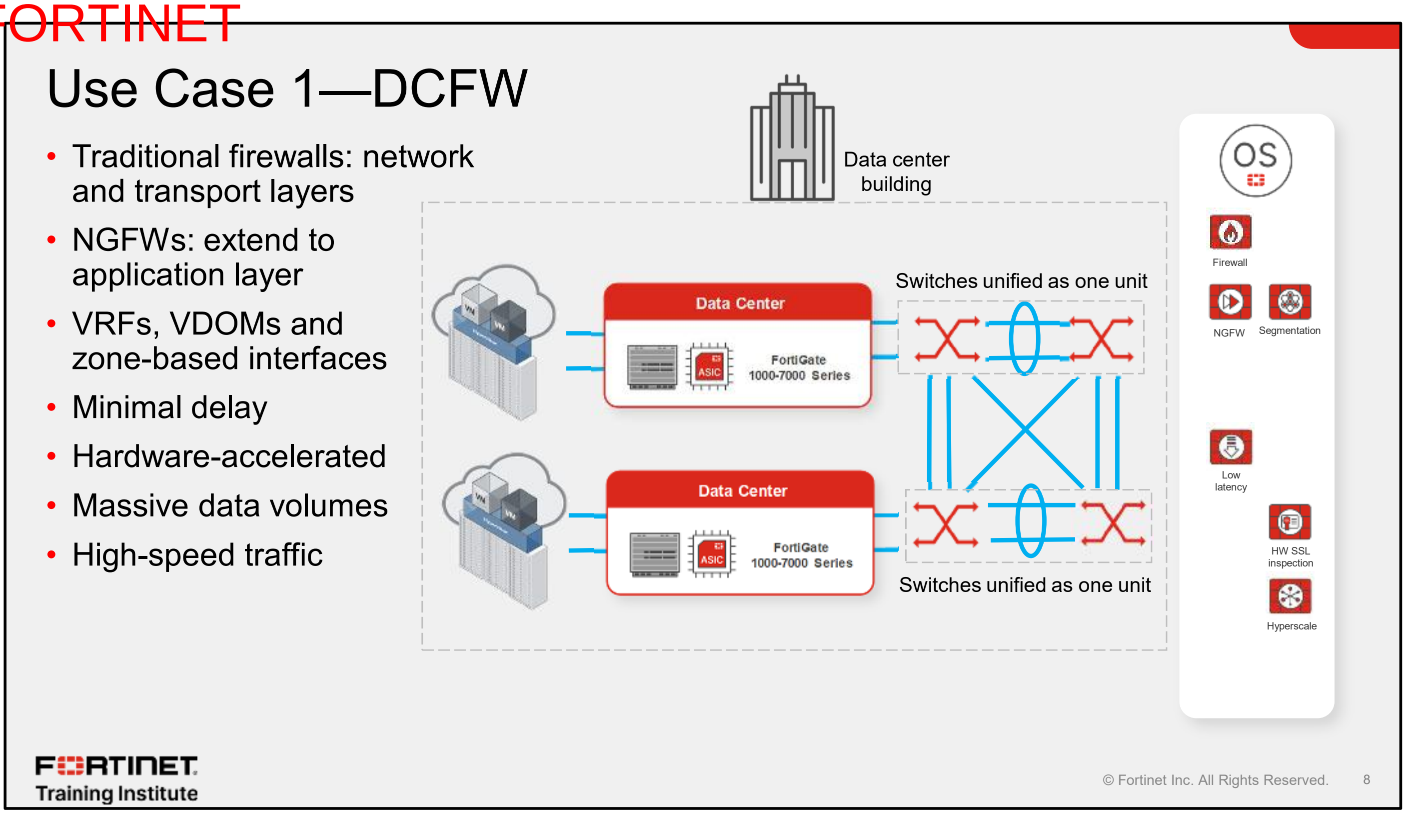
A key feature of the solution is the connection to FortiGuard Labs AI-powered threat intelligence, which leverages AI and machine learning (ML) algorithms to analyze and identify potential threats in real time.

Some examples of where AI and ML is used include SSL inspection, antimalware, malicious file detection, blocking malicious URLs, and preventing data exfiltration.

FortiGuard has an extensive database, which is not limited to what is shown on this slide. To see more details about FortiGuard databases, visit `https://www.fortiguard.com`.



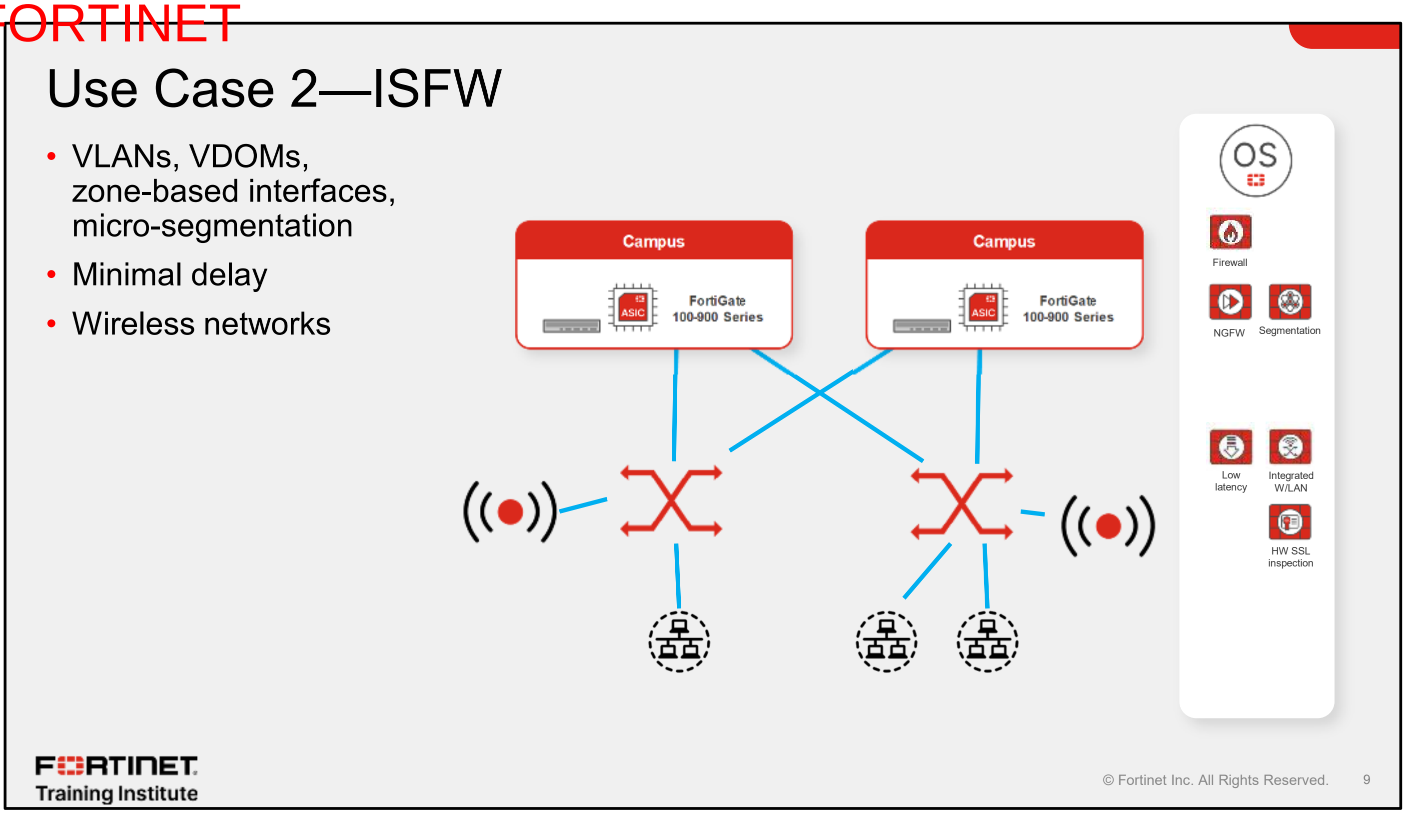
In this section, you will learn about designing secure networks using Fortinet solutions at a high level.



Fortinet DCFWs have the following characteristics:

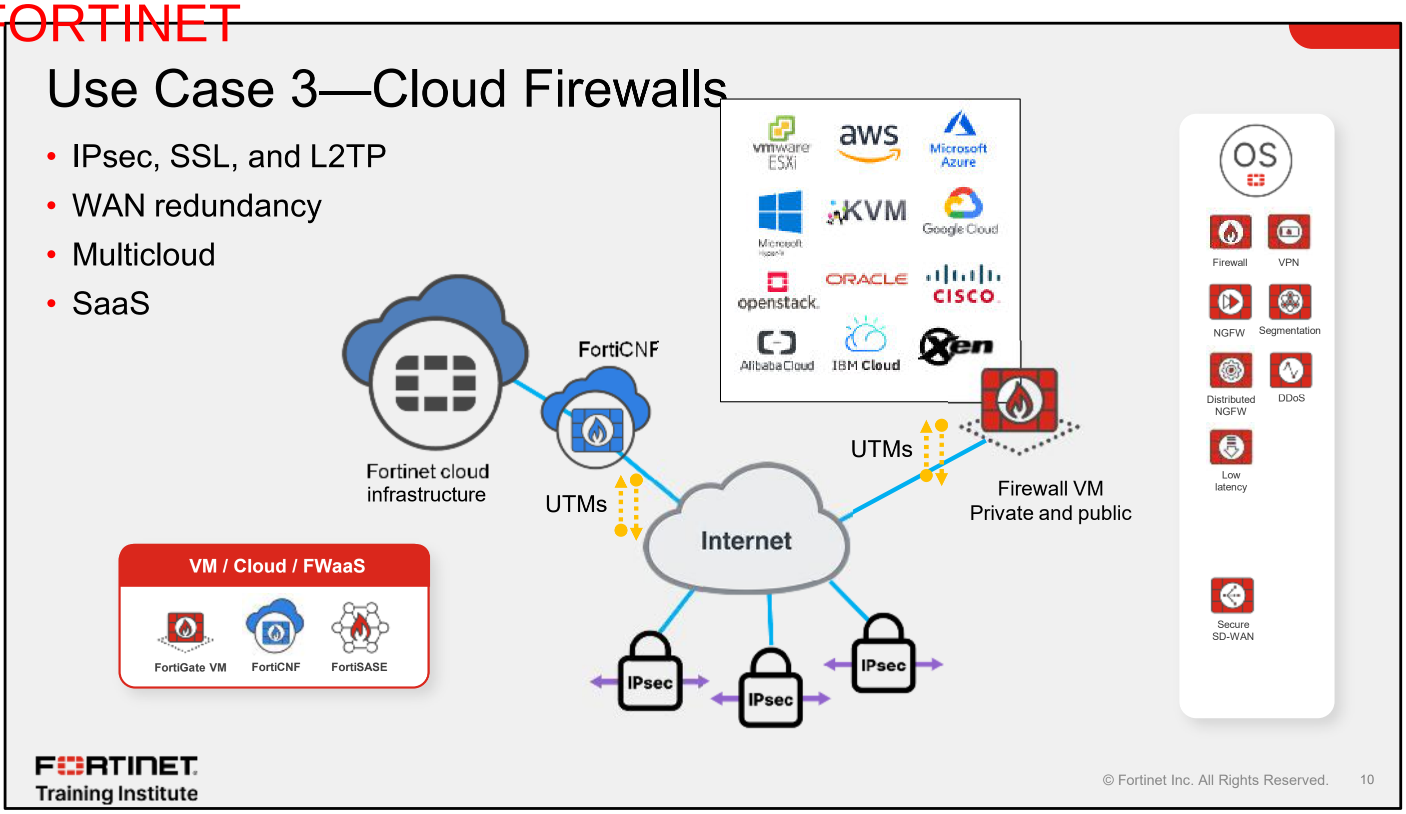
- A traditional firewall filters traffic based on IP addresses and ports, operating at the network and transport layers of the OSI model.
- A NGFW includes additional features such as application awareness, intrusion prevention, and advanced threat protection, operating at the application layer.
- Segmentation divides a network into smaller, isolated segments using methods such as virtual routing and forwarding (VRF) and VDOMs. Additionally, zone-based segmentation groups interfaces assigned into security zones for consistent policies.
- Low latency ensures minimal delay in data processing and transmission, crucial for real-time applications.
- Hardware SSL inspection uses specialized security processors to offload tasks from the main CPU, enhancing encryption, decryption, and packet inspection speeds, thereby managing more traffic and improving network performance and scalability.
- Hyperscale can scale to handle massive data volumes and high-speed traffic in data center firewalls, which typically support network speeds from 10 Gbps to 1000 Gbps.

Typically, these firewalls connect to multiple physical switches, combining them into a single logical interface for redundancy and high-speed data transfer by bundling multiple physical interfaces.



Fortinet ISFWs have the following characteristics:

- Traditional firewall
- NGFW
- Segmentation
- Low latency
- Hardware SSL inspection
- Fortinet's integrated wireless LAN in Fortinet enables FortiGate firewalls to manage FortiAPs directly, ensuring seamless and secure connectivity across wired and wireless networks. This solution spans FortiGate firewalls with wireless controllers, FortiAP access points for secure wireless connections, and FortiWLC controllers for large-scale wireless deployments.



Fortinet cloud firewall solutions have the following characteristics:

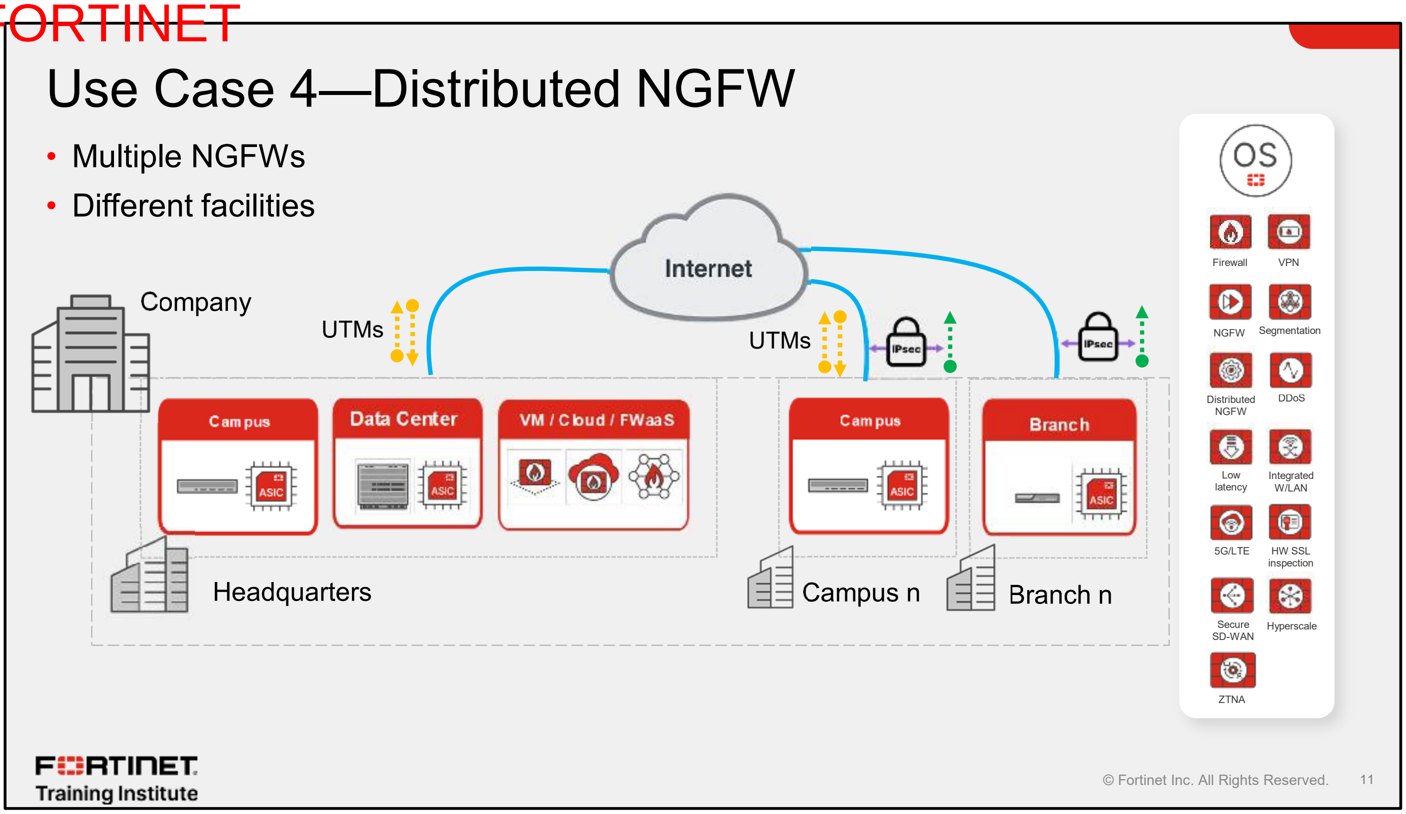
- Traditional firewall
- NGFW
- Segmentation
- DDoS
- Low Latency
- VPNs for secure and encrypted remote access and site-to-site connectivity using protocols like IPSec, SSL, and L2TP.
- Secure SD-WAN that combines advanced SD-WAN capabilities with integrated security features to optimize network performance and ensure secure, reliable connectivity across multiple locations and multiple internet connections within each location.

Virtual firewalls have a wide range of options.

Fortinet offers a diverse range of FortiGate virtual firewalls that are compatible with both private and public cloud platforms, including VMware ESXi, AWS, Microsoft Azure, Hyper-V, KVM, Google Cloud, Openstack, Oracle, Cisco, Alibaba Cloud, IBM Cloud, and Xen.

Additionally, FortiGate Cloud-Native Firewall (Fortinet CNF) provides robust security within the Fortinet cloud infrastructure. Fortinet's cloud infrastructure offerings include, besides FortiGate VM for network security, FortiWeb Cloud for web application protection, FortiCWP for workload security, FortiCASB for cloud application control, FortiSandbox Cloud for threat analysis, FortiMail Cloud for email security, and FortiManager Cloud and FortiAnalyzer Cloud for centralized management and analytics.

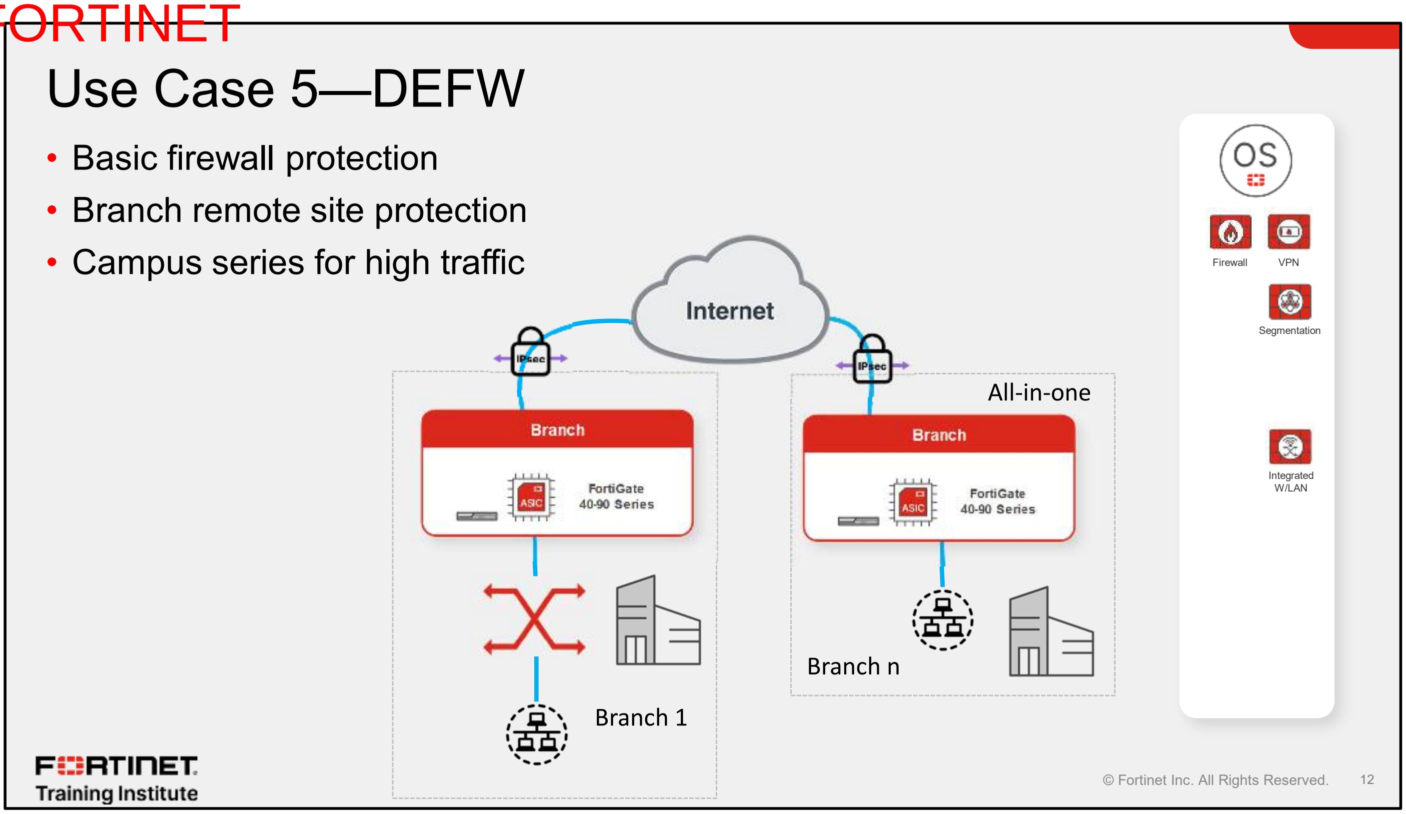
And finally, but not least, you can use FortiSASE (Secure Access Service Edge), which is a SaaS (Software-as-a-Service) solution that delivers security services from the cloud to protect users, devices, and data in distributed environments.



Fortinet Distributed NGFW has the following characteristics:

- Traditional firewall
- VPNs
- NGFW
- Segmentation
- Distributed NGFW
- Low latency
- Integrated wireless LAN
- HW SSL inspection (on-premise devices)
- Secure SD-WAN
- Hyperscale
- DDoS protection to detect and mitigate malicious traffic aimed at overwhelming network resources, ensuring continuous availability and service
- 5G/LTE for secure wireless WAN connectivity for remote and mobile deployments, supported by specific models and the FortiExtender series support this feature
- ZTNA to enforce strict identity verification for network access, adhering to the "never trust, always verify" principle. Key elements include FortiGate, FortiClient, FortiAuthenticator, FortiToken, and FortiNAC

This topology is commonly used in large organizations with various facilities such as buildings, sites, schools, branches, or factories. In these scenarios, a single NGFW falls short. Instead, a more extensive system is necessary to ensure comprehensive security across all company locations.

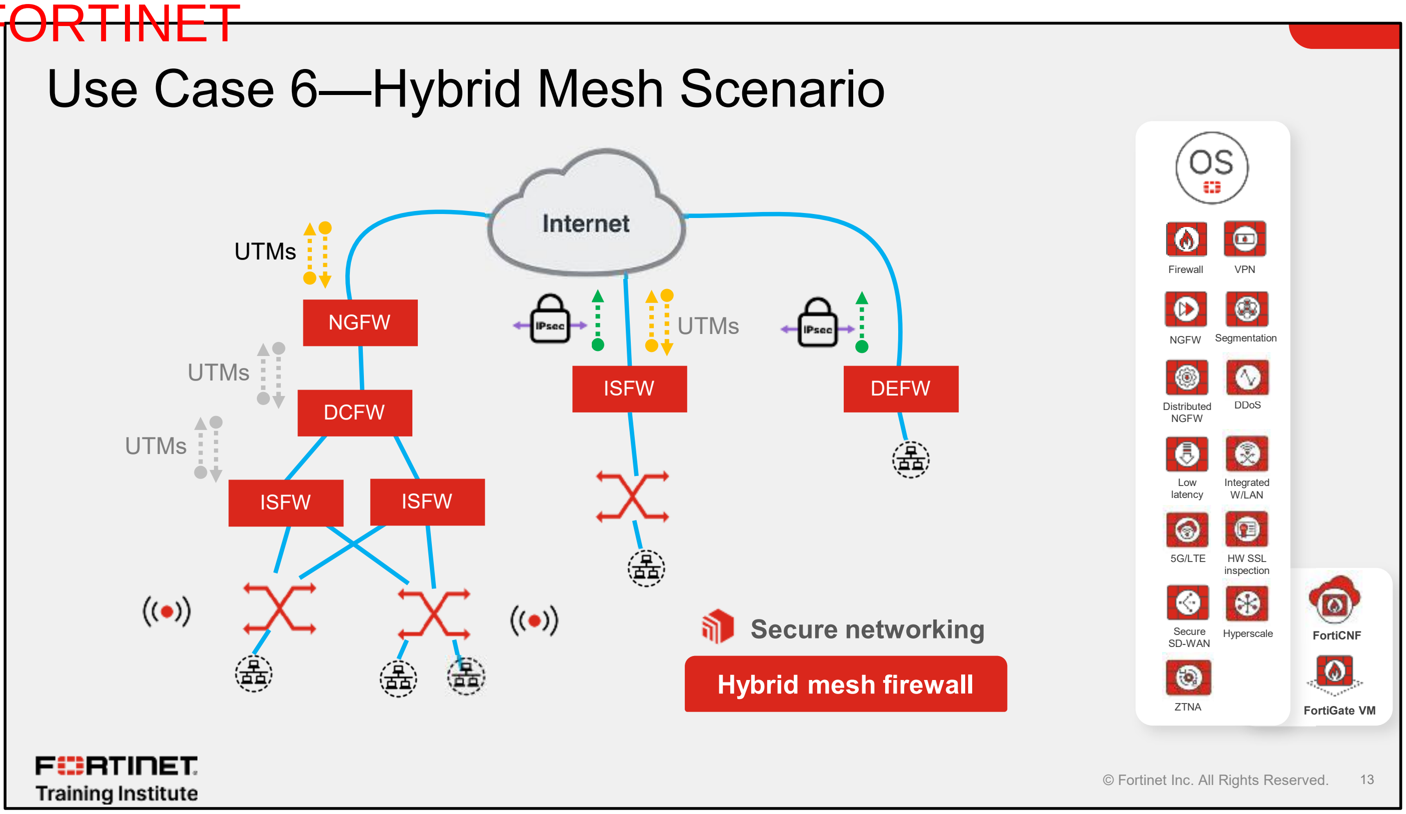


Fortinet DEFW has the following characteristics:

- Traditional firewall
- VPNs
- Segmentation
- Integrated wireless LAN

DEFWs provide basic firewall protection for remote sites or branches. Although not designed for in-depth security analysis, they primarily support small office connections to a central site, often using an IPsec VPN to connect to the main firewall.

Be aware that if a branch requires more than basic security analysis for your small site, you might need to consider larger models, such as those in the 100-900 Series typically used in campus environments. This decision depends on your specific needs and the size of your network.



The hybrid mesh scenario, which is common, includes a collection of various firewall models.

Hybrid mesh firewall secure networking distributes features such as internal segmentation, data centers, virtual NGFW, and D-NGFWs, across firewalls. This method is extremely effective at addressing the challenges presented by the complex network and security infrastructures of today.

The role of a firewall does not necessarily match its model. For instance, while 100-900 series is considered an ISFW or a campus firewall, using it with security features or a UTM profile categorizes it as a NGFW. The technology can adapt to your needs, so it's crucial to understand all the features that Fortinet devices offer.

DCFW vs. ISFW vs. NGFW vs. DEFW

Acronym	Average throughput	Models	Firewall type	Purpose	Features
DCFW	10-1000 Gbps	FortiGate 1000-7000 series	Data center	High-speed network protection	Firewall, NGFW, segmentation, HW SSL inspection, low latency, hyperscale
ISFW	1-100 Gbps	FortiGate 100-900 series	Campus	Network microsegmentation	Firewall, NGFW, segmentation, low latency, integrated W/LAN, HW SSL inspection
NGFW	1-40 Gbps	FortiGate 100-7000 series	VM / Cloud / FWaaS Data center Campus	Multilayered security	Firewall, VPN, D-NGFW, segmentation, DDoS, low latency, integrated wireless LAN, 5G/LTE, HW SSL inspection, secure SD-WAN, hyperscale, ZTNA
DEFW	Up to 1 Gbps	FortiGate 30-90 series	Branch	Remote site protection	Firewall, VPN, segmentation, HW SSL inspection, integrated wireless LAN

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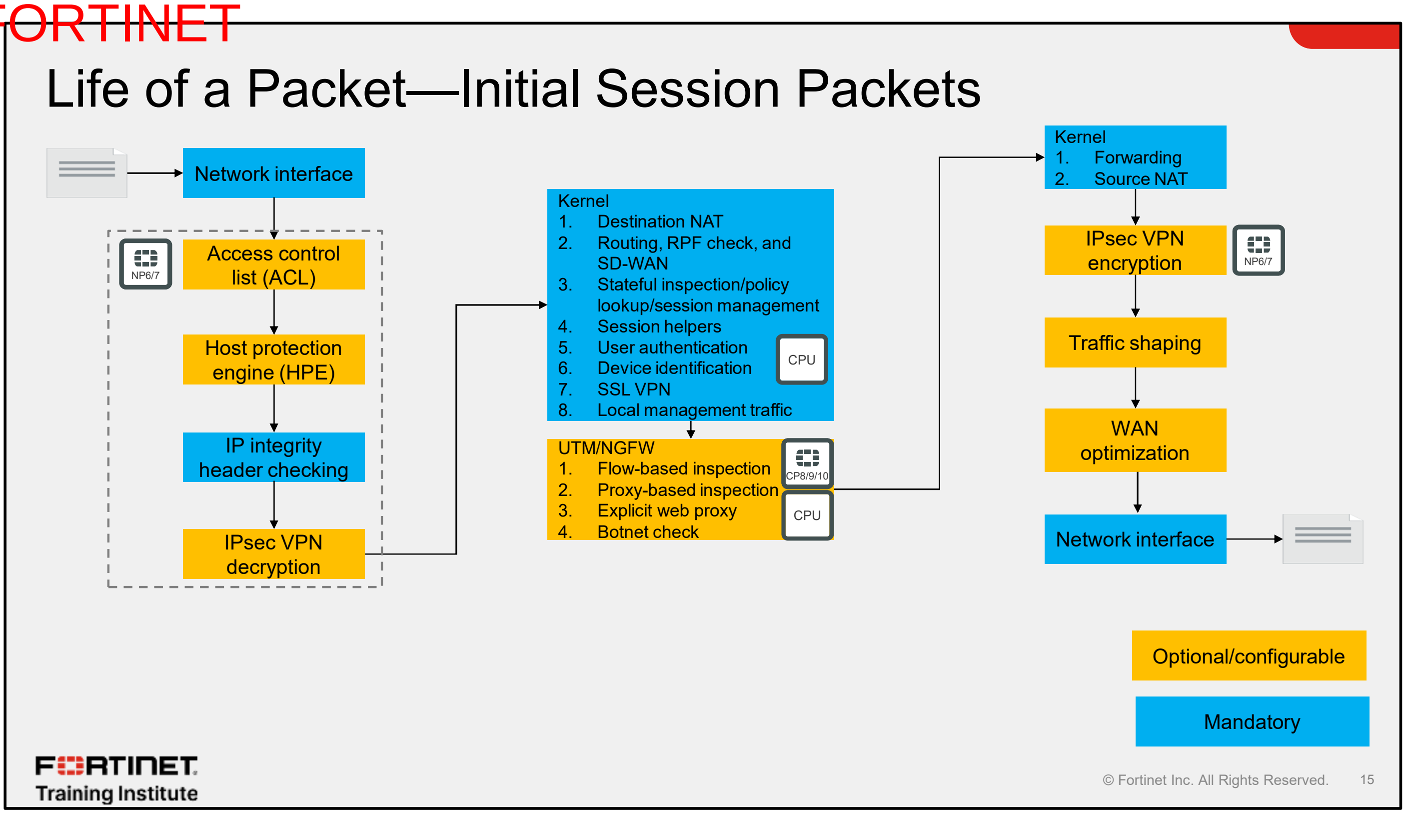
This slide shows a summary of the main differences between DCFWs, ISFWs, NGFWs, and DEFWs.



This slide shows the steps that the first packets of a session go through as they enter, pass though, and exit FortiGate.

FortiGate first performs some security inspections such as ACL, HPE, and IP integrity header checking, to make sure the packets are within acceptable parameters before allowing them to move through the rest of the processes. These inspections are handled by the network processor (NP) to minimize impact on the FortiGate CPU.

Each version of the NP has criteria that define which traffic can be offloaded. The NP enhances overall performance by allowing offloaded sessions to bypass the FortiGate CPU after the session is established and the session key is installed in the NP. The NP can also handle IPsec VPN encryption and decryption operations, where the configured encryption and hashing algorithms are supported in the hardware.

The content processor (CP) functions like a coprocessor for the FortiGate CPU to improve overall system performance by offloading certain tasks, such as pattern matching for flow-based UTM inspection with the intrusion prevention system (IPS) engine, SSL/TLS decryption and encryption for deep SSL inspection, and IPsec encryption and decryption operations for supported algorithms.

Note that the packet processing for virtual FortiGate devices is identical with the only difference being that the CPU handles all processes instead of being able to offload some of them to NPs and CPs.

Review

- ✓ Understand the enterprise firewall solution
- ✓ Design secure networks using Fortinet solutions

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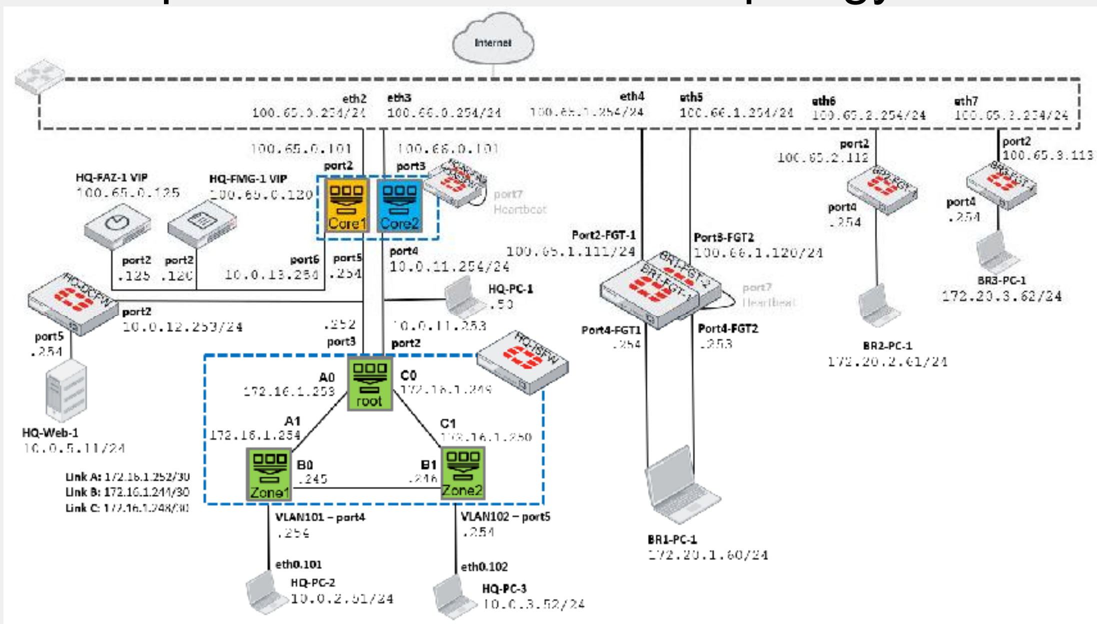
This slide shows the objectives that you covered in this lesson.

By mastering the objectives covered in this lesson, you learned about the enterprise firewall solution and the network security reference architecture. You also learned about the roles of firewalls and their placement in an enterprise network.

Lab—Enterprise Firewall Network Topology



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This slide shows the enterprise network topology used in the labs.

In this lesson, you will learn about central management.

Objectives

- Examine FortiManager architecture
- Explore dynamic objects on FortiManager
- Use scripts on FortiManager

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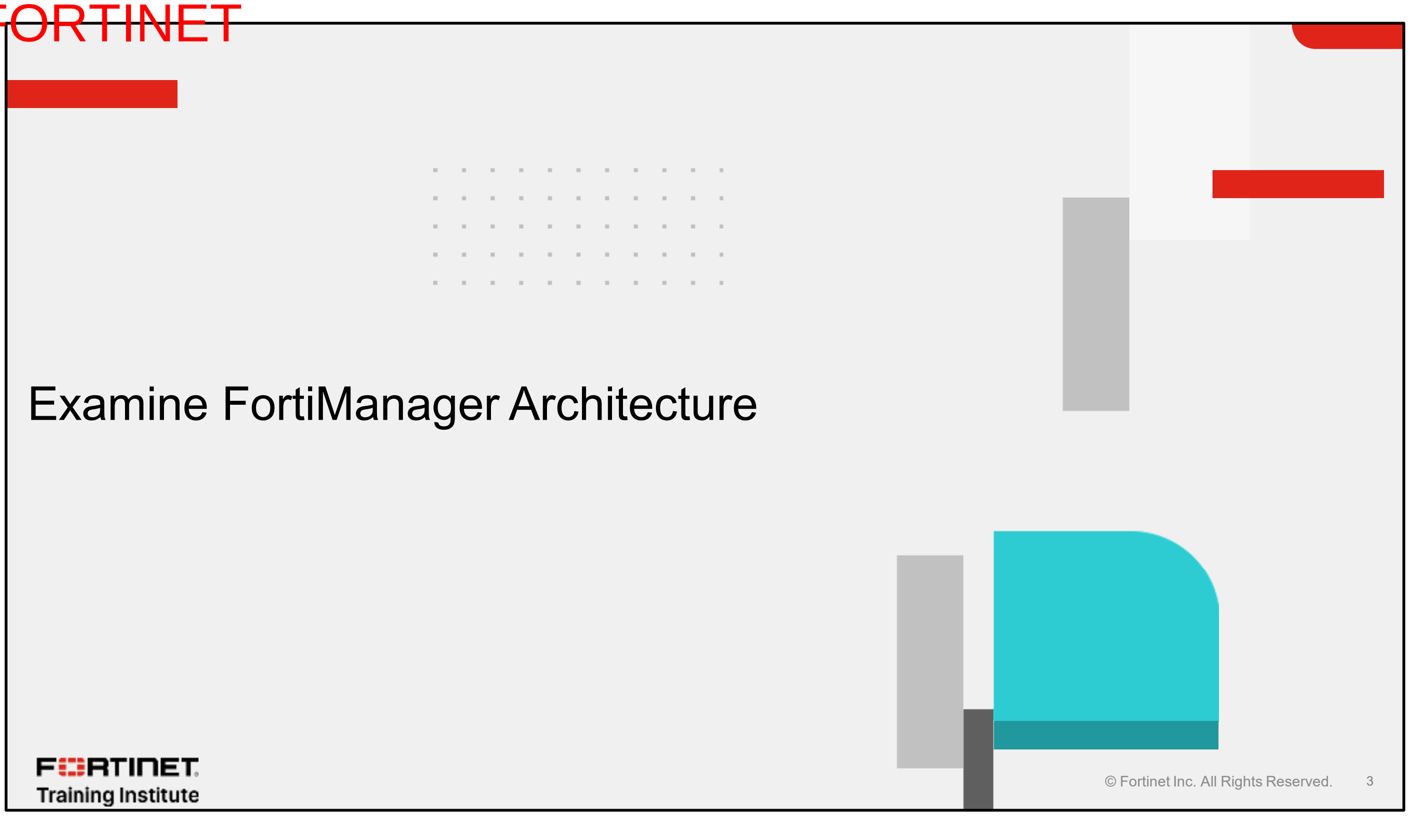
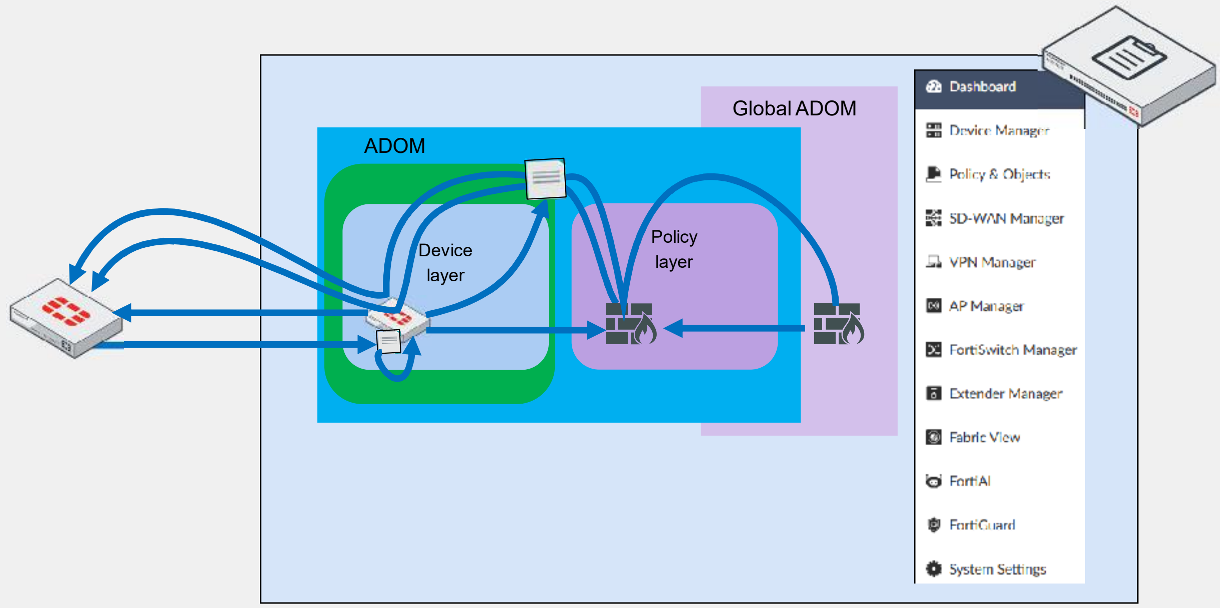
After completing this lesson, you should be able to achieve the objectives shown on this slide.



In this section, you will learn about central management using the architecture of FortiManager.

Exploring FortiManager Core Functions



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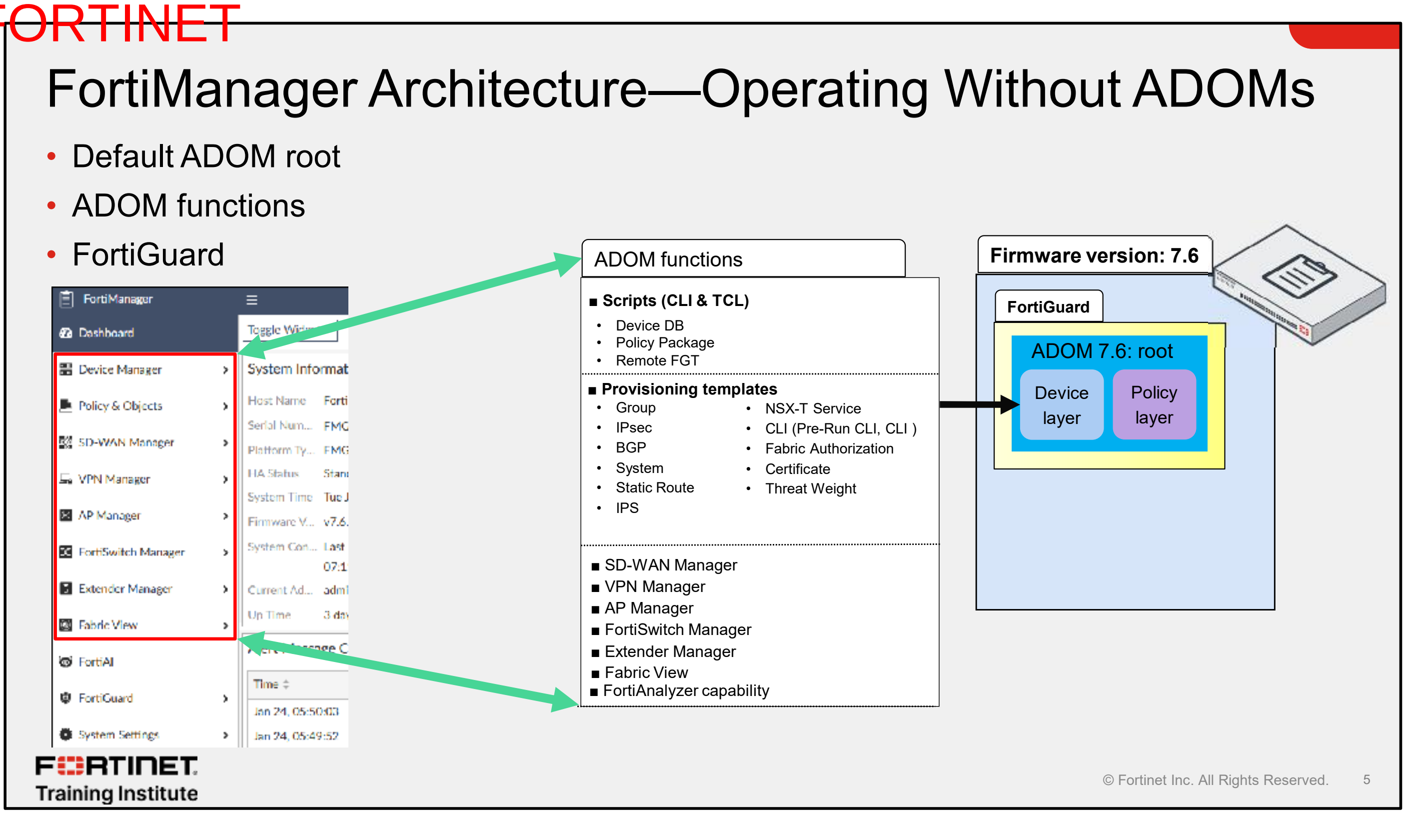
FortiManager is a centralized network management solution featuring ADOMs, device databases, policy layers, and provisioning templates.

It offers global ADOM capabilities and additional features to streamline configuration and administration across multiple Fortinet devices.

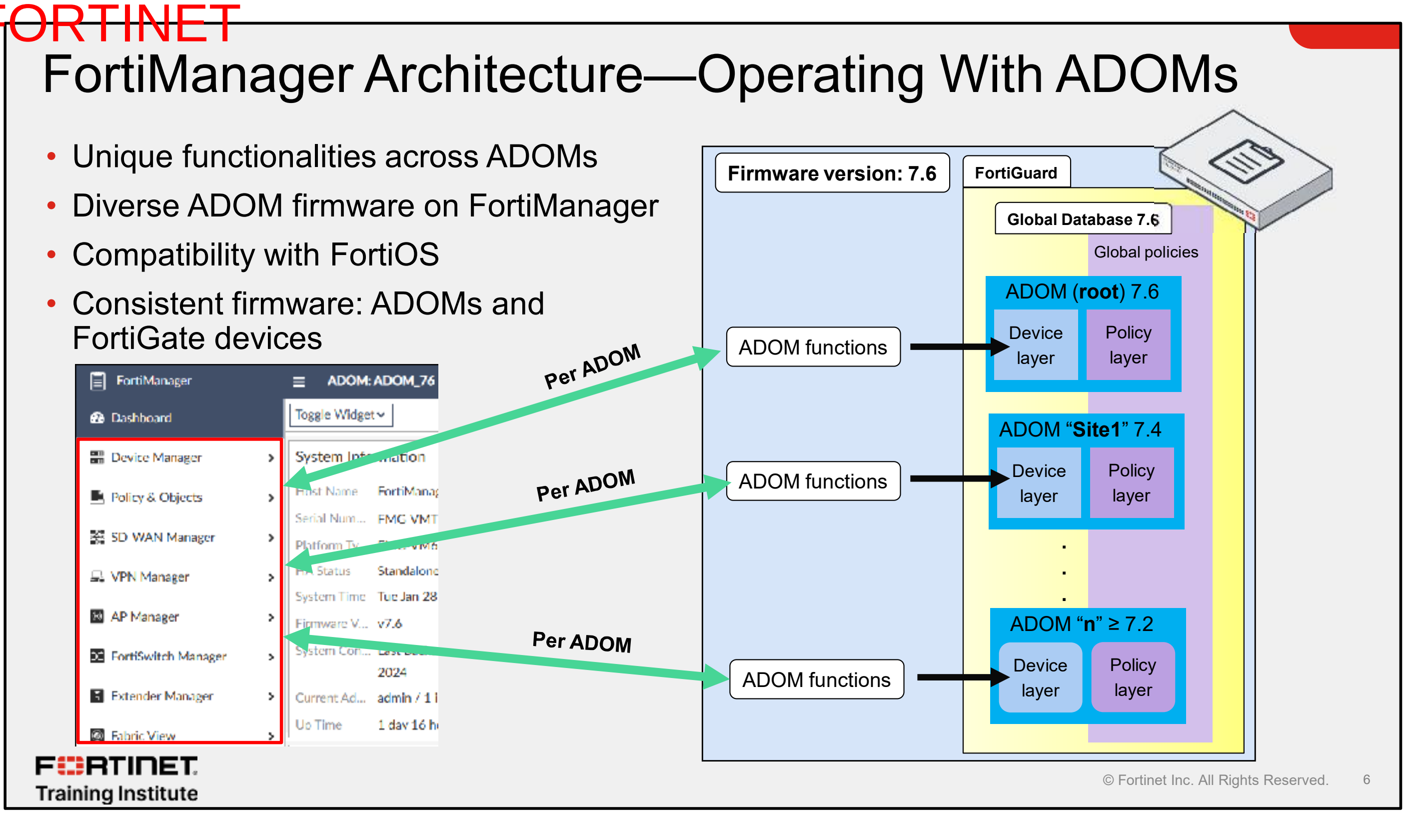


FortiManager allows you to enable multiple virtual instances known as ADOMs.

When ADOMs aren't enabled, you still work in a default ADOM named root, which contains two databases:

- Device layer
- Policy layer

It's important to note that the root ADOM has its own functions, as shown on the slide. Notably, you can see that the FortiGuard database serves the root ADOM.



This slide shows the FortiManager architecture with ADOMs enabled.

Each ADOM has unique functions, such as scripts, provisioning templates, SD-WAN Manager, VPN Manager, AP Manager, FortiSwitch Manager, Extender Manager, Fabric View, and FortiAnalyzer capability.

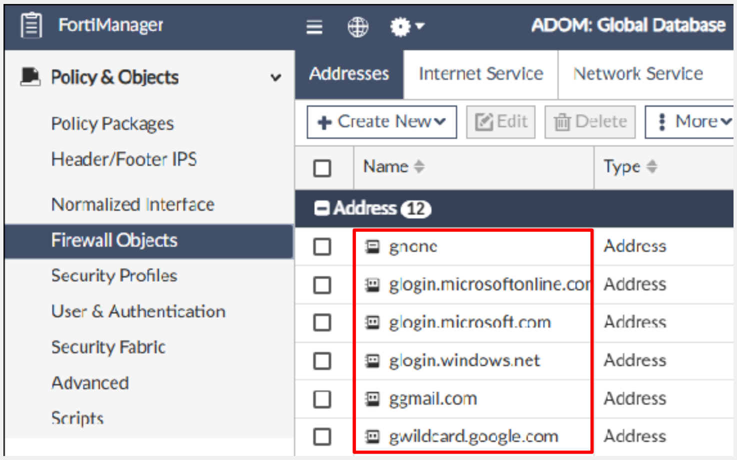
The FortiGuard database serves all ADOMs (7.6, 7.4, 7.2, and the global database).

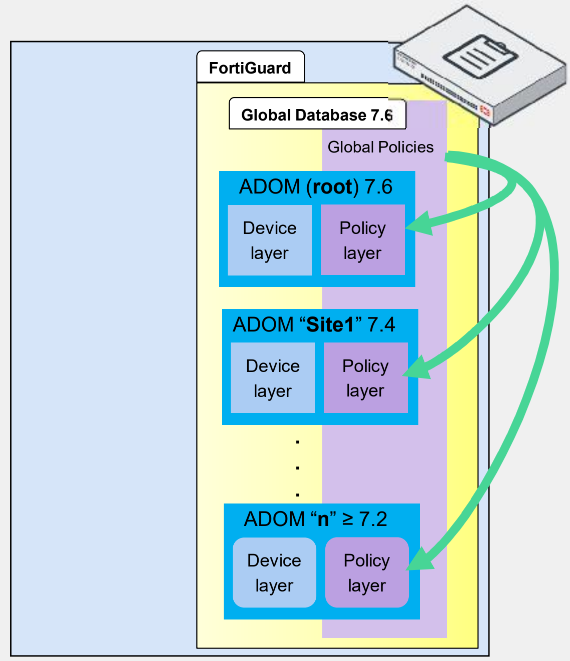
Compatibility with FortiOS is crucial for both ADOMs and FortiGate devices.

Mixing FortiGate versions in ADOMs is permitted, especially for migration purposes, but you should use matching firmware for ADOMs and FortiGate devices.

Global Database

- Global-to-local ADOM compatibility
- Reserved prefix g





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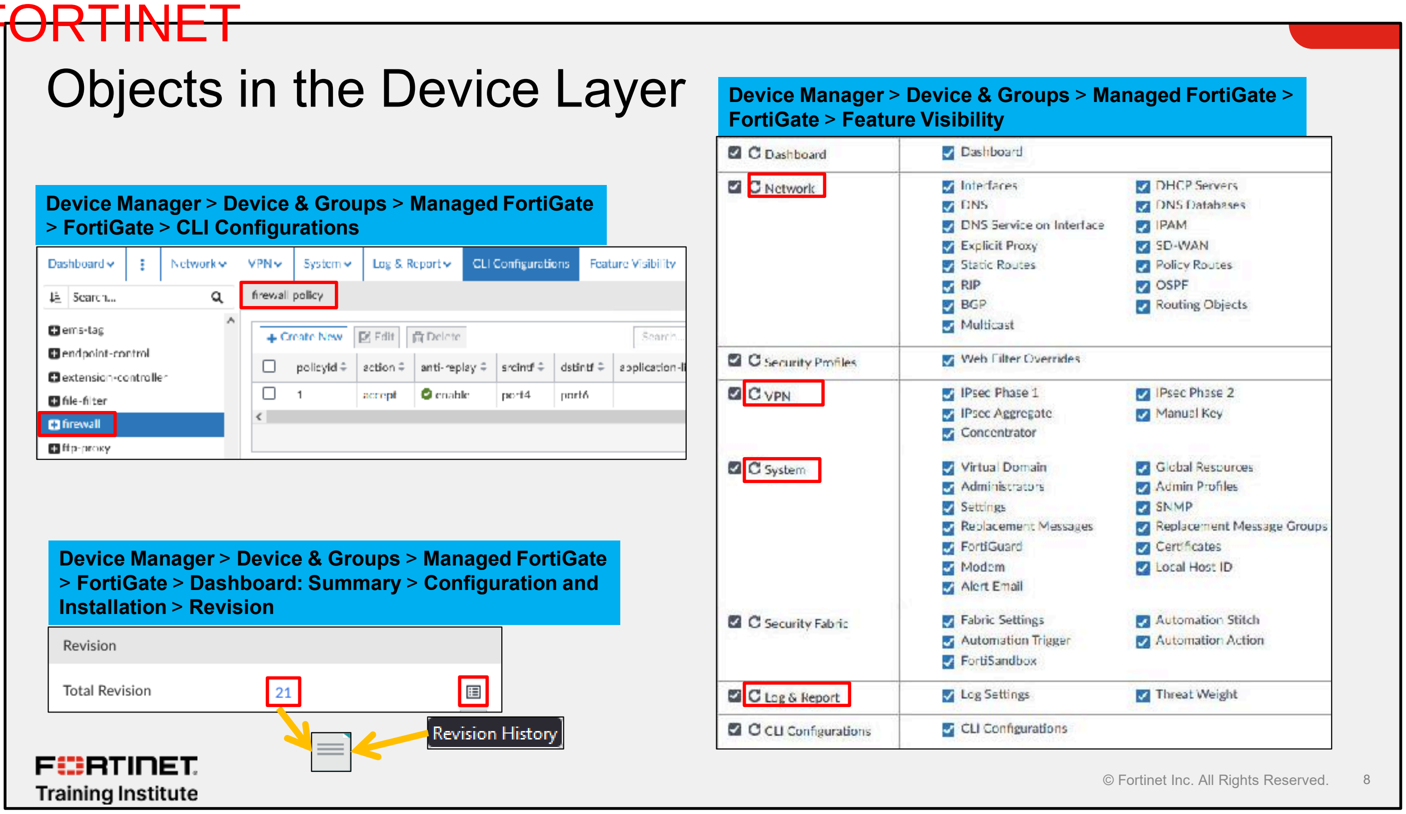
The global database stores common objects and policies for use across multiple ADOMs, enabling the creation and management of global security policies, objects, and configurations that can be simultaneously applied to several ADOMs.

The global ADOM and local ADOMs are compatible with ADOMs at the same version and two versions earlier. For example, a global ADOM at version 7.6 supports ADOMs at both versions 7.2 and 7.4. You should ensure that firmware versions match for optimal synchronization.

Global objects and configurations on FortiManager are prefixed with the letter “g”, which is reserved to prevent the creation of custom objects with this prefix.

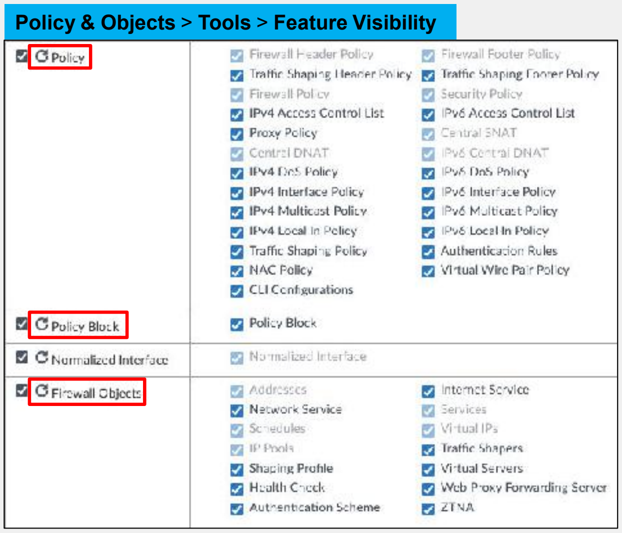


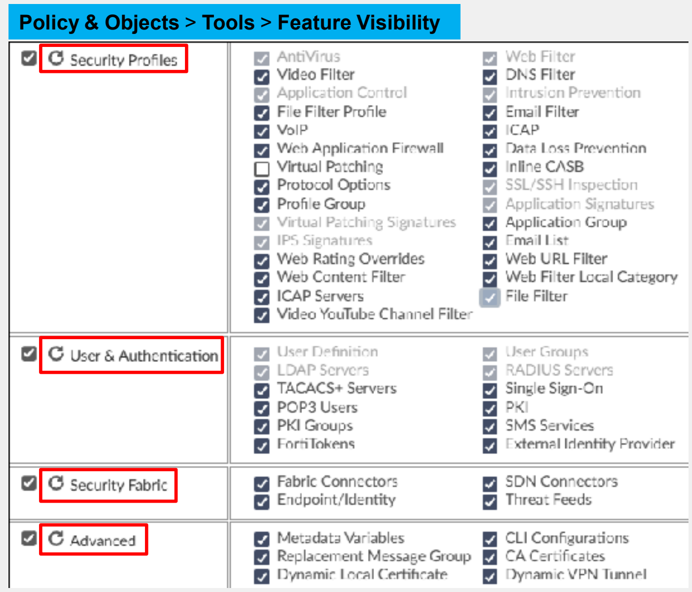
In the device layer, you configure multiple device settings.

Although it is possible to configure policies in the device layer, it is not recommended. It is best to restrict configuration to device settings, including **Network**, **VPN**, **System**, and **Log & Report**.

Additionally, each FortiGate you add to the device layer has its own device database, which is recorded in the revision history of each firewall.

Objects in the Policy Layer





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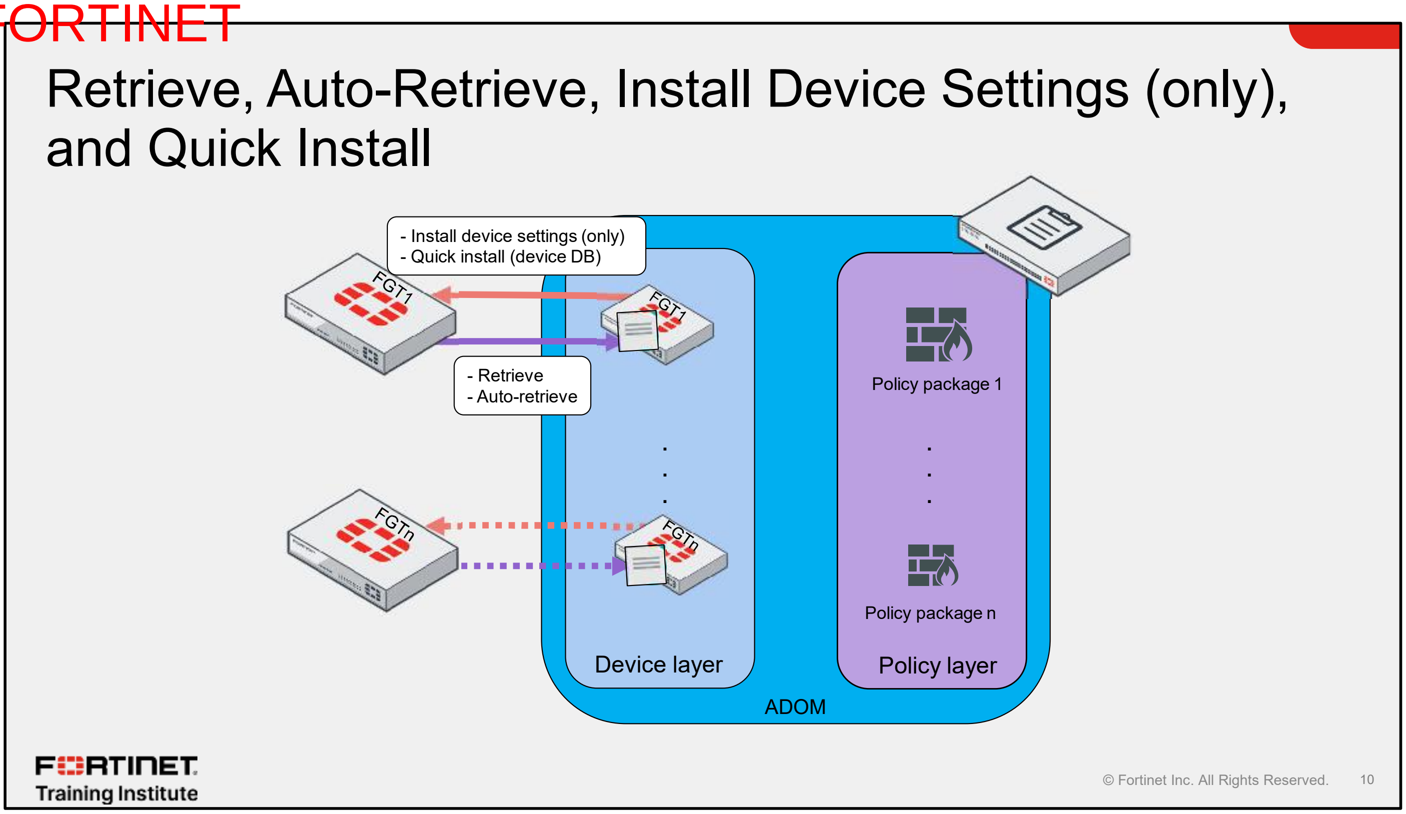
The policy layer is where you configure all objects related to firewall policies.

Configurations in the policy layer and the assignment of objects to a policy package overwrite existing settings in the selected revision history ID. Therefore, even though you can make configuration changes in the device layer, you should make them in the policy layer instead, to update the values in the device database accordingly.

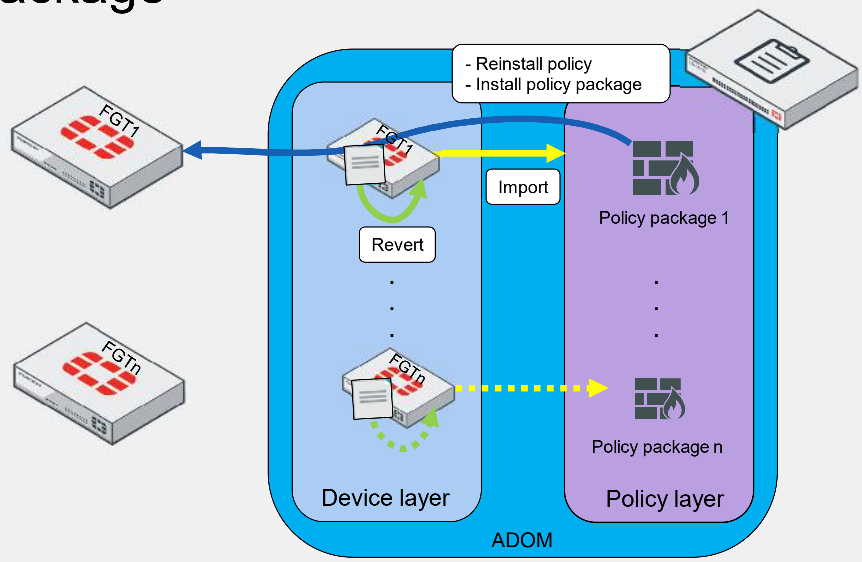
Changes in the device layer directly affect FortiGate, while the effects of changes made in the policy layer depend on the included policy package. Changes made in the policy layer are applied only after policy deployment and installation approval.



This slide outlines how FortiManager uses the device and policy layers for FortiGate configuration, highlighting key operations:

- **Retrieve**: captures the current FortiGate configuration and saves it as a new revision without updating the policy package.
- **Auto-retrieve**: is initiated when a FortiGate device is added to the device layer, capturing any changes made directly on the FortiGate into its revision history. Auto-retrieve doesn't update the policy package; an import operation is required for that.
- **Install device settings**: applies the device layer configuration to FortiGate, offering a preview before installation to review changes.
- **Quick install**: is similar to **Install device settings**, but without a preview. It applies the device layer configuration to FortiGate immediately. This operation is ideal for zero-touch provisioning (ZTP) or when you are confident about the changes.

Revert, Import, Install Policy Package, and Reinstall Policy Package



This slide outlines how FortiManager uses the device and policy layers for FortiGate configuration, highlighting key operations:

- **Revert**: chooses an earlier revision from the history for potential import or installation on FortiGate. To synchronize the device and policy layers afterward, importing is required.

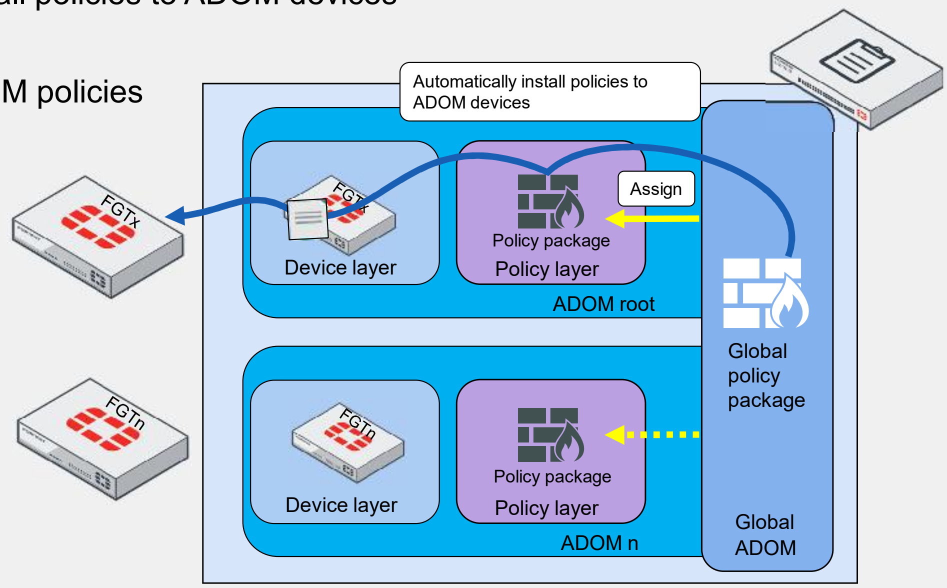
- **Import**: moves policies from a selected revision in the history into the policy package, which can then be renamed or left with the default name. Care is needed when importing to prevent conflicts, especially with objects sharing the same name in the policy layer.

- **Install policy package**: transfers the policy package from the policy layer to both the device layer and FortiGate, displaying a step-by-step wizard and installation preview with an option to cancel.

- **Reinstall policy**: bypasses the wizard because the FortiGate device and the policy package are already selected. It offers an installation preview with an option to cancel. Also, reinstall applies not only policy package settings, but also any modifications, directly to FortiGate in the device database.

Global Policy Package Installation

- Automatically install policies to ADOM devices
- Assign
- Share global ADOM policies



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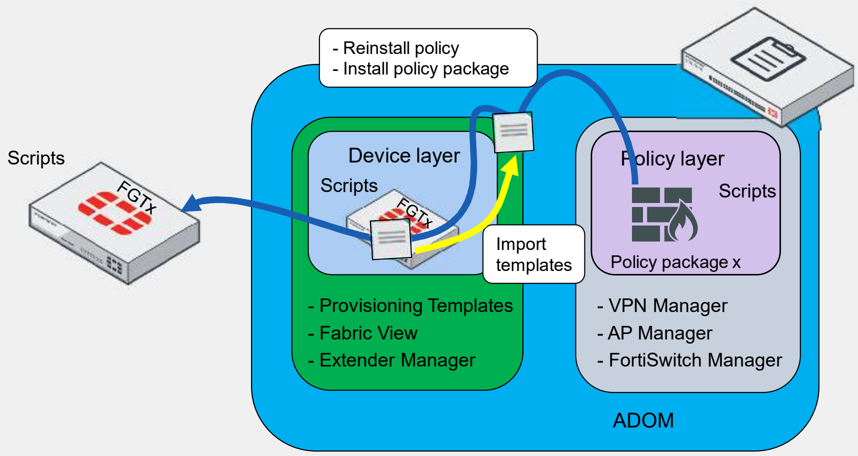
This slide explains the assignment and installation of global policy packages to FortiGate devices.

- **Automatically Install Policies to ADOM Devices**: The FortiManager global ADOM directly installs assigned policy packages on three targets: the policy package in the policy layer, the device database, and FortiGate. No additional steps are required for this installation. However, this method bypasses the installation preview and proceeds directly with the installation.
- **Assign**: The process starts at the FortiManager global ADOM and updates the policy layer, indicating a **modified** status. Next, you must install or reinstall the policies to apply them to the device layer and FortiGate.

It's important to note that the same global ADOM can distribute identical global policies to different ADOMs.

ADOM Capabilities—Device Layer or Policy Layer?

- Provisioning templates override device database
- Normalized interfaces: zones, VLANs, and SSIDs

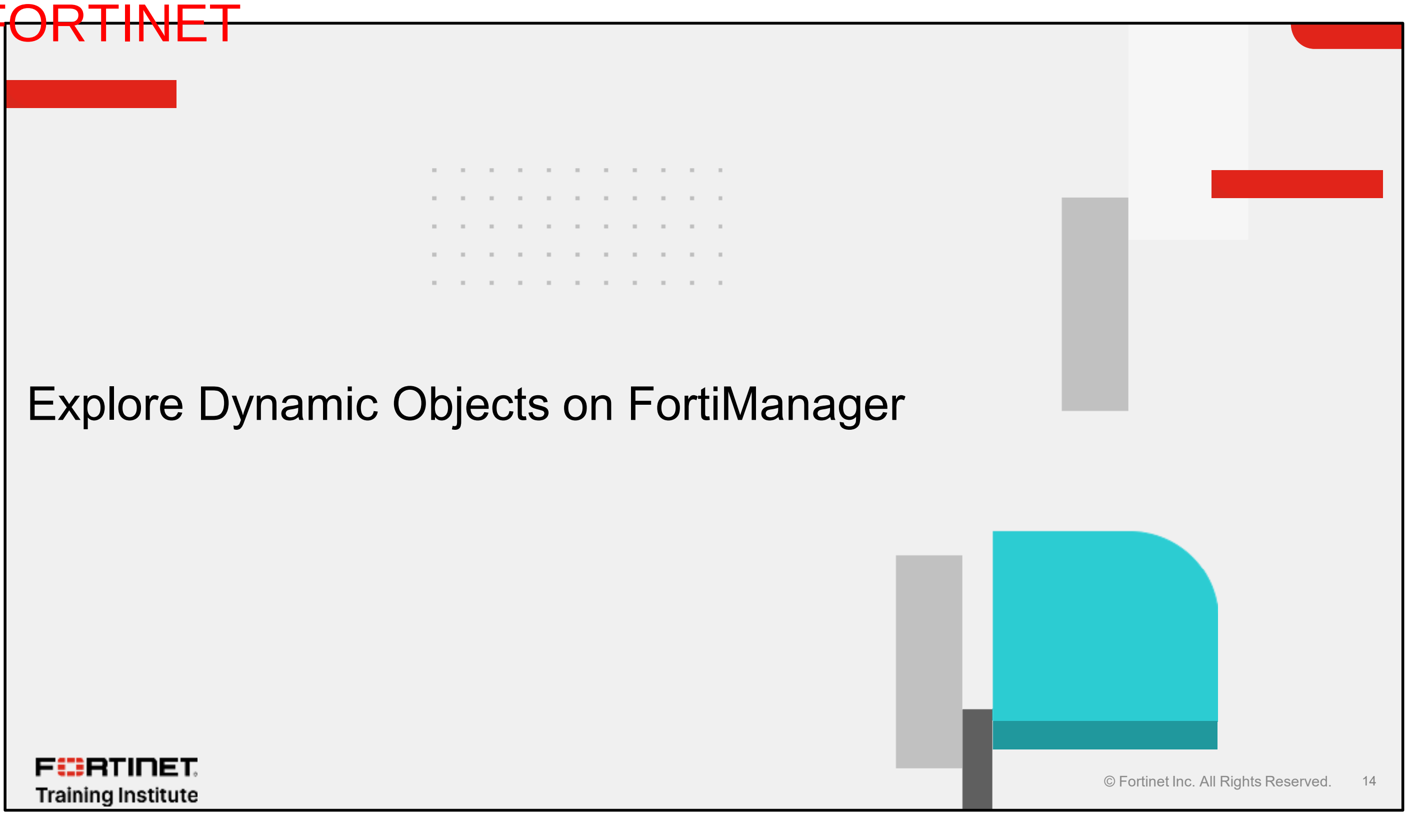


This slide presents the essential elements of ADOM functions and their connection to the device and policy layers.

The provisioning templates, fabric view, and Extender Manager are integral to the device layer configuration. Changes made here will automatically update the device layer database for each FortiGate device. Notably, provisioning templates have precedence and can supersede other device layer settings. After you configure these features, the standard device installation procedures apply.

The features related to the policy layer include VPN Manager, AP Manager, and FortiSwitch Manager. These features help to set up normalized interfaces such as zones, VLANs, or SSID interfaces. After configuring these features, the standard policy installation procedures apply, requiring either a reinstall policy or install policy package operation to apply changes to FortiGate devices.

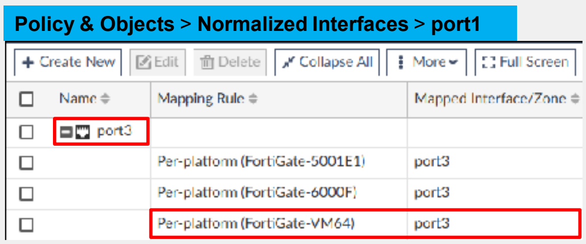
Lastly, you can run scripts using the CLI or Tcl directly on FortiGate, the device layer, or the policy packages. This is a streamlined method to create multiple objects without relying on the GUI.

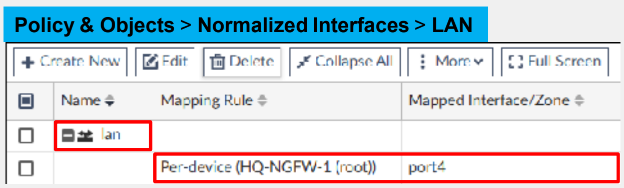


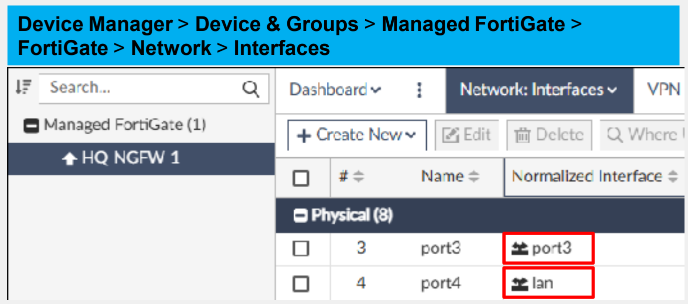
In this section, you will learn about dynamic objects on FortiManager.

Standardizing FortiGate Interfaces

- Easier management
- Per-platform mapping: consistency
- Per-device mapping: individual needs
- FortiManager-to-FortiGate mapping
- Avoiding naming confusion
- Case-sensitive normalized interfaces
- Dual-layer interface assignment







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Use the **Normalized Interfaces** feature to standardize configurations across devices for easier management and consistent settings.

There are two configuration methods:

- Per-platform mapping: applies uniform settings across similar device models.
- Per-device mapping: allows customized configurations for individual devices.

For example, port3 on a FortiGate VM could be standardized across similar models, while port4 might be customized to specific needs by linking to the LAN interface in device policies.

Using normalized interfaces does not automatically install the interface names assigned on FortiManager; for instance, **lan** defined on FortiManager becomes **port4** on FortiGate after you approve it in the install preview.

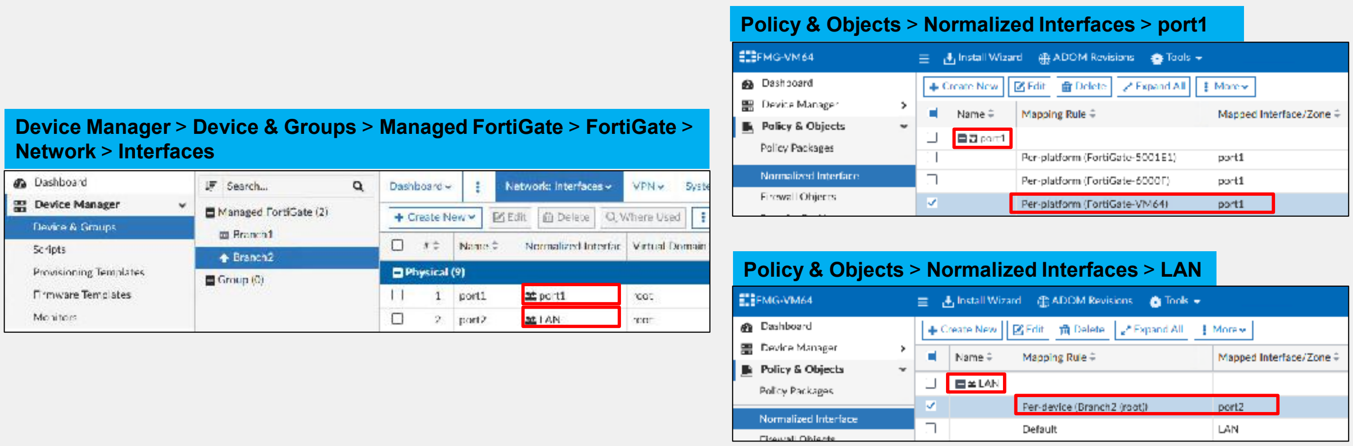
You should use consistent interface naming between normalized interfaces and per-device mapping to avoid confusion in large scenarios.

Remember that normalized interfaces are case sensitive; *LAN* and *lan* are not treated in the same way.

You can assign normalized interfaces in both the policy layer and the device layer.

Standardizing FortiGate Interfaces (Contd)

- FortiGate interfaces in device database
- Normalized interfaces in policy layer



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In the example shown on the slide, **port1** is associated with **port1** in the device layer and is set as **Per-platform** for the FortiGate VM in the policy layer.

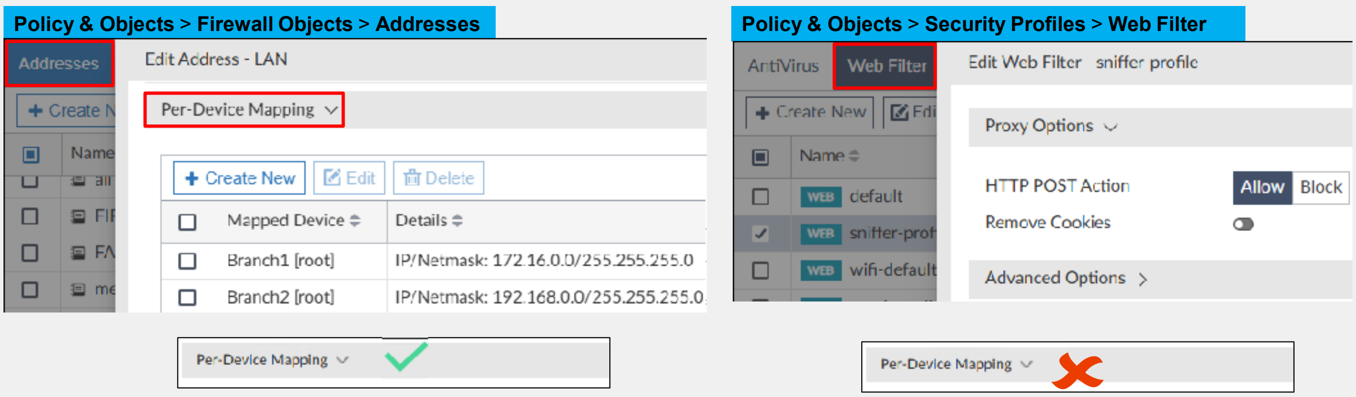
Similarly, **port2** is linked to the **LAN** normalized interface in the device layer. In the policy layer, the **LAN** interface is matched with **port2** as **Per-device**, meaning that whenever FortiManager references the **LAN** interface in policies for this firewall, it substitutes it with port2 based on the per-device mapping.

Key considerations include:

- Using a normalized interface to change an original firewall interface name means FortiManager will use the interface specified in per-device mapping. For example, it replaces **LAN** with **port2**.
- You should keep interface names consistent between the device layer and the policy layer. If necessary, per-device mapping allows standardized names for different ports across firewalls, but remember that FortiManager will replace these with the specified normalized interface in policies.

Dynamic Configurations on FortiManager

- Multiple dynamic object configurations available
- Not all objects support mappings



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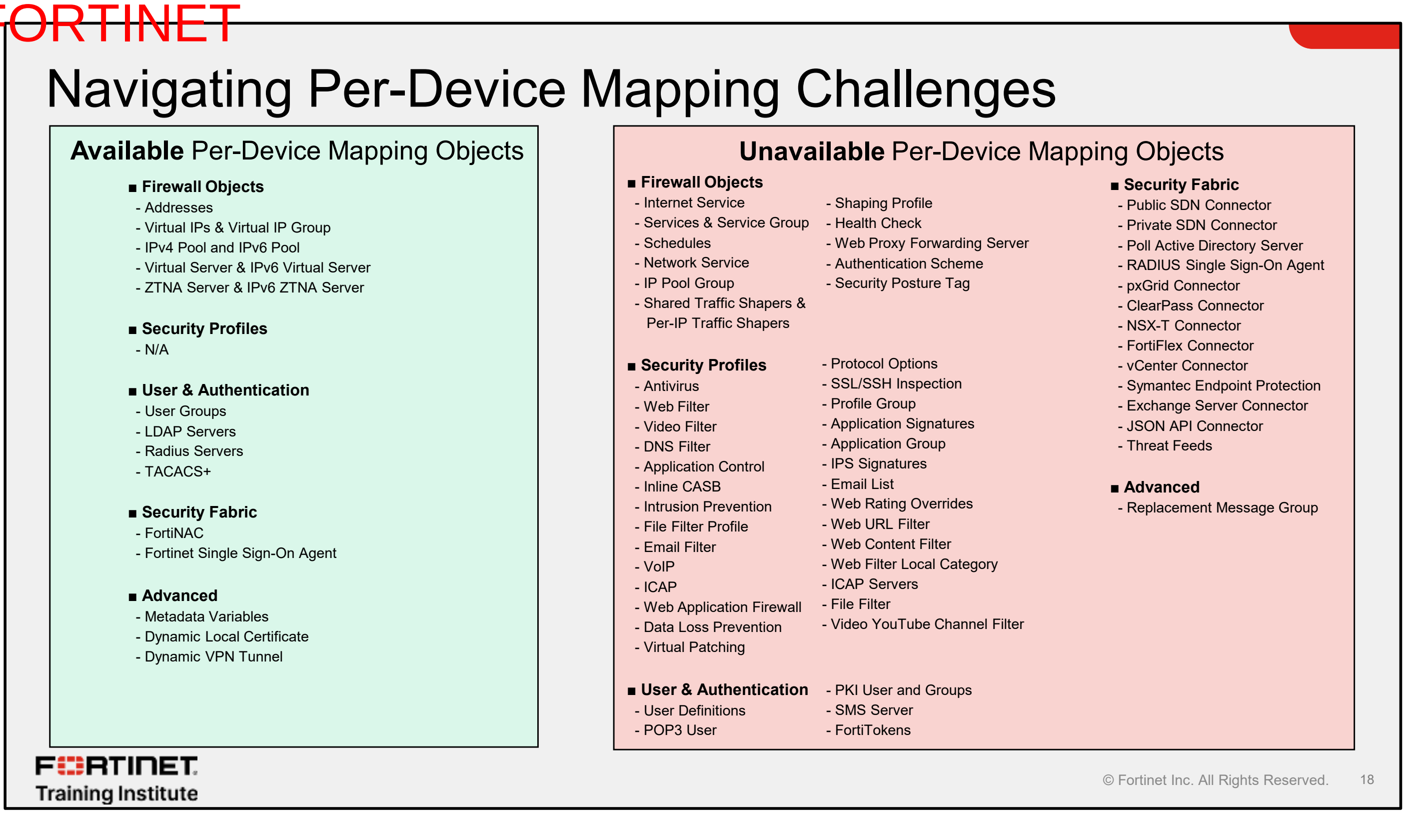
On FortiManager, you can use dynamic mapping to enable specific configurations for each device, applied in:

- Interface mapping (as previously described)
- Object configurations
- Metadata variable mapping

Not all object configurations support per-device mapping. This restriction prevents you from using objects with the same name for different configurations at the policy layer.

The example on the slide shows the **LAN** firewall address, where **Branch1** uses the network segment **172.16.0.0/24**, and **Branch2** uses **192.168.0.0/24**. Per-device mapping differentiates these segments under the same object name.

However, some configurations, like web filter profiles, don't support per-device mapping. If different settings for the same profile name are needed across multiple FortiGate devices, you must create unique profile names for each variation.



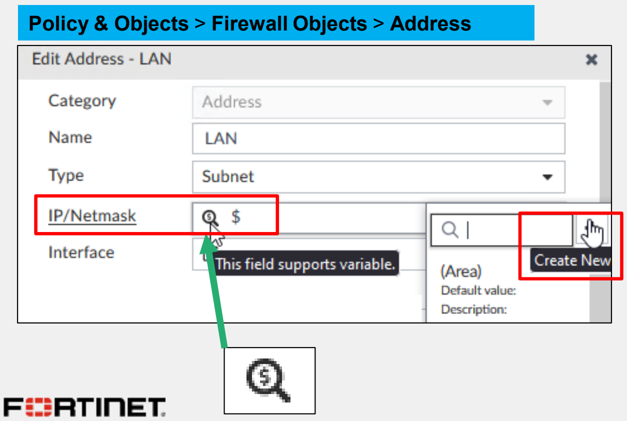
Why is it important to recognize object configurations that are eligible for per-device mapping? If you import a configuration from the device layer that the policy layer doesn't support, it causes a conflict, forcing FortiManager to choose between the FortiGate value and its own.

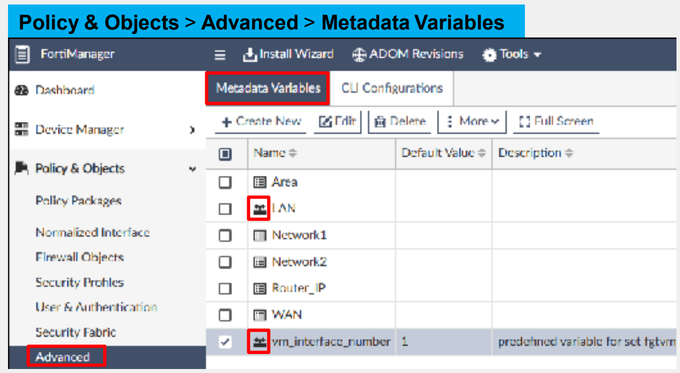
It's beneficial to memorize or identify objects eligible for per-device mapping so that you don’t import objects with duplicate names into the FortiManager policy layer database.

If a conflict occurs, the solution is to create a uniquely named object on FortiGate and import it again to the policy.

Templates and Policy—Metadata Integration

- Created and edited in the policy layer
- Used in provisioning templates or policy layer
- Magnifying glass symbol
- Supports per-device mapping





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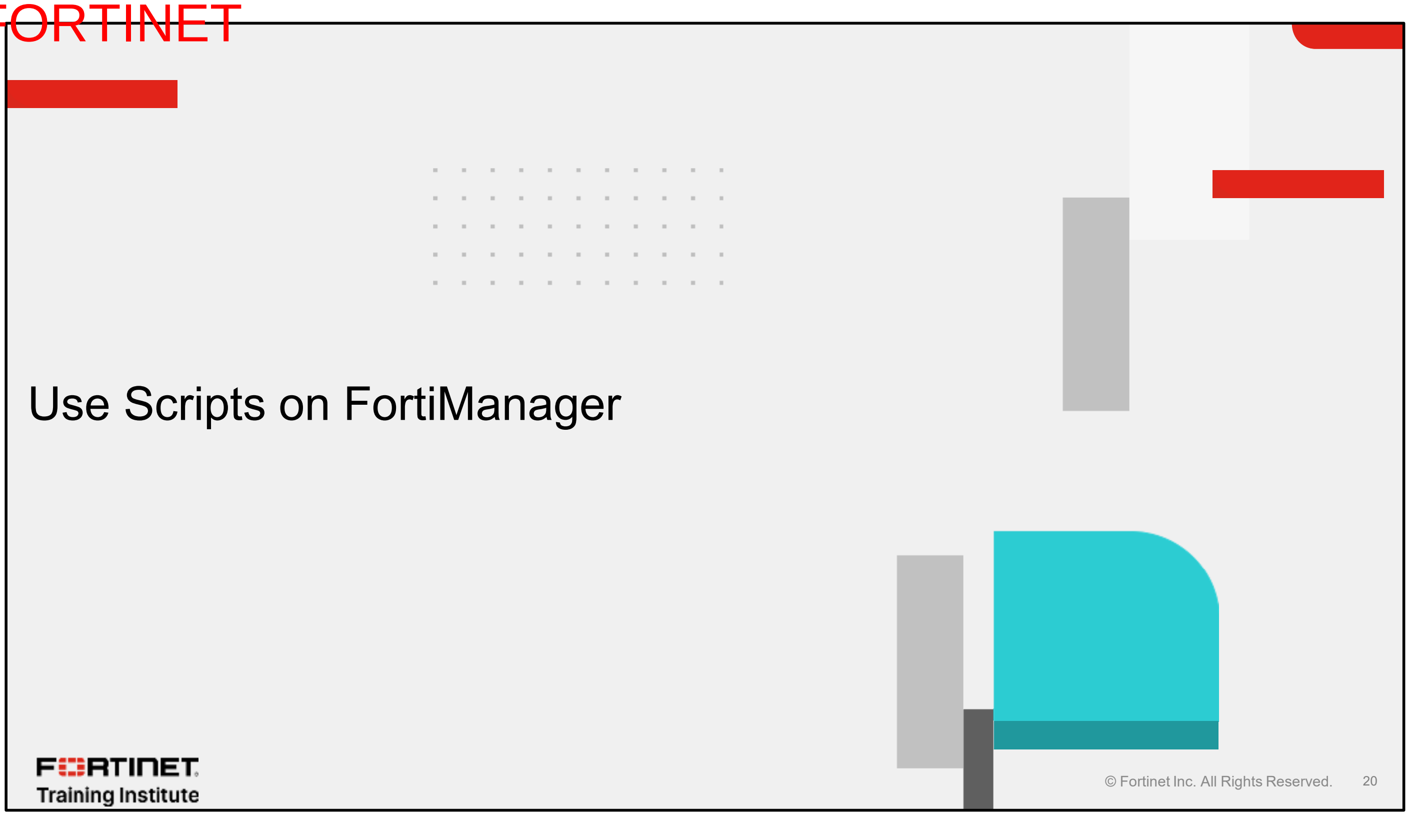
The GUI indicates metadata variables by an icon of a magnifying glass with the $ symbol.

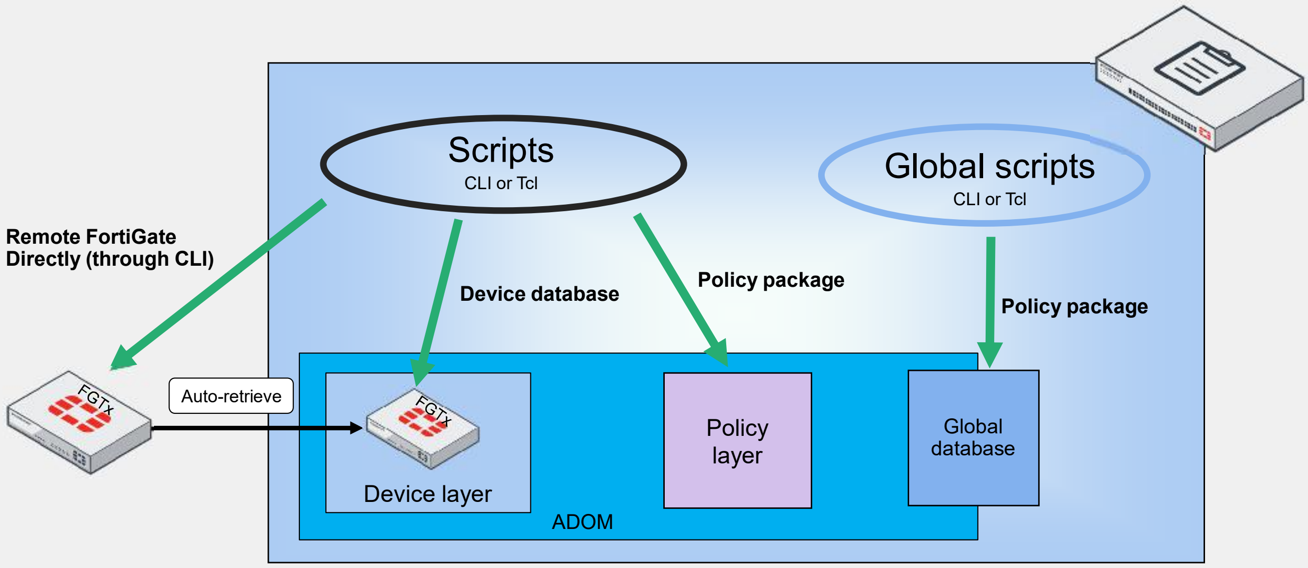
To use a metadata variable, type $, and a pop-up window will appear. If the variable hasn't been configured, click +. For example, you can add a metadata variable for **IP/Netmask** to a firewall address named **LAN**.

You can also create metadata variables by clicking **Policy & Objects** > **Advanced** > **Metadata Variables**. Supported per-device mapping allows for default settings across FortiGate devices and custom values for specific firewalls.



In this section, you will learn about scripts on FortiManager.

Remote, Device, and Policy Scripts



You can execute scripts in three different contexts: directly on FortiGate, in the device database, and on the policy package.

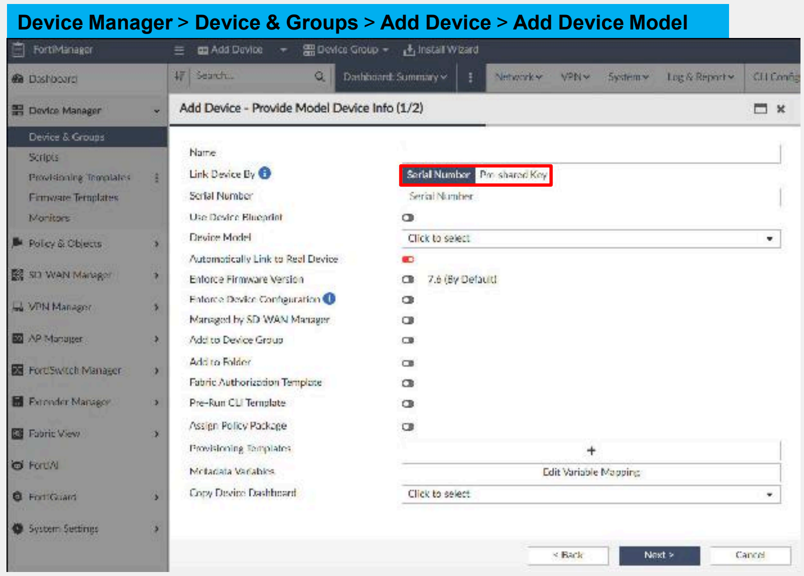
Key points to note:

1. Executing a script directly on FortiGate allows for automatic synchronization of the device database through auto-retrieve, provided the changes do not involve firewall policy configurations. Changes to firewall policies require manual intervention to sync with the policy layer.
2. When scripts modify policy settings in the device database, these changes cannot be automatically imported into the policy layer.
3. For scripts executed in the policy layer, changes do not automatically update the device database and FortiGate. You must install or reinstall the policy package to apply these changes.
4. Scripts run in the global database also require manual action to apply changes to global policies, either through assignment or automatic installation.

Scripts can be written in either the CLI or Tcl.

ZTP and LTP

- Model device
- Serial number or a pre-shared key
- Auto-link process
- Zero-touch provisioning (ZTP)
- Low-touch provisioning (LTP)
- Large-scale deployments
- Various ZTP/LTP methods



FortiManager uses model devices to support the ZTP and LTP of FortiGate devices. You configure a model device for a FortiGate device before you add it to FortiManager. FortiManager allows you to add FortiGate models using either a serial number or a pre-shared key. The FortiManager administrator configures the model device, and when a real FortiGate device connects, the auto-link process installs the settings and policies from the model device on it. Once auto-linking is complete, the real device is configured, connects to FortiManager, and replaces the model device for central management.

How the FortiGate devices discover and connect to FortiManager determines if the provisioning is ZTP or LTP.

ZTP: FortiGate automatically connects to the WAN or the internet and links to FortiManager for central management without any preconfiguration.

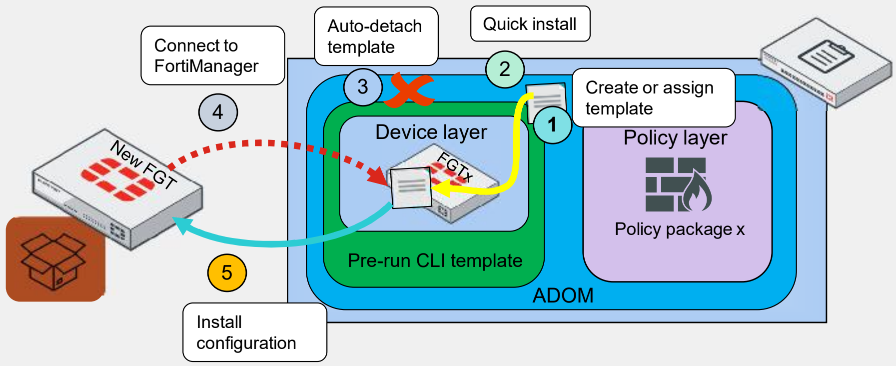
LTP: You must do some initial setup on FortiGate, such as specifying network settings and FortiManager location, before it can discover and connect to FortiManager.

Both methods are useful for large-scale deployments or remote locations where it is not practical to have IT staff manually configure each device.

Besides FortiManager, other methods for configuring ZTP devices include FortiSOAR with FortiManager, Ansible, Terraform, custom API calls, FortiProvision, FortiZTP, and FortiDeploy. The method you choose depends on your expertise, budget, ability to preconfigure FortiGate, and scope of deployment.

Pre-Run CLI Templates

- Feature of ZTP and LTP
- One-time tasks
- Quick install
- Automatically link to real device



Pre-run CLI templates are a feature of ZTP and LTP.

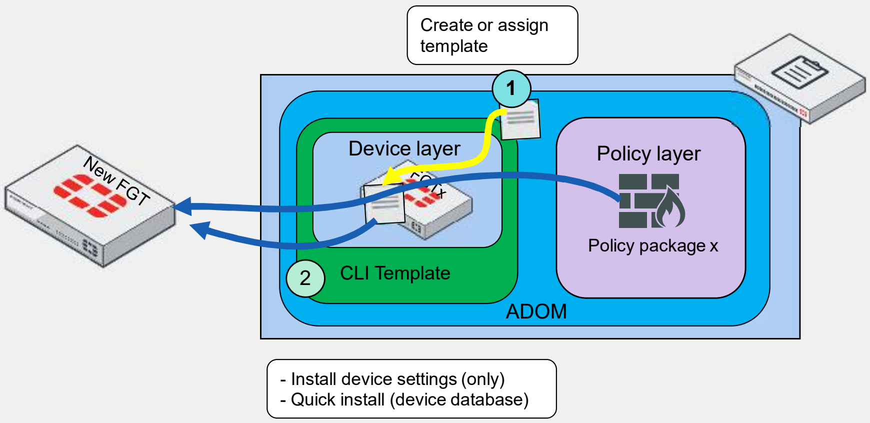
These templates, intended for specific, one-time tasks, do not remain in the FortiGate device database after installation.

The diagram on the slide shows the following process:

1. A FortiManager administrator creates and assigns a pre-run CLI template for a new FortiGate added as a model method to FortiManager.
2. The administrator uses the quick install process to deploy configurations from the pre-run CLI template directly to the FortiGate device database.
3. FortiManager automatically unassigns the pre-run CLI template from the FortiGate device database after the quick install process.
4. When you connect a new FortiGate to the network and establish a connection to FortiManager, FortiManager automatically applies the configurations if **Automatically Link to Real Device** is enabled.
5. If **Automatically Link to Real Device** is disabled, you must manually push configuration changes after the FortiGate- FortiManager communication protocol (FGFM) tunnel is active.

CLI Templates

- Continuous CLI template assignment
- CLI templates overwrite the device database
- Avoid setting changes with CLI templates
- Metadata variables available
- CLI and Jinja compatibility



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CLI templates are designed for repeated use and are not automatically unassigned after the initial setup, unlike pre-run CLI templates.

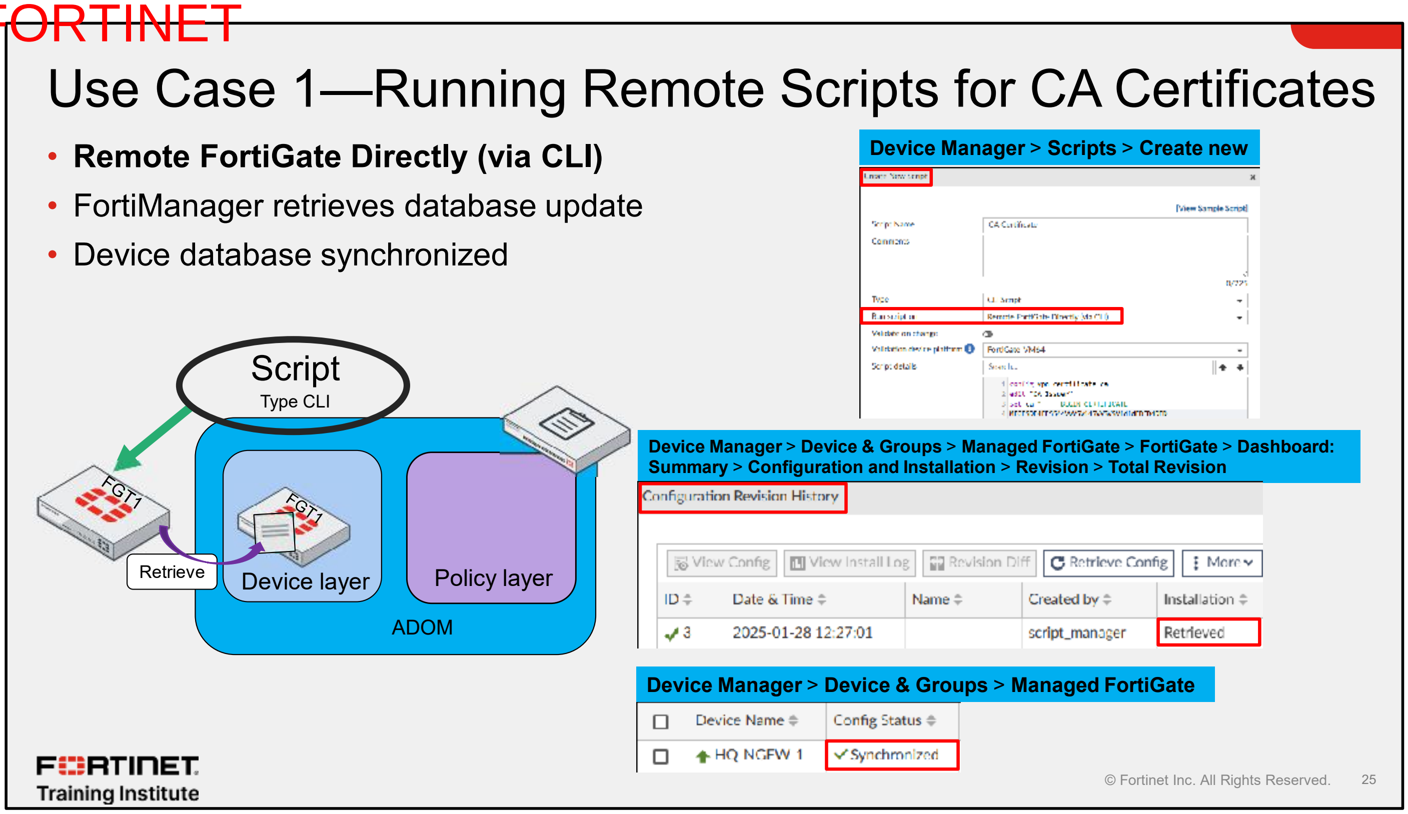
As key components of provisioning templates, CLI templates overwrite the device database.

The setup process involves:

1. A FortiManager administrator assigns a CLI template to new or existing FortiGate devices.
2. The administrator installs policy packages or device settings directly from the device layer.

CLI templates offer practical alternatives for scenarios where adjusting policy layer settings or current database configurations isn't required.

Additionally, both CLI and pre-CLI templates support the use of metadata variables for dynamic configuration and include support for CLI and Jinja scripting, enhancing their versatility.



The example on this slide shows how to run a remote script to create and push a remote CA certificate.

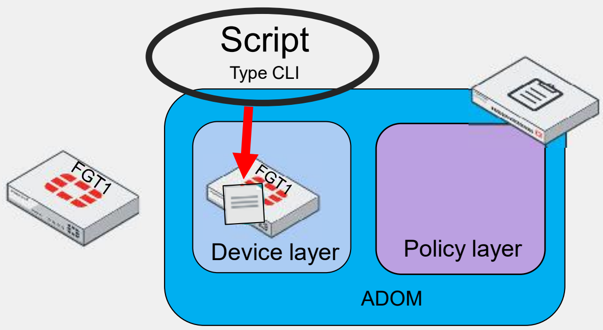
To create this type of script, click **Device Manager** > **Scripts**, and then click **Create New**. Select the CLI type, and then select **Remote FortiGate Directly (via CLI)** as the run location. Copy the script to create a remote CA certificate.

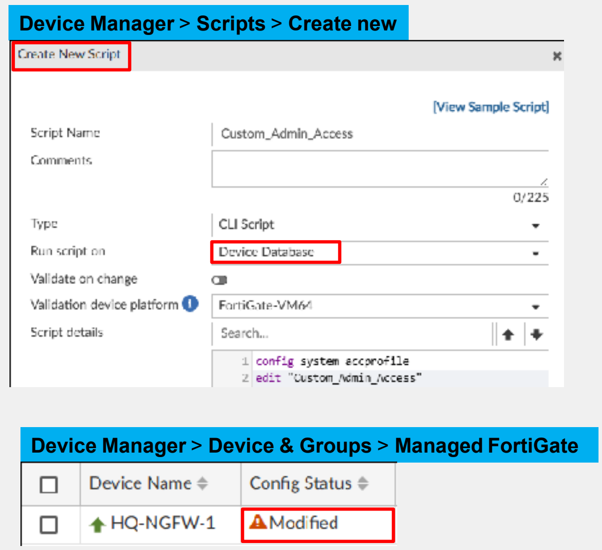
FortiManager executes the script, and then requests a retrieval to update the FortiGate device database, which is reflected in the **Configuration Revision History**. The **Config Status** under **Managed FortiGate** displays **Synchronized**.

To update device settings, use configurations like those shown on the next three slides. Avoid modifying objects used in firewall policies to prevent conflicts in the policy layer status.

Use Case 2—Scripting in the Device Database

- **Device Database**
- Script changes status to **Modified**
- Finish by installing settings





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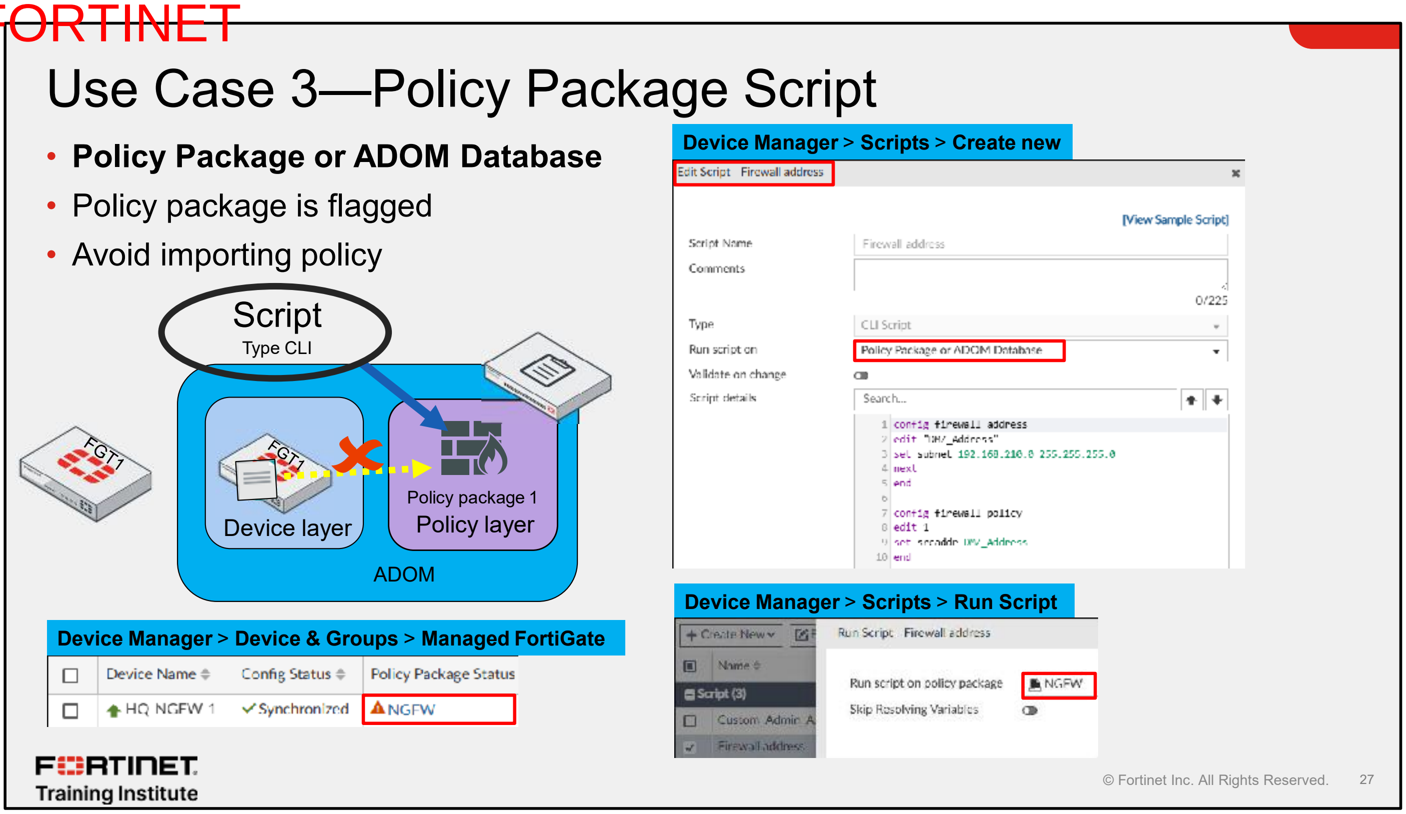
Now, we will learn about creating a custom administrator profile using a device database script.

It's important to note that running this script does not push changes on FortiGate. Instead, the device database changes the **Config Status** to **Modified**.

To complete the changes, install device settings or a policy package if policy changes are also required.



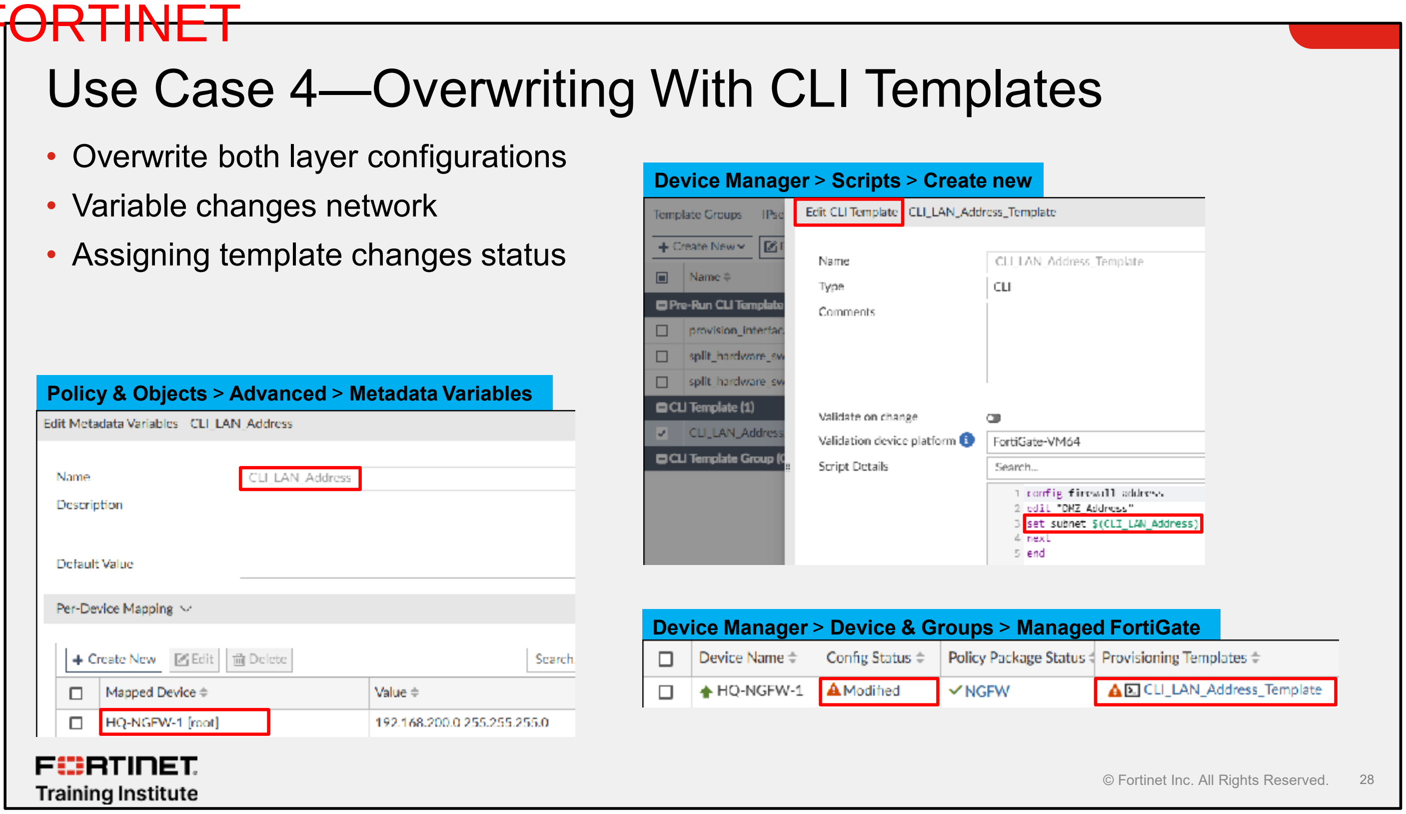
The example on this slide shows how to run a policy package or ADOM database script.

This script creates a firewall address and includes it within a policy. Both objects are created in the policy layer, and running the script flags the policy package as **Modified**.

To apply changes, install the policy package to push the new firewall address and policy to the device database and firewall.

Such scripts are beneficial for centralizing management of multiple objects across firewalls without importing policy packages. They allow the creation of objects using the FortiGate CLI, which you can then add to the required policy package.

Objects you add to the policy package are installed; those not added are not reinstalled and remain in the ADOM database.



The example on this slide shows how a CLI template can overwrite configurations in both the policy and device layers.

Use a `$` to configure a metadata variable. This variable changes the network segment each time the address object `DMZ_Address` is used.

After you assign the CLI template, you'll immediately notice two changes: the device layer status and the provisioning templates status are marked as **Modified** with an exclamation mark inside a triangle.

This behavior triggers FortiManager to detect that changes need to be pushed to the firewall to synchronize the firewall, device database, policy package, and the statuses of provisioning templates.

Compare CLI vs. Tcl. vs. Jinja Scripts

Feature	CLI scripts	CLI templates	Tcl scripts	Jinja templates	API
Language	FortiOS	FortiOS	Tcl	Python	JSON
Function	Interact with FortiOS	Interact with FortiOS	Automation tasks	Dynamic configuration schema	Interface software with FortiManager
Dynamic variables	✗	✓ Metadata variables	✓	✓ Metadata variables	✓
Complexity	Varies	Varies	Moderate	Moderate	High
Software examples	FortiOS	FortiOS	Tclsh, Tkinter, ActiveTcl	Flask, Jinja2, Pelican	Fortinet Developer Network, Terraform, Postman, Insomnia, Swagger UI

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FortiManager scripts can be written with a broad range of languages, including API scripting. API scripting facilitates communication between software applications and FortiManager.

If you have a solid foundation in FortiGate and FortiManager, you're well-positioned to effectively leverage CLI scripts and CLI templates. This strategy enables you to use the established nomenclature and syntax of traditional FortiOS, maximizing your capabilities within this environment.

However, if you possess programming knowledge in languages such as Tcl, Python, and JSON, you should consider leveraging these languages to automate tasks, taking advantage of their powerful capabilities.

Additionally, if you're interested in deepening your understanding of Tcl, Jinja, and API, explore the various software tools available to learn more about the language and nomenclature you need to know.

Review

- ✓ Examine FortiManager architecture
- ✓ Explore dynamic objects on FortiManager
- ✓ Use scripts on FortiManager

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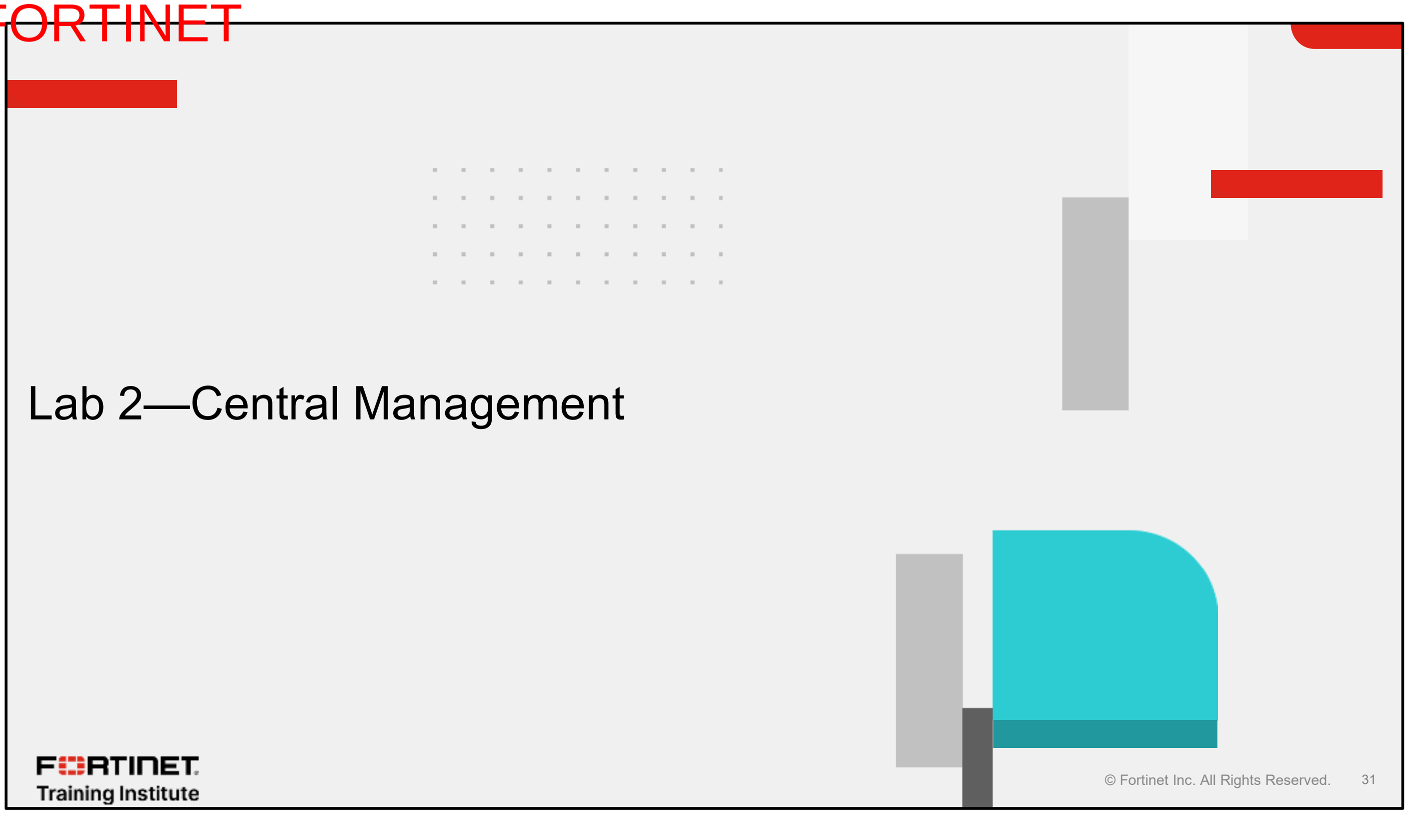
This slide shows the objectives that you covered in this lesson.

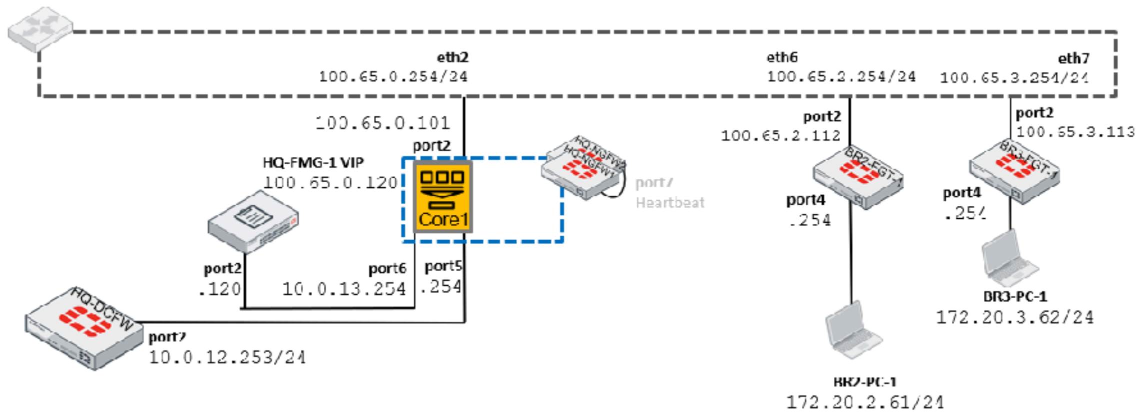
By mastering the objectives covered in this lesson, you learned how to use central management.



Now, you will work on *Lab 2—Central Management*.

Lab 2—Central Management

- Scripts and LTP



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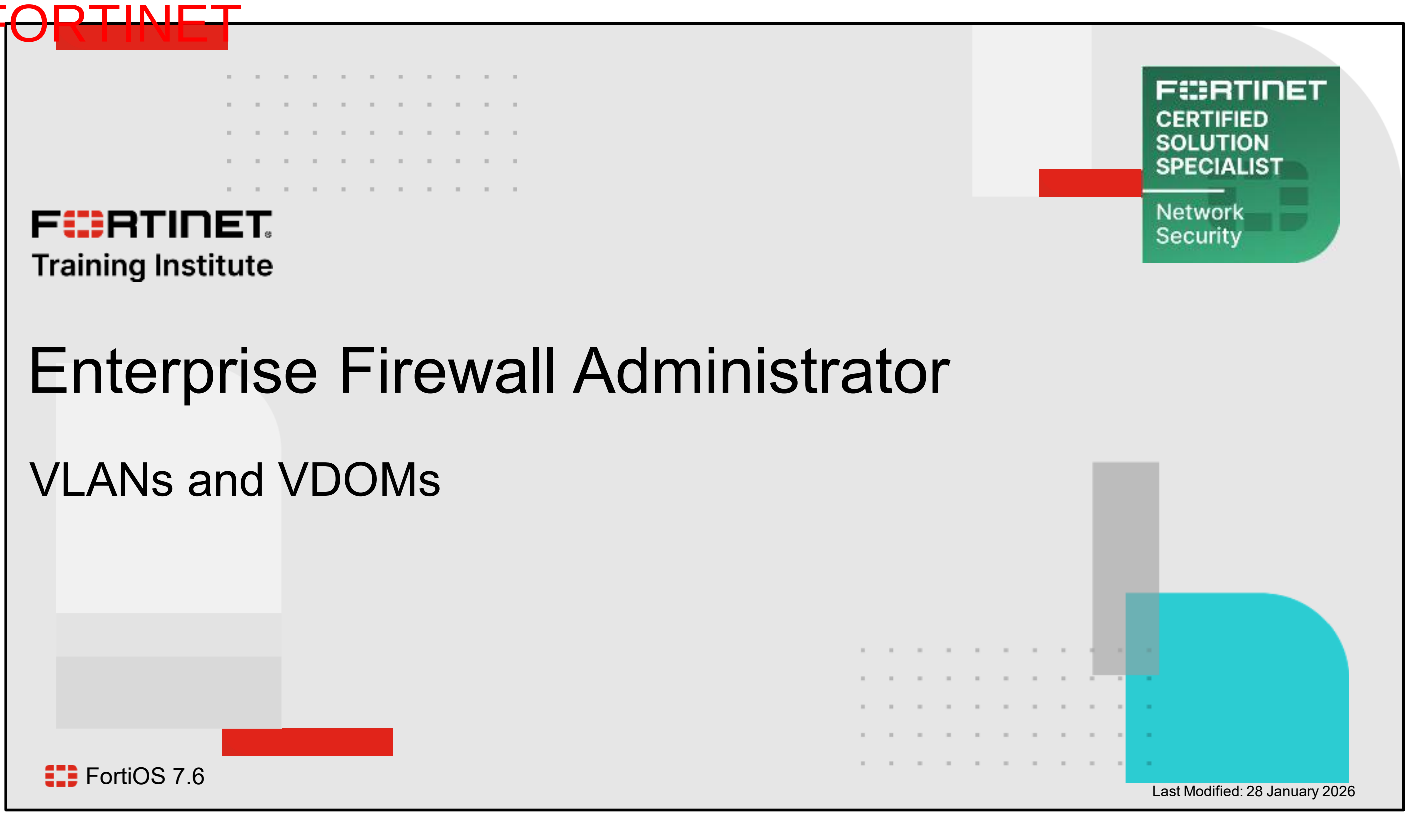
In this lab, you will configure different FortiManager scripts in the data center firewall (DCFW) and you will add FortiGate devices using the LTP feature to deploy a pre-run CLI template.



In this lesson, you will learn about VLANs and VDOMs on FortiGate.

Objectives

- Understand how to configure VLANs on FortiGate
- Understand how to implement an enterprise network using VDOMs

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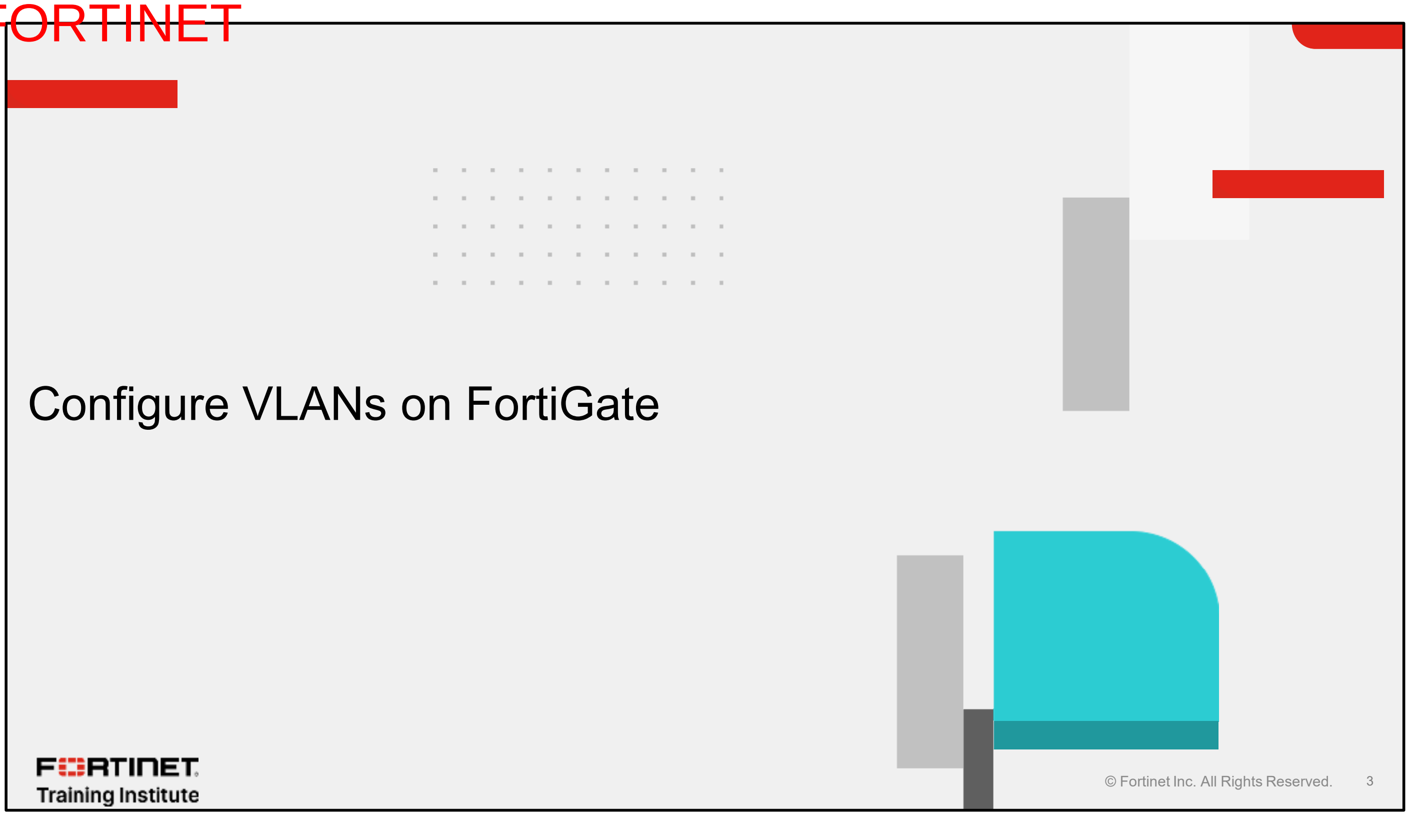
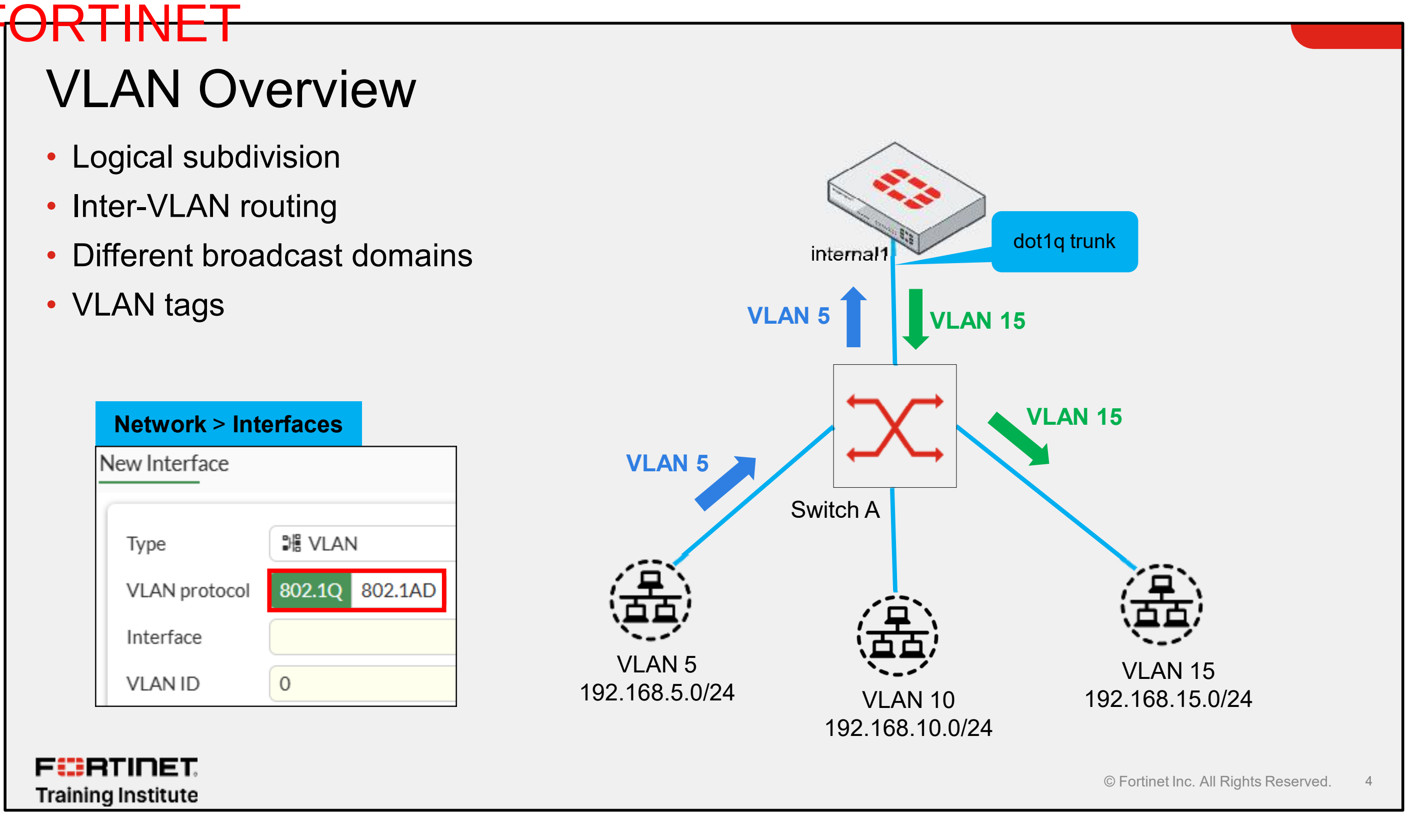
After completing this lesson, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in VLANs and VDOMs, you will be able to configure VLANs and VDOMs on FortiGate.



In this section, you will learn how to configure VLANs on FortiGate.



A VLAN is a logical subdivision of a network that segregates devices into separate broadcast domains.

Devices in the same broadcast domain can communicate directly with each other. Otherwise, inter-VLAN routing or policies enable communication between different VLANs.

When more than one VLAN shares the same interface, it is called a dot1q trunk link.

The example on this slide shows an instance of inter-VLAN routing where a host on VLAN 5 sends a frame to a host on VLAN 15. Switch A receives the frame on the untagged VLAN 5 interface and then adds the VLAN 5 tag to the frame for the tagged trunk link between Switch A and FortiGate.

FortiGate receives the frame on the VLAN 5 interface. It then routes the traffic from VLAN 5 to VLAN 15, rewriting the VLAN ID to VLAN 15 in the process.

Switch A receives the frame on the VLAN trunk interface and removes the VLAN tag before forwarding the frame to its destination on the untagged VLAN 15 interface.

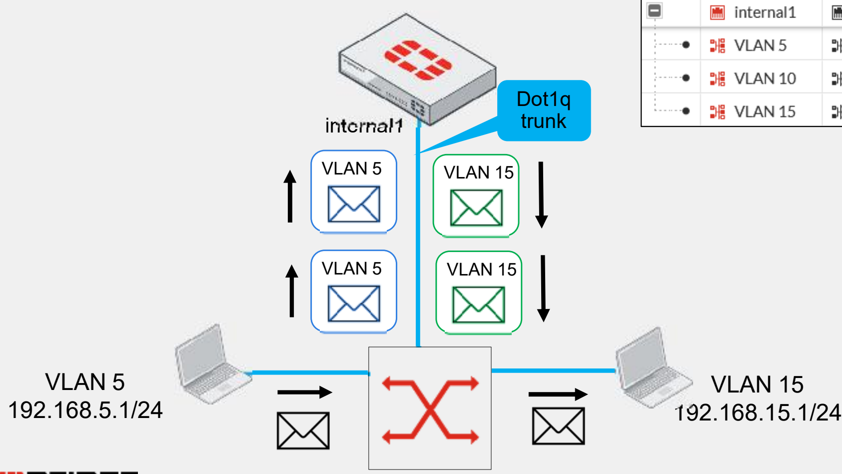
Using the FortiGate GUI, a VLAN interface can use the IEEE 802.1Q protocol, the most common protocol for VLAN tagging, or IEEE 802.1ad, a provider bridging protocol.

VLAN Tagging—Egress Addition and Ingress Removal

- Layer 2 devices add or remove tags
- Layer 3 devices modify tags
- Tagging based on a routing decision

Network > Interfaces

	Name	Type	IP/Netmask	VLAN ID
	Physical Interface 6			
	internal1	Physical Interface	0.0.0.0/0.0.0.0	
	VLAN 5	VLAN	192.168.5.254/255.255.255.0	5
	VLAN 10	VLAN	192.168.10.254/255.255.255.0	10
	VLAN 15	VLAN	192.168.15.254/255.255.255.0	15



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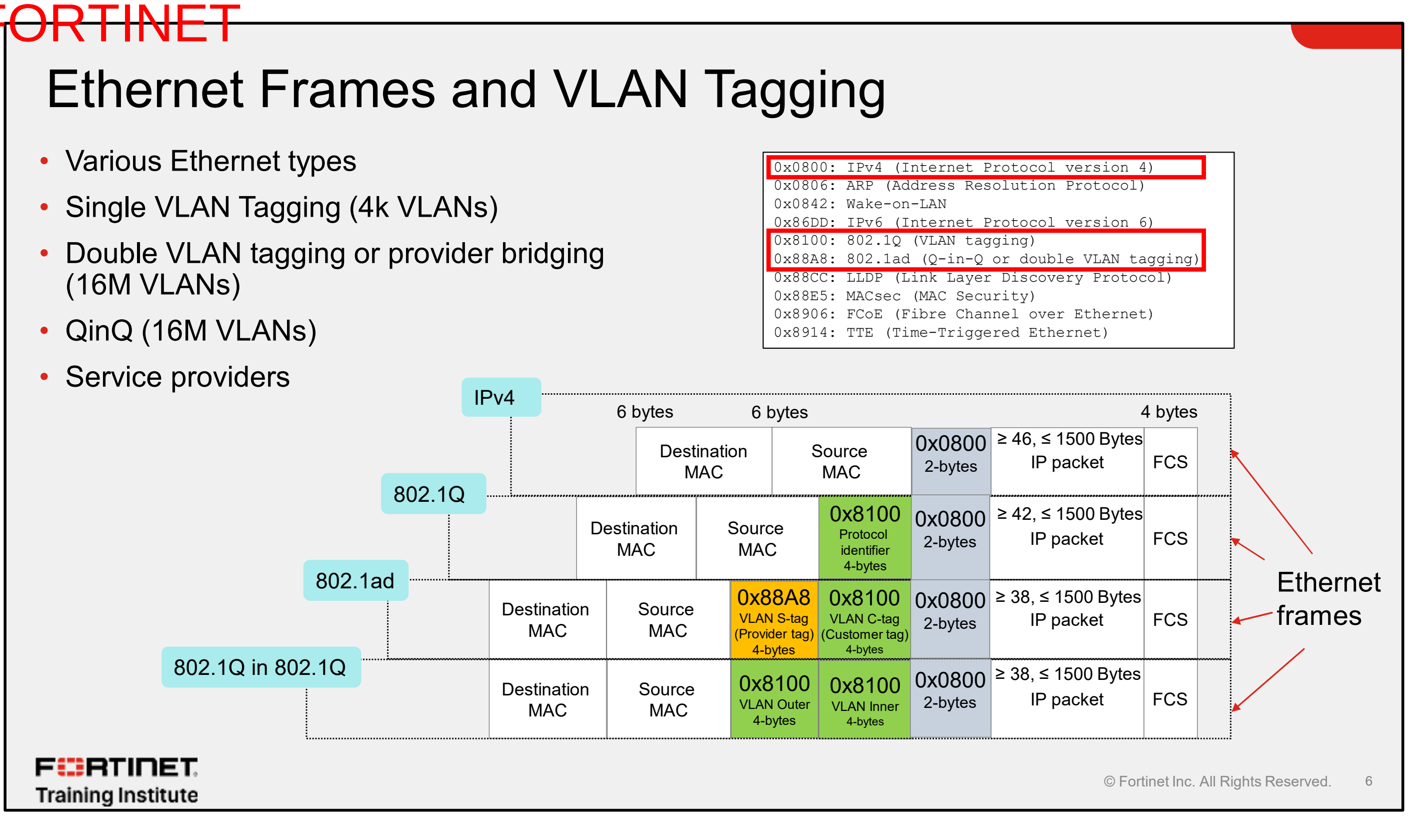
Layer 2 devices, like a switch, can add or remove these tags but not alter them.

Layer 3 devices, such as a router or FortiGate, can modify the VLAN tag to facilitate routing between VLANs. FortiGate does not add VLAN tags on ingress; this is the responsibility of the preceding device.

In the example on this slide, FortiGate removes the VLAN 5 tag in the packet coming from the switch. After the routing operation, FortiGate adds the VLAN 15 tag to the packet.

On the GUI, you have the option to add a VLAN ID column to quickly view all the VLANs.

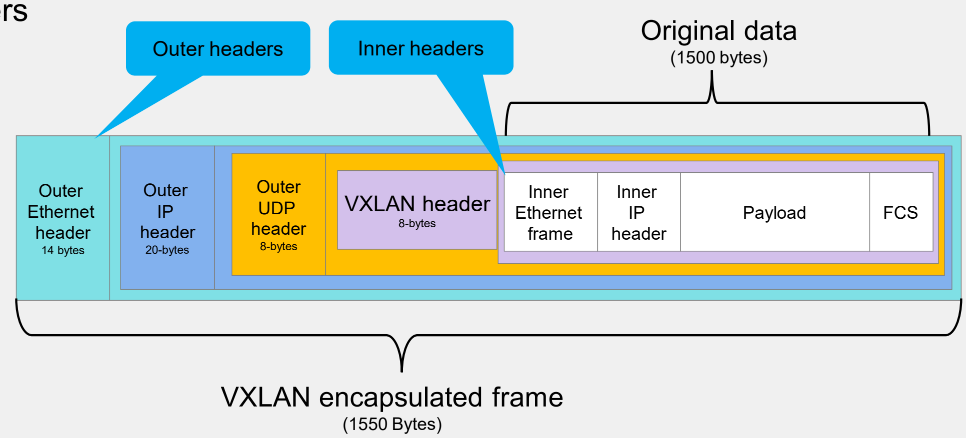


The layer 2 domain of the OSI model has various Ethernet types, as shown on the slide. You will specifically explore those associated with VLANs.

- A standard Ethernet frame uses the type number 0x0800, indicating that it carries an IPv4 payload. These frames lack VLAN segmentation and are considered typical packets for the IPv4 protocol.
- 802.1Q: This standard enables VLAN tagging in Ethernet frames, using an EtherType of 0x8100. It allows for a single VLAN tag per frame, supporting up to 4,094 VLANs.
- 802.1ad (double VLAN tagging or provider bridging): This standard is designed for VLAN stacking with additional features tailored for service providers. It uses an EtherType of 0x88A8 for the provider VLAN tag (S-tag) and 0x8100 for the customer VLAN tag (C-tag). It supports up to 16 million VLAN combinations, with 4,094 possible S-VLANs and 4,094 possible C-VLANs. This protocol is typically used by service providers to segregate customer traffic and manage services efficiently.
- 802.1Q-in-802.1Q (QinQ): This method enables stacking two VLAN tags (an outer and an inner tag) in an Ethernet frame, often referred to as double tagging. Both the outer and inner tags typically use an EtherType of 0x8100. QinQ can support up to 16 million VLAN combinations, with 4,094 possible outer VLANs and 4,094 possible inner VLANs. It is commonly used in service provider networks and large enterprise networks for enhanced VLAN scalability and segregation.

VXLAN—Network Virtualization and Scalability

- Encapsulate original data within UDP
- UDP port 4789
- Outer and inner headers
- 16 million VLANs
- Data centers
- Cloud environments



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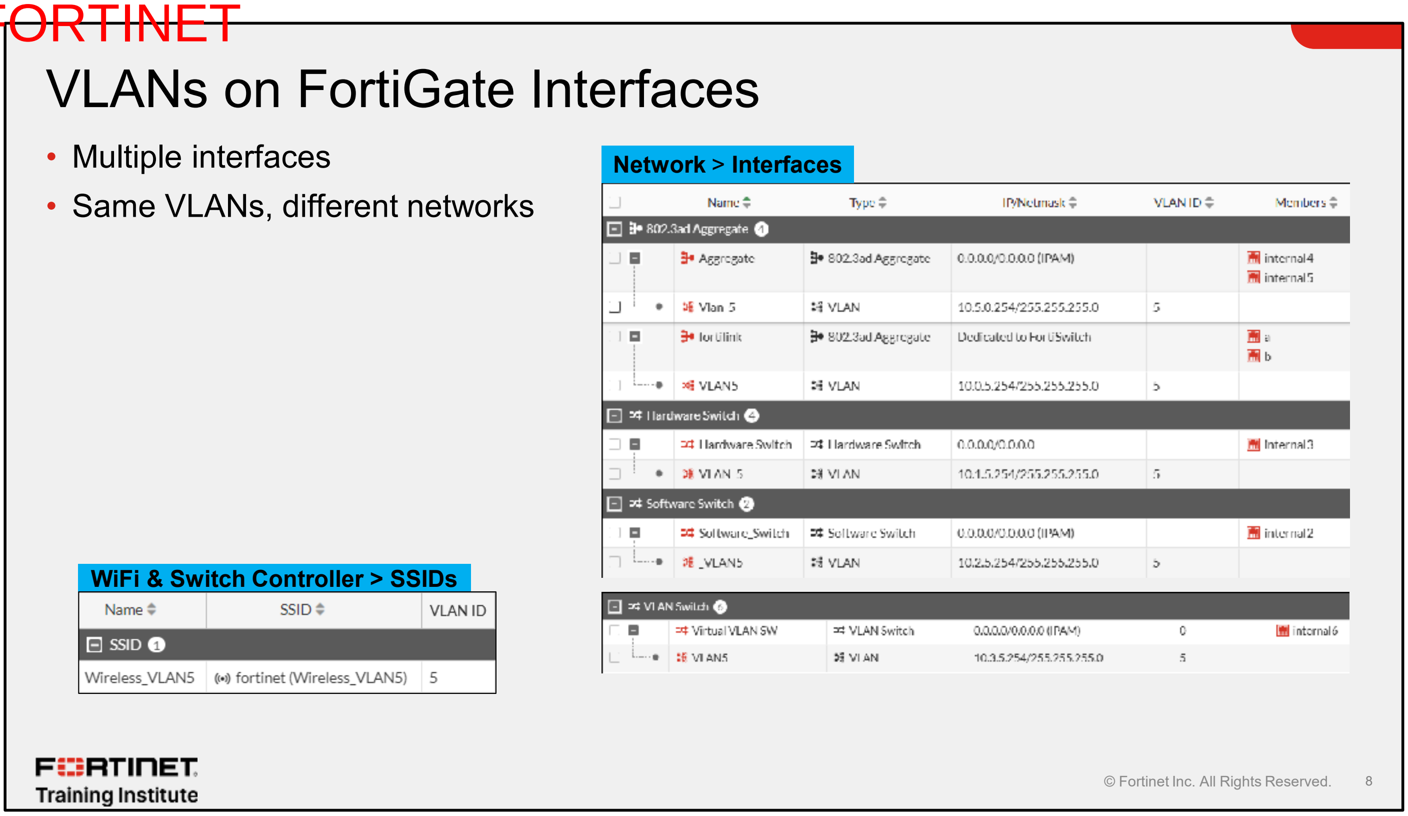
VXLAN (Virtual extensible LAN) is a network virtualization protocol that, unlike traditional VLAN standards like 802.1Q, 802.1ad, and 802.1Q-in-802.1Q, which use VLAN tags inserted directly into the Ethernet frame, encapsulates original data within a UDP packet for tunneling over an IP network.

VXLAN employs UDP encapsulation with a default port number of 4789.

As you can see in this diagram, the original Ethernet frame is encapsulated by outer headers (Ethernet, IP, UDP) for tunneling across the network, while the inner Ethernet/IP header and payload remains as original frame's header. This additional layer of encapsulation for tunneling distinguishes VXLAN from the traditional VLAN headers used in the 802.1Q standards.

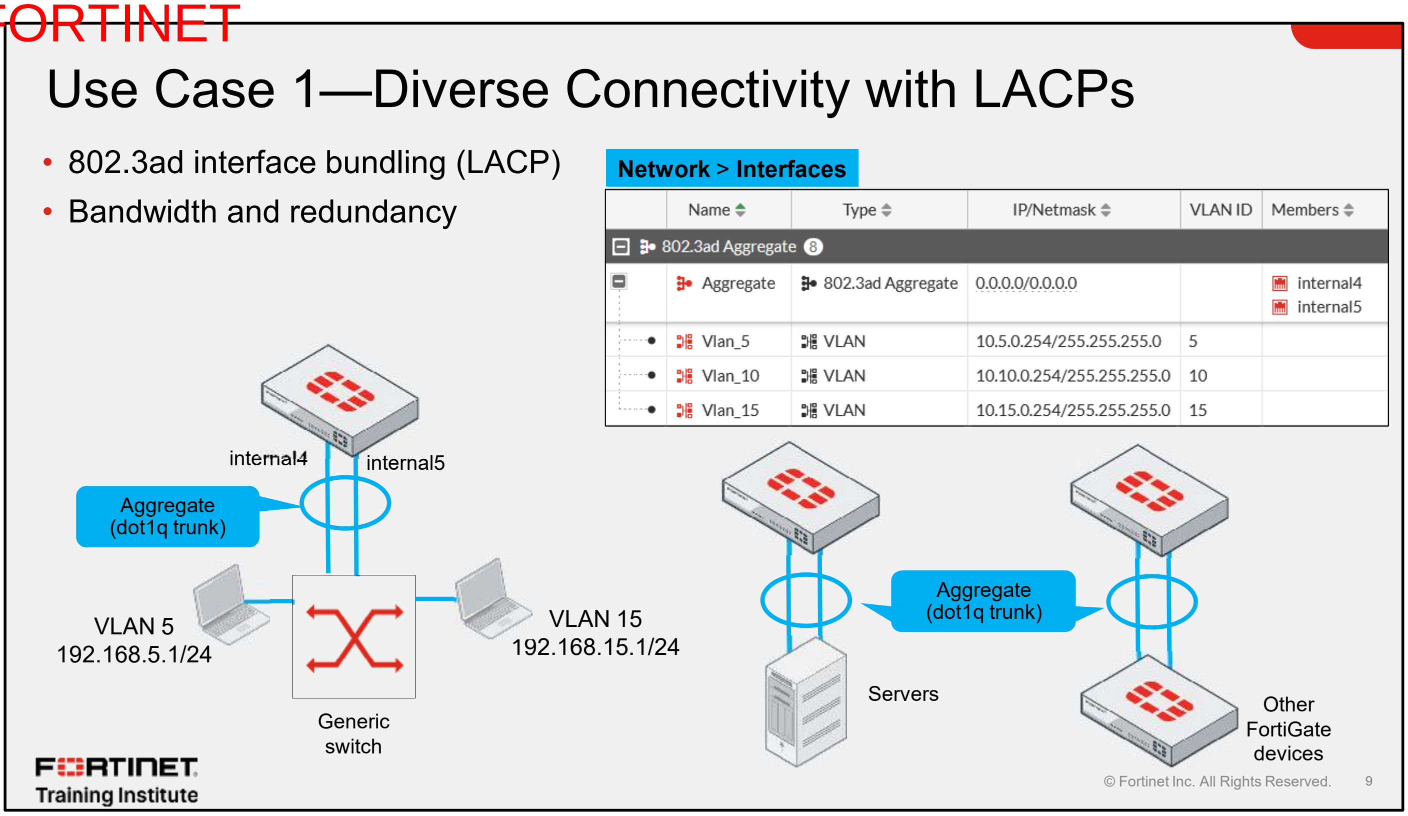
VXLAN supports up to 16 million virtual networks using a 24-bit VXLAN network identifier (VNI), offering significantly more scalability than the 4,094 VLANs supported by 802.1Q. VXLAN can be used in data centers and cloud environments to overcome VLAN scalability limitations and to provide flexible, large-scale network segmentation.



Now that you know the standards and protocols, you need to know that you can configure VLANs on different types of interfaces, not just physical ones. You can configure VLANs on the following interface types:

- Aggregate interfaces
- FortiLink interfaces
- Hardware switches
- Virtual VLAN switches
- Software switches
- Wireless SSID interfaces

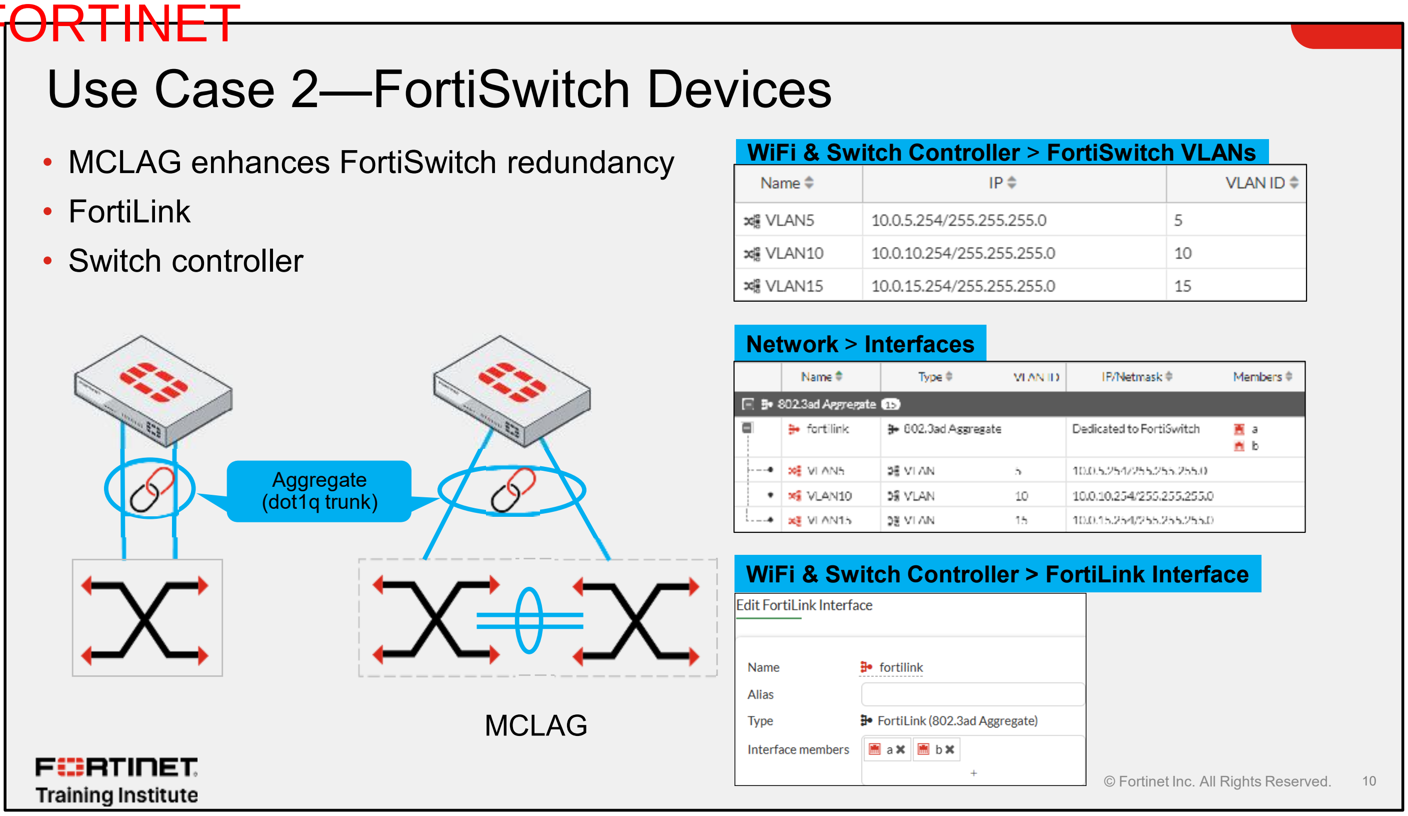
You can even use the same VLAN ID on different interfaces with different names, resulting in completely different broadcast domains.



Standard 802.1ad is not the same as Standard 802.3ad. The first is considered double VLAN tagging, and the second one is known as Link Aggregation Control Protocol (LACP), which is used for bundling multiple network connections to increase bandwidth and provide redundancy.

LACP interfaces, often called aggregate interfaces, allow for connections to switches, servers, and other FortiGate devices. This setup maximizes bandwidth by distributing traffic across all member interfaces. The recommendation is to use either two, four or eight physical ports in the aggregate.

In the example on this slide, FortiGate members internal4 and internal5 form one aggregate interface connected to a generic switch. Note that this method requires you to connect multiple ports to the same switch, creating one aggregate interface, so in this case, the multiple interfaces are considered to be one virtual interface.



You can connect FortiGate to a single FortiSwitch using an aggregate interface. However, using Multichassis Link Aggregation (MCLAG) with multiple FortiSwitch devices enhances layer 2 redundancy, allowing two FortiSwitch devices to act as one unit, as shown on the slide.

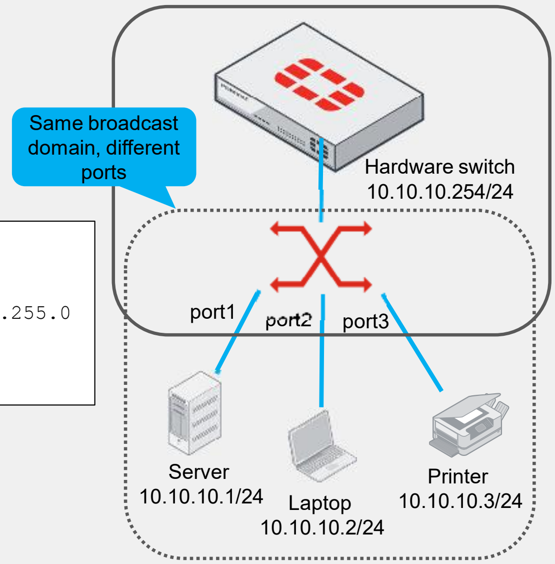
Both scenarios typically use FortiLink with a standard 802.3ad protocol to connect the firewall and switches.

You can manage your FortiSwitch via FortiGate using the path **WiFi & Switch Controller** > **FortiSwitch VLANs**, the recommended method for VLAN creation. Once you create a VLAN, it appears under the FortiLink interface in the **Network** > **Interfaces** section.

You can add FortiLink members under **Network** > **Interfaces** or **WiFi & Switch Controller** > **FortiLink Interface**. In the example on this slide, interfaces a and b are shown as members of the FortiLink interface.

Use Case 3—Hardware Switch

- Same broadcast domain in multiple ports
- Intraswitch traffic allowed




```
config system virtual-switch
edit "Hardware Switch"
set physical-switch "sw0"
config port
    edit port1
    next
    edit port2
    next
    edit port3
    next
  end
end
```

```
config system interface
edit "Hardware Switch"
set vdom "root"
set ip 10.10.10.254 255.255.255.0
set allowaccess ping https
next
end
```

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The advantage of using a hardware switch with FortiGate is that it allows multiple ports to share the same network segment, overcoming the FortiGate default allowance of one single broadcast domain per interface.

The network topology on this slide illustrates that the hardware switch interface has an IP address acting as the default gateway for devices connected to a server (port1), laptop (port2), and printer (port3).

Note that the CLI configuration for a hardware switch uses the `system virtual-switch` command, and you add ports within a subcommand line configuration as command `config port`. Similarly, you can set up an IP address and access services like ping and HTTPS, much like configuring a standard system interface.

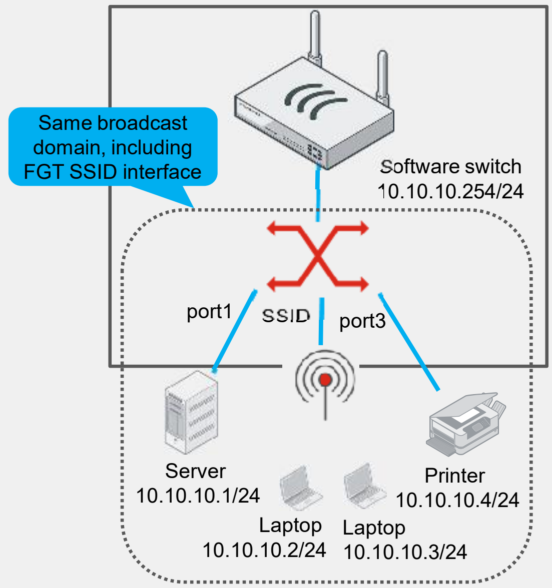
Devices connected to these ports can communicate because they share the same broadcast domain and because the hardware switches support intraswitch traffic.

Use Case 4—Software Switch

- Shared broadcast domain for ports and SSIDs
- Intraswitch traffic can be disabled

```
config system switch-interface
edit "Software Switch"
set vdom "root"
set member "port1" "SSID" "port3"
set intra-switch-policy <implicit or explicit>
next
end
```

```
config system interface
edit "Software Switch"
set vdom "root"
set ip 10.10.10.254 255.255.255.0
set allowaccess ping https
next
end
```



A software switch offers the same benefits as a hardware switch, including the default intraswitch feature using the command `intra-switch-policy implicit`. However, you can also disable this feature with the command `intra-switch-policy explicit`. If you disable intraswitch traffic, you must then configure firewall policies to allow traffic from one member software switch to another.

The network topology on this slide shows that the software switch interface, serving as the default gateway, connects devices to a server, laptops, and a printer using port1; a FortiGate wireless SSID interface; and port3, respectively.

Note that the CLI configuration for a software switch uses the `system switch-interface` command, and you add interfaces as members at the same configuration level, not as ports, like you do for a hardware switch. Similarly, you can set up an IP address and enable access services such as ping and HTTPS, much like configuring a standard system interface.

Hardware vs. Software Switches

Feature	CLI configuration	Intraswitch traffic	Wireless SSIDs	STP	Supported models	Processing

Hardware switch	`config system virtual-switch`
	`edit "Hardware Switch"`
	`config port`
	`edit port1`
	`end`
	`end`
		✓	✗	✓	Specific models	Offloaded to hardware

Software switch	`config system switch-interface`
	`edit "Software Switch"`
	`set member "port1" "port2"`
	`set intra-switch-policy implicit`
	`end`
		✓ (can be blocked)	✓	✗	All models	Processed in software by CPU

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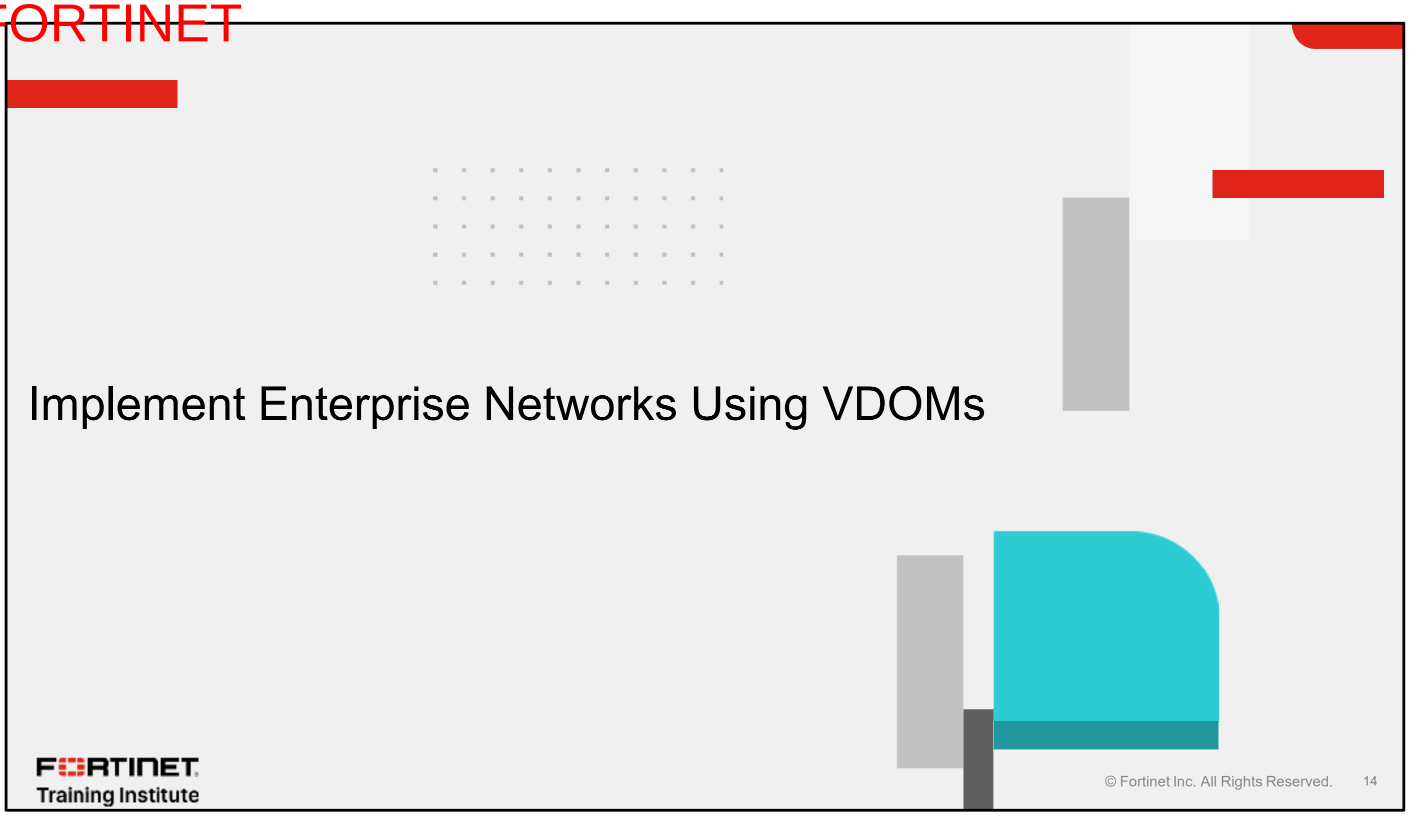
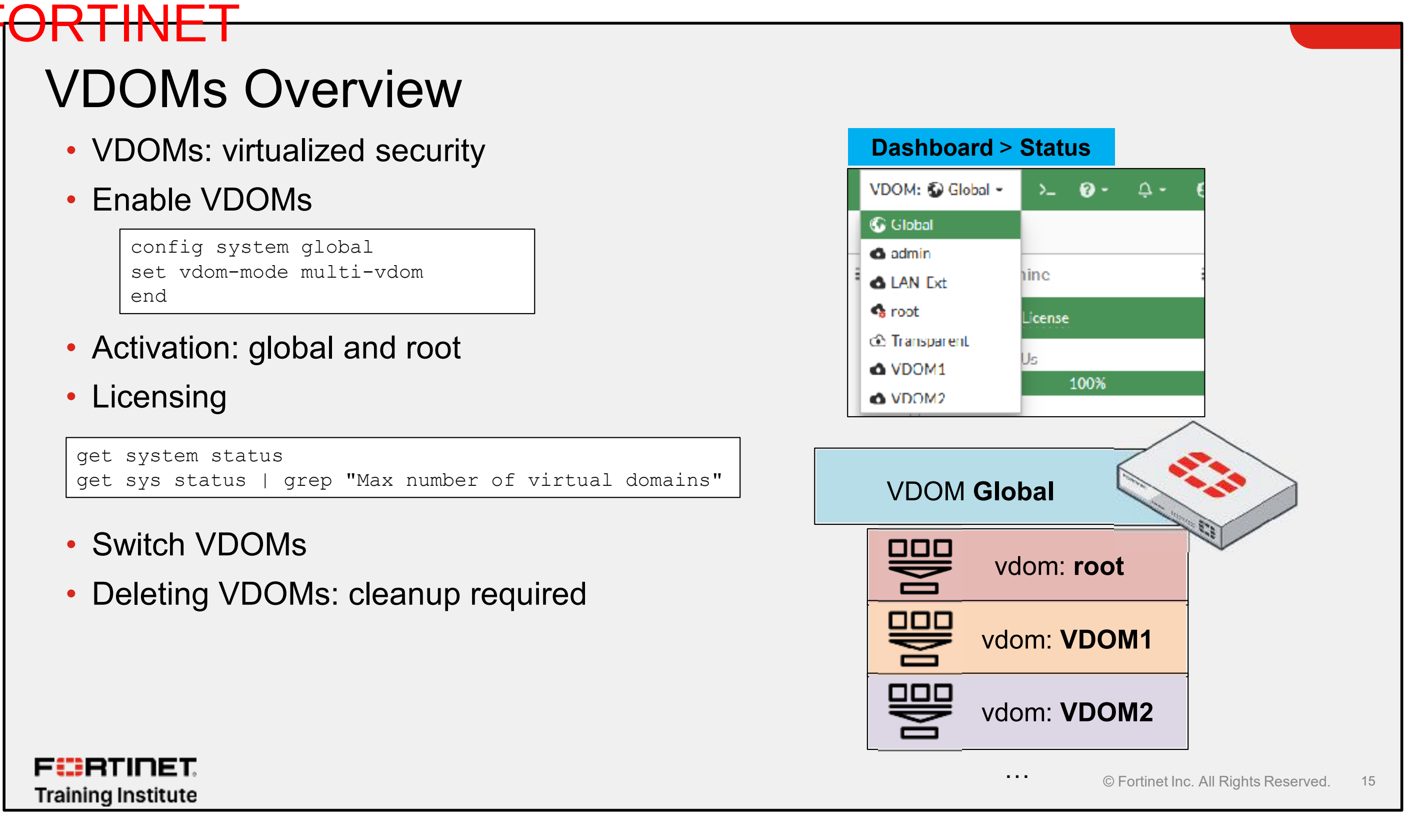
Even though software switches are somewhat equivalent to hardware and virtual VLAN switches in terms of creating the same broadcast domain and VLANs across multiple ports, there are some slight differences to consider:

- The CLI configuration is different —the hardware switch has a subconfiguration and the software switch does not.
- Both types of switches permit intraswitch traffic, but you can explicitly set software switches to require a policy for traffic.
- Hardware and VLAN switches do not support wireless SSIDs, but software switches do.
- Hardware and VLAN switches support Spanning Tree Protocol (STP), but software switches do not.
- Only some models support hardware and VLAN switches, but all models can enable software switches.
- Hardware controllers or SPUs handle packet processing for hardware and VLAN switches. Software switches rely on the CPU, potentially increasing CPU usage and impacting performance in resource intensive modes.

Consider extending the VLANs in your network with these switch options only if acquiring additional FortiSwitch devices, FortiAP devices, or general switches is not feasible. The hardware and software switches play a crucial role in managing layer 2 broadcast traffic, and are better suited for handling tagged packets than performing layer 3 device tasks.



In this section, you will learn about using VDOMs in enterprise networks.



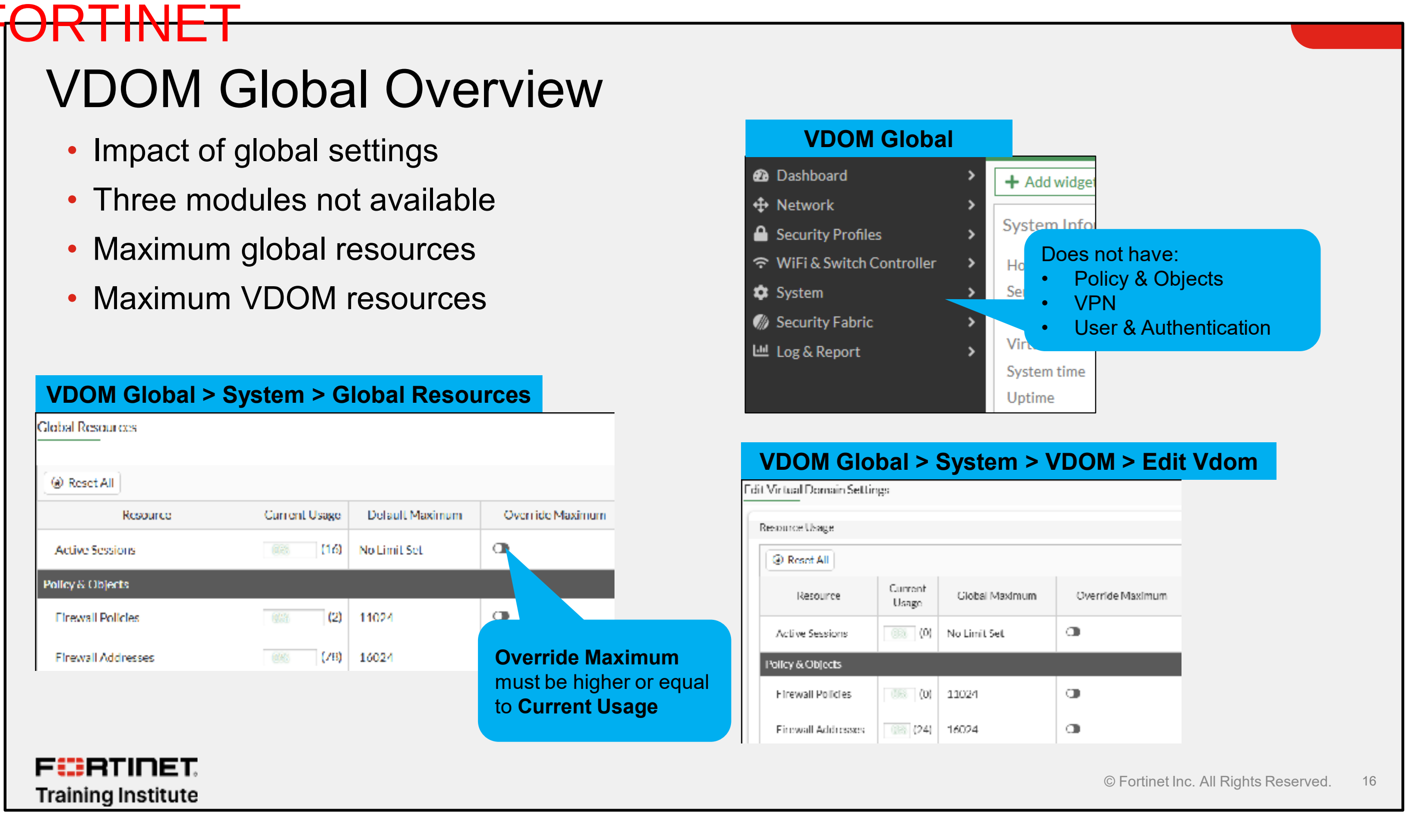
FortiGate allows you to partition a single device into multiple VDOMs, each with its own security configuration and network segmentation.

You enable the multi-VDOM feature using the GUI or the CLI commands shown on this slide. Enabling this feature creates the Global VDOM and the root VDOM. You cannot delete either of these VDOMs.

Most FortiGate models support up to 10 VDOMs. If you require more than 10 VDOMs, you can obtain a license to configure additional VDOMs. To identify how many VDOMs your FortiGate supports, use the command `get system status` and look for the option `Max number of virtual domains`. You can also use the command `get sys status | grep "Max number of virtual domains"`.

You can switch between VDOMs using the icon located at the top right of the FortiGate GUI, which, by default, displays the Global and root VDOMs. Only Super Admin users have access to the Global VDOM.

To delete a VDOM, you must first remove all references used by the VDOM. This includes deleting policies, removing firewall objects associated with interfaces, and reassigning interfaces that are linked to nonroot VDOMs back to the root VDOM, among other tasks.



Changes you make to **VDOM Global** settings affect the entire FortiGate device, including configurations for interfaces, firmware, DNS, some logging, and sandboxing options. Given this wide-reaching impact, only top-level administrators should modify **VDOM Global** settings.

The following options are unavailable for **VDOM Global** due to its inherent functionality:

- **Policy & Objects**
- **VPN**
- **User & Authentication**

You can modify resources that affect the entire FortiGate device, or are specific to one VDOM:

- Global resources: Affect the entire device and can be edited by clicking **VDOM Global** > **System** > **Global Resources**.
- VDOM resources: Allocated to each specific VDOM and can be edited by clicking **VDOM Global** > **System** > **VDOM** > **Edit VDOM**.

You can manage resources such as active sessions, firewall policies, and firewall objects for both the global VDOM and individual VDOMs.

Global Objects

- Consistent UTM across VDOMs
- Automatic update of global profiles
- Sync FortiGate with FortiManager
- Reserved prefix **g-** objects
- Multiple global objects available

VDOM Global > Security Profiles > Antivirus

Name	Comments	Scope
AV g-default	Scan files and block viruses.	Global
AV g-wifi-default	Default configuration for offloading WiFi traffic.	Global

VDOM Global > Security Profiles > Web Filter

Name	Comments	Scope
WEB g-default	Default web filtering.	Global
WEB g-wifi-default	Default configuration for offloading WiFi traffic.	Global

VDOM Global > Security Profiles > Application Control

Name	Comments	Scope
APP g-default	Monitor all applications.	Global
APP g-wifi-default	Default configuration for offloading WiFi traffic.	Global

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The purpose of global objects is to ensure consistent unified threat management (UTM) profiles across various VDOMs.

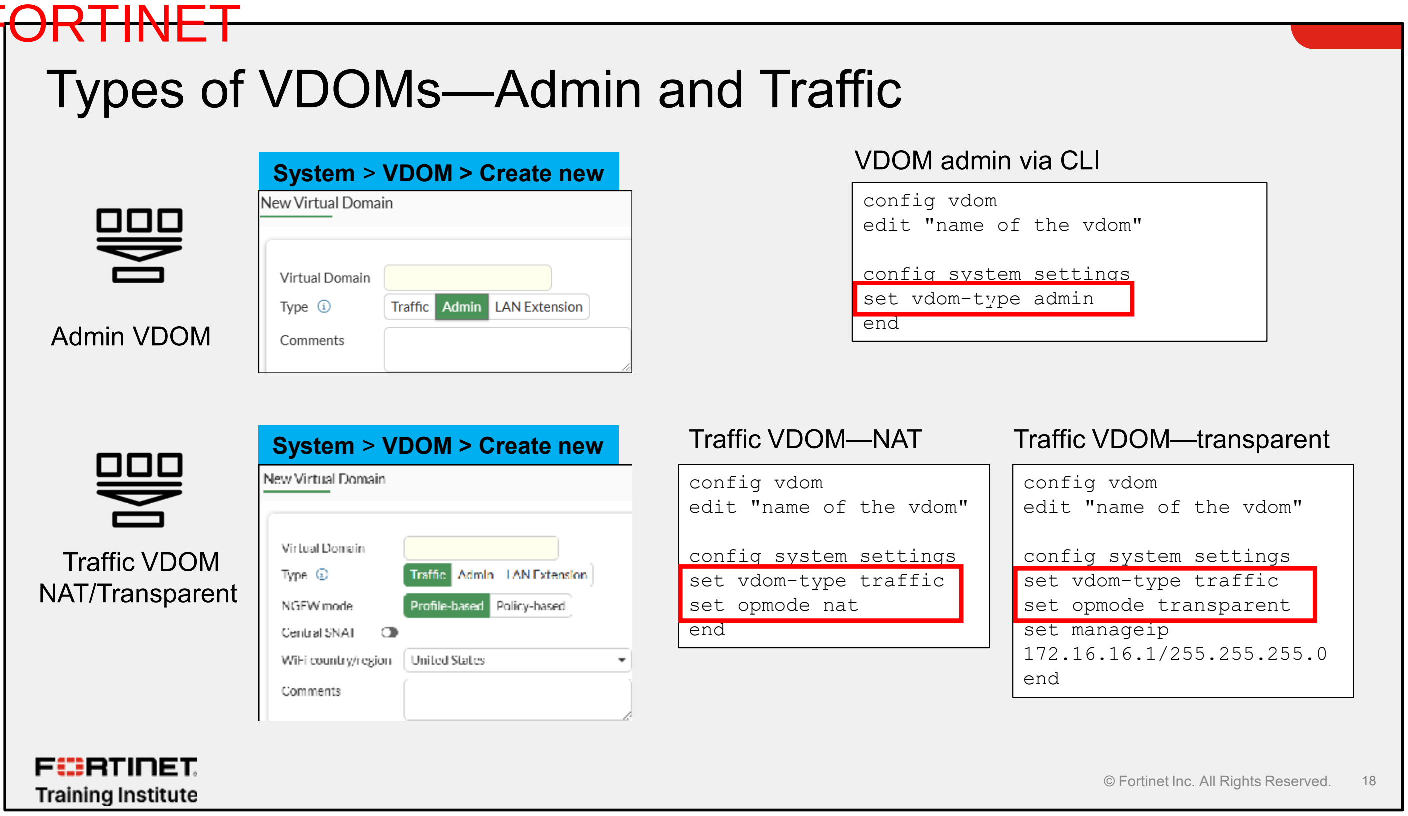
Modifying a global profile automatically updates the same profile across all VDOMs, aiding in the uniform management of multiple VDOM firewalls.

When you import configurations from FortiGate to FortiManager, you should switch from FortiGate's global objects to FortiManager's global objects, for improved synchronization between FortiManager and FortiGate.

Global objects are identifiable by the prefix **g-**, followed by the name of the object. These values are reserved, and you should avoid using the same nomenclature on your FortiGate VDOMs and FortiManager ADOMs.

You can create or edit the following global objects:

- Antivirus
- Web Filters
- Application Control
- Intrusion Prevention
- File Filter
- Data Loss Prevention
- Virtual Patching

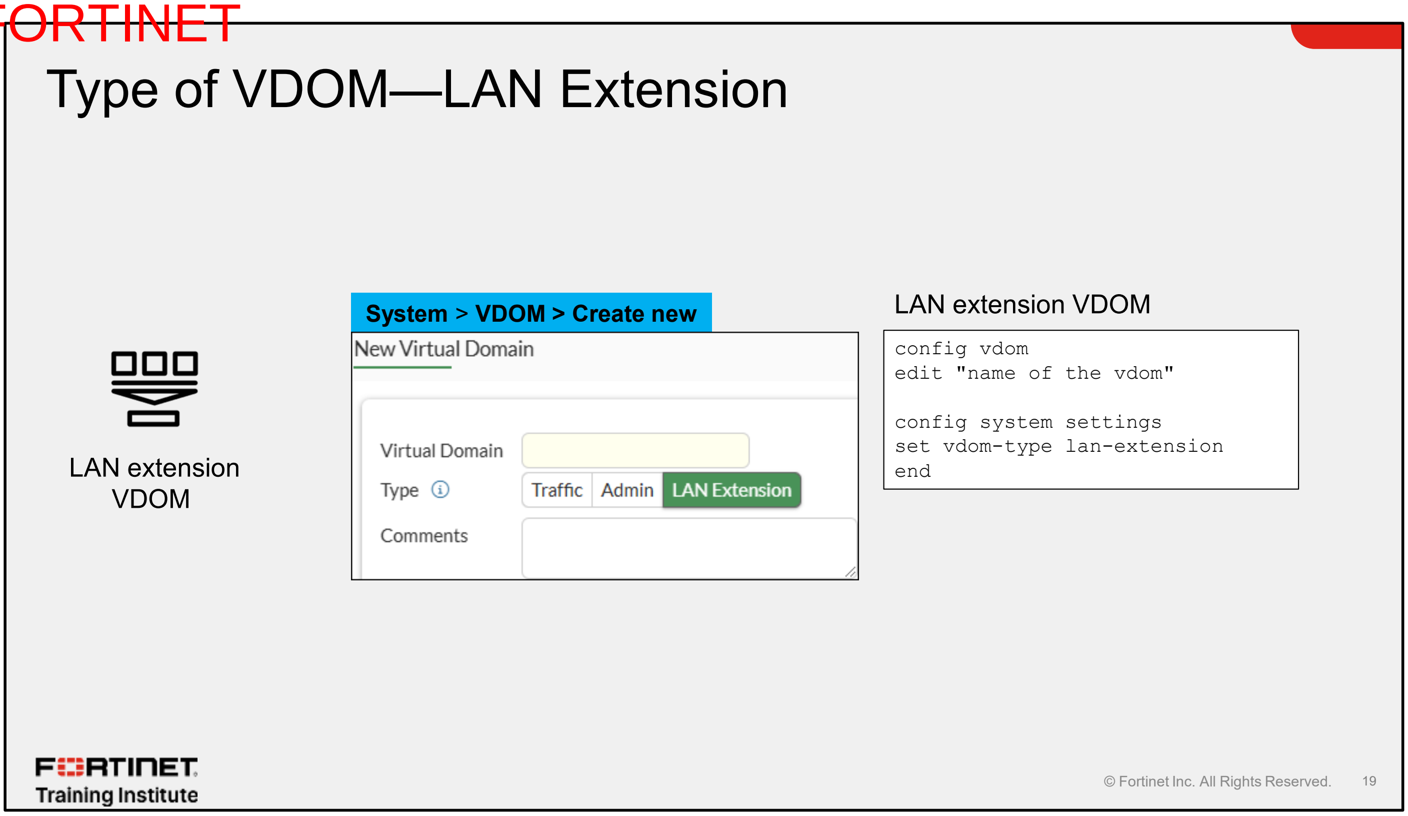


In FortiGate, you can configure two types of VDOM:

- Admin VDOM: This type is dedicated to the management of FortiGate, handling no user traffic.

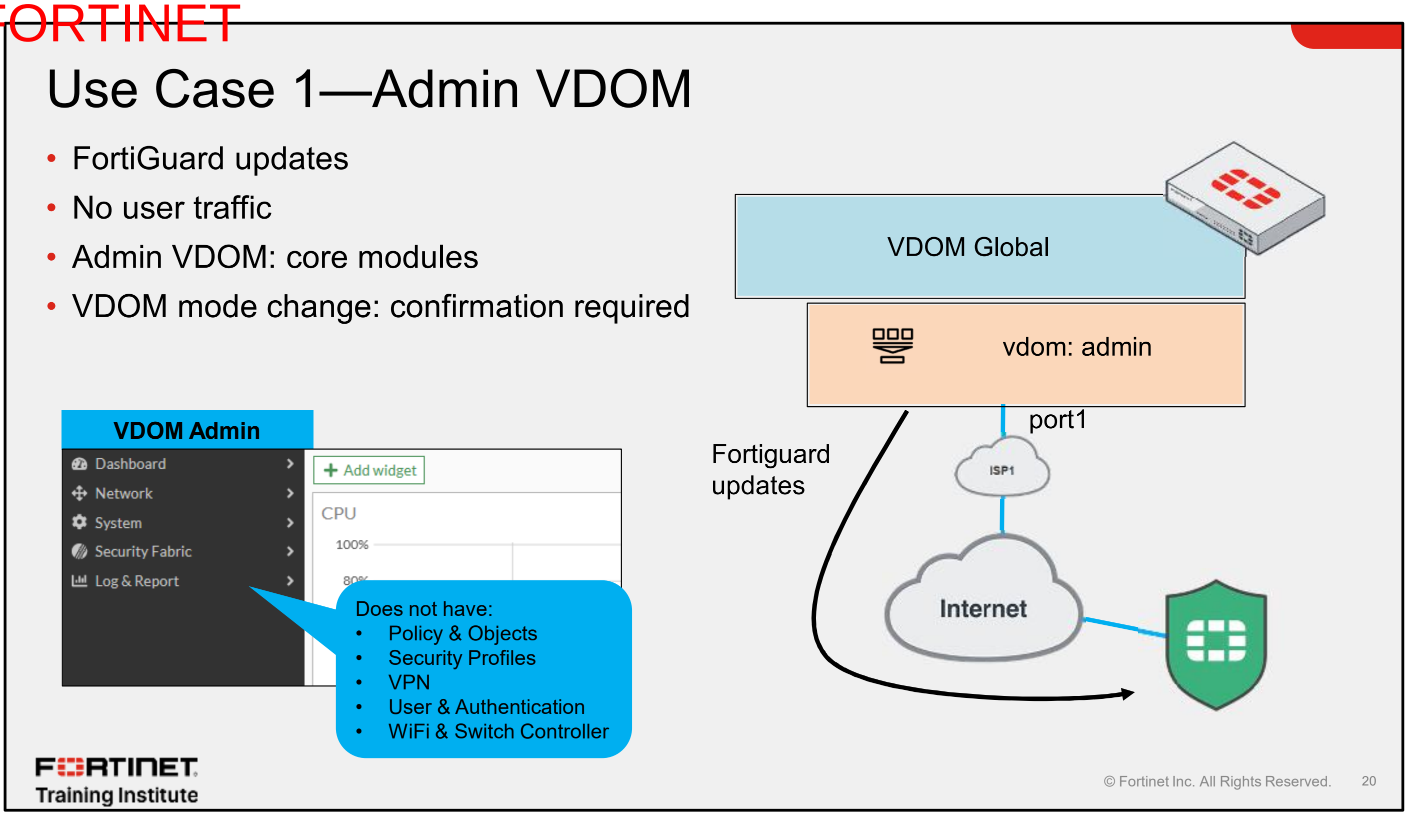
- Traffic VDOM: This type has two operation modes:
  - Network address translation (NAT) operation mode: This is the most common type of VDOM, dedicated to passing user traffic and analyzing it as a firewall.
  - Transparent operation mode: This mode allows for security scanning of traffic without the need for routing or NAT, used only on layer 2 of the OSI model, and offers the benefits of a firewall.

Configure a Traffic VDOM under **System** > **VDOM** > **Create new**. However, to choose NAT or transparent mode, you must use the CLI commands shown on this slide.



A LAN extension VDOM is dedicated to expanding your LAN using VPN IPsec and VXLAN to connect to a FortiGate LAN extension.

Each VDOM type serves distinct functions, from managing the FortiGate device, to handling traffic with or without NAT, to expanding the LAN, thus offering flexibility in how networks are segmented and managed.

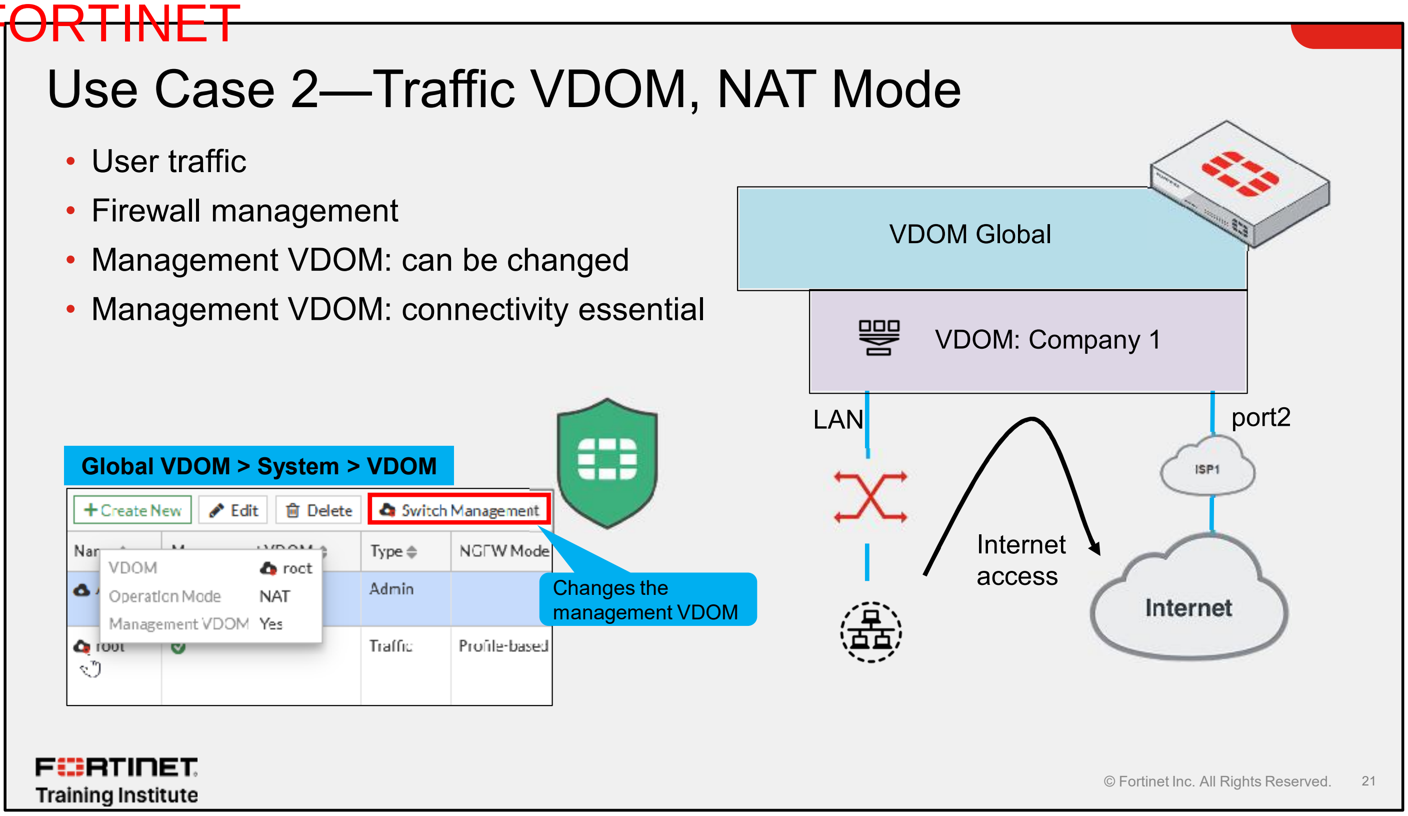


The admin VDOM does not handle internet access or user traffic, except for allowing FortiGate to access the internet to download FortiGuard updates, which will affect all VDOMs.

This type of VDOM has only the following FortiGate modules:

- Dashboard
- Network
- System
- Security Fabric
- Log & Report

If you change a VDOM from traffic to admin, you will receive a message informing you that certain settings such as firewall **Policy and Objects**, **Security Profiles**, **Wifi & Switch Controllers**, **User & Authentication**, devices, and dashboards in this VDOM will be deleted; you need to confirm to proceed with the change.



A traffic VDOM operating in NAT mode functions like a regular firewall, passing traffic and being accessible to all network segments you manage.

You can also manage the firewall by configuring the administrative access of the interfaces belonging to this VDOM. In the example on this slide, the LAN and port2 interfaces can be accessed using protocols like SSH or HTTPS.

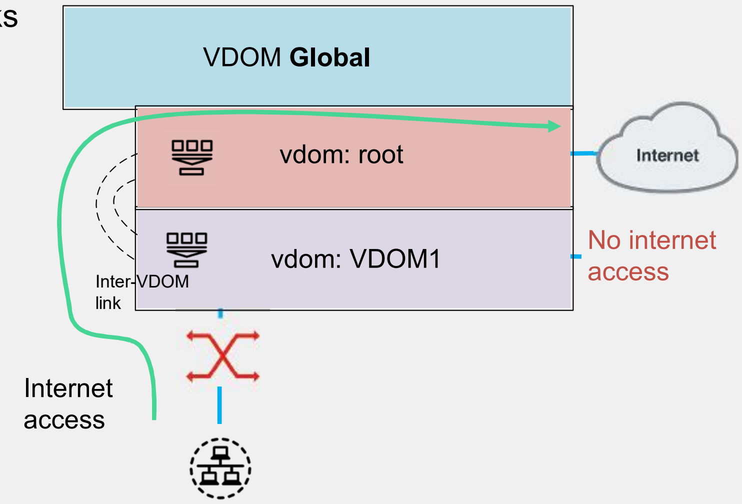
It's also vital to understand that a traffic VDOM can be used to download FortiGuard updates and validate the firewall license, if you select it as the **Management VDOM**. By default, the root VDOM serves as the management VDOM, but you can change it as needed.

In the screenshot displayed on the slide, a red dot indicates the management VDOM. The **Admin** VDOM type is selected. You can click **Switch Management** to switch the management VDOM.

If your management VDOM does not have internet access or is not connected to a FortiManager serving as a FortiGuard Distribution Server (FDS) in a closed network, then the firewall and all VDOMs may experience issues with the FortiGuard databases, potentially leading to firewall malfunctions.

Use Case 3—Internet Access via Inter-VDOM Routing

- Internet from root to VDOM1
- Connectivity to different VDOM networks



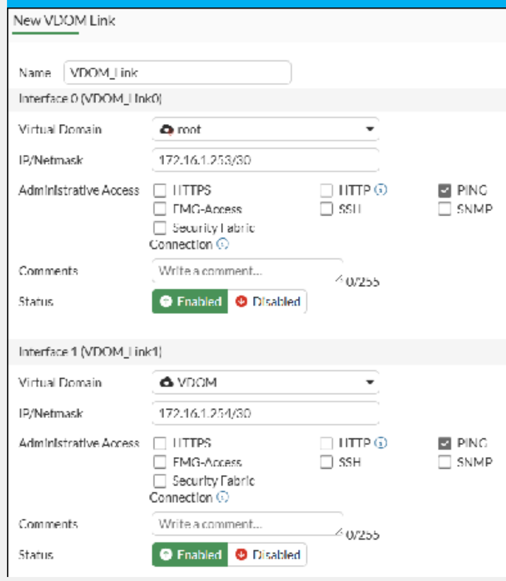
If your VDOM (VDOM1, for instance) lacks internet access, and you need to connect through another firewall VDOM (root) that has internet, an inter-VDOM Link enables VDOM1 to access the internet via the root VDOM.

An inter-VDOM link facilitates not only internet access between VDOMs but also connectivity for networks across different VDOMs.

Understanding VDOM Link Configuration

- Virtual interfaces
- How to create a vdom link
- 0 and 1 on interface names
- Static or dynamic routing
- Policies

VDOM Global > Network > Interfaces > VDOM Link



VDOM Global > Network > Interfaces

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VDOM links, or inter-VDOM links, enable VDOMs to communicate via virtual interfaces, eliminating the need for physical cables or interfaces.

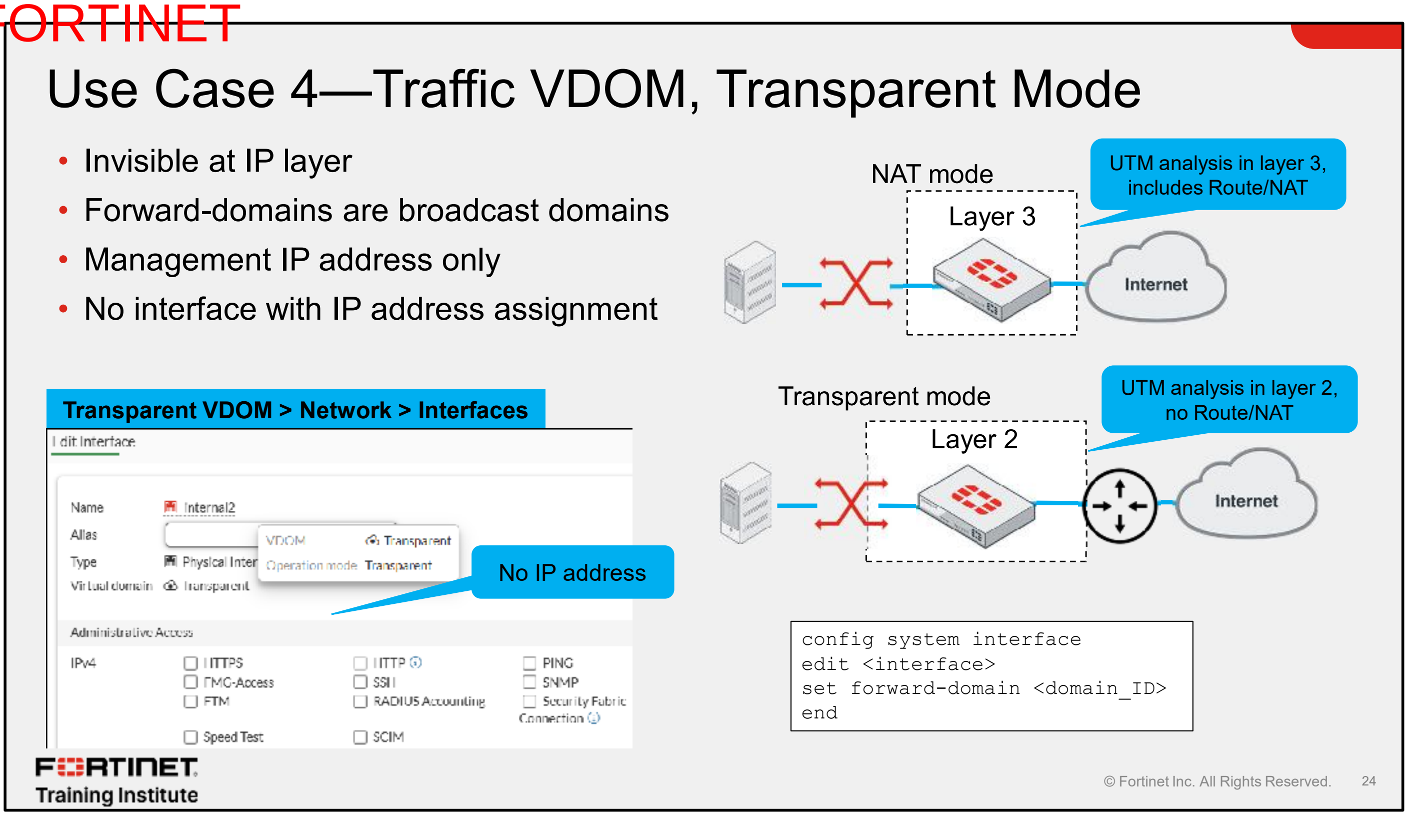
To create a VDOM link, click: **Global VDOM** > **Network** > **Interfaces** > **Create New** > **VDOM Link**.

FortiGate automatically assigns unique names to VDOM links by appending 0 and 1 to your interface names. For example, an inter-VDOM link named **VDOM_Link** results in:

VDOM root: Interface 0 (**VDOM_Link0**)
VDOM: Interface 1 (**VDOM_Link1**)

This convention helps easily identify and manage interfaces across VDOMs, streamlining the configuration of inter-VDOM communication.

Additionally, to fully enable communication through inter-VDOM links, you must configure both static or dynamic routing and traffic management policies for these virtual interfaces.



A VDOM in transparent mode operates invisibly at the IP layer, unlike NAT mode where interfaces are assigned IP addresses.

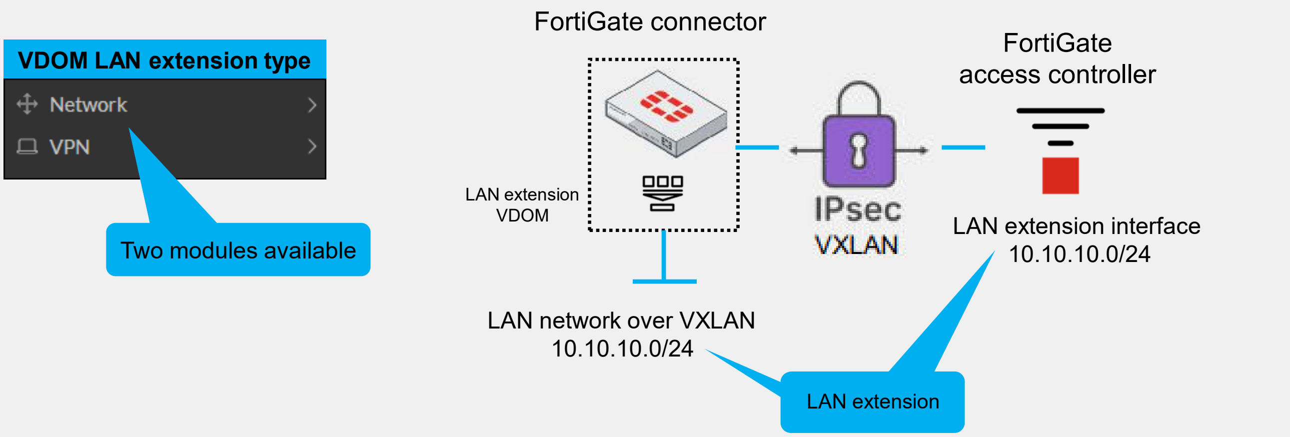
Transparent VDOMs enable UTM profile implementation for enhanced security and traffic analysis without extra layer 3 devices. Place the VDOM in transparent mode between network devices to boost security via FortiGate without changing layer 3 settings like gateways.

All interfaces in a transparent VDOM share the same broadcast domain. If your configuration includes more than two VLAN IDs, you should use the `forward-domain` command, as shown on the slide, to subdivide a VDOM into multiple broadcast domains using the VLAN ID as the domain ID for easy recognition.

Additionally, when you configure a transparent VDOM, you must assign a management IP address. If you don't, you will see an error message about a failed node check object for 'manageip'. This attribute is essential for administrative access and managing the VDOM without affecting its capability to handle traffic transparently.

Use Case 5—LAN Extension VDOM (FortiExtender)

- Connector and controller
- VXLAN/UDP encapsulation
- Modules available



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The LAN extension VDOM is established between a connector (such as FortiGate or FortiExtender) and a FortiGate access controller to facilitate network expansion using the VPN IPsec and VXLAN protocols.

The LAN extension VDOM has only the network and VPN modules.

Comparing VDOMs

VDOM Type	Enable VDOM	Purpose	Traffic	Default VDOMs per FortiGate	Usage
Admin	GUI/CLI	Administration and management Validate and download FortiGuard updates	✗	1	Management network in ISFW, DCFW, NGFW, DEFW
Traffic NAT	GUI/CLI	Pass traffic like a regular firewall	✓	10 *	Traffic network ISFW, DCFW, NGFW, DEFW
Traffic transparent	GUI/CLI	Layer 2 with UTM firewall analysis	✓	10 *	Anywhere in layer 2
LAN extension	GUI/CLI	Provide remote connectivity to a local FortiGate	✓	10 * (* more through licensing)	ISFW, NGFW, DEFW

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Now you will learn more about the benefits and characteristics of each type of VDOM available on FortiGate.

The primary functions of an admin VDOM are centered around administration, management, and the validation and downloading of FortiGuard updates. In contrast, the traffic VDOM, depending on its configuration, either passes traffic similar to a conventional firewall in NAT mode, or operates at layer 2 with unified threat management (UTM) firewall analysis in transparent mode. The LAN extension VDOM specializes in providing direct and secure connectivity to the central FortiGate network.

Of the three types, only the admin VDOM is not designed for traffic. The traffic VDOM, in both its operational modes, and the LAN extension VDOM, facilitate the flow of network traffic.

Both traffic type and LAN extension VDOMs support up to 10 VDOMs by default, with the option to expand this number through additional licensing. However, the admin VDOM is limited to a single VDOM per FortiGate.

Each VDOM type has a typical deployment scenario:

- Admin VDOMs are usually found within management networks, ensuring centralized control and oversight.
- Traffic VDOMs in NAT mode often serve roles in various firewall types, such as internal segmentation firewall (ISFW), data center firewall (DCFW), next-generation firewall (NGFW), and distributed enterprise firewall (DEFW).
- Transparent mode is ideal for environments requiring LAN analysis with firewall benefits, without the need for NAT or routing.
- LAN extension VDOMs are well-suited for extending network capabilities, often in settings where NGFW and DEFW functionalities are required.

Review

- ✓ Understand how to configure VLANs on FortiGate
- ✓ Understand how to implement enterprise networks using VDOMs

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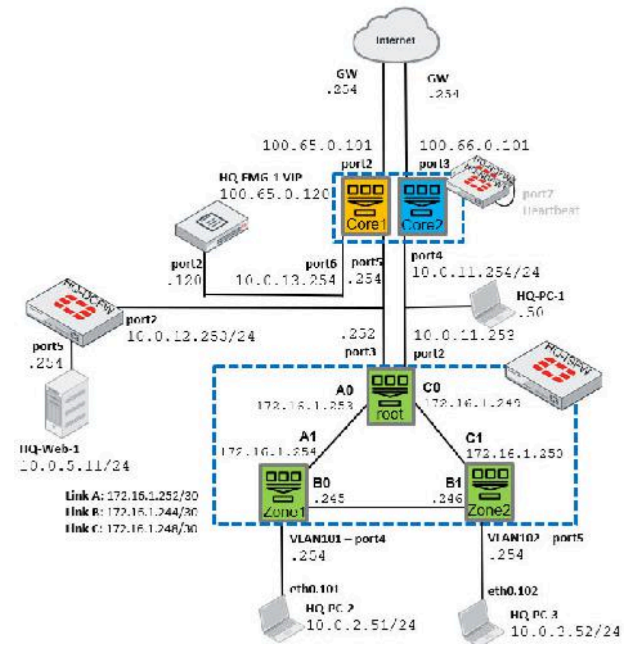
This slide shows the objectives that you covered in this lesson.

By mastering the objectives covered in this lesson, you learned how to use VLANs and VDOMs.

Lab 3—VLANs and VDOMs



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In this lab, you will enable VDOMs and create two additional VDOMs using FortiManager. You will configure two VLANs and assign them to VDOM Zone1 and VDOM Zone2. Then, you will configure inter-VDOM routing to allow each VLAN to have internet access using the **root** VDOM.

To achieve this, you will need to configure:

- VDOM links
- VLANs
- Static routes
- Firewall policies

In this lesson, you will learn about high availability (HA) on FortiGate.

Objectives

- Explore the FortiGate Clustering Protocol (FGCP)
- Examine the FortiGate Session Life Support Protocol (FGSP)
- Investigate the Virtual Router Redundancy Protocol (VRRP)
- Compare HA protocols

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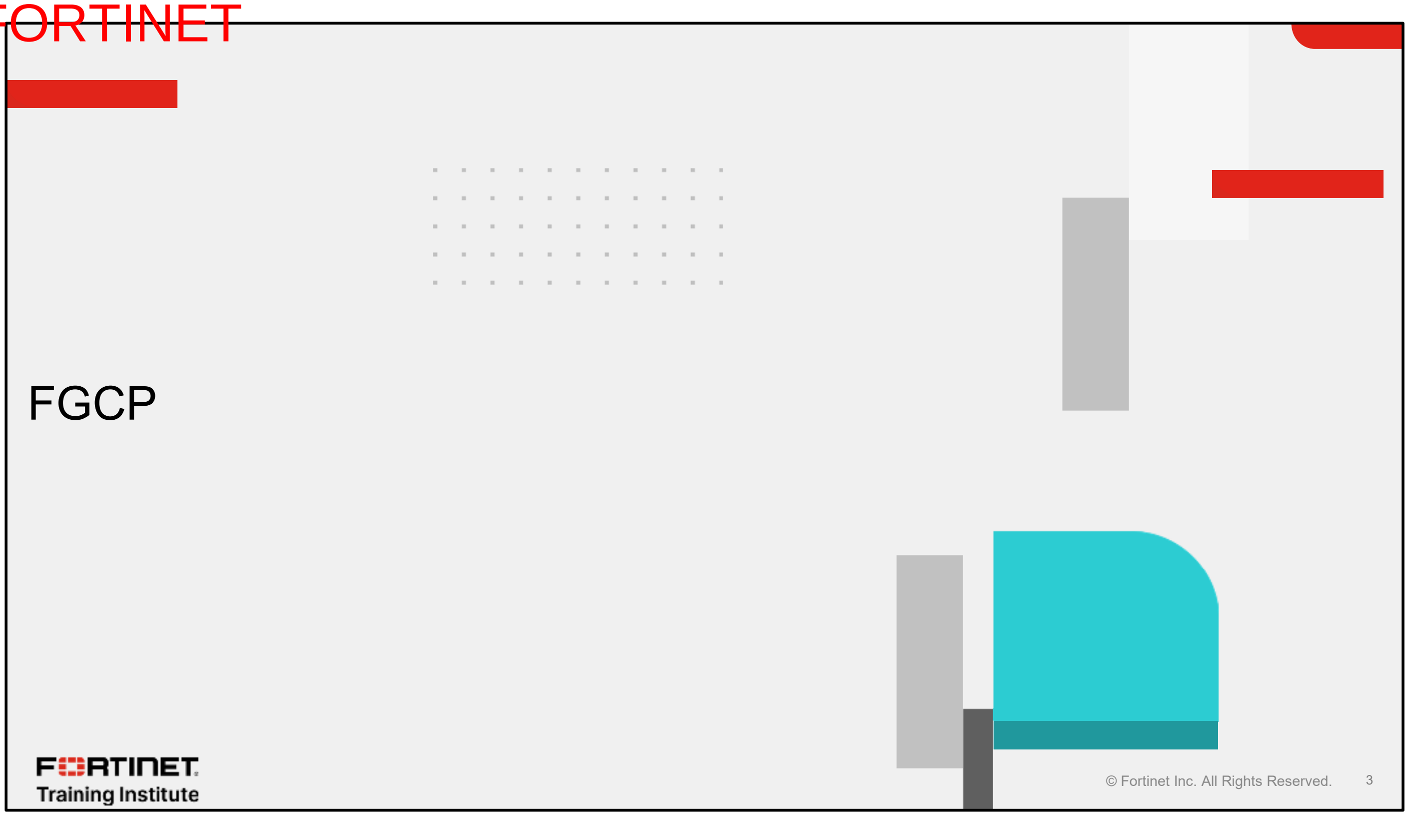
After completing this lesson, you should be able to achieve the objectives shown on this slide.

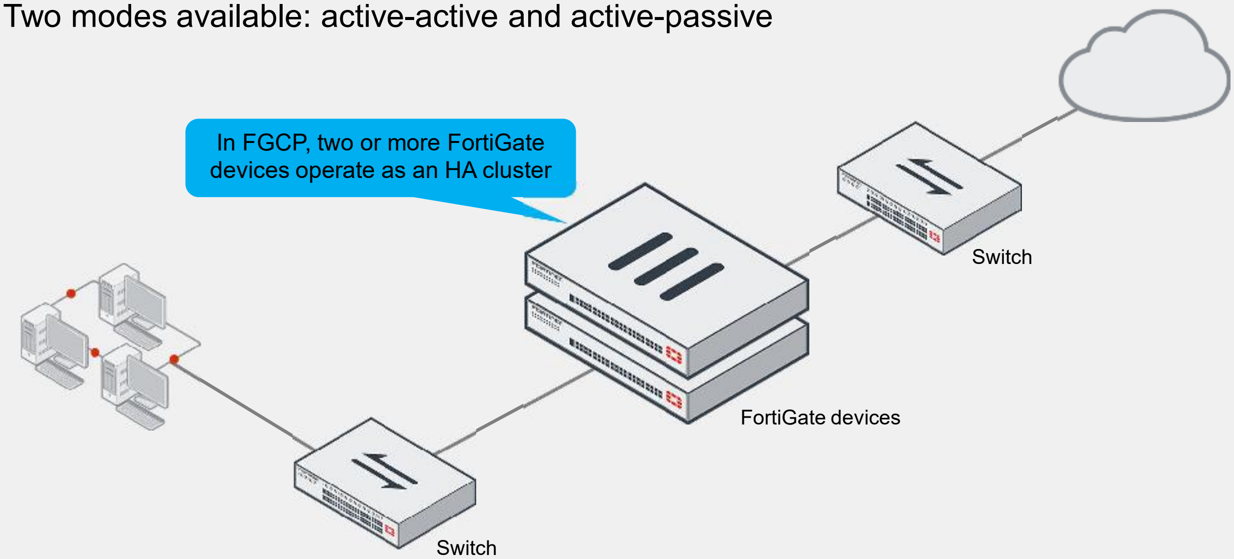
By demonstrating competence in HA, you will be able to understand and compare the different HA protocols, including FGCP, FGSP, and VRRP.



In this section, you will learn about FGCP.

FGCP Modes Overview

- Two modes available: active-active and active-passive



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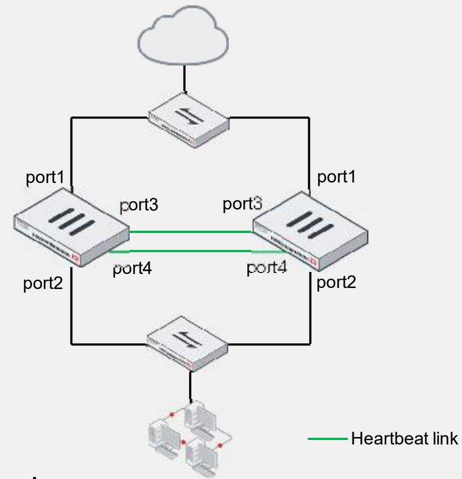
FGCP provides two HA operation modes: active-active and active-passive.

Active-passive mode provides a working FortiGate while the other is on standby. Active-active mode provides two active FortiGate devices. After learning HA communication and virtual MAC addresses, you will discover how FortiGate performs HA load balancing.

HA Requirements

- All members must have the same:
  - Model
  - Firmware version
  - Licensing
    - If different, the cluster uses the lowest-level license
  - Hard drive configuration
  - Operating mode (management VDOM)
- Setup:
  - Same HA group ID, group name, password, and heartbeat interface settings
- Best practice:
  - Use at least two heartbeat interfaces
  - Initially, switch DHCP and PPPoE interfaces to static configuration



Example:

```
config system ha
    set mode a-p
    set group-id 10
    set group-name "Training"
    set password <password>
    set hbdev "port3" 10 "port4" 20
end
```

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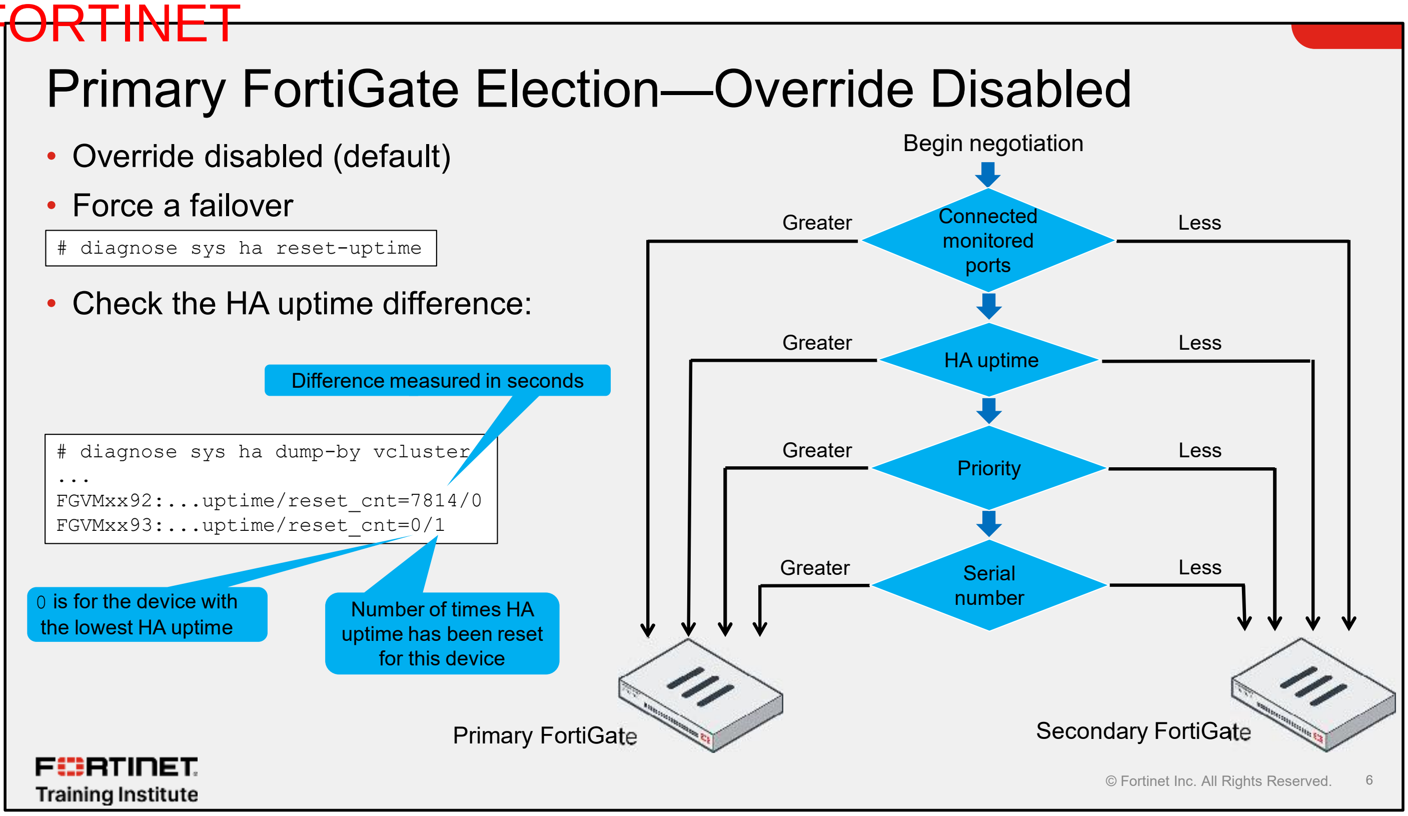
To successfully form an HA cluster, you must ensure that the members have the same:

- Model: the same hardware model or VM model
- Firmware version
- Licensing: includes the FortiGuard license, virtual domain (VDOM) license, FortiClient license, and so on
- Hard drive configuration: the same number and size of drives and partitions
- Operating mode: the operating mode—NAT mode or transparent mode—of the management VDOM. VDOMs divide a FortiGate device into two or more virtual units, essentially dividing one physical firewall into additional logical devices.

If the licensing level among members isn't the same, the cluster resolves to use the lowest licensing level among all members. For example, if you purchase FortiGuard Web Filtering for only one of the members in a cluster, none of the members will support FortiGuard Web Filtering when they form the cluster.

From a configuration and setup point of view, you must ensure that the HA settings on each member have the same group ID, group name, password, and heartbeat interface settings. Try to place all heartbeat interfaces in the same broadcast domain or, for two-member clusters, connect them directly. It's also a best practice to configure at least two heartbeat interfaces for redundancy purposes. This way, if one heartbeat link fails, the cluster uses the next one, as indicated by the priority and position in the heartbeat interface list. This slide shows how FortiGate defines the priority—a higher value means higher priority.

If you are using DHCP or Point-to-Point Protocol over Ethernet (PPPoE) interfaces, use static configuration during the initial setup of the cluster to prevent incorrect address assignment. After the cluster is formed, you can revert to the original interface settings.



This slide shows the criteria that a cluster considers during the primary FortiGate election process. The order in which the cluster evaluates the criteria depends on the HA override setting. This slide shows the order when the HA override setting is disabled, which is the default behavior. Note that the election process stops at the first matching criteria that successfully selects a primary FortiGate in a cluster.

1. The cluster compares the number of monitored interfaces that have a status of up. The member with the most available monitored interfaces becomes the primary.
2. The cluster compares the HA uptime of each member. The member with the highest HA uptime by at least five minutes becomes the primary.
3. The member with the highest priority becomes the primary.
4. The member with the highest serial number becomes the primary.

When HA override is disabled, the HA uptime has precedence over the priority setting. This means that if you must manually fail over to a secondary device, you can do so by reducing the HA uptime of the primary FortiGate. You can do this by running the `diagnose sys ha reset-uptime` command on the primary FortiGate, which resets its HA uptime to 0.

Note that the `diagnose sys ha reset-uptime` command resets the HA uptime and not the system uptime. Also note that if a monitoring interface fails, or a member reboots, the HA uptime for that member is reset to 0.

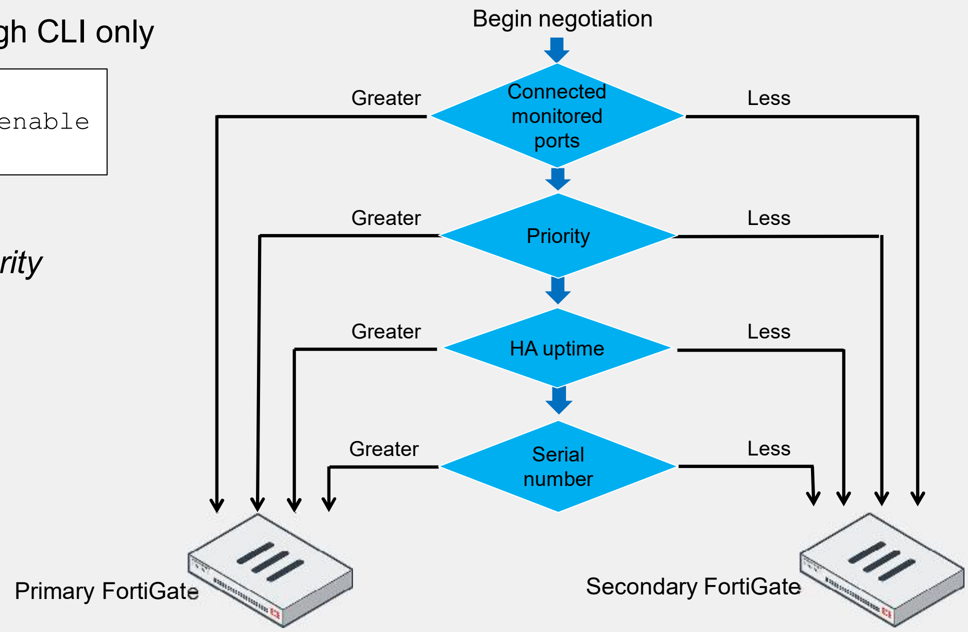
This slide also shows how to identify the HA uptime difference between members. The member with `0` in the `uptime` column indicates the device with the lowest uptime. The example shows that the device with the serial number ending in `92` has an HA uptime that is `7814` seconds higher than the other device in the HA cluster. The `reset_cnt` column indicates the number of times the HA uptime has been reset for that device.

Primary FortiGate Election—Override Enabled

- Configuration through CLI only

```
config system ha
    set override enable
end
```

- To force a failover:
  Change the HA *priority*



If the HA override setting is enabled, the cluster considers the priority before the HA uptime.

The advantage of this method is that you can specify which device is the preferred primary every time (as long as it is up and running) by configuring it with the highest HA priority value. The disadvantage is that a failover event is triggered when the primary fails and also when the primary is available again. That is, when the primary becomes available again, it takes its primary role back from the secondary FortiGate that temporarily replaced it.

When override is enabled, the easiest way of triggering a failover is to change the HA priorities. For example, you can either increase the priority of one of the secondary devices, or decrease the priority of the primary.

The override setting and device priority values are not synchronized with cluster members. You must manually enable override and adjust the priority on each member.

Ethernet Types and Synchronization Optimization

- Ethernet type 0x8890 and 0x8891
  - Heartbeats
  - Discover other FortiGate devices in the same HA group
  - Elect the primary
  - Synchronize other data
  - Detect when a device fails
  - Traffic redistribution
- Ethernet type 0x8893
  - Configuration synchronization
- For example, to sniffer HA heartbeat packets for a NAT/route mode cluster:

```
# diagnose sniff packet any "ether proto 0x8890" 4
```

- Dedicated session synchronization interfaces
  - `set session-sync-dev <interface1> <interface2>`

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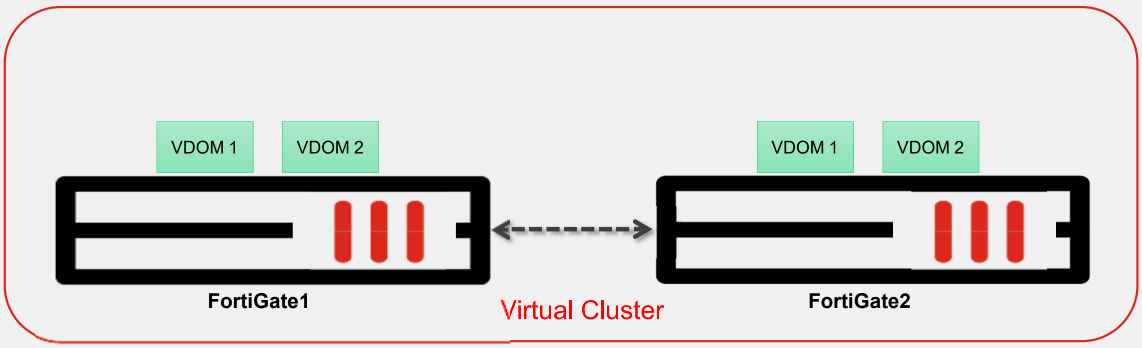
FortiGate HA uses FGCP for HA-related communications. FGCP travels among the clustered FortiGate devices over the links that you have designated as the heartbeats.

FGCP traffic uses a different Ethernet type than the IP protocol.

When session synchronization is enabled, you can also select dedicated interfaces in the HA setup for synchronizing sessions. By default, session synchronization occurs over the HA heartbeat link. Dedicated synchronization links reduce the bandwidth requirements of the HA heartbeat interface and improve the efficiency and performance of the cluster. Specifically, it is recommended that you use dedicated synchronization links because session synchronization requires a higher bandwidth with an impact on heartbeats when the links are not separated. When they are mixed, it can result in the potential loss of heartbeats and false failover when there is a very large number of sessions to be synchronized. If you select multiple interfaces, session synchronization traffic is load balanced among the selected interfaces. If all the session synchronization interfaces become disconnected, session synchronization reverts to using the HA heartbeat link.

What Is Virtual Clustering?

- Extension of FGCP for a cluster of FortiGate devices with multiple VDOMs enabled
- Virtual clustering operates in active-passive mode
- FortiGate virtual clustering is limited to 30 virtual clusters



For active-passive load balancing, you can consider virtual clustering for further redundancy and resource optimization in your enterprise network. Virtual clustering is essentially a cluster of FortiGate devices operating with multiple VDOMs enabled.

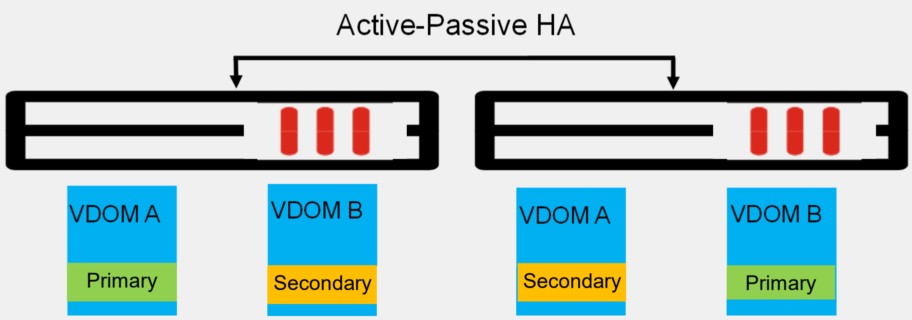
In active-active mode, you can load balance sessions between two cluster devices. In active-passive mode, you can configure a virtual cluster to provide standard failover protection between two instances of a VDOM operating on two different devices. Another way to load balance sessions within a virtual cluster is through VDOM partitioning.

Virtual clustering operates on a cluster of FortiGate devices. You can create up to 30 virtual clusters. If you want to create a cluster of more than two FortiGate devices operating with multiple VDOMs, you can consider other solutions that either do not include multiple VDOMs in one cluster, or employ a feature, such as standalone session synchronization with FGSP.

Other requirements for configuring virtual clustering are the same as those for a standard HA configuration, except that the maximum HA group ID value is seven.

Use Case—Virtual Clustering

- If you have two VDOMs with high traffic volume, then you can configure each cluster device to be the primary device for each VDOM
  - VDOM A and B with high traffic volume
  - Two FortiGate devices in a cluster, FortiGate1 and FortiGate2
  - For VDOM A, FortiGate1 is the primary device
  - For VDOM B, FortiGate2 is the primary device



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As an extension of virtual clustering, VDOM partitioning allows one FortiGate device to process all traffic for a VDOM, while the other FortiGate processes all traffic for the other VDOM. During a failover, one device in the cluster processes all traffic for all VDOMs.

Use Case—Virtual Clustering Configuration

- You must set HA mode to active-passive
- Uses VDOM partitioning to distribute traffic between both cluster devices
- Control the distribution of traffic between the devices in the cluster by adjusting which cluster device is the primary device for each VDOM

```
NGFW-1 #
config system ha
    set mode a-p
    ...
    set vcluster-status enable
    config vcluster
        edit 1
            set override disable
            set priority 200
            set vdom "VDOM1" "root"
        next
        edit 2
            set override disable
            set priority 50
            set vdom "VDOM2"
        next
    end
end
```

NGFW-1 firewall is primary device for VDOM1 and root VDOMs

NGFW-1 firewall is secondary device for VDOM2

```
NGFW-2 #
  config system ha
    set mode a-p
    ...
    set vcluster-status enable
    config vcluster
        edit 1
            set override disable
            set priority 100
            set vdom "VDOM1" "root"
        next
        edit 2
            set override disable
            set priority 200
            set vdom "VDOM2"
        next
    end
end
```

NGFW-2 firewall is secondary device for VDOM1 and root VDOMs

NGFW-2 firewall is primary device for VDOM2

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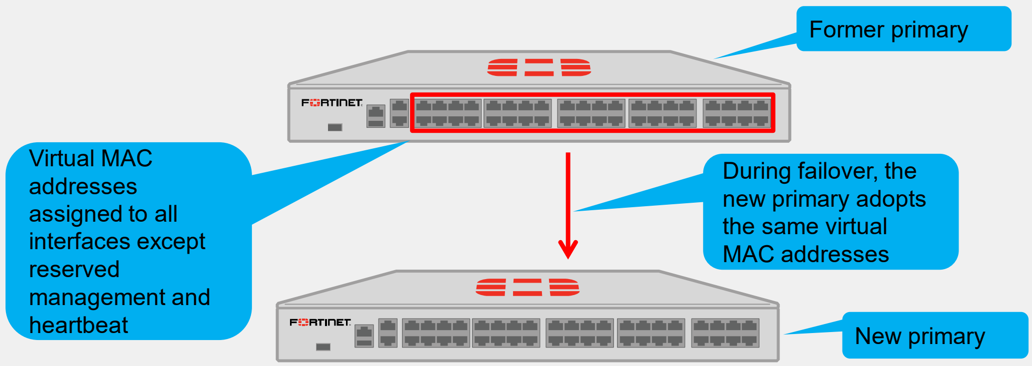
To configure VDOM partitioning, set one cluster device as the primary device for some VDOMs, and set the other cluster device as the primary device for other VDOMs. You can control the distribution of traffic between cluster devices by adjusting which cluster device is the primary device for each VDOM.

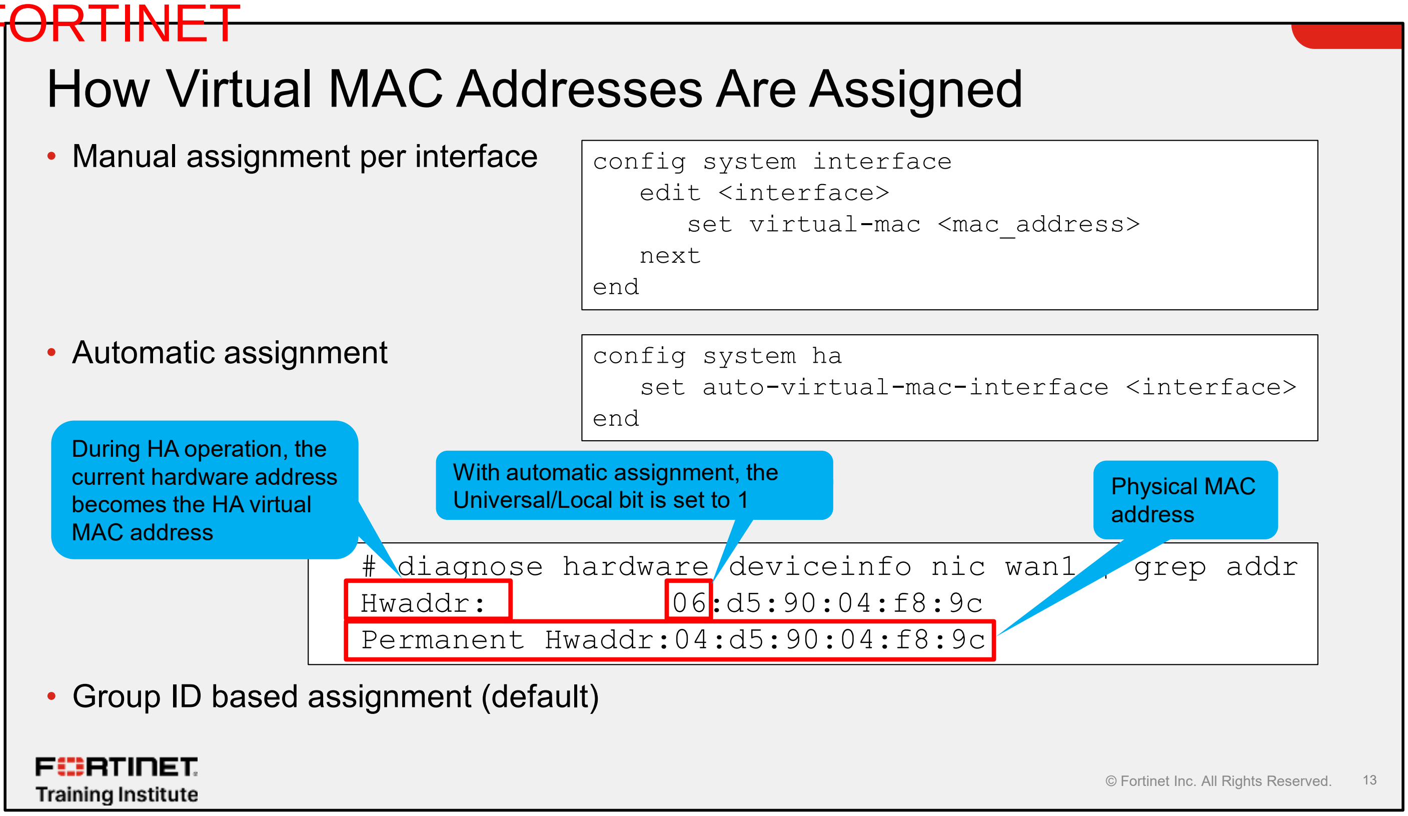
In the example shown on this slide, NGFW-1 is configured as the primary cluster device for VDOM1 and the root VDOMs, and NGFW-2 is the primary cluster device for VDOM2. The `set priority` command determines the role of the cluster device for each VDOM. The cluster device with the highest priority becomes the primary device for individual VDOMs in a VDOM partitioning setup.

Virtual MAC Addresses and Failover

- On the primary device, each interface—except HA heartbeat interfaces and reserved management interfaces—is given a virtual MAC address
- During a failover, the newly elected primary adopts the same virtual MAC addresses as the former primary



In addition to HA communication, another feature of the FortiGate HA solution is the use of virtual MAC addresses to forward traffic correctly. When a primary device joins an HA cluster, FortiGate gives each interface a virtual MAC address. The primary device informs all secondary devices about the assigned virtual MAC addresses. During a failover, a secondary device adopts the same virtual MAC addresses for equivalent interfaces.



FortiGate supports three methods of assigning virtual MAC addresses, in order of highest priority to lowest:

- Manual assignment per interface
- Automatic assignment per interface
- Group-ID-based assignment for all interfaces

When you set an interface to automatic assignment, it uses the hardware MAC address of the primary device with the second bit of the first octet set to 1. In the example shown on this slide, the first octet of the virtual MAC address is 00000110 (06 in hexadecimal) while the first octet of the hardware MAC address is 00000100 (04 in hexadecimal).

Default Virtual MAC Address Assignment

- Virtual MAC addresses are assigned using the following logics:

Logic	Vcluster ID	Group ID	Physical index
Logic 1	0 or 1	Greater than 255	Less than 128
Logic 2	0 or 1	Less than 256	Less than 128
Logic 3	0 or 1	Less than 256	Greater than 127
Logic 4	Greater than 1	0-7	Less than 1024

  - `Vcluster ID` is the vcluster number that the interface belongs to, minus one
  - `Group ID id` is the HA group ID
  - `Physical index` is the interface index number

```
config vcluster
  edit 1
```

Vcluster ID is 0

```
# diagnose sys ha dump-by debug-zone
...
Port1 ifindex=3 phyindex=0 mac=.. .
```

Interface index number

- Therefore, two or more HA clusters in the same LAN segment should use different HA group IDs to prevent virtual MAC address conflicts

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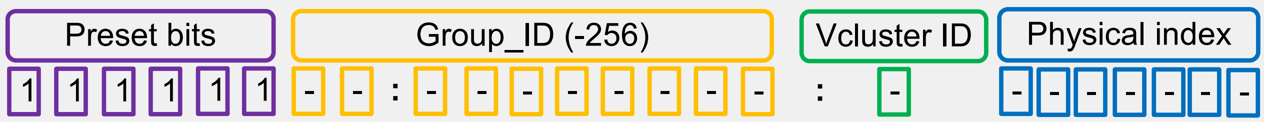
By default, FortiGate determines the HA virtual MAC addresses assigned to each interface by the virtual cluster ID, group ID, and interface index. Depending on these parameters, you must apply one of the four virtual MAC address assignment logics.

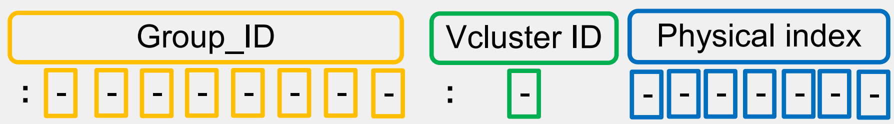
Because the group ID is a discriminator, Fortinet strongly recommends that you assign different HA group IDs to each cluster. If you have two or more HA clusters in the same broadcast domain using the same HA group ID, you might experience MAC address conflicts.

Default Virtual MAC Address Assignment (Contd)

- Logic 1: The virtual MAC address starts with `e0:23:ff` and then



- Logic 2 : The virtual MAC address starts with `00:09:0f:09` and then



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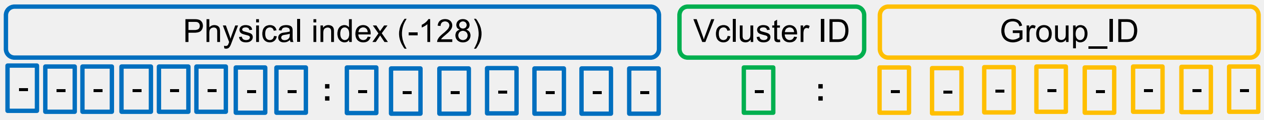
Each logic applies a different start of the virtual MAC address. The end of the virtual MAC address requires you to convert the group ID, virtual cluster ID, and physical index numbers to binary.

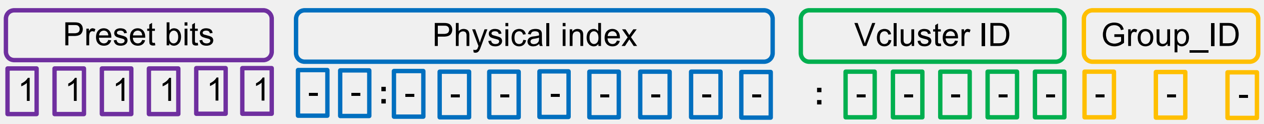
For logic 1, you must subtract 256 from the group ID before you convert the number to binary.

Default Virtual MAC Address Assignment (Contd)

- Logic 3: The virtual MAC address starts with `70:4c:a5` and then



- Logic 4: The virtual MAC address starts with `e0:23:ff` and then



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For logic 3, you must subtract 128 from the physical index before you convert the number to binary.

After you fill the table with the corresponding twenty-four bits, you can convert them in hexadecimal to obtain the virtual MAC address as shown with the command `diagnose hardware deviceinfo nic <port>`.

Virtual MAC Address—Examples

- Logic 2:
  - `Group ID` is set to `24` (`11000` in binary and `18` in hexadecimal)
  - Virtual cluster is disabled so `Vcluster ID` is `0`
  - Port2 `physical index` is `1`

Group ID	Vcluster ID	Physical index
:0 0 0 1 1 0 0 0:	0	0 0 0 0 0 0 1

```
# diag hardware deviceinfo nic port2 | grep addr
HWaddr:            00:09:0f:09:18:01
Permanent Hwaddr:02:09:0f:00:0a:01
```

Virtual MAC address start for logic 2

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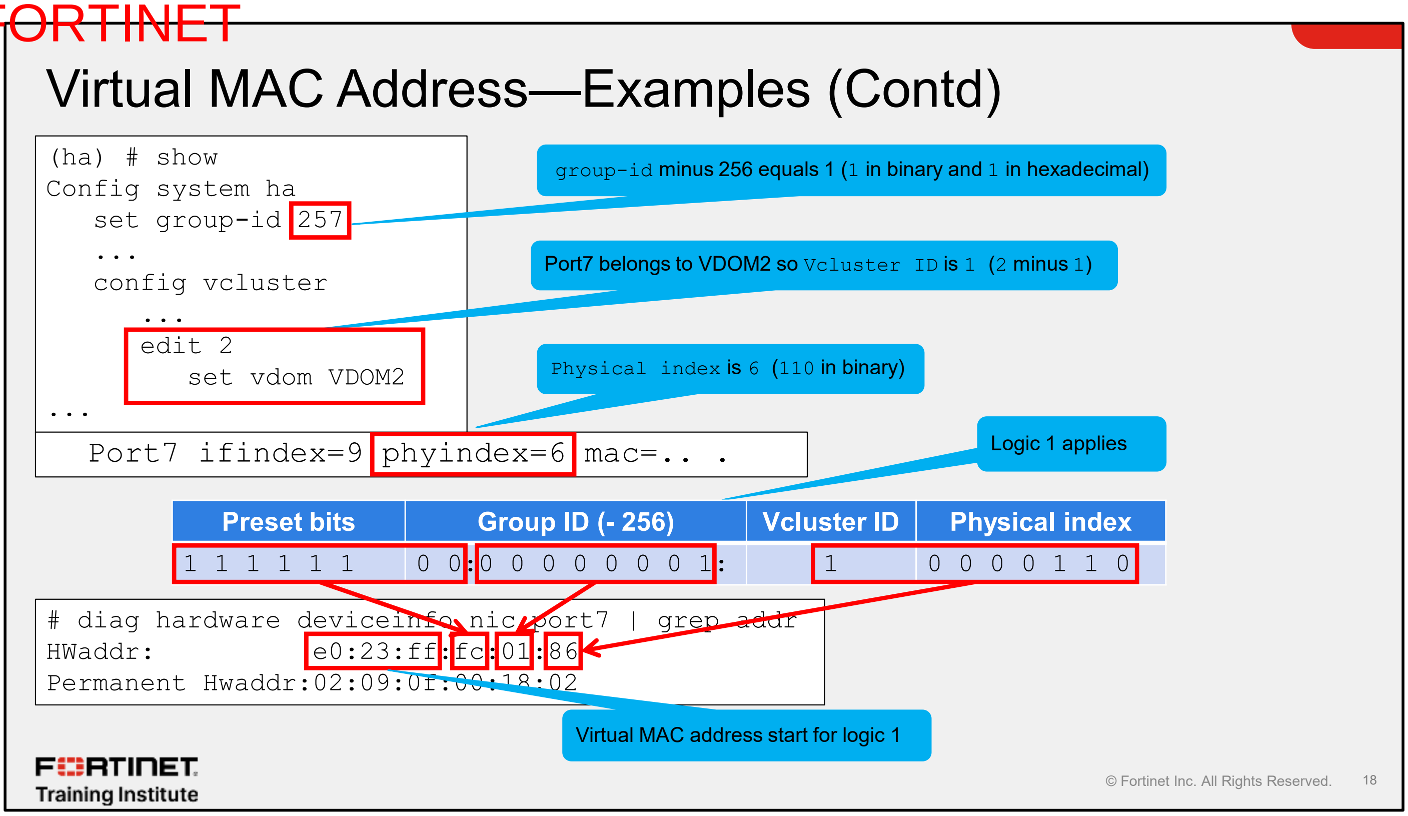
The example on this slide shows the virtual MAC address assignment when the parameters refer to logic 2.



The example on this slide shows the virtual MAC address assignment when the parameters refer to logic 1.

Virtual MAC Addresses and Failover

- After a failover, gratuitous ARP informs the network that the virtual MAC addresses are now reachable through a different device
- Some switches might not clear their MAC tables fast enough, so they would keep sending packets to the former primary device
- To shut down the interfaces of the former primary FortiGate (except the heartbeats and reserved management) for one second during failover, use the following commands:

```
config system ha
      set link-failed-signal enable | disable
end
```

Default value

- Because of the link outage, all switches detect the failure and clear their MAC tables

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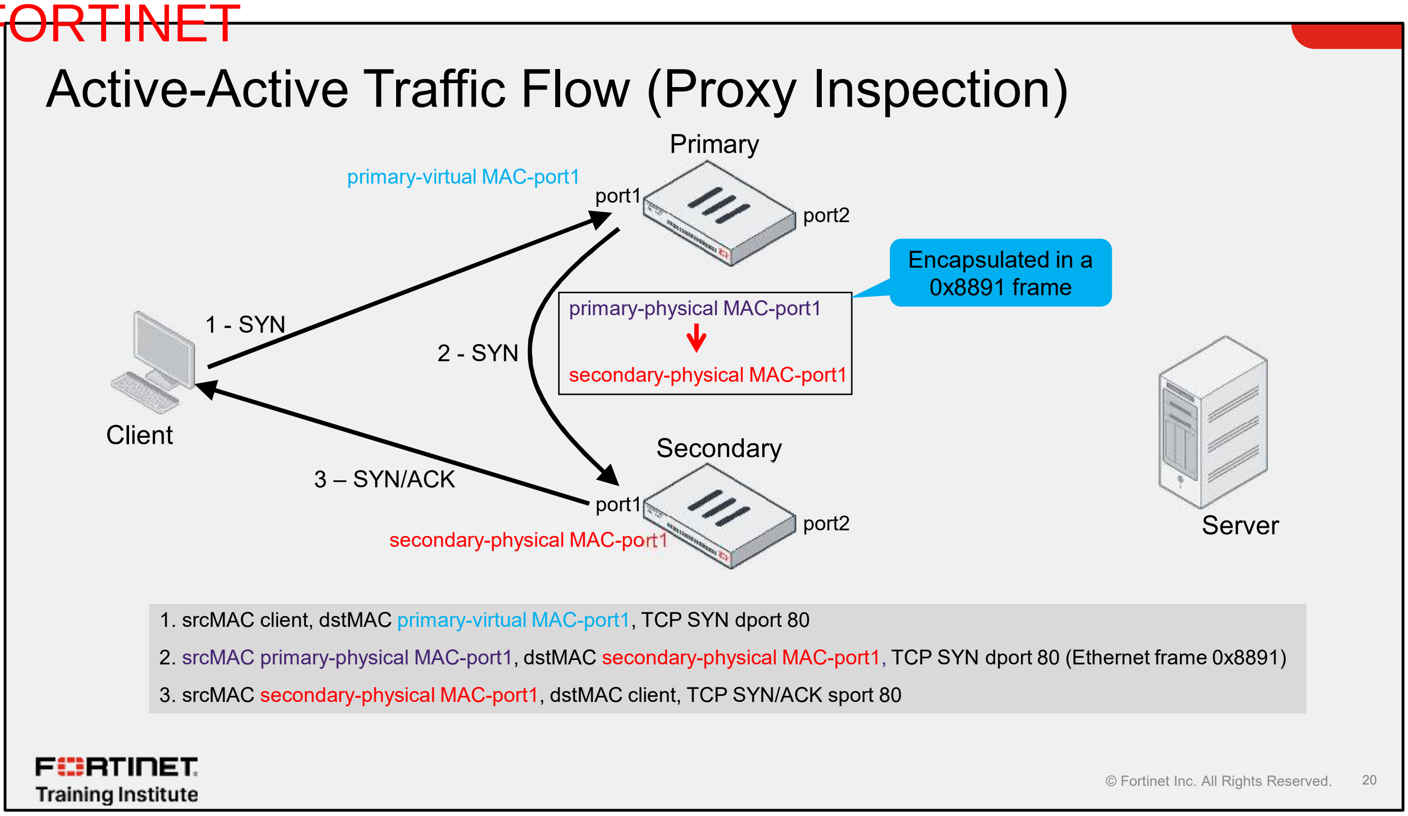
After a failover, the new primary device broadcasts gratuitous ARP packets, notifying the network that each virtual MAC address is now reachable through a different switch port.

In most networks, that's enough for the switches to update their MAC forwarding tables with the new information. However, some high-end switches might not clear their MAC tables correctly after a failover. So, they keep sending packets to the former primary device even after receiving the gratuitous ARPs. In these cases, you should use the command shown on this slide to force the former primary device to shut down all its interfaces for one second when the failover happens, excluding heartbeat and reserved management interfaces. This simulates a link failure that clears the related entries from the MAC table of the switches.



Besides failover, the virtual MAC is also important for HA load balancing. In active-active mode with proxy inspection enabled, the following occurs:

- The traffic destined to the cluster is sent to the primary. Because all network ports on the primary—except the heartbeat ports—are assigned a virtual MAC address, the traffic is destined to the virtual MAC address of the receiving port on the primary FortiGate.

- For traffic that is distributed to the secondary, the traffic destined to the endpoints is sent by the secondary. The traffic is sourced from the physical MAC address of the egressing port on the secondary.

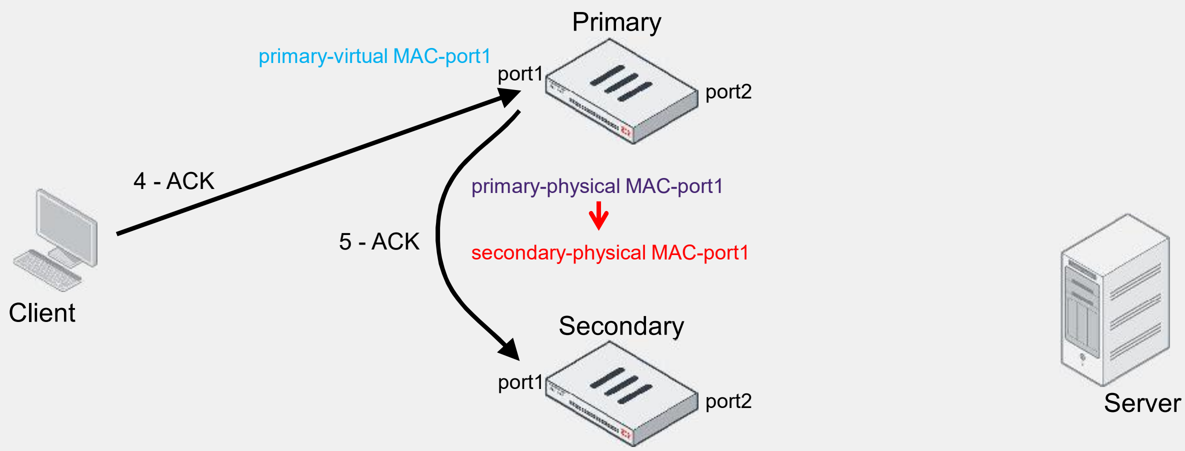
This slide shows the flow for distributed traffic that is subject to proxy inspection:

1. The client sends a SYN packet, which is forwarded to port1 on the primary. The packet destination MAC address is the virtual MAC address on port1.

2. The primary forwards the SYN packet to the selected secondary. In this example, the source MAC address of the packet is changed to the physical MAC address of port1 on the primary and the destination MAC address to the physical MAC address of port1 on the secondary. This is also known as MAC address rewrite. In addition, the primary encapsulates the packet in an Ethernet frame type 0x8891. The encapsulation is done only for the first packet of a load balanced session. The encapsulated packet includes the original packet plus session information that the secondary requires to process the traffic.

3. The secondary responds to the client with a SYN/ACK packet that contains the physical MAC address of port1 on the secondary as the source and the MAC address of the client as the destination.

Active-Active Traffic Flow (Proxy Inspection) (Contd)



4. srcMAC client, dstMAC primary-virtual MAC-port1, TCP ACK dport 80
5. srcMAC primary-physical MAC-port1, dstMAC secondary-physical MAC-port1, TCP ACK dport 80

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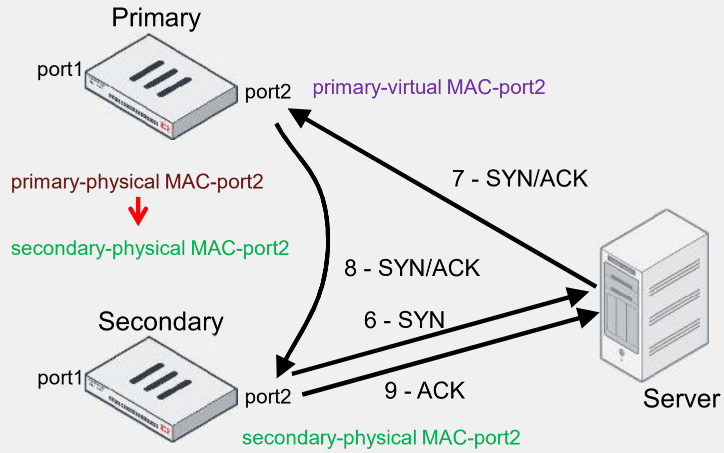
4. The client acknowledges the SYN/ACK by sending an ACK to the cluster. The ACK packet is destined to port1 on the primary.

5. The primary receives the packet and knows that it matches a session that was previously distributed to the secondary. As a result, the primary forwards the ACK packet to the corresponding secondary FortiGate. The packet is sourced from the physical MAC address of port1 on the primary and destined to the physical MAC address of port1 on the secondary. The three-way handshake on the client side is complete.

Active-Active Traffic Flow (Proxy Inspection) (Contd)

Limitation: FGCP active-active mode can't load balance traffic that traverses inter-VDOM links



6. srcMAC secondary-physical MAC-port2, dstMAC server, TCP SYN dport 80
7. srcMAC server, dstMAC primary-virtual MAC-port2, TCP SYN/ACK sport 80
8. srcMAC primary-physical MAC-port2, dstMAC secondary-physical MAC-port2, TCP SYN/ACK sport 80
9. srcMAC secondary-physical MAC-port2, dstMAC server, TCP ACK dport 80

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6. The secondary starts the connection with the server by sending a SYN packet using the physical MAC address of port2 as the source. Note that FortiGate contacts the server after it finishes the three-way handshake to the client, not before. The same behavior is seen when FortiGate operates in standalone mode and performs proxy-based inspection.

7. The SYN/ACK packet from the server is sent to port2 on the primary. The destination MAC address is the virtual MAC address of port2.

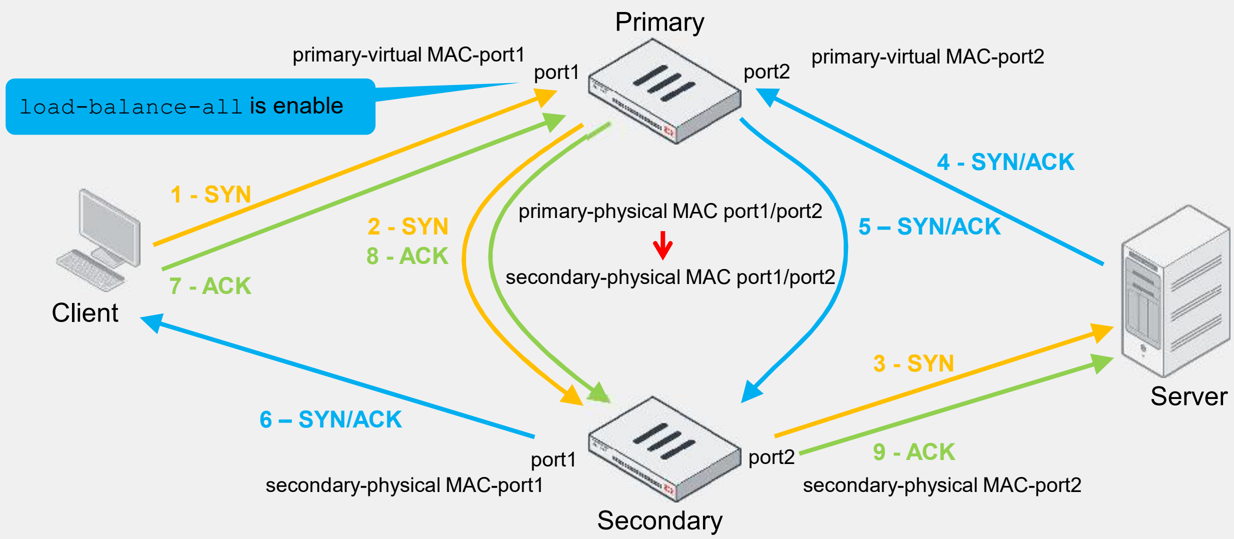
8. The primary receives the packet and knows that it matches a session that was previously distributed to the secondary. The primary forwards the SYN/ACK packet to the corresponding secondary FortiGate. The packet is sourced from the physical MAC address of port2 on the primary and destined to the physical MAC address of port2 on the secondary.

9. The secondary responds to the server with an ACK packet that contains the physical MAC address of port2 on the secondary as the source and the MAC address of the server as the destination.

The three-way handshake on the server side is also complete. From now on, packets that the client sends follow the same flow. For example, an HTTP GET request packet from the client is first received by the primary, which then forwards it to the secondary for proxy-based inspection. If the packet is allowed, the secondary forwards the packet to the server. Any server response packets to the client HTTP GET request are sent to the primary, which then forwards the packets to the secondary for inspection, and so on.

Note that the goal of active-active mode is to leverage unused CPU and memory resources on secondary devices. The intention is not really to load balance traffic. In fact, because the traffic from endpoints is always sent to the primary, you usually see more traffic on the primary than any secondary devices.

Active-Active Traffic Flow (No Proxy Inspection)



When there is no proxy inspection, that is, when traffic is either subject to flow inspection or no inspection at all, sessions are distributed to the secondary FortiGate only if you enable the `load-balance-all` setting (which is disabled by default) under HA configuration. In addition, as in proxy inspection, you will also see the following behavior:

- Traffic sourced from the client or server and destined to the FortiGate cluster is sent to the primary FortiGate. The source and destination MAC addresses are the endpoint (client or server) and the primary FortiGate virtual MAC address, respectively.
- The primary FortiGate may, in turn, forward the traffic to the secondary if the session is to be load balanced.
- When distributing the traffic to the secondary, FortiGate uses the physical MAC addresses of the primary and secondary device interfaces as the source and destination MAC addresses, respectively.
- If traffic is load balanced to the secondary FortiGate, any traffic sourced from the cluster and destined to the endpoint is sourced from the secondary FortiGate. This means that the source MAC address is the physical address of the secondary egress interface.

When compared to proxy inspection, the difference is that FortiGate does not reply to packets on behalf of the client or server. For example, instead of replying to the SYN packet that the client sends, FortiGate forwards the packet to the server through the secondary. Similarly, FortiGate forwards packets that the server sends to the client through the secondary.

Active-Active Load Balancing Methods

```
config system ha
(ha)# set schedule <none|leastconnection|round-robin|weight-round-robin|random|ip|ipport>
```

Default value

Method	Description
`none`	The primary device handles all sessions
`leastconnection`	Sessions are sent to the member with the least number of sessions
`round-robin`	Default method. Sessions are distributed equally across members
`weight-round-robin`	The more weight a member is assigned, the more sessions it handles
`random`	Sessions are distributed randomly across members
`ip`	Sessions with the same source and destination IP pair are handled by the same member
`ipport`	Distribution based on source address, source port, destination address, and destination port information

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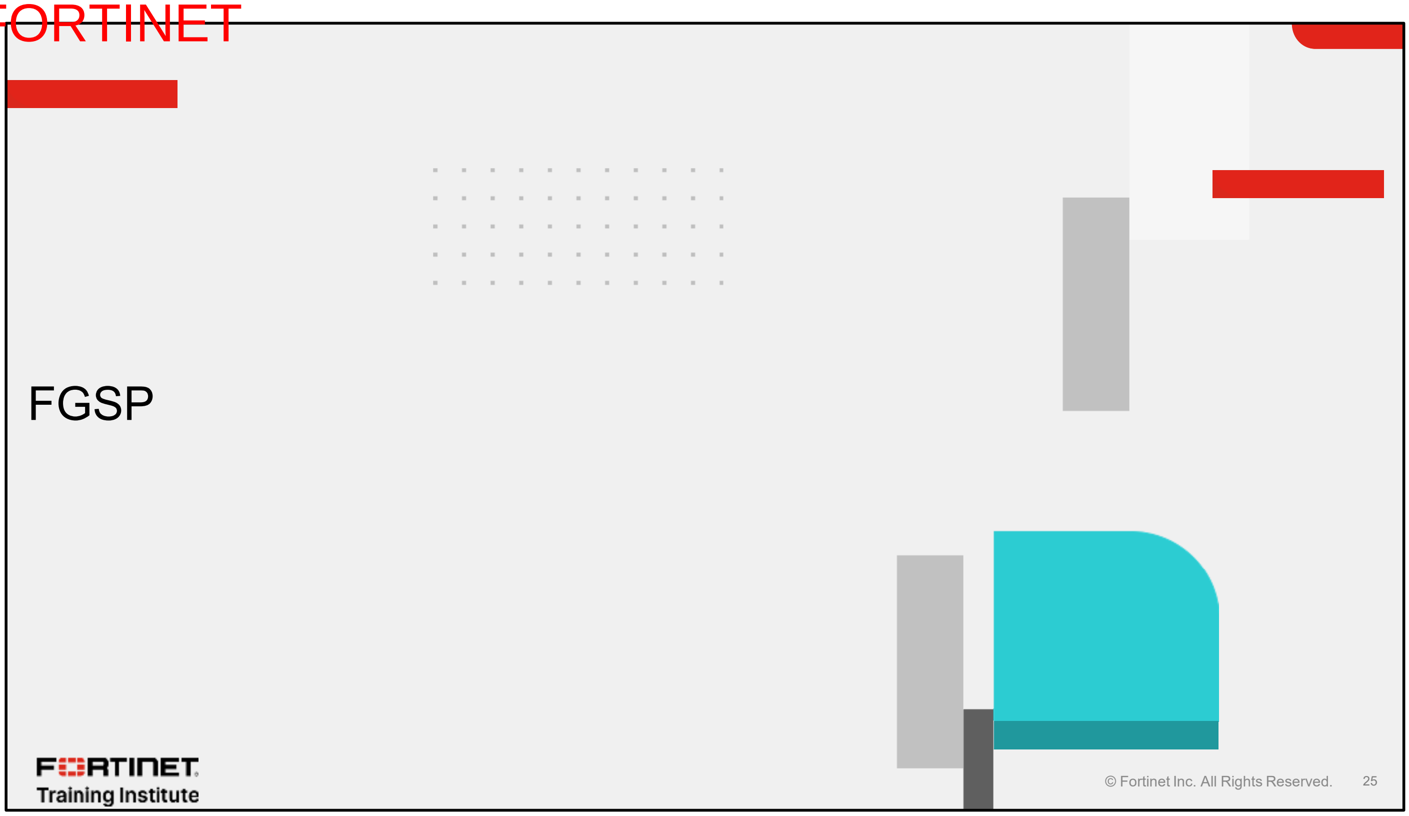
More generally on HA load balancing, only the primary device distributes sessions. Active subordinate devices do not make load balancing decisions. The primary device uses one of the following load balancing methods available for the parameter `schedule` in `config system ha`:

- `none`: Load balancing is turned off. The primary handles all sessions.
- `leastconnection`: The primary distributes sessions to the member with the least number of sessions.
- `round-robin`: This is the default method. The primary distributes sessions equally across members.
- `weight-round-robin`: The primary distributes sessions across members based on the member weight. The higher the member weight, the more sessions are distributed to that member.
- `random`: The primary distributes sessions randomly across members.
- `ip`: The primary distributes sessions with the same source and destination IP pair to the same member.
- `ipport`: The primary distributes sessions based on the source address, source port, destination address, and destination port information. The more diverse the traffic is, the more evenly the traffic is distributed across members.

By default, only sessions subject to proxy inspection are distributed to secondary devices. If you want to force the distribution of sessions that are subject to flow inspection or no inspection at all, then you must enable the `load-balance-all` setting under HA configuration.



In this section, you will learn about FGSP.

FGSP Overview

- HA solution only for sharing sessions
- Peer-to-peer communication between:
  - Standalone FortiGate
  - FGCP cluster
- Load balancing and failover is performed by external network devices
- All IPv4 and IPv6 TCP sessions, and IPsec tunnels are synchronized by default
- Configuration required to synchronize other types of sessions

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FGSP is another proprietary HA solution used only for sharing sessions between standalone FortiGate devices or an FGCP cluster. Session sharing is based on peer-to-peer communications between FortiGate devices where traffic is load balanced by an upstream load balancer and scanned by downstream FortiGate devices. FGSP is also compatible with FortiGate VRRP.

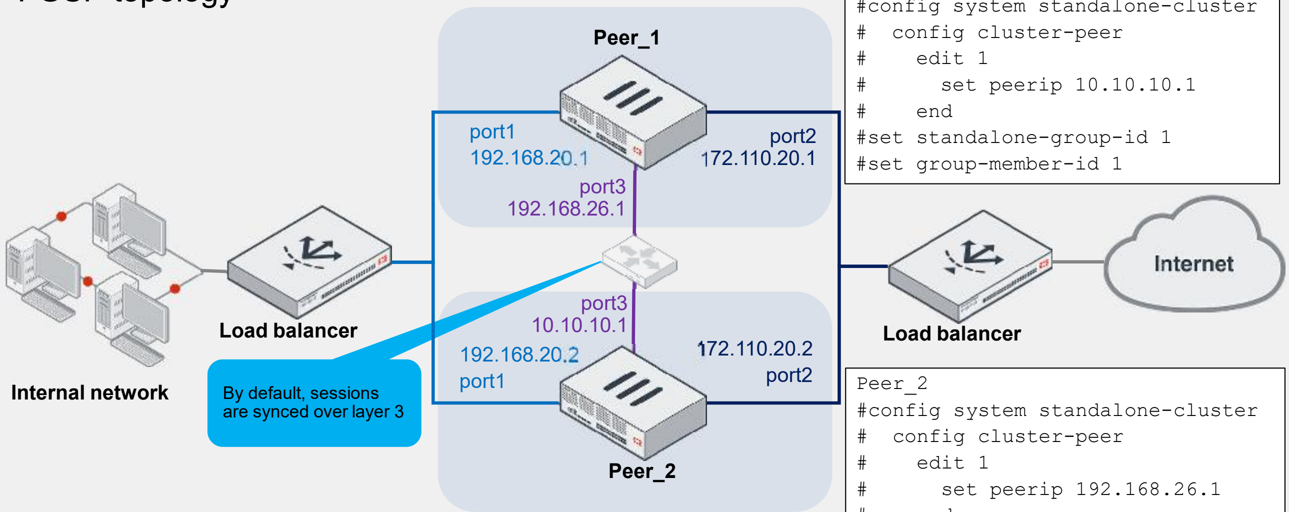
FortiGate devices in FGSP operate as peers that process traffic and synchronize sessions. An FGSP deployment can include 2-16 standalone FortiGate devices, or 2-16 FortiGate FGCP clusters of two members each. Adding more FortiGate devices increases the CPU and memory required to keep all the FortiGate devices synchronized, and increases network synchronization traffic.

FGSP can perform session synchronization of IPv4 and IPv6 TCP, SCTP, UDP, ICMP, expectation, and NAT sessions, to keep the session tables synchronized on all FortiGate devices. If one of the FortiGate devices fails, the upstream load balancer should detect the failed member and stop distributing sessions to it. Session failover occurs and active sessions fail over to the peers that are still operating. Traffic continues to flow on the new peer without data loss, because the sessions are synchronized.

By default, FGSP synchronizes all IPv4 and IPv6 TCP sessions, and IPsec tunnels.

Basic Topology

- FGSP topology




```
Peer_1
#config system standalone-cluster
#  config cluster-peer
#    edit 1
#      set peerip 10.10.10.1
#    end
#set standalone-group-id 1
#set group-member-id 1
```

```
Peer_2
#config system standalone-cluster
#  config cluster-peer
#    edit 1
#      set peerip 192.168.26.1
#    end
#set standalone-group-id 1
#set group-member-id 2
```

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FGSP is primarily used instead of FGCP, when external load balancers are part of the topology and are responsible for distributing traffic among the downstream FortiGate devices. FGSP provides the means to synchronize sessions between the FortiGate peers, without needing a primary member to distribute the sessions, like you do in FGCP active-active mode. If the external load balancers direct all sessions to one peer, the effect is similar to active-passive FGCP HA. If external load balancers balance traffic to both peers, the effect is similar to active-active FGCP HA. The load balancers should be configured so that all packets for any given session are processed by the same peer, including return packets, whenever possible.

This slide shows a basic FGSP setup. This example uses two peer FortiGate devices. The load balancer is configured to send all sessions to Peer_1 and, if Peer_1 fails, all traffic is sent to Peer_2. By default, FGSP peers use layer 3 connectivity to synchronize their sessions.

Session Synchronization

Session Types	Syntax

To sync connectionless sessions (UDP and ICMP)	`config system ha`
	`  set session-pickup enable`
	`  set session-pickup-connectionless enable`

To sync expectation sessions (pinholes opened by firewall like SIP or FTP traffic)	`  set session-pickup-expectation enable`
To sync NAT sessions	`  set session-pickup-nat enable`

To filter the session synchronization from a particular source, destination, or source addresses/subnets, or a destination interface in a configured cluster peer	`config system standalone-cluster`
	`  config cluster-peer`
	`    edit <sync-id>`
	`    config session-syn-filter`
	`      set srcintf <interface>`
	`      set dstintf <interface>`
	`      set srcaddr <IPv4_address/subnet>`
	`      set dstaddr <IPv4_address/subnet>`
	`      set srcaddr6 <IPv6_address/subnet>`
	`      set srcaddr6 <IPv6_address/subnet>`

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By default, FGSP synchronizes all IPv4 and IPv6 TCP sessions, and IPsec tunnels. However, you can add other sessions to synchronize between peer FortiGate devices.

The table on this slide shows the CLI commands that you can use to enable additional types of sessions to be synchronized. It shows also the available filters that you can configure on a cluster peer to synchronize only specific sessions.

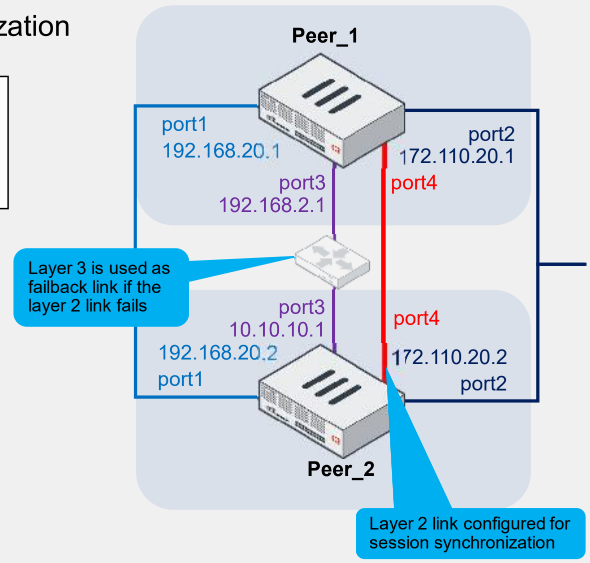
Session Synchronization Over Layer 2

- Layer 2 connectivity can be used for session synchronization

```
config system standalone-cluster
    set session-sync-dev port4
end
```

Configuration for the diagram shown

- Session synchronization will happen over layer 2 if:
  - Session synchronization interface is configured
  - Peer session synchronization interfaces are directly connected



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When peering over FGSP, by default, the FortiGate devices or FGCP clusters share information over layer 3 between the interfaces that are configured with peer IP addresses. You can also specify the interfaces used to synchronize sessions in layer 2 instead of layer 3 using the `session-sync-dev` setting. When a session synchronization interface is configured and FGSP peers are directly connected on this interface, then session synchronization is done over layer 2, only falling back to layer 3 if the session synchronization interface becomes unavailable.

If you are doing only session synchronization, and not configuration synchronization, no layer 2 connectivity is required.

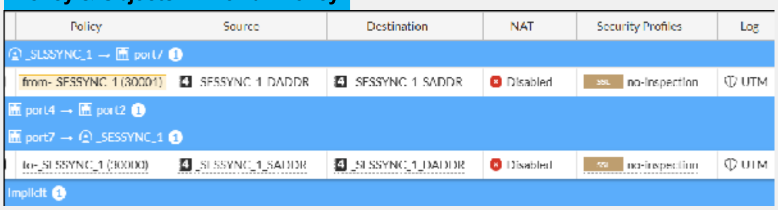
Use Case 1—Session Synchronization Encryption

- Session synchronization can be encrypted using an IPsec tunnel
  - Layer 3 connections
- Enable `encryption` and set the `psksecret`

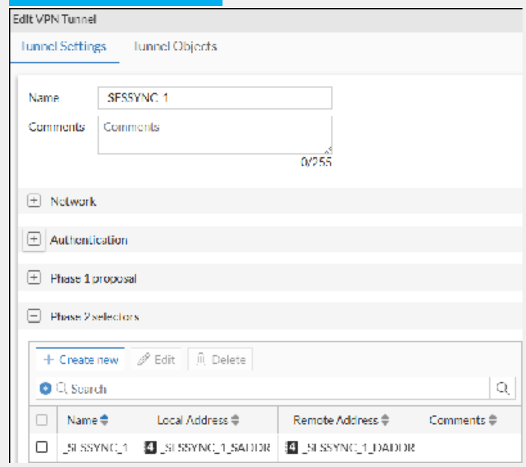
```
config system standalone-cluster
    set encryption enable
    set psksecret fortinet
end
```

- IPsec vpn and policies are automatically created

Policy & Objects > Firewall Policy



VPN> VPN Tunnels



- Useful for securing session sync data when sent between data centers

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You can enable encryption in the form of an IPsec tunnel for session synchronization. This is achieved by enabling `encryption` and setting the `psksecret` under `config system standalone-cluster`.

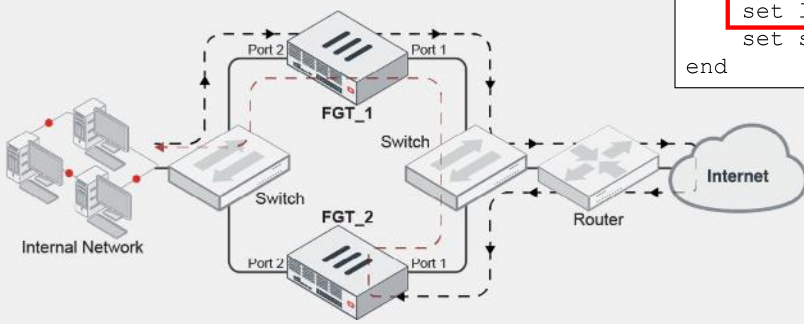
When enabled, the IPsec VPN and policies are automatically created. This is useful for securing the session synchronization data, especially if it is travelling between data centers. Note that, encryption is supported for layer 3 connections only.

Use Case 2—Inspection With Asymmetric Traffic—Layer 2

- Security inspection is supported with asymmetric traffic by passing all packets for a session back to the device that processed the first packet
- This requires that both FortiGate devices in the FGSP pair have:
  - Internal and outgoing interfaces in the same subnet
  - Layer 2 access with each other

```
config system standalone-cluster
    set standalone-group-id 1
    set group-member-id 1
    set layer2-connection available
    set session-sync-dev port1
end
```

Configuration required in this scenario



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When traffic passes asymmetrically through FGSP peers, UTM inspection can be supported by always forwarding traffic back to the session owner for processing. The session owner is the FortiGate device that receives the first packet of the session.

The layer 2 connection setting is for forwarded traffic between FGSP peers. Set it to `available` if the peer interface user for traffic forwarding is directly connected and supports layer 2 forwarding.

In the example shown on this slide, traffic from the internal network first hits FGT_1, but the return traffic is routed to FGT_2. Consequently, traffic bounces from FGT_2 port1 to FGT_1 port1 using the MAC address of FGT_1. FGT_1 then inspects the traffic.

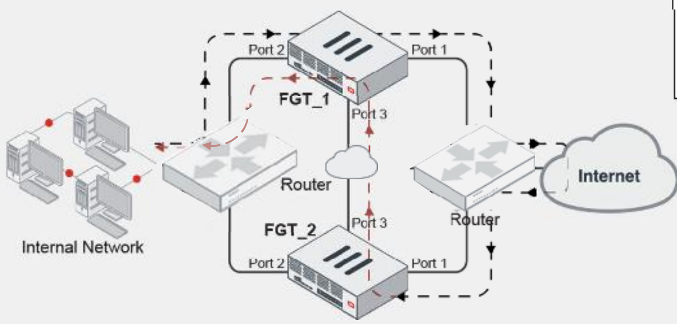
This example requires that the internal and outgoing interfaces of both FortiGate devices in the FGSP pair are in the same subnet, and that both peers have layer 2 access with each other.

Use Case 3—Inspection With Asymmetric Traffic—Layer 3

- Some environments do not have layer 2 connectivity
  - Cloud environments
- Traffic is forwarded through the peer interface
- Traffic is forwarded using a UDP connection

```
config system standalone-cluster
    set standalone-group-id 1
    set group-member-id 1
    set layer2-connection unavailable
    unset session-sync-dev
end
```

Default value



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For networks where layer 2 connectivity is not available, such as cloud environments, traffic bound for the session owner is forwarded through the peer interface using a UDP connection.

In the example shown on this slide, traffic from the internal network first hits FGT_1, but the return traffic is routed to FGT_2. Consequently, return traffic is packed and sent from FGT_2 to FGT_1 using UDP encapsulation between two peer interfaces (port 3). FGT_1 then inspects the traffic.

Configuration Synchronization

- FGSP configuration synchronization is also known as standalone configuration synchronization
- Configuration and external files are synced over heartbeat connections from one device to another in the same way as in FGCP
  - Layer 2 connectivity required
  - Configure multiple heartbeat ports for redundancy
- Standalone configuration synchronization is basically the same configuration synchronization feature that is available in FGCP
  - A primary device is elected by following the same election process
  - The primary and secondary device roles are only for config sync purposes and *not* for traffic
  - The same two-layer approach that is used for config sync in FGSP, is used in FGCP

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In addition to enabling session synchronization for FGSP deployments, you may want to enable configuration synchronization as well. Configuration synchronization, also known as standalone configuration synchronization, is basically the same configuration synchronization feature that is available in FGCP deployments, but it can be enabled separately, in case you want to synchronize the configuration between two devices operating in standalone mode. Remember, FGSP is just a cluster formed by two devices or FGCP clusters operating in standalone mode, but they can synchronize sessions and configuration between them.

Standalone configuration synchronization allows for configuration and external files to synchronize from one device to another, in the same way it happens for FGCP. The configurations from **Policy & Objects** that are synchronized between peers include policies, objects, users, and security profiles. For the election, a primary device is elected by the same election process that is used in FGCP. Just keep in mind that in standalone configuration synchronization, the primary and secondary device roles are useful only for configuration synchronization purposes and *not* for traffic processing purposes.

Finally, the approach used by standalone configuration synchronization is the same two-layer approach taken by FGCP.

HA Settings for Configuration Synchronization

- Device 1

```
config global
config system ha
    set group-id 1
    set group-name "Cluster1"
    set mode standalone
    set password fortinet1
    set hbdev "port2" 50 "port3" 50
    set standalone-config-sync enable
    set override disable
    set priority 200
end
```

- Device 2

```
config global
config system ha
    set group-id 1
    set group-name "Cluster1"
    set mode standalone
    set password fortinet1
    set hbdev "port2" 50 "port3" 50
    set standalone-config-sync enable
    set override disable
    set priority 100
end
```

Settings to enable (standalone) configuration synchronization

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If you want to enable standalone configuration synchronization, you can apply a configuration like the one shown on this slide. The key is to set `mode` to `standalone`, and after that, enable the `standalone-config-sync` option. The rest of the settings have the same purpose as they do in active-passive mode; therefore, they are used for primary device election, heartbeat communication, and so on.

FGSP—Coverage and Limits

FGSP	Coverage	Limits
Redundancy	Through FortiGate devices	Same model, firmware version, and licensing (temporary exceptions for example during upgrades)
Load balancing	Through external load balancer	Each FortiGate device must have a unique IP
Sessions synchronization	By default, IPv4 and IPv6 TCP sessions and IPsec tunnels Configuration required for SCTP, UDP, ICMP, expectation, and NAT sessions	Configurations related to session tables must match
Standalone configuration synchronization	For policies and objects	Layer 2 adjacencies required
Performances with security profiles	Available in any case	Better performances when UTM is enabled with symmetric traffic

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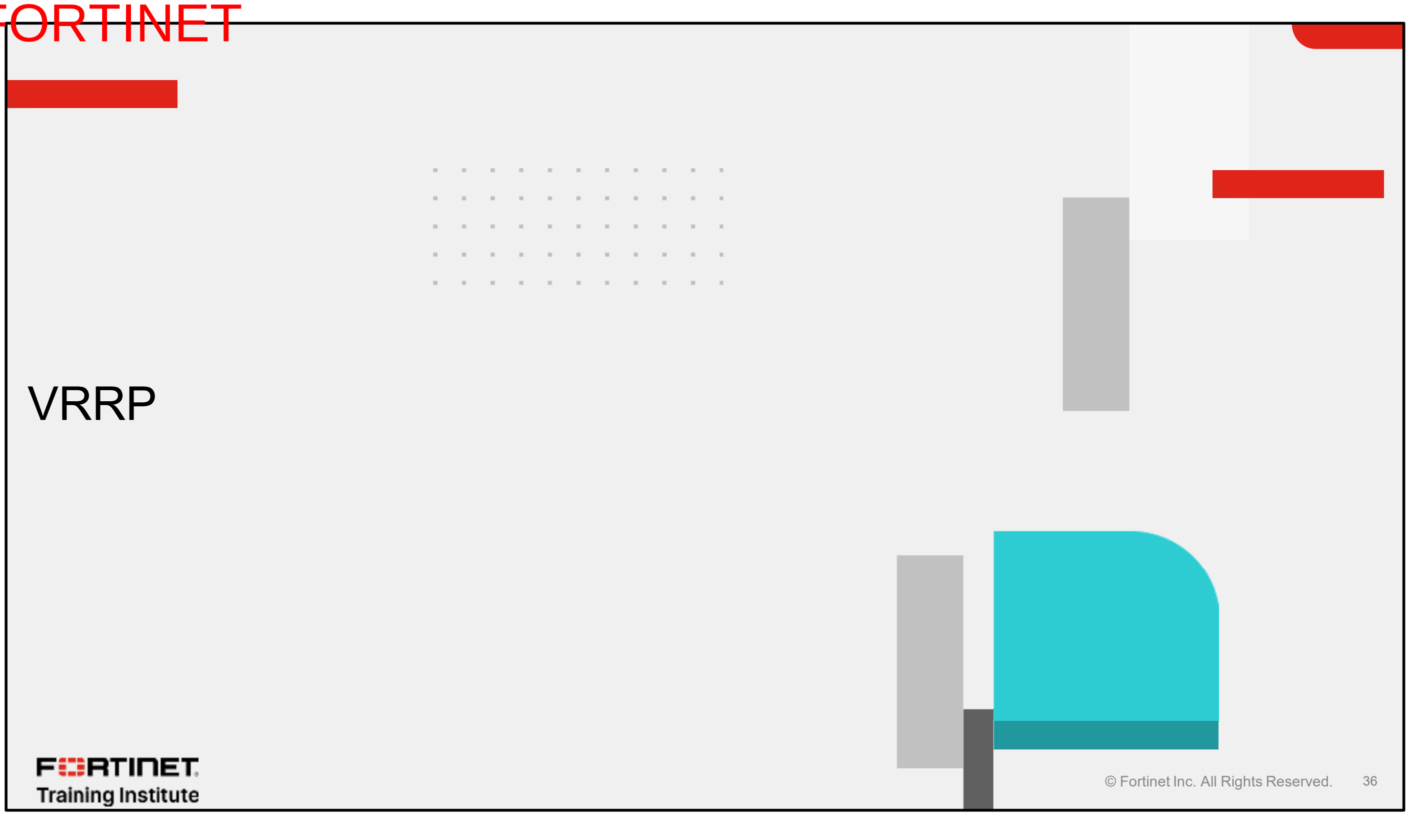
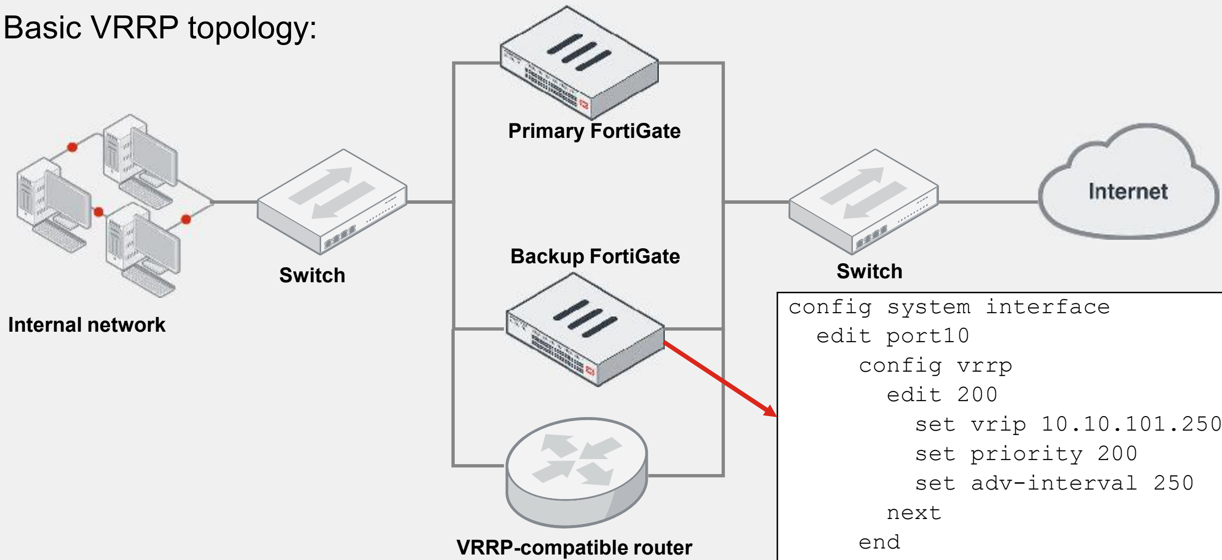
This slide summarizes what is included in FGSP and the corresponding limits.



In this section, you will learn about VRRP.

Virtual Router Redundancy Overview

- Basic VRRP topology:




```
config system interface
  edit port10
    config vrrp
      edit 200
        set vrip 10.10.101.250
        set priority 200
        set adv-interval 250
      next
    end
  next
end
```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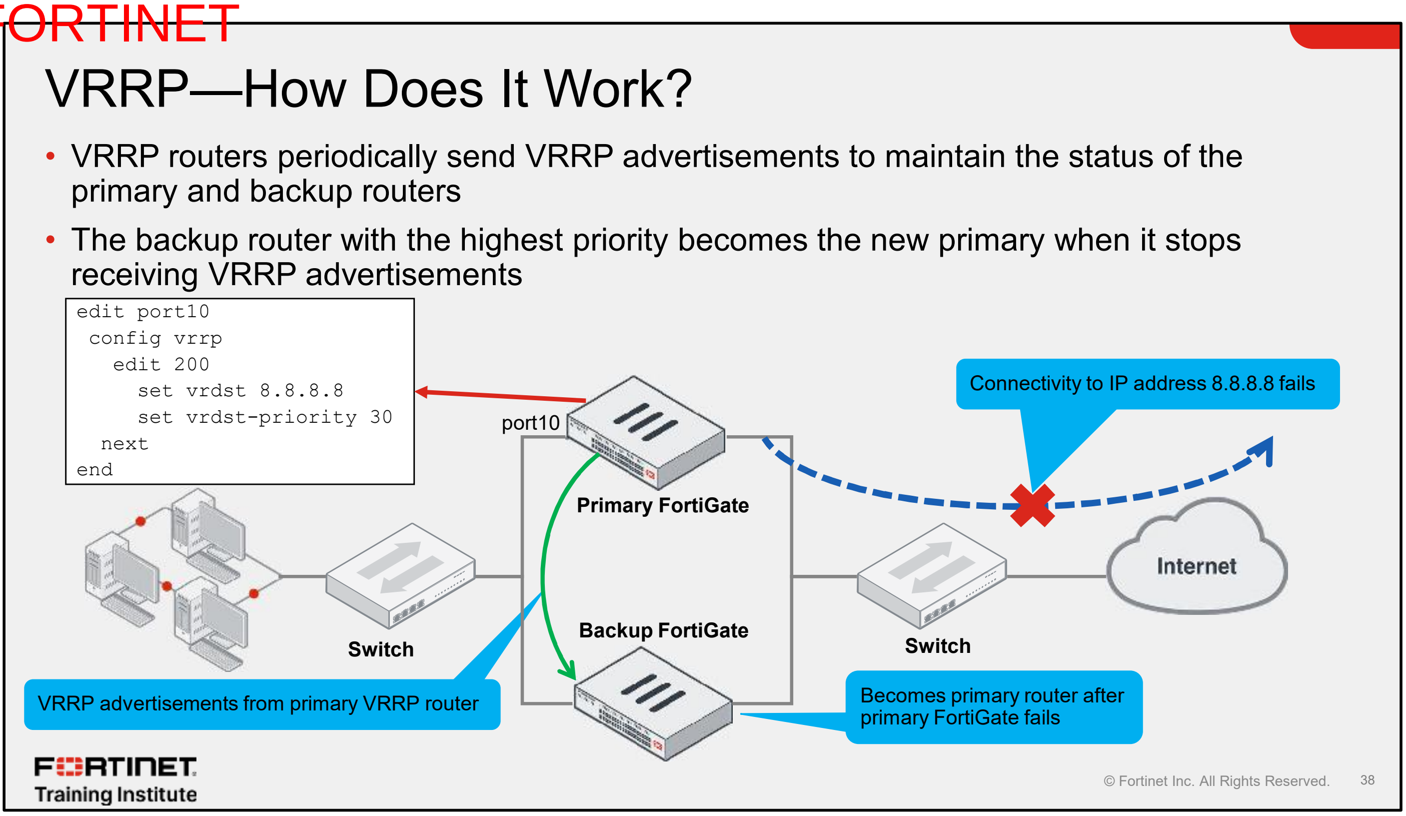
This slide shows the basic single-domain VRRP topology and the configuration required at the physical or VLAN interface level. You can use a VRRP configuration as a high availability solution to ensure that a network maintains connectivity with the internet (or with other networks), even if the default router for the network fails. If a router or a FortiGate device fails, all traffic to this device transparently fails over to another router or FortiGate that takes over the role of the failed device. If the failed device is restored, it will take over processing the network traffic.

FortiOS supports VRRP versions 2 and 3. You can create VRRP domains, which can include multiple FortiGate devices and other VRRP-compatible routers. You can add different FortiGate models to the same VRRP domain. FortiOS supports IPv4 and IPv6 VRRP, so you can add IPv4 and IPv6 VRRP virtual routers to the same interface. FortiGate devices can quickly and easily integrate into a network that has already deployed VRRP.



VRRP FortiGate devices in a VRRP domain periodically send VRRP advertisement messages to all FortiGate devices in the domain to maintain one device as the primary router and the others as backup routers. The primary FortiGate has the highest priority. If the backup FortiGate stops receiving these packets from the primary FortiGate, the backup FortiGate with the highest priority becomes the new primary.

The primary FortiGate stops sending VRRP advertisement messages if it fails or becomes disconnected from a configured remote connection. You can configure up to two VRRP destination addresses to be monitored by the primary device. As a best practice, the destination addresses should be remote addresses, as shown on this slide, to check the access to internet. If the primary router is unable to connect to the IP address `8.8.8.8`, it stops sending VRRP advertisement messages so the backup FortiGate can become the new primary VRRP router.

FGCP, FGSP, and VRRP Comparison

Features	FGCP	FGSP	VRRP
Redundancy capacity	Two to four FortiGate devices	2–16 standalone FortiGate devices or 2–16 FGCP clusters	Unlimited to devices compliant to VRRP open standard
Load balancing	Native active-active HA load balancing In active-passive mode, no load balancing (VDOM partitioning available for load sharing)	Through external load balancers	Not available
Session synchronization	Available through configuration	Available through configuration	Not available
Configuration synchronization	Native	Available if required	Not available

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In summary, this slide shows the features of the three HA protocols, to help you select the one most suitable for an enterprise network. The table does not include the main features that are common to all of them, including redundancy, failover mechanism, advertisements or heartbeat communication, and link failure detection.

Review

- ✓ Explore FGCP
- ✓ Examine FGSP
- ✓ Investigate VRRP
- ✓ Compare HA protocols

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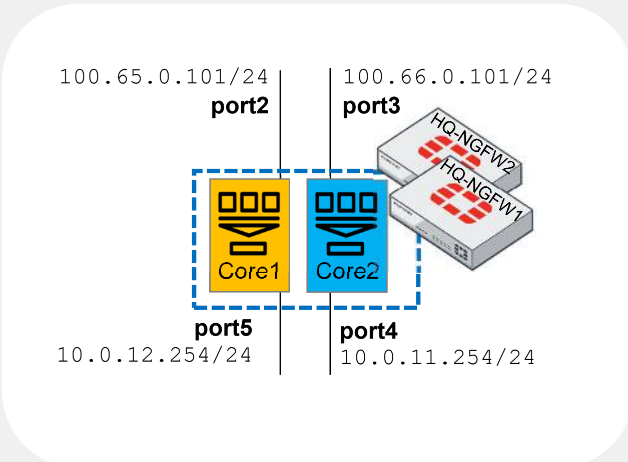
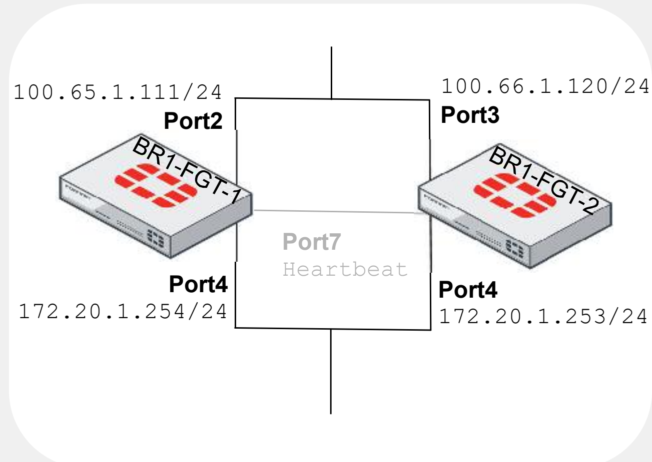
This slide shows the objectives that you covered in this lesson. By mastering the objectives covered in this lesson, you learned about and compared the features of the three HA protocols.

Lab 4–High Availability

- On HQ-NGFW-1 and HQ-NGFW-2:
  - VDOM partitioning
- On BR1-FGT-1 and BR1-FGT-2:
  - Asymmetric traffic—layer 3
  - Session synchronization encryption using IPsec tunnel





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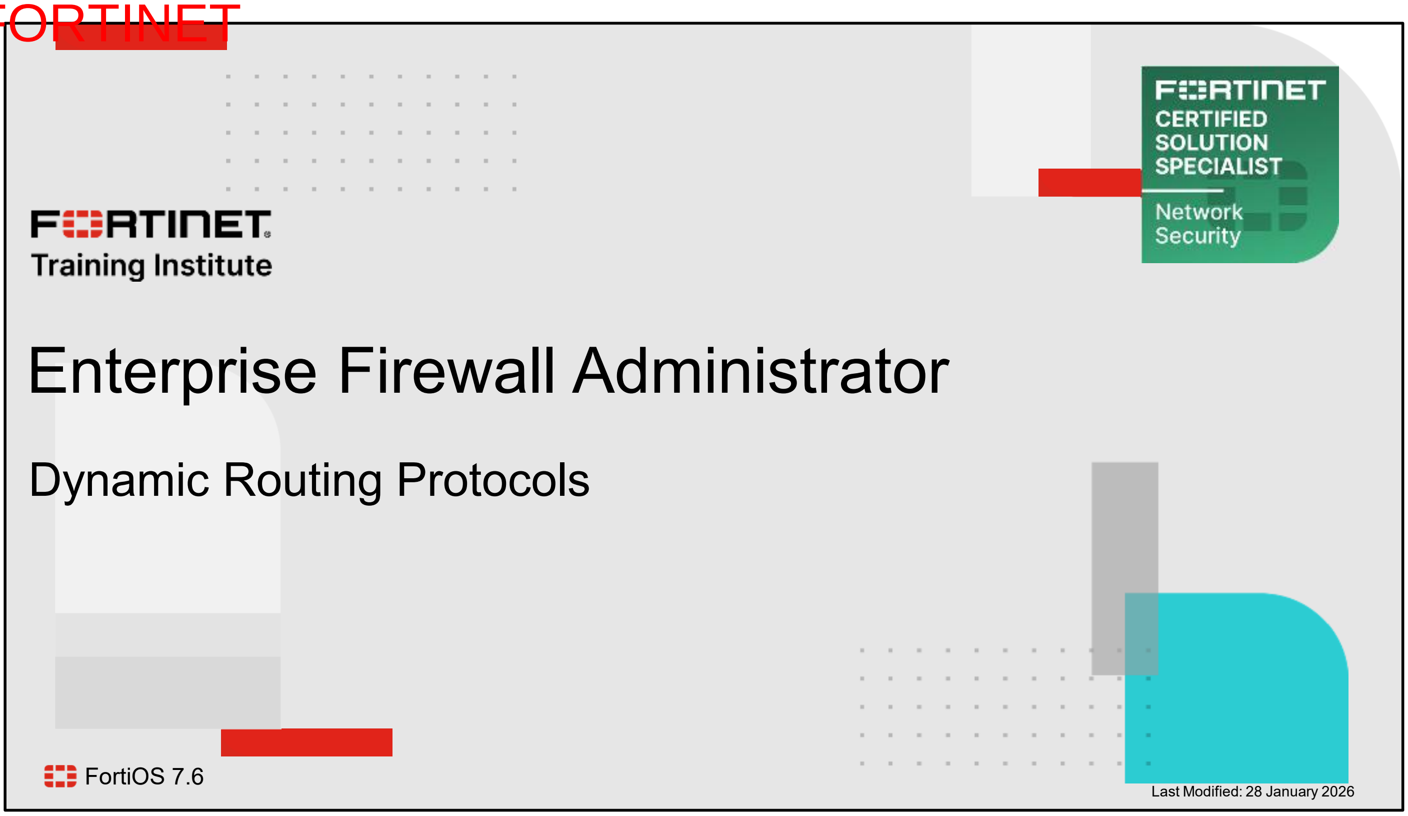
Now, you will work on *Lab 4–High Availability*.

In this lab, you will configure an FGCP cluster with VDOM partitioning and two FortiGate devices in an FGSP cluster with asymmetric traffic. You will then add encryption using an IPsec tunnel for the session synchronization of the FGSP cluster.



In this lesson, you will learn how to handle OSPF and BGP on FortiGate.

Objectives

- Implement OSPF features
- Deploy BGP in enterprise networks
- Optimize BGP for rapid convergence
- Deploy BGP using FortiManager

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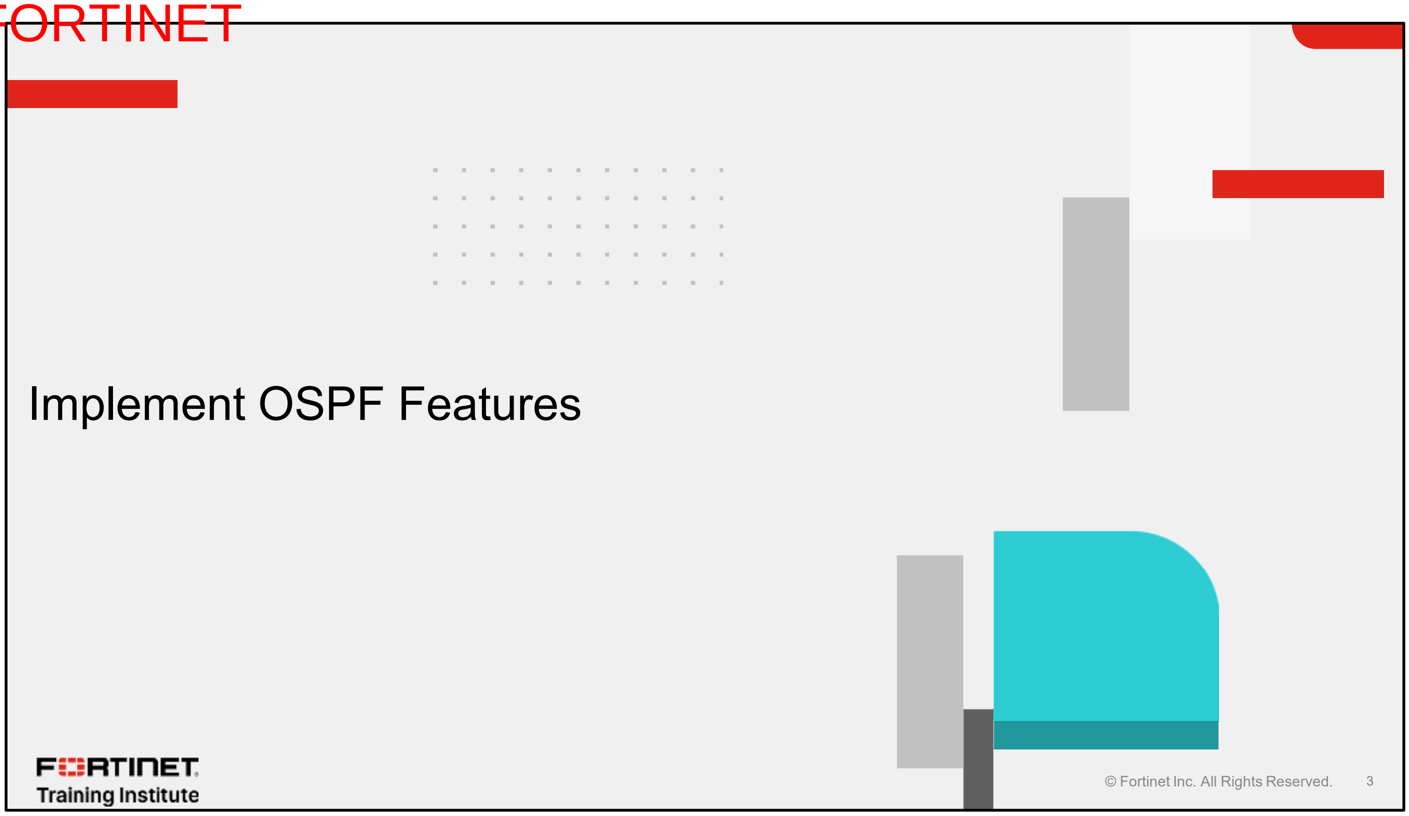
After completing this lesson, you should be able to achieve the objectives shown on this slide.

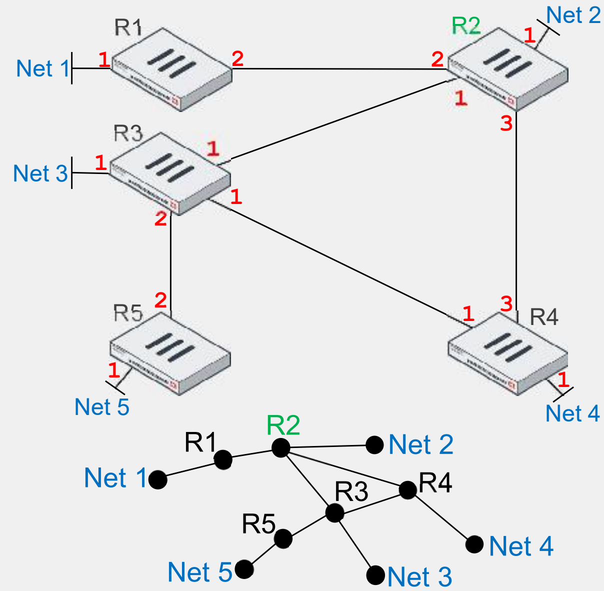
By demonstrating a competent understanding of OSPF and BGP, you will be able to configure these dynamic routing protocols and understand the features they include.



In this section, you will learn about the features of the OSPF routing protocol on FortiGate.

OSPF Overview

- Link state protocol—each router maintains identical databases describing the network topology
- It uses Dijkstra's algorithm—each router builds a tree with the shortest paths
- The tree gives a route to each destination
- Advantages:
  - Scalable to large networks
  - Faster convergence than distance-vector routing protocols
  - Relatively quiet during steady-state conditions
    - Periodic refresh every 30 minutes
    - Otherwise, updates sent only when there are changes

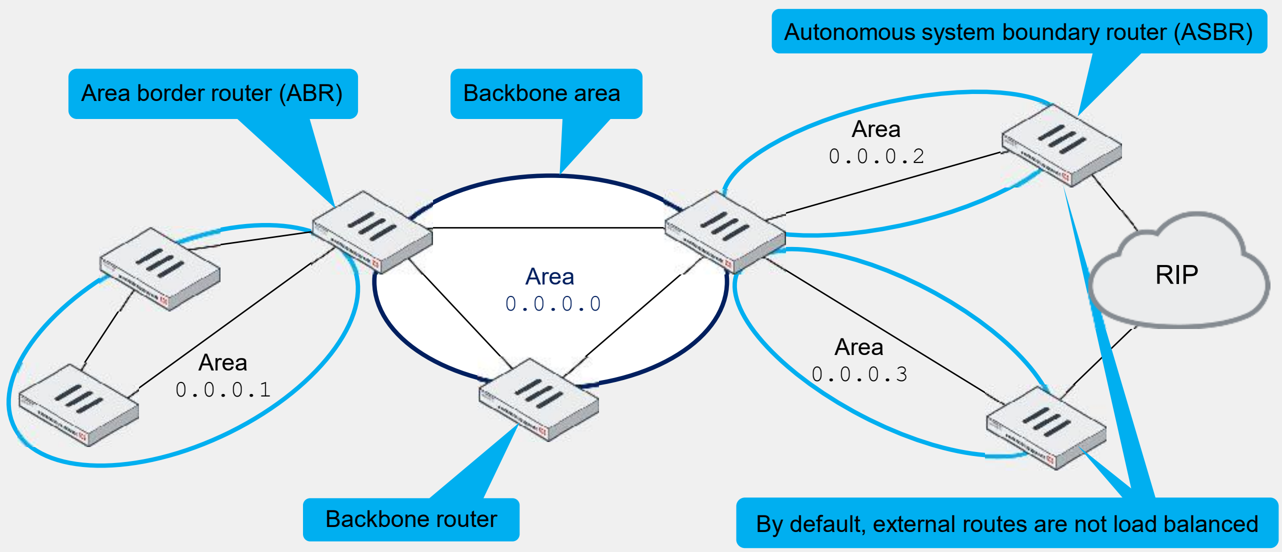


OSPF is a link-state routing protocol, which maintains a synchronized database. All routers within the same OSPF area have the same copy of the database, unlike distance vector protocols. Distance vector protocols require periodic updates to maintain accurate routing information across all routers, and these updates must be distributed within the domain, as in the case of RIP.

Each router in the same area has identical and synchronized databases. An OSPF router uses the information in the link state database (LSDB) and Dijkstra's algorithm to determine the best route to each destination. The best routes can be represented as a tree with the local router at the root. Dijkstra's algorithm is a recursive process that the router repeats multiple times until it finds the best routes. For example, this slide shows the OSPF tree for router R2. It indicates that the best route to Net 5 is through R3, and that Net 2 is locally connected.

In a link-state routing protocol like OSPF, every router has a complete view of the network topology. Advantages of OSPF include scalability and fast convergence. Every 30 minutes, the routers readvertise their OSPF information. During those 30-minute intervals, the routers send updates when they detect a topology change. So, it is a relatively quiet protocol, as long as the network topology is stable.

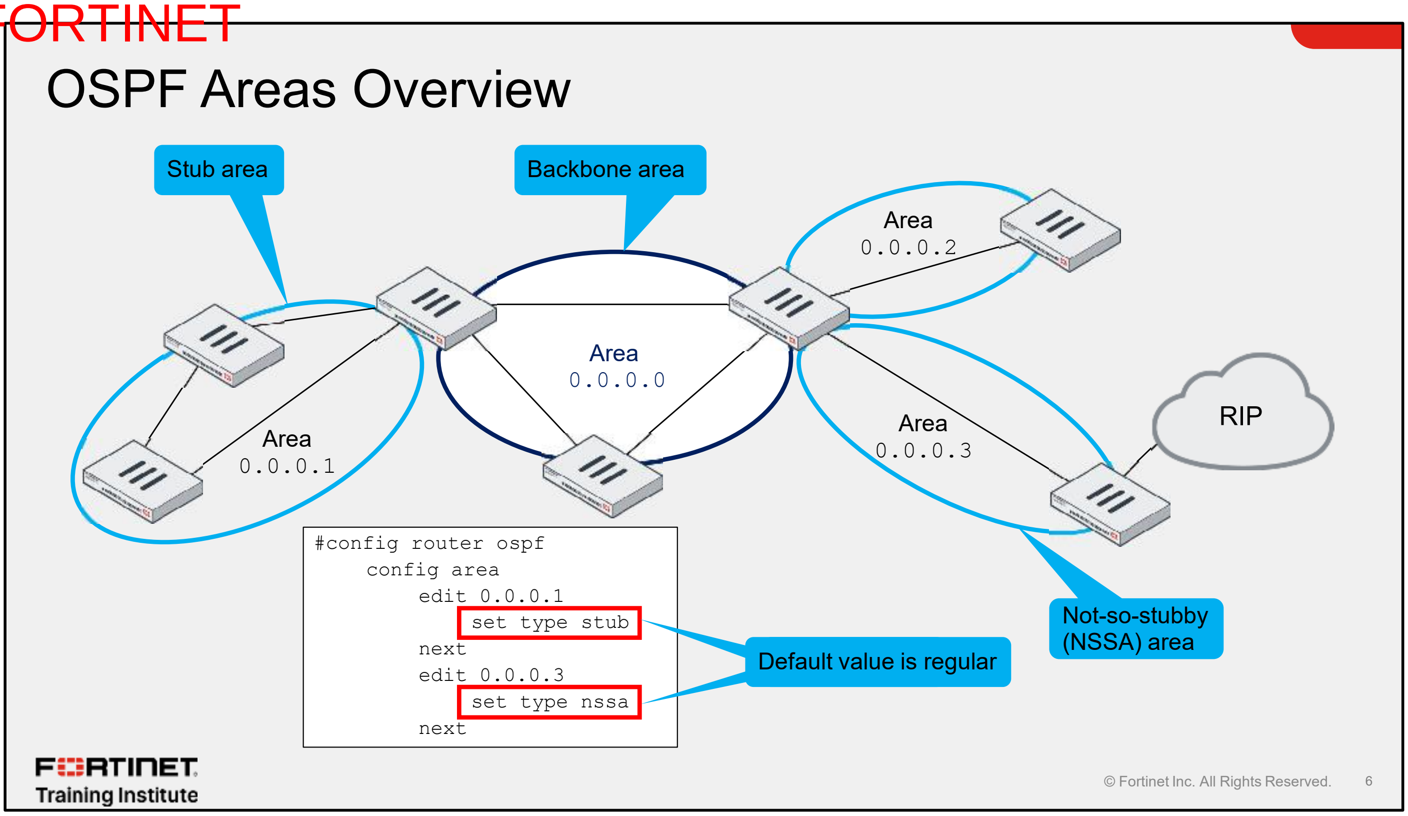
OSPF Routers Overview



All OSPF networks must have at least one area—the backbone area. The backbone area is the core of the network and routers with at least one interface connected to it are backbone routers.

Area border routers (ABR) are connecting other areas to the backbone area. To simplify the route table, the command `config range` is available for ABRs to configure route summarization. For example, in a hub and spoke scenario, by configuring the prefix `10.10.10.0 255.255.255.0` in `config range`, the hub advertises only this subnet instead of multiple 10.10.10.X 255.255.255.252 subnets from the spokes.

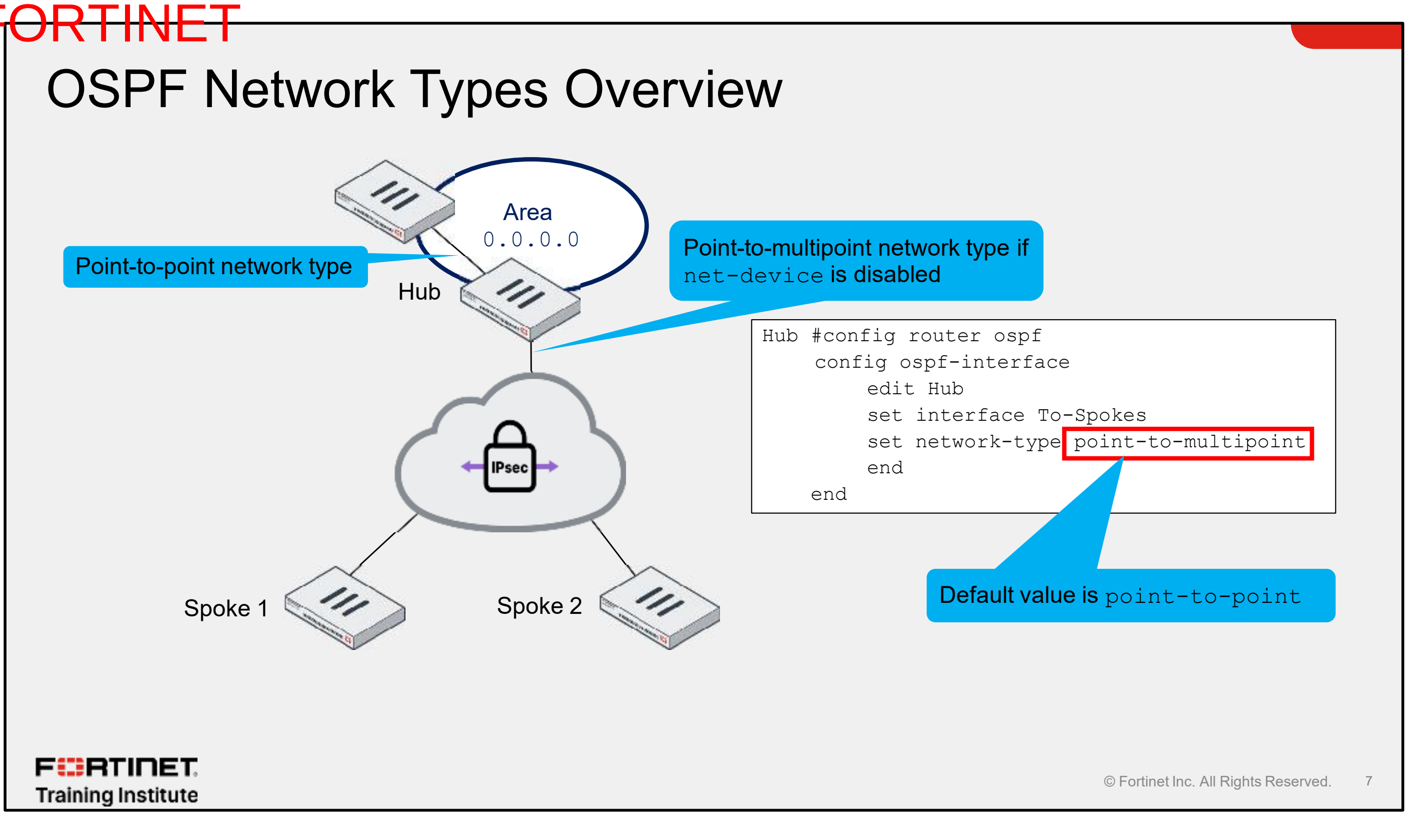
Autonomous system boundary routers (ASBR) allow you to import external non-OSPF routes into an OSPF network. You can summarize these routes on ASBRs with the command `config summary-address`.



By default, the OSPF area type is `regular` but you can configure `stub` and `nssa` area types.

An ABR can advertise routes from a stub area to the backbone area while it advertises external routes to the stub area only through a default route. In the case of a totally stub area, the ABR replaces the inter-area prefixes advertised to the stub area by a default route.

NSSA areas are similar to stub areas except that they can have an ASBR, allowing the ABR to advertise external routes to the backbone area.



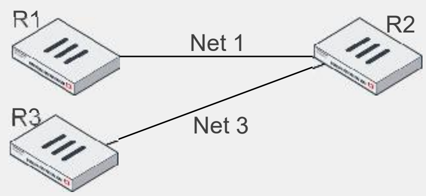
Usually, routers are connected directly and the configuration of the interface network type is `point-to-point`. In the case of multiple dialup connections with net-device disabled, the hub creates a single IPsec virtual interface that is shared by all IPsec clients connecting to the same dial-up VPN. As this slide shows, you must configure the network type to `point-to-multipoint` on the hub.

OSPF Protocol Redistribution

- OSPF redistribution control over other protocols

```
R2 # config router ospf
R2 (ospf) # config redistribute ?
          *name    Redistribute name.
          bgp           disable    0
          connected     disable    0
          isis          disable    0
          rip           disable    0
          static        disable    0
```

By default, R2 does not advertise connected network Net 1 to R3



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In OSPF, you can also manually filter routes by setting the redistribution of the other routing protocols. This is particularly useful for ASBRs to specify the redistribution of other protocols, like RIP or static routes, to other areas. By default, the setting is disabled on all the routing protocols within the OSPF router settings.

Comparative Filtering—Route Maps, and Access and Prefix Lists

<table>
<tr><th>Routes Filtering</th><th>Type</th><th>Action</th><th>Options</th></tr>
<tr><td>Access lists</td><td>Simple (filter based on prefixes)</td><td>Prevent injecting routes into the routing table with the parameter distribute-list-in</td><td>Can also filter routes distributed from other protocols (connected, static, or RIP) in distribute-list configuration</td></tr>
<tr><td>Prefix lists</td><td>Simple (filter based on prefixes with logical operator added)</td><td>Applies only to distribute-list-in</td><td>ge and le parameters added in the rule configuration for more granularity on the prefix match</td></tr>
<tr><td>Route map</td><td>Advanced (access and prefix lists can be route map objects)</td><td>Filter incoming external routes with the parameter distribute-route-map-in</td><td>Can also filter routes redistributed by other protocols in config redistribute <bgp|connected|isis|rip|static></td></tr>
</table>

More generally, you can filter routes on OSPF with the following route filtering rules options:

1. Access lists
2. Prefix lists
3. Route maps

The route filtering methods help to control or prevent advertising routes. This slide shows a summary of the main differences between the route filtering options for OSPF.

Route maps provide advanced features that can modify OSPF attributes and provides more granularity than access and prefix lists.

OSPF Commands Comparison

Command	Information
`get router info ospf neighbor`	Statuses of all the OSPF neighbors and their adjacency states (DR, BDR, or DROther)
`get router info ospf interface`	Status per interface, including number of adjacencies, DR and BDR router IDs
`get router info ospf status`	Full status, including the router ID, features negotiated, number of areas attached to this router, and number of fully adjacent neighbors
`get router info ospf database brief`	Complete link state database (LSDB) ordered by link state advertisement (LSA) types
`get router info ospf database router lsa`	Further details on type 1 LSAs
`get router info bfd neighbor`	BFD negotiation status when BFD is enabled on OSPF routers

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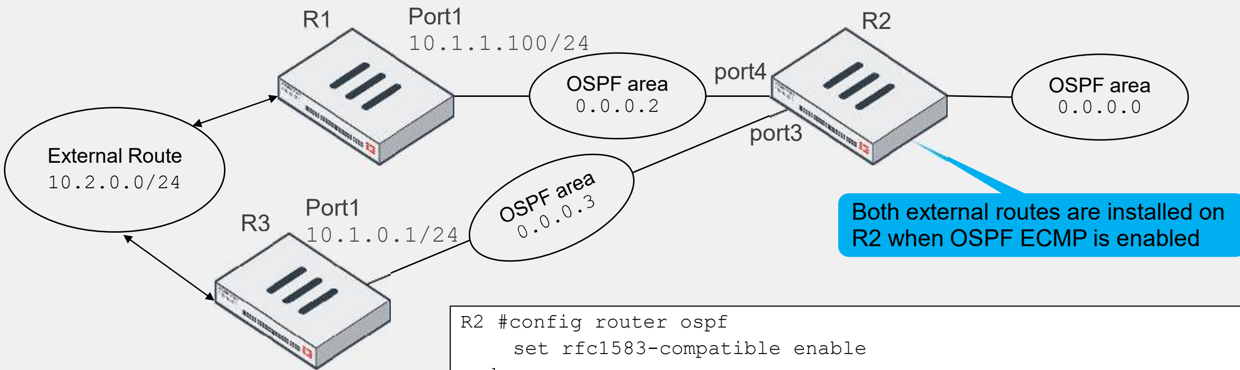
This slide shows a summary of the main OSPF commands.

Use Case 1—OSPF ECMP

- OSPF ECMP allows FortiGate to install external routes of the same cost in the routing database



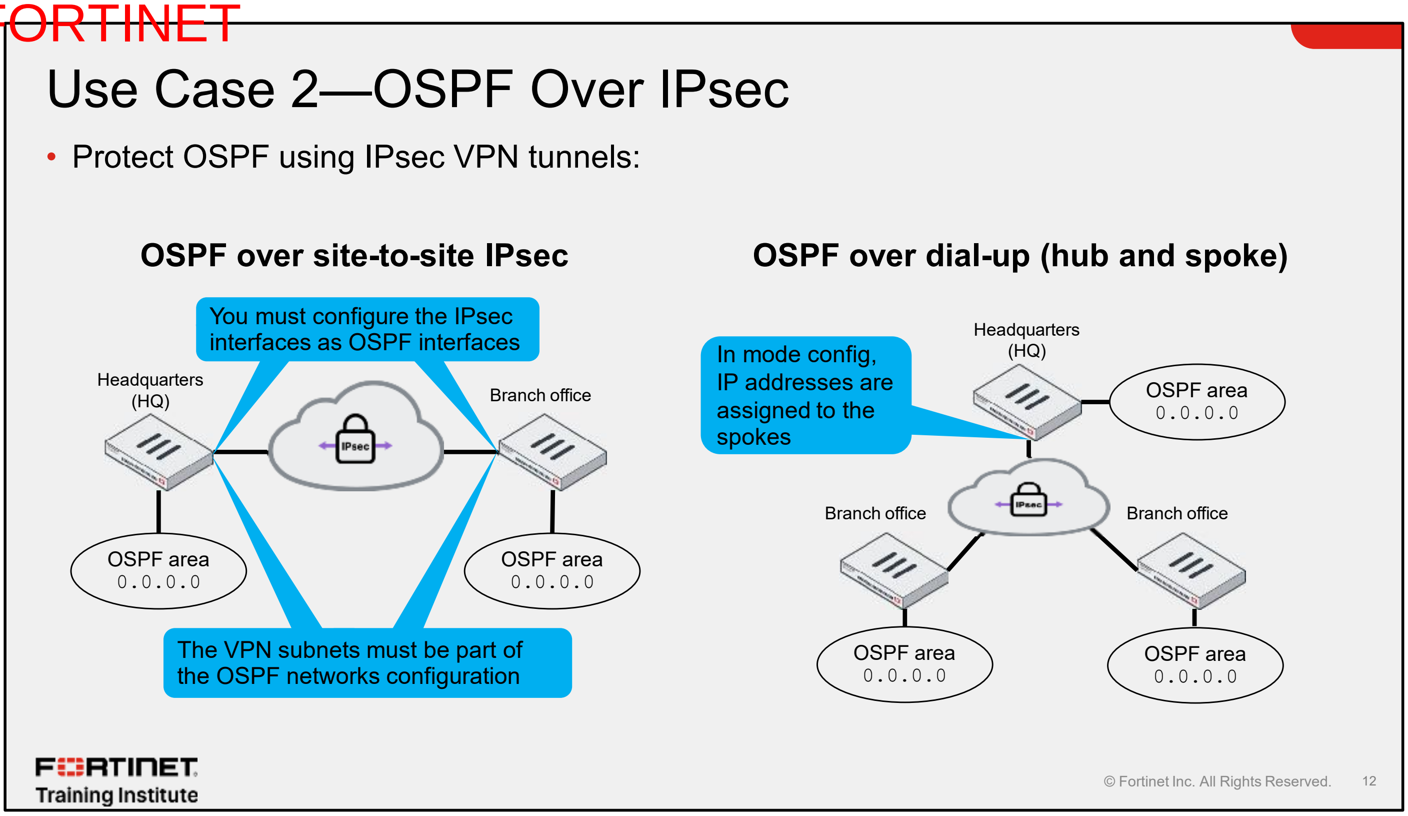

```
R2 #config router ospf
    set rfc1583-compatible enable
end
R2 # get router info routing-table all
...
O E2    10.2.0.0/24 [110/10] via 10.1.1.100, port4, 00:00:21, [1/0]
                    [110/10] via 10.1.0.1, port3, 00:00:21, [1/0]
...
```


OSPF by default works with ECMP RFC 2328.

RFC 2328 introduces new path preference rules:

- Intra-area paths using non-backbone areas are most preferred.
- Intra-area backbone paths and inter-area paths have equal preference.
- Intra-area routes in non-backbone areas are preferred to reduce backbone overhead.

However, you can enable ECMP RFC 1583, where the path selection is based solely on cost. It is recommended to enable it when you want to reach external routes.

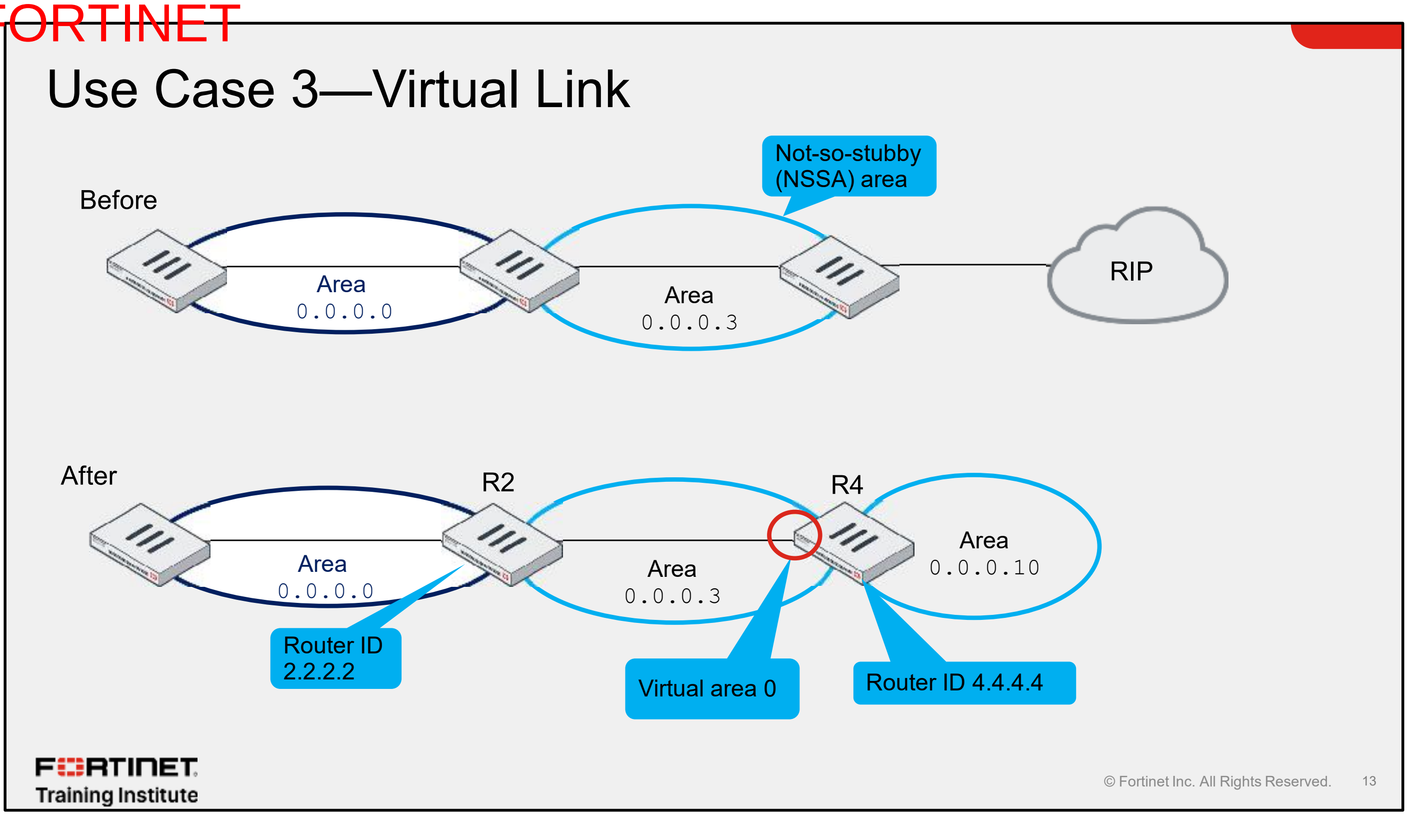


OSPF can be protected using IPsec VPN tunnels. The two most commonly used implementations of OSPF over IPsec VPN are:

- Site-to-site
- Dial-up

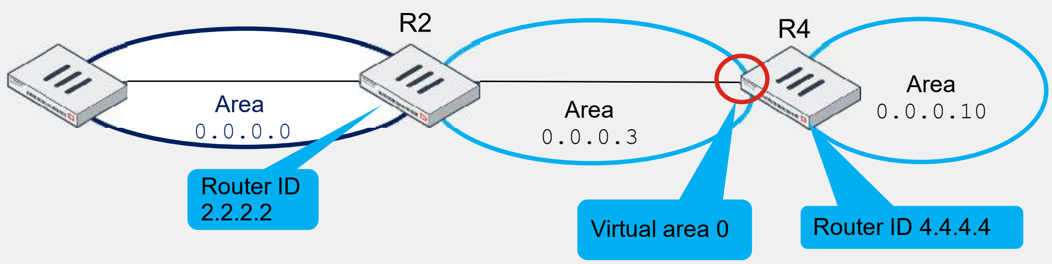
Site-to-site OSPF requires route-based IPsec VPN tunnels with complete phase1 and phase2 configuration. On both FortiGate devices, the phase1 configuration must have tunnel-end IP addresses assigned on tunnel interfaces. OSPF must be defined on the VPN tunnel interfaces as part of the OSPF interface configuration.

In a hub-and-spoke implementation, OSPF over dial-up uses the generic VPN setup for FortiGate as a dial-up client. The hub must have an IPv4 range specified in the VPN phase1 configuration to ensure that spokes are assigned an IP address. The hub also must have OSPF redistributing other routing protocols, especially static and directly connected routes.



In an NSSA area, you can integrate it into the OSPF network with a virtual link. When you configure a virtual link on both routers, you allow a remote area to virtually connect directly to the backbone area.

Use Case 3—Virtual Link (Contd)




```
R2#config router ospf
    set router-id 2.2.2.2
    config area
        edit 0.0.0.3
        config virtual-link
            edit VLINK_area
            set peer 4.4.4.4
            next
        end
    next
end
```

```
R4#config router ospf
    set router-id 4.4.4.4
    config area
        edit 0.0.0.3
        config virtual-link
            edit VLINK_area
            set peer 2.2.2.2
            next
        end
    next
end
```

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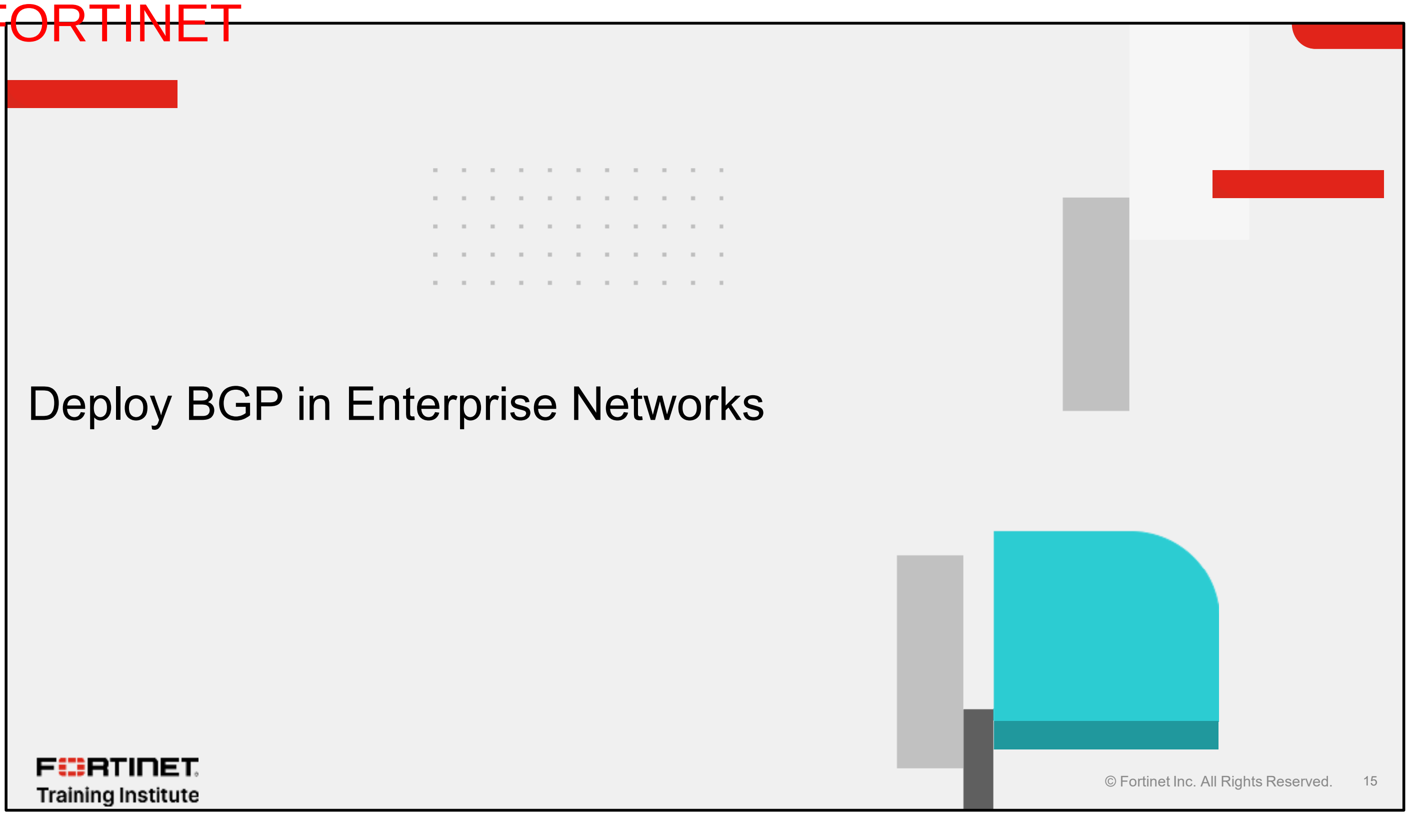
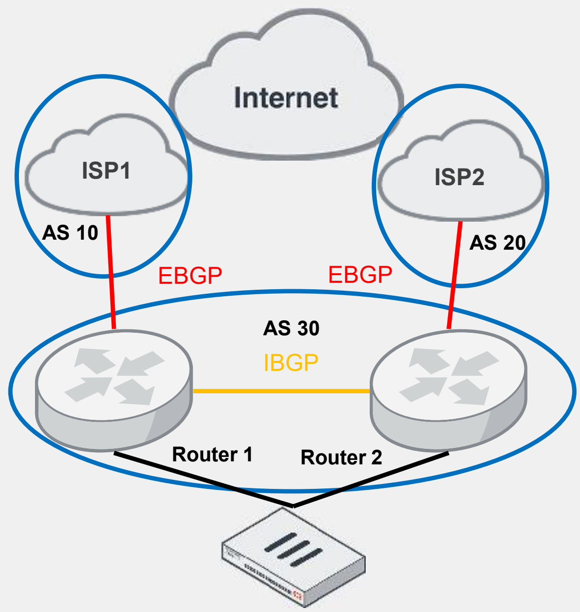
This slide shows the configuration on R2 and R4 so the area 0.0.0.10 is virtually connected to the backbone area.



In this section, you will learn about deploying BGP in enterprise networks.

BGP Overview

- Basic BGP network design



BGP is the protocol underlying the global routing system of the internet. BGP is, therefore, widely and increasingly used to exchange routing and reachability information.

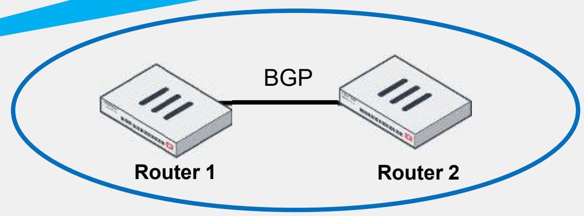
BGP uses autonomous systems (AS), as the basic design shows on this slide. Routers 1 and 2 are in the same AS and advertise routes that follow IBGP. They are connected to ISP1 and ISP2, which are on different autonomous systems. They therefore establish a connection and advertise routes through EBGP.

Protocol Redistribution

- A non-BGP route can be redistributed into BGP

```
# config router bgp
(bgp) # config redistribute ?
        *name    Redistribute name.
        bgp           disable    0
        connected     disable    0
        isis          disable    0
        rip           disable    0
        static        disable    0
```

By default, FortiGate BGP doesn't advertise prefixes from other protocols



By default, FortiGate BGP doesn't advertise prefixes. You can use the redistribution command to configure FortiGate to advertise prefixes. You can redistribute connected and static routes, and routes learned from other routing protocols, into BGP.

Comparative Filtering—Route Maps, and Access and Prefix Lists

Routes Filtering	Type	Action	Options
Access lists	Simple (filter based on prefixes)	Prevent injecting routes into the routing table with the parameter `distribute-list-in` or advertising routes to neighbors with the parameter `distribute-list-out`	
Prefix lists	Simple (filter based on prefixes with logical operator added)	Prevent injecting routes into the routing table with the parameter `prefix-list-in` or advertising routes to neighbors with the parameter `prefix-list-out`	`ge` and `le` parameters added to the rule configuration for more granularity on the prefix match
Route map	Advanced (access and prefix lists can be route map objects)	Filter injecting and advertising routes with the parameters `route-map-in` and `route-map-out`	Can also filter routes redistributed by other protocols in `config redistribute <bgp|connected|isis|rip|static>` and can modify BGP attributes

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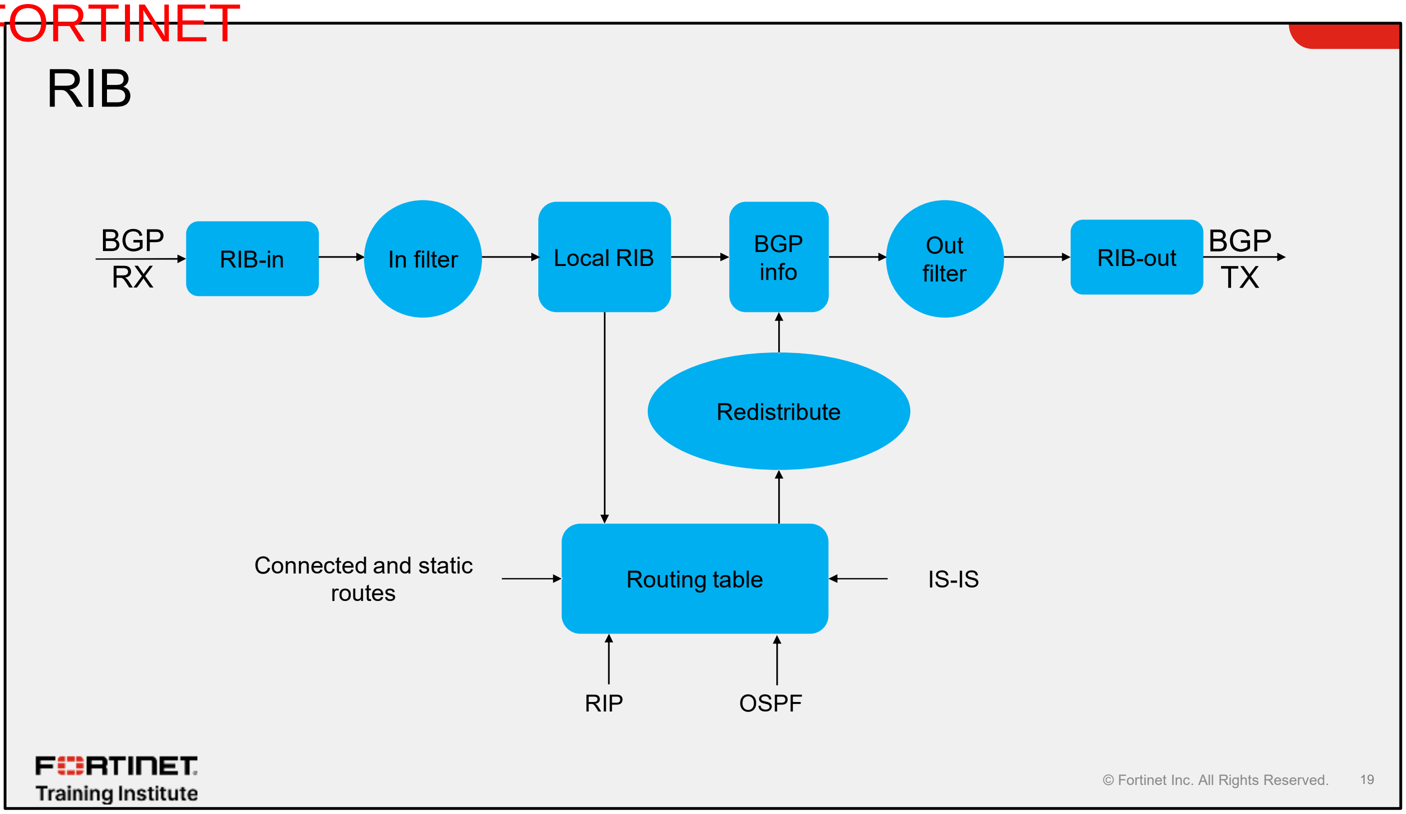
More generally, you can filter routes on BGP with the following route filtering rules options:

1. Access lists
2. Prefix lists
3. Route maps

The route filtering methods help to control or prevent advertising. This slide shows a summary of the main differences between the route filtering options for BGP.

Route maps provide advanced features that can modify BGP attributes and provides more granularity than access and prefix lists.



This slide shows a flowchart that summarizes the BGP process and where route filtering can occur. The BGP router stores the BGP routes it receives from other routers in the RIB-in table. The BGP router applies a filter, and the resulting routes are stored in the local routing information base (RIB) table. Then, the BGP router adds routes that were redistributed from the routing table and applies another filter (outbound). The BGP router advertises the resulting routes.

BGP Commands Comparison

Command	Information
`get router info bgp summary`	BGP status of the router and all its neighbors, including: the AS, packet counters, and the length of time the neighbor has been up
`get router info bgp neighbors`	Details of the neighbors, including: peer IP address and router ID, remote AS, BGP state, and negotiated capabilities
`get router info bgp network`	Displays the BGP database
`get router info routing-table bgp`	Displays the BGP routing table

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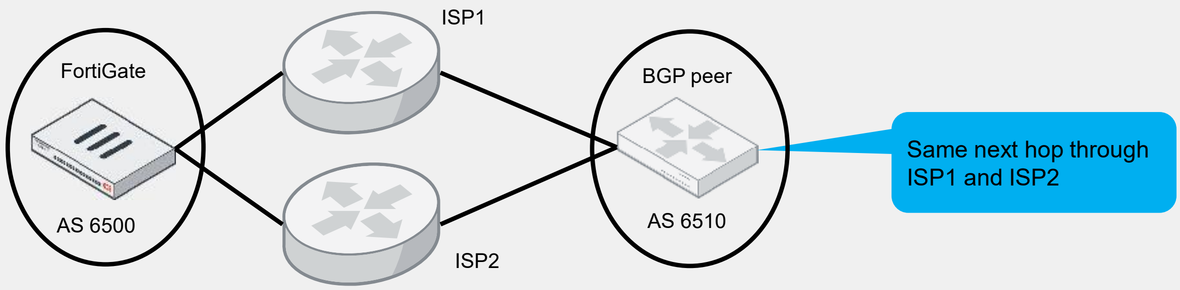
This slide shows a summary of the main BGP commands.

Use Case 1—Configure ECMP With BGP Routes

- You must enable EBGP or IBGP multipath

```
config router bgp
     set ebgp-multipath enable
end
```

Allows FortiGate to load balance the outgoing traffic



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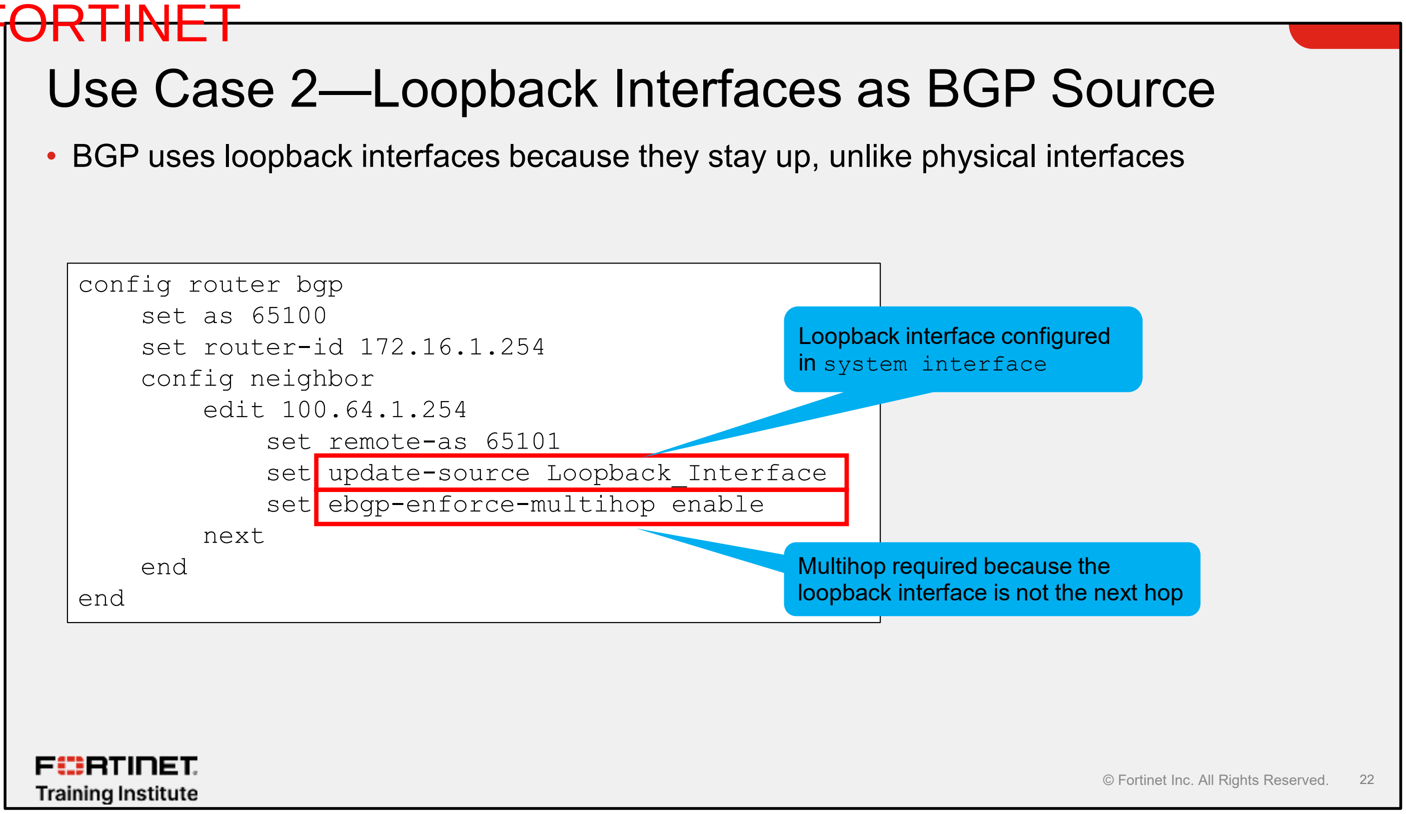
When there are multiple equal-cost multi-path (ECMP) routes to a BGP next hop, FortiGate can now consider all the ECMP routes for the next-hop resolution by enabling EBGP or IBGP multipath, allowing load balancing.



In a production environment, a loopback interface as a source interface for BGP sessions is used mostly to ensure continuous BGP peering through redundant links to the same BGP peer. This allows the BGP session to work uninterrupted.

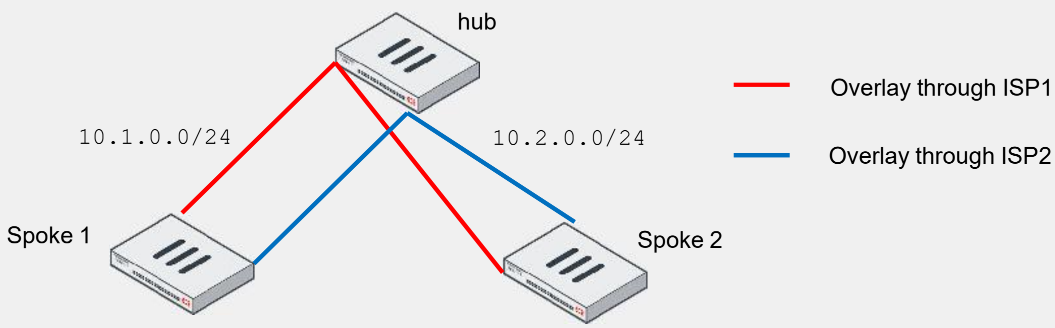
This slide shows the required FortiGate configuration in BGP:

- You must explicitly configure the loopback interface in the same way as the source interface in the `config neighbor`.
- You must enable multihop because the loopback interface adds one hop.

The BGP session on TCP port 179 may be traveling through a physical interface. Therefore, you must configure a corresponding firewall policy to allow the traffic between the loopback interface and the physical interface, so that the BGP connection can be established.

Use Case 3—The `neighbor-group` Command

- Allows FortiGate to apply common settings in the neighbor group for each BGP peer relationship
- Is useful in an SD-WAN overlay design



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In a standard SD-WAN design, the spokes, which are the remote BGP speakers, are dial-up clients. By defining a neighbor group for overlays established over each underlay, FortiGate can apply the common settings in the neighbor group for each BGP peer relationship. You can use the `neighbor-range` command to define the IP address range that characterizes the neighbor group.

Use Case 3—The `neighbor-group` Command (Contd)

- This use case is within the same AS, meaning IBGP

```
config router bgp
    set as 65100
    set router-id 172.16.1.254
    set ibgp-multipath enable
    config neighbor-group
        edit SpokeISP1
            set interface ISP1
            set remote-as 65100
        next
    end
    config neighbor-range
        edit 1
            set prefix 10.1.0.0 255.255.255.0
            set neighbor-group SpokeISP1
        next
    end
end
```

Parameters applied to the neighbor group SpokeISP1

Defines the peers to assign the corresponding neighbor group

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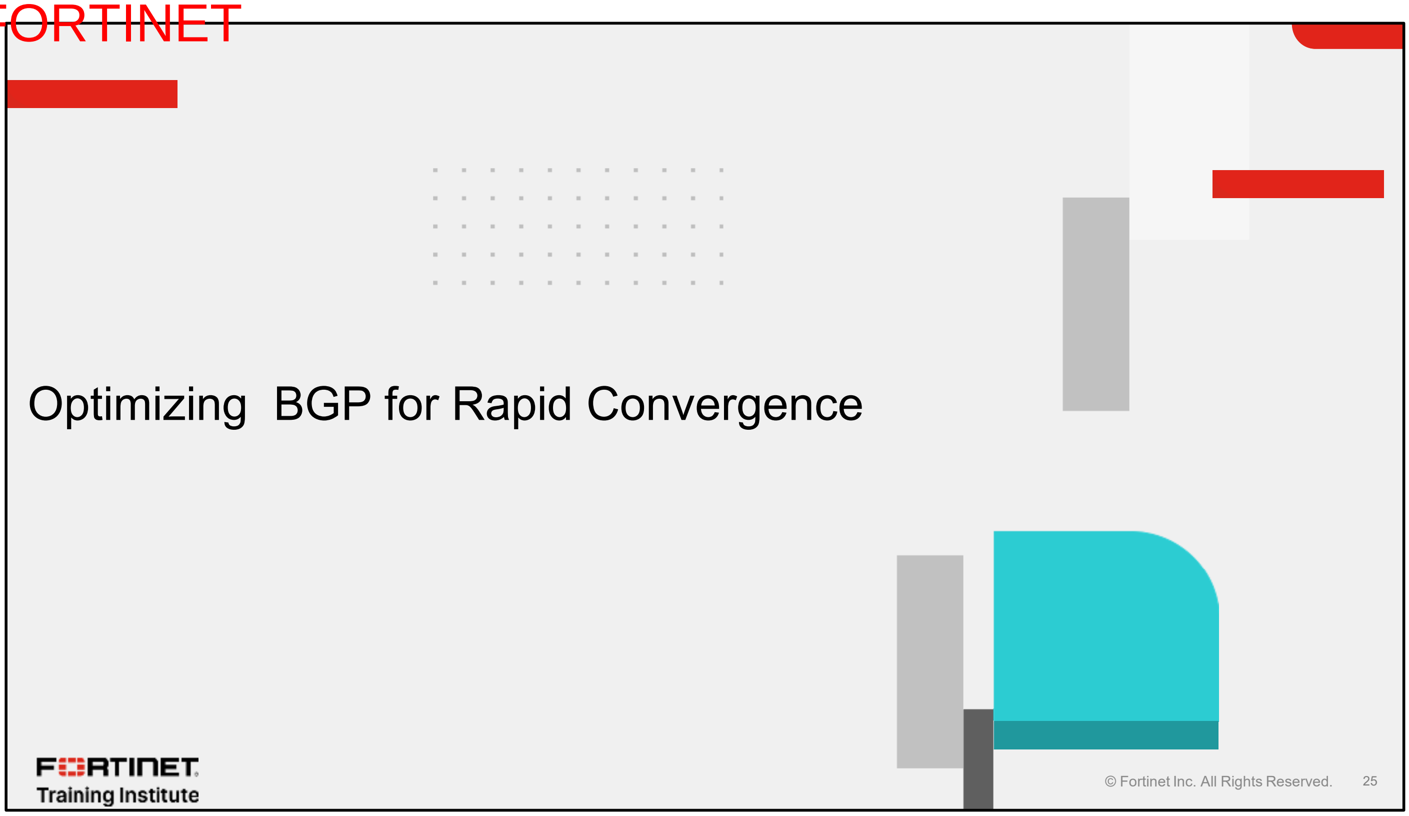
This slide shows the BGP configuration on the hub.

The prefix in the neighbor range defines that peers with an IP address in the `10.1.0.0/24` subnet (the ISP1 subnet) are included in the SpokeISP1 neighbor group.

These peers then receive the same settings configured in the SpokeISP1 neighbor group, meaning interface ISP1 and remote-as 65100.

Because this remote AS is the same as the local AS, `ibgp-multipath` is enabled to support ECMP for IBGP routes.



In this section, you will learn about BGP advanced features for rapid convergence.

BGP Convergence

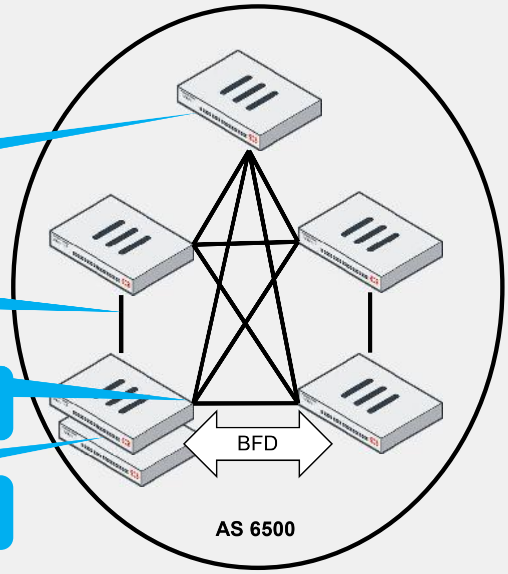
- BGP convergence includes multiple steps:
  - Installing a new path in the routing and forwarding tables
  - Processing and finding an alternate path
  - Fast failure detection and information propagation

Optimize the RIB size by adding in and out filters

Improve processing by implementing route reflectors

Failure detection by fine-tuning BGP keepalive timers or enabling BFD

Prevent route flapping during HA failover by configuring `graceful-restart`



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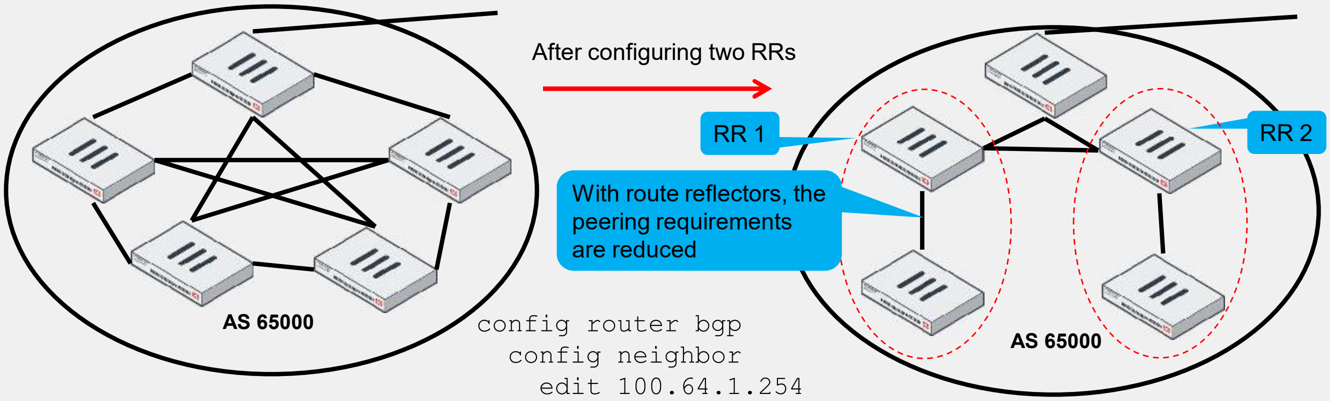
BGP is a distance vector protocol that sends its entire routing table to directly connected neighbors. This routing table is regularly updated depending on new routing paths or failure detection. You can modify the timing of the update using parameters like the size of the RIB, the number of hops multiplied by the advertisement interval for each of them, and the delay involved in failure detection.

To improve BGP convergence, you must have a stable network, without port flapping, which creates BGP update messages that require RIB processing and adds to the CPU load. You can reduce RIB size using filters. In IBGP, you can implement route reflectors to concentrate the updates and avoid a full mesh between all BGP-speaking routers. For faster failure detection, you can reduce the BGP keepalive timers or enable BFD. And for HA failover, you can prevent unnecessary BGP updates by configuring `graceful-restart`, as explained on the next slides.

Route Reflectors

- Router reflectors (RR) act as concentrators for IBGP-speaking routers




```
config router bgp
  config neighbor
    edit 100.64.1.254
      set next-hop-self enable
      set route-reflector-client enable
    next
  end
end
```

The RR defines which neighbors are the clients

When running IBGP, you usually need to configure full mesh peering between all the routers. In large networks, full mesh peering between routers can be difficult to administer and is not scalable.

RRs help to reduce the number of IBGP sessions inside an AS. An RR forwards the routes learned from one peer to the other peers. If you configure RRs, you don't need to create a full mesh IBGP network. RRs pass the routing updates to other RRs and border routers within the AS, improving the BGP convergence.

BFD Parameter

- BGP has a keepalive timer configured in seconds
- Enable BFD for faster failure detection (in less than 1 second)
- Configure BFD on both connected routers and their corresponding interfaces
- Configure BFD for each neighbor and for multihop paths

```
config router bgp
  config neighbor
    edit 100.64.1.254
      set bfd enable
      set ebgp-enforce-multihop
enable
    next
  end
end
```

For BFD multihop path, the check is performed using a template

```
config router bfd
  config multihop-template
    edit 1
      set src 100.64.2.0 255.255.255.0
      set dst 100.64.1.0 255.255.255.0
      set auth-mode md5
      set md5-key password
    next
  end
end
```

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Bidirectional Forwarding Detection (BFD) is like a probe that checks the status of the BGP peer. It is independent of the type of media and detects a one-way device failure in less than a second, allowing faster BGP convergence. BFD was initially supported for two routers directly connected on the same subnet. Now, FortiGate can also support neighbors with BFD connected over multiple hops.

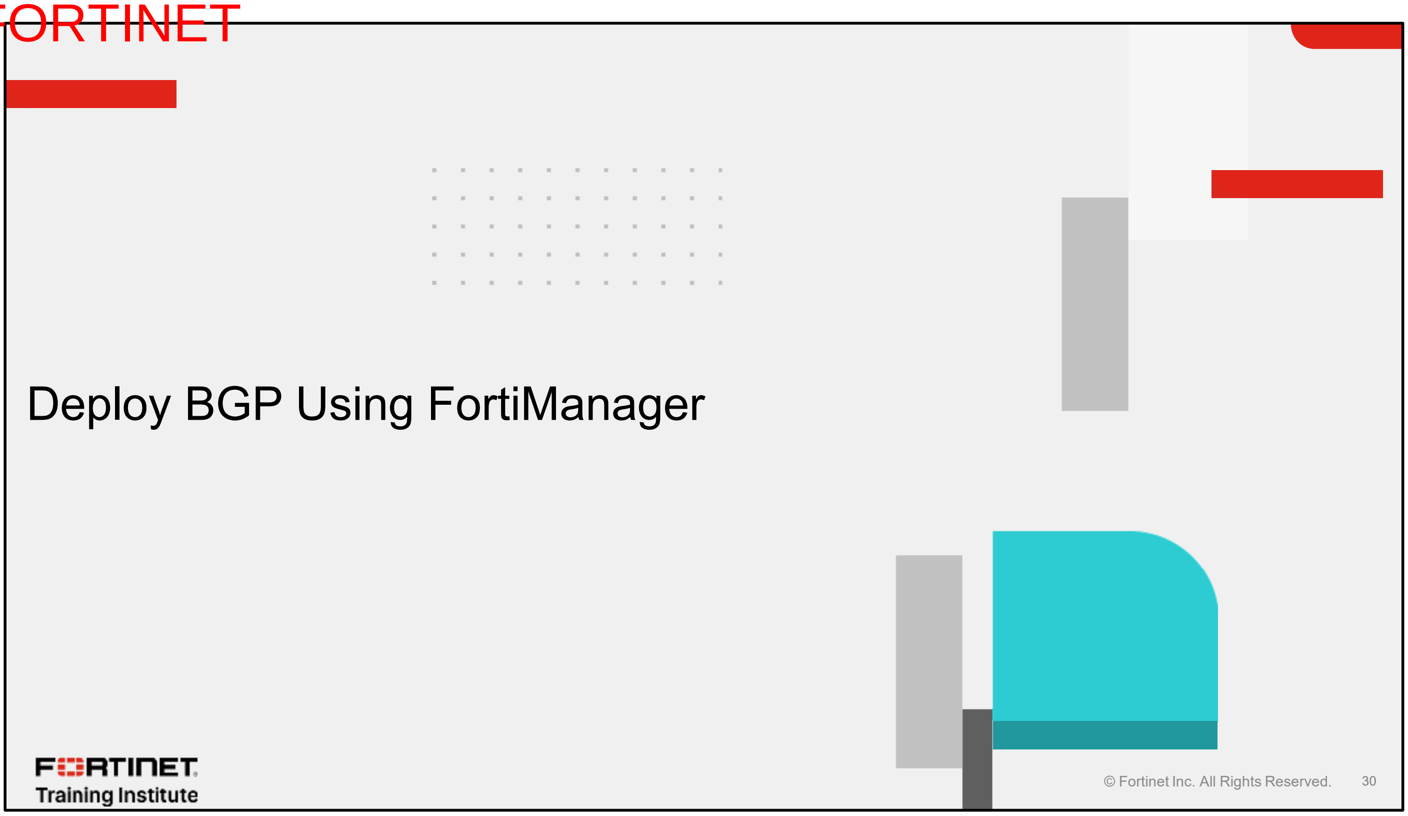
The `graceful-restart` Command

- Use the BGP `graceful-restart` command to prevent traffic interruption during HA failover
- The HA cluster advertises that it is going offline
- Enable the parameter on both connected routers

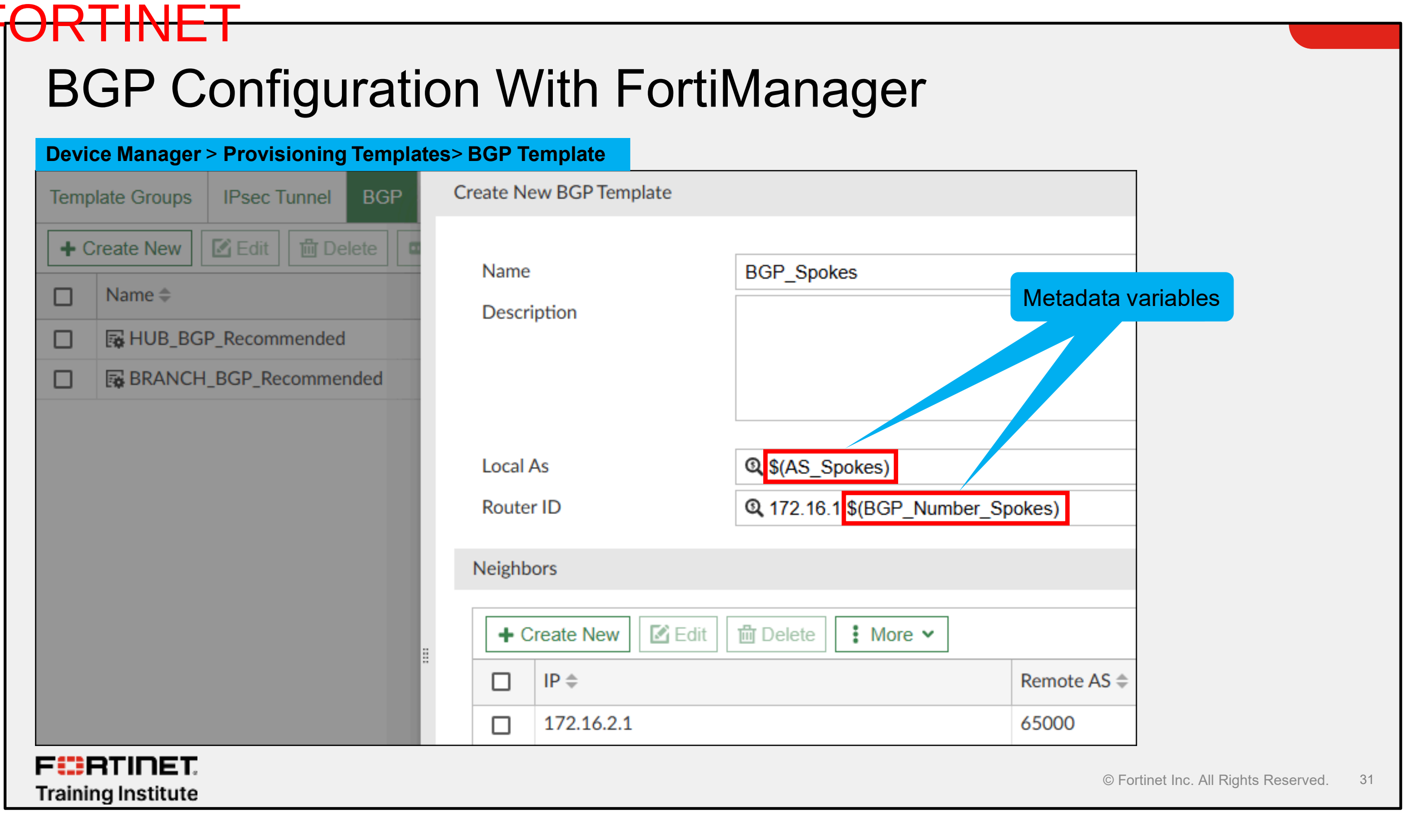
```
config router bgp
    set as 65100
    set graceful-restart enable
    set router-id 172.16.1.254
    config neighbor
        edit 100.64.1.254
            set capability-graceful-restart enable
            set remote-as 200
        next
    end
end
```

FortiGate advertises the capability to the corresponding neighbor

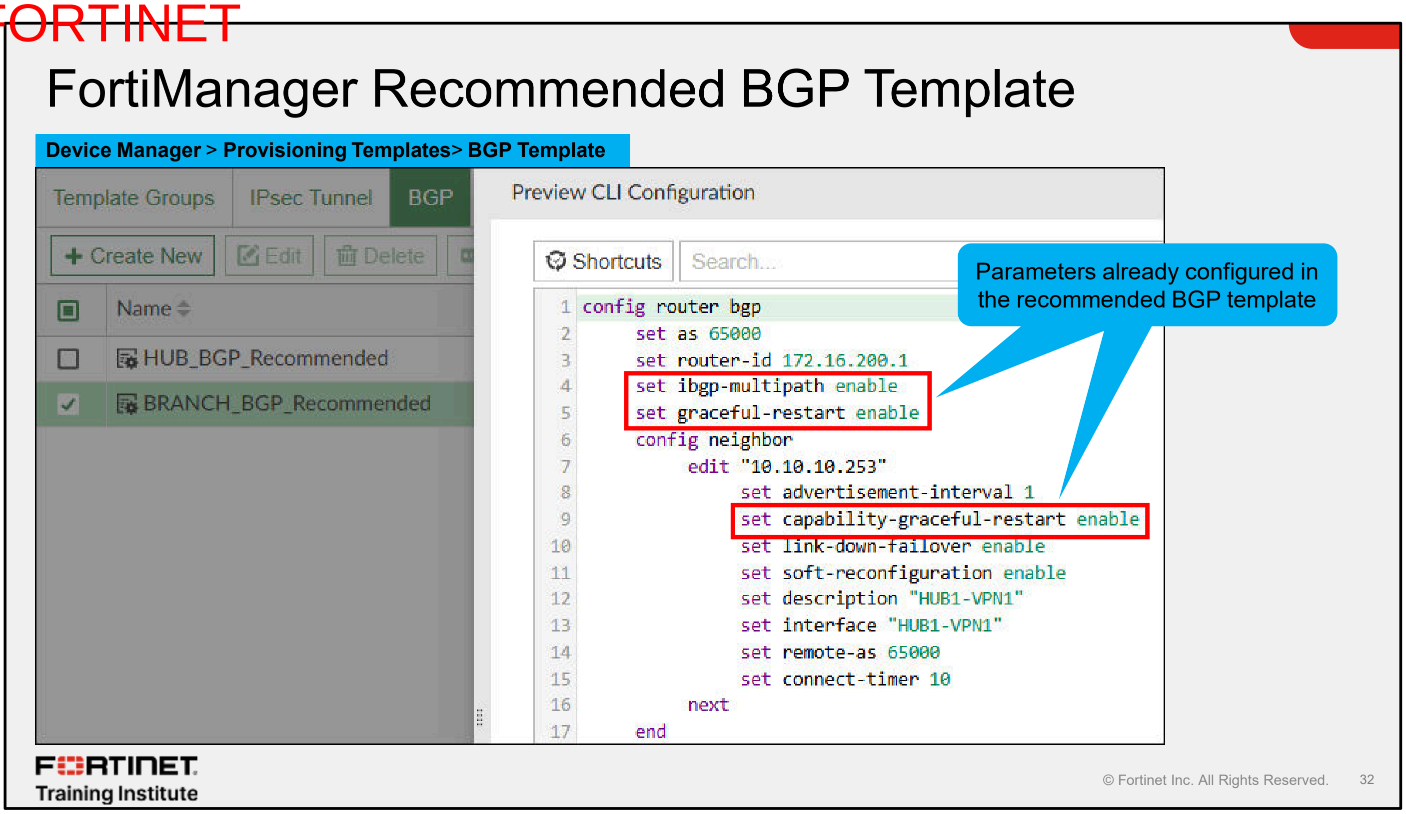
As the BGP router daemon process is only running on the primary unit, BGP peering needs to be reestablished upon HA failover. In a cluster, the FortiGate `graceful-restart` command allows BGP routes to remain in the routing tables. This is particularly important during a reboot or an upgrade maintenance window to avoid potential new BGP convergence and traffic interruption. In this situation, the HA cluster advertises that it is going offline, and does not appear as a route flap. It also enables the new HA main unit to come online with an updated and usable routing table.



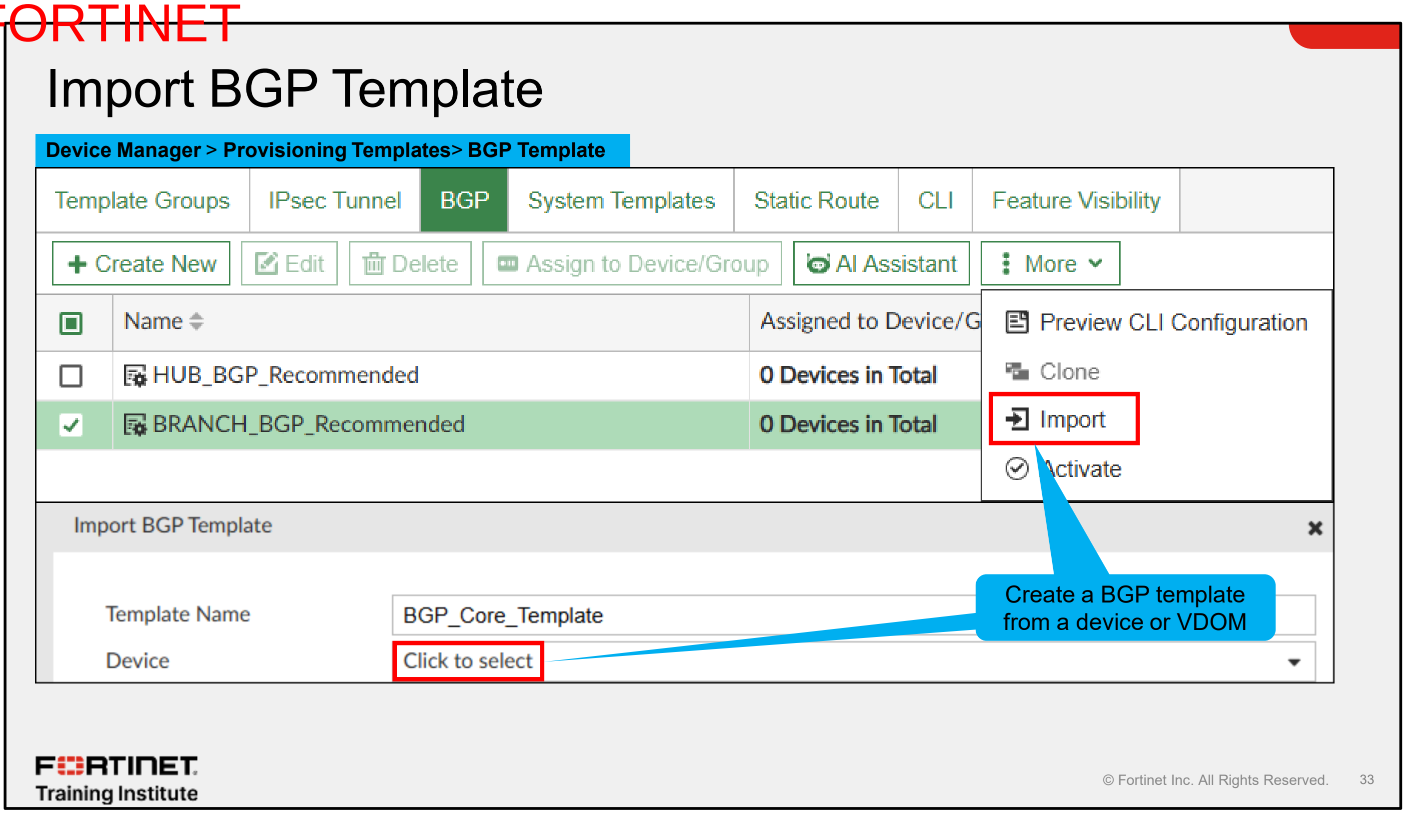
In this section, you will learn about deploying BGP using FortiManager.



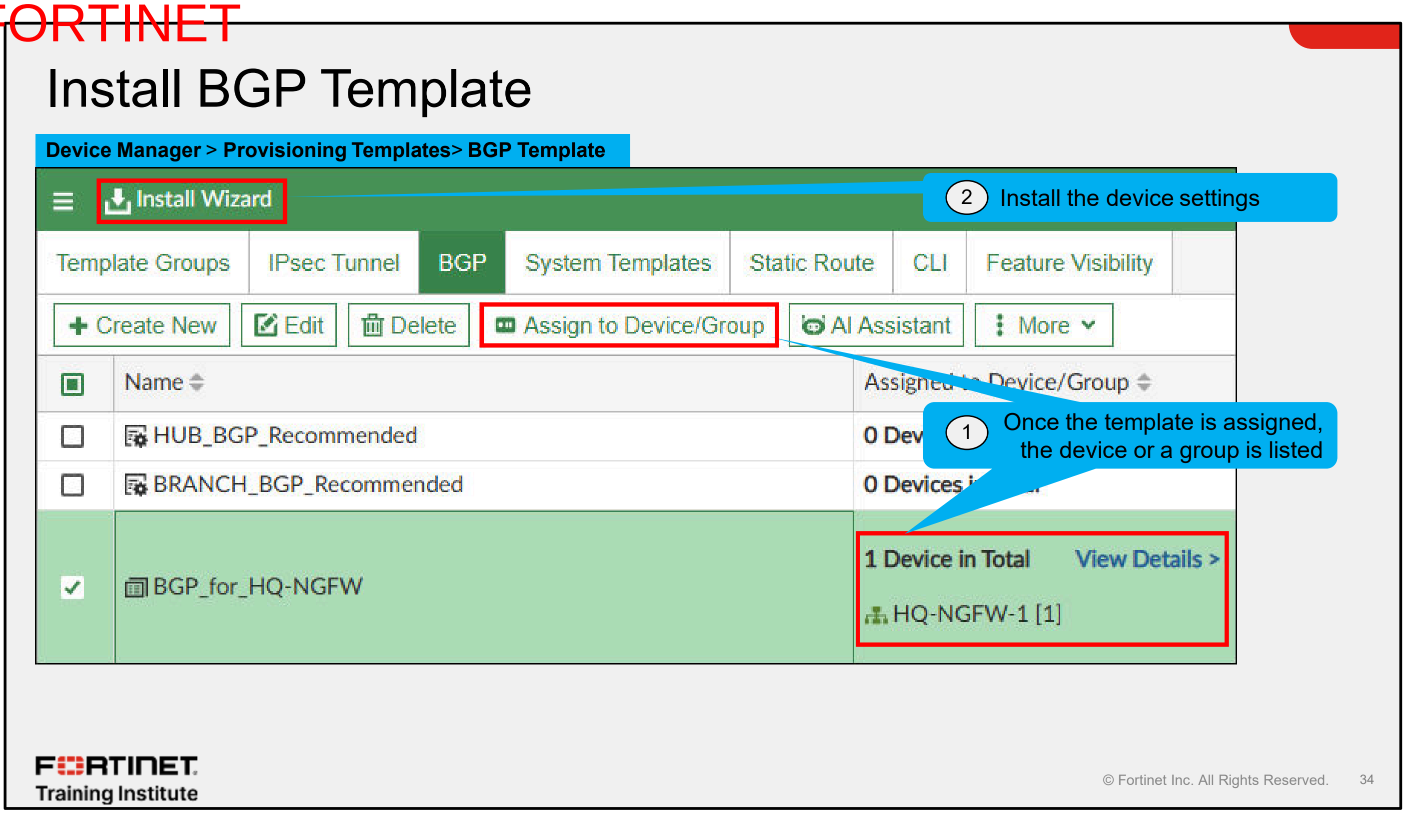
On the FortiManager GUI, you can create BGP templates and use the **Metadata Variables** option to help configure a large BGP environment. For example, in the configuration shown on this slide, a metadata variable allows you to implement the same BGP template on different FortiGate spokes. FortiManager automatically configures the AS number and the last byte of the `router-id` individually according to the metadata variable mapped to each device.



In the recommended BGP template, the configuration follows the best practices recommendations. For example, in the configuration shown on this slide, `ibgp-multipath`, `graceful-restart` and `capability-graceful-restart` are preconfigured for a branch office FortiGate.



Another option is to import the BGP configuration from a device or VDOM. You can then create a BGP template that you can adjust and install on other devices.



After you create the BGP template, you can then assign it to a device or group. Eventually, you must install the device settings with the wizard to get the BGP configuration on the corresponding devices. Metadata variables will be automatically transposed to the appropriate values during the installation on the devices.

Review

- ✓ Implement OSPF features
- ✓ Deploy BGP in enterprise networks
- ✓ Optimize BGP for rapid convergence
- ✓ Deploy BGP using FortiManager

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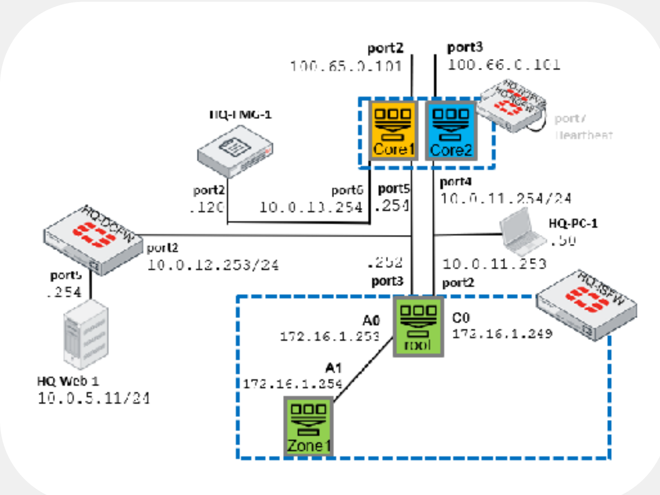
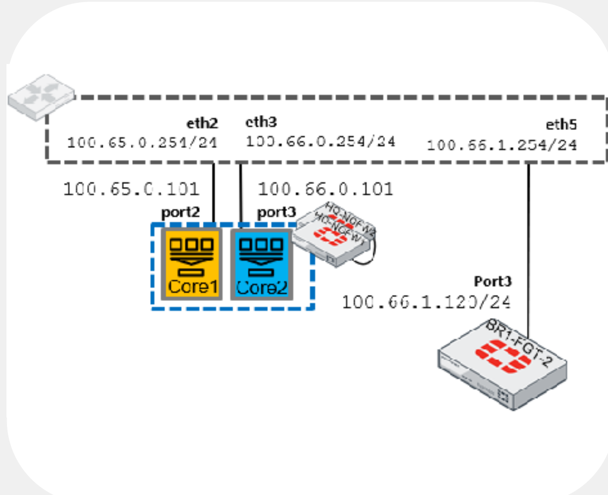
This slide shows the objectives that you covered in this lesson.

By mastering the objectives covered in this lesson, you learned how to configure OSPF and BGP on FortiGate, and about the features they include.

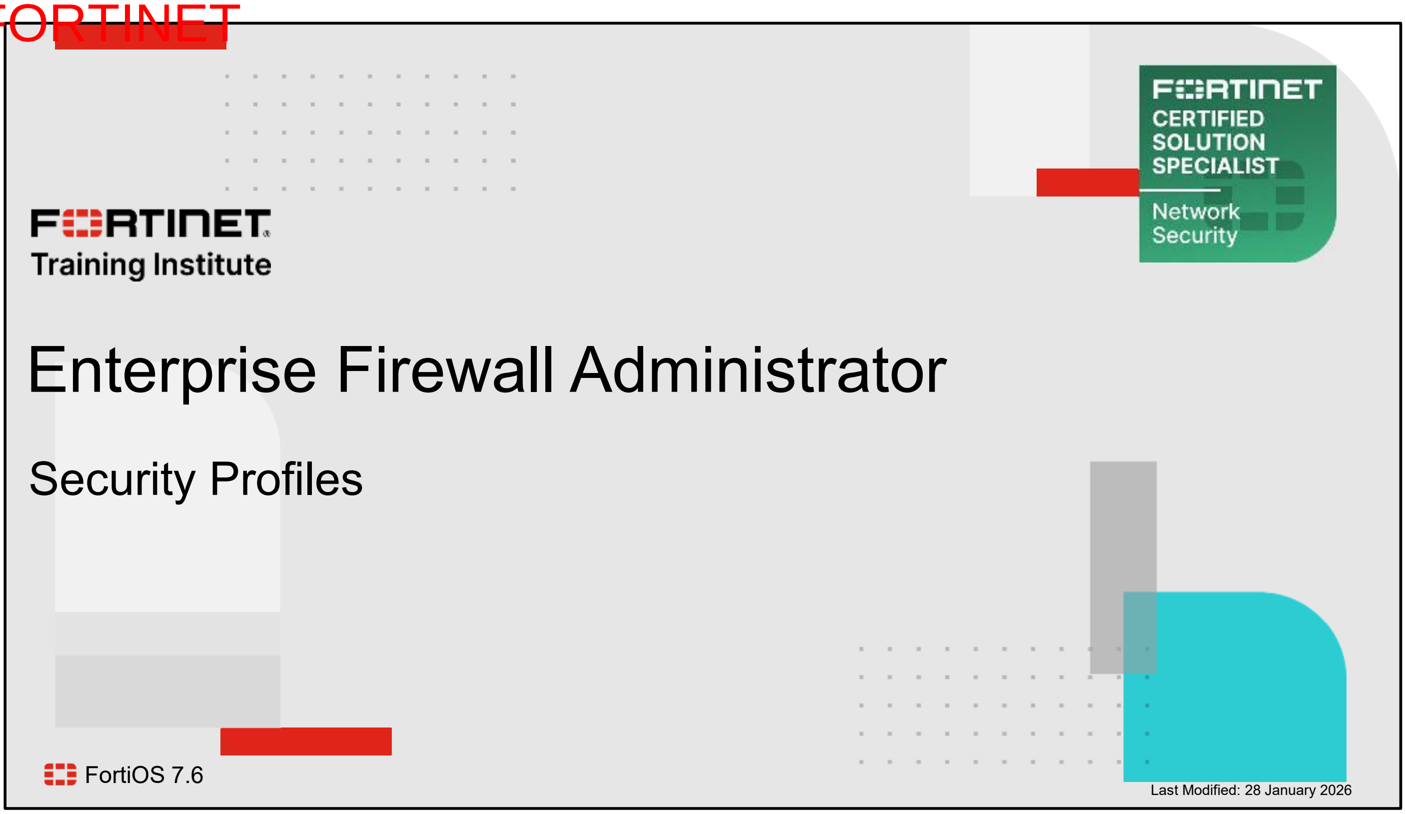
Lab 5—Dynamic Routing Protocols

- On HQ-DCFW, HQ-ISFW, and HQ-NGFW:
  - Configure OSPF ECMP
- On Core1, Core2, and BR1-FGT-2:
  - Configure BGP and a loopback interface as the BGP source





In this lab, you will configure FortiGate devices using FortiManager to provide OSPF ECMP. You will configure BGP as the dynamic routing protocol for the network using BGP templates, as well as a loopback interface as the BGP source.



In this lesson, you will learn advanced concepts of security profiles on FortiGate.

Objectives

- Explain SSL/SSH inspection profiles
- Secure networks: web filter, application control and internet service database (ISDB)
- Integrate intrusion prevention system (IPS) to perform security checks in enterprise networks
- Implement UTM profiles in enterprise networks

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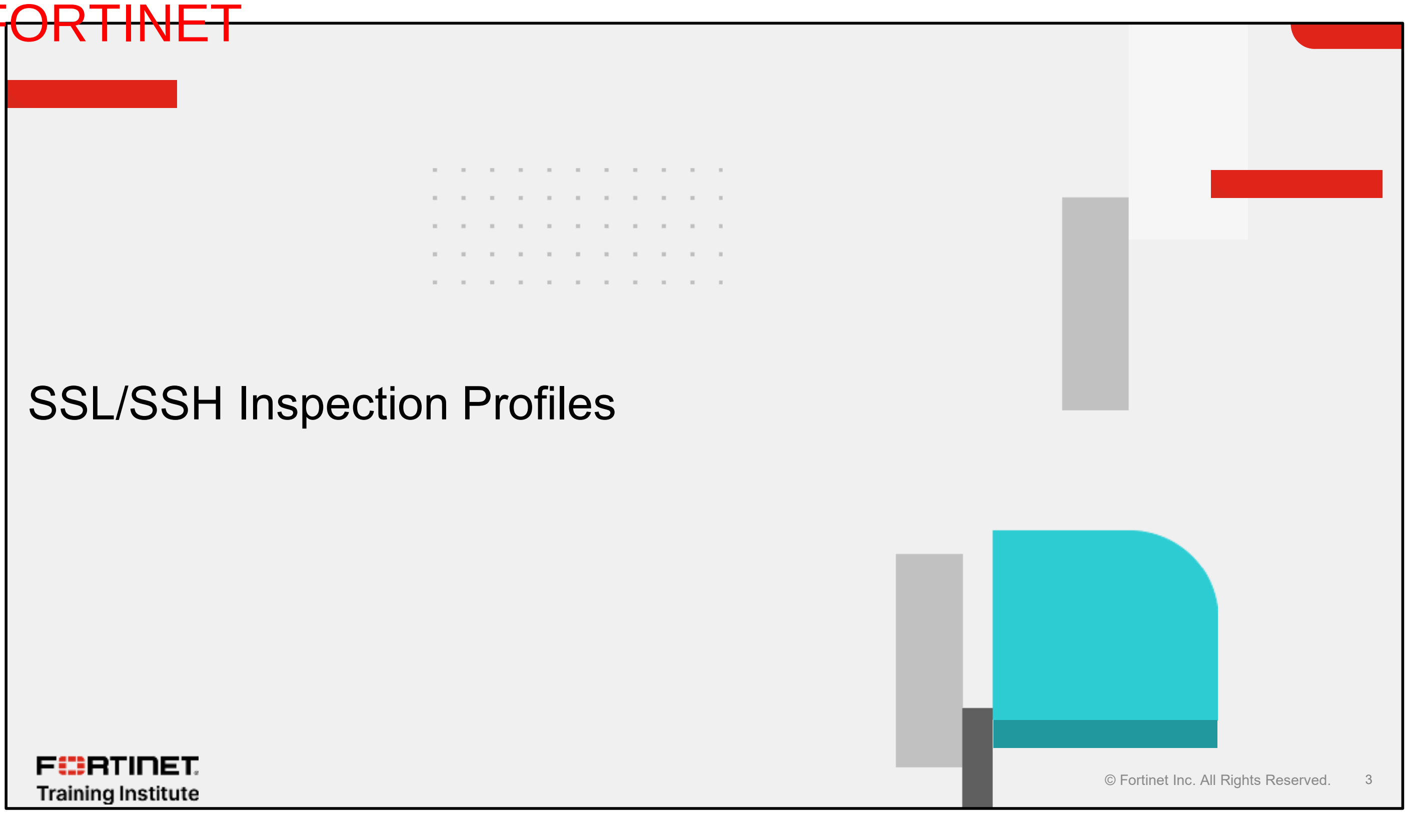
 2

After completing this lesson, you should be able to achieve the objectives shown on this slide.

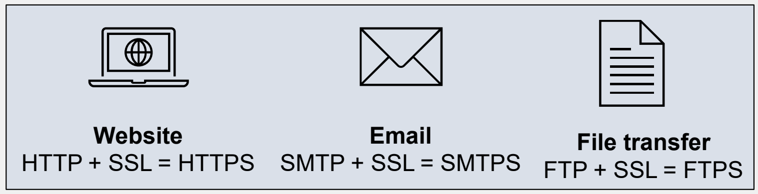
By demonstrating competence in understanding of SSL/SSH inspection profiles, web filters, application control, ISDBs, IPS, and their impact on firewall performance, you will be able to implement and maintain security profiles and policies on FortiGate.



In this section, you will learn about SSL/SSH inspection profiles.

SSL to TLS for Improved Security

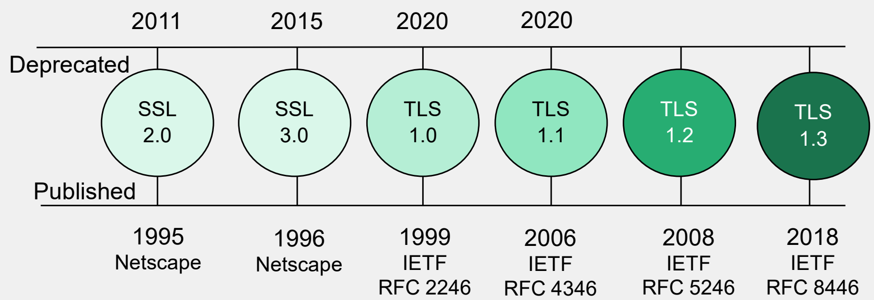
- HTTPS, SMTPS, and FTPS
- SSL (old) vs TLS (new)
- TLS 1.2 and TLS 1.3
- Admin HTTPS SSL




```
config system global
    set admin-https-ssl-versions tlsv1-2 tlsv1-3
end
```

- Find TLS in the configuration

```
show full | grep -f tls
```



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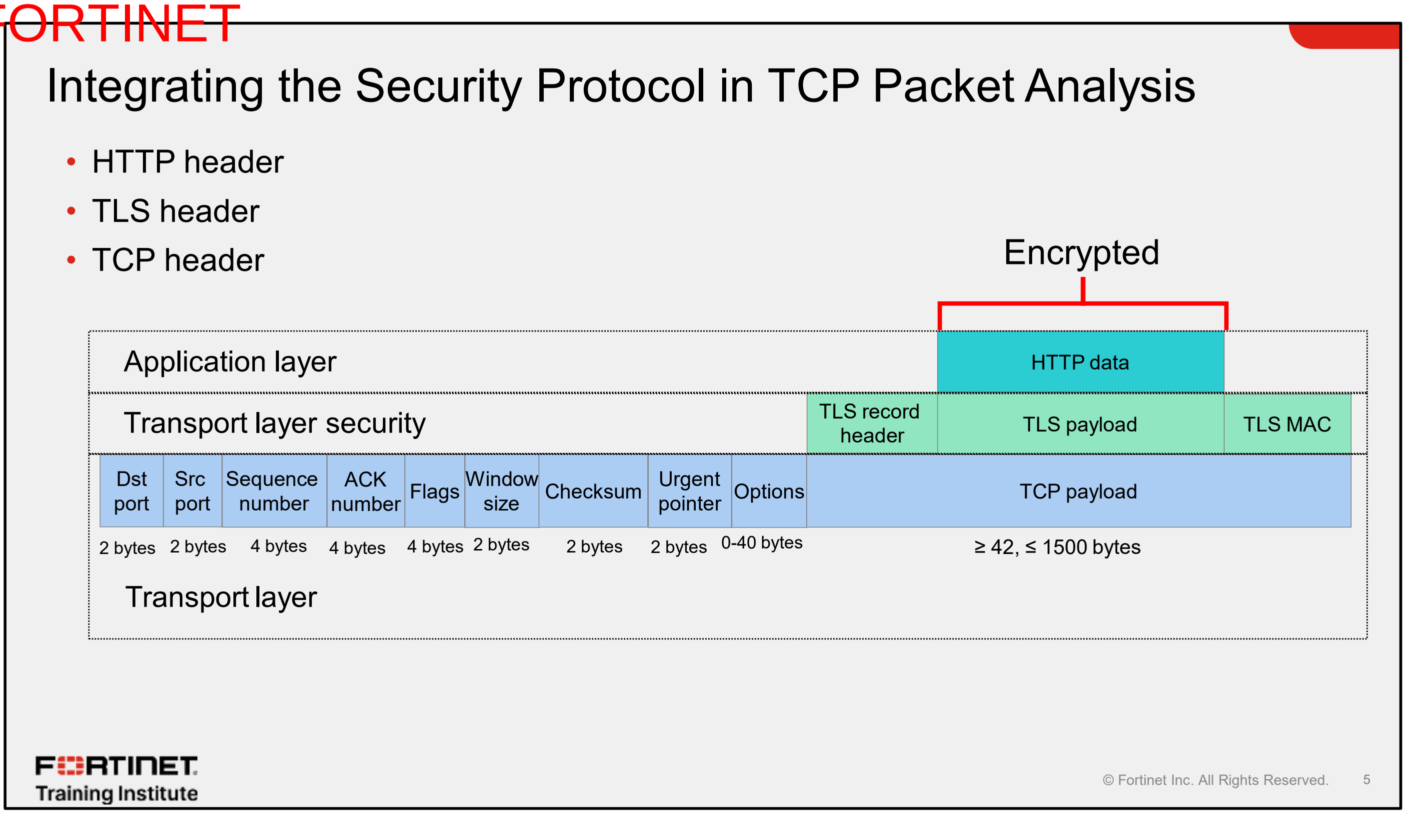
One of the most important protocols for protecting your data is encryption. Common protocols like HTTP, SMTP, and FTP have secure versions that utilize the TLS protocol. These secure versions are known as HTTPS, SMTPS, and FTPS where the "S" stands for "secure".

Previously, information technology environments used SSL. The primary purpose of SSL is to establish a secure and encrypted connection between two devices or applications over any network. However, SSL is an older technology and has several security vulnerabilities. Transport Layer Security (TLS) is a more efficient and upgraded version of SSL that addresses these vulnerabilities. It's important to note that SSL 2.0, SSL 3.0, TLS 1.0, and TLS 1.1 have been deprecated and should not be used. For broad compatibility with older systems, TLS 1.2 is recommended, while TLS 1.3 offers improved security and faster connection times.

You can configure FortiGate to use the SSL protocol that best suits your network configuration. For example, you can change the supported versions for GUI access using the following commands in the CLI: `config system global` and `admin-https-ssl-versions`.

You can display all the TLS commands by entering the command `show full | grep -f`. Make sure that both devices support the same version, otherwise, you will have issues with interoperability.

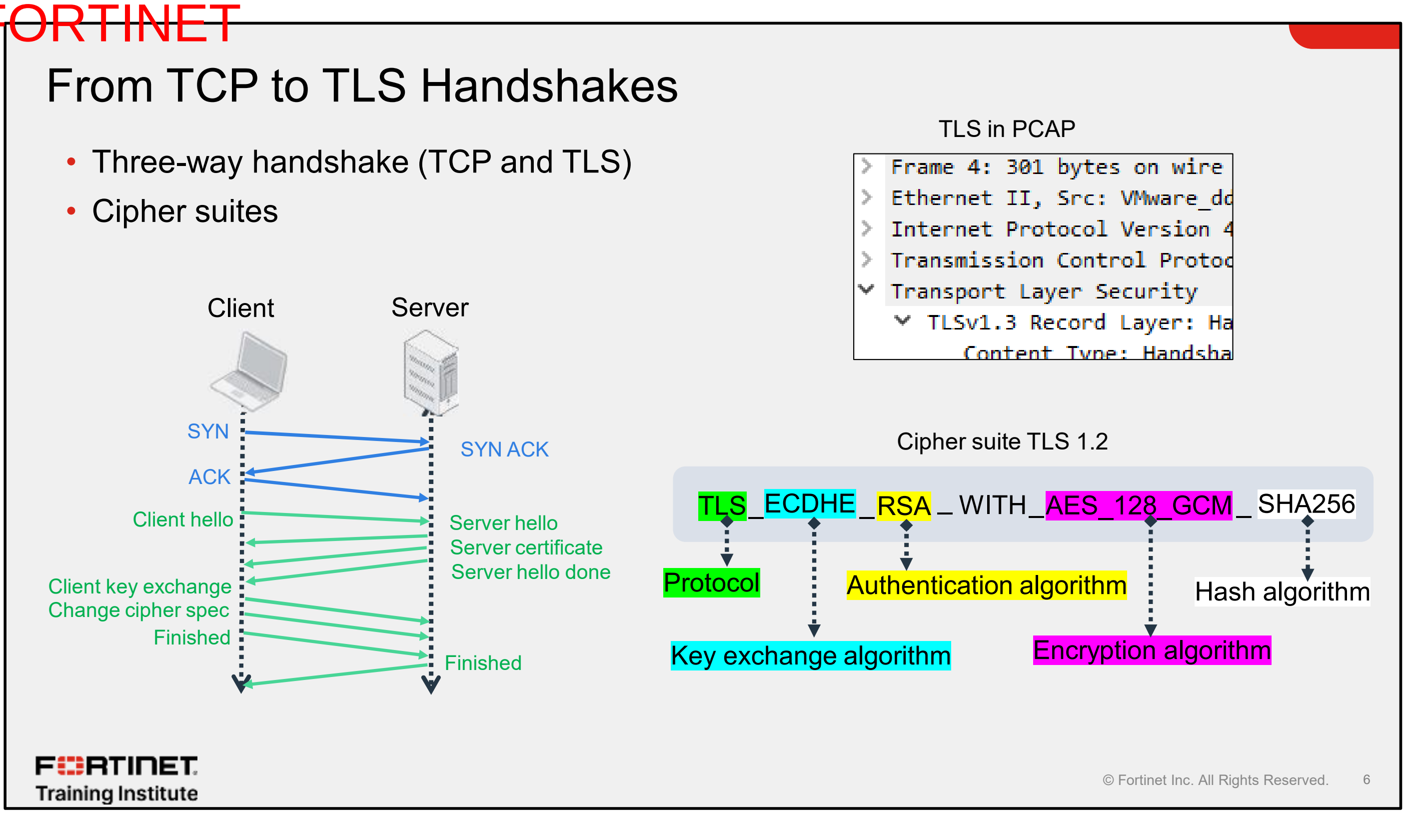


This slide shows how a TCP packet relates to TLS with the transport layer and the application layer of the TCP/IP model.

The TCP header, which is shown at the base, contains essential fields for data transmission, such as destination port, source port, acknowledgment number, and window size, among others. These fields control the connection, flow, and integrity of the data.

Above the TCP header, the payload opens in the transport security layer, which contains essential fields for data transmission.

After the encrypted information passes through the authentication of the transport security layer, then the header of the application layer appears. In this case, that is the HTTP header.



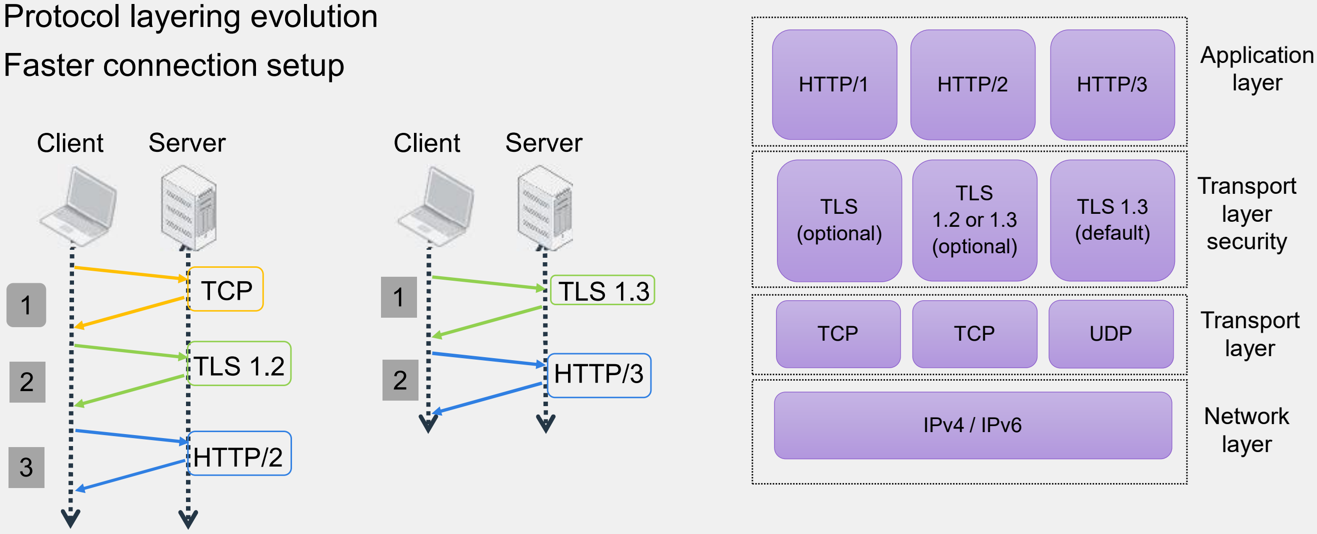
The TCP three-way handshake (SYN, SYN-ACK, ACK) must be completed before the TLS handshake can begin. Once the TCP connection is established between the client and the server, the TLS handshake follows the sequence shown on the slide.

The client hello and server hello phases of the handshake include cipher suites, which are essential for ensuring that both the client and server support the same suites; otherwise, communication will fail. The TLS handshake adds the security layer to the connection. If you examine the packet capture (PCAP) files, you can observe the three-way handshake that is shown on this slide.

The cipher suite consists of several components: the protocol, key exchange algorithm, authentication algorithm, encryption algorithm, and hash algorithm.

Enhancing Security With TLS in HTTP/3

- HTTP
- Protocol layering evolution
- Faster connection setup



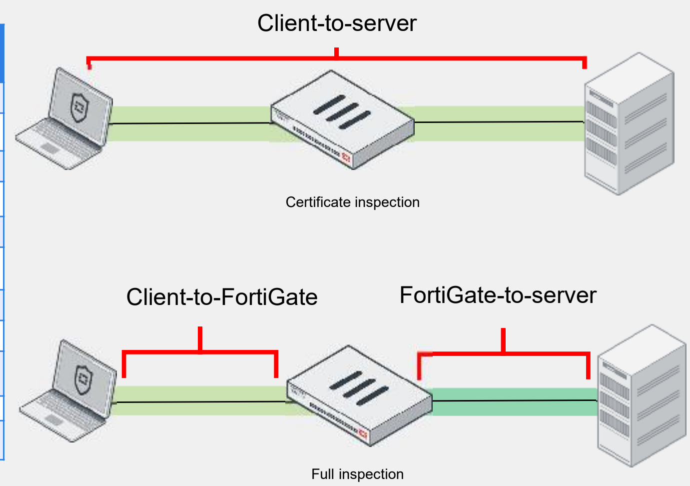
HTTP/2 and HTTP/3 are protocols used for web communication. While they often reference TLS to secure HTTP connections, TLS is not limited to HTTP. It can also be used to secure various other protocols and applications, such as email, messaging, and other forms of communication.

In a layered protocol model, the structure is organized from the bottom to the top. This slide shows at the base the network layer, which operates on the IP layer. The transport layer indicates whether TCP is being used. Above that, the transport layer security shows the different versions of TLS. Finally, the application layer displays the various versions of HTTP.

HTTP/1, the oldest version, treats security as optional, which is not recommended. In contrast, HTTP/3 employs TLS 1.3 by default, providing enhanced security. Both HTTP/3 and TLS 1.3 support faster connection setup times, which reduces the number of round trips required between web servers and clients.

Certificate Inspection Versus Full Inspection

Feature	SSL certificate inspection	Full SSL inspection
Traffic Decryption	No	Yes
SSL Handshake Inspection	Yes	Yes
SNI Parsing	Yes	Yes
Server Certificate Validation	Yes	Yes
URL Extraction from SNI	Yes	Yes
URL Extraction from Certificate	Yes (if SNI not present)	Yes
Web Filtering	Yes	Yes
Application Control	Yes	Yes
Inspection of Encrypted Payload	No	Yes
SSL Proxying	No	Yes
Protection from HTTPS-based Attacks	Limited	Yes



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Two devices that use the same security layer or protocol to encrypt and decrypt data, can transfer data without the intervention of FortiGate. In the example shown on this slide, a laptop connected to a server transfers data without another user or device intercepting the connection.

However, if you must analyze the packet to check it for infected or malicious data, FortiGate must open the file. Otherwise, the firewall won't be able to inspect and verify the data traveling across the network.

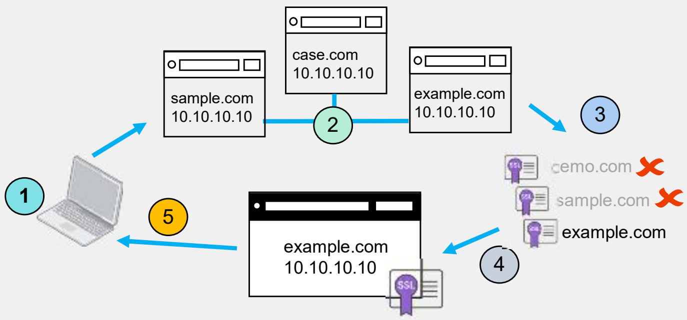
When implementing FortiGate analysis and firewall inspection, it’s important to understand the differences between certificate inspection and full inspection.

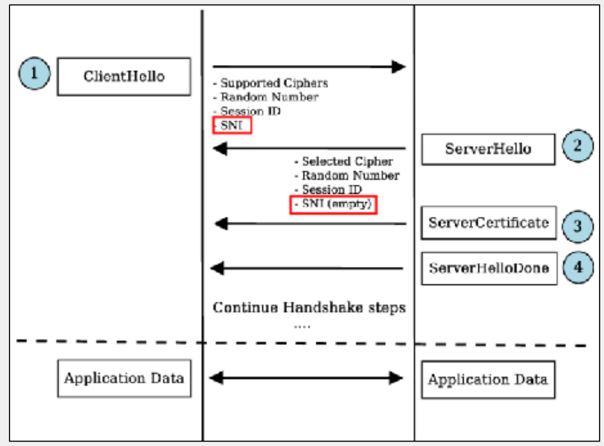
To conserve CPU or RAM resources in your firewall, SSL certificate inspection mode should be sufficient. It covers several functions, such as SSL handshake inspection, server name indication (SNI) parsing, server certificate validation, URL extraction from SNI, URL extraction from certificate, web filtering, and application control. However, if you require FortiGate to perform traffic decryption, inspection of encrypted payload, SSL proxying, and protection from HTTPS-based attacks, you will need to use full SSL inspection.

Keep in mind that full inspection mode uses more CPU and RAM than certificate inspection, but it ensures a deeper and more thorough packet analysis. In the certificate inspection example shown on the slide, FortiGate cannot open and analyze packets. In contrast, during full inspection, where FortiGate splits the communication, it can perform this analysis.

Implement Server Certificate SNI Check Feature

- Extension of TLS
- TLS handshake
- Same server IP, multiple hosted domains
- Cloud computing environments





SNI was first introduced in 2003 as an extension to the TLS protocol. It was developed in response to the growing need for more efficient use of IP addresses and the increasing popularity of virtual hosting.

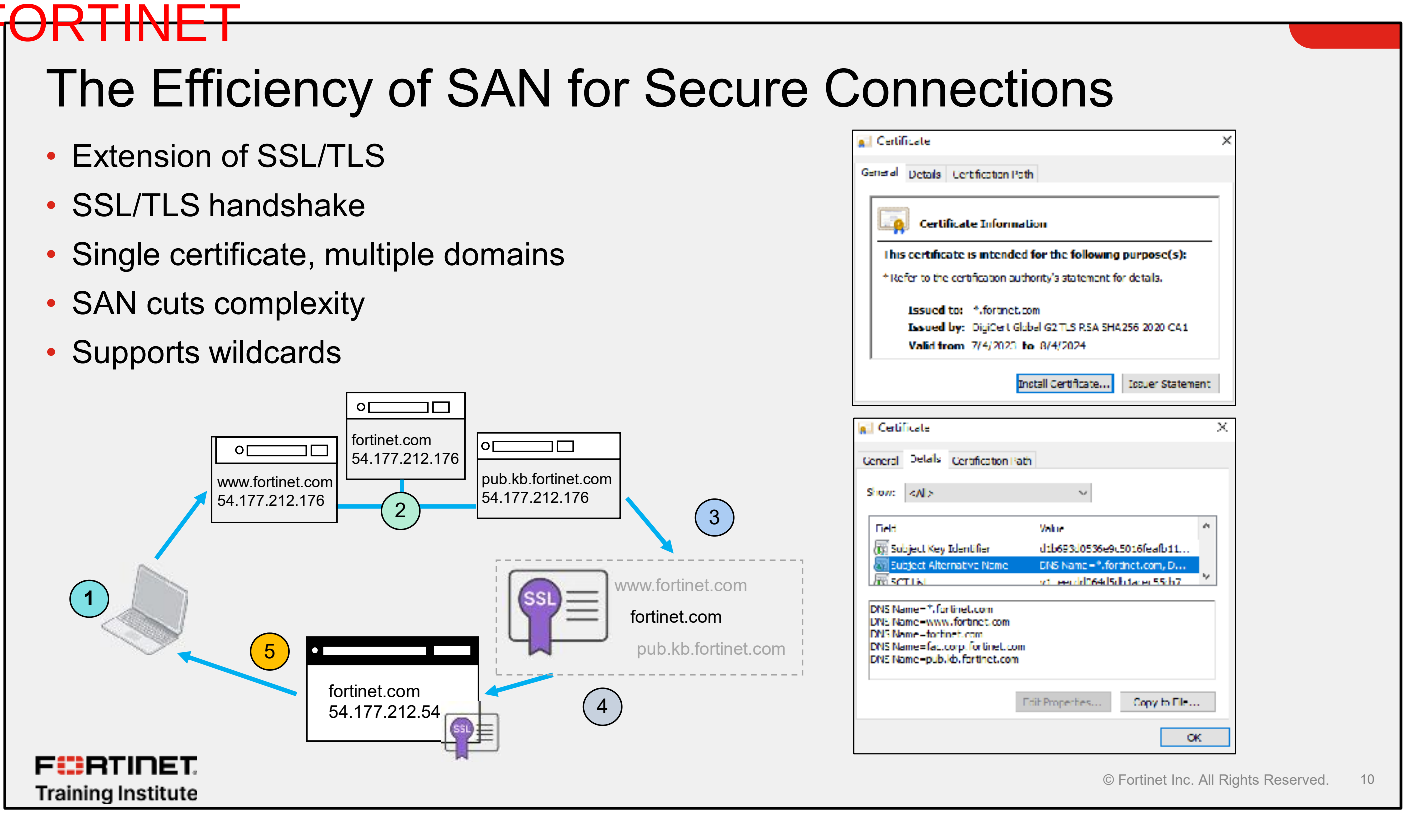
SNI enables the use of different certificates for different domains hosted on the same server during the TLS handshake process.

The stages of the TLS handshake process are as follows:
1. The client sends a Client Hello message, which includes SNI, to the server.
2. The server receives the packet from the client and replies with Server Hello, which may initially include empty information in the SNI response.
3. The server sends the authentication details required by the SNI request.
4. Finally, the server sends a Server Hello Done message.

SNI allows servers to host multiple certificates on a single IP address, enabling various secure (HTTPS) websites or services to use that address without sharing a common certificate.

The slide also shows a user access request and response using SNI:
1. A user initiates a request to access example.com.
2. The server (`10.10.10.10`), which hosts sample.com, case.com, and example.com, receives the packet.
3. Upon analyzing the SNI, the server rejects access to other domains, such as demo.com and sample.com, and it selects example.com.
4. The server sends the SSL certificate for example.com.
5. Consequently, the user can trust the website associated with the domain example.com.

SNI plays a crucial role in cloud computing environments, where multiple customers may host their secure websites on shared infrastructure. It allows cloud providers to deliver HTTPS services more effectively and at a reduced cost.



Subject alternative name (SAN) was introduced in 1999 as an extension in SSL/TLS certificates. Unlike traditional certificates that cover a single domain name or hostname, SAN allows a single certificate to cover multiple domain names or host names. SAN, like SNI, also operates during the handshake process.

In the example shown on this slide, you can see that five alternative names are assigned to the fortinet.com certificate.

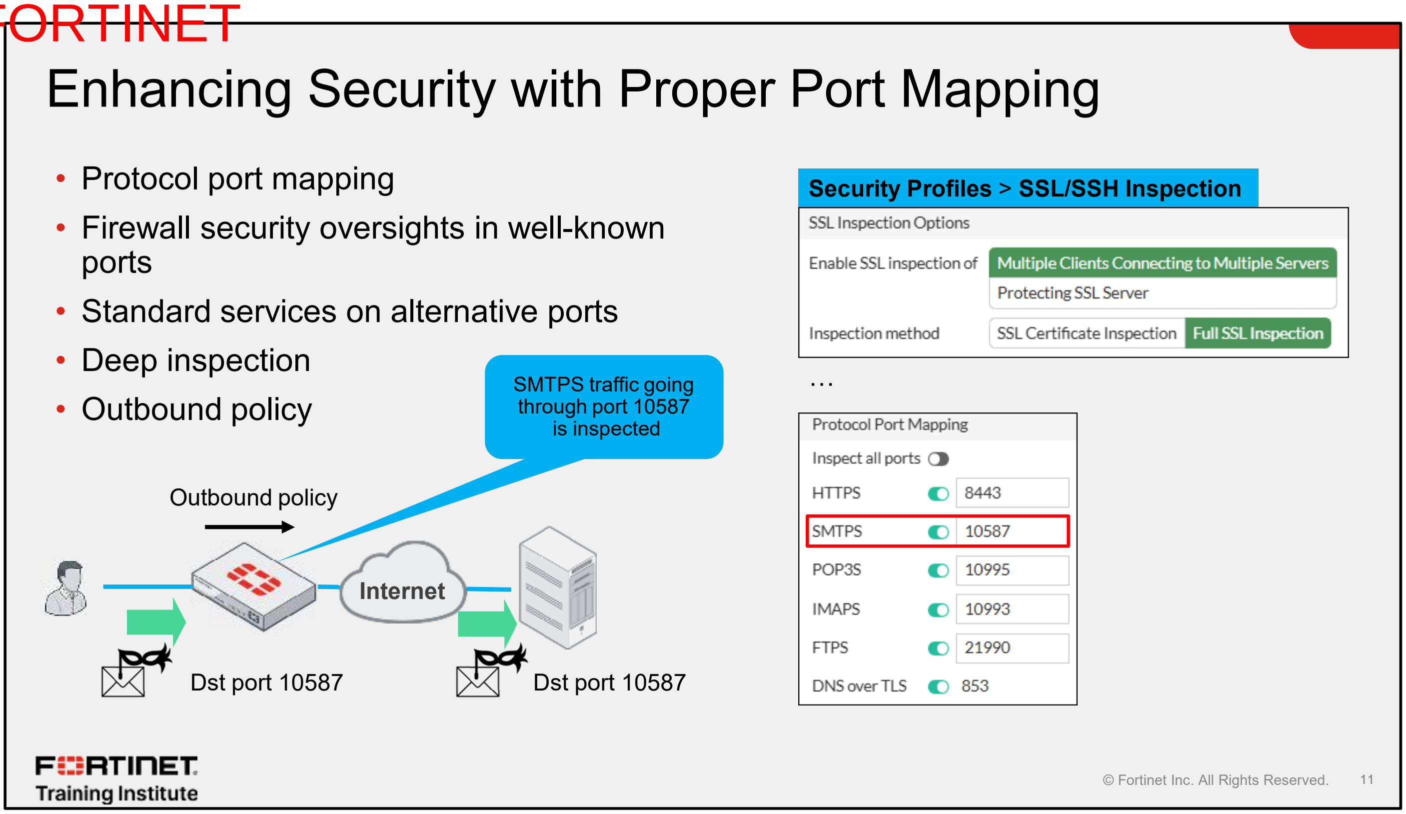
This process can be replicated for any website to identify how many domains are secured under the same certificate, establishing trust based on the provided certificate.

The slide also shows a user access request and response using SAN:

1. A user initiates a request to access the website fortinet.com.
2. The server (`54.177.212.176`) receives the request and determines which TLS certificate to use in response, based on the SAN.
3. After analyzing the SAN, the server recognizes that it can support requests for other domains such as www.fortinet.com and pub.kb.fortinet.com, in addition to fortinet.com. This is possible because the server uses the same certificate, which includes SAN entries for these additional domains. If the SAN matches, the server can provide the appropriate certificate.
4. The server then sends the SSL certificate, which is fortinet.com in this case.
5. As a result, the user can trust the website associated with any of the domains covered by the SAN.

Managing a single SAN certificate is simpler than managing multiple individual certificates, which reduces administrative overhead and potential configuration errors.

SAN certificates can include wildcard entries, allowing them to secure an entire domain and its subdomains. This feature enhances the versatility of SAN certificates and is something that SNI certificates don't have.



Understanding protocol port mapping is vital for preventing firewall security oversights.

Blocking well-known ports like HTTP (80), HTTPS (443), SMTPS (587), SSH (22), FTPS (990), and DNS (53) in your firewall is not the same as blocking nonstandard ports, such as HTTP (8080), HTTPS (8443), SMTPS (10587), SSH (2222), or FTPS (21090). This concept refers to using standard services on alternative ports, or deep inspection.

In the example on this slide, a user is trying to use the SMTPS service on a nonstandard port, port 10587. Because the firewall in this example has enabled protocol port mapping for SMTPS and port 10587, FortiGate inspects this traffic. Otherwise, by default, FortiGate inspects only SMTPS port 587 and the user, using the nonstandard port 10587, would bypass the firewall inspection to reach the SMTPS server.

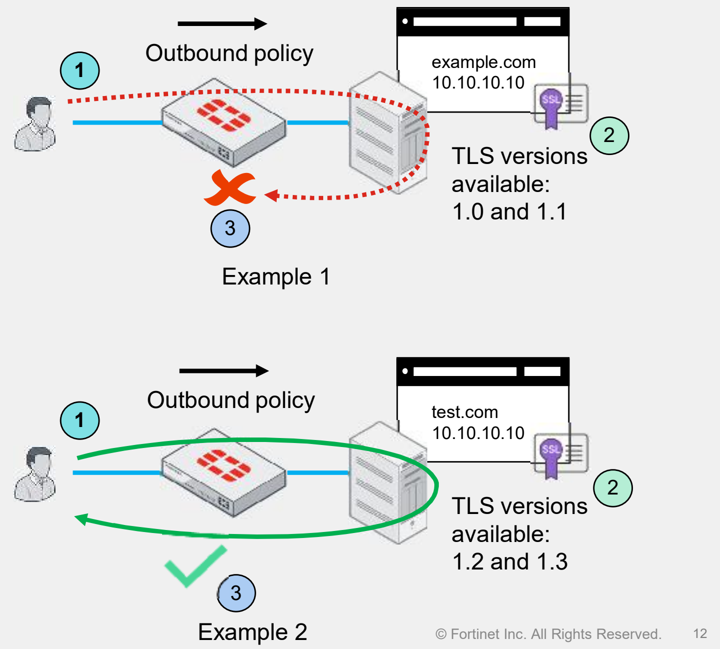
Protocol port mapping can be enabled with an outbound policy only if the following conditions are met: SSL inspection must be activated for multiple clients connecting to multiple servers, full SSL inspection must be employed, and the appropriate protocol port-mapping settings must be configured. Additionally, protocol port mapping is applicable only to proxy-based inspections, while flow-based inspections assess all ports, regardless of protocol port-mapping settings.

Enabling these settings may increase your firewall's resource usage, so it’s recommended not to activate them all at once. However, it’s important to recognize the significance of these settings and to closely monitor logs, particularly for nonstandard ports or ports that are not typically used on your network.

Optimizing TLS and HTTPS Security

- Minimum TLS version supported
- HTTPS example
- Various protocols applicable

```
config firewall ssl-ssh-profile
edit "ssl_ssh_profile"
config https
set unsupported-ssl-version block
set min-allowed-ssl-version tls-1.2
end
end
```



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FortiGate can detect the minimum TLS version allowed in an outbound policy. In the example shown on this slide, the configurations `unsupported-ssl-version` and `min-allowed-ssl-version` block TLS versions 1.0 and 1.1 while accepting newer versions.

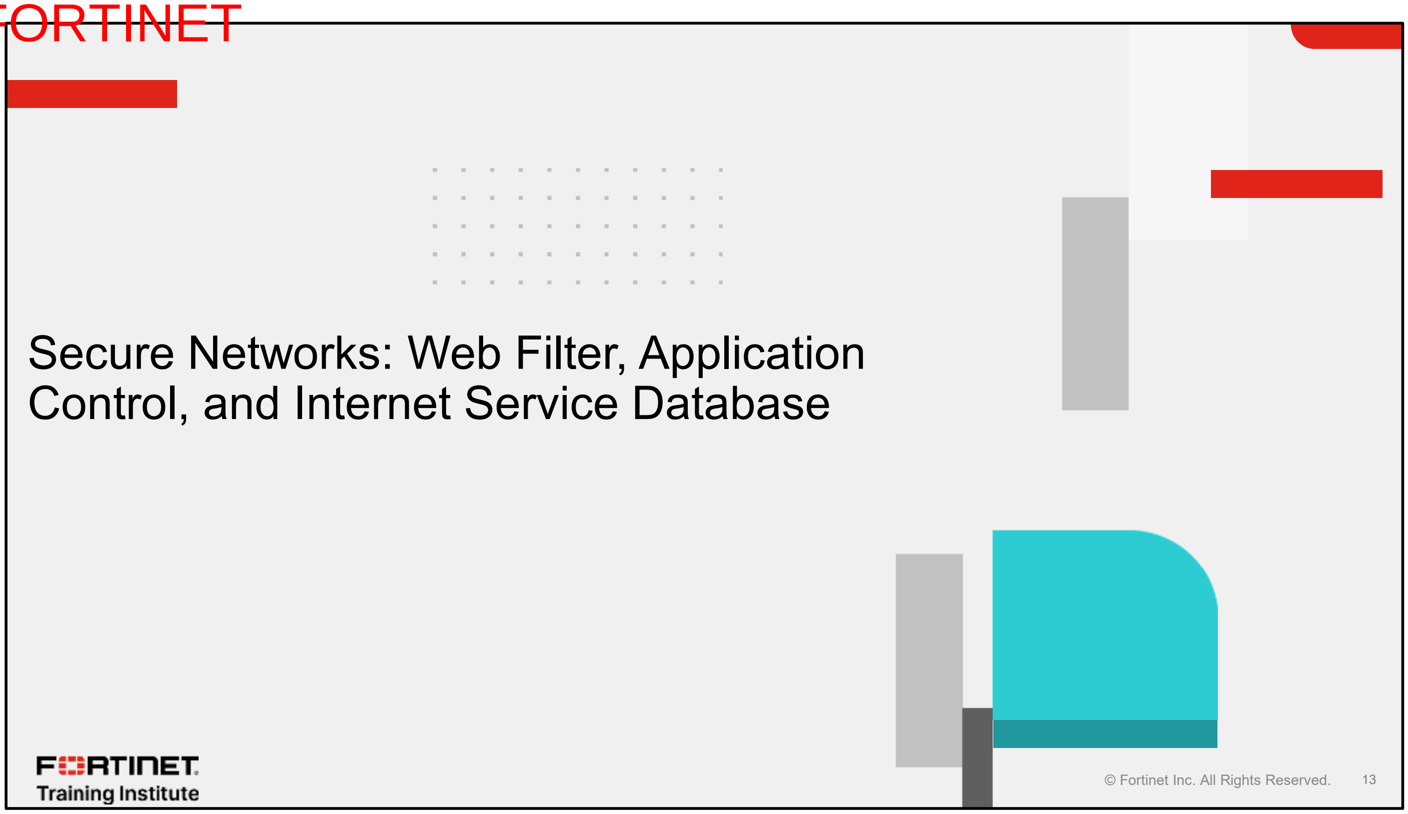
Here are two example scenarios:

Example 1
1. A user attempts to access example.com.
2. The web server offers TLS versions 1.0 and 1.1.
3. FortiGate blocks the connection because it is instructed to block these unsupported TLS versions.

Example 2
1. A user attempts to access test.com.
2. The web server offers TLS versions 1.2 and 1.3.
3. FortiGate allows the connection because these TLS versions are supported.

By using FortiGate to specify which versions of TLS should be blocked or accepted, you can enhance the security of your enterprise network's HTTPS protocol. Always ensure optimal protection by understanding that, while you cannot control the TLS version a website uses, you can mitigate the risks posed by weaker versions.

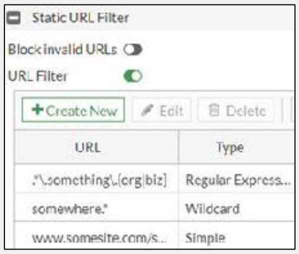
You can configure the setting related to TLS version control only through the CLI. You can also apply these version control settings to other protocols, such as SMTPS, POP3S, IMAPS, and FTPS.



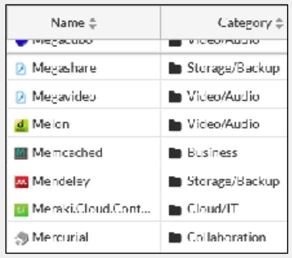
In this section, you will learn about the main differences among web filter, application control, and internet service database (ISDB).

Comparing Web Filter, Application Control, and ISDB

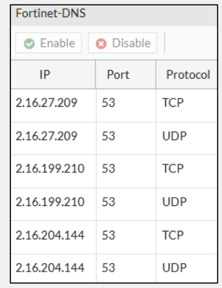
Web filter
- Domains and URLs
- Flow and proxy-based



Application control
- Layer 7 OSI
- Flow-based



ISDB
- Layer 3 and 4 layer OSI
- No inspection



- CLI

```
conf webfilter profile
edit "profile"
end
```

- CLI

```
conf application list
edit "profile"
end
```

- CLI

```
conf firewall internet-service-name
edit "service"
end
```

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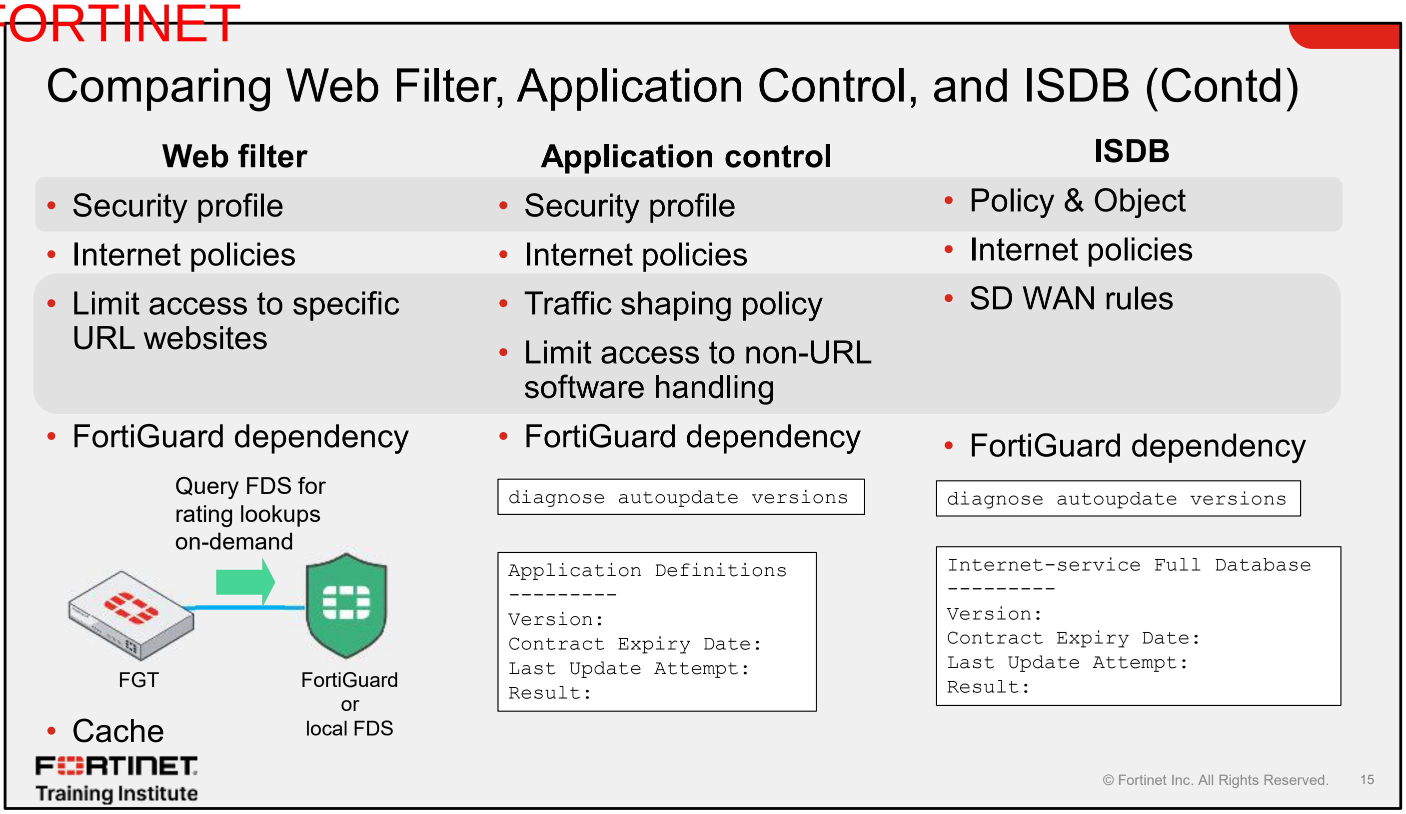
In FortiGate firewalls, web filter, application control, and ISDB are three different features used to manage and control network traffic. It is important that you understand the main differences between these features so that you can implement them effectively in your enterprise firewalls.

Web filter is a security profile that manages website access by inspecting content, URLs, domains, server name indications, or SANs in either flow-based or proxy-based inspection modes.

Application control operates solely with flow-based inspection and identifies applications through their transmission patterns, using application signatures and protocol decoders, as well as rate-based IPS signatures to spot anomalies.

ISDB serves as a policy and object tool, holding a database of IP addresses for layer 3 use and TCP/UDP ports for layer 4 usage, and operates without inspecting the traffic.

You configure each feature through the CLI differently, and they each have different options. For example, web filter has the subcommand `urlfilter-table`, that can significantly extend the configuration of the web filter.



While web filter and application control are security profiles, ISDB is a policy and an object and is part of the firewall addresses section.

All three features can be used in normal firewall policies, like internet policies.

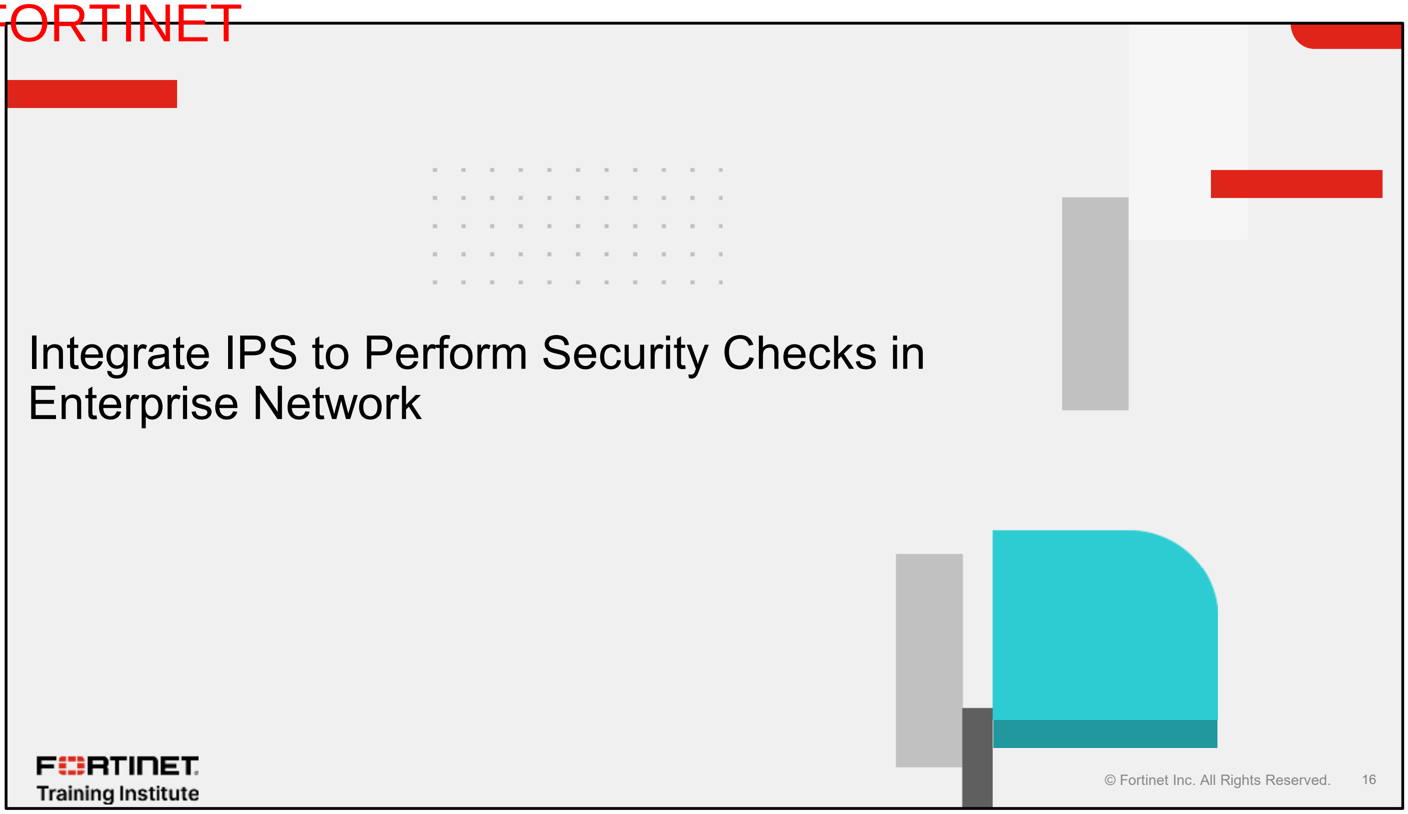
Web filters are primarily used to restrict access to specific URLs. Application control is used to manage traffic shaping, SD-WAN priority rules, and access to non-URL-based software. ISDB plays a crucial role in SD-WAN rules by limiting traffic to certain vendors or geographical zones. To use application control within SD-WAN rules, enable the **Application Detection-Based SD-WAN** feature under **System** > **Feature Visibility**.

To ensure that these features work properly, all FortiGuard updates must be applied to the firewall; otherwise, you can expect malfunctioning.

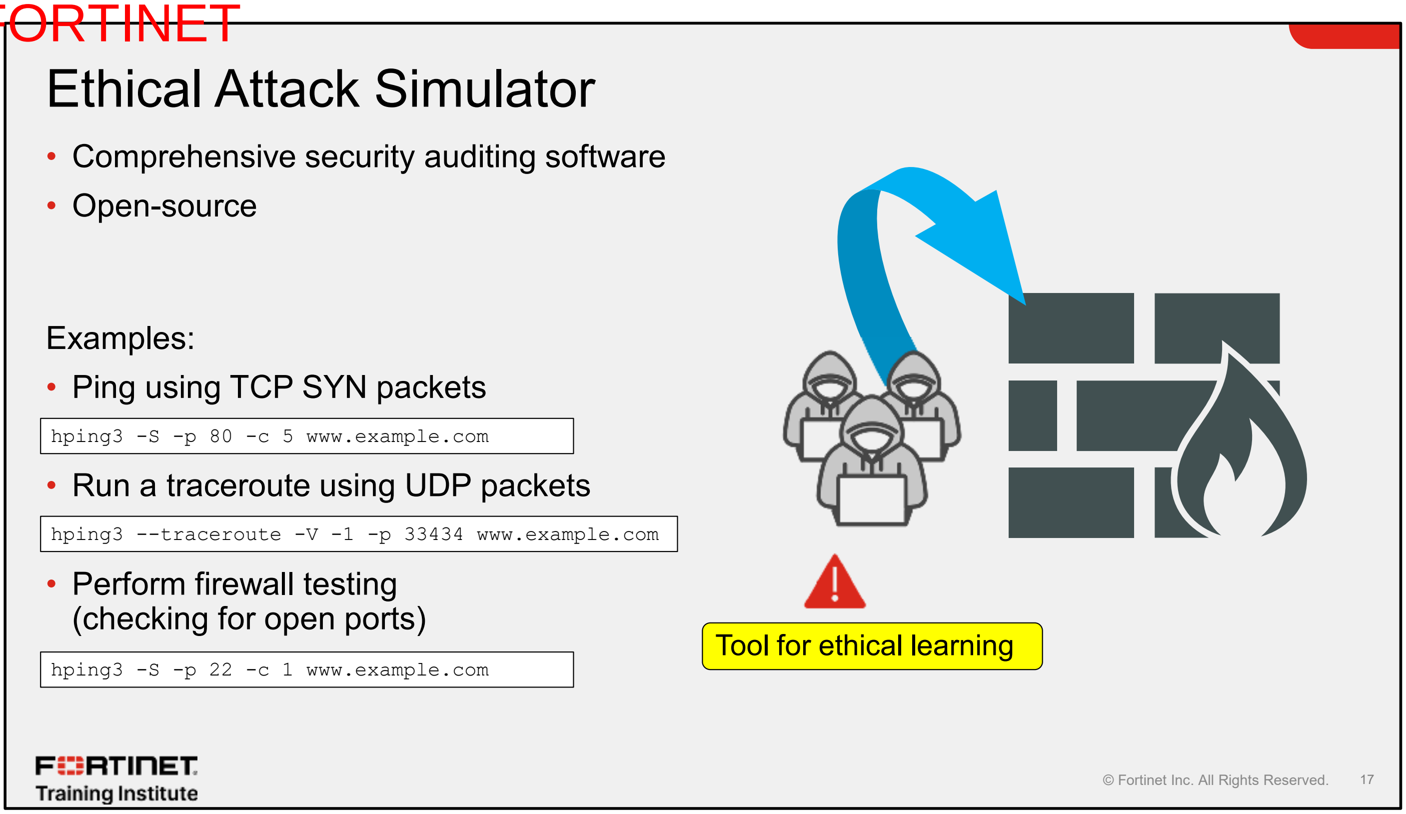
To find the version of the database for each feature:

- For application control and ISDB, enter the command `diagnose autoupdate versions`. For application control, search the `Application Definitions` section. For ISDB, search for `Internet-service Full Database`.
- The web filter database isn't hosted locally on the firewall. FortiGate must query FortiGuard or FortiManager, which act as a FortiGuard Distribution Center (FDS) for rating lookups on-demand.

To view the web filter database that FortiGate uses to categorize a website, look at the cache of recent web filter lookups.



In this section, you will learn about how to integrate IPS to perform security checks in an enterprise network.



To validate and simulate IPS attacks, you can use hping3.

hping3 is a network tool used for various purposes, including network security auditing, firewall testing, network performance analysis, probing and testing network devices, conducting denial-of-service attacks for ethical testing, simulating traffic patterns, and crafting custom TCP/IP packets for protocol analysis.

hping3 is free, open-source software that is distributed under the GNU general public license (GPL), which allows users to freely use, modify, and distribute software.

This slide shows the following three examples of using hping3 via the CLI:

- Ping using TCP SYN packets: This command sends five TCP SYN packets (`-S`) to port 80 (`-p 80`) of www.example.com. It's similar to a regular ping but uses TCP instead of ICMP.

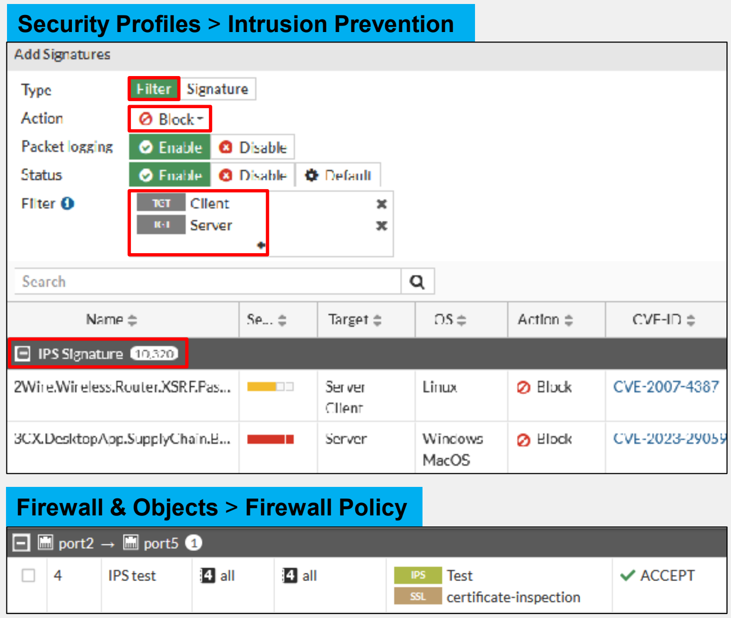
- Traceroute using UDP packets: This command performs a traceroute (`--traceroute`) to `www.example.com` using UDP packets (`-1`) with a destination port of 33434 (`-p 33434`). The `-V` option enables verbose mode, providing detailed output.

- Firewall testing (checking for open ports): This command sends a single TCP SYN packet (`-S -c 1`) to port 22 (`-p 22`) of `www.example.com`. If the port is open, you'll receive a `SYN/ACK` response; if it's closed, you'll get a `RST` (reset) response.

You should use this tool responsibly and ethically. The main objective is to deal with false positives in your enterprise firewall network.

Use Case 1—Protect Server and Client Targets

- IPS profile
- **Type: Filter**
- **Action: Block**
- **Target: Server and Client**
- All severity levels, protocols, and operating systems are blocked
- This strategy comes with a risk of false positives
- Know your network protocols and workflows



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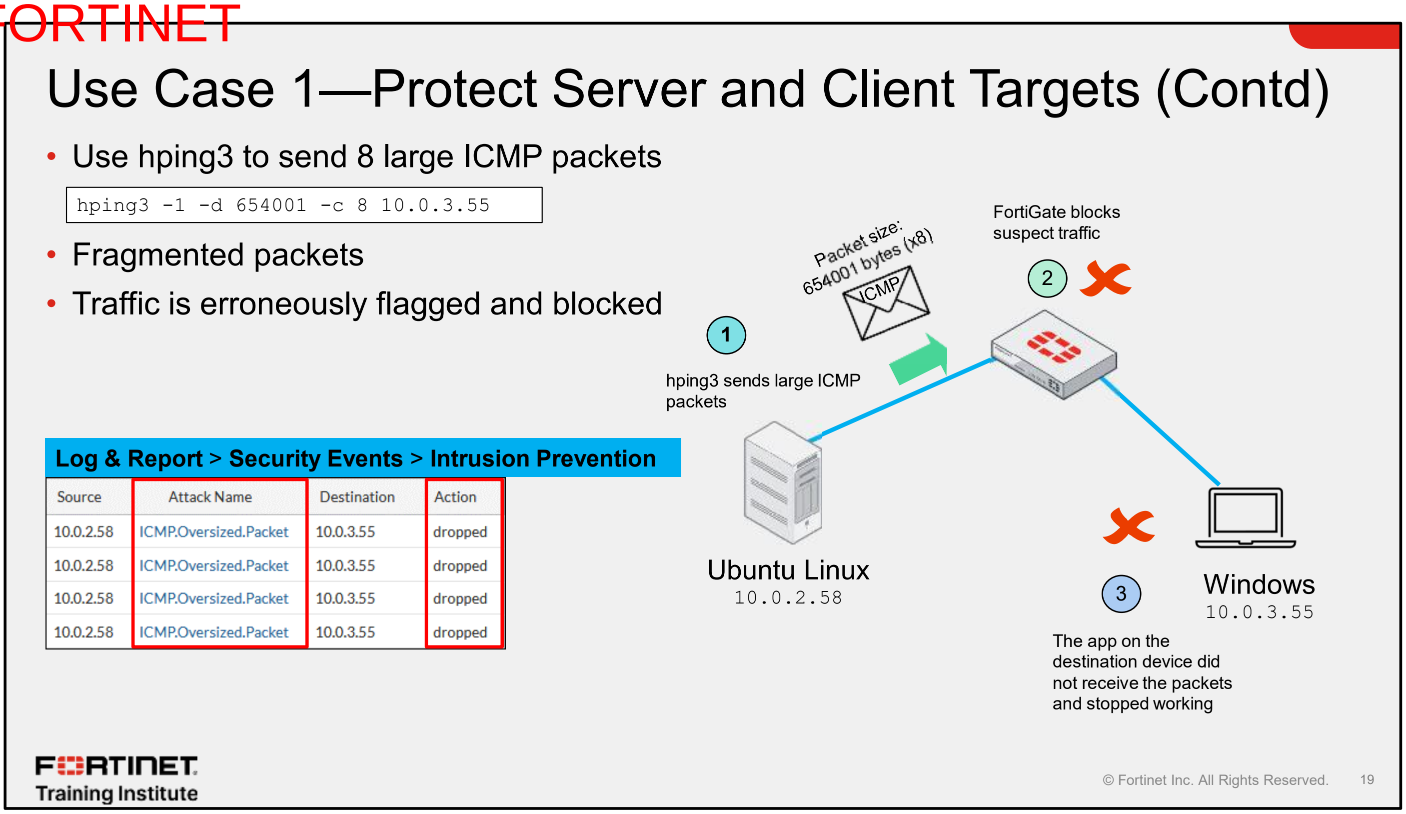
Let's imagine you have configured an IPS profile with the client and server as targets, to protect your network resources.

To create a profile for this purpose, set the **Type** to **Filter**, set the **Action** to **Block**, enable **Packet logging**, and add the target **Client** and **Server** to the **Filter**. To ensure that this filter blocks everything that it might consider an attack, you include all available severities, protocols, and operating systems. In the example shown on this slide, these settings result in 10,320 IPS signatures being delivered.

With these settings, this filter will block all possible attacks on your network; however, it could also produce false positives. For example, it would block an app that is trying to use a different packet size, potentially causing a disruption to your day-to-day business or operations. So, when you are configuring an IPS filter, it is important that you have a clear understanding of your network's protocols and workflows. You want to block activity that is excessive, intrusive, or harmful, without inadvertently blocking activity that is vital to your business.



The command shown on the slide sends eight large ICMP packets (654001 bytes) to `10.0.3.55`.

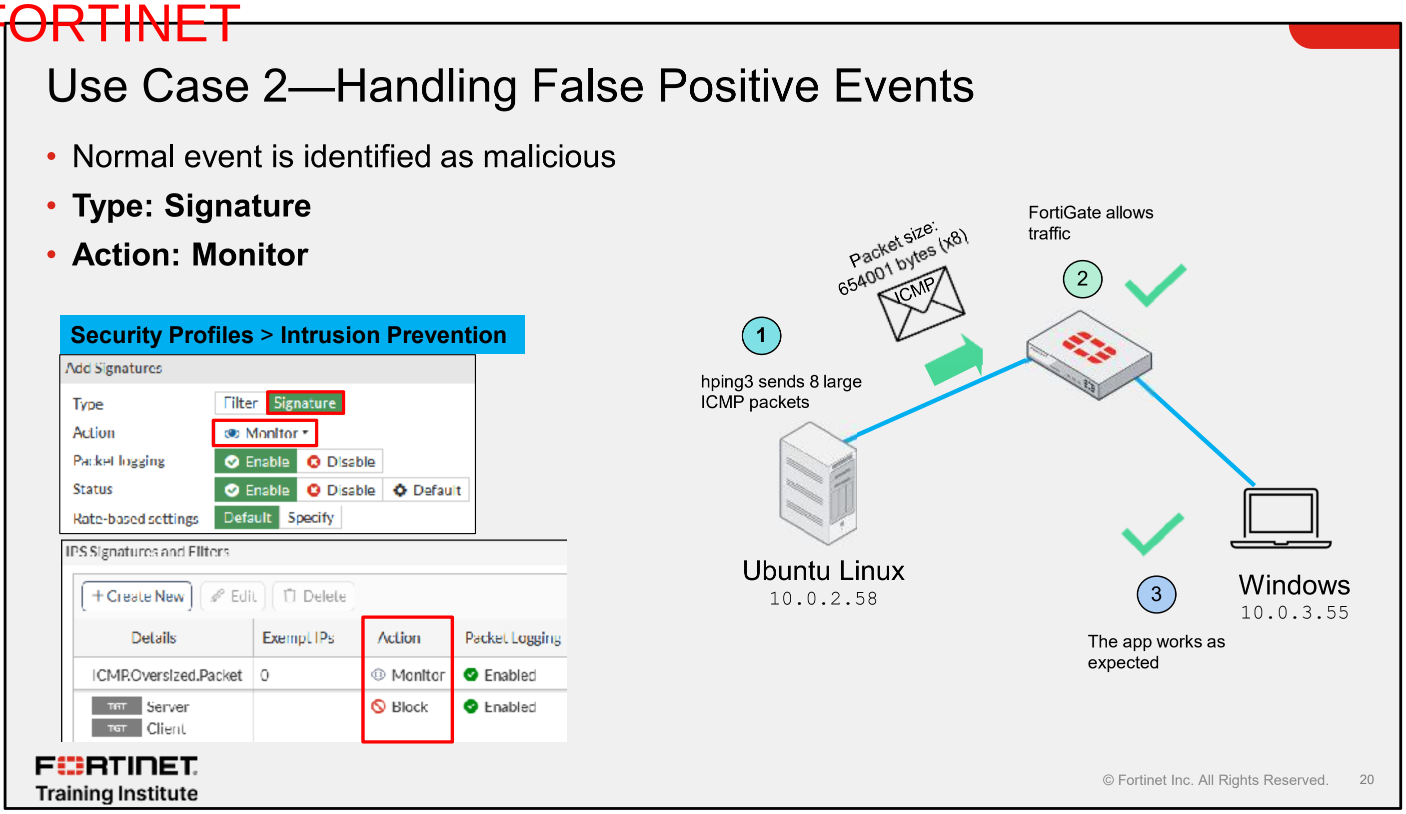
The details of the command used in this use case are as follows:

- hping3 is the program used to generate and send the ICMP packets.
- `-1` tells hping3 to use ICMP for sending packets.
- `-d 654001` sets the ICMP packet size to 654001 bytes, a size that exceeds the maximum transmission unit (MTU), causing fragmentation.
- `-c 8` tells hping3 to send 8 ICMP packets.
- `10.0.3.55` is the destination IP address for the ICMP packets.

The diagram on the slide illustrates:

1. A user on the Linux server uses hping3 to send large ICMP packets to a Windows PC.
2. The firewall, perceiving possible risky traffic, blocks the traffic according to the assigned IPS profile.
3. The Windows PC fails to receive packets from the Linux server, disrupting the functionality of the app.

This use case illustrates a hypothetical scenario where FortiGate mistakenly flags large ICMPs from Linux to Windows as intrusive. Such occurrences are common when using IPS profiles aimed at blocking; however, not all flagged traffic is genuinely malicious. It's important to consider the possibility of false positives when blocking IPS traffic.



If you experience a situation where FortiGate incorrectly identifies a benign or normal event as malicious or problematic, you should take the following steps:

1. Add another signature using the **Signature** as the **Type**.
2. Select **Monitor** as the **Action** and enable packet logging to allow traffic and generate a log.

You need to make sure the new signature has a higher priority than the filter signatures; if it is ranked lower, it won't work as expected.

The flow of events in this scenario is as follows:
1. A user on the Linux server uses hping3 to send large ICMP packets to a Windows PC.
2. The firewall detects that the new signature oversized packet should have the action Monitor, so it allows the traffic and generates a log.
3. The Windows PC receives the packet from the Linux server, making the app work as expected.

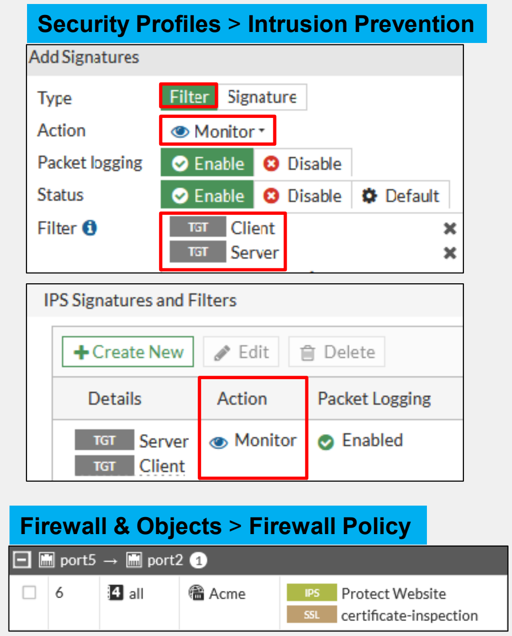
If you encounter a scenario where an IPS signature blocks your native traffic and an application that you are using is considered hostile, you should set the **Action** to **Monitor**, at the top of your rules.

Use Case 3—PHP Code Injection in HTTP

- **Type: Filter**
- **Action: Monitor**
- Detect before blocking
- Certificate-inspection mode
- FortiGuard encyclopedia

Log & Report > Security Events > Intrusion Prevention

Source	Attack Name	Destination	Action
10.0.3.55	PHP.URI.Code.Injection	10.0.2.58	detected
10.0.3.55	PHP.URI.Code.Injection	10.0.2.58	detected
10.0.3.55	PHP.URI.Code.Injection	10.0.2.58	detected



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This use case shows the reverse approach: start by monitoring all traffic and then act against potential malicious activity.

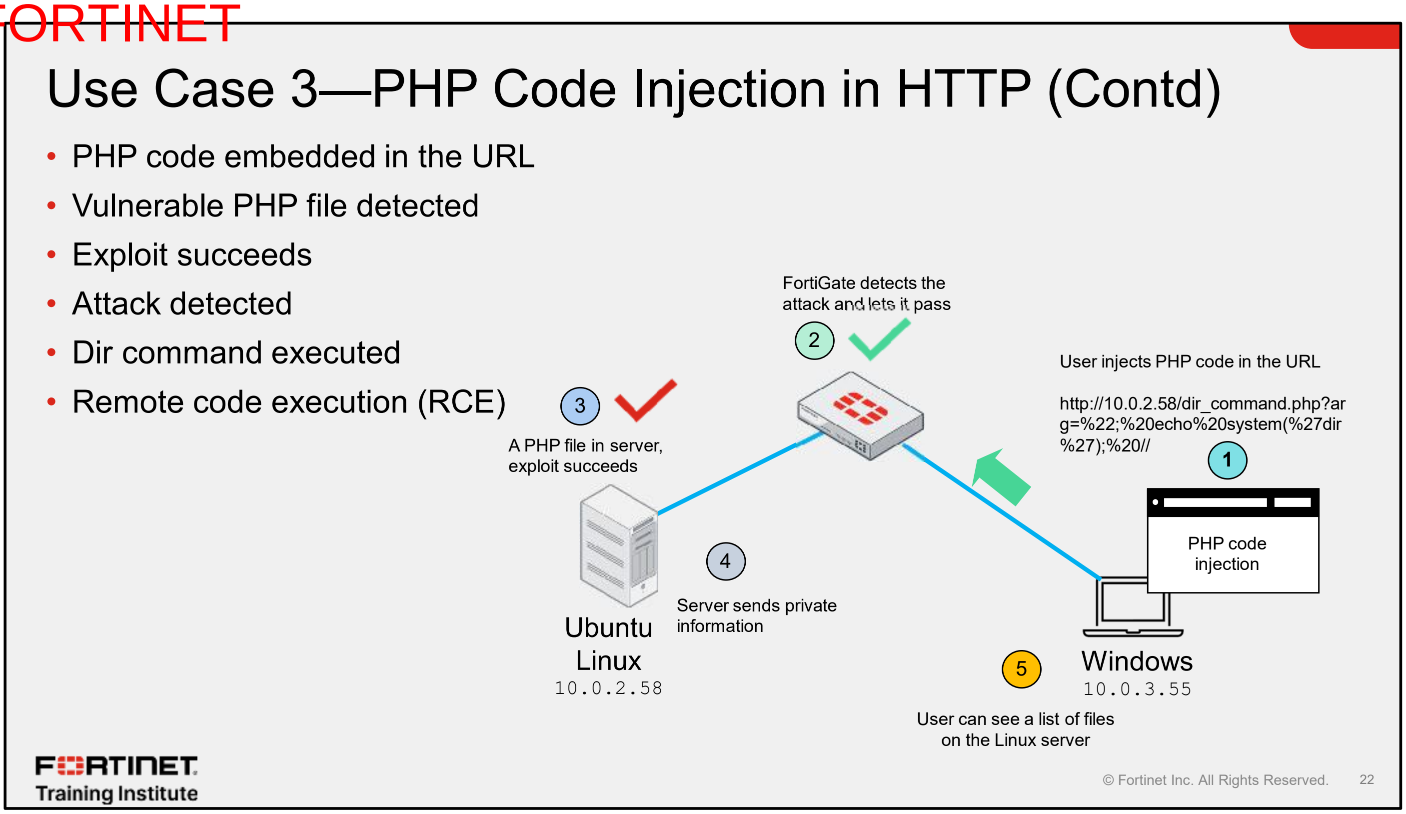
Set the **Type** to **Filter**, set the **Action** to **Monitor**, enable **Packet logging**, enable **Status**, and add the target **Client** and **Server** to the **Filter**. It is important that you set action to **Monitor** and enable **Packet logging.** Use packet logging only for addressing false positives or creating specific signatures. Avoid its long-term or production use, unless you account for it during system sizing.

This use case has a firewall policy with the virtual IP (VIP) address **Acme**. This policy is used to convert external IP addresses to internal IP addresses using the **Protect Website** IPS profile. This traffic travels from the Windows server to the Linux server.

Because this use case doesn't use HTTPS traffic or protect an SSL server, you don't need to decrypt and encrypt the file with a full inspection SSL/SSH profile. The policy requires only flow mode with certificate-inspection mode.

When FortiGate detects an attack as a PHP code injection, click on the name of the attack to view more information in the FortiGuard Encyclopedia, such as:

- Affected products
- Impact
- Recommended actions
- Description



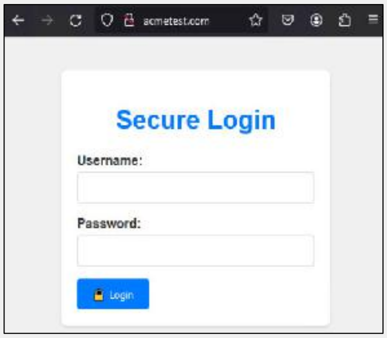
In this portion of the use case, your firewall has correctly detected a PHP code injection from a Windows PC to an Ubuntu server.

1. A Windows device (IP: `10.0.3.55`) accesses http://acmetest.com, which is hosted on an Ubuntu server (IP: `10.0.2.58`), embedding PHP code in the URL.
2. A firewall detects an attempted PHP vulnerability exploit from the laptop and logs the incident.
3. The Ubuntu server processes the request, and due to an existing vulnerable PHP file, the exploit succeeds.
4. The server responds by executing a directory (dir) command through the exploited PHP file, sending the results back.
5. The Windows user receives a list of files from the server's directory, where the vulnerable PHP file (`dir_command.php`) resides.

This PHP command, or instruction, clearly demonstrates how, without having access to the remote server, there could potentially be an attempt to exploit a security vulnerability known as remote code execution (RCE). Specifically, the URL includes a query parameter that seems to inject PHP code. This type of code injection could be maliciously used to execute arbitrary commands on a web server, but only if the server is susceptible to this kind of attack.

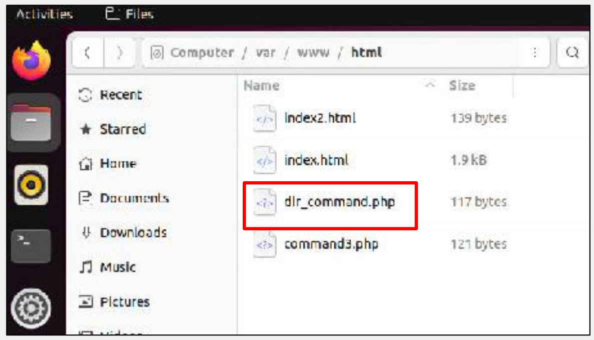
Use Case 3—PHP Code Injection in HTTP (Contd)

- Normal website



- PHP code in the URL

- Linux web file storage directory



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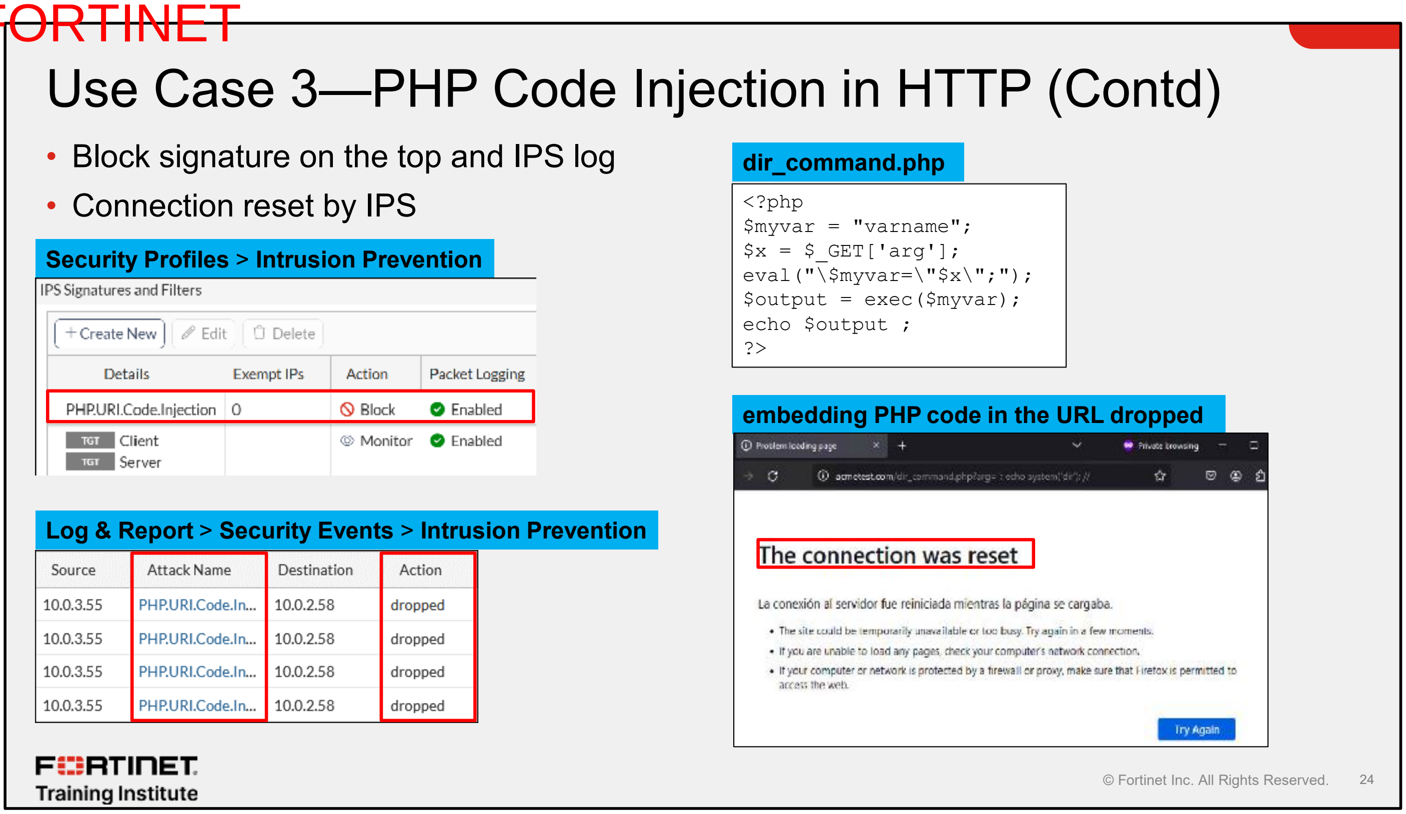
This slide shows the progression of PHP code as it is embedded in a file. The screenshots show a normal website, the URL of that website after a user embeds PHP code in the URL, and the third shows the default web file storage directory on an Apache web server, where you can see the PHP code.



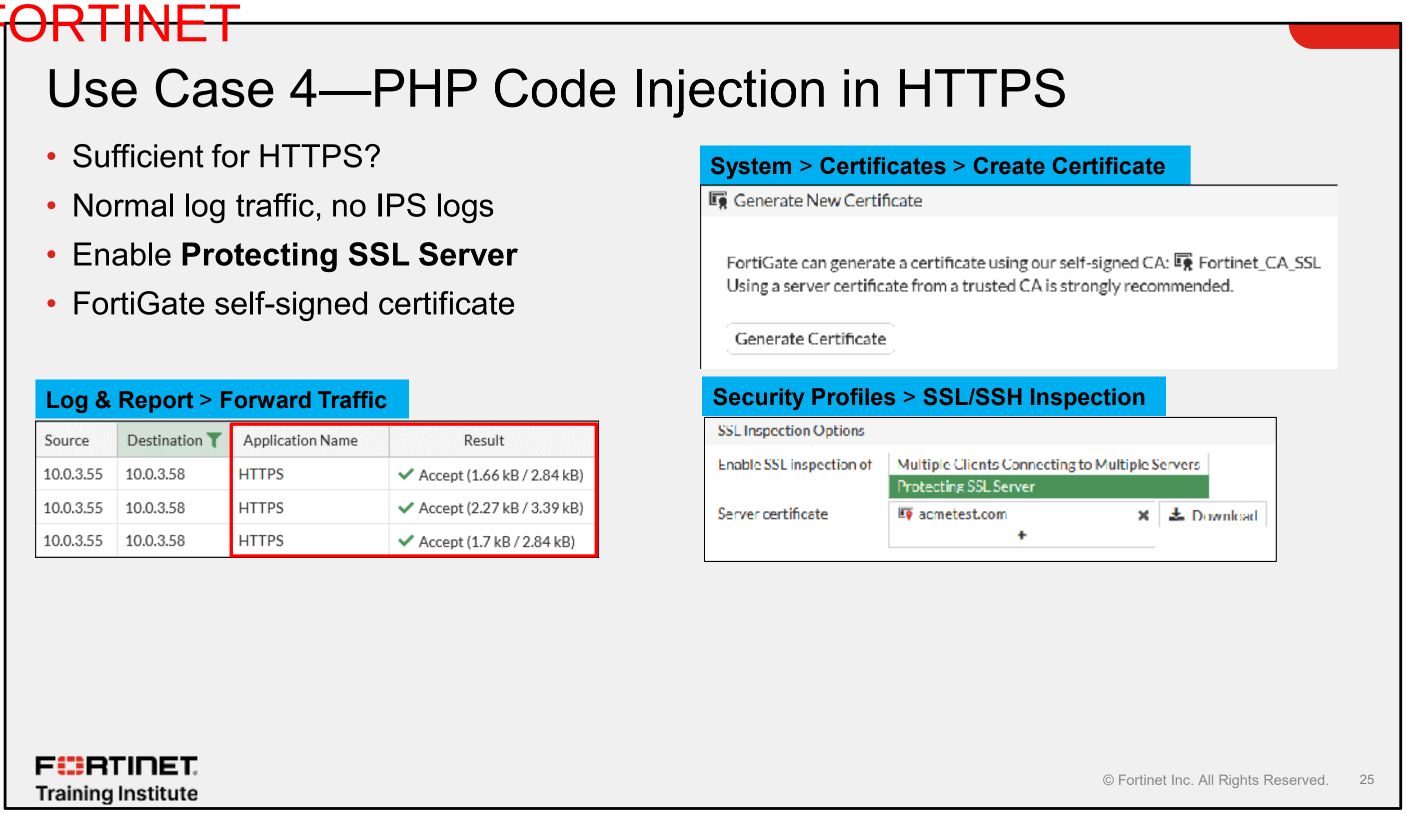
When an attack like this occurs, one recommendation from the FortiGuard Encyclopedia is to block it. You must add a signature at the top of the signature list to prioritize blocking this attack.

When the PHP URI code is considered malicious, the firewall will drop packets, and the web browser will show a "connection reset" error because the code can no longer pass through the firewall.

Note that this code is vulnerable to PHP code injection and command injection, which could allow attackers to execute arbitrary code on the server. You should not implement it in real-world applications without proper input validation and security measures.

If you're not a PHP expert, focus on using an IPS profile to identify and block attack signatures. However, note the following about how this PHP code operates:

1. Retrieves user input: Stores URL parameter `arg` in `$x`.
2. Processes input: Constructs and executes a command using `eval()` and `exec()`.
3. Outputs result: Displays the result of the executed command on the web page.



But what happens if a code injection occurs over an encrypted protocol? Will a policy with an IPS profile and certificate inspection be sufficient to analyze HTTPS traffic? In this scenario, FortiGate will detect HTTPS access, but it won't detect an attack.

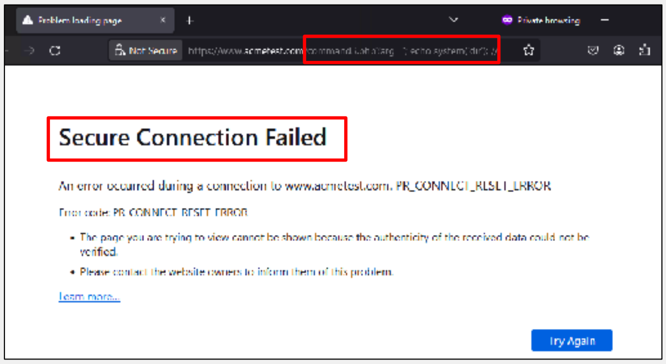
So, the answer is no. FortiGate won't decrypt packets to analyze if there is a possible attack. You must create an SSL/SSH inspection profile to use the **Protecting SSL Server** option.

You need to have a certificate that validates the authenticity of your website. In the example on this slide, a self-signed certificate from FortiGate is used, but you can use any of the following:

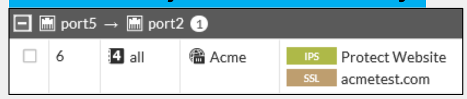
- A certificate signed by an internal CA, which is typically used for lab environments.
- A firewall with a self-signed certificate, also for lab environments, as demonstrated in this use case.
- A firewall with a certificate signed by a global CA, which is strongly recommended for daily operations.

Use Case 4—PHP Code Injection in HTTPS (Contd)

- HTTPS injections will be blocked



Firewall & Objects > Firewall Policy



Log & Report > Security Events > Intrusion Prevention

Source	Attack Name	Destination	Service	Action
10.0.3.55	PHP.URI.Code.I...	10.0.2.58	HTTPS	dropped
10.0.3.55	PHP.URI.Code.I...	10.0.2.58	HTTPS	dropped
10.0.3.55	PHP.URI.Code.I...	10.0.2.58	HTTPS	dropped

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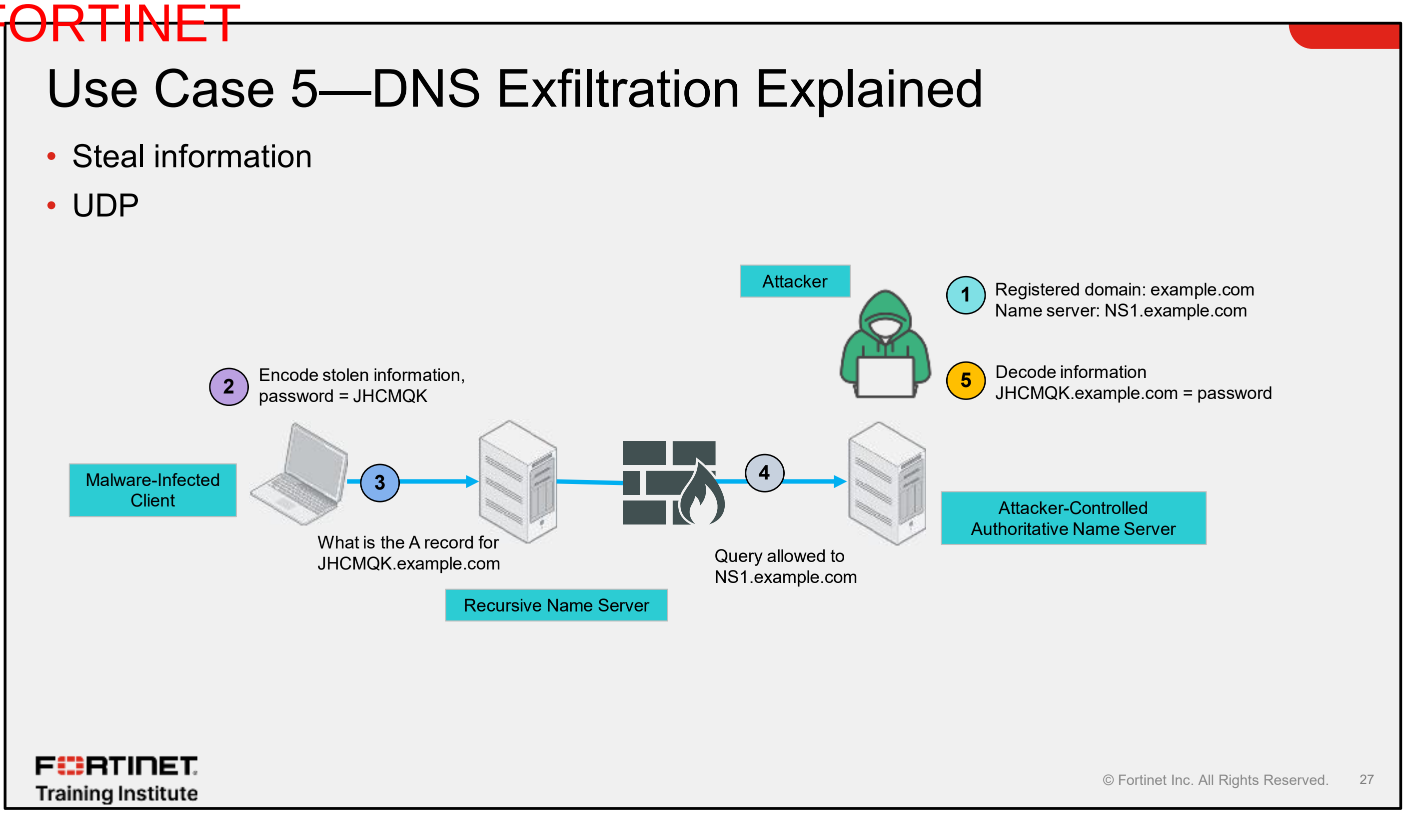
Once you change the SSL/SSH profile in your policy, if a user attempts to insert PHP code, FortiGate will be able to detect and drop the packet, as stated in your IPS profile.



DNS converts IP addresses to text.

DNS uses UDP, which can be seen by anyone because it is not encrypted.

Why is it important to understand how protocol options work? DNS exfiltration is a technique which basically steals information in a very smart way using a domain, DNS request/response, and the most important, an infected client.

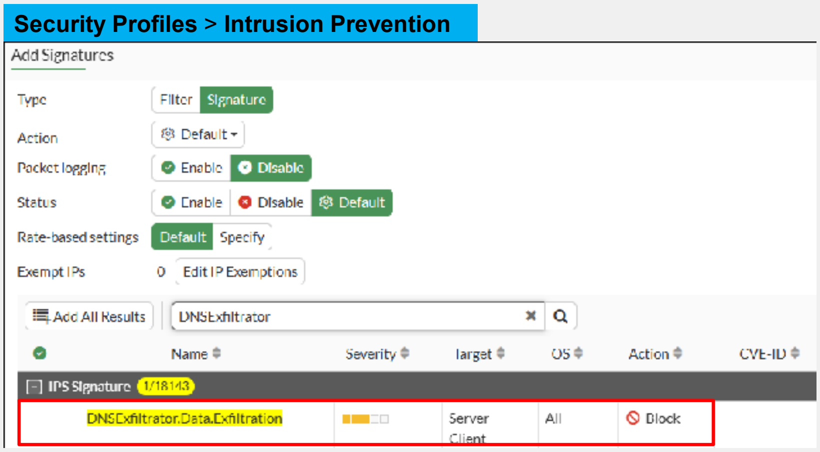
In the example shown on this slide, you see the following:

1. The attacker registered a domain with the name `NS1.example.com`.
2. The infected client encodes a stolen password as an encoded sequence.
3. The infected client requests to resolve the encoded sequence `.example.com` using the `NS1.example.com` server.
4. The attacker receives the stolen password to access the IP address; however, the main intention is not to use the DNS, but to receive coded information.
5. The attacker can decode the encoded sequence to learn the password.

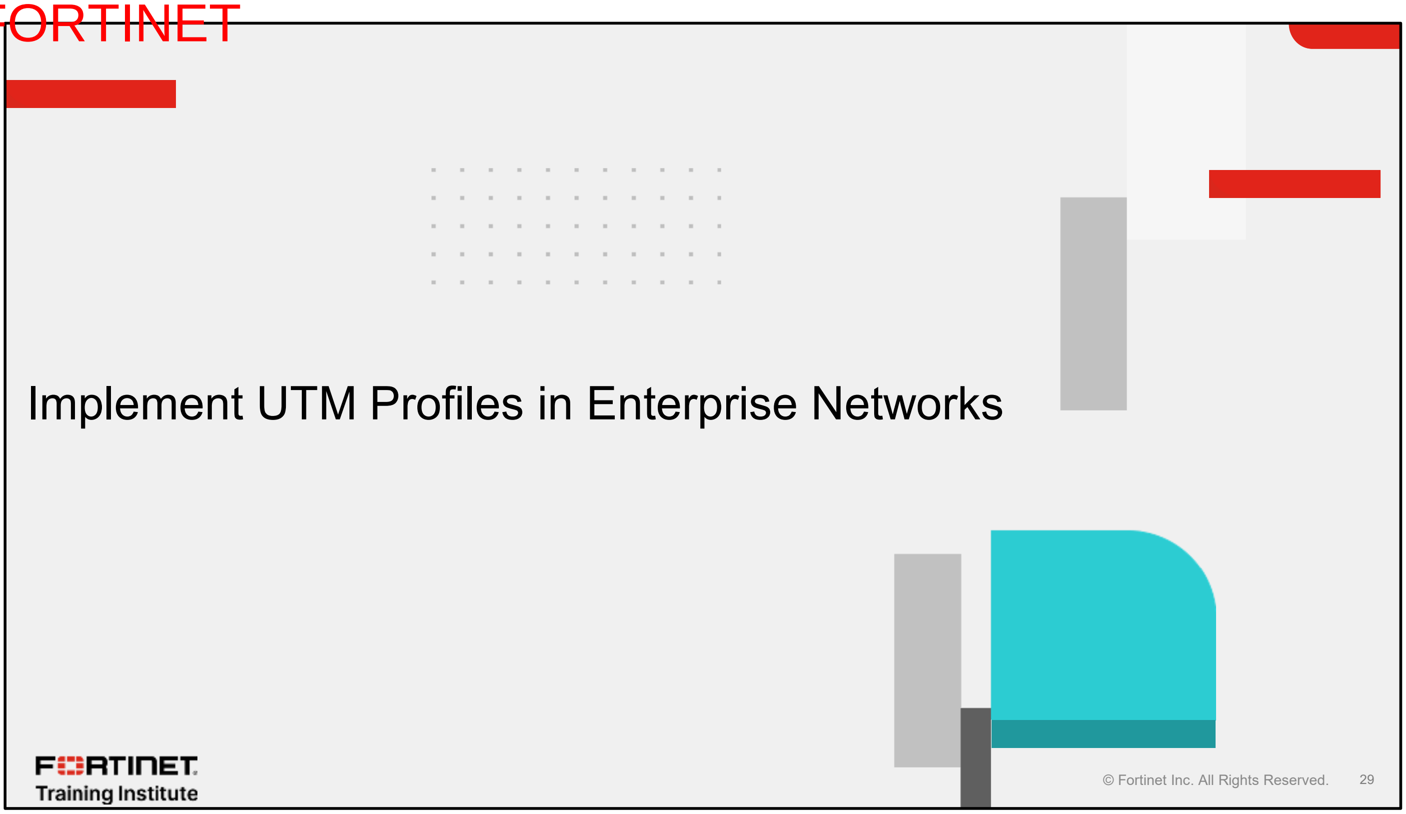
Using DNS exfiltration, an infected client, encoded information, and a simple protocol like unencrypted UDP have stolen a password without alerting the user.

Use Case 5—DNS Exfiltration Explained (Contd)

- SSL/SSH inspection
- DNS over TLS



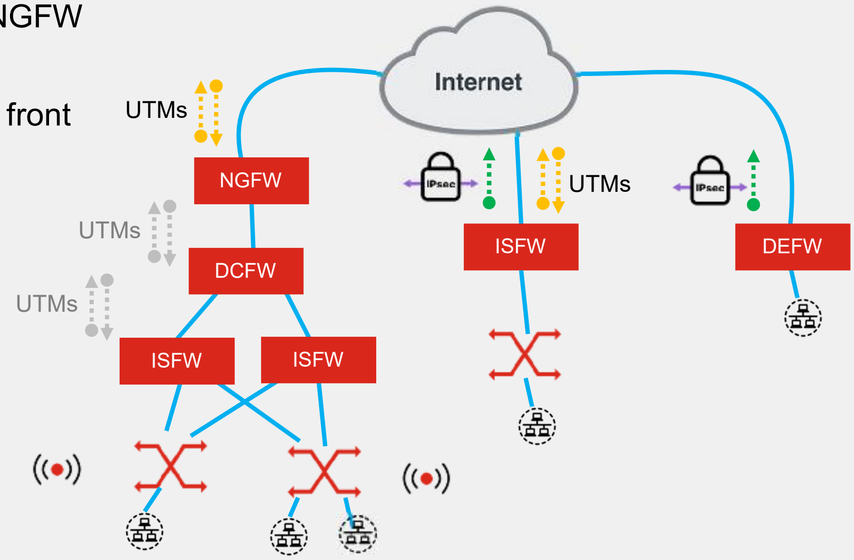
FortiGate includes a signature named **DNSExfiltrator.Data.Exfiltration** that you can use to detect and prevent DNS exfiltration. To use the signature, you must enable SSL/SSH inspection with multiple clients connecting to multiple servers and using full inspection method. With these settings, FortiGate will be able to encrypt, decrypt, and see the information traveling across the network.



In this section, you will learn about implementing unified threat management (UTM) profiles in enterprise networks.

UTM Profiles—Varying Roles and Implementations

- UTMs profiles are deployed in NGFW
- ISFW may serve as NGFW
- DEFW does not need NGFW in front under native function
- DCFW may serve as NGFW
- UTM profiles are based on operational needs



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The optimal placement of UTM profiles depends largely on your network topology and the specific role of each firewall. UTM profiles are standard features in next-generation firewalls (NGFW), but their implementation can vary based on their functions.

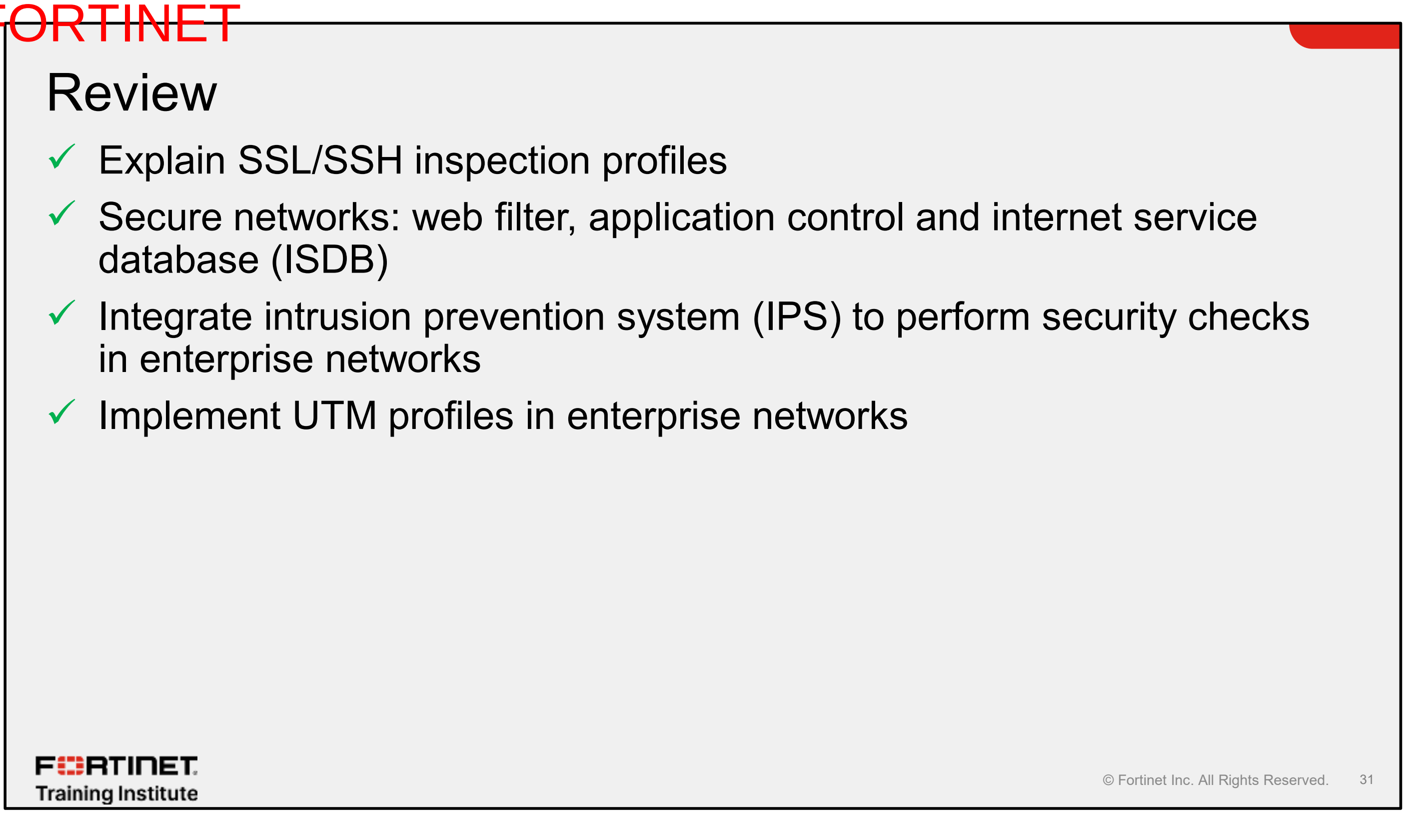
This lesson outlines the differences among web filtering, application control, ISDB, and IPS implementations. Typically, content filtering is configured across all NGFWs. However, in the network diagram shown on this slide, the internal segmentation firewall (ISFW) has content filtering capabilities but is configured primarily for segmentation, low latency, and integrated wireless LAN, as discussed in the *Introduction to Network Security Architecture* lesson.

An ISFW may serve a dual role as an ISFW and an NGFW. For example, an ISFW might use VPNs to manage sensitive traffic and require traffic filtering to enable secure internet access or to prioritize SD-WAN and traffic-shaping policies.

Additionally, distributed enterprise firewalls (DEFW), designed solely for connecting to headquarters, may not necessarily require to use UTM profiles. If it becomes necessary to activate UTM profiles on a DEFW, consider using a midrange firewall, which offers greater capabilities than firewalls that function only as remote connection points, such as those in the 100-900 series typically used for ISFWs.

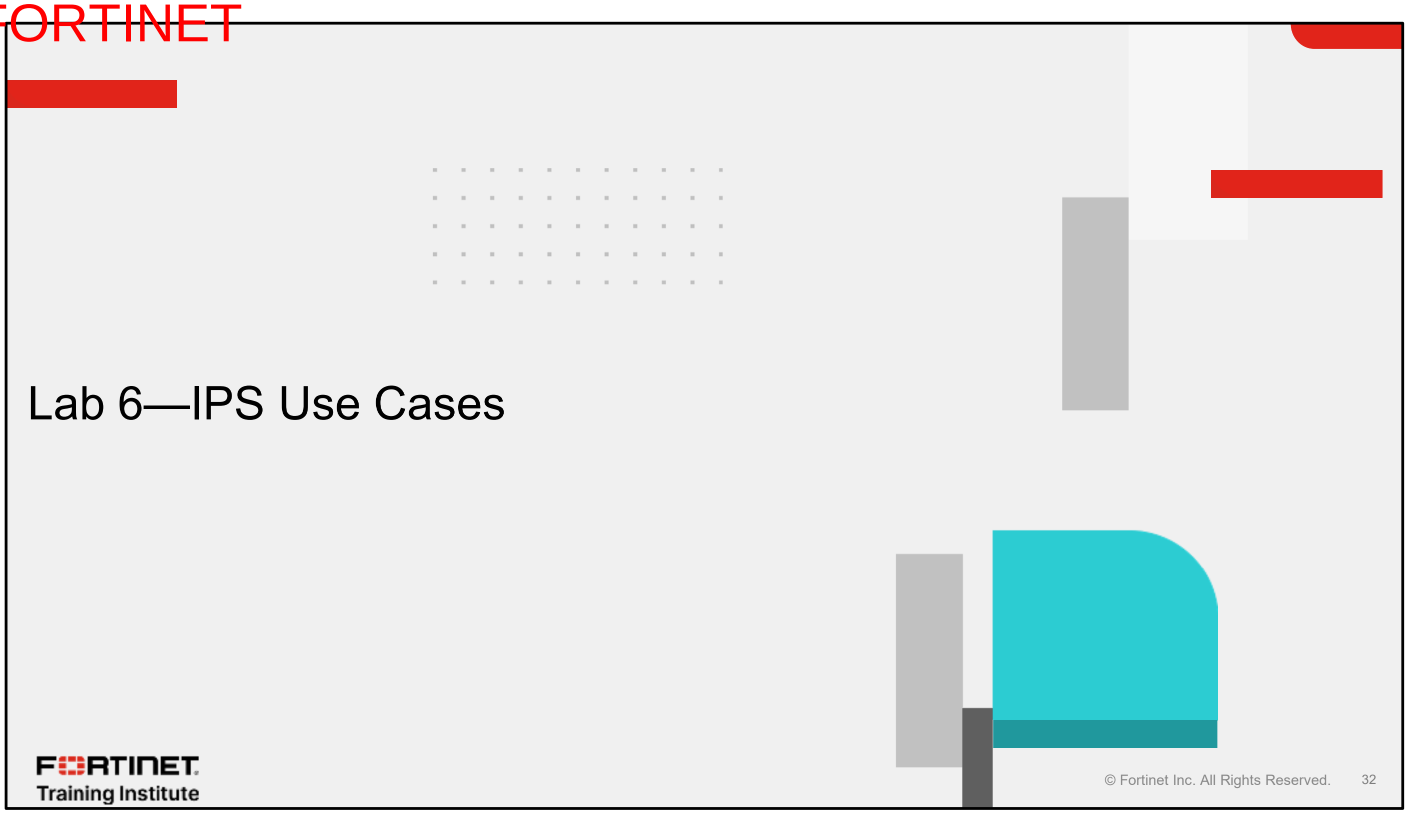
Furthermore, the topology shows two FortiGate devices, the data center firewall (DCFW) and NGFW, situated above the ISFWs. Usually, a DCFW is protected by another firewall that acts as a gateway for external traffic. This gateway often performs NGFW functions and is a critical node for connecting internal and external traffic. In this context, the NGFW is ideal for applying content filters and IPS, though there may be redundancy between the NGFW and DCFW. Additionally, a DCFW might connect directly to the internet, utilizing segmentation methods like VDOMs to create individual virtual firewalls.

The placement of UTM profiles should be strategically determined based on these configurations and operational priorities.



This slide shows the objectives that you covered in this lesson.

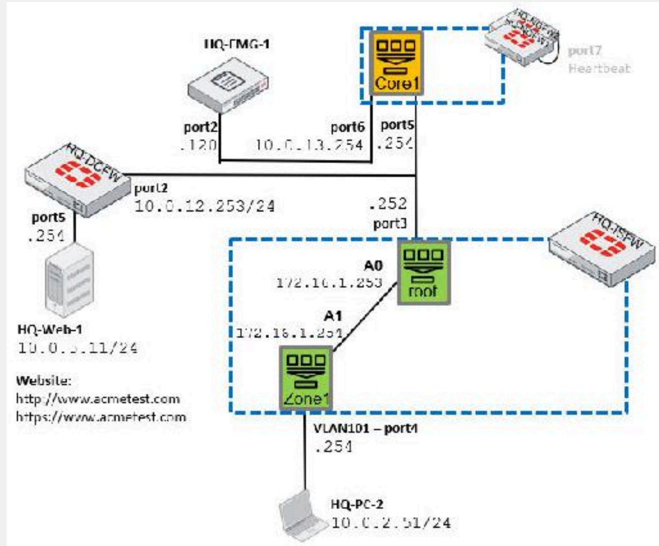
By mastering the objectives covered in this lesson, you learned how security profiles work on FortiGate.



Now, you will work on *Lab 6—IPS Use Cases.*

Lab 6—IPS Use Cases

- On DCFW:
  - Solve IPS false positives
  - Protect against unencrypted attacks
  - Protect against encrypted attacks



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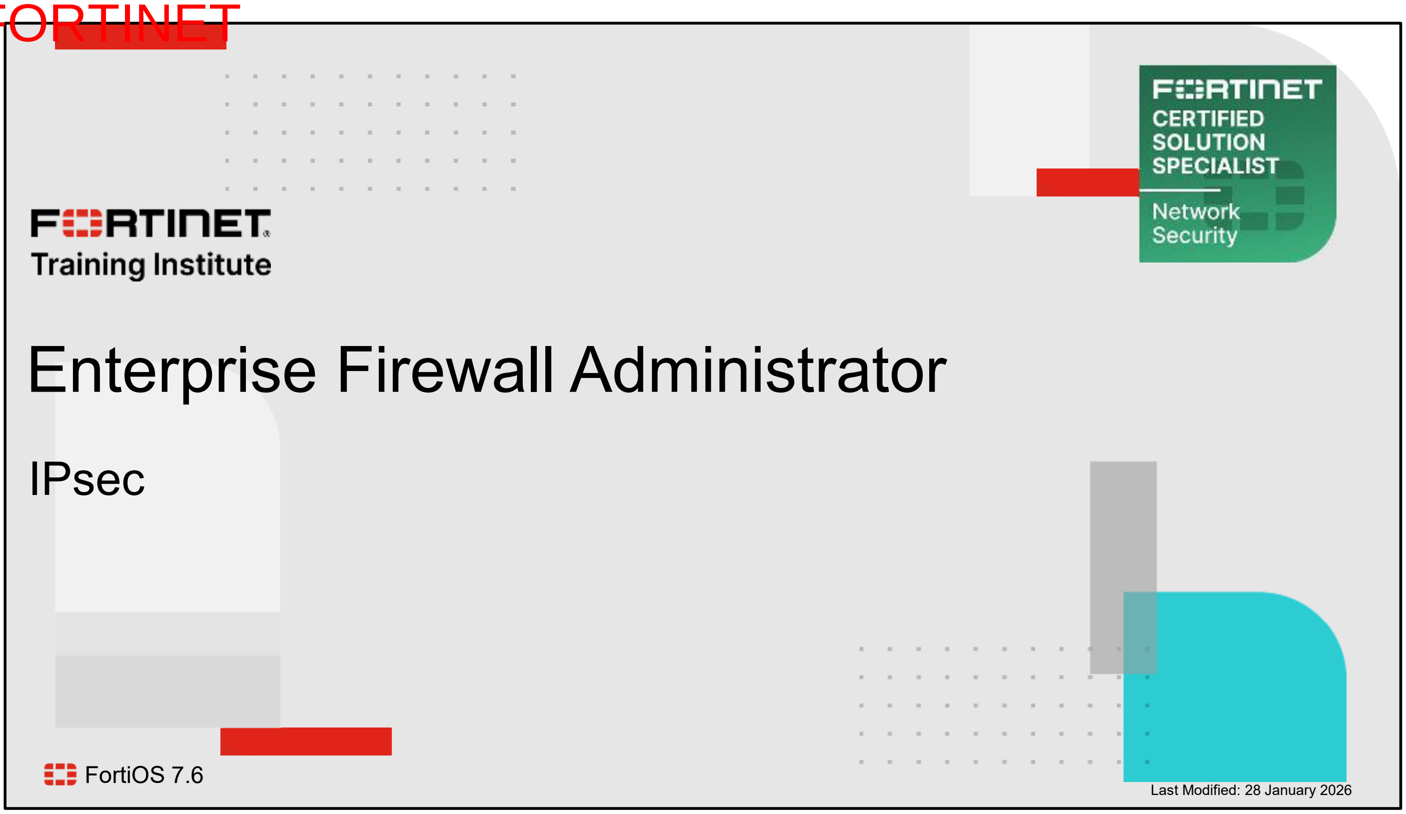
In this lab, you will configure the following IPS use cases:
- Solve IPS false positives
- Protect against unencrypted attacks
- Protect against encrypted attacks



In this lesson, you will learn about IPsec.

Objectives

- Master IPsec topology design
- Implement IPsec networks using FortiManager IPsec templates

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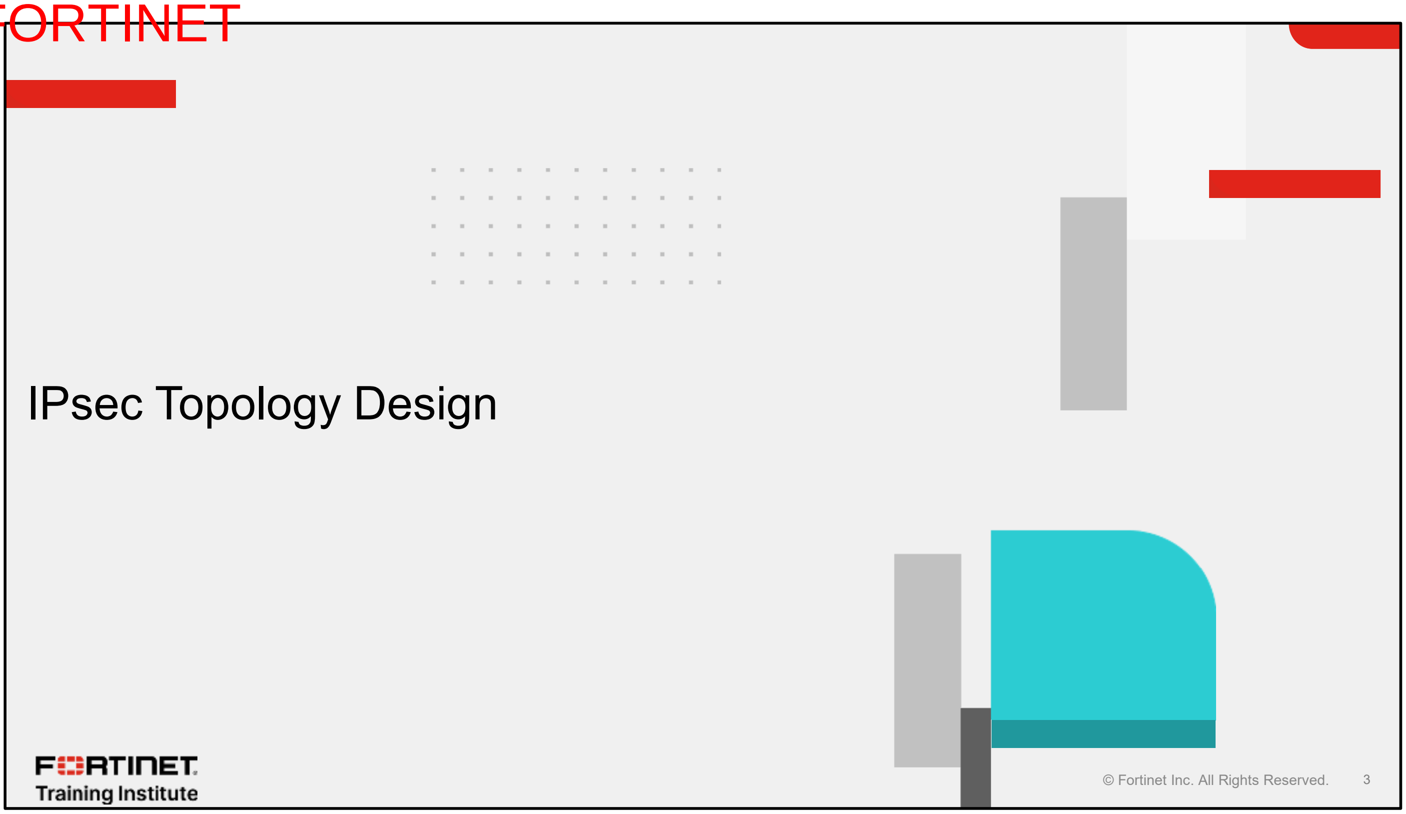
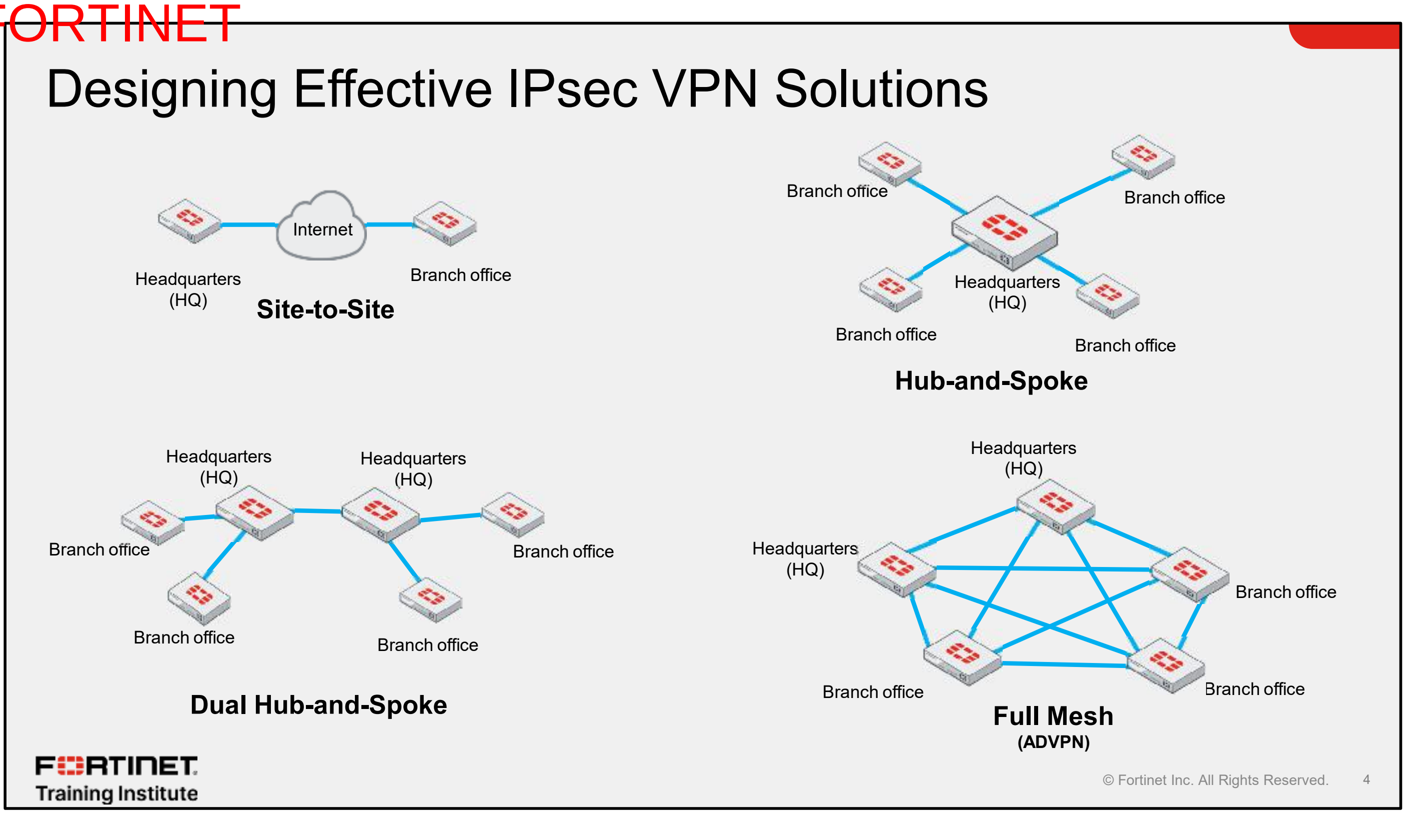
After completing this lesson, you should be able to achieve the objectives shown on this slide.



In this section, you will learn how to select an IPsec topology design.



When you design IPsec VPN topologies, you should consider multiple factors to enhance communication between devices.

It is crucial that you select protocols that use reliable algorithms, ensuring solid and secure communication across all devices.

Additionally, you should consider the scalability of the network for future growth and the security measures needed to protect data integrity and confidentiality. Take into account the type of firewall used at your sites; a branch model may support fewer tunnels compared to a campus model. The number of tunnels also depends on the type of traffic within the IPsec tunnel.

Implementing robust authentication methods and choosing appropriate encryption standards is also vital to safeguarding the network against unauthorized access and potential threats.

The Superior Security Features of IKEv2

Features	IKEv1	IKEv2
Encryption algorithms	DES, 3DES, AES-CBC	AES-GCM, ChaCha20P
Diffie-Hellman groups	Group 1 (768-bit), Group 2 (1024-bit), Group 5 (1536-bit)	Group 14 (2048-bit), Group 15 (3072-bit), Group 16 (4096-bit), Group 19 (256-bit elliptic curve), Group 20 (384-bit elliptic curve)
Authentication	Pre-shared Key, Digital Signatures	EAP, Pre-shared Key, Digital Signatures
Integrity	HMAC-MD5, HMAC-SHA1	HMAC-SHA2, AES-XCBC
Mode Config	✓	✓

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When discussing IPsec VPN, it's essential to consider internet key exchange (IKE) and its versions. The table on this slide provides an overview of IKEv1 and IEv2. Consider the following features when selecting which version of IPsec you use in your network topology.

- Encryption algorithm: IKEv2 supports advanced encryption algorithms like Advanced Encryption Standard with Galois/Counter Mode (AES-GCM) for authenticated encryption, enhancing security with minimal performance impact.
- Diffie-Hellman groups: IKEv2 includes stronger Diffie-Hellman (DH) groups, such as Elliptic Curve (ECP) groups, which offer improved security with lower computational demands.
- Authentication: IKEv2 is compatible with EAP, which facilitates integration with various external authentication services, boosting the flexibility of security without significant performance drawbacks.
- Integrity: In IKEv2, algorithms like AES-XCBC, deliver robust integrity checks efficiently, balancing security and computational efficiency.
- Mode Config: Integrated directly into the IKE exchange, IKEv2 mode config simplifies client configuration delivery, reducing message count and the associated CPU and memory load.
- Performance: IKEv2 remains efficient despite supporting higher-security algorithms. It is optimized to perform well even on less capable hardware, sometimes enhancing security with little performance loss.

It is recommended that you consider migrating from IKEv1 to IKEv2, if all your devices support IKEv2 and the conversion will not not impact network operability.

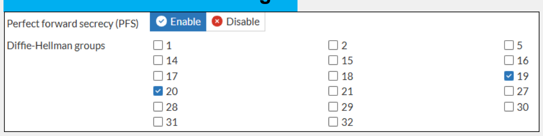
Note that all examples in this lesson specifically pertain to IKEv2, not only because of the improved security posture but also to leverage the enhanced features of IKEv2.

Understanding MODP and ECP Key Exchanges

- Mathematical theories
- Large prime numbers and exponents
- Discrete logarithm problem
- ECP uses shorter keys
- PFS enhances security in phase 2

DH groups	Group name	Key length	Encryption type
1	MODP 768	768-bit	Weak
2	MODP 1024	1024-bit	Moderate
5	MODP 1536	1536-bit	Strong
14	MODP 2048	2048-bit	Very Strong
15	MODP 3072	3072-bit	Stronger
16	MODP 4096	4096-bit	Very Strong, Secure
17	MODP 6144	6144-bit	Extremely Strong
18	MODP 8192	8192-bit	Maximum Security
19	ECP 256	256-bit	Strong, Efficient
20	ECP 384	384-bit	Very Strong, Efficient

IPsec VPN Phase 2 Settings



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Modular Exponential (MODP) and Elliptic Curve (ECP) are encryption protocols used in Diffie-Hellman key exchange methods:

- MODP uses large prime numbers and exponents for key generation. Its security relies on the complexity of solving the discrete logarithm problem in modular arithmetic. As prime sizes increase, so do security and computational demands.
- ECP uses elliptic curve points for key generation, leveraging the challenge of solving the elliptic curve discrete logarithm problem. ECP offers security comparable to MODP with shorter keys, enhancing computational efficiency.

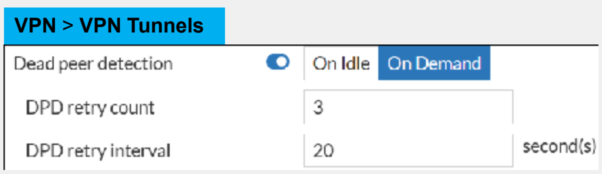
Diffie-Hellman groups (like those in the table on the slide) use these methods with varying key lengths and mathematical properties to enhance security and efficiency. The appropriate group depends on the desired balance between computational overhead and security level.

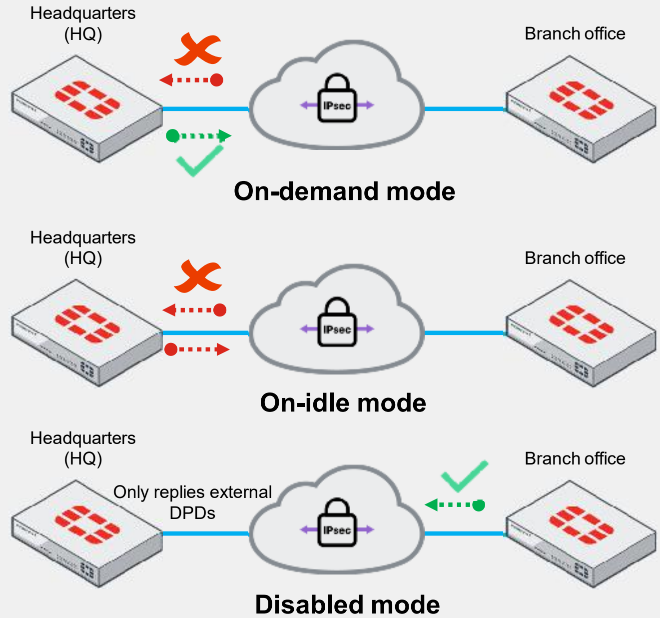
Consider that enabling perfect forward secrecy (PFS) in phase 2 of IPsec VPN settings means using Diffie-Hellman groups to generate new, independent keys for each session, enhancing security.

Disabling PFS omits this step using a simpler key derivation and increasing potential security risks. Enabling PFS with a specific Diffie-Hellman group improves security against attacks on compromised keys.

Choosing DPD Modes—Best Practices

- DPD detects offline VPN peers
- On-demand mode is best for unpredictable traffic
- On-idle mode is best for regular traffic patterns
- Disable mode is best for in stable networks
- Proper DPD boosts network reliability



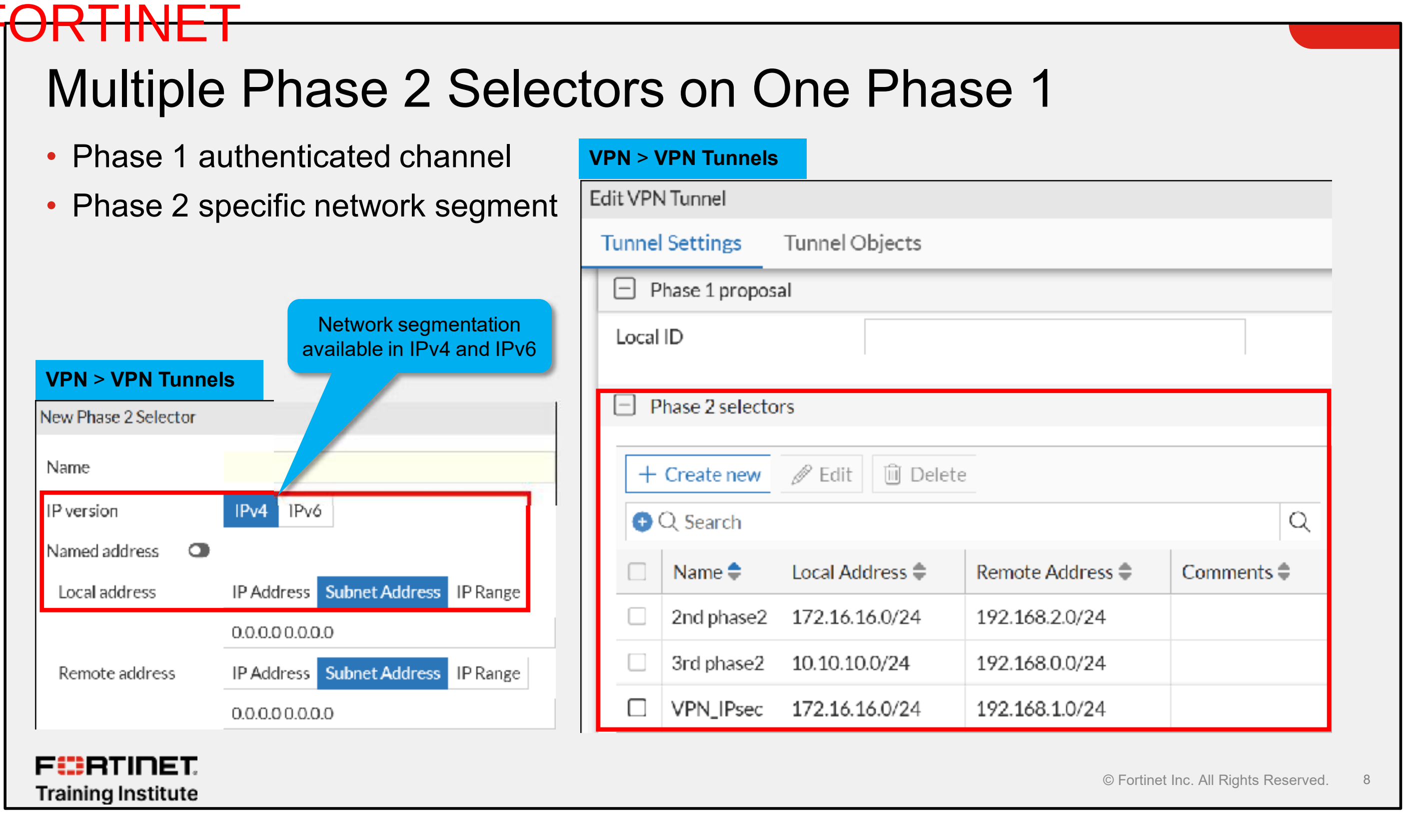


Dead peer detection (DPD) is essential for ensuring VPN stability by detecting unreachable peers.

FortiGate supports three DPD modes: on-demand, on-idle, and disable.

- On-demand mode is best for environments where traffic patterns are unpredictable, and immediate response to connectivity issues is crucial.
- On-idle mode is best for networks with regular traffic intervals, providing a balance between connectivity assurance and resource utilization.
- Disable mode is suitable in highly stable environments where DPD overhead is unwarranted.

In any network topology—whether hub and spoke, dual hub, or full mesh—the DPD mode can significantly impact the performance and reliability of your VPN connections. For instance, in a full mesh topology where multiple nodes are interconnected, using an efficient DPD mode like on-demand, can prevent unnecessary traffic and processing overhead while ensuring quick reactivity to any link failures.



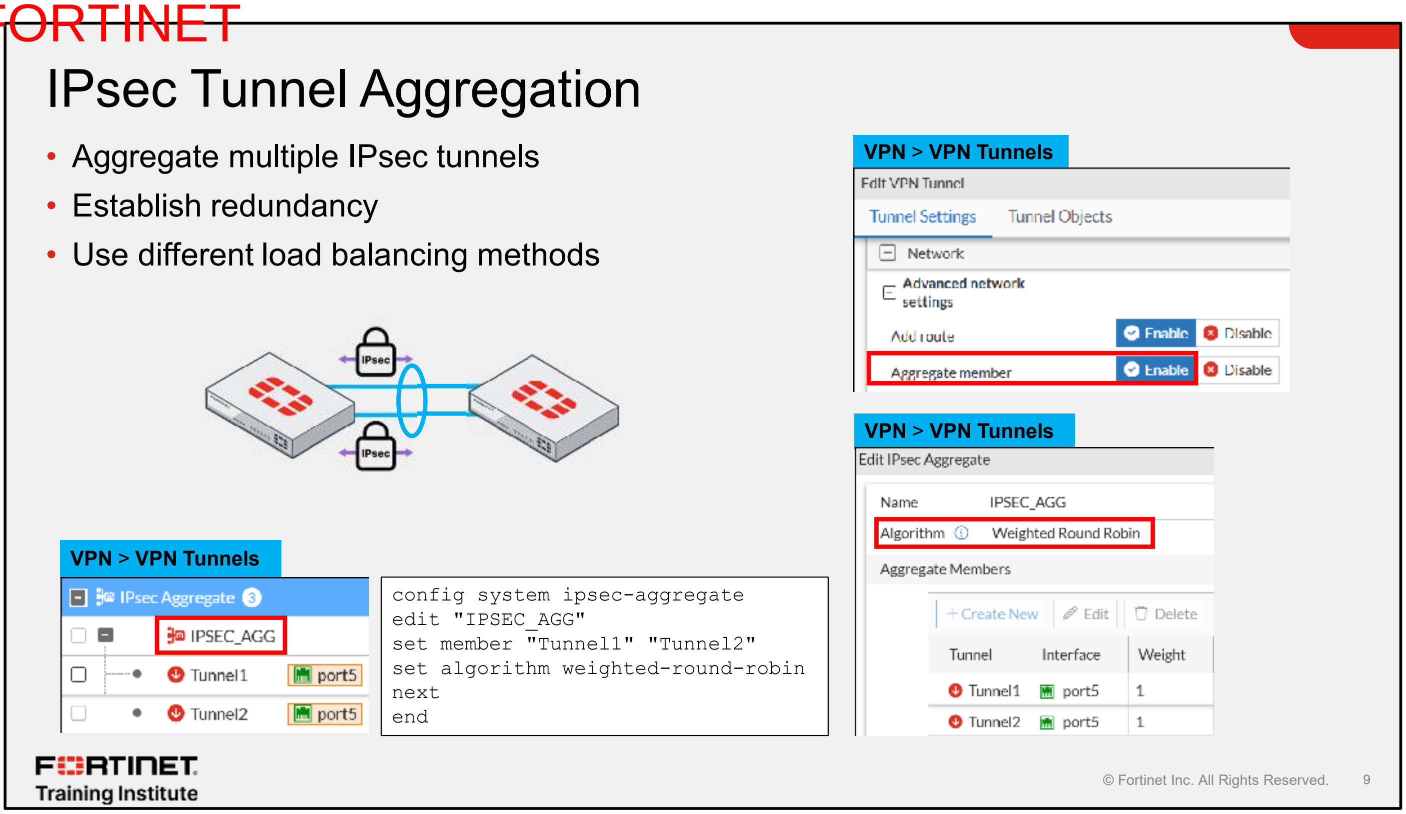
You can allow multiple phase 2 selectors to operate over a single phase 1 connection.

- Phase 1 (IKE phase): This phase establishes a secure, authenticated channel between two endpoints. It sets up a protected connection for exchanging key information and negotiating the policies to be used by the IPsec security associations (SA).
- Phase 2 (IPsec phase): This phase uses the secure channel established in phase 1 to negotiate the settings for specific network segments, facilitating the establishment of IPsec tunnels.

In this example, you can see different network segments, each with editable names in phase 2. The purpose of specifying network segments is to clearly indicate to the VPN which segments will be encrypted. Also, this approach ensures that only the network segments designated will be allowed on the IPsec VPN tunnel.

You can configure these network segments in multiple ways. The most used and flexible method is the address subnet, which allows for easy modifications. The full list is shown on the slide.

In addition, phase 2 configurations that consist of all zeros in phase selectors during migrations, particularly when advertising many routes over tunnels, can introduce the risk of invalid paths. These can disrupt operations but are crucial when partnering with corporations that have overlapping subnets. Use routing protocols to specify allowed subnets over the tunnel to mitigate these issues.



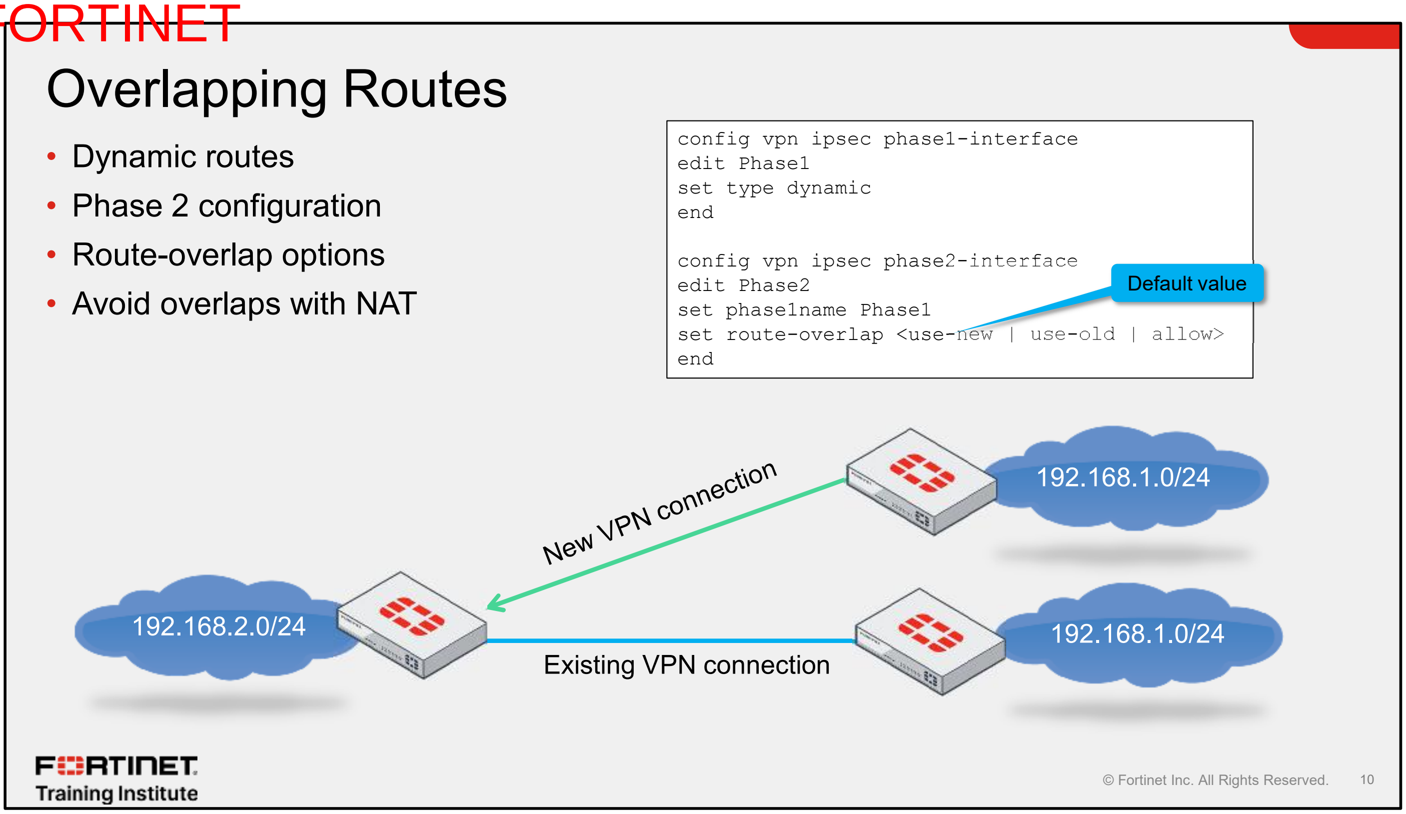
In the same way you can create LACP in layer 2, you can create virtual binding interfaces within IPsec VPNs using IPsec tunnel aggregation.

In this example, when you create a VPN tunnel, you have the option to include it as an aggregate member. For instance, configuring at least two tunnels as aggregate members allows you to form an IPsec aggregate, as demonstrated in this example where **Tunnel1** and **Tunnel2** comprise the same aggregate interface.

Using IPsec aggregate interfaces offers redundancy benefits. If one tunnel fails, another takes over to maintain continuous traffic flow.

Additionally, various load balancing methods are available:

- Round-robin: Traffic is balanced per packet.
- L3: Traffic is balanced based on Layer 3 header information.
- L4: Traffic is balanced based on Layer 4 header information.
- Redundant: All traffic is sent through the first active tunnel, with others serving as backups.
- Weighted-round-robin: Traffic is balanced in a round-robin fashion, adjusted by link weights configured for each tunnel.



If two remote sites have the same subnets, they may create overlapping static routes in the central FortiGate device.

In phase 2, the setting you use with the `route-overlap` command determines the action FortiGate will take when a new remote site connects with an overlapping subnet.

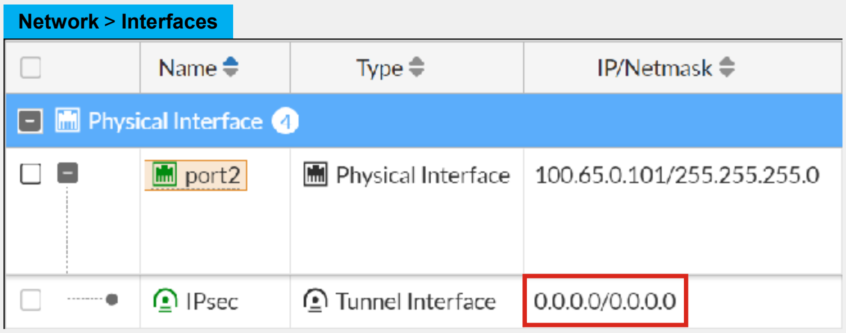
The options available for route-overlap are:
- `use-new` (default): Disconnects the existing dialup VPN and accepts the new VPN.
- `use-old`: Maintains the existing dialup VPN and rejects the new one.
- `allow`: Keeps the existing dialup VPN active and accepts the new one, with traffic from the central FortiGate load balanced, equal-cost multi-path (ECMP) between both VPNs.

In SD-WAN setups with multiple egress interfaces, not enabling `set route overlap allow` on the hub can lead to instability, with interfaces falsely toggling between active and dead states. This often results in only one interface being recognized consistently by the hub, especially if you don't adjust the defaults during upgrades.

Additionally, using a network address translation (NAT) solution to manage overlapping networks by hiding one network segment behind virtual IPs allows for both dial-up and static IP address connections on your remote gateway.

IPsec Interfaces and IP Addresses

- NAT uses the IP address of an IPsec interface
- IPsec interfaces do not require an IP address, but you can assign one to use the following features:
  - Identify and route traffic for tunnel establishment
  - Facilitate ADVPN discovery
  - Help manage and troubleshoot



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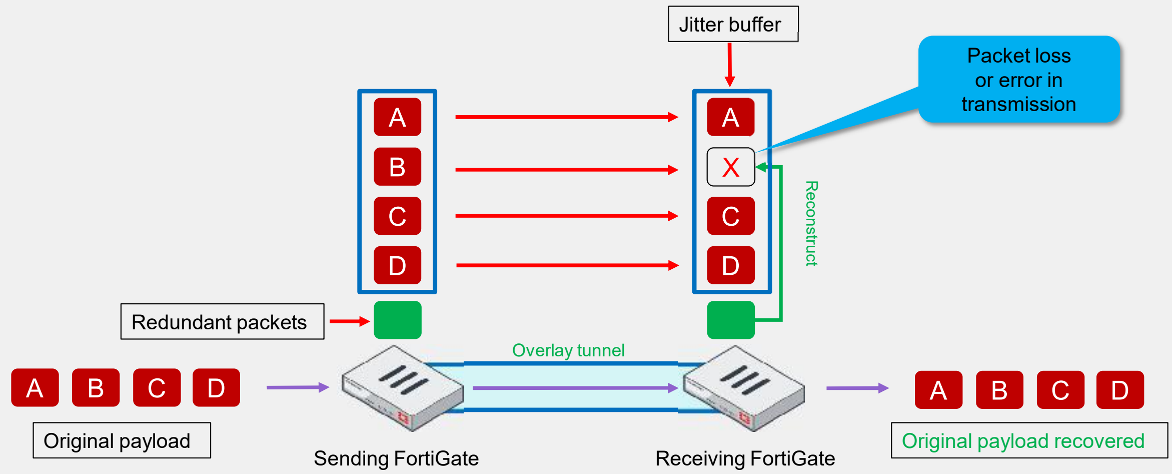
Using NAT on an IPsec interface means packets will use the IP address configured on the IPsec interface, not the IP address of your internet connection.

By default, typical IPsec VPN interfaces do not require an IP address. However, assigning IP addresses to IPsec interfaces can:

- Identify and route traffic for tunnel establishment.
- Enable dynamic discovery of VPN peers in auto-discovery VPN (ADVPN) setups.
- Assist in network management and troubleshooting.

FEC

- Adds redundant data
- Reconstructs lost packets
- Improves reliability over IPsec tunnels
- Boosts voice and video quality

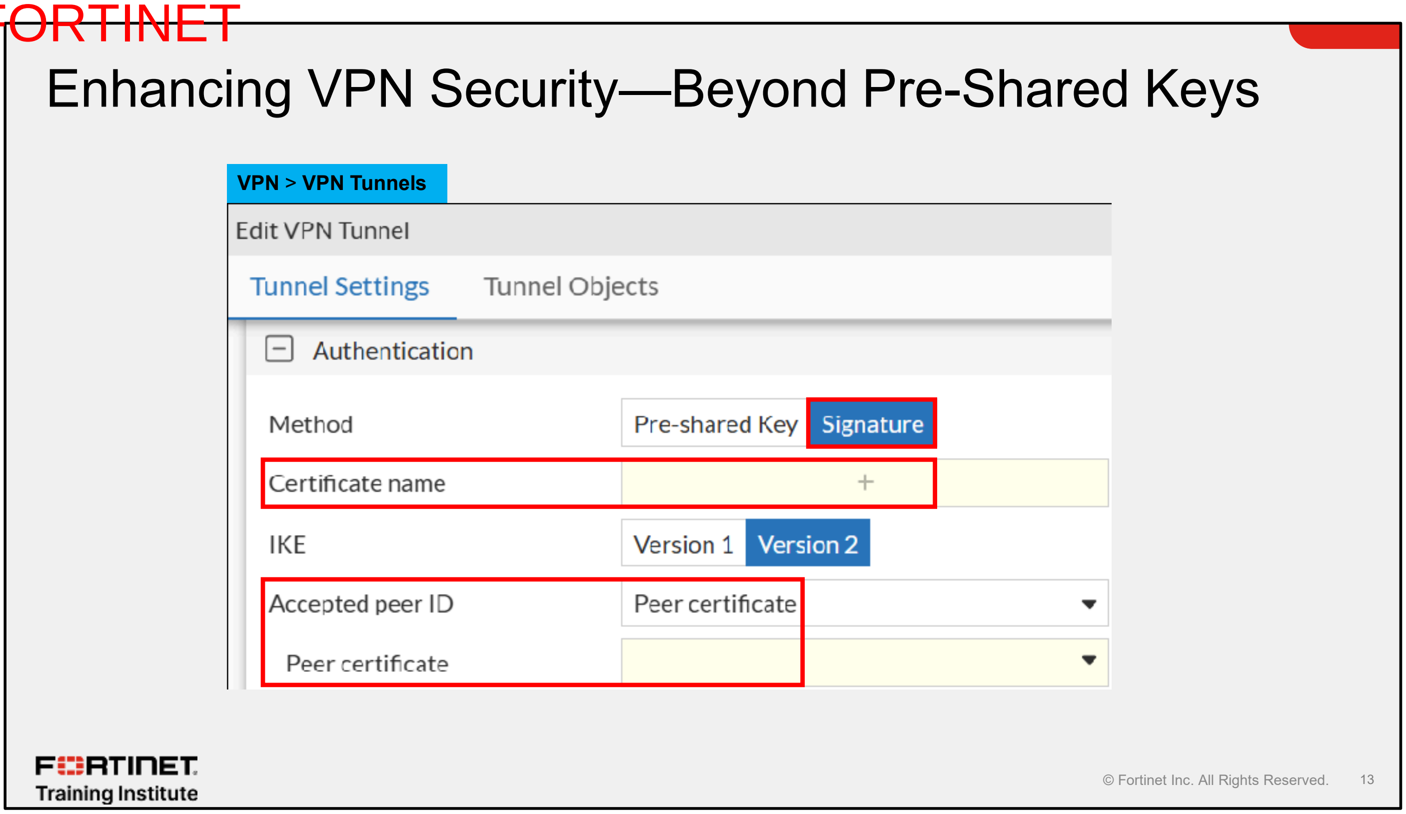


Forward error correction (FEC) is a Phase 1 setting that, when enabled, adds additional packets with redundant data.

The recipient can use this redundant information to reconstruct any lost packets, or any packet that arrived with errors.

Although this feature increases the bandwidth usage, it improves reliability that can overcome adverse WAN conditions such as lossy or noisy links.

FEC can be critical for delivering a better user experience for business-critical applications like voice and video services.

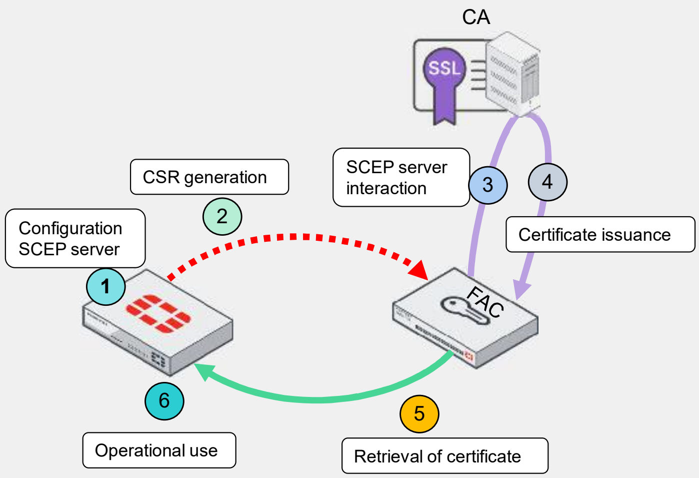


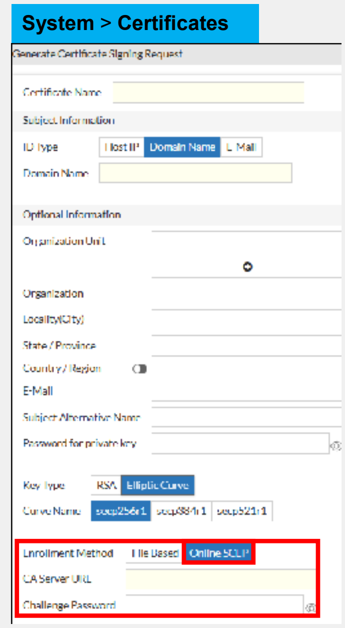
You are not limited to using a pre-shared key for Phase 1 authentication in your IPsec VPN.

You have the option to use digital signatures or certificates to enhance the security of your connection. Select the certificate and your peer certificate as displayed on the screen.

Ensure that you use a valid certificate authority (CA). You can also add your own certificates based on your preferences.

Automated Digital Certificate Enrollment Through FortiAuthenticator and SCEP





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To simplify your certificate management, especially in large deployments, consider using a FortiAuthenticator along with the online Simple Certificate Enrollment Protocol (SCEP) process on FortiGate.

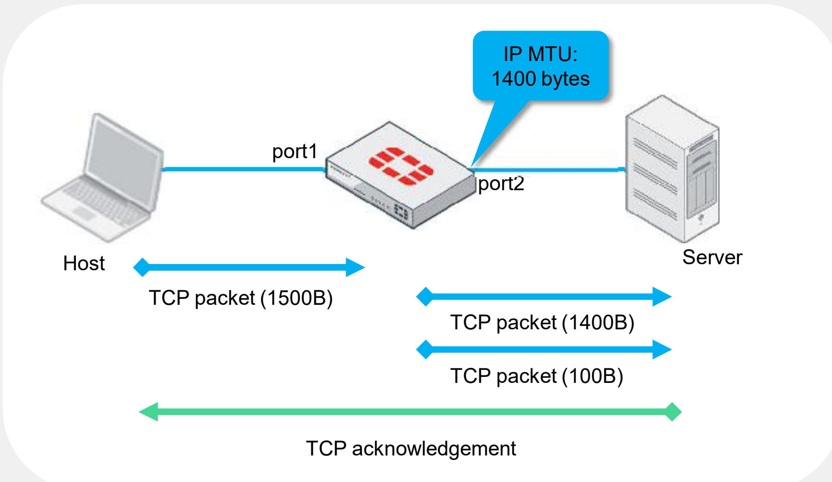
This approach automates the enrollment and installation of digital certificates, reducing manual configuration and ensuring secure communications.

Here's a concise overview:

1. Configure FortiGate to use SCEP by specifying the URL of the SCEP server and any necessary credentials or parameters required by the SCEP server.
2. FortiGate generates a certificate signing request (CSR), which includes the public key and identification information of the FortiGate device. This request is prepared to be sent to the SCEP server.
3. The SCEP server (FortiAuthenticator) acts as a proxy, forwarding the CSR to the CA. The server handles the communication between the FortiGate device and the CA, facilitating the certificate signing process.
4. The CA processes the CSR and, if all the verification checks are satisfied, signs the certificate, thus vouching for the identity and authenticity of the FortiGate device.
5. The signed certificate is sent back to the FortiGate device through the SCEP server. FortiGate then installs this certificate and uses it for various security functions, such as establishing VPNs or securing management interfaces.
6. With the certificate properly installed, FortiGate can use it to secure communications, authenticate itself to other devices, and establish secure channels like VPN tunnels.

The Consequences of IP Fragmentation

- Exceeding MTU causes fragmentation
- Fragmentation impacts network performance
- Fragmentation can cause data loss



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IP fragmentation occurs when packets exceed the network's maximum transmission unit (MTU) size, necessitating their division for transit and subsequent reassembly upon arrival. This process ensures that the packets are delivered, but can decrease efficiency, reduce throughput, and risk data loss, if fragments are lost.

In this example:

1. The host sends a TCP packet of 1500 bytes.
2. The firewall, with an MTU of 1400 bytes, fragments the TCP packet into two packets: one of 1400 bytes and another of 100 bytes.
3. The server receives both packets and sends a TCP acknowledgment.

Typically, the MTU is 1500 bytes.

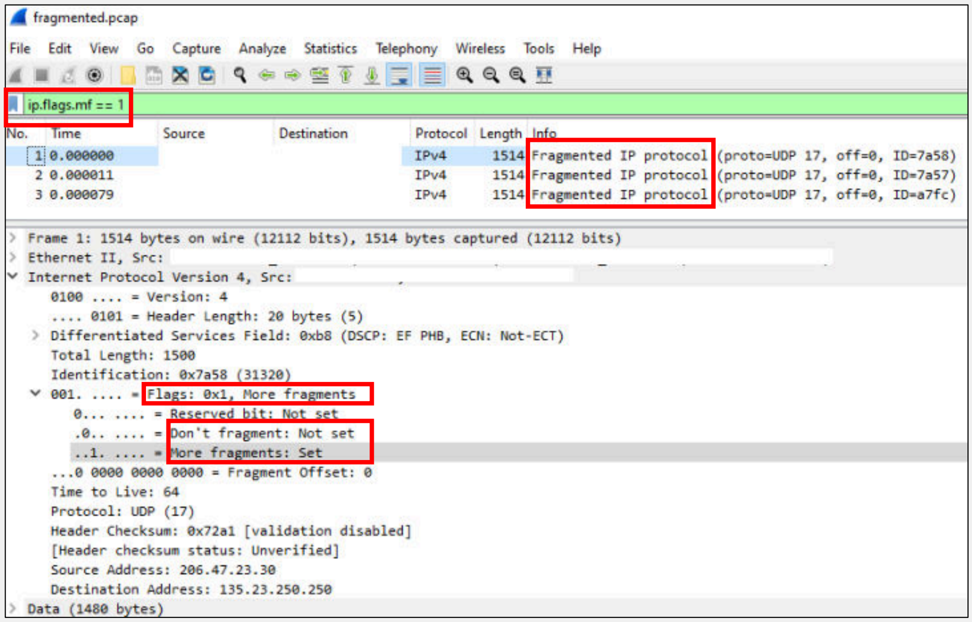
Understanding IP Flag Roles in Fragmentation

- Filter on PCAP files

```
ip.flags.mf == 1
```

- First bit always zero
- Second bit prevents fragmentation
- Third bit indicates more fragments
- Honors DF command

```
config system global
set honor-df enable
end
```



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You can determine if a packet has been fragmented using a PCAP file and the filter displayed on the slide.

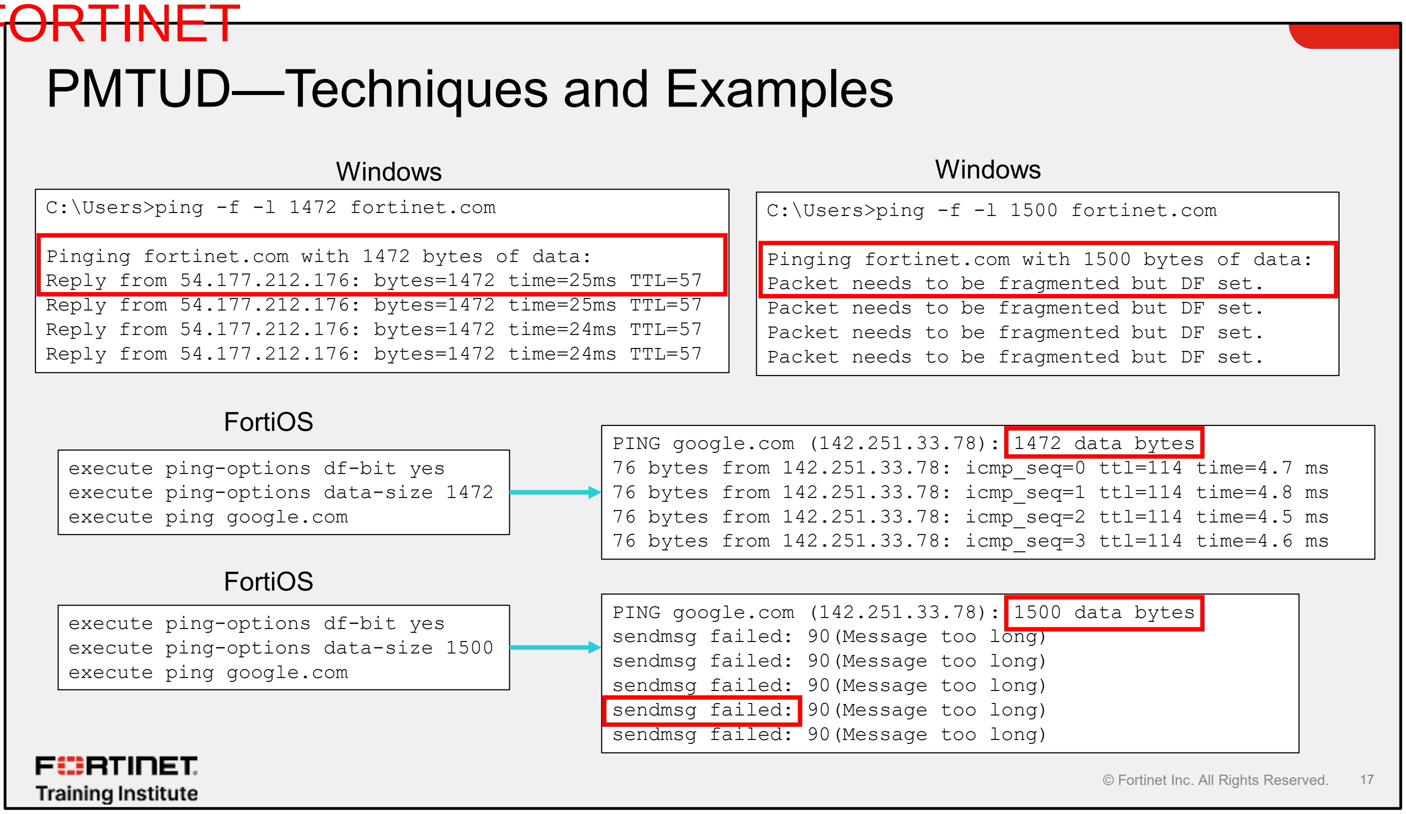
IP flags, which belong to the IP layer and consist of three bits, manage fragmentation as follows:

- The first bit is always set to 0.
- The second bit, the DF (don't fragment) bit, indicates the packet should not be fragmented.
- The third bit, the MF (more fragments) bit, is set on all fragments except the last one.

Fragmentation occurs when a packet's size exceeds the MTU along its path.

FortiOS defaults to honoring the DF bit, meaning FortiGate won't fragment IP packets larger than the interface MTU.

For example, a 1600 byte packet on an 1500 byte MTU interface, will be dropped



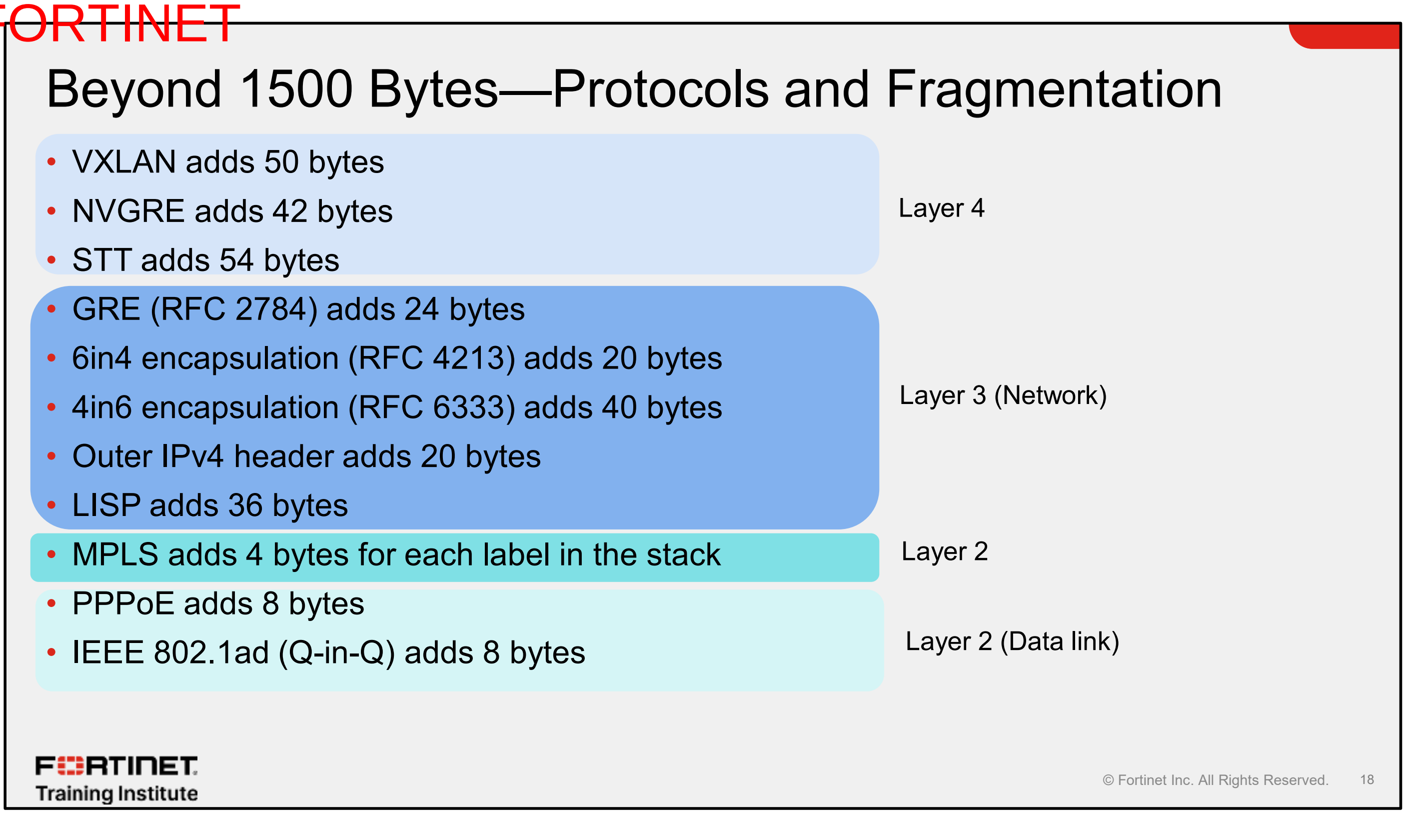
Path MTU discovery (PMTUD) is a technique used to determine the MTU size on the network path between two IP hosts, ensuring that IP packets are transmitted without the need for fragmentation.

PMTUD works by setting the DF flag on outgoing packets and then listening for ICMP "fragmentation needed" messages from devices along the path.

There are multiple ways you can use PMTUD. This slide shows two examples.

The Windows examples use the `-f` option to prevent packet fragmentation, and `-l` sets payload sizes of 1472 and 1500 bytes. The 1472 byte payload, with 28 bytes added for the header, matches the standard MTU of 1500 bytes, while the 1500 byte payload, with the same header addition, exceeds the standard MTU, indicating a need for fragmentation.

In contrast, the FortiOS example uses the `df-bit` option to set the DF bit in the IP header and data size to specify the datagram size in bytes. Here, a payload of 1472 bytes succeeds, but a payload of 1500 bytes fails, suggesting that the latter exceeds MTU limits.

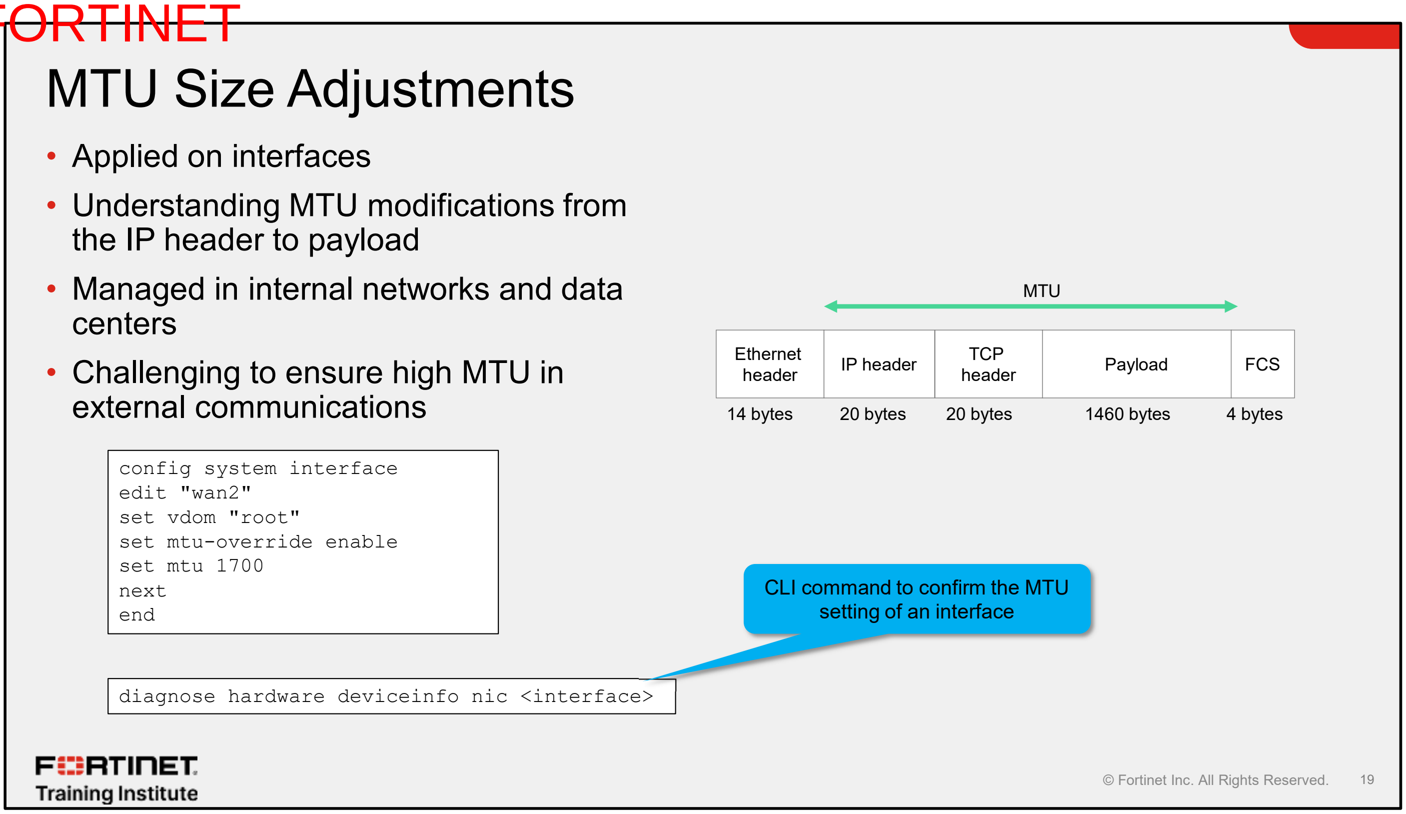


Data packets around 1500 bytes typically transmit without issues. However, certain protocols can create conditions that require fragmentation. The following protocols are noteworthy:

- Virtual eXtensible LAN (VXLAN): extends layer 2 networks, adding 50 bytes.
- Network virtualization using Generic Routing Encapsulation (NVGRE): creates virtual networks over layer 3 using GRE encapsulation, adding 42 bytes.
- Stateless Transport Tunneling (STT): tunnels VM traffic using a TCP-like header for overlay networks, adding 54 bytes.
- Generic Routing Encapsulation (GRE): tunnels various protocols over IP networks, adding 24 bytes.
- 6in4: encapsulates IPv6 packets within IPv4, adding 20 bytes.
- 4in6: encapsulates IPv4 packets within IPv6, adding 40 bytes.
- Outer IPv4 header: adds an additional IPv4 header, adding 20 bytes.
- Locator/ID Separation Protocol (LISP): separates IP addresses into endpoint identifiers (EID) and routing locators (RLOC) for better routing scalability, adding 36 bytes.
- Multiprotocol label switching (MPLS): directs packets using labels, with each label adding 4 bytes.
- Point-to-Point Protocol over Ethernet (PPPoE): is primarily used for digital subscriber line (DSL) connections that encapsulate PPP frames within Ethernet frames, adding a PPPoE header of 6 bytes plus a PPP protocol field of 2 bytes.
- IEEE 802.1ad tag (Q-in-Q): carries VLAN information, 8 bytes.

If your packet flow involves one or more of the protocols shown in this slide, you may need to consider the following options:

- Increase the MTU on the interface (the most common approach).
- Adjust the TCP maximum segment size in the firewall policy.
- Set the fragmentation MTU in IPsec phase 1.



When modifying the MTU size, it's important to consider the range from the IP header to the payload. Changing MTU sizes on internet-facing interfaces is nearly impossible, as control is limited beyond that point. However, experts can modify the MTU in controlled environments, such as internal networks or data centers.

In external communications, such as those across the internet, ensuring a higher MTU from end to end is generally not feasible.

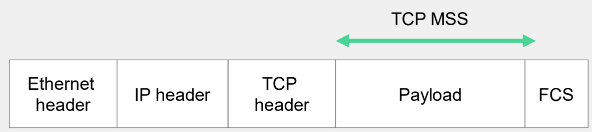
To adjust the MTU size, use the command shown on this slide. You can also review the current MTU settings on your FortiGate device.

- For a physical interface, the MTU can be configured between 68 and 65535 bytes, with the default set to 1500 bytes.
- For an IPsec VPN interface, the MTU can be configured between 68 and 1700 bytes, with a default of 1420 bytes.

TCP MSS in Network Packet Transmission

- TCP segment payload limit
- Applied in firewall policies
- Smaller MSS increases header count
- Optimizing performance

```
config firewall policy
edit <policy id>
set tcp-mss-sender <mss value>
set tcp-mss-receiver <mss value>
end
```



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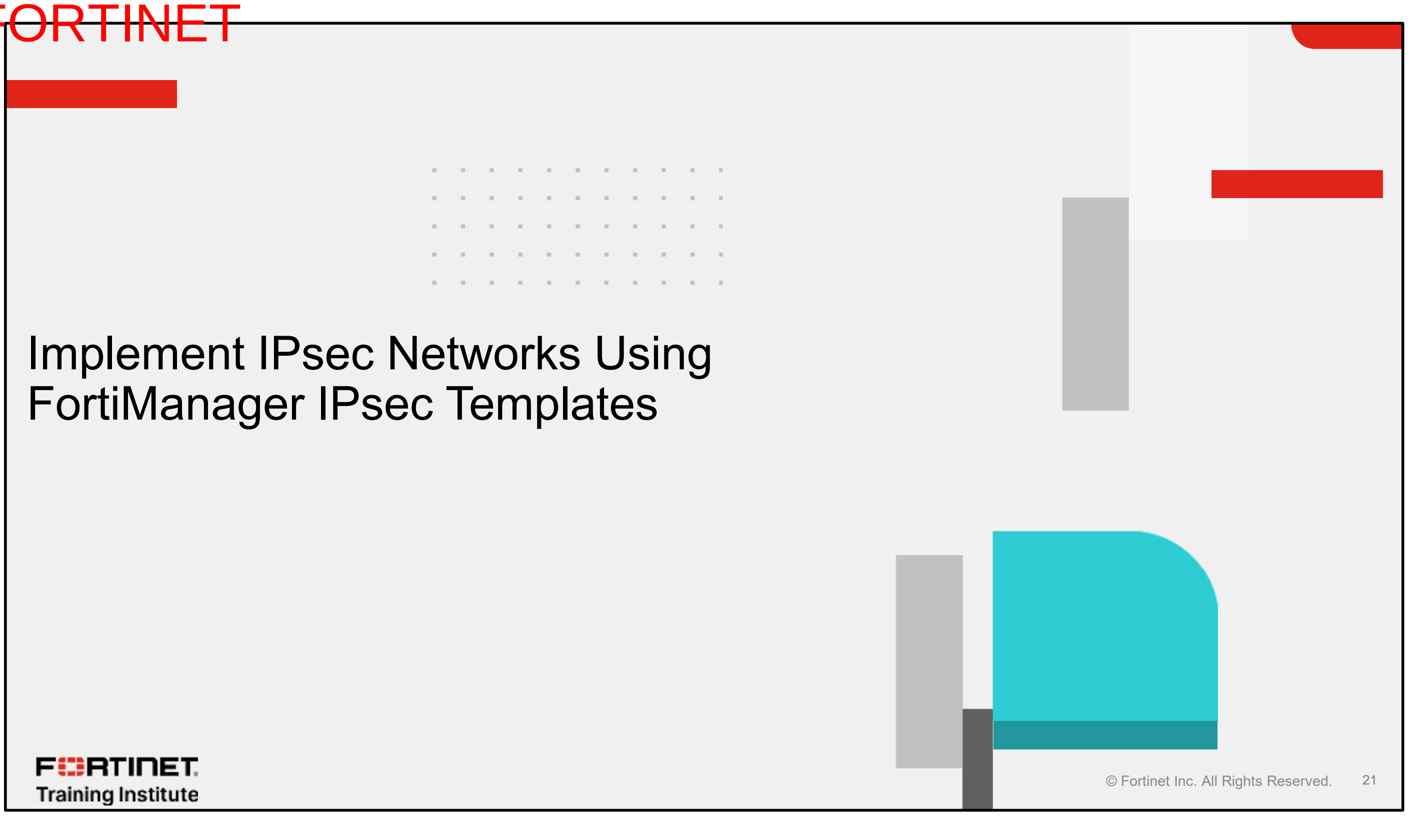
The maximum segment size (MSS) is only the payload size and does not include the TCP headers.

You configure this parameter within a firewall policy and it determines the largest payload a device can handle in a single TCP segment.

- Lowering the MSS increases packet count and adds unnecessary TCP headers, reducing network efficiency by consuming bandwidth with extra headers that were not originally required.
- More packets mean more processing, which can decrease overall data throughput.
- Properly setting MSS requires detailed knowledge of the network path, making it error-prone and complex.
- Incorrect MSS settings can lead to mismatches and data flow problems between devices.

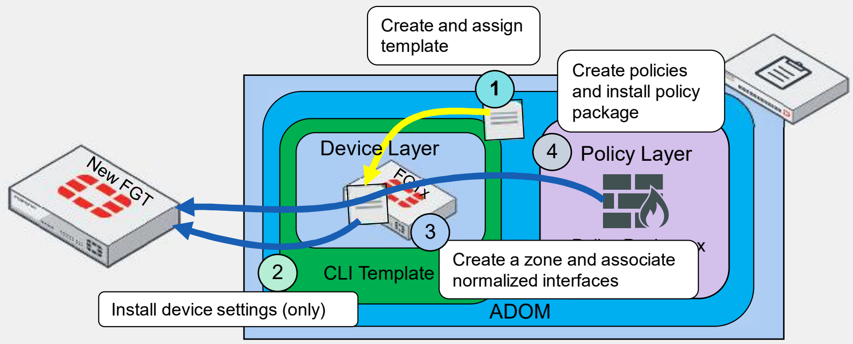
However keep on mind that proper MSS settings enhance data flow efficiency and reduce network latency, which is crucial for real-time applications.



In this section, you will learn how to implement IPsec networks using FortiManager IPsec templates.

How to Install IPsec Templates for IPsec VPN

- Keep a diagram
- Identify interfaces
- Identify remote networks
- Metadata variables: remote gateway (IP address), outgoing interface, local ID, remote subnet, tunnel interface setup, IP, remote IP
- Choose IKEv1 or IKEv2
- Identify Phase 1 and Phase 2 DH groups
- Recognize Phase 1 and Phase 2 proposals



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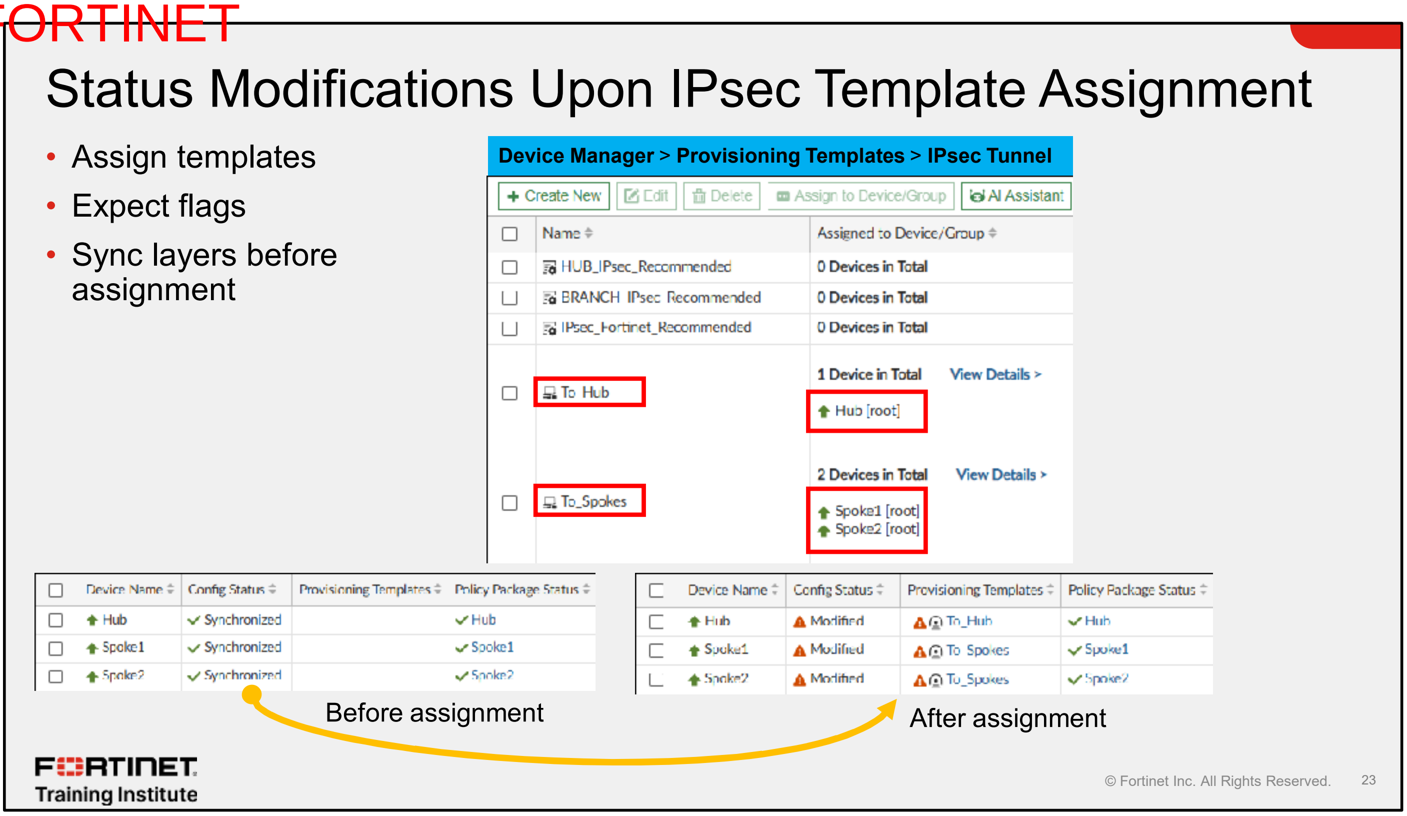
When you use IPsec VPN templates you must:

- Keep a diagram of your connection handy as a reference and double-check to ensure the settings are correct.
- Identify which interfaces are participating in the process.
- Identify any remote networks that must be connected through IPsec.
- Determine the metadata variable options you will be using. The available options are shown on this slide.
- Decide whether you will use IKEv1 or IKEv2.
- Identify your Phase 1 and Phase 2 Diffie-Hellman (DH) groups.
- Recognize your Phase 1 and Phase 2 proposals.

Installing IPsec templates for the first time is straightforward:

1. Create and assign IPsec templates for hub and spokes.
2. Install device settings to enable IPsec VPN interfaces.
3. Create a zone for the hub and normalized interfaces.
4. Create policies and the install policy package.



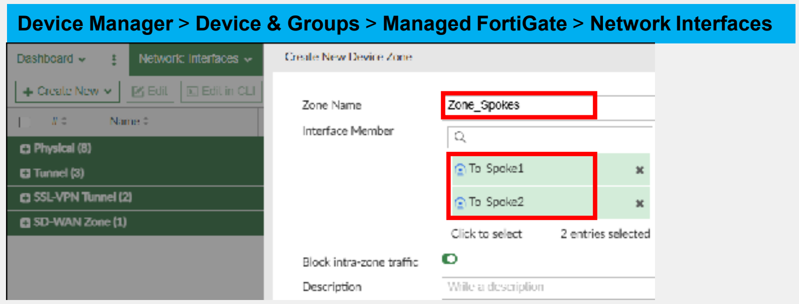
When you assign an IPsec template, the status of each firewall device database and the provisioning template status are flagged as modified, indicated by an orange warning.

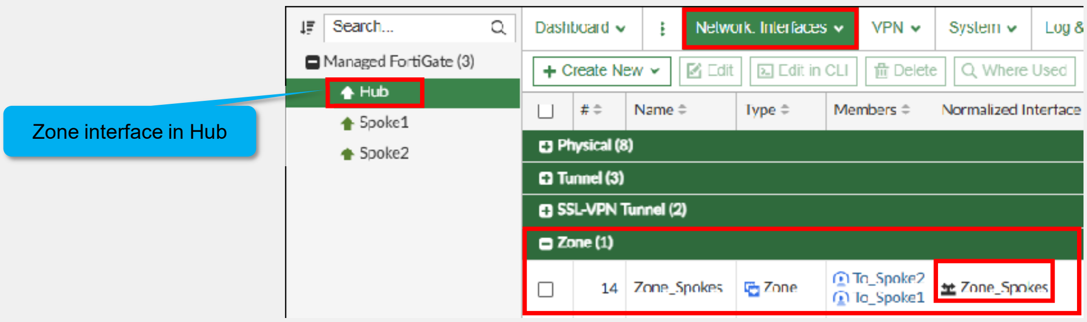
This flagging is expected, because template assignment does not equate to installation on either FortiGate or the device database.

Additionally, it's ideal to have the device layer and policy layer synchronized before assigning provisioning templates. If they are not synchronized, you may encounter additional configurations in pending status as you begin to set up IPsec.

Creating and Using a Zone Interface in a Hub

- Create a unified zone interface
- Link the zone to normalized interface
- Be cautious when using per-platform mapping





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If you need to manage the traffic of all your spokes under the same interface, you can create a zone interface in the device database as shown on this slide.

Notice that two IPsec VPN interfaces are included in the zone named Zones_Spokes, and the associated normalized interface that you created is visible.

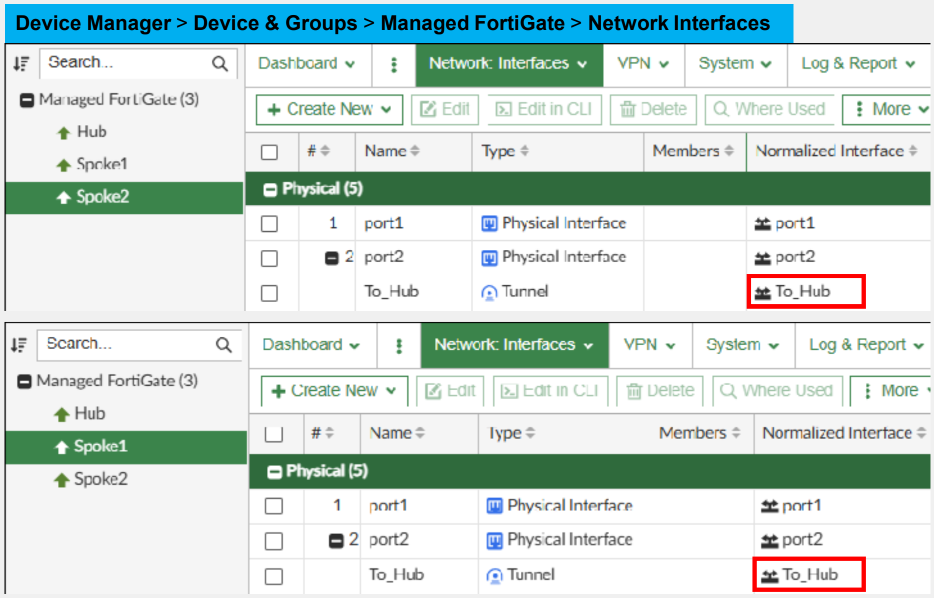
After you create your zone interface, you must associate it with a normalized interface, to use it in the policy layer, since creating or having interfaces in the device layer does not automatically mean they can be used in the policy layer.

Although per-device mapping is recommended, per-platform mapping is easier to manage because the same names are used in both layers. However, using a per-platform approach to mapping can sometimes lead to issues in finding the correct interface, because the per-platform type is case-sensitive.

How you create your zone and map your interfaces will depend on your skills and plans for deploying your network. Keep in mind that if you use a per-platform approach, you must be careful with case-sensitive names.

Create Normalized Interfaces for Spokes

- Verify normalized interfaces
- Device layer: quick identification



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On the other hand, you do not need to create a zone interface for spokes because they use only one IPsec VPN connection to the hub. However, if you have more than one interface, configuring a zone interface may be necessary, depending on your network needs.

You must manually create and map a normalized interface to your IPsec VPN interfaces.

After you create and associate normalized interfaces in both the hub and spokes, you should create policies, install them on devices, and ensure your tunnels are ready to go, assuming there are no communication or configuration issues.

The Benefits of Using IPsec Templates

Feature / capability	VPN manager	IPsec templates
Central management	Centralized management interface for VPNs	Templates are applied from Device Manager
Routing setup	Manual or automatic	Manual or automatic
Normalized zones	Manual or automatic	Manual
VPN creation	Assisted	Manual and Assisted
Configuration	Community and Nodes	Phase1 and Phase2
Import configuration	✗	✓
Dynamic configurations	✗	Metadata variables
Editable IPsec names	✗	✓

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The table on this slide compares the benefits of using IPsec templates with using VPN Manager in FortiManager.

VPN Manager, a central management module, automatically creates normalized interfaces when you select it in the policy layer. In contrast, IPsec templates which are found in Device Manager, require manual configuration of normalized interfaces.

In VPN Manager, you can configure communities and nodes using an assistant or wizard. When you use IPsec templates you must manually set up Phase 1 and Phase 2.

Importing existing IPsec VPN configurations to FortiManager is not possible, but IPsec templates allow for the importation and reuse of configurations across different firewalls.

IPsec templates support dynamic values, a feature that is not available in VPN Manager. Furthermore, IPsec templates enable the naming of VPNs in both phases, which VPN Manager does not allow.

The Benefits of Using IPsec Templates (Contd)

Feature / capability	VPN manager	IPsec templates
Reusability	✗	✓
SSL-VPN settings	✓	✗
Non-managed GWs	✓	✗
Use of two interfaces	✓	✓
Recommended	Large-scale and complex deployments	Preprovisioning and zero-touch provisioning (ZTP)

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You can reuse IPsec templates such as those assigned to branches, which typically have the same destination IP address as the hub firewall.

While VPN Manager offers SSL VPN configuration—a feature absent in IPsec templates—it also allows you add nodes not managed within your FortiManager administrative domain (ADOM), including those from firewalls outside Fortinet devices. IPsec templates, however, must be assigned to firewalls managed within the same FortiManager ADOM.

Both options support multiple interfaces, but assigning the same template to more than one firewall requires correct interface association with metadata variables.

Ultimately, the choice depends on the scale and complexity of your scenario: FortiManager is suited for large-scale and complex environments, while IPsec templates are ideal for preprovisioning and ZTP.

Review

- ✓ Master IPsec topology design
- ✓ Implement IPsec networks using FortiManager IPsec templates

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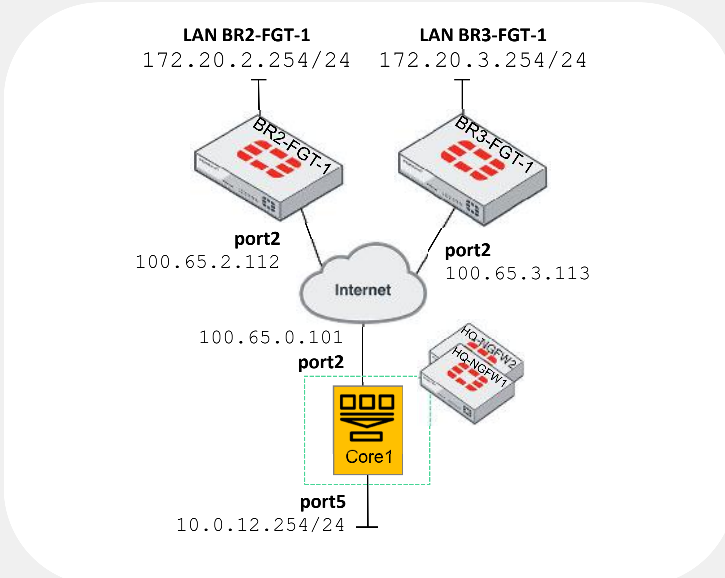
This slide shows the objectives that you covered in this lesson.

By mastering the objectives covered in this lesson, you learned how to use IPsec.

Now, you will work on *Lab 7–IPsec*.

Lab 7—IPsec

- IPsec templates:



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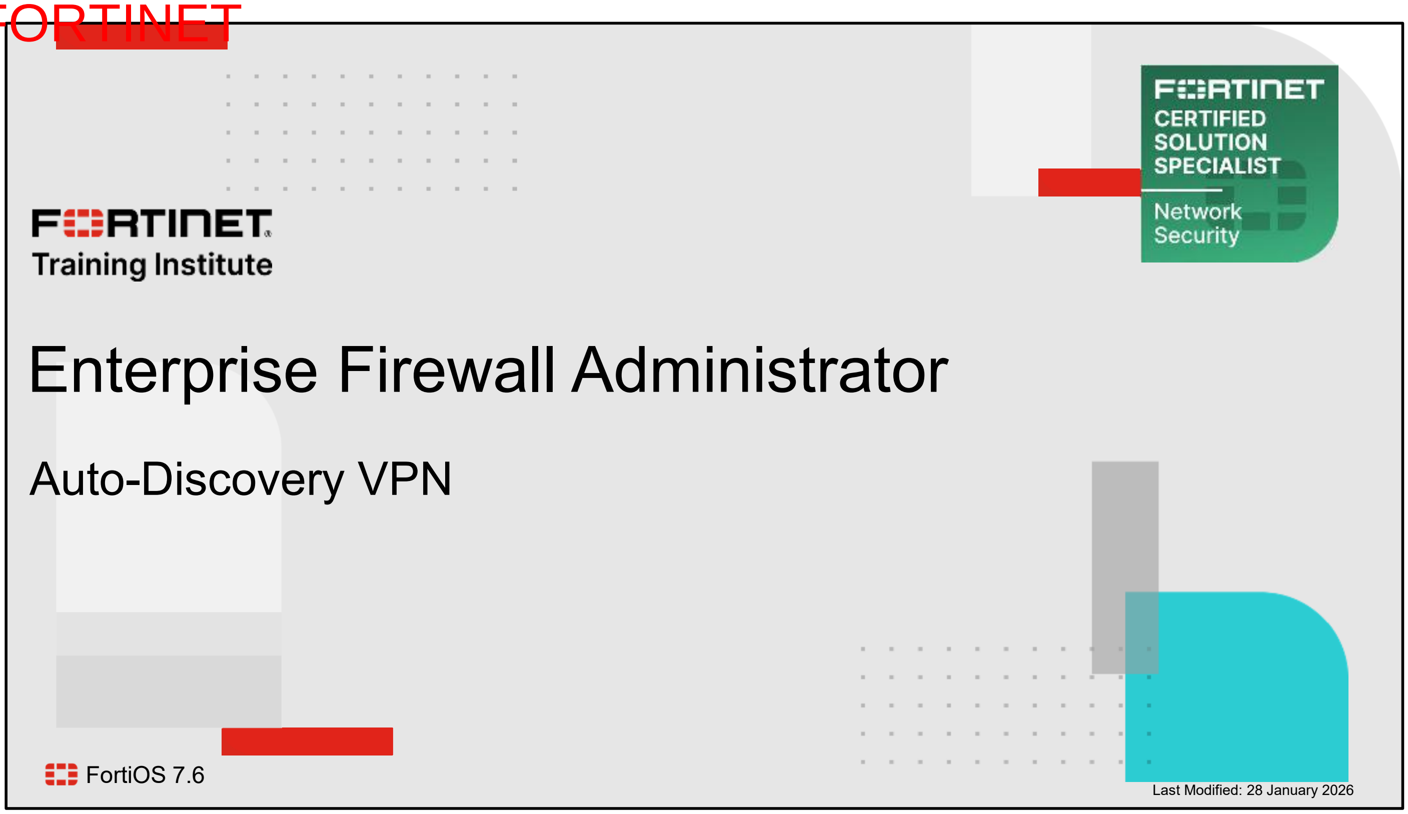
You will configure IPsec templates to set up a hub-and-spoke topology.



In this lesson, you will learn about auto-discovery VPN (ADVPN).

Objectives

- Implement Fortinet ADVPN using FortiGate or FortiManager

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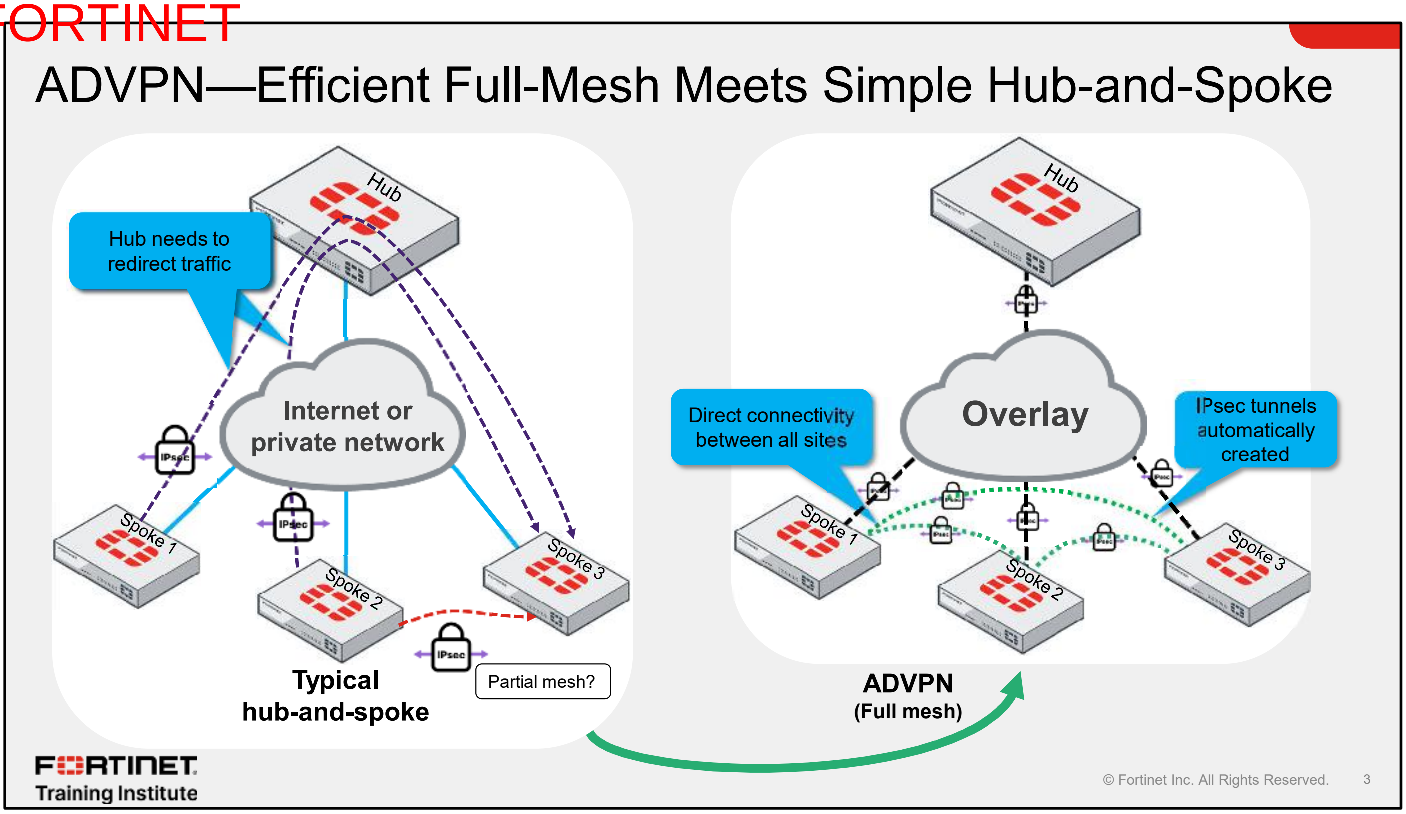
After completing this lesson, you should be able to achieve the objective shown on this slide.

By demonstrating competence in Fortinet ADVPN, you will be able to configure and test ADVPN.



In traditional hub-and-spoke topologies, branches are connected through the hub firewall, introducing potential bottlenecks and increased latency. Alternatively, creating manual IPsec connections between branches for a partial mesh topology can be resource-intensive because it not only complicates network management but also escalates maintenance requirements.

ADVPN addresses these challenges. This innovative solution combines the benefits of hub-and-spoke and full-mesh topologies because spoke-to-spoke tunnels are dynamically created on demand. After a shortcut tunnel is established between two spokes and routing has converged, spoke-to-spoke traffic no longer needs to flow through the hub.

ADVPN provides direct connectivity between spokes, and it supports:

- Single or multiple-hub architectures
- Network address translation (NAT) for on-demand tunnels
- Both IPv4 and IPv6
- The use of a dynamic routing protocol like BGP, OSPF, RIPv2 or RIPng

Advantages of IKEv2 in ADVPN

Phase/details	IKEv1 aggressive mode	IKEv1 main mode	IKEv2
Phase 1 messages	3 messages (identity protection)	6 messages (identity protection)	2 messages (IKE_SA_INIT request and response)
Phase 2 messages	3 messages (quick mode)	3 messages (quick mode)	2 messages (IKE_AUTH request and response)
Total messages	6 messages	9 messages	4 messages
Renegotiations	Extensive	Extensive	Simplified process
Security proposals	AES, SHA1, DH Group 14	AES, SHA1, DH Group 14	AES-GCM, SHA-256, DH elliptic curves groups (modern algorithms)
Security	Simpler, less secure	More complex, secure	Superior safety
Peer IDs	Clear text	Encrypted	Encrypted
Network ID	N/A	N/A	Yes
Purpose	Faster, identities exposed	Secure, identity-protected negotiation	Faster, safer

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For ADVPN, IKEv2 is preferred because of its efficiency and flexibility, using just four messages across two streamlined exchanges to negotiate security protocols.

IKEv2 effectively adjusts to optimal settings without extensive renegotiations, demonstrating better efficiency and adaptability than IKEv1.

It supports a wide range of cryptographic algorithms and features like AES-GCM and Diffie-Hellman elliptic curves groups, making it ideal for dynamic environments where rapid and secure VPN reconfigurations are necessary.

In IKEv1 aggressive mode, peer IDs are unencrypted and exposed, creating a security risk. Conversely, IKEv1 main mode and IKEv2 ensure peer IDs are encrypted.

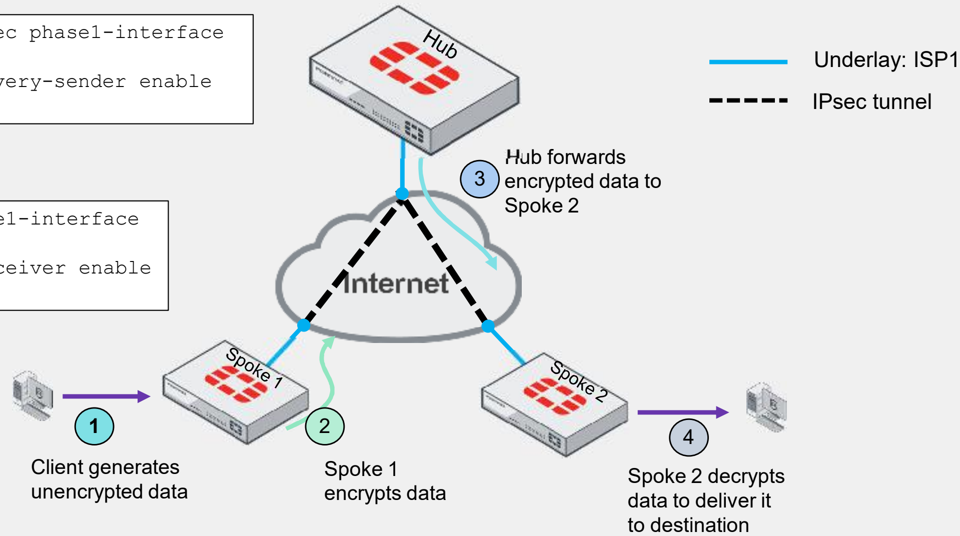
The network ID is a Fortinet-proprietary attribute that is used to select the correct phase 1 between IPsec peers, so that multiple IKEv2 tunnels can be established between the same local/remote gateway pairs.

Compared to IKEv1, IKEv2 reduces complexity and enhances security, making it better suited for automatically discovered VPN setups.

Trigger On-Demand IPsec Tunnel

```
config vpn ipsec phase1-interface
edit "VPN1"
set auto-discovery-sender enable
end
```

```
config vpn ipsec phase1-interface
edit "Hub1-VPN1"
set auto-discovery-receiver enable
end
```



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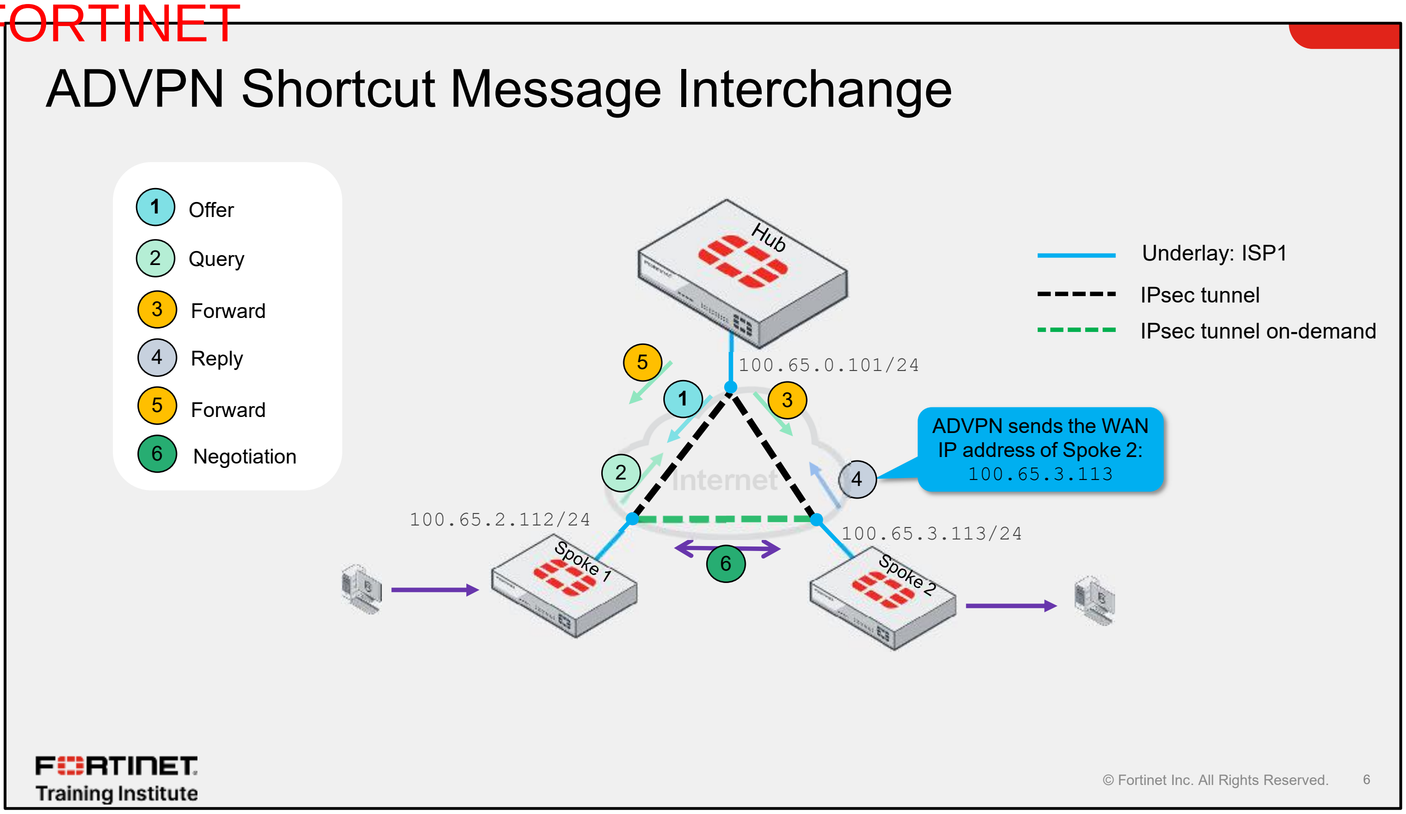
Before ADVPN shortcut messages can begin, a client must first generate traffic to trigger the hub to initiate an on-demand tunnel.

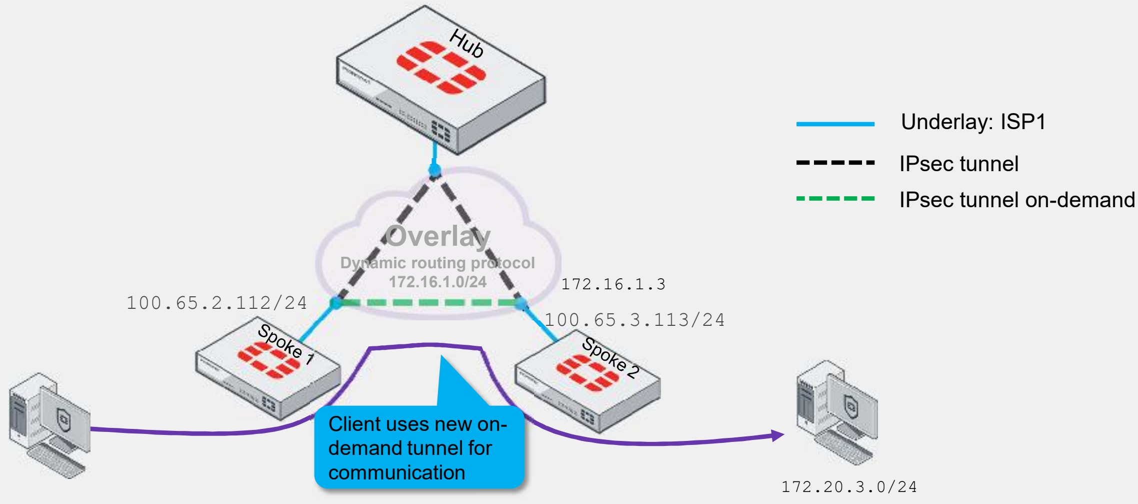
1. The client behind Spoke 1 generates traffic for devices located behind Spoke 2.
2. Spoke 1 receives the packet, encrypts it, and sends it to the hub.
3. The hub receives the packet from Spoke 1 and forwards it to Spoke 2.
4. Spoke 2 receives the packet, decrypts it, and forwards it to the destination device.



As shown on this slide, once the traffic travels from Spoke 1 to Spoke 2, the three FortiGate devices are involved in the shortcut message interchange.

1. The hub sends a shortcut offer message to Spoke 1 for a more direct tunnel from Spoke 1 to Spoke 2.
2. Spoke 1 acknowledges the shortcut offer by sending a shortcut query to the hub.
3. The hub forwards the shortcut query to Spoke 2.
4. Spoke 2 acknowledges the shortcut query and sends a shortcut reply to the hub, including the WAN IP address of Spoke 2, in this case `100.65.3.113`.
5. The hub forwards the shortcut reply to Spoke 1.
6. Spoke 1 and Spoke 2 initiate the tunnel IKE negotiation with the IP address of Spoke 2 sent to ADVPN.

On-Demand Established and Overlay Network




```
Spoke 1 # get router info routing-table details 172.20.3.62
Routing entry for 172.20.3.0/24
  Known via "bgp", distance 200, metric 0, best
  * vrf 0 172.16.1.3 priority 1 (recursive is directly connected, HUB1-VPN1_0_0)
```

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After ADVPN establishes the on-demand VPN IPsec tunnel between Spoke 1 and Spoke 2, the client behind Spoke 1 can communicate directly with Spoke 2 without routing through the hub.

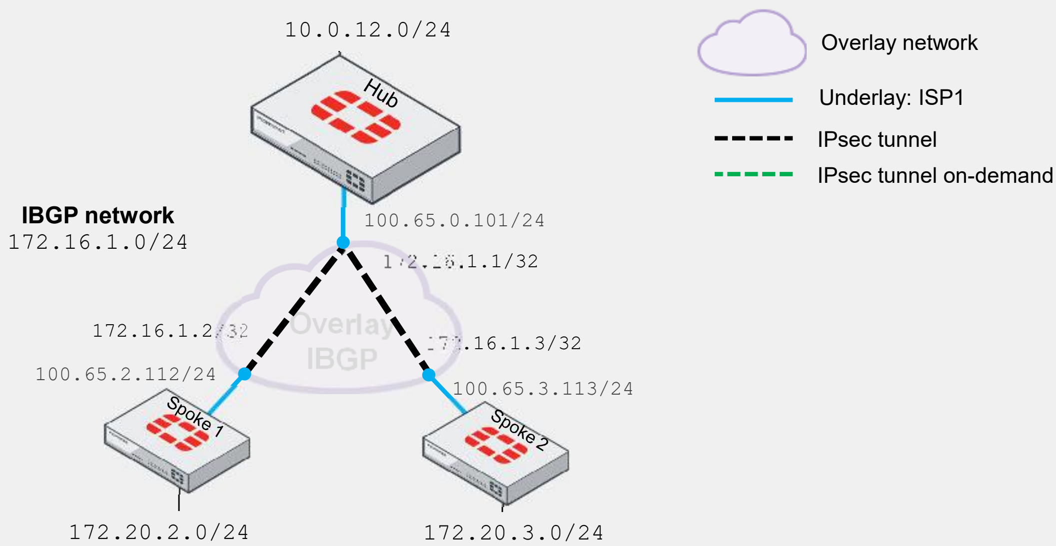
In addition, the overlay network is crucial for routing with a dynamic protocol and allows peers to advertise their local networks. Unless you include the overlay network, firewalls can't correctly advertise local networks within the hub-and-spoke topology.

Overlay network `172.16.1.0/24`, which already existed between the hub and spokes, now adds a shortcut between Spoke 1 and Spoke 2.

It's important to identify that IPsec tunnels are created over underlay IP addresses, and the dynamic routing protocol is running using the overlay network.

In the example shown on this slide, Spoke 1 at IP `100.65.2.112` connects the VPN to Spoke 2 at `100.65.3.113`. However, the next hop from Spoke 1 to Spoke 2 is through the Spoke 2 IPsec virtual interface IP address, `172.16.1.3`.

Use Case 1—ADVPN and IBGP



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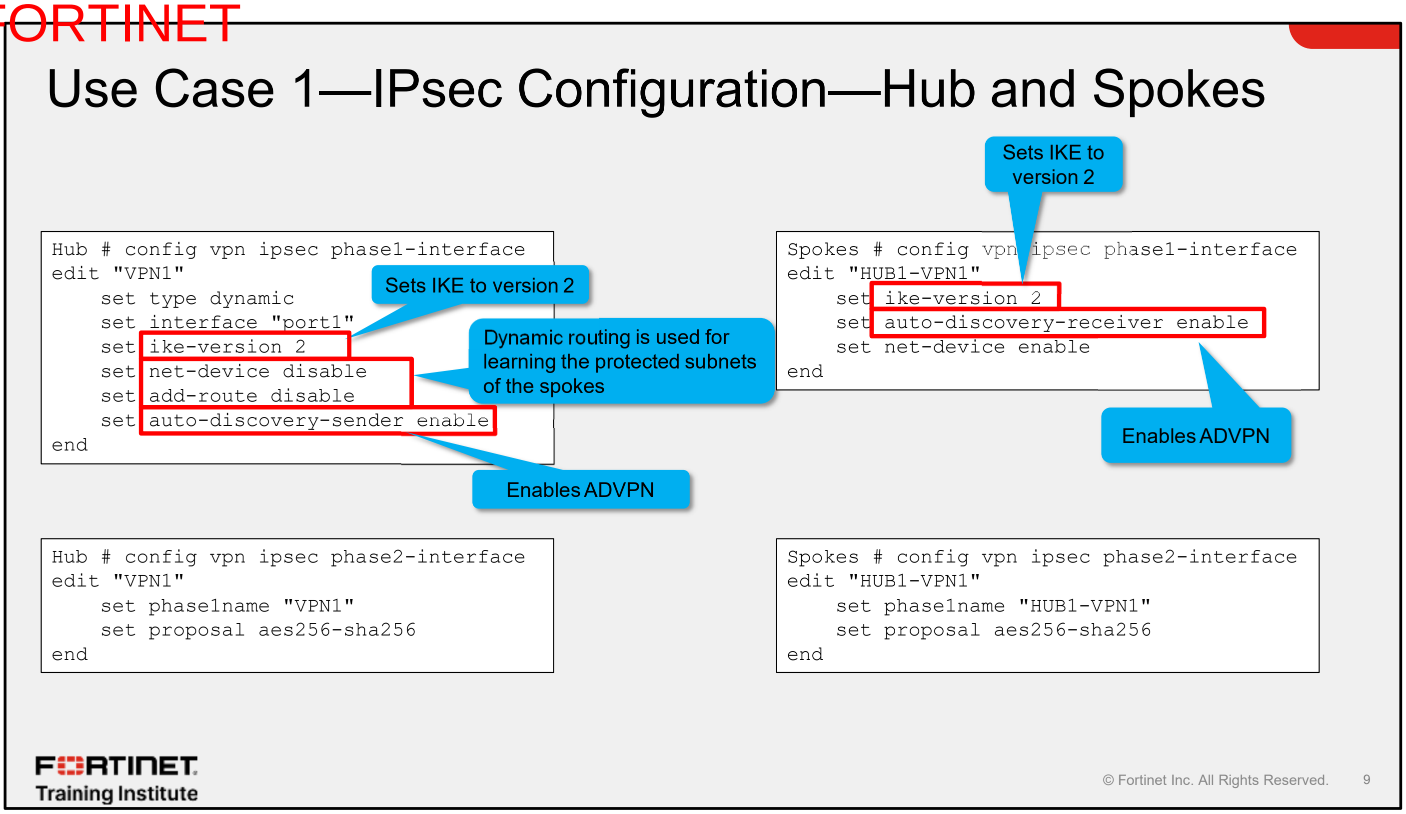
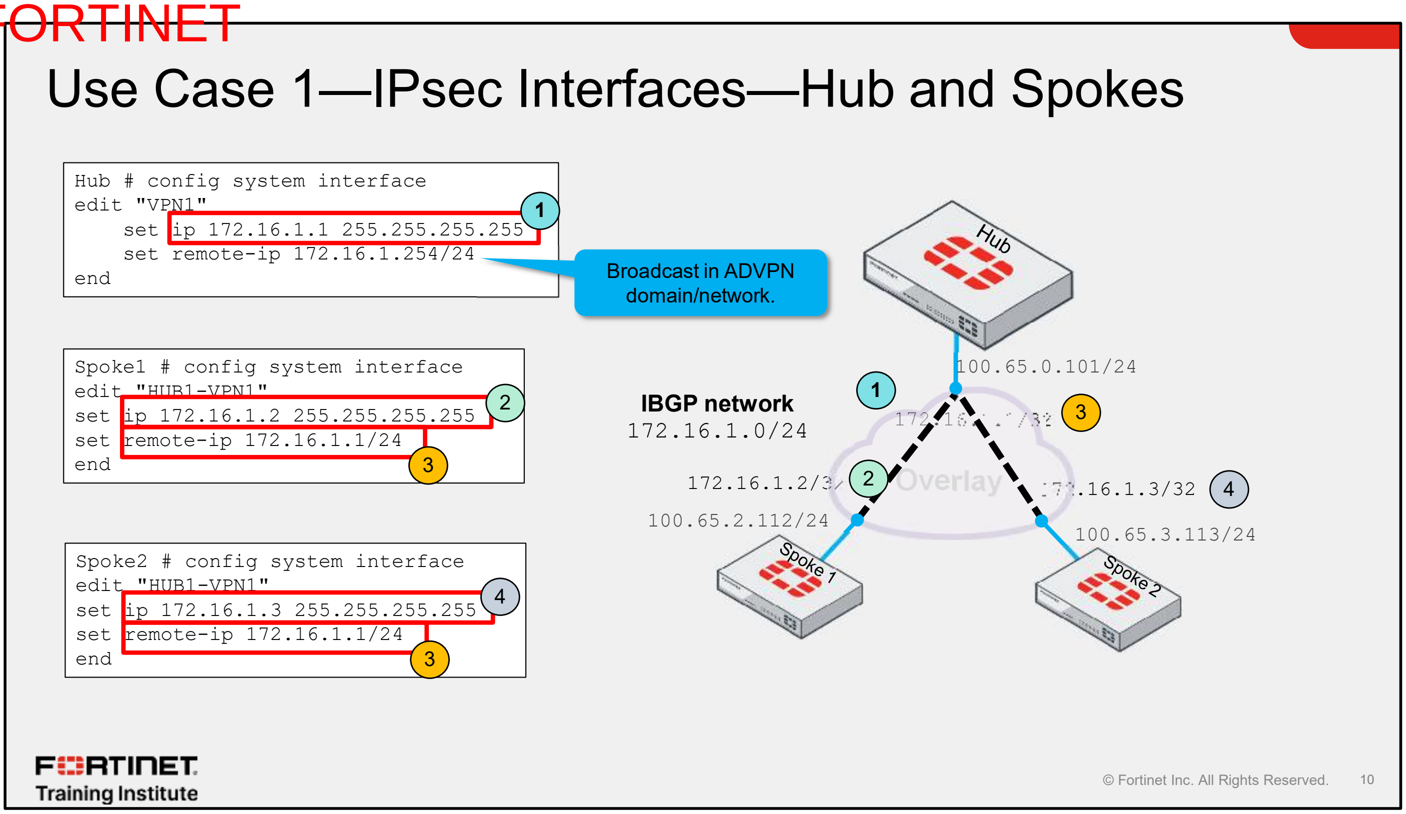
Now, you will examine the use case of ADVPN and IBGP.

This topology has only one internet connection for each firewall, creating only one overlay network. In this case, it uses IBGP to interconnect peers and share information across the IPsec topology.



This slide shows the following ADVPN configuration:

- Set IKE to version 2.
- Disable add-route to ensure that dynamic routing is used for learning the protected subnets of the spokes.
- Disable net-device to ensure that FortiGate does not create a dynamic interface.
- Enable auto-discovery-sender to use ADVPN. This setting indicates that when IPsec traffic transits the hub, it should send a shortcut offer to the initiator of the traffic to indicate that it can establish a more direct connection (shortcut).
- For the phase 2 configuration, ensure that quick modes are set to all (`0.0.0.0/0.0.0.0`).
- Set a firewall policy to allow traffic from the spokes to the hub, from the hub to the spokes, and between spokes through the hub.



This slide highlights that all internet members of the hub, Spoke 1, and Spoke 2 are configured with IP addresses on their IPsec interfaces using the overlay network, in this case `172.16.1.0/24`. This configuration enables them to participate in the routing protocol to dynamically share routing table databases.

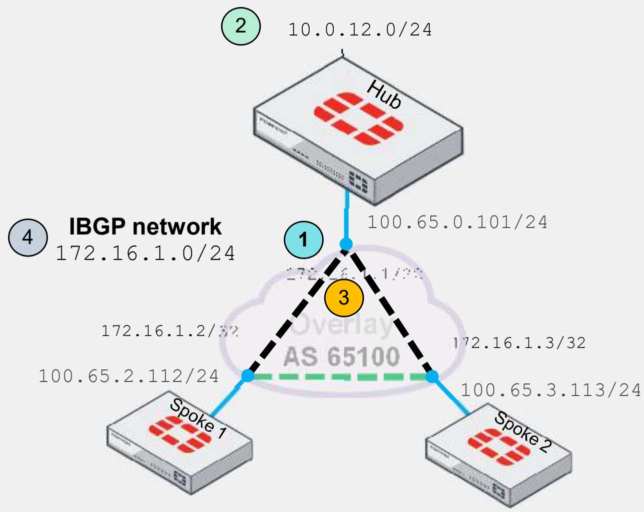
Remember, the phase 1 interface, phase 2 interface, and IPsec virtual interface are three distinct concepts.

The phase 1 interface establishes a secure channel to negotiate IPsec keys. In contrast, the phase 2 interface uses this secure channel to negotiate encryption and tunneling methods for the data traffic, allowing multiple phase 2 interfaces per secure phase 1 channel.

An IPsec virtual interface is automatically created when you set up a phase 1 interface. In the example shown on this slide, you can see that each phase 1 interface or IPsec interface has only one IP address configured from the overlay network.

Use Case 1—BGP Configuration—Hub

```
Hub # config router bgp
set as 65100
set router-id 172.16.1.1
config neighbor-group
    edit "Overlay1"
    set remote-as 65100
    set route-reflector-client enable
    end
config neighbor-range
    edit 1
    set prefix 172.16.1.0 255.255.255.0
    set neighbor-group "Overlay1"
    end
config network
    edit 1
    set prefix 10.0.12.0 255.255.255.0
    end
end
```



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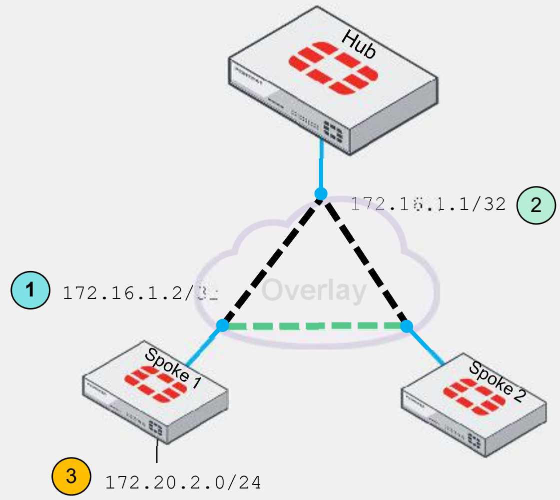
This slide shows the following IBGP configuration in the hub:

1. Configure the autonomous system and the router ID for hub.
2. Add the local networks behind the hub to be advertised over BGP, in this case `10.0.12.0/24`.
3. Set up a BGP neighbor group. If you are using IBGP for ADVPN, you must configure the hub as a `route-reflector-client`. So, routes learned from one spoke are forwarded to the other spokes.
4. Create a neighbor range with a prefix that includes all the spokes. When configured this way, you don't need to define each spoke individually as a neighbor.

Use Case 1—BGP Configuration—Spoke1

```
Spoke1 #config router bgp
set as 65100
set router-id 172.16.1.2
config neighbor
    edit "172.16.1.1"
    set remote-as 65100
    end
config network
    edit 1
    set prefix 172.20.2.0 255.255.255.0
    end
end
```



This slide shows the following IBGP configuration on Spoke 1:

1. Configure the autonomous system and the router ID for Spoke 1.
2. Configure the hub as a BGP neighbor.
3. Define the internal network that will be advertised over the BGP, in this case `172.20.2.0/24`.

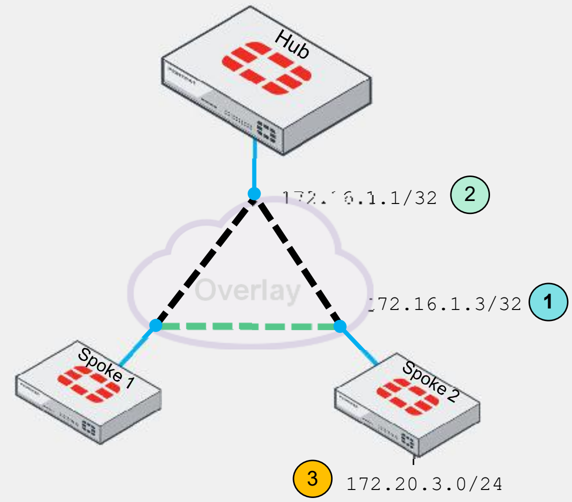
Use Case 1—BGP Configuration—Spoke2

```
Spoke2 #config router bgp
set as 65100
set router-id 172.16.1.3

config neighbor
    edit "172.16.1.1"
    set remote-as 65100
    end

config network
    edit 1
    set prefix 172.20.3.0 255.255.255.0
    end
end
```

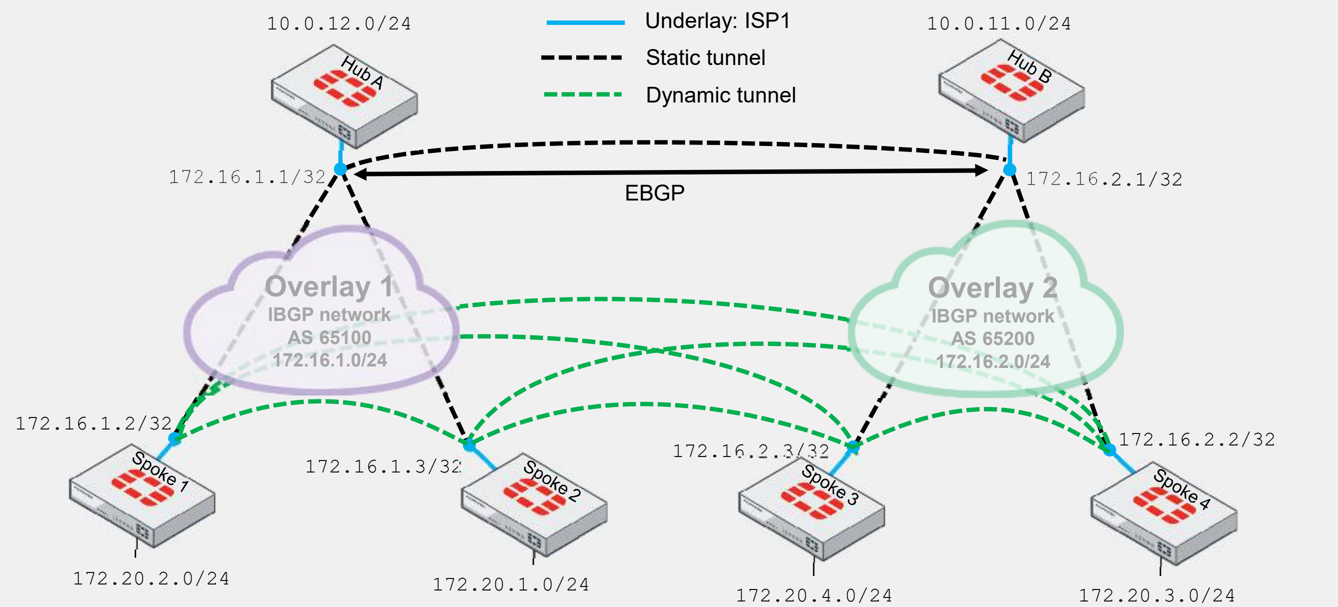
1 2 3



This slide shows the following IBGP configuration on Spoke 2:

1. Configure the autonomous system and the router ID for Spoke 2.
2. Configure the hub as a BGP neighbor.
3. Define the internal network that will be advertised over the BGP, in this case `172.20.3.0/24`

Use Case 2—ADVPN and IBGP/EBGP



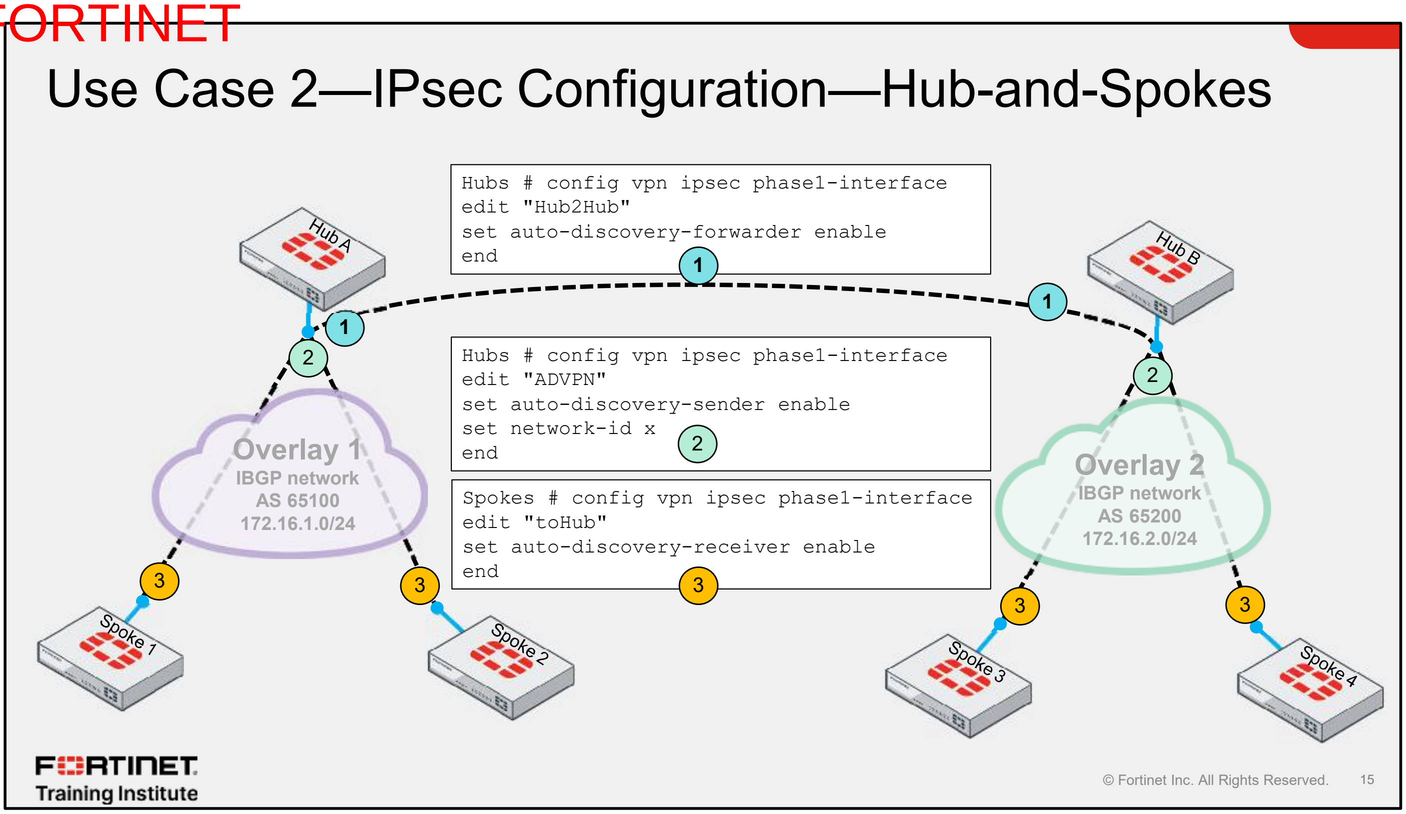
You can implement ADVPN across multiple regions, as the example on this slide shows where each region represents a different autonomous system.

This autonomous system is not connected to the internet but to the overlay network you create in your VPN IPsec topology.

The dual-region ADVPN topology shown on this slide employs IBGP for intraregion routing and EBGP for interregion routing.

Each overlay network has different network segments.

Dynamic IPsec tunnels connect the spokes.

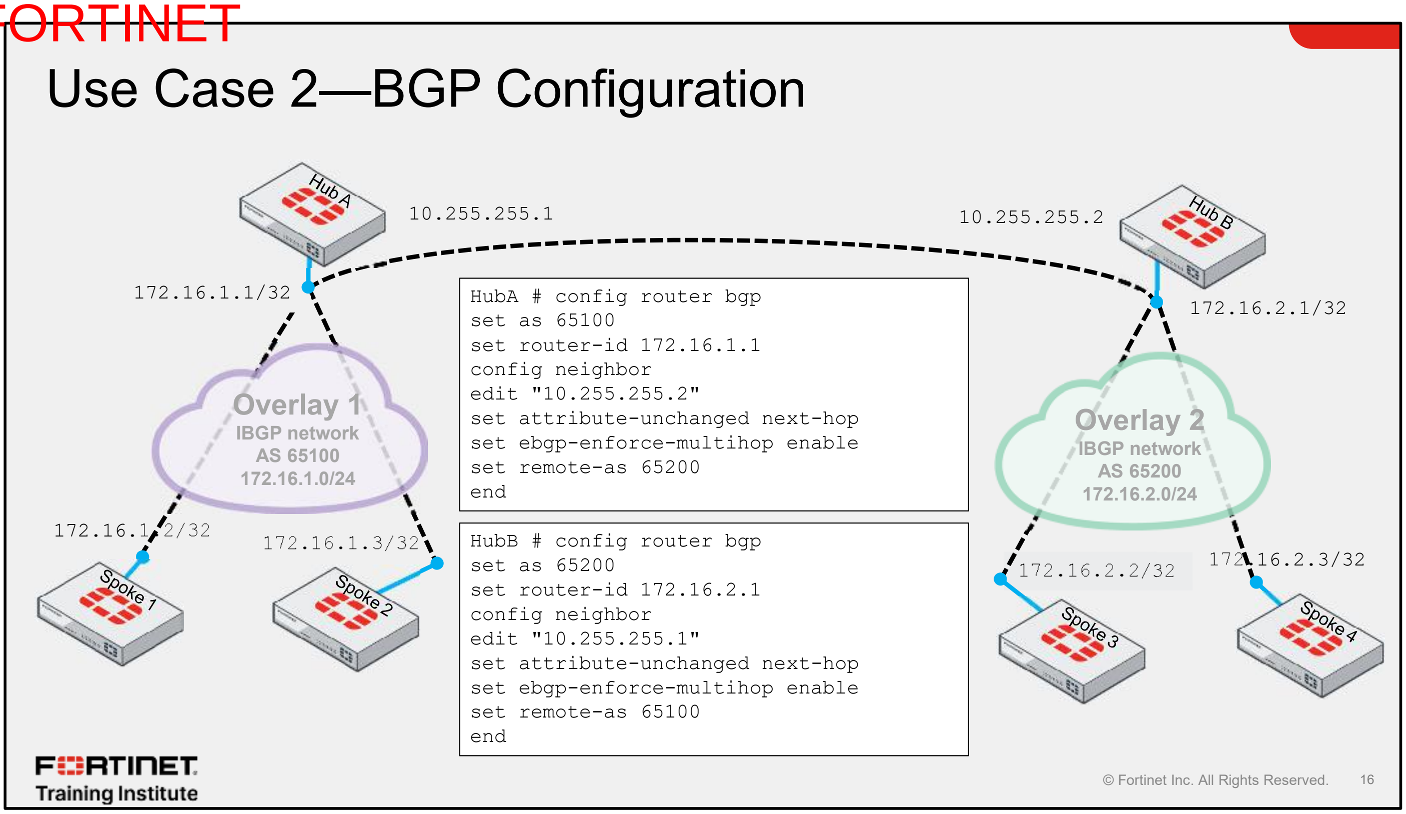


You must set up ADVPN on each face VPN IPsec interface depending on its role.

If your hub is connected to another hub through IPsec tunnels, set up `auto-discovery-forwarder` to send shortcut messages across overlay networks.

If a hub connects to a spoke, configure an `auto-discovery-sender` to send shortcut messages to the spokes, allowing them to opt for a better shortcut if available. You must use IKEv2 to select the correct phase 1 between IPsec peers, so that multiple IKEv2 tunnels can be established between the same local/remote gateway pairs.

If a spoke connects to a hub, enable `auto-discovery-receiver` to ensure the spoke is ready to receive shortcut messages and start negotiations for dynamically creating an IPsec tunnel.



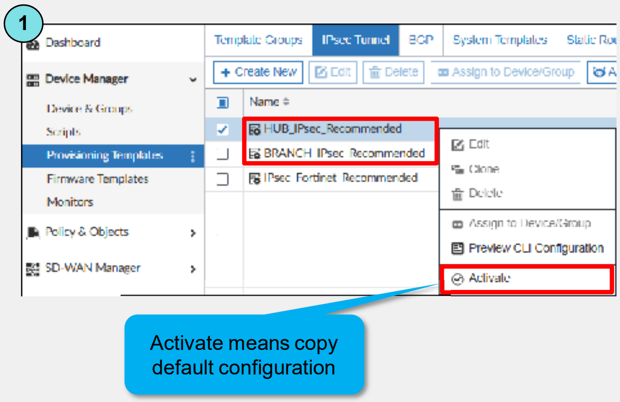
The EBGP configuration uses the commands shown on the slide.

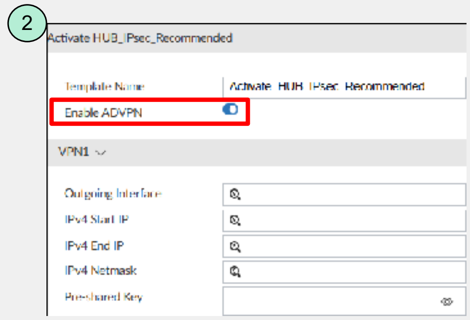
These commands will enable neighbors in the `172.16.1.0/24` network to exchange routing tables over BGP with other neighbors in the `172.16.2.0/24` network segment.

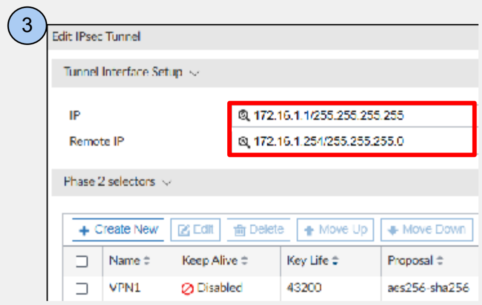
You must also add `neighbor-group` with `route-reflector-client` enable

Use Case 3—ADVPN and IPsec Templates

- IPsec templates simplify ADVPN setup
- Set up tunnel IP addresses in phase 1
- Metadata variables available







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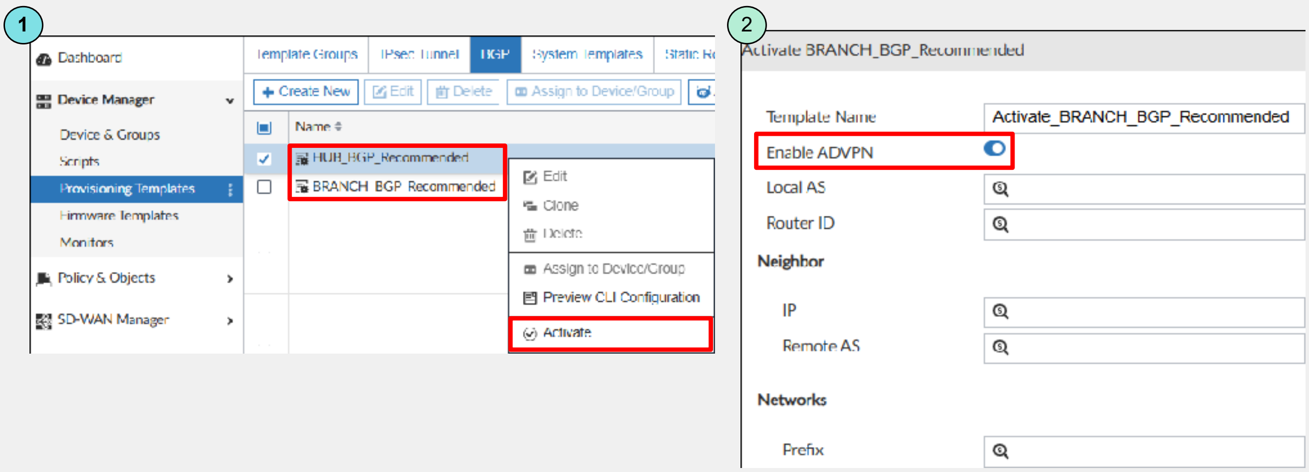
You can set up ADVPN in FortiManager effortlessly by using IPsec templates. Activating either the hub or branch IPsec provides an option to enable ADVPN. Remember, this feature requires configuring dynamic routing and policy implementation.

Additionally, you can set up tunnel IP addresses in phase 1 settings instead of using the IPsec interfaces.

IPsec templates allow the use of metadata variables, enabling dynamic value configuration tailored to specific needs.

Use Case 3—ADVPN and IPsec Templates (Contd)

- BGP templates with ADVPN options
- Preview CLI before template installation



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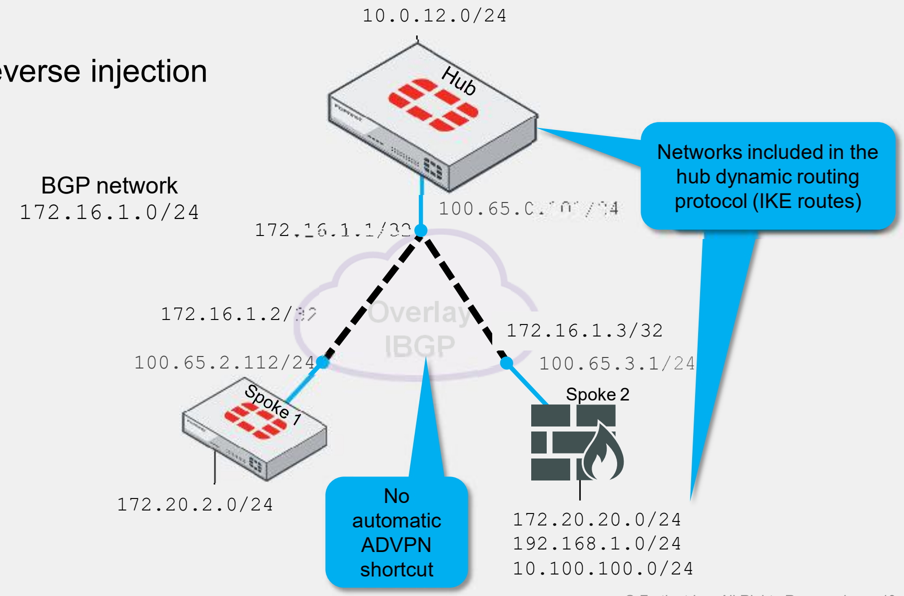
You can easily enable ADVPN using BGP templates, facilitating the configuration of your topology.

You can integrate both hub templates and spoke templates into your ADVPN topology.

Similar to the IPsec template, the BGP template has a feature that allows you to preview the CLI commands before assigning the template to any FortiGate device. This preview helps you understand the changes that will be implemented after the template is assigned and installed.

Use Case 4—ADVPN and Non-ADVPN Spokes

- Third-party vendors excluded from ADVPN
- Diverse spoke integration
- Phase 2 hub routing using reverse injection



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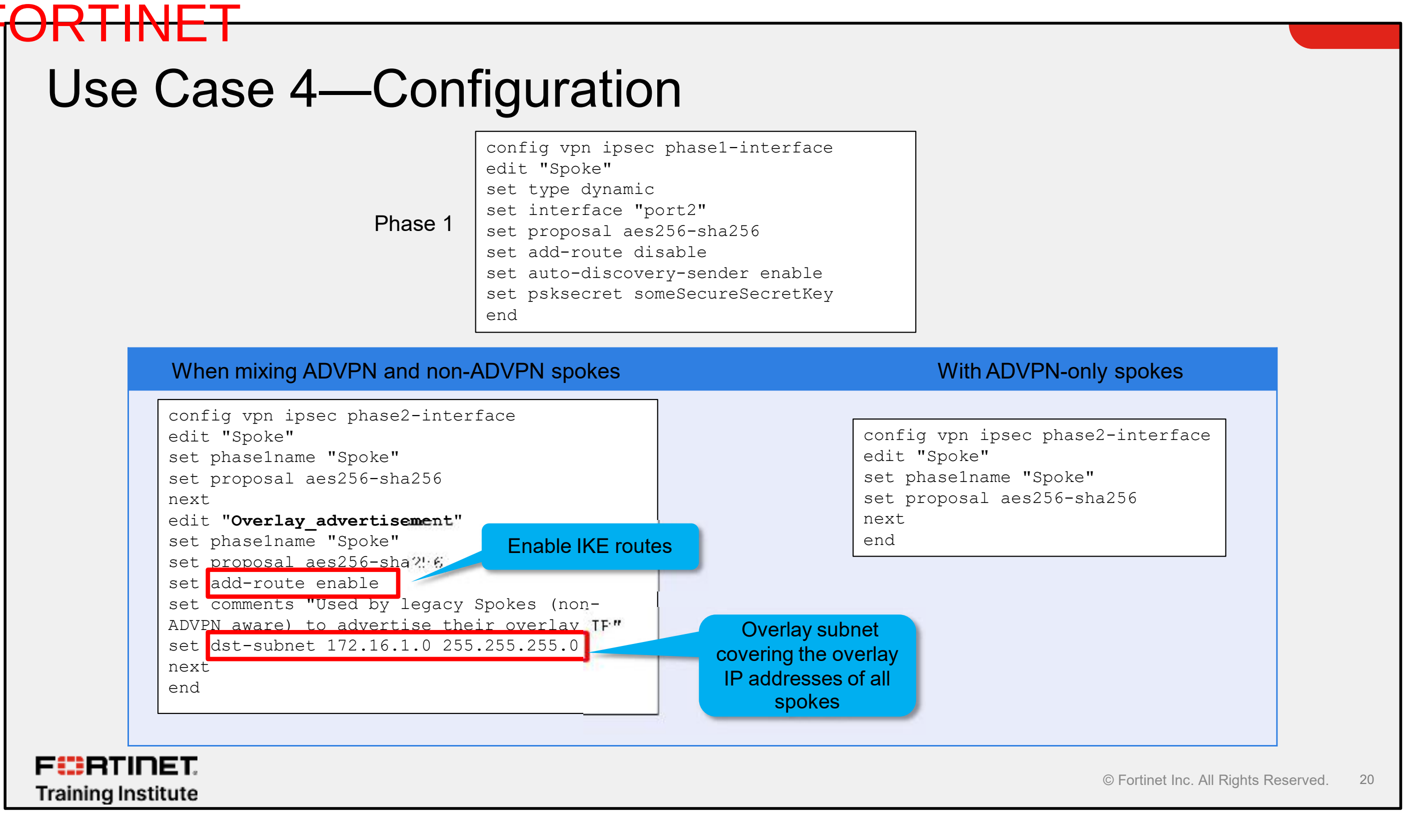
If a spoke is an IPsec gateway from a different vendor, the ADVPN feature won't be available because ADVPN is a proprietary protocol not supported by other vendors.

However, a third-party vendor can still be part of the ADVPN hub-and-spoke architecture but won't be able to negotiate ADVPN shortcuts with other spokes.

If non-ADVPN spokes want to participate in route exchange using any dynamic routing protocol, it is necessary for them to announce their network segments using an IP address from the overlay network.

The non-ADVPN spokes must create the overlay network segment in their phase 2. Although these devices can't perform dynamic IPsec with shortcuts using ADVPN, they can share networks using dynamic routing protocols, because the overlay network is part of the IPsec SA negotiation (quick-mode/phase 2). The hub can then add a return route to the Spoke 2 for these subnets.



This slide shows that phase 1 configuration remains unchanged, while phase 2 introduces a new phase. It highlights the `add-route` option for IKE routes and `dst-subnet` for the overlay subnet covering the IP addresses for all spokes.

Use Case 4—Phase 2 Alphabetical Lookup

```
Hub # diag vpn ike config list
...
phase2s:

  Overlay_advertisement proto 0 src 0.0.0.0/0.0.0.0:0 dst 172.16.1.0 255.255.255.0:0
dhgrp 14 5 add-route

  Spoke proto 0 src 0.0.0.0/0.0.0.0:0 dst 0.0.0.0/0.0.0.0:0  dhgrp 14 5
```

Overlay_advertisement alphabetically precedes Spoke

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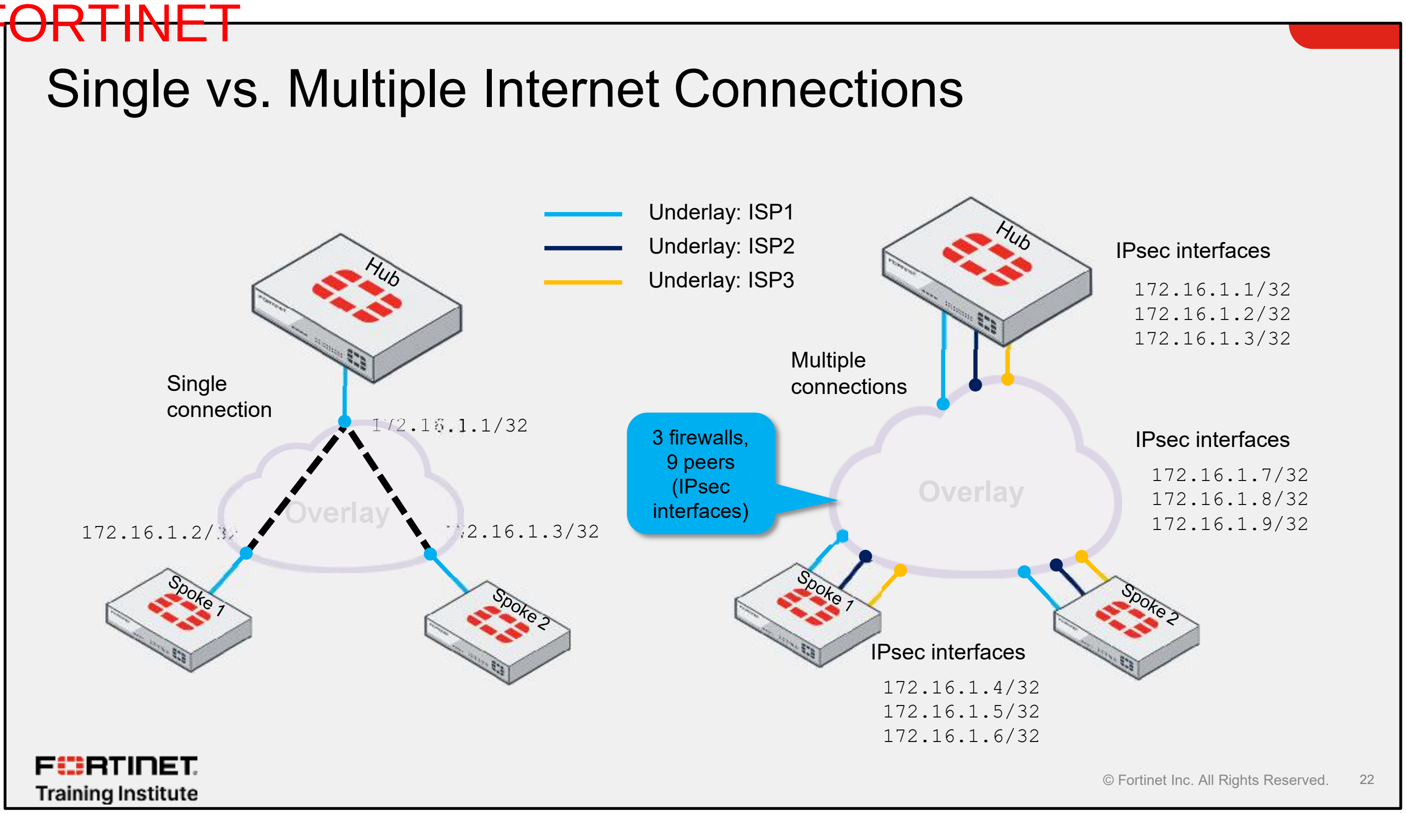
It is mandatory for the name of the additional phase 2, `Overlay_advertisement`, to alphabetically precede the regular phase 2 name, `Spoke`, because phase 2 lookup occurs in alphabetical order, and the more specific configuration in 'Overlay_advertisement' must be matched first.

In this example, you can see that the command `diag vpn ike config list` shows `Overlay_advertisement` in the first position for looking up network segments. This phase includes the network segment `172.16.1.0/24`, which belongs to the overlay, to advertise local networks using a dynamic routing protocol.

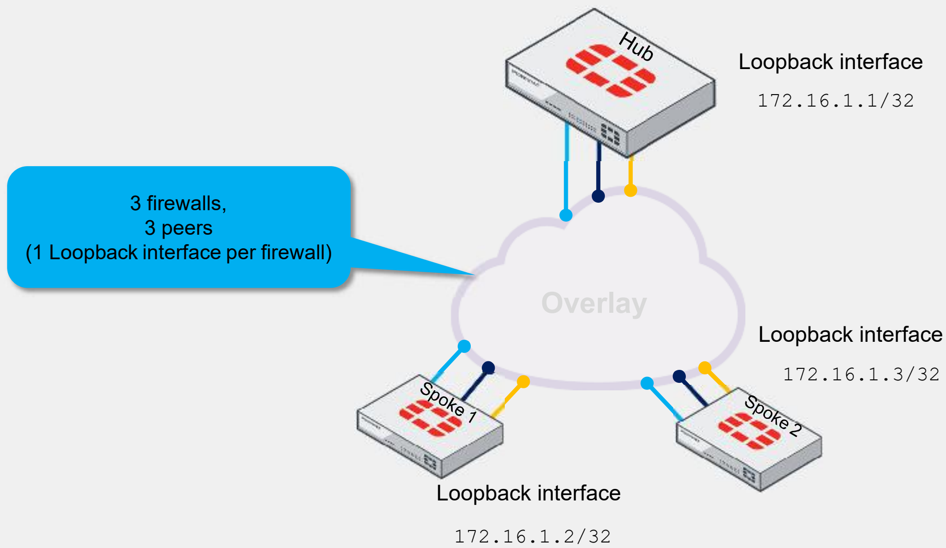


If your FortiGate devices have multiple internet connections, assign an IP address to each IPsec interface, as this slide shows.

The left side shows a hub-and-spoke topology with one internet connection and three IPsec virtual interfaces. On the right is a hub-and-spoke topology with three internet connections per firewall and nine IPsec virtual interfaces.

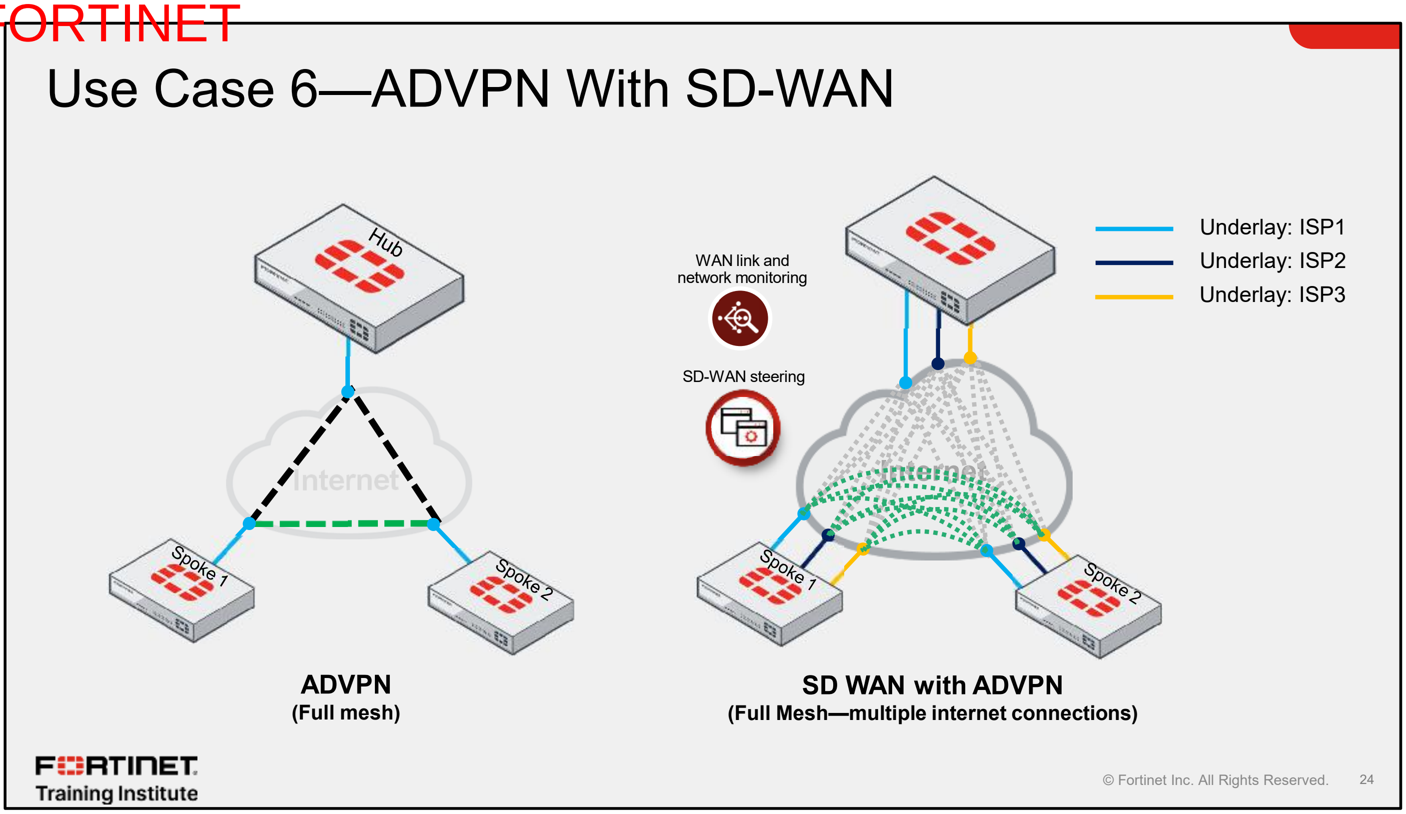
Each internet connection requires a separate phase 1 interface, necessitating an IP address from the overlay network to enable dynamic sharing of the networks behind each firewall

Use Case 5—ADVPN With BGP on Loopback



In this scenario, you can simplify the process by using a loopback approach. This method helps reduce the number of routes and peers at hubs.

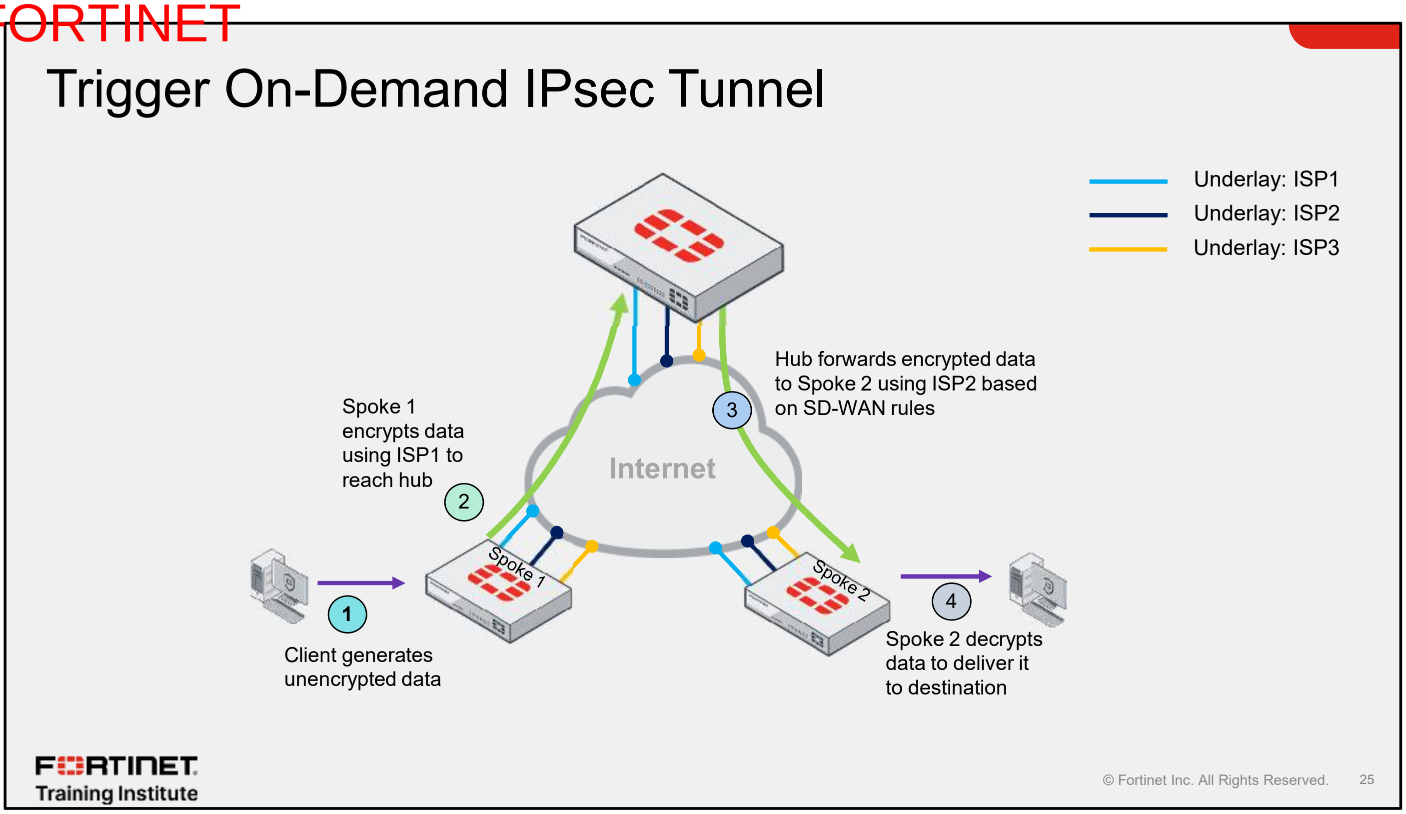
This approach is particularly effective for sites with multiple or varying numbers of internet links and is ideal for large-scale or multiregional deployments because it simplifies the complex task of route reflection in BGP, for example.



You have configured ADVPN in FortiGate and FortiManager. Now, it's important to prepare for SD-WAN environments.

You should consider using ADVPN with SD-WAN, especially when you have:

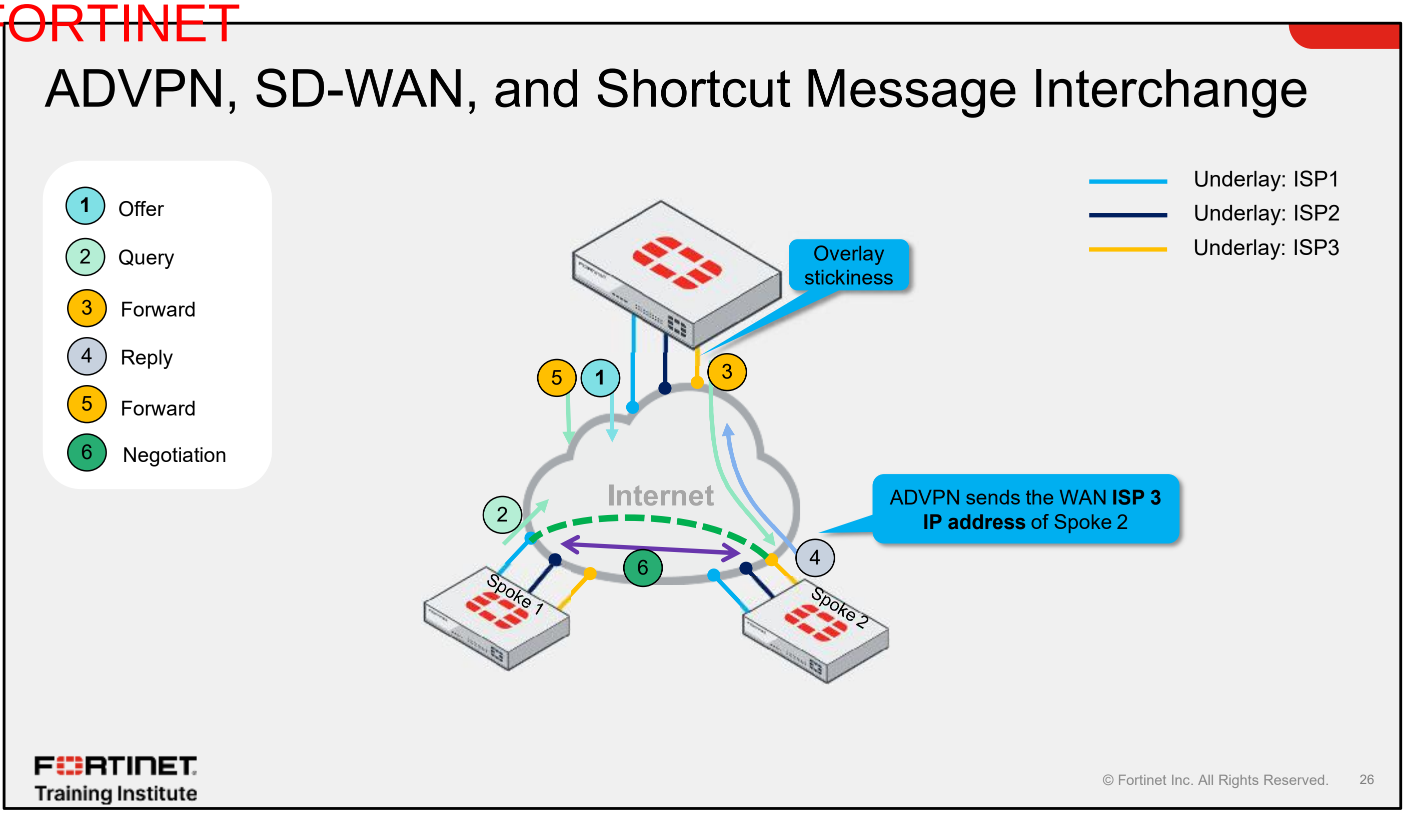
- Multiple internet connections on your firewalls
- A need to monitor interfaces and respond to their statuses, particularly if one goes down



Using the same process as ADVPN, ADVPN with SD-WAN shortcut messages begins when the user generates traffic to trigger the on-demand tunnel that the hub initiates.

1. The user behind Spoke 1 generates traffic for devices located behind Spoke 2.
2. Spoke 1 receives the packet, encrypts it, and sends it to the hub.
3. The hub receives the packet from Spoke 1 and forwards it to Spoke 2 using ISP3 based on the SD-WAN rules.
4. Spoke 2 receives the packet, decrypts it, and forwards it to the destination device.

ADVPN, SD-WAN, and Shortcut Message Interchange



The VPN IPsec shortcuts are determined by the path of the initial data traffic flowing through the hubs, which are usually defined by SD-WAN rules (or policy routes).

However, you need to consider that SD-WAN rules select an ISP for initial data transmission, and this choice persists based on shortcut messaging, despite potentially faster alternatives. ADVPN does not indicate connection quality, so the selection remains unchanged. This is also known as overlay stickiness.

Additionally, in shortcut message exchanges, only the information related to the internet connection receiving the packet is sent on the reply shortcut (number 4 on the slide), without considering details like other internet connections and the status of those links for Spoke 2.

ADVPN 2.0—A New Era of Shortcuts in SD-WAN

- Simplifies connectivity
- Improves resiliency
- Enhances routing
- Edge discovery: spokes share link details, WAN link updates every 5 seconds
- Path management: IKE establishes optimal shortcuts

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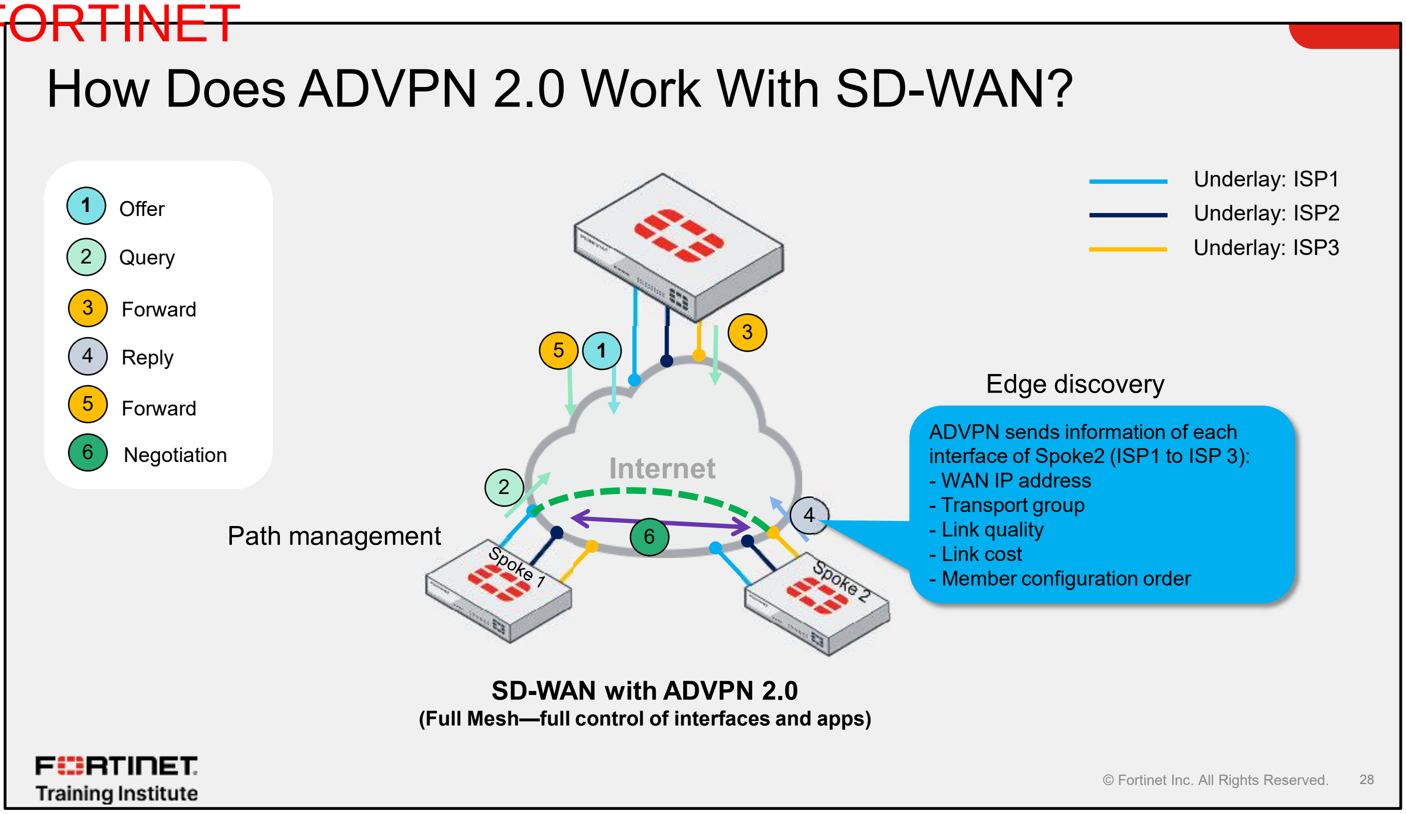
SD-WAN with ADVPN 2.0 establishes shortcut tunnels using underlay availability and link health, and it also prevents reliance on specific BGP routing designs.

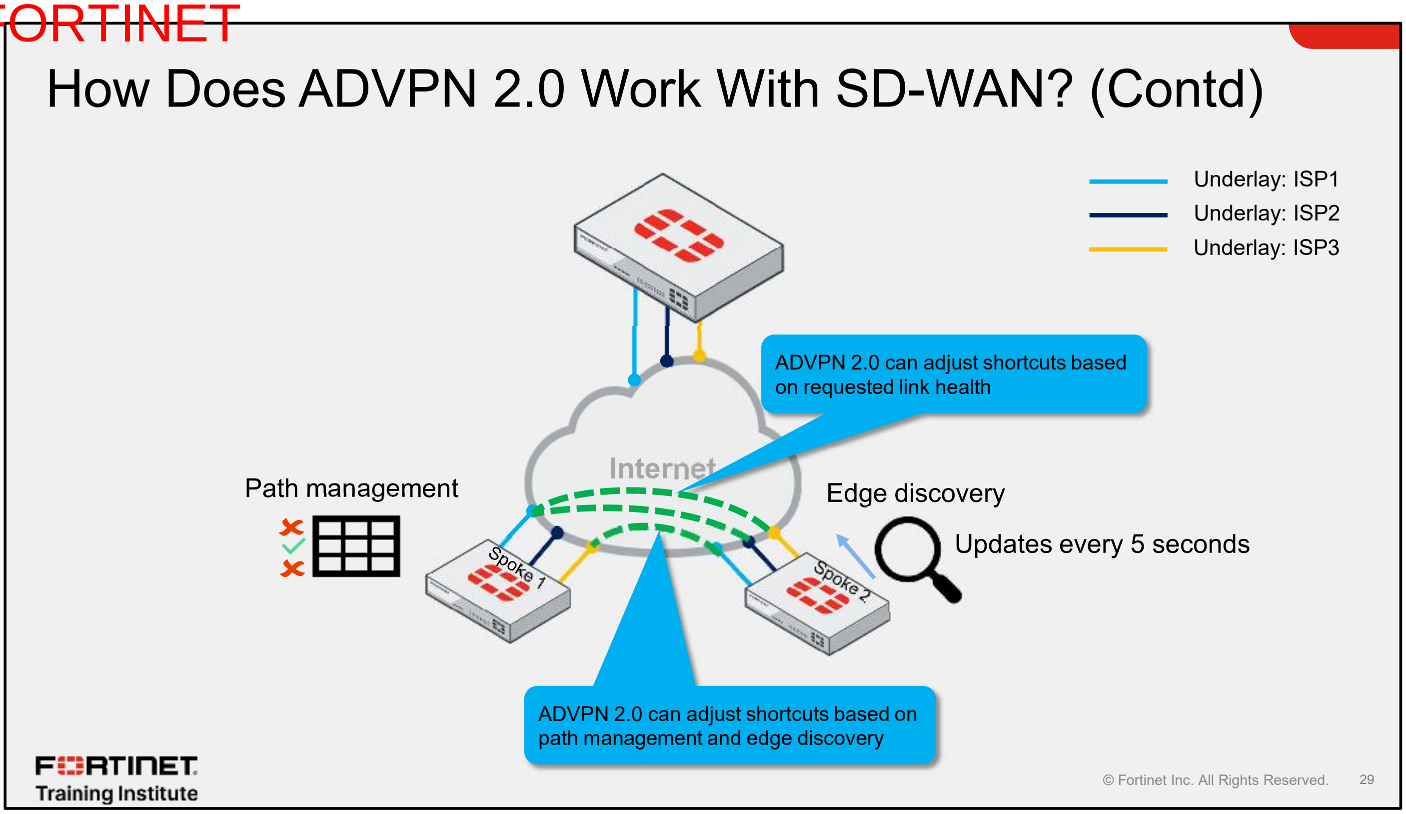
Additionally, ADVPN 2.0 includes major updates like edge discovery and path management for spokes, enhancing flexibility and resiliency:

- Edge discovery: Local spokes obtain remote WAN link info via IKE messages, and they exchange WAN link info every 5 seconds to update the best path.
- Path management: Local spokes calculate and direct IKE to establish the best shortcut path based on SD-WAN rules.



ADVPN 2.0 uses the same process as ADVPN 1.0. Shortcut messages begin when the user generates traffic to trigger the on-demand tunnel that the hub initiates.

ADVPN 2.0 by default includes more information, not only about the internet connection receiving the encrypted packet but also about other internet members of the firewall. This slide shows that after Spoke 2 generates the shortcut, it provides multiple parameters valuable for Spoke 1 to maintain a table and identify the best time to switch.



In ADVPN 2.0, shortcut selection is independent of the initial data traffic path through the hubs, avoiding overlay stickiness and dynamically finding a new path between spokes based on SD-WAN and ADVPN 2.0 parameters.

ADVPN 2.0 can adjust shortcuts not only if an ISP is down, but also if another ISP has a better link status.

There are multiple advantages of using ADVPN 2.0 in a use case, one of them would be spoke-to-spoke connections connecting different branches to use VoIP and selecting the best path to interconnect remote sites.

Review

- ✓ Implement Fortinet ADVPN using FortiGate or FortiManager

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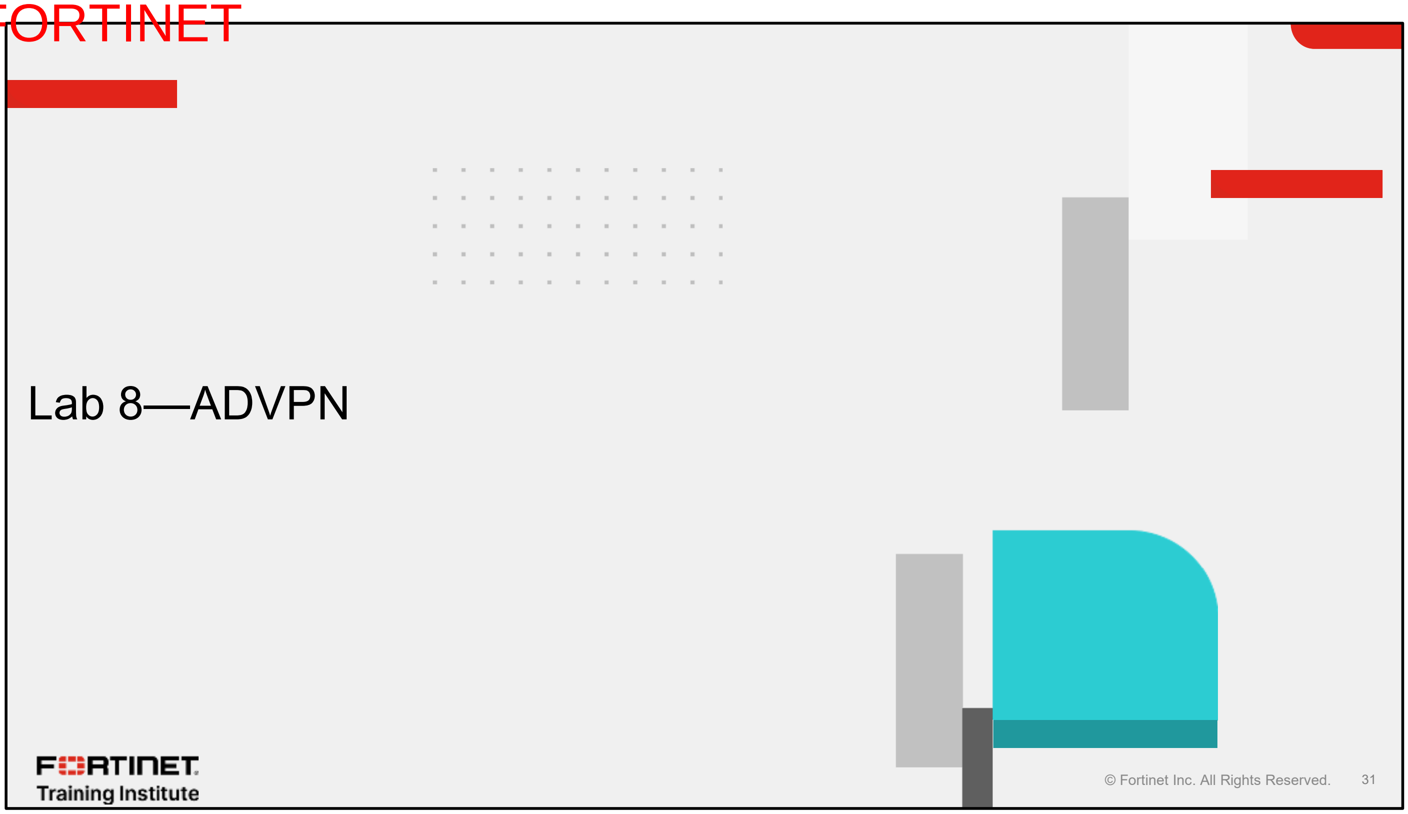
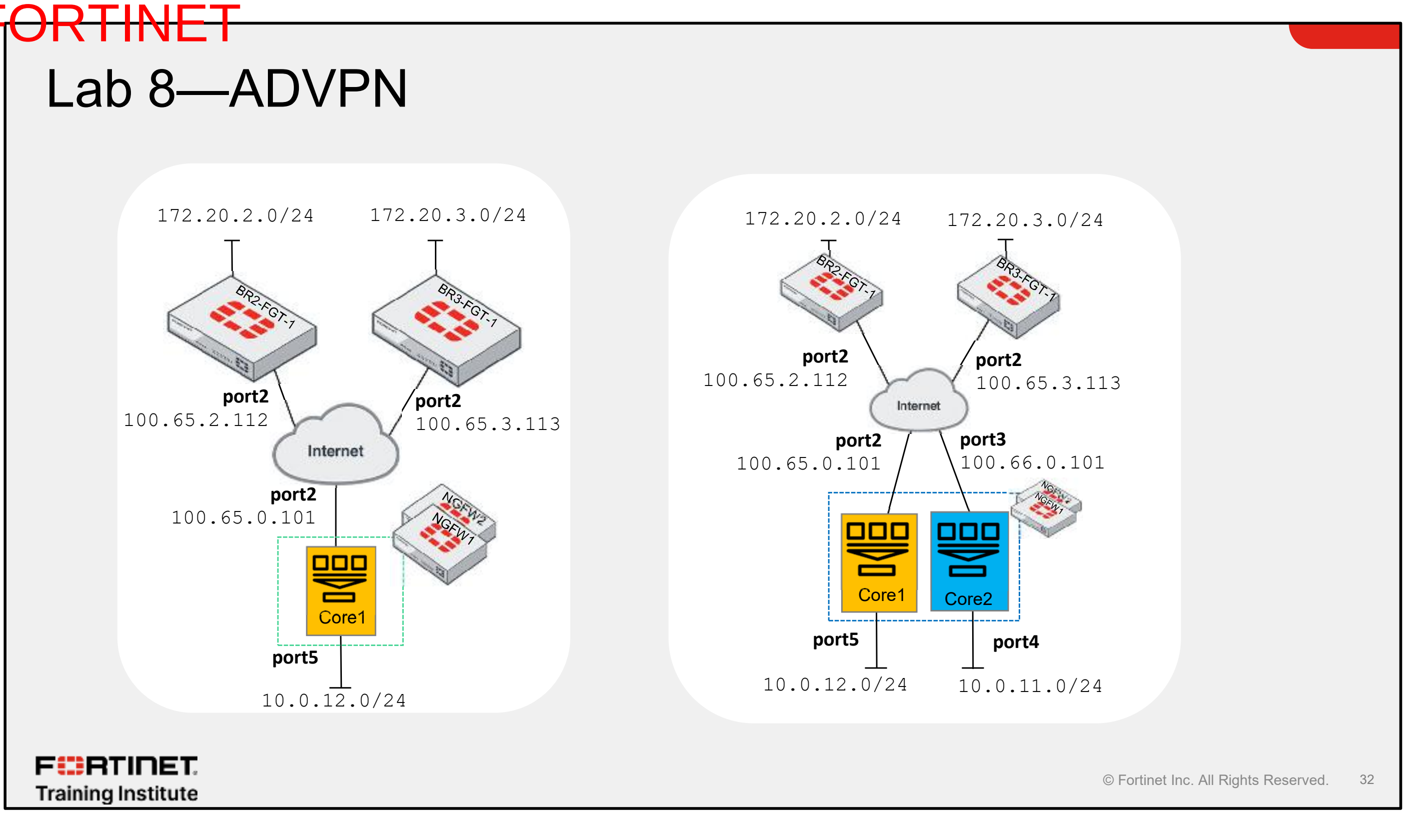
This slide shows the objective that you covered in this lesson.

By mastering the objective covered in this lesson, you learned how to implement Fortinet ADVPN.



Now, you will work on *Lab 8–ADVPN*.



In this lab, you will run ADVPN.

In the first exercise, you will configure a hub-and-spoke topology with only one region. In the second exercise, you will set up a dual hub-and-spoke topology with two different regions.

In this lesson, you will learn about Fortinet Security Fabric use cases for an enterprise environment.

Objectives

- Understand fabric connectors and external connectors
- Identify Security Fabric use cases based on different scenarios
- Apply automation stitches

 2

After completing this lesson, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in the Fortinet Security Fabric, you will be able to understand the differences between fabric connectors and external connectors, implement the Fortinet Security Fabric, and configure automation stitches in use cases.

Understand Fabric Connectors and External Connectors

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 3

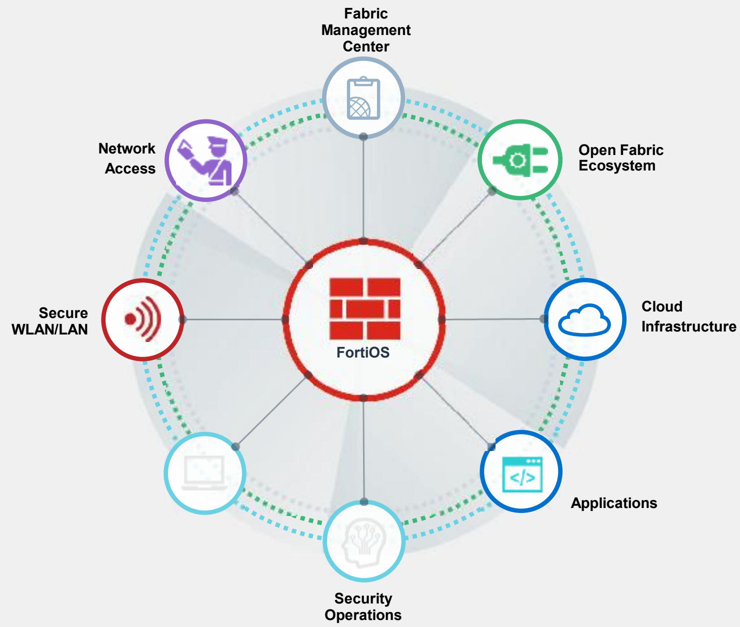
In this section, you will learn about fabric connectors and external connectors.

Security Fabric Overview

BROAD
Detect threats and enforce security everywhere

INTEGRATED
Close security gaps and reduce complexity

AUTOMATED
Enable faster time-to-prevention and efficient operations



The Fortinet Security Fabric is a high-performance cybersecurity mesh platform powered by FortiOS. It comprises a whole ecosystem of security products to secure a network. It brings together the concepts of convergence and consolidation to provide comprehensive, real-time cybersecurity protection from users to applications.

The Security Fabric is built on three key concepts: broad, integrated, and automated.

- Broad, to detect threats and enforce security everywhere. The Fortinet portfolio includes converged networking and security offerings across endpoints, networks, and clouds. It enables high-performing connectivity and coordinated real-time threat detection and policy enforcement across the entire digital attack surface and lifecycle.
- Integrated, to close the security gaps and reduce complexity. Integrated device operation within the Security Fabric allows simplified operation to deliver cohesive and consistent security across different technologies, locations, and deployments.
- Automated, to enable faster time-to-prevention and efficient operations. The Security Fabric allows a context-aware, self-healing network. The security posture leverages the cloud-scale and advanced AI to automatically deliver near-real-time, user-to-application coordinated protection. Process automation simplifies operations for large-scale deployments and frees up IT teams from repetitive and time-consuming tasks.

The Security Fabric is open and allows the communication between Fortinet and third-party devices through connectors.

Fabric Connectors and External Connectors

	Fabric connectors	External connectors
Root FortiGate GUI location	**Security Fabric** > **Fabric Connectors**	**Security Fabric** > **External Connectors**
Extensions	• **Core network security connectors** (FortiGate, FortiAnalyzer, and FortiClient Enterprise Management Server (EMS)) • **Security fabric connectors** (Central Management, Sandbox, and additional supported connectors)	• **Public SDN** (like AWS and GCP) • **Private SDN** (like Kubernetes, VMware, and SAP) • **Endpoint/identity** (like FSSO and RADIUS) • **Threat feeds** (like FortiGuard category and Malware Hash)
Interactions	• EMS connectors ensure automatic actions (like quarantine) • Additional supported connectors allow interaction with FortiDeceptor, FortiMail, FortiMonitor, FortiNAC, FortiNDR, FortiPolicy, FortiTester, FortiVoice, and FortiWeb	• SDN connectors ensure automatic update about cloud environment attribute changes (like address objects) • Single sign-on (SSO) connectors allow users to enter their credentials only once • Threat feed connectors allow FortiGate to import external dynamic lists

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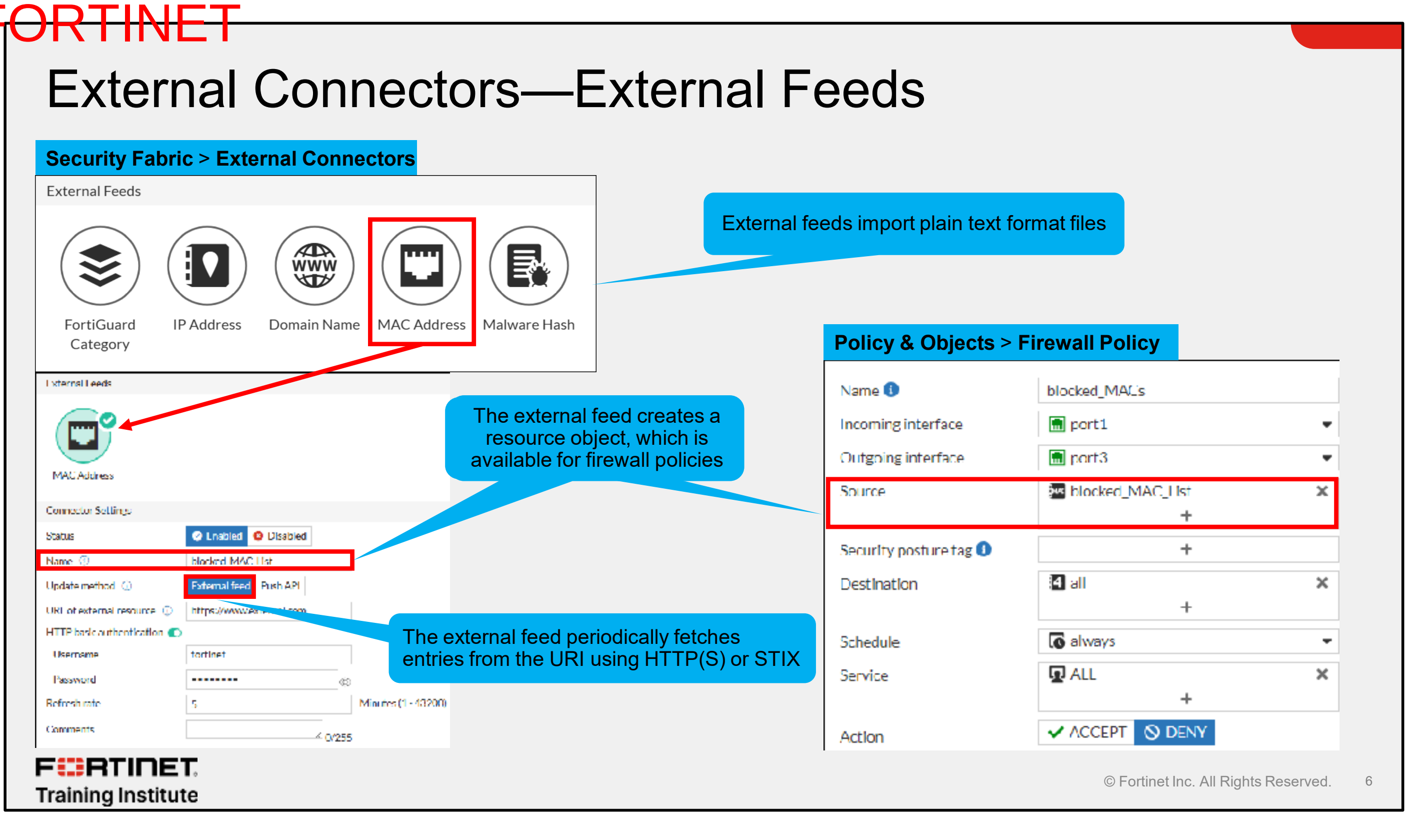
This slide shows the main differences between fabric connectors and external connectors, that you can configure in the **Security Fabric** section of the root FortiGate.

While fabric connectors are focused on Fortinet technologies, external connectors allow you to integrate Fortinet and other technologies into a multivendor ecosystem.

Using APIs, a variety of scripts, or specific code, the Security Fabric provides automatic updates in dynamic environments, such as public and private clouds.



External feeds allow you to enforce special security requirements, defined in an external list, on FortiGuard categories, IP addresses, domain names, MAC addresses, and malware hashes. External feeds are dynamically synchronized and updated periodically so that FortiOS immediately imports any changes. There are two methods you can use to update the external feeds:

- **External feed**, where you must configure the URL of the external resource using HTTP, HTTPS, or STIX.
- **Push API**, where the external feed receives entry updates from webhook requests to the FortiGate REST API.

When you create an external feed, a resource object corresponding to the dynamic list becomes available. You can then configure a firewall policy with this new resource object.

In this section, you will learn about multiple Fortinet Security Fabric scenarios in enterprise network environments.

Use Case 1—Security Fabric Logging on FortiAnalyzer

- FortiAnalyzer Fabric connector consolidates the traffic logs
- The Security Fabric, as a whole, logs each session once
  - The first FortiGate that handles a session in the Security Fabric logs the session
  - Any upstream FortiGate that is a member of the Security Fabric does not create duplicate traffic logs for sessions coming from another member's MAC address with the following exceptions:
    - If an upstream FortiGate performs NAT, FortiGate generates another log on that device
    - Upstream FortiGate devices still log UTM events, if configured
- FortiAnalyzer performs UTM and traffic log correlation, so that session details, UTM events, reporting, and automation in the Security Fabric work correctly

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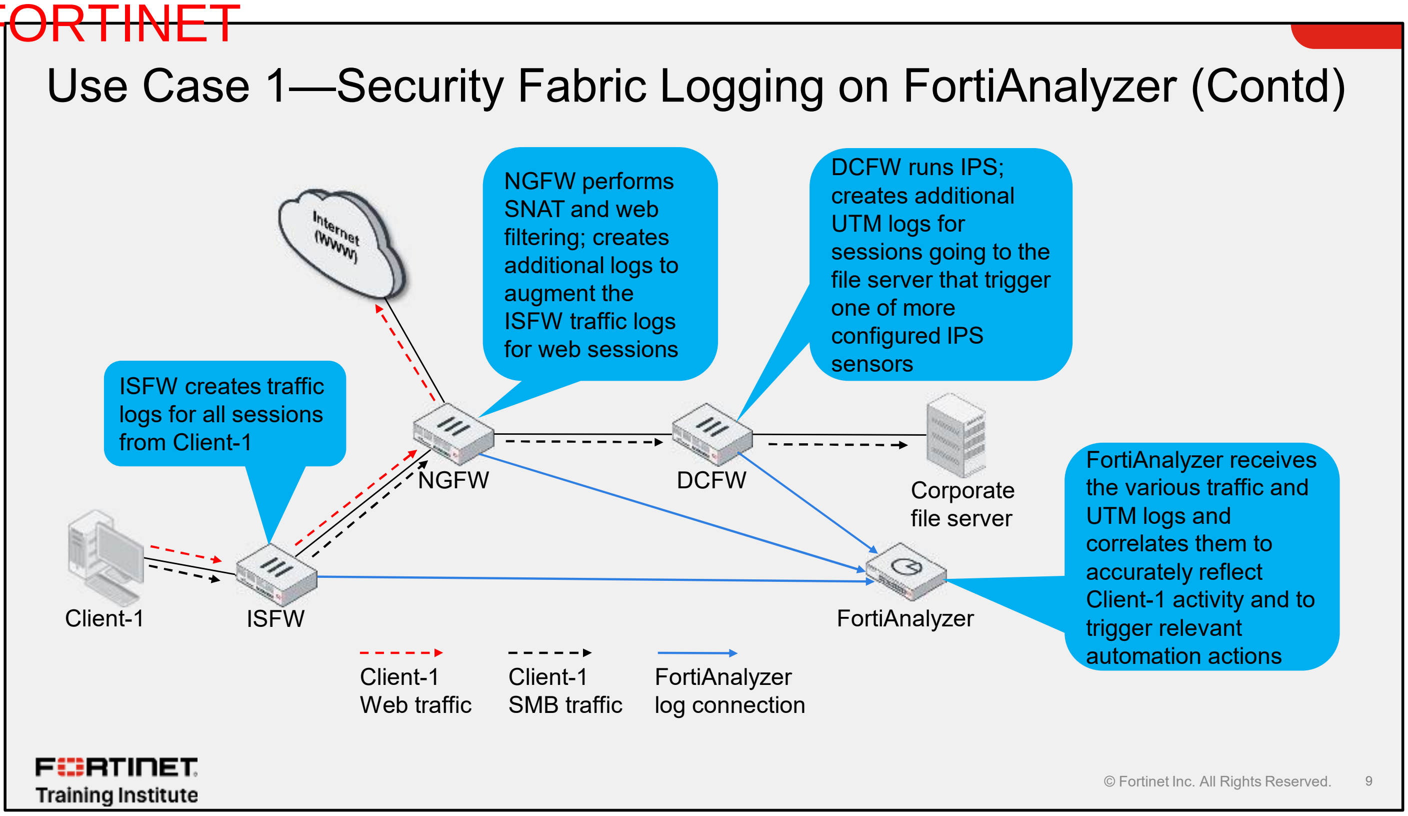
The FortiAnalyzer Fabric connector allows you to consolidate the traffic logs. A session's traffic is always logged by the first FortiGate that handled it in the Security Fabric. FortiGate devices in the Security Fabric know the MAC addresses of their upstream and downstream peers. If FortiGate receives a packet from a MAC address that belongs to another FortiGate in the Security Fabric, it does not generate a new traffic log for that session. This helps to eliminate the repeated logging of a session by multiple FortiGate devices. One exception to the behavior is that if an upstream FortiGate performs network address translation (NAT), then another log is generated. The additional log is needed to record NAT details such as translated ports and addresses.

Upstream devices complete unified threat management (UTM) logging, if configured, and FortiAnalyzer performs UTM and traffic log correlation for the Security Fabric, in order to provide a concise and accurate record of any UTM events that may occur. No additional configuration is required for this to take place because FortiAnalyzer performs this function automatically.

Note that each FortiGate in the Security Fabric logs traffic to FortiAnalyzer independent of the root or other leaf devices. If the root FortiGate is down, logging from leaf FortiGate devices to FortiAnalyzer continues to function.



This slide shows how logging functions in the Security Fabric with FortiAnalyzer. This functionality provides full visibility and eliminates duplicate logs throughout the environment. The topology shows three FortiGate devices and one FortiAnalyzer configured in a Security Fabric.

- An internal segmentation firewall (ISFW) in the access layer provides device detection, breach isolation, and basic DoS protection from the attached end-user LANs.
- A next-generation firewall (NGFW) installed between the corporate network and the internet service provider (ISP) performs SNAT on outbound communications for RFC-1918 hosts, as well as web filtering for HTTP/HTTPS sessions.
- A data-center firewall (DCFW) installed in the data center runs the intrusion prevention system (IPS) for all inbound communications to the servers behind it.

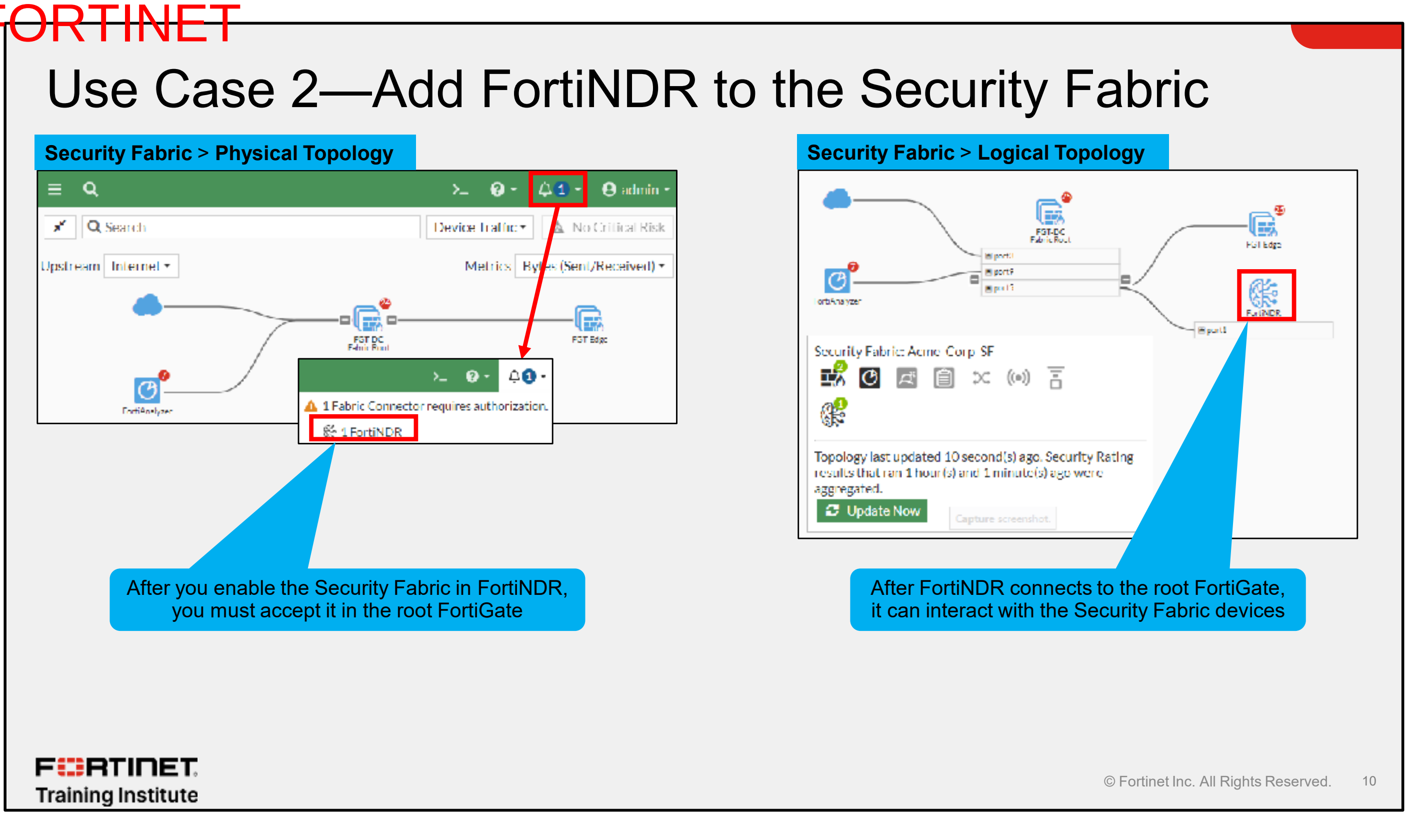
The ISFW receives all traffic from Client-1 and creates traffic logs for the initial session.

The ISFW forwards the web session to the NGFW, which doesn't duplicate the initial traffic log but does generate a traffic log because it applies SNAT to the session.

Additionally, the NGFW applies a web filtering policy to this session and generates the relevant UTM logs, if appropriate.

The ISFW forwards the SMB session to the NGFW, which doesn't duplicate the initial traffic log. The NGFW doesn't need to perform NAT or apply web filtering, so it forwards the traffic to the DCFW. The DCFW also doesn't generate a duplicate traffic log, but it performs IPS inspection based on its configuration and if a signature match results in an action generating a log, logs the event.

FortiAnalyzer receives the various traffic and UTM logs and automatically correlates them so that they are linked for proper viewing, reporting, and potential automation actions.

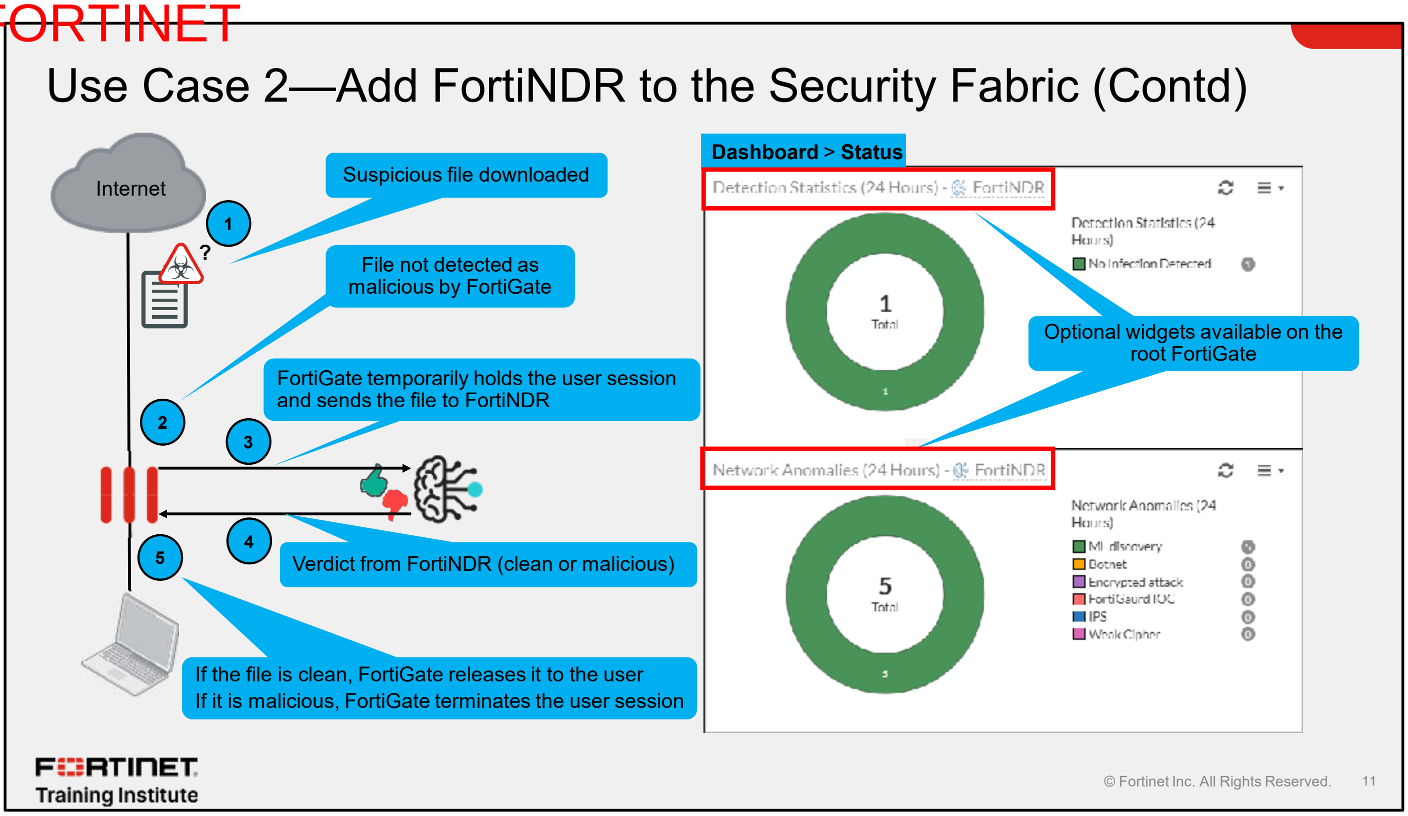


After you enable the Security Fabric in a device, you must accept the fabric connector configuration on the root FortiGate. This slide shows an example of the FortiNDR device when it submits a connection request to the Security Fabric root device. On the GUI of the Security Fabric root device, the request appears as a number next to the alarm icon on the top menu bar.

The Security Fabric administrator can review and accept the connection request. Once accepted, FortiNDR joins the Security Fabric. This is visible in the **Logical Topology** and **Physical Topology** windows of the root FortiGate.

In the **Logical Topology** example shown on this slide, you can see that FortiNDR connects using its port1 interface to the **Fabric Root** on port5.

Any FortiGate devices from the Security Fabric can then send files for analysis to FortiNDR. For downstream devices, the files transit through the Security Fabric root FortiGate device.

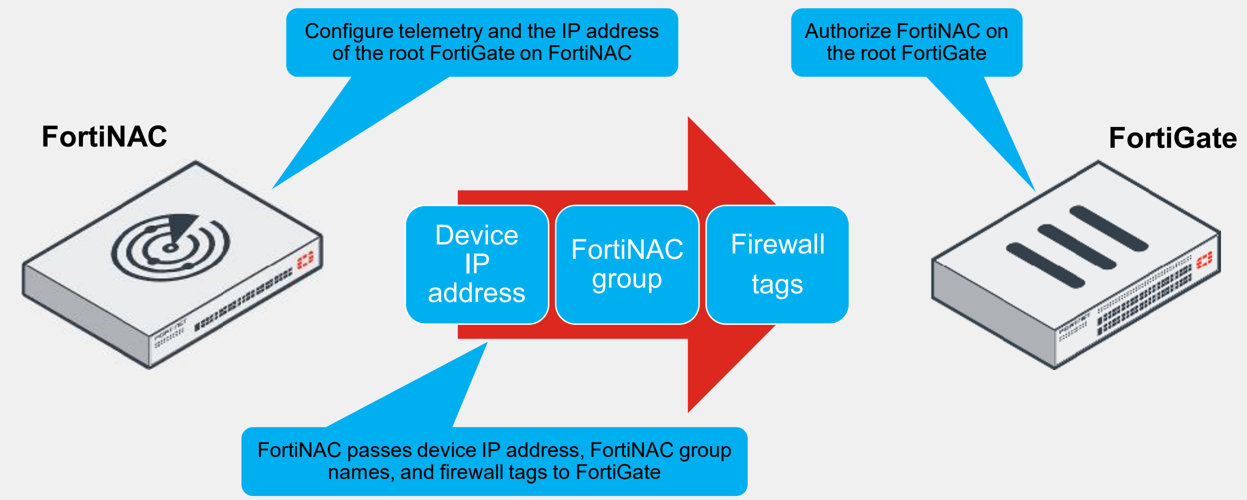


When integrated in a Security Fabric group, FortiNDR can perform on-the-fly analysis of files that FortiGate devices submit. This is called inline blocking mode. On reception of a file, FortiGate performs the usual antivirus analysis. If the file matches a malware signature present in the antivirus database, FortiGate blocks the flow and doesn't involve FortiNDR. The user doesn't receive the file.

If FortiGate does not detect the file as malicious, it places the user session on hold while it forwards the file to FortiNDR. Within seconds, FortiNDR performs the next level of analysis, leveraging machine learning (ML) and artificial neural network (ANN) capabilities, and then returns its verdict to FortiGate. Then, FortiGate acts accordingly to block malicious content or deliver clean files.

On the FortiGate Security Fabric root device, along with FortiNDR system information, you can view the statistics information shared by FortiNDR with the following optional widgets: **System Information**, **Detection Statistics (24 Hours)**, and **Network Anomalies (24 Hours)**.

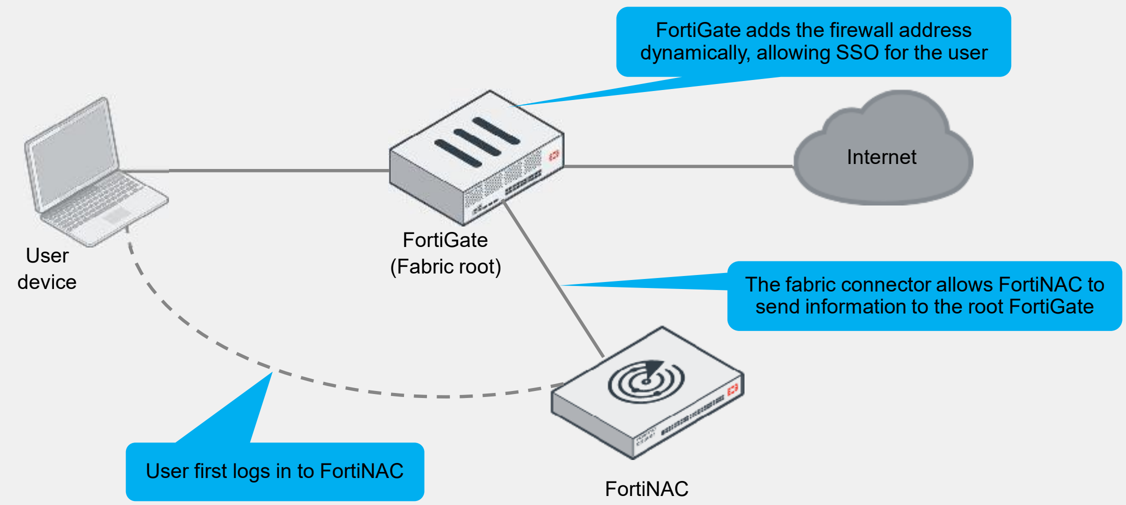
Use Case 3—FortiNAC and Dynamic Firewall Addressing



FortiNAC communicates with infrastructure devices, such as wireless controllers, autonomous APs, switches, routers, and others. Because these infrastructure devices are inline, they can detect connected devices and connecting endpoints. They send this information back to FortiNAC, or FortiNAC gathers this information from them.

You can add a FortiNAC device to the Security Fabric environment on the root FortiGate. The FortiNAC tag dynamic firewall address includes the device IP address, FortiNAC firewall tags, and FortiNAC group information sent to the root FortiGate using the REST API when FortiNAC registers user login and logout events.

Use Case 3—FortiNAC and Dynamic Firewall Addressing (Contd)



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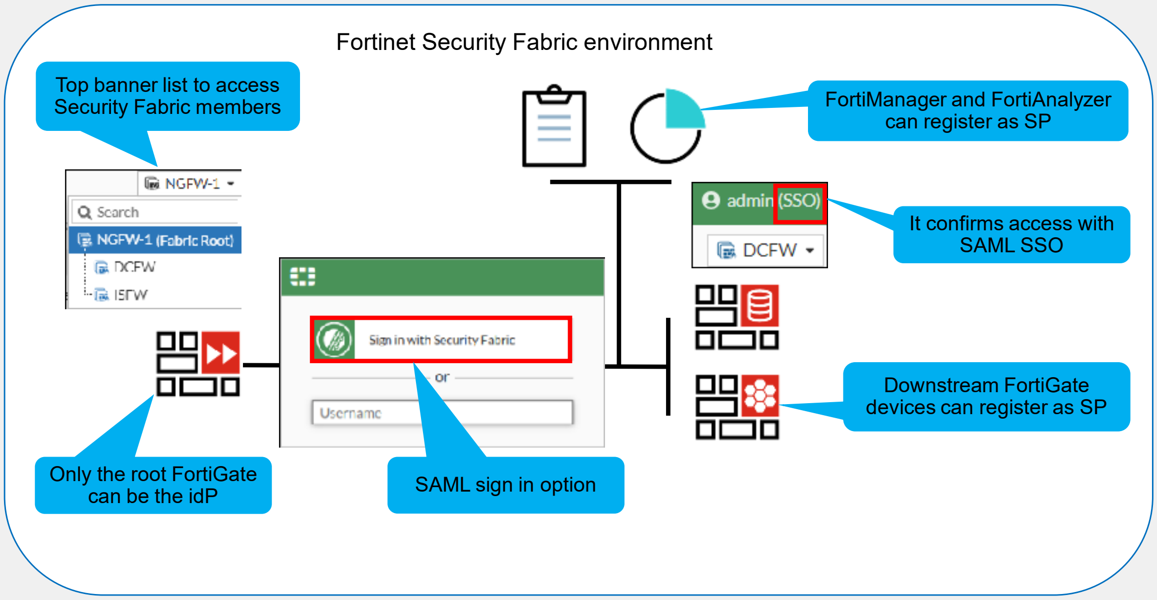
In the example shown on this slide, the user connecting to the network must first log in to FortiNAC. When the login succeeds, the REST API synchronizes the login information with FortiGate. FortiGate updates the dynamic firewall address object with the user and IP address information of the user device. This firewall address is used in firewall policies to dynamically allow network access to authenticated users, thereby allowing SSO for the end user.

FortiNAC requires two user login events to trigger the use of the tag dynamic firewall addresses. You can view the newly created dynamic firewall address in FortiOS, in the **FortiNAC Tag (IP Address)** section in **Policy & Objects** > **Addresses**. The dynamic firewall addresses that match the current user login status on FortiNAC have the current IP address of the user devices.

The firewall policies can use the FortiNAC tag dynamic firewall address as the source or destination addresses.

Use Case 4—Security Fabric with SAML SSO



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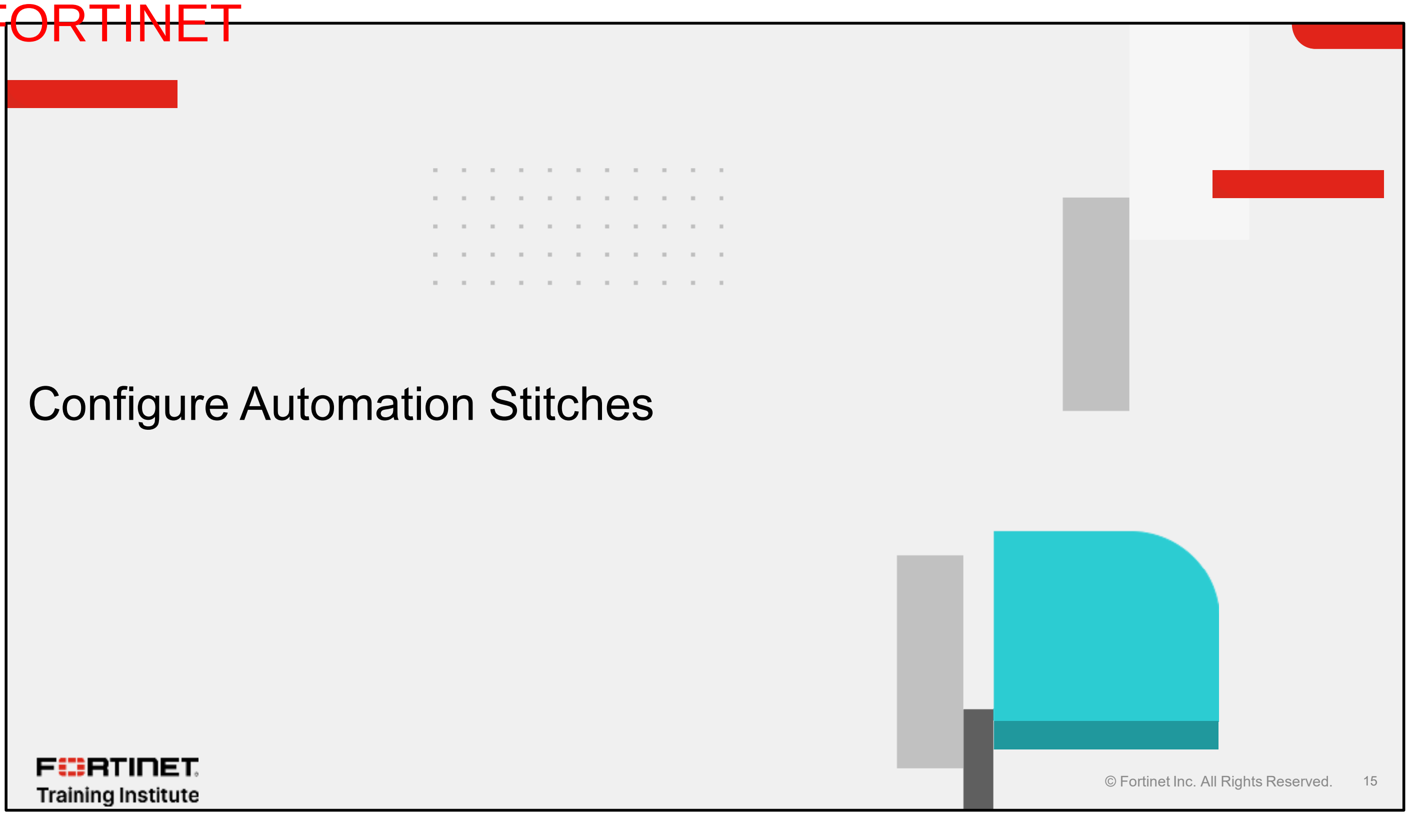
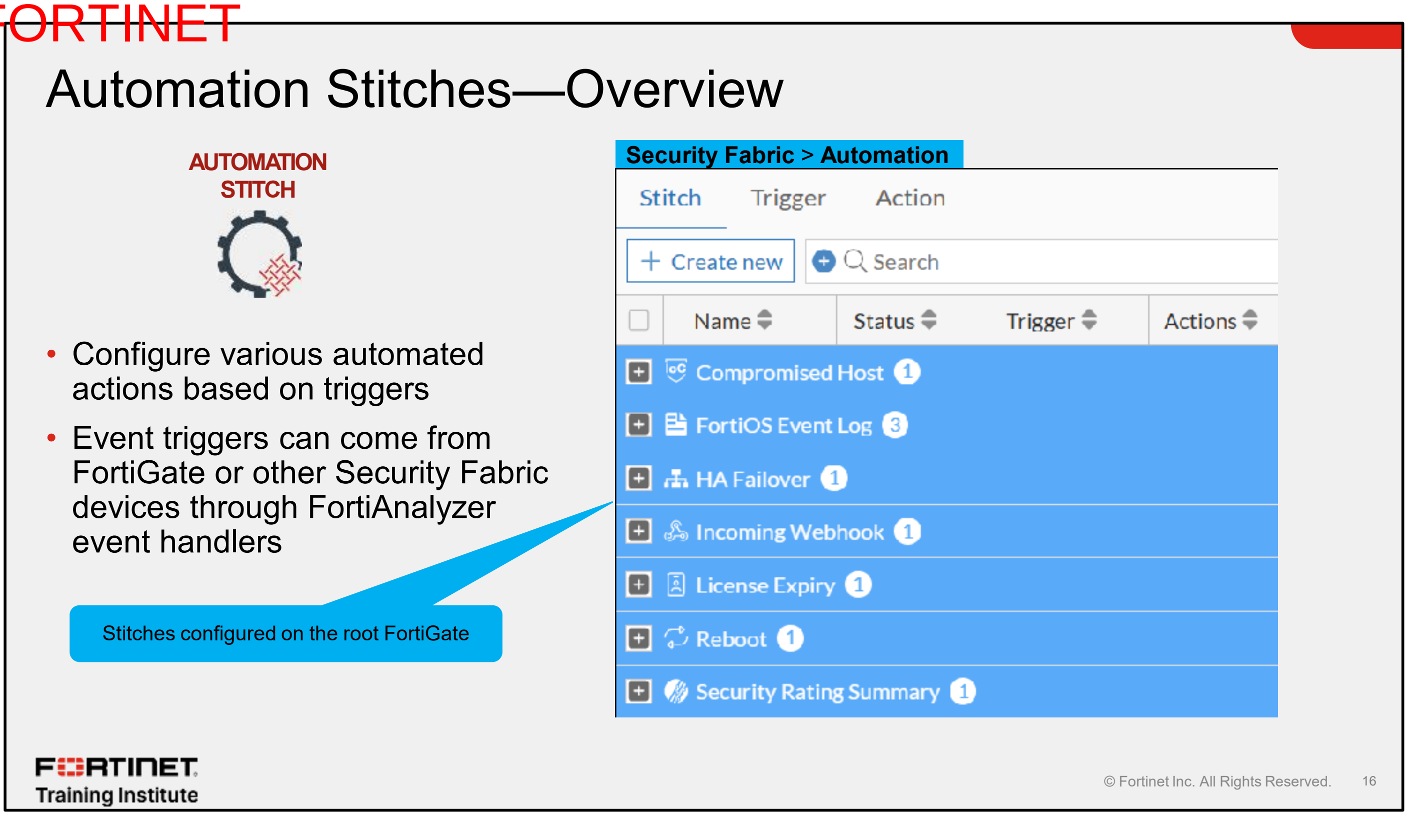
In a Security Fabric environment, you can enable SAML SSO authentication. It allows you to access the downstream FortiGate devices, the FortiAnalyzer, and the FortiManager without needing to provide credentials again.

The root FortiGate acts as the identity provider (IdP) and you configure the other devices as service providers (SP). Once authorized, an administrator can then move between Fabric devices without logging in again, as long as it is within the same browser session.

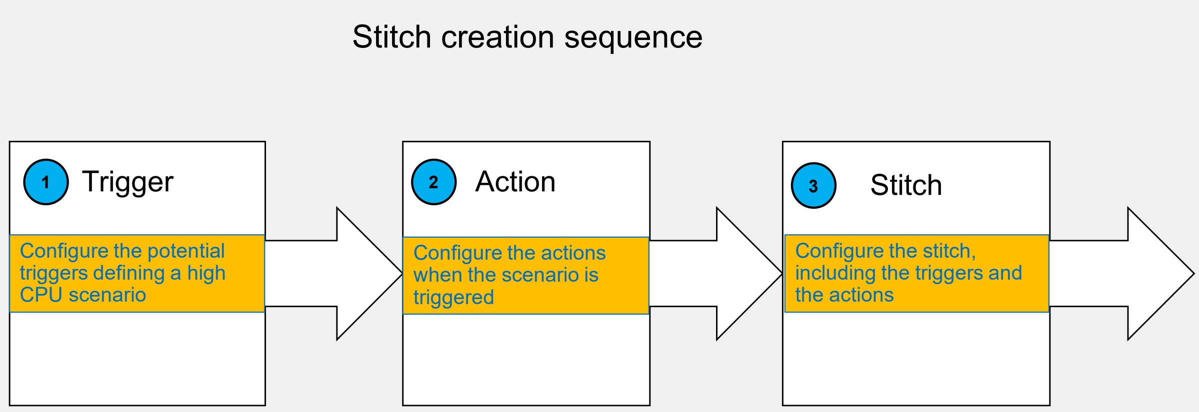


In this section, you will learn about automation stitches.

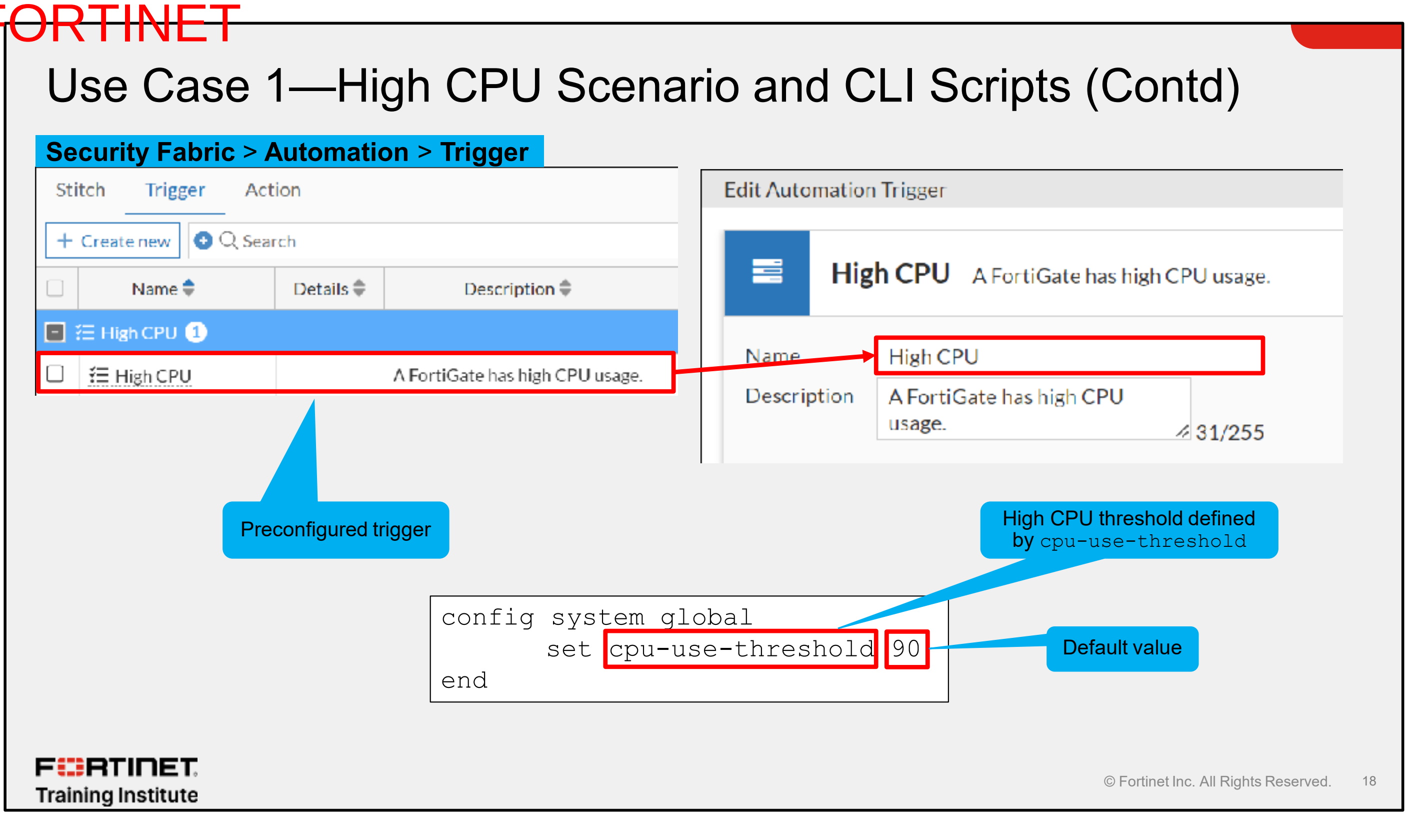


Administrator-defined automated workflows (called stitches) use if/then logic to cause FortiOS to automatically respond to an event in a preprogrammed way. Because this workflow is part of the Security Fabric, you can set up stitches for any device in the Security Fabric. However, the Security Fabric is not a requirement to use stitches. If you configure stitches in the Security Fabric, you must configure them on the root FortiGate. You configure stitches to run on all FortiGate devices in the Security Fabric or a subset of FortiGate devices. Stitches configured on the root FortiGate are pushed to the relevant leaf FortiGate devices. The root FortiGate does not need to be operational for previously configured stitches to function on leaf FortiGate devices.

Use Case 1—High CPU Scenario and CLI Scripts



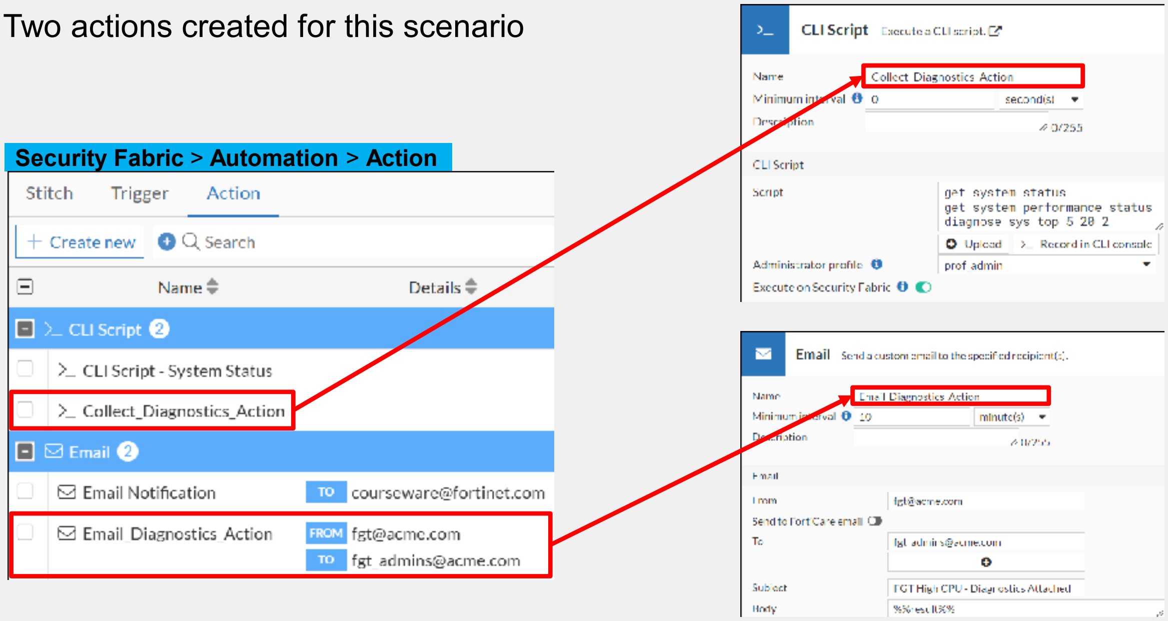
Each automation stitch pairs an event trigger and one or more actions, which allows you to monitor your network and take appropriate action when the Security Fabric detects a threat or other actionable event. This slide shows the three steps to create a stitch. The following slides show each step for this use case.



For a high CPU scenario, the recommended preconfigured trigger is **High CPU**. It identifies when FortiGate exceeds the CPU use threshold configured with the `cpu-use-threshold` parameter. By default, the threshold is 90%.

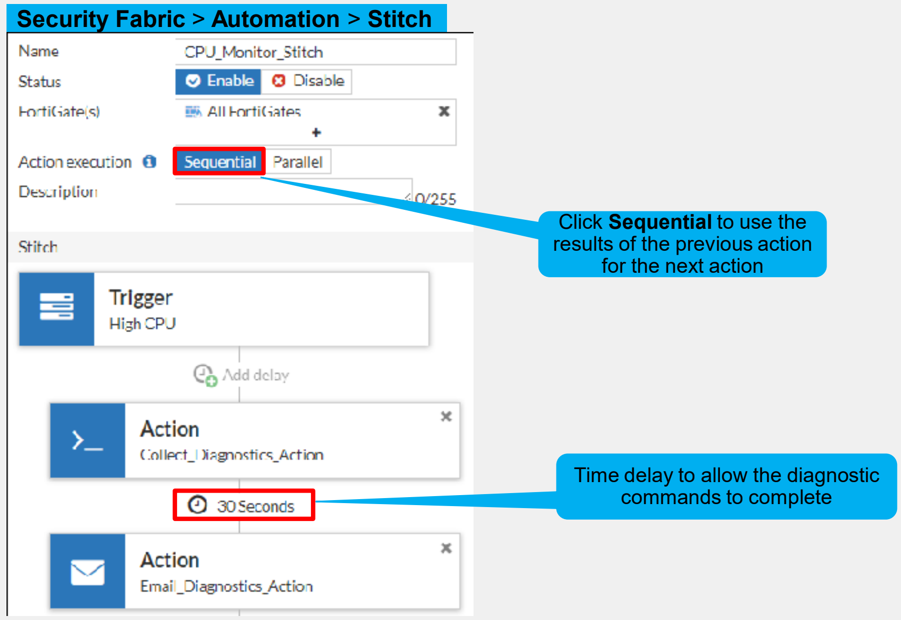
Use Case 1—High CPU Scenario and CLI Scripts (Contd)

- Two actions created for this scenario

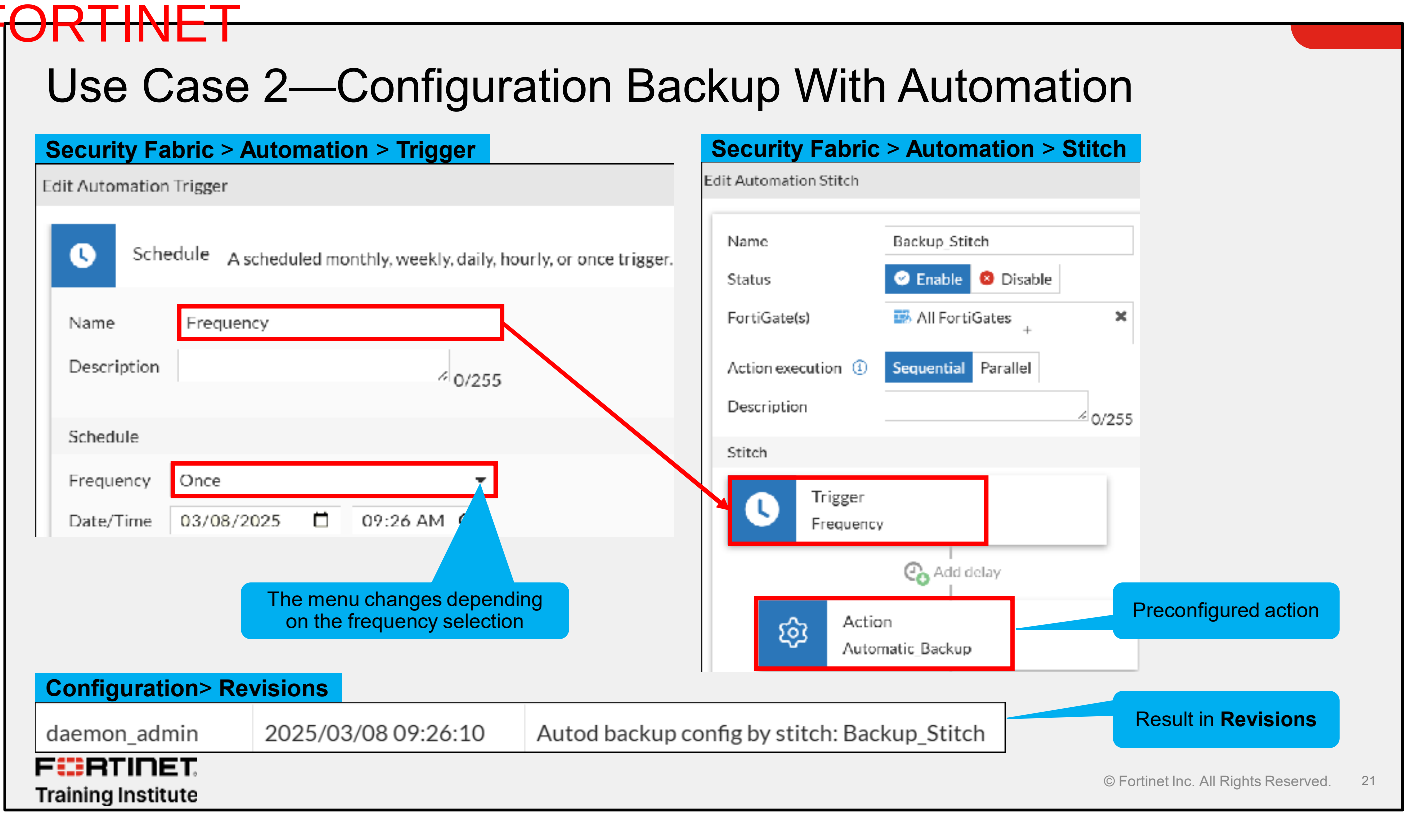


The next step is to configure the actions. The scenario on this slide shows two custom automation actions. **Collect_Diagnostics_Action** tells FortiOS to run a custom CLI script consisting of various diagnostic commands that help to identify the cause of performance issues on FortiGate. **Email_Diagnostics_Action** sends an email message to staff to notify them that a FortiGate device has experienced a period of high CPU use, and provides the output of **Collect_Diagnostics_Action** in the email body so they have the relevant information for troubleshooting.

Use Case 1—High CPU Scenario and CLI Scripts (Contd)



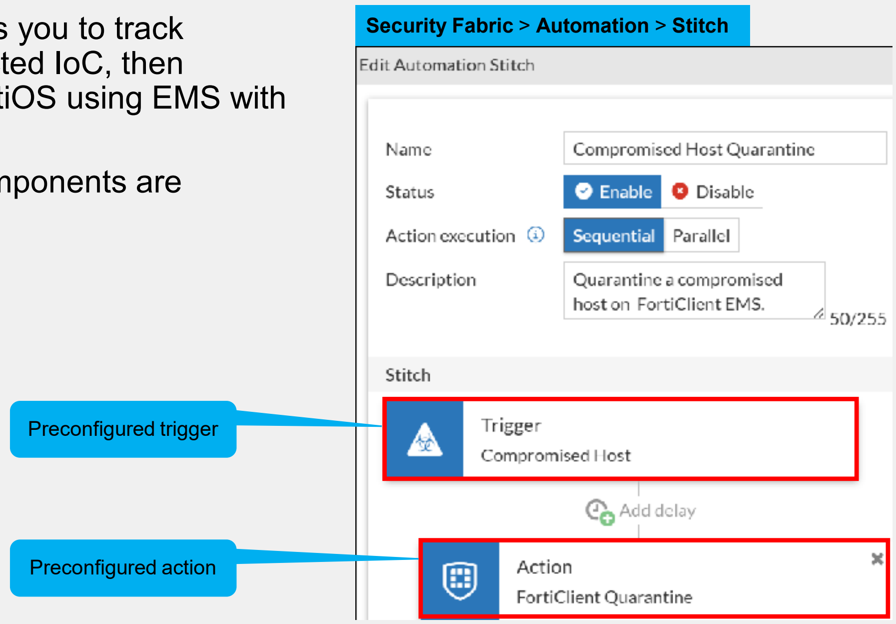
In the last step, you configure the automation stitch, which is going to execute actions sequentially. When the **High CPU** trigger occurs, the **Collect_Diagnostics_Action** runs, followed by a 30-second delay before proceeding to the **Email_Diagnostics_Action**. The **Collect_Diagnostics_Action** runs a series of CLI diagnostic commands so the 30-second delay allows time for the commands to finish. **Email_Diagnostics_Action** uses the output of these commands as a parameter, `%%results%%`, which forms the body of the email message to be sent.



An external event can also trigger a stitch. For configuration backup at a defined time, you configure the **Schedule** trigger. The frequency can be once, hourly, daily, weekly, or monthly according to the time system used by the root FortiGate. The action can be applied to all the FortiGate devices in the Security Fabric, or to specific FortiGate devices defined in the automation stitch. The triggering of this stitch can result in a one-time configuration backup or a regular configuration backup. You can use a configuration backup to roll back the corresponding FortiGate device, if needed.

Use Case 3—IoC Detection and Automated Quarantine

- The Security Fabric allows you to track endpoints that have detected IoC, then quarantine them from FortiOS using EMS with an API
- The following network components are required:
  - FortiGate
  - FortiAnalyzer
  - FortiClient EMS
  - FortiClient



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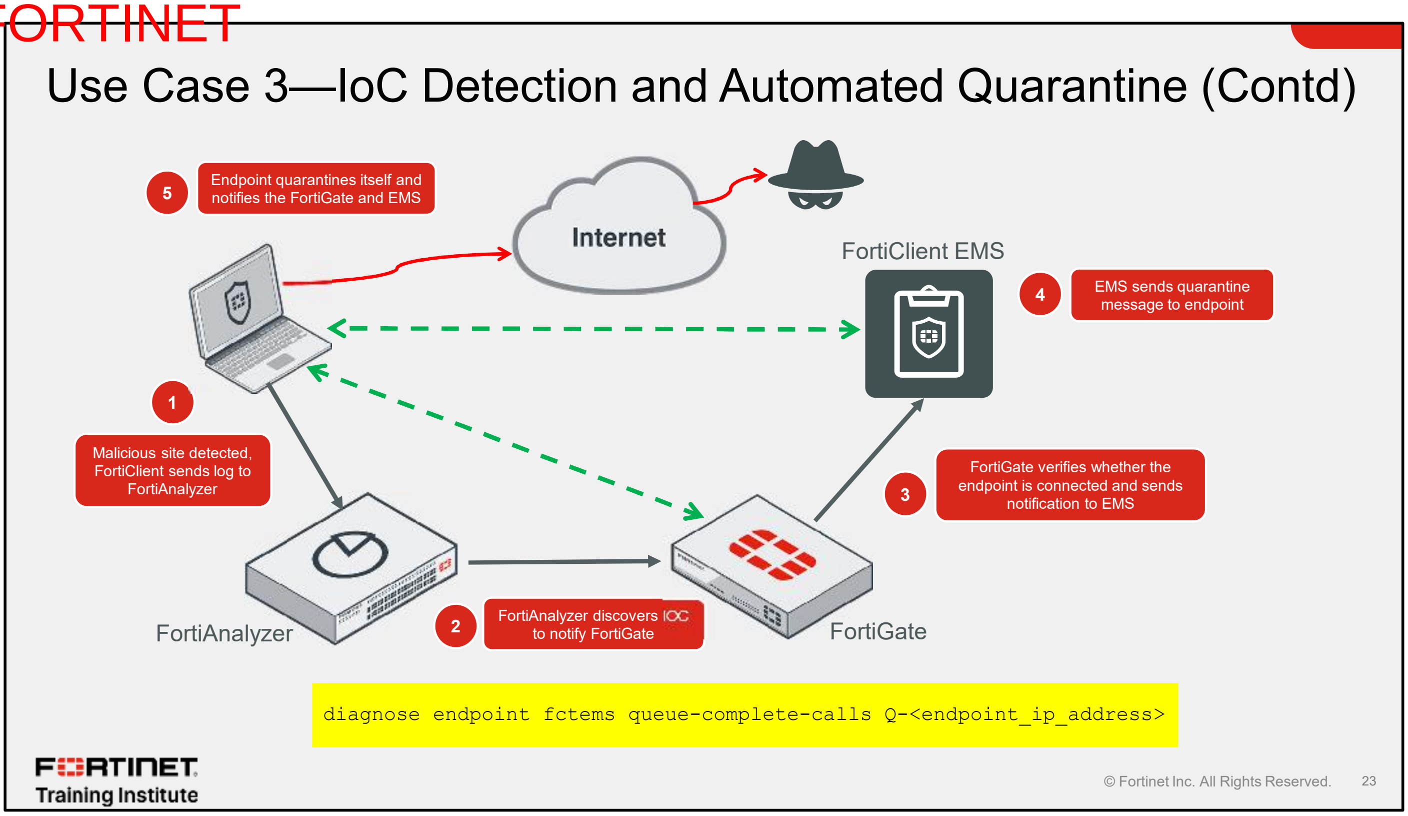
You can expand the automation using a combination of Fabric connectors and stitches. With the Security Fabric network devices listed here, you can configure the system to automatically quarantine an endpoint on which an indicator of compromise (IoC) is detected. This requires the following network devices:

- FortiGate
- FortiAnalyzer
- FortiClient EMS
- FortiClient

You must connect FortiClient to both the Endpoint Management Server (EMS) and FortiGate. FortiGate and FortiClient must both be sending logs to FortiAnalyzer. You must configure the EMS IP address on FortiGate, as well as administrator login credentials.



This slide shows the IoC detection and automated quarantine flow:

1. FortiClient sends logs to FortiAnalyzer.
2. FortiAnalyzer discovers IoCs in the logs and notifies FortiGate.
3. FortiGate identifies whether FortiClient is a connected endpoint, and whether it has the login credentials for the FortiClient EMS device that FortiClient is connected to. With this information, FortiGate sends a notification to FortiClient EMS instructing it to quarantine the endpoint.
4. FortiClient EMS searches for the endpoint and sends a quarantine message to it.
5. The endpoint receives the quarantine message and quarantines itself, blocking all network traffic. The endpoint notifies FortiGate and EMS of the status change.

The CLI command on the slide triggers the quarantine action on the endpoint at `endpoint_ip_address`.

This feature is not supported on FortiClient (Linux).

Review

- ✓ Understand the fabric and external connectors
- ✓ Identify Security Fabric use cases based on different scenarios
- ✓ Apply automation stitches

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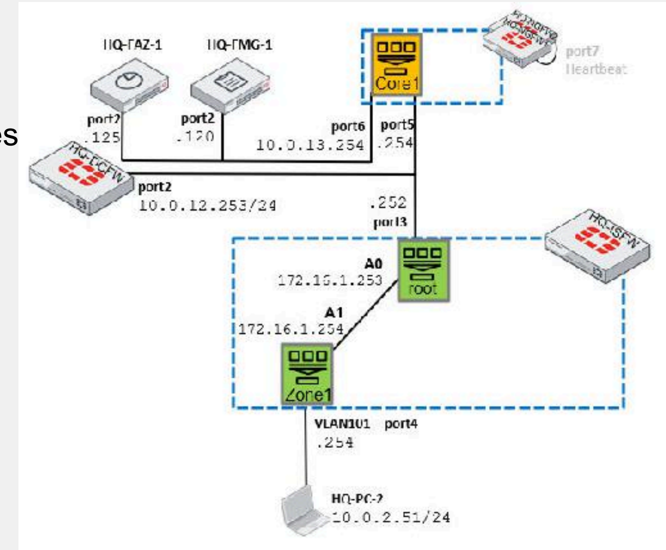
This slide shows the objectives that you covered in this lesson.

By mastering the objectives covered in this lesson, you learned about fabric connectors and external connectors, and use cases for automation stitches and the implementation of the Fortinet Security Fabric in enterprise environments.

Lab 9—Security Fabric

- The Security Fabric uses a tree topology
- HQ-NGFW is the root of the tree HQ-DCFW and HQ-ISFW are branches
- On HQ-NGFW, HQ-DCFW, and HQ-ISFW you will:
  - Enable SAML SSO
  - Configure automation for configuration backups
  - Run CLI scripts for configuration changes and email the results



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In this lab, you will configure the Security Fabric with SAML SSO on HQ-NGFW, HQ-ISFW, and HQ-DCFW. You will then configure automation on the root FortiGate.

In this lesson, you will learn about hardware acceleration on FortiGate.

Objectives

- Describe the architecture of hardware offload on FortiGate
- Describe how traffic is offloaded from the kernel to an SPU
- Identify how hardware offloading accelerates FortiGate features
- Configure hardware acceleration

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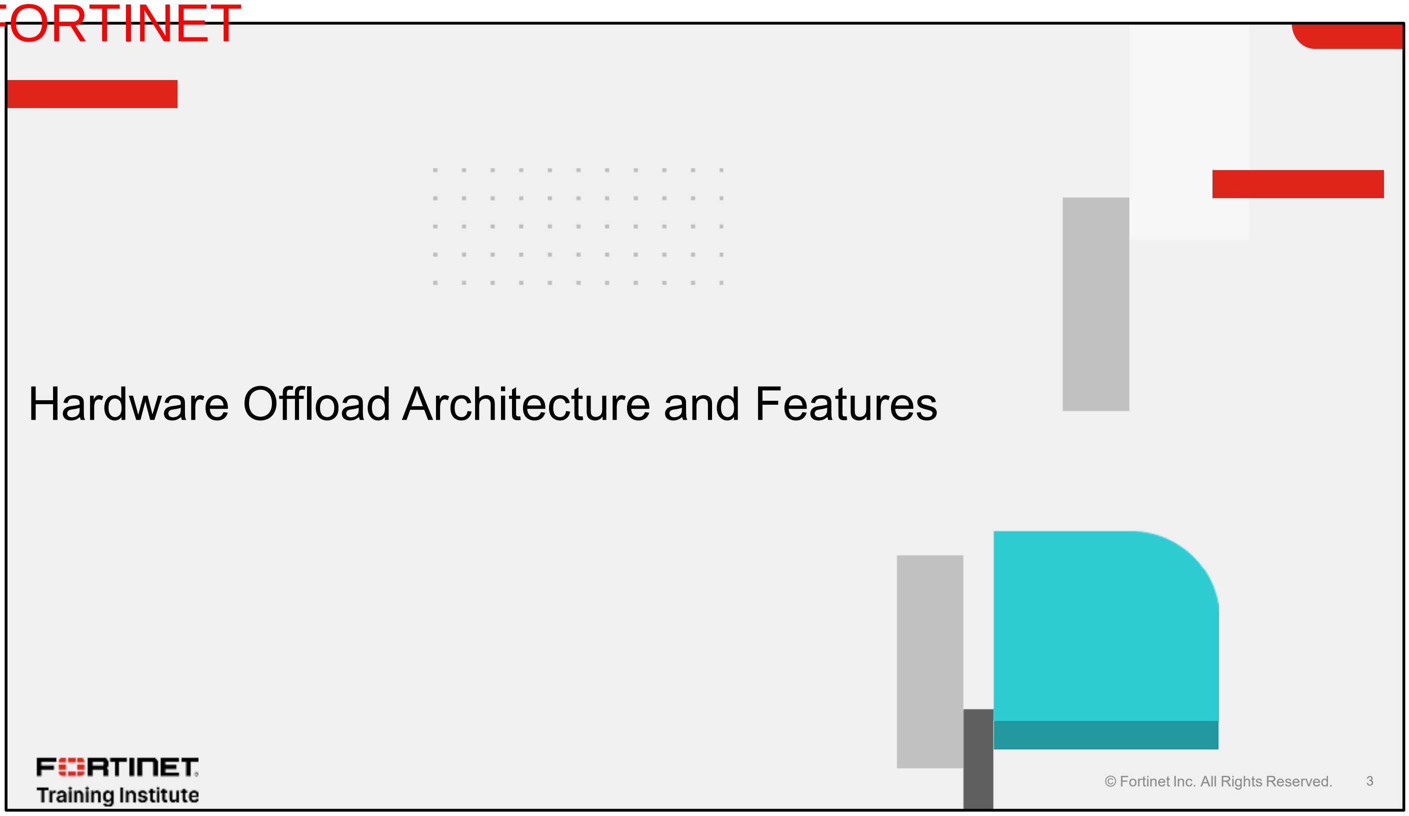
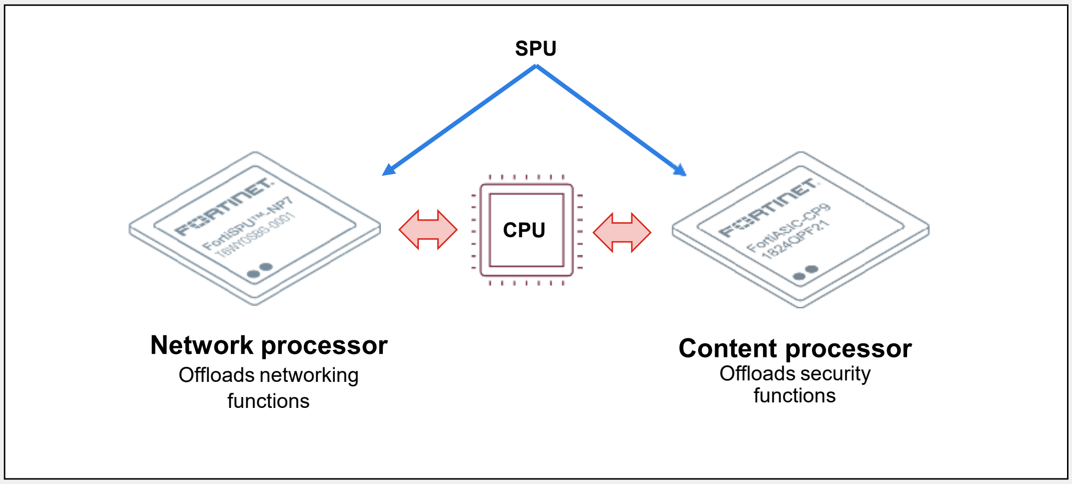
After completing this lesson, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in hardware acceleration on FortiGate, you will understand the design aspect of hardware acceleration on FortiGate and be familiar with security processing units (SPU) available on FortiGate, as well as their features.



In this section, you will learn about the architecture of hardware offload on FortiGate and the features of the processors.

SPU



Fortinet hardware acceleration technology refers to the use of specialized hardware devices, such as network security processors, to offload certain security-related tasks from the main CPU in Fortinet security appliances. This allows for improved performance and higher processing speeds for certain functions, such as encryption and decryption, packet inspection, and others. By using hardware acceleration, Fortinet devices can handle more traffic and provide more efficient processing of network security functions, resulting in better network performance and scalability.

The specialized acceleration hardware on most FortiGate models, also referred to as an SPU or an application-specific integrated circuit (ASIC), can offload resource-intensive processing from the main processing unit (CPU) on FortiGate. On FortiGate, an SPU can be a network processor (NP), or a content processor (CP), or both. Depending on the FortiGate model, both processors can work simultaneously while traffic is passing through to be offloaded and/or accelerated.

NPs, CPs, and SPs

SPU	Features	Latest version	FortiGate example
NP	Works at the interface level Accelerates traffic by offloading traffic from the CPU Can encrypt and decrypt IPsec VPN Performs IP integrity header check	NP7	FortiGate 900G
CP	Focused on the application Accelerates common resource-intensive security processes, such as application identification, IPS, and antivirus in flow-based mode Can decrypt and encrypt for SSL deep inspection	CP10	FortiGate 200G
SP	Includes light versions of NP and CP For entry-level and mid-range FortiGate devices	SP5	FortiGate 90G

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This slide shows a summary of the main characteristics of the FortiGate SPUs.

The NP works at the interface level to accelerate network traffic by offloading it from the CPU. The latest model, the NP7 processor, can support up to 12 million sessions. This number is limited by processor memory. Once an NP7 processor hits its session limit, sessions over the limit are sent to the CPU. You can prevent this problem by distributing incoming sessions evenly among multiple NP7 processors. To do this, you must be aware of which interfaces connect to which NP7 processors and distribute incoming traffic accordingly.

The CP accelerates common resource-intensive security processes and offloads them from the main CPU. The CPU determines which security tasks are accelerated and offloaded to the CP.

For entry-level and mid-range FortiGate devices, the SPU, named SoC4 and SP5 for the latest version, combines NP and CP components on one chip with lower throughput.

Identify SPU Information

- To check the NP model available on FortiGate

```
# diagnose hardware lspci | grep 1a29
...
9b:00.0 Class 1000: Device 1a29:NPID
```

`1a29` is the vendor ID for Fortinet

  - If NPID is 4e36, it is NP6
  - If NPID is 4e37, it is NP7

- To determine the CP model available on FortiGate

```
# get hardware status
Model name: FortiGate-900G
ASIC version: CP9
...
```

`ASIC version` represents CP version available on FortiGate

- To confirm SP5 on FortiGate

```
# get hardware status
Model name: FortiGate-91G
ASIC version: SOC5
...
```

`ASIC version` confirms SP5 available on FortiGate

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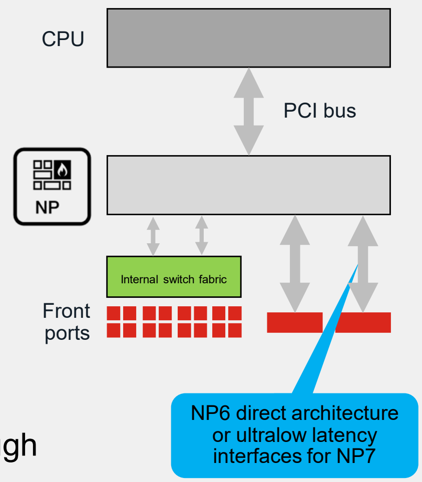
To check the SPU information on FortiGate, you can run the CLI command `lspci` and review the hexadecimal value that comes after the vendor ID for Fortinet, which is `1a29`.

For information about CP and SP on FortiGate, you can run the command `get hardware status` and review the `ASIC version` value.

NP Direct Architecture

- Available on FortiGate devices that have two or more NP6 processors, like:
  - FortiGate 200E
  - FortiGate 2000E
  - FortiGate 2500E
- Available on FortiGate devices that have ports X5 to X8 directly connected to the NP7 processor, like:
  - FortiGate 400F
  - FortiGate 600F
  - FortiGate 900G
- Hardware architecture that has a FortiASIC network processor directly connected to the interfaces, eliminating ISF
- Reduces forwarding latency
- Physical topology must ensure that all traffic passes through the offloaded interfaces of one NP6/NP7 processor



An additional characteristic is available for FortiGate devices with two or more NP6 processors: NP direct architecture. Concerning NP7, the option is available on FortiGate devices with ultralow latency interfaces. It offers access to the NP by removing the use of the internal switch fabric (ISF), which is the hardware chip switch that is attached to the FortiGate physical interfaces. This allows access between the interfaces directly to the NP, for the lowest-latency forwarding.

Part of the requirement for implementing NP direct architecture is to ensure that the offloaded traffic enters and exits FortiGate through interfaces connected to the same NP6 or NP7 processor.

NTurbo

- NTurbo offloads traffic when flow security profiles are used in flow-based mode
  - Usual NP offloading cannot be used with security profiles
  - Anything that is handled by the IPS engine can use NTurbo
- NTurbo creates a special data path to redirect traffic from the ingress interface to IPS
- For a new session, the NTurbo driver sends packets to the IPS engine
  - The IPS engine processes the required security action
  - The IPS engine then sends the packet back to the NTurbo driver
- NTurbo continues to send processed packets back to the NP
- As a consequence, Nturbo improves IPS performance

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NTurbo offloads traffic that the NP cannot offload because of the security profiles used to inspect the traffic.

FortiGate runs the NTurbo driver on a CPU that is dedicated to processing traffic that the NP sends and that requires a security inspection in flow-based mode. The FortiGate CPU processes proxy-based inspections.

When packets from the NP and CPU pass through the NTurbo driver, it sends the packets to the IPS engine to complete the security task required. Then, the NTurbo driver sends the processed packets to the NP.

NTurbo and IPSA

- Configure NTurbo to distribute sessions with flow-based security profile to different IPS engine processes
- Configure IPSA to offload pattern matching to CPs

```
config ips global
    set np-accel-mode {none | basic}
    set cp-accel-mode {none | basic | advanced}
end
```

Default value providing acceleration mode for IPS processing

Default value to offload more pattern matching

Default value for devices with one CP8

- Disable NTurbo on a firewall policy for testing purposes

```
config firewall policy
    edit <policy_id>
        set np-acceleration {disable | enable}
end
```

Default value

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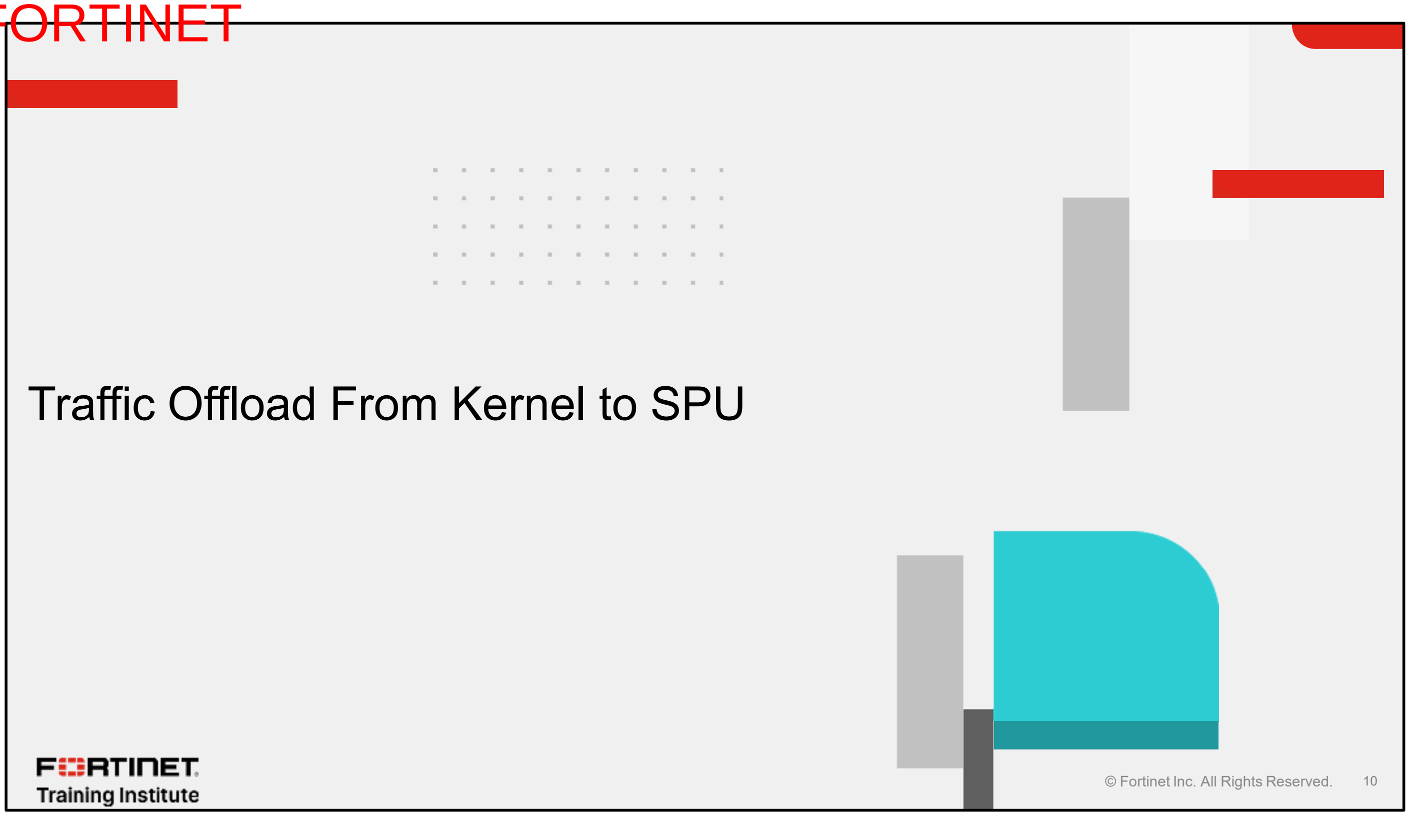
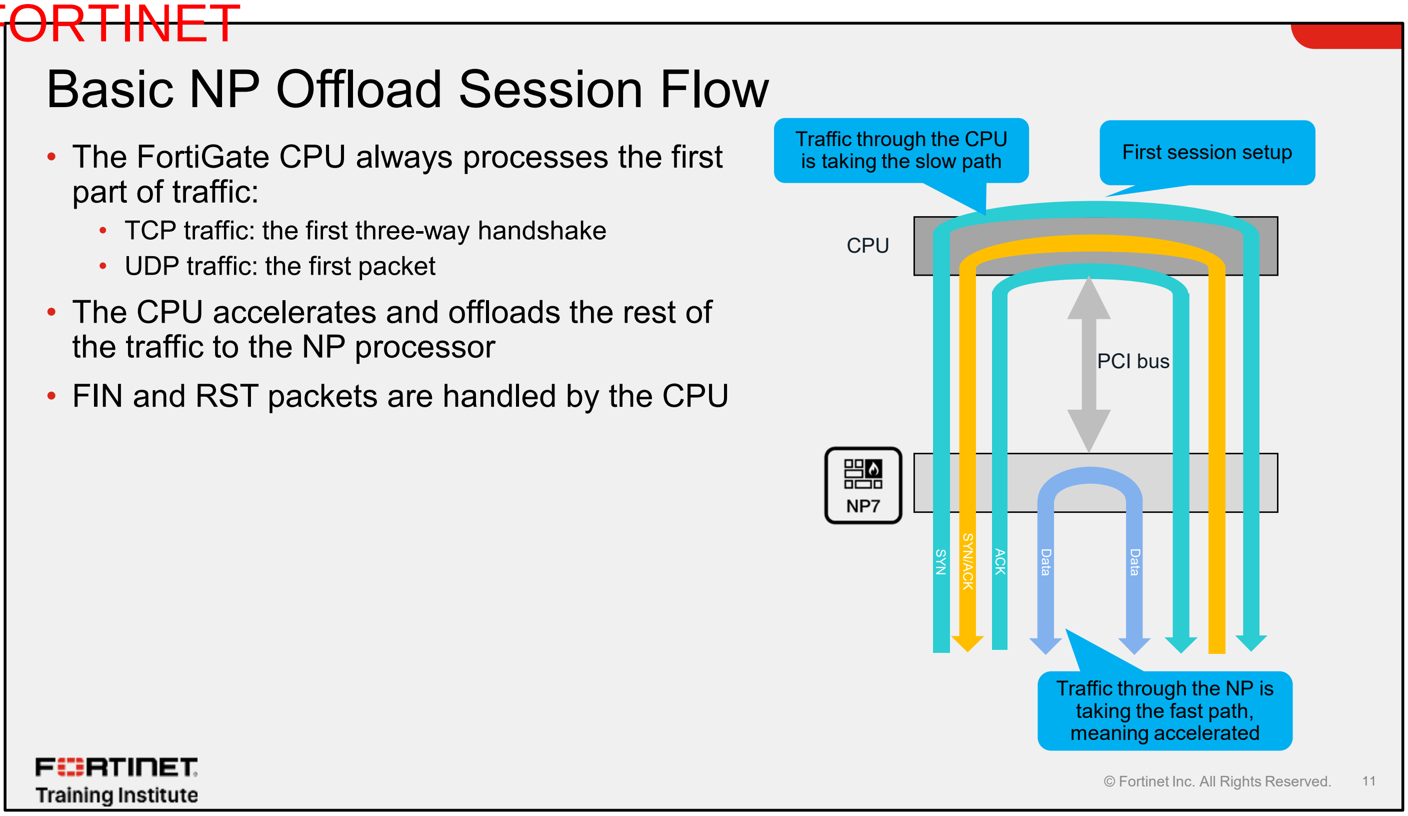
By default, NTurbo is enabled to ensure a load-balance approach for handling IPS signature and pattern matching tasks. For specific purposes like troubleshooting, you can disable it globally, by setting `np-accel-mode` to `none`, or you can disable it per firewall policy, by setting `np-acceleration` to `disabled`.

For traffic handled by IPS engines, IPS acceleration (IPSA) is involved along with Nturbo.

By default, IPSA allows flow-based pattern matching requests to be redirected to the CP hardware, reducing the load on the FortiGate CPU and accelerating pattern matching. On FortiGate devices with one CP8, the only mode is `basic`, while other FortiGate devices have the `advanced` setting by default, to offload more types of pattern matching.



In this section, you will learn how FortiGate offloads traffic from the kernel to an SPU.

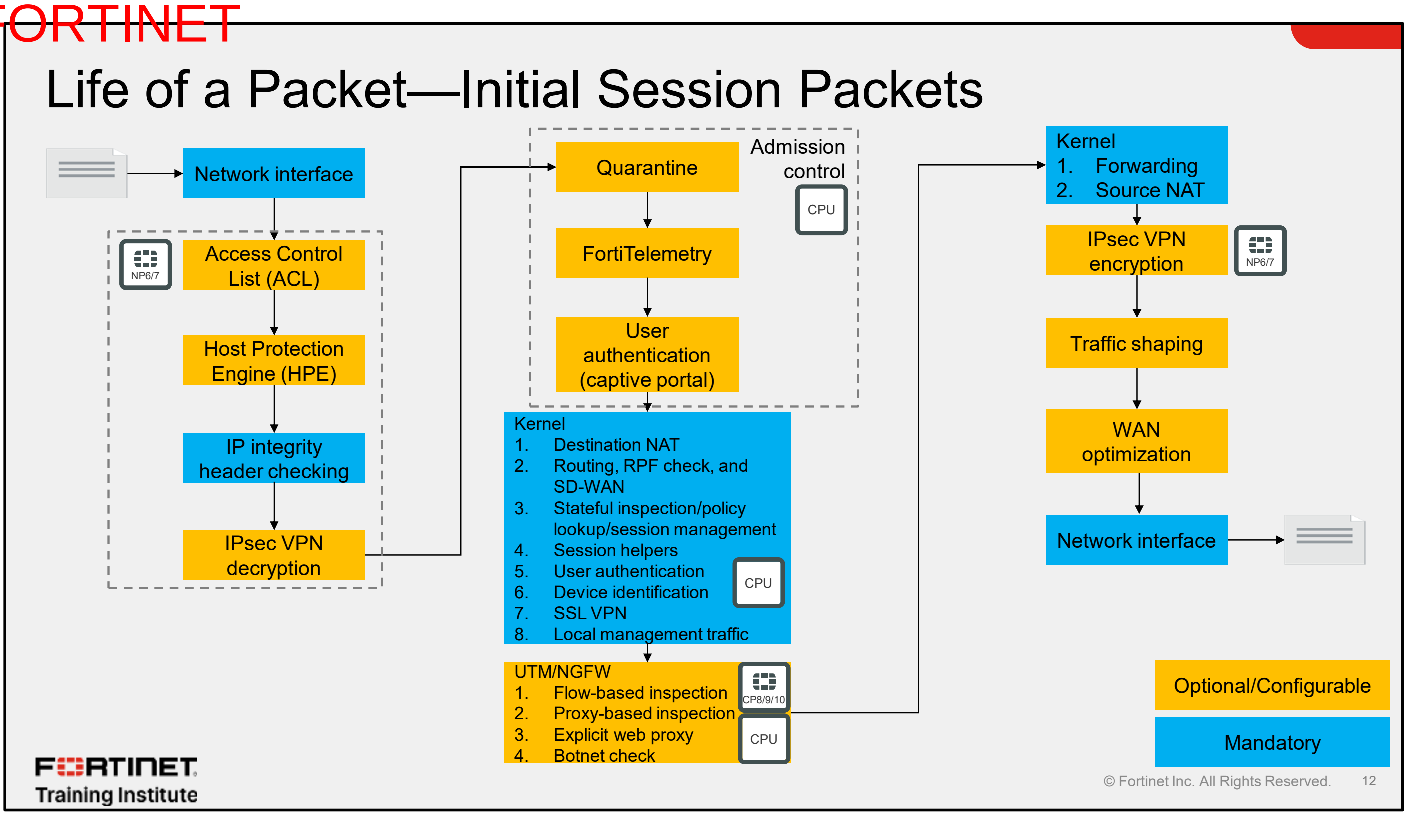


For NP processing, the traffic first passes through the FortiGate CPU, which performs the checks required to determine whether the packets can be accelerated and offloaded to the NP processor.

For TCP traffic, the first packets of the three-way handshake must pass through the FortiGate CPU. For UDP traffic, only the first packet must pass through the FortiGate CPU.

If the FortiGate CPU determines that the conditions are met for the first part of the traffic, then the CPU offloads the rest of the traffic to the NP processor. The path through the CPU is considered a slow path, while the path through the NP is considered a fast path, or accelerated path.

Now with different SPUs, FortiGate uses parallel path processing to choose from a group of parallel options and identify the optimal path for processing a packet. The next few slides provide flowcharts displaying examples of packet processing in several scenarios. Note that these examples do not cover all possible scenarios, nor do they show every step or process packets are subjected to during inspection. These slides provide common examples of packet flow on FortiGate.



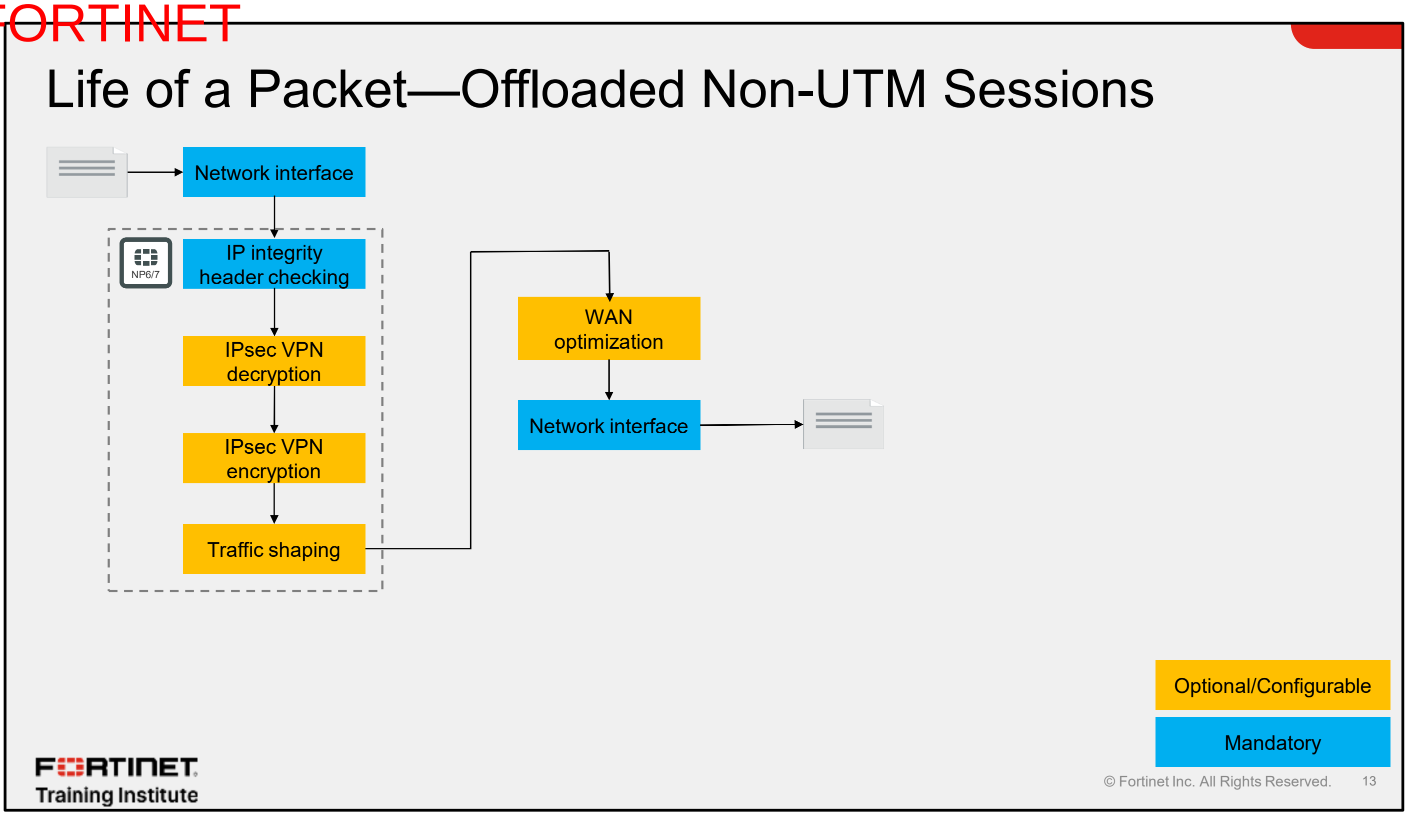
This slide shows the steps that the first packets of a session go through as they enter, pass though, and exit FortiGate. This scenario is for FortiGate with SPUs.

FortiGate performs some security inspections early in the life of the packet, such as ACL, HPE, and IP integrity header checking. FortiGate does this to make sure the packets are within acceptable parameters before allowing the packet to move through the rest of the processes. These inspections are handled by the network processor in order to minimize impact on the FortiGate CPU.

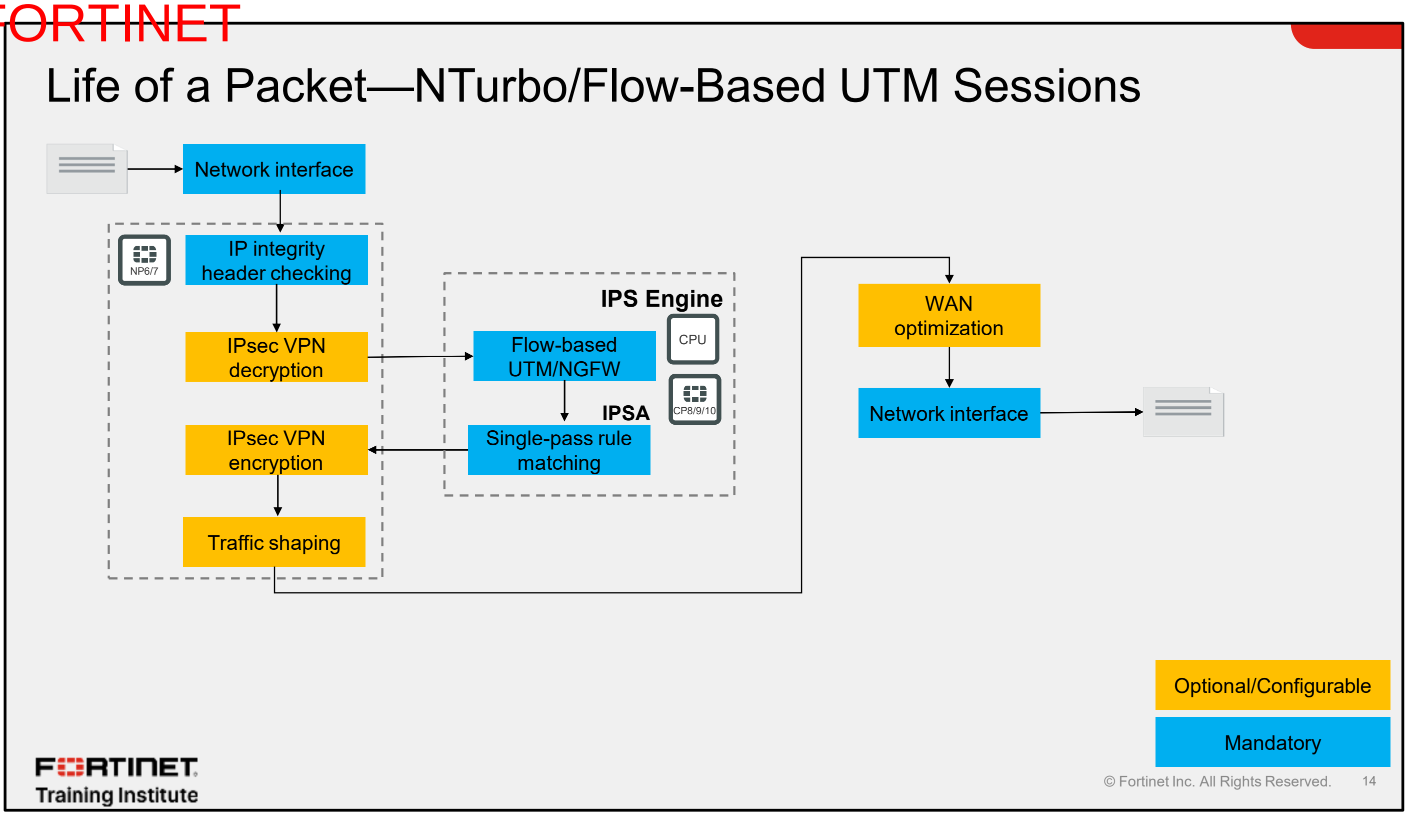
Each version of the network processor has criteria that defines which traffic can be offloaded. The network processor enhances overall performance by allowing offloaded sessions to bypass the FortiGate CPU after the session is established and the session key is installed in the network processor. The network processor can also handle IPsec VPN encryption and decryption operations, where the configured encryption and hashing algorithms are supported in hardware.

The content processor functions like a coprocessor for the FortiGate CPU to improve overall system performance by offloading certain tasks, such as pattern matching for flow-based UTM inspection with the intrusion prevention system (IPS) engine, SSL/TLS decryption and encryption for deep SSL inspection, and IPsec encryption and decryption operations for supported algorithms.

Note that the packet processing for virtual FortiGate devices is identical except that the CPU handles all processes instead of being able to offload some of them to the NP and CP.



This slide shows how subsequent packets in an established session where security profiles are not configured are handled by FortiGate after being offloaded to a network processor. After the session key is installed in the network processor, subsequent packets for the offloaded session skip routing and kernel processors, bypassing the FortiGate CPU, and are forwarded out the egress interface by the network processor.



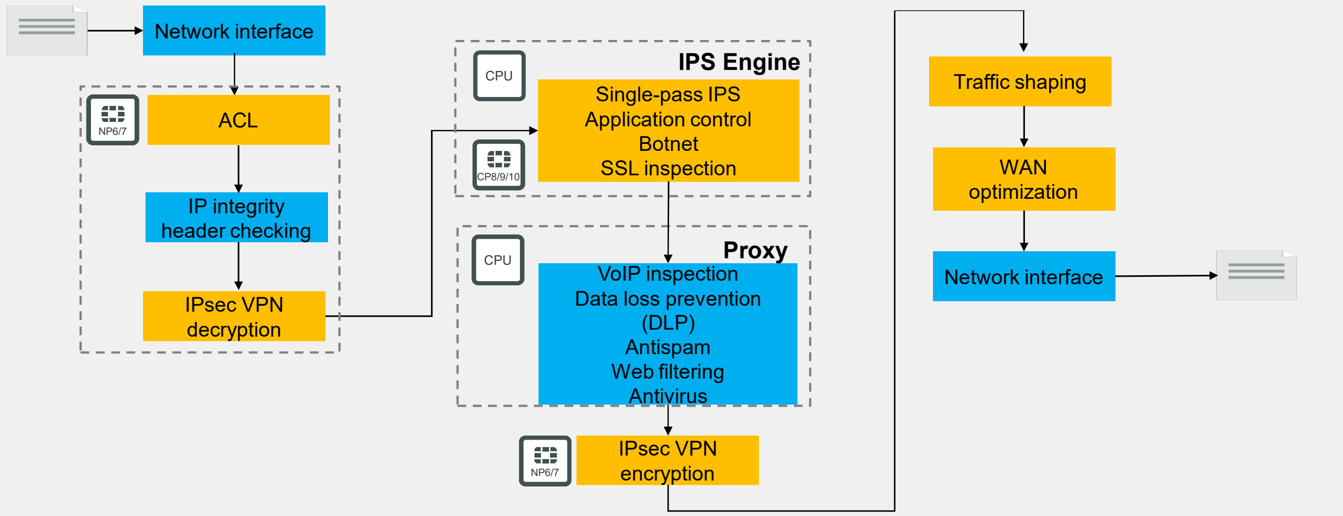
This slide shows how subsequent packets in an established session with flow-based security profiles configured are handled by FortiGate, where NTurbo and IPSA are supported and enabled.

NTurbo is a feature that enables flow-based unified threat management (UTM)/next-generation firewall (NGFW) sessions to be accelerated by NP6 or NP7 network processors to and from the IPS engine.

IPSA is a feature that allows basic or advanced pattern matching operations required for flow-based security profile inspection to be offloaded to the content processors. When IPSA is enabled, flow-based pattern matching databases are compiled and downloaded to the content processors from the IPS engine and IPS database.

After the session key is installed in the network processor, subsequent packets for the accelerated session skip routing and kernel processors. However, UTM/NGFW operations are still handled by the CPU with IPSA offloading pattern matching to the content processors.

Life of a Packet—Proxy-Based UTM Sessions

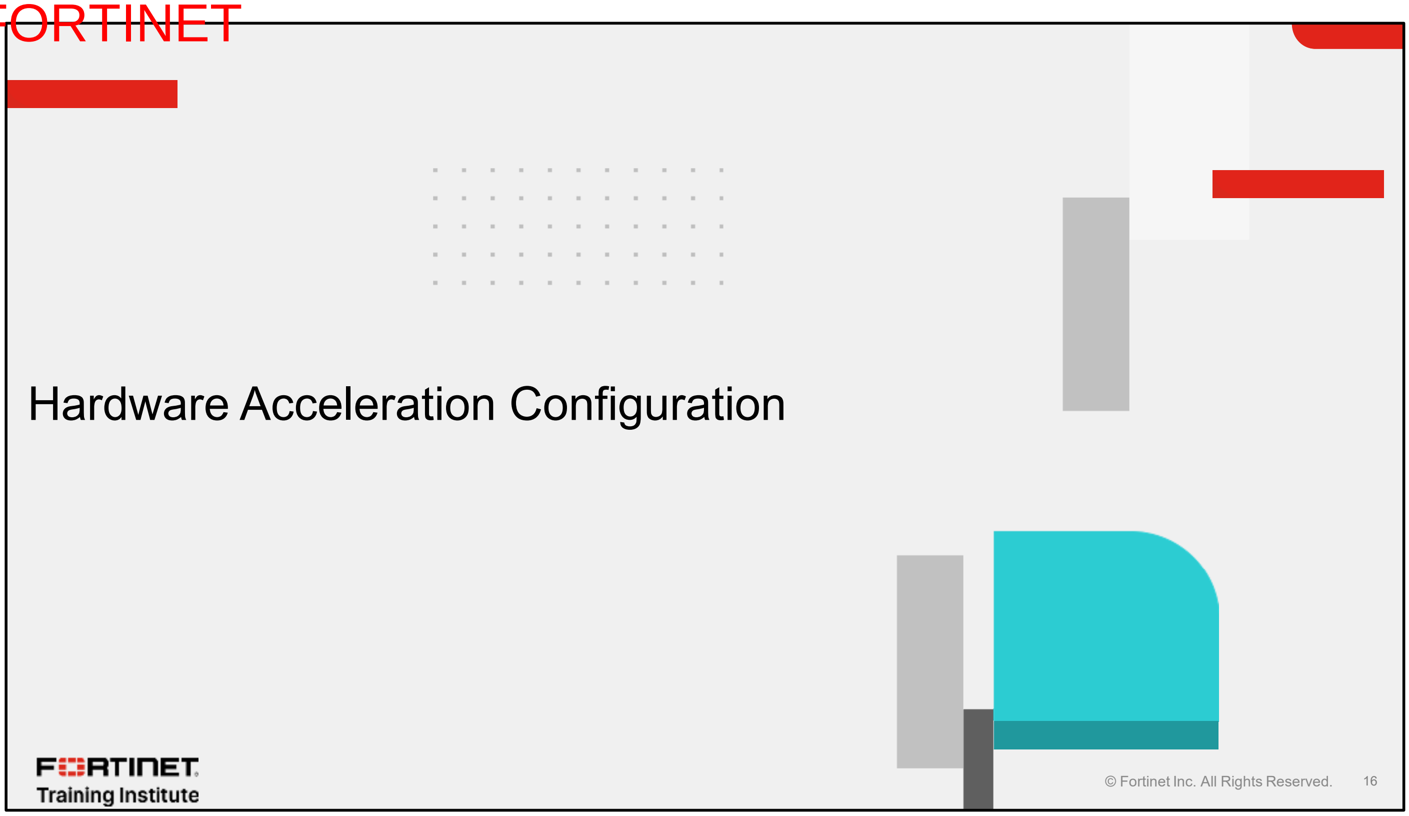


This slide shows how subsequent packets in an established session with proxy-based security profiles configured are handled by FortiGate.

The network processor does not offload sessions where proxy-based features are configured, so the FortiGate CPU handles packet processing and inspection through a combination of the IPS engine and FortiOS proxy. The network processor still conducts some early security inspections before handing off the packets to the IPS engine and can still be leveraged for IPsec tunnel decryption and encryption.

The FortiGate CPU can leverage the content processors to handle SSL/TLS encryption and decryption—if SSL deep inspection is configured. Note that the packet flow is modified when SSL deep inspection is configured.

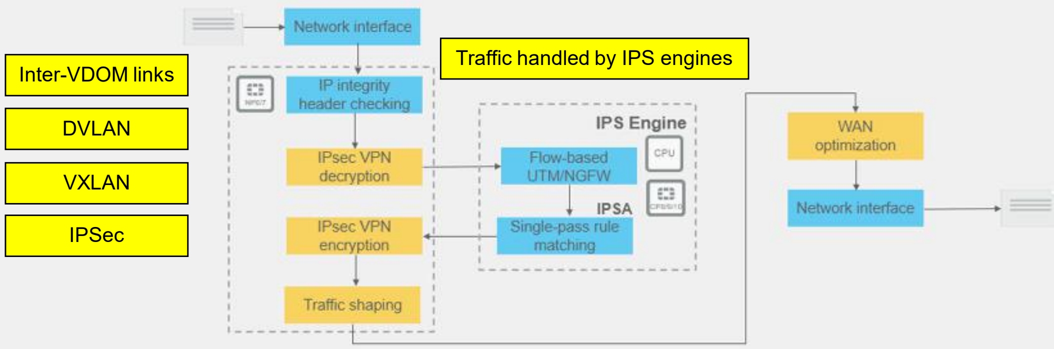
If the IPS engine determines that the session needs to be decrypted:
1. The IPS engine sends the packet to the proxy for decryption.
2. The proxy decrypts the SSL/TLS packet by offloading the operation to the content processor.
3. The proxy sends the decrypted packet back to the IPS engine for IPS and application control inspection.
4. The IPS engine sends the decrypted packet back to the proxy for the configured proxy-based inspection.
5. The proxy encrypts the SSL/TLS packet by offloading the operation to the content processor.



In this section, you will learn about hardware acceleration configuration.

Configuring at Different OSI Model Layers

- Physical layer with inter-VDOM links
- Data link layer with DVLAN and VXLAN
- IP layer with IPsec
- Application layer with traffic handled by IPS engines



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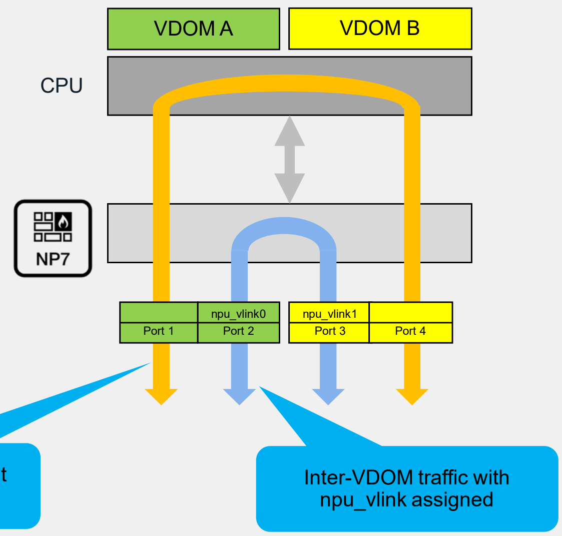
From the life of a packet diagram, you now understand that hardware acceleration occurs at different OSI layers. Now, you will investigate the configuration of these elements, from lower to higher OSI model layers:

- Inter-VDOM links
- Dynamic virtual LAN (DVLAN) and Virtual eXtensible LAN (VXLAN)
- IPsec
- Traffic handled by IPS engines

Configuring VDOM Link Acceleration

- VDOM links interfaces associated with an NP processor are accelerated
- The naming convention is:
  - npu_vlink0 and npu_vlink1
- Assign each VDOM link to offload traffic between two VDOMs



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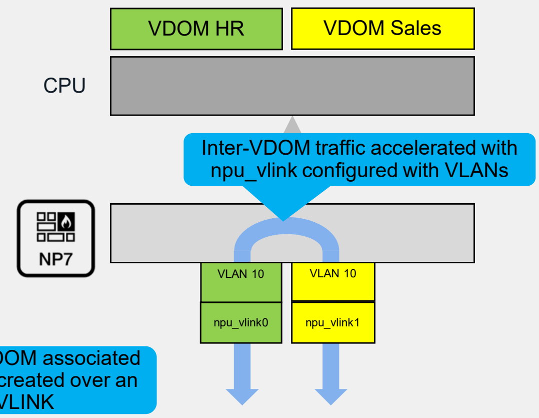
By default, NP processors have VDOM links and they are all assigned to the VDOM root. To offload the traffic between two VDOMs, you must assign one VDOM link to one VDOM and the other VDOM link to the other VDOM. The inter-VDOM traffic is then accelerated through the NP, offloading the CPU.

Use Case—VLAN Over NPU VDOM Link

- VLANs allow you to create more accelerated inter-VDOM links

Network > Interfaces

NPU VDOM Link 5			
npu0_vlink	NPU VDOM Link	root	
npu0_vlink0	NPU VDOM Link Interface	root	0.0.0.0/0.0.0.0
VLAN_vlink0_HR	VLAN	HR	10.10.10.9/255.255.255.252
npu0_vlink1	NPU VDOM Link Interface	root	0.0.0.0/0.0.0.0
VLANvlink1Sales	VLAN	Sales	10.10.10.10/255.255.255.255




```
FortiGate-901G (VLANvlink1Sales) # show
config system interface
    edit "VLANvlink1Sales"
        set vdom "Sales"
        set ip 10.10.10.10 255.255.255.252
        ...
        set interface "npu0_vlink1"
        set vlanid 10
    next
end
```

Name of the VDOM associated with the VLAN created over an NPU VLINK

NPU VLINK associated with the configured VLAN

VLAN must be the same on both NPU VLINKS

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The number of VDOM links per NP is limited, but you can add VLAN interfaces to NPU VDOM link interfaces to create additional accelerated links between VDOMs.

When you create a new VLAN interface, you associate it with an NPU VLINK and you must configure the same VLAN ID on both NPU VLINKS. The remaining configuration, including the firewall policies, is the same except that the interface objects are VLAN over NPU VDOM links instead of the limited VDOM links per NP.

VDOM Link Acceleration Validation

- The command `diagnose sys session list` confirms acceleration

```
FortiGate-901G # diagnose sys session list
...
npu info: flag=0x81/0x91, offload=9/9  ips_offload=0/0, epid=128/166, ipid=166/128, vlan=0x0000/0x0000
...
```

Code 9 represents NP7

- To configure acceleration in a firewall policy

```
FortiGate-901G # config firewall policy
FortiGate-901G (policy) # edit 1
FortiGate-901G (1) # set auto-asic-offload
enable     Enable auto ASIC offloading.
disable    Disable ASIC offloading.
```

Default value

- When acceleration is disabled in a firewall policy, it is confirmed in the session list

```
FortiGate-901G # diagnose sys session list
...
npu_state=0x000001 no_offload
no_ofld_reason:  disabled-by-policy
```

`auto-asic-offload` is set to `disable` in the firewall policy

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Like any traffic going through a firewall, you must create a firewall policy for the corresponding traffic to be allowed on inter-VDOM links. If the firewall policy includes npu_vlink or vlan over npu_link interfaces, you can confirm that the traffic is accelerated with the command `diagnose sys session list`. This command provides further details at the line `npu_info`.

If you must sniff the traffic on npu_vlink interfaces for troubleshooting purposes, you can temporarily disable acceleration on the corresponding firewall policy by disabling `auto-asic-offload`. The session is then handled by the CPU, as shown in the session list. Because this command affects all the traffic going through the firewall policy, you must set it back to its default value, `enable`, as soon as possible to manage resources and improve performance with acceleration.

DVLAN and VXLAN

- NP7 and SP5 (named also NP7lite) offloads 802.1ad and 802.1Q double VLAN
- To view the current DVLAN mode setting

```
FG3K2F-1 # diagnose npu np7 dvlan-mode-list
Device_name      Dvlan mode
np7_0            802.1AD-0x88a8
np7_1            802.1AD-0x88a8
```

```
FG-91G # diagnose npu np7lite dvlan-mode-list
Device_name      Dvlan mode
np7lite_0        802.1AD
```

Default value is 802.1AD

- To change the DVLAN mode

```
FG3K2F-1 # diagnose npu np7 dvlan-mode ?
802.1Q      Set dvlan mode to 802.1Q [Take 0-1 arg(s)]
802.1AD     Set dvlan mode to 802.1AD [Take 0-1 arg(s)]
Disable     Set dvlan mode to disable [Take 0-1 arg(s)]
FG3K2F-1 # diag npu np7 dvlan-mode 802.1.Q <dev-id>
```

The mode can be defined per NP7 ID or for all

DVLAN modes

```
FG-91G # diagnose npu np7lite dvlan-mode ?
802.1Q      Set dvlan mode to 802.1Q [Take 0-1 arg(s)]
802.1AD     Set dvlan mode to 802.1AD [Take 0-1 arg(s)]
FG-91G # diag npu np7lite dvlan-mode 802.1.Q <dev-id>
```

- NP7 and SP5 offload VXLAN

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DVLAN and VXLAN offloading is available with NP7 and SP5.

By default, NP7 handles double VLAN with the provider VLAN tag (S-Tag with ethertype of 0x88A8) and the customer VLAN tag (C-Tag with ethertype 0x8100). The command `diagnose npu np7 dvlan-mode` allows NP7 to handle Q-in-Q VLAN tags for all NP7 IDs or for a specific NP7 ID.

IPsec Encryption and Decryption Offloading

- You can offload IPsec encryption and decryption to hardware on some FortiGate models
- Hardware offloading capabilities and supported algorithms vary by processor type and model
- By default, offloading is enabled for supported algorithms
  - You can manually disable offloading:

```
config vpn ipsec phase1-interface
        edit <tunnel_name>
                set npu-offload enable | disable
        next
end
```

Default value

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Concerning IPsec traffic, you can offload the encryption and decryption to hardware on some FortiGate models. The supported algorithms depend on the model and type of processor on the unit that is offloading the encryption and decryption.

By default, hardware offloading is enabled for the supported algorithms. This slide shows the commands you can use to disable NP hardware offloading per tunnel, if necessary.

Session NPU-Flag Field

- IPsec SAs have an NPU-flag field

```
# diagnose vpn tunnel list name Hub2Spoke1
list ipsec tunnel by names in vd 0
...
npu_flag=03 npu_rgwy=10.10.2.2 npu_lgwy=10.10.1.1
npu_selid=1
```

npu_flag field

NPU-flag Value	Description
npu_flag= 00	Both IPsec SAs loaded to the kernel
npu_flag= 01	Outbound IPsec SA copied to NPU
npu_flag= 02	Inbound IPsec SA copied to NPU
npu_flag= 03	Both outbound and inbound IPsec SA copied to NPU
npu_flag= 20	Unsupported cipher or HMAC, IPsec SA cannot be offloaded

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All IPsec SAs have an `npu_flag` field indicating offloading status. In the case of IPsec traffic, the FortiGate session table also includes that field.

First, when phase 2 goes up, the IPsec SAs are created and loaded to the kernel. As long as there is no traffic crossing the tunnel, the SAs are not copied to the NPU, and the `npu_flag` shows `00`. The value of that field also remains `00` when IPsec offloading is disabled.

Second, if the first IPsec packet that arrives is an outbound packet that can be offloaded, the outbound SA is copied to the NPU and the `npu_flag` changes to `01`. However, if the first IPsec packet is inbound and can be offloaded, the inbound SA is copied to the NPU and the `npu_flag` changes to `02`.

After both SAs are copied to the NPU, the `npu_flag` changes to `03`.

The value `20` in the `npu_flag` field indicates that hardware offloading is unavailable because of an unsupported cipher or HMAC algorithm.

IPsec Diffie-Hellman Offloading

- FortiGate accelerates the Diffie-Hellman key exchange for IPsec ESP traffic
- you can disable ASIC offloading globally using the following command (especially for troubleshooting purposes):

```
config system global
    set ipsec-asic-offload disable
end
```

Default value is `enable`

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You can also run the CLI command on this slide in the system global settings on FortiGate to disable CP offloading to accelerate the IPsec Diffie-Hellman key exchange for IPsec ESP traffic. By default, FortiGate uses IPsec Diffie-Hellman hardware offloading.

For debugging purposes or other reasons you can disable `ipsec-asic-offload` for this function to be processed by software.

Strict Header Checking

- By default, FortiGate performs basic header checking with the NPs including:
  - L4 header length
  - IP header length
  - IP version
  - IP checksum
  - IP options
- You configure strict header checking to add ESP packets checking including:
  - ESP sequence number
  - SPI
  - Data length

Default value

```
config system global
    set check-protocol-header [loose strict]
end
```

- `check-protocol-header strict` disables all NPs, CPs, and SPs

More generally, you can disable all hardware acceleration for NPs, CPs and SPs, to cause FortiGate to apply strict header checking and verify that traffic is part of a session that should be processed.

By default, FortiGate checks the packet header and verifies the following parameters:

- Layer-4 protocol header length
- IP header length
- IP version
- IP checksum
- IP options

By setting `check-protocol-header` to `strict`, FortiGate checks also the ESP packets and verifies the following parameters:

- ESP correct sequence number
- SPI
- Data length

If the packet fails header checking, FortiGate drops it.

You may also set temporally `check-protocol-header` to `strict` for troubleshooting purposes because it disables hardware acceleration globally on all NPs, CPs, and SPs.

Hardware Acceleration—Best Practices

- To be considered especially for entry-level models in enterprise network
- Selecting FortiGate inspection modes has an impact on resources:
  - Flow-based
    - Default mode and has hardware offload possibilities
    - Optimizes performance
  - Proxy-based
    - Processed by FortiGate proxy or CPU
    - Provides thorough inspection
    - Supports advanced features like *web-profile override*
- Encrypted traffic that requires SSL inspection can also impact throughput
  - SSL deep inspection can cause high CPU and memory usage
  - All traffic needs to decrypt, inspect, and re-encrypt
  - Use hardware acceleration to offload SSL processing

This slide summarizes best practices for hardware acceleration in an enterprise network environment. You must take them into account when defining an enterprise network strategy that includes multiple FortiGate types—especially when entry-level models are positioned as hubs.

Enabling security profiles on FortiGate impacts firewall resources and throughput. FortiOS supports flow-based and proxy-based inspection in firewall policies and security profiles.

When you select the inspection mode, you must consider your requirements, the differences between modes, and how those differences can impact firewall performance. Flow-based inspection identifies and blocks threats in real time as FortiOS identifies them, and typically requires lower processing resources than proxy-based inspection. Flow-based inspection is recommended for policies that prioritize traffic throughput.

Proxy-based inspection involves buffering traffic and examining it as a whole before determining an action. Having complete data to analyze enables inspection of more data points than flow-based inspection. Proxy-based inspection also supports advanced features like category usage quota and web-profile override.

Also, FortiGate requires SSL inspection to apply web filtering. As a result, enabling overall SSL deep inspection can reduce overall FortiGate performance. This is because all traffic must be decrypted, inspected, and re-encrypted. These are some best practices to reduce the impact:

1. Limit the number policies that allow encrypted traffic.
2. Use the flexibility of a FortiGate device security policy to gradually deploy SSL inspection, rather than enabling it all at once.
3. Apply SSL deep inspection only where it is needed. For example, exempt known and trusted traffic from SSL inspection.
4. Use hardware acceleration, such as a content processor (like CP9), to offload SSL content scanning.

Hardware Acceleration—Best Practices (Contd)

- NPs help offloading resource-intensive processing from main CPU
  - Offload high volume of network traffic
  - Offloads firewall sessions that include flow-based security profiles
  - Does not support proxy-based profile
- Firewall policy with mixed (flow-based and proxy-based) security profiles does not offload
- There are some cases where sessions may not be offloaded even if NP acceleration is available:
  - NP acceleration is disabled on the policy
  - Firewall policy includes proxy-based security profiles
  - Firewall session requires FortiOS session helpers
  - Tunnelling is enabled (except IPSec VPN sessions)

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If network processors (like NP7) are available on the FortiGate device, the firewall sessions that require flow-based security features can be offloaded to them. Network processors reduce the workload on the FortiGate CPU and improve overall throughput.

Note that a firewall policy that includes a mix of flow-based and proxy-based profiles never offloads to NP and is always processed by the FortiGate CPU.

There are some cases where firewall sessions may not offload even when NP acceleration is enabled and sessions are handled by the FortiGate CPU:
1. NP acceleration is disabled on the policy.
2. The firewall policy includes proxy-based security profiles.
3. The firewall session requires a FortiOS session helper.
4. Tunneling is enabled, including traffic to or from a tunnelled interface (SSL VPN, GRE, and so on), except for IPsec VPN sessions.

Review

- ✓ Describe the architecture of hardware offload on FortiGate
- ✓ Describe how traffic is offloaded from the kernel to an SPU
- ✓ Identify how hardware offloading accelerates FortiGate features
- ✓ Configure hardware acceleration

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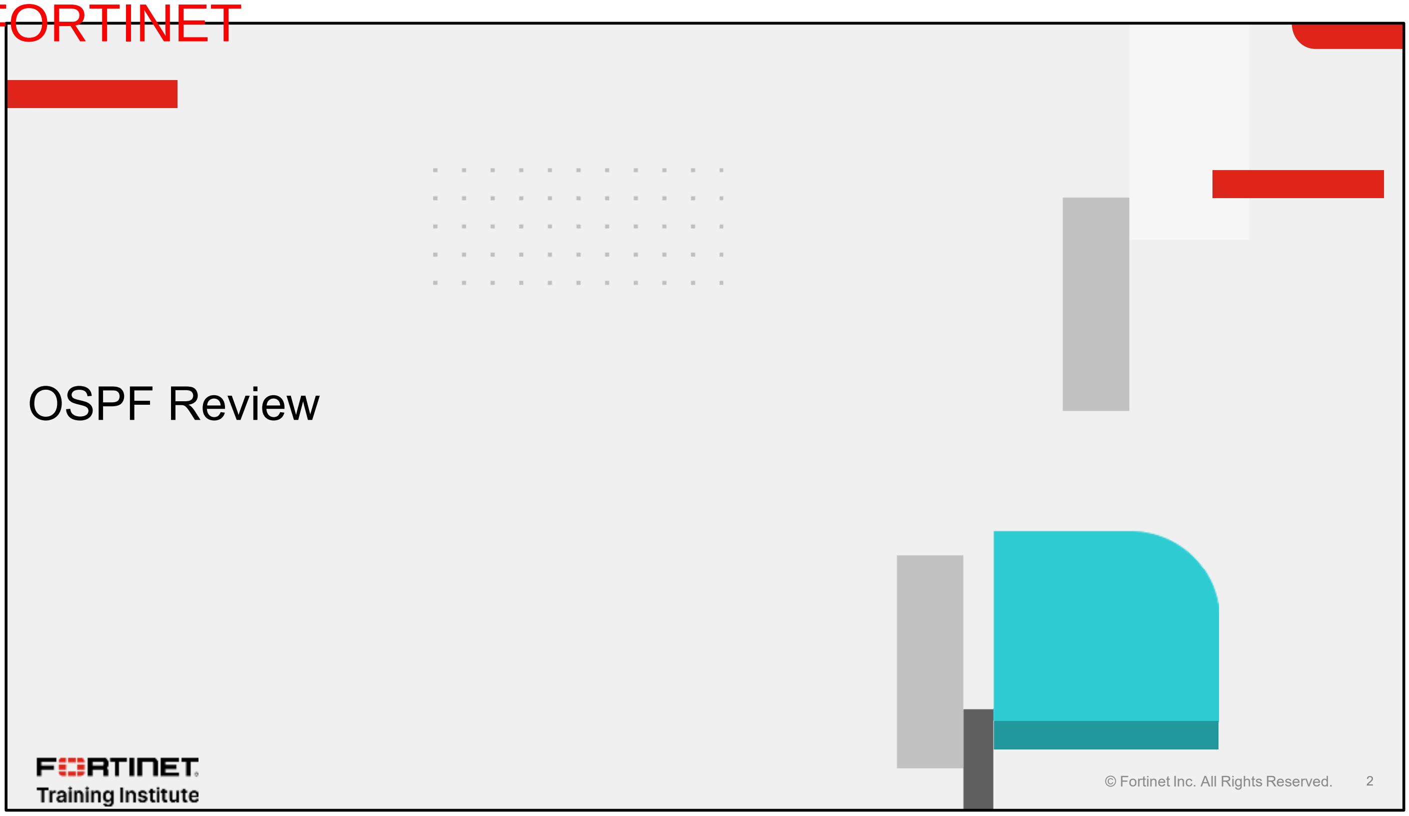
This slide shows the objectives that you covered in this lesson.

By mastering the objectives covered in this lesson, you learned about the design aspect of hardware acceleration on FortiGate and the SPUs available on FortiGate.

In this lesson, you will review OSPF and BGP concepts.



In this section, you will review OSPF concepts.

OSPF Overview

- Link state protocol
- Advantages:
  - Scalable to large networks
  - Faster convergence than distance-vector routing protocols
  - Relatively quiet during steady-state conditions
    - Periodic refresh every 30 minutes
    - Otherwise, updates sent only when there are changes
- Disadvantages:
  - May require planning and tuning to optimize performance
  - May be difficult to troubleshoot in large networks

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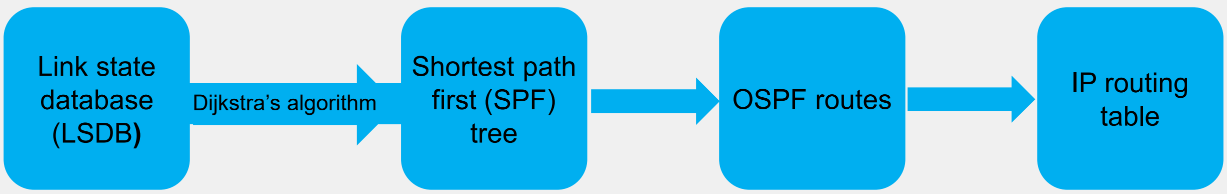
With a link state protocol like OSPF, every router has a complete view of the network topology. Advantages of OSPF include scalability and fast convergence. Every 30 minutes, routers advertise their OSPF information again. During these 30-minute intervals, routers send updates only if they detect topology changes.. So, it is a relatively quiet protocol, as long as the network topology is stable. In large networks, using OSPF requires good planning and may be difficult to troubleshoot.

OSPF Components

- Link state database (LSDB): Each router maintains identical databases describing the network topology
- From the LSDB, and using Dijkstra's algorithm, each router builds a tree with the shortest paths
- The tree gives an OSPF route to each destination
- OSPF can inject routes into the IP routing table



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Each router in the same area has identical and synchronized databases. You will learn about OSPF areas later in this lesson. An OSPF router uses the information in the LSDB and Dijkstra's algorithm to generate an OSPF tree, which contains the shortest path from the local router to each other router and network. This tree gives the best route to each destination, which is the information that OSPF can inject into the device routing table.

Link State Advertisement

- Routers exchange link state advertisements (LSAs) to maintain consistent databases
  - Network changes generate LSAs
  - LSDBs are composed of all received LSAs
- A link state update consists of an OSPF header and a string of LSAs
  - Each LSA has its own header
  - Routers acknowledge the receipt of LSAs

IP header	OSPF header	Number of LSAs	LSA 1	LSA 2	…	LSA *n*

The topology information exchanged by OSPF peers is contained in LSAs. The LSDB of a router is populated with information from the local LSAs and all the LSAs received from other routers.

OSPF Cost

- Metric in OSPF is cost:
  - 16-bit positive number (1 to 65,535)
- Routes with lowest cost are most preferred
- Routing decisions made on total cost of a path to the final destination

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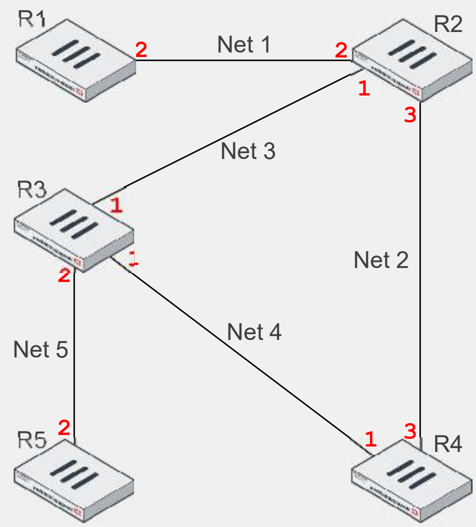
If there are multiple OSPF routes to the same destination subnet, OSPF selects the route with the lowest cost. Each router interface is associated with an interface cost, which is usually related to how fast or preferable that interface is. An OSPF route cost is the sum of the costs of all interfaces to the final destination.

LSDB

- Each router advertises its locally connected subnets
- LSDBs in all routers contain information about the network topology

```
Link State Database (network/cost)
R2    Net 1/2, Net 2/3, Net 3/1
R1    Net 1/2
R3    Net 3/1, Net 4/1, Net 5/2
R4    Net 2/3, Net 4/1
R5    Net 5/2
```



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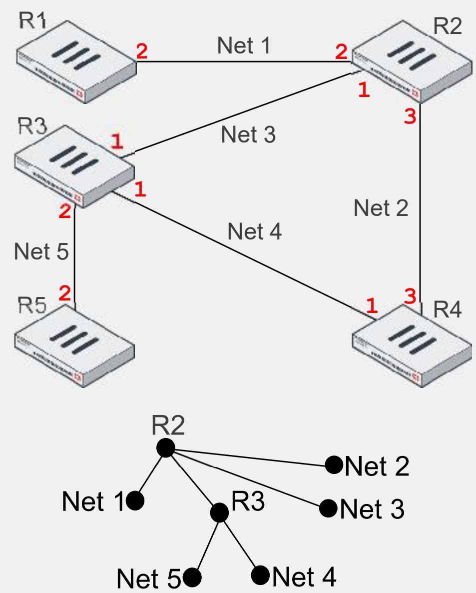
This slide and the next explain how an OSPF router builds its OSPF tree. The initial information for each router is the locally connected networks, together with the OSPF cost for each interface. In the example shown on this slide, the router `R2` has three locally connected subnets: subnet `Net 1` with a cost of `2`, subnet `Net 2` with a cost of `3`, and subnet `Net 3` with a cost of `1`. Router `R1` has only one subnet connected: `Net 1` with a cost of `2`, and so on.

Each router starts advertising its locally connected subnets by sending LSAs.

OSPF Tree

- Routers use Dijkstra's algorithm to determine the best path to each destination
  - The best path has the lowest overall cost
  - The destination is either a network or a router
- Routers build an OSPF tree
  - During each iteration, the router maps all known paths to a destination and chooses the lowest path
  - The router repeats the process until it discovers the best path to each destination



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OSPF routers use Dijkstra’s algorithm to determine the best route to each destination. The best routes can be represented as a tree with the local router at the root. Dijkstra’s algorithm is a recursive process that the router repeats multiple times until it finds the best routes. For example, this slide shows the OSPF tree for router R2. It indicates that the best route to Net 5 and Net 4 is through R3, and that Net 1, Net 2, and Net 3 are locally connected.

OSPF Areas

- A logical collection of OSPF networks and routers
- Defined with a 32-bit number
  - IP address format
    `0.0.0.10`
  - Single decimal format
    `10`

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You can segment an OSPF network into areas. Each area is identified by a unique number, which you can define in decimal or IP address formats.

OSPF Areas (Contd)

- When a network is broken up into areas, routers maintain separate LSDBs for each area
- Advantages:
  - Smaller LSDB tables
  - Impact of a topology change is minimized outside the area
  - You can summarize routes on the area borders
- Disadvantages:
  - More complex to troubleshoot
  - Network design considerations

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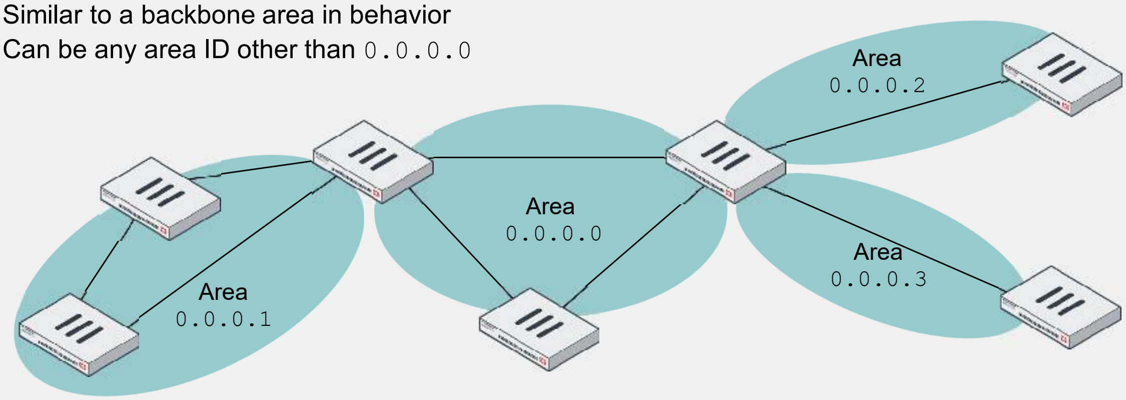
Each area has its own separate LSDB. All routers in the same area maintain an identical copy of the LSDB of an area. As you will learn in this lesson, a router can belong to more than one area. In those cases, the router maintains multiple LSDBs—one LSDB for each area connected to it.

Segmenting big OSPF networks into areas reduces the sizes of the LSDB tables. Additionally, a topology change does not impact the whole network, but only the area where the change happens.

Using OSPF areas requires good planning and may complicate the troubleshooting process.

Types of Areas

- Backbone area
  - Must have the area ID `0.0.0.0`
  - All other areas must connect to the backbone area by physical or virtual links
  - Distributes information between areas
- Normal area
  - Similar to a backbone area in behavior
  - Can be any area ID other than `0.0.0.0`



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All OSPF networks must have at least one area—the backbone area. The backbone area is the core of the network, and all the other areas connect to it in a hub-and-spoke topology.

OSPF Router Types

- Internal router
  - All connected interfaces belong to the same area
  - Maintains one LSDB and one OSPF tree
- Area border router (ABR)
  - A router with interfaces in multiple areas
  - One LSDB and one OSPF tree for each connected area
  - Always connected to the backbone
- Backbone router
  - Has at least one interface in the backbone area
- Autonomous system boundary router (ASBR)
  - Imports external (non-OSPF) routes into OSPF
  - Has at least one source of routing information that is not of OSPF origin

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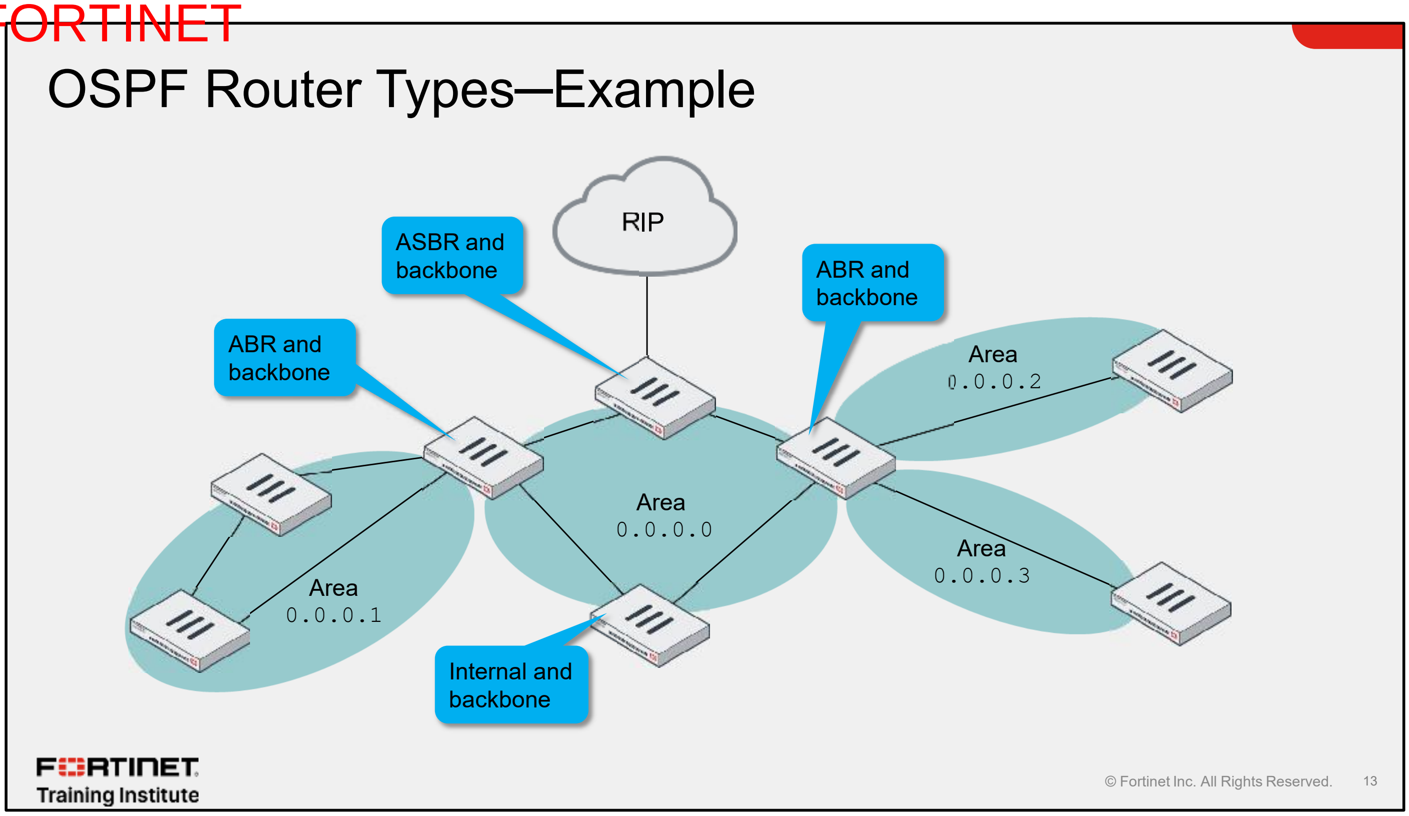
An internal OSPF router has all its interfaces connected to the same area. So, it maintains only one LSDB.

On the other hand, an ABR is connected to multiple areas, so it keeps multiple LSDBs.

A backbone router has at least one interface connected to the backbone area.

An ASBR redistributes non-OSPF routes into the OSPF network.



This slide shows an example of each router type.

Network Types

- Point-to-point
  - A pair of routers connected through a point-to-point link
- Broadcast (multiaccess)
  - Supports more than two attached routers
  - Supports the sending of a single message to all routers (multicast)
  - Example: Ethernet networks
- Point-to-multipoint
  - Supports more than two attached routers
  - Does not support multicast

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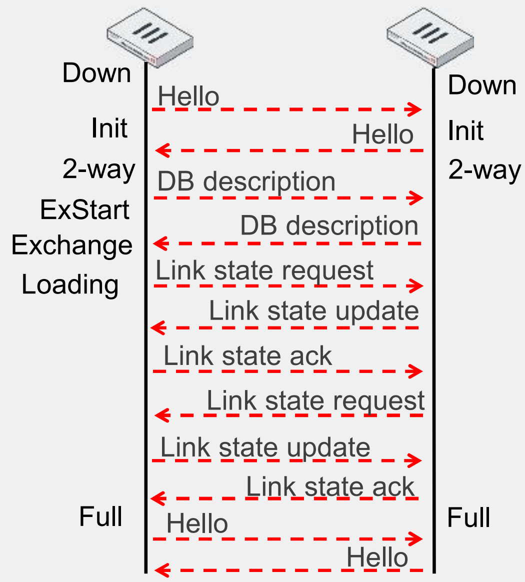
There are three types of OSPF networks:

- Point-to-point networks contain only two peers, one at each end of a point-to-point link.
- Broadcast networks support more than two attached routers. They also support sending messages to multiple recipients (broadcasting).
- Point-to-multipoint networks support more than two attached routers. However, they do not support broadcasting.

Forming an Adjacency

- **Down**: Initial state
- **Init**: A hello packet was seen from a non-adjacent neighbor
- **2-way**: Communication is bidirectional between the two routers
- **ExStart**: A primary and secondary relationship is negotiated
- **Exchange**: Database description packets are exchanged
- **Loading**: LSA information is exchanged
- **Full**: LSDBs are identical



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An OSPF session between two OSPF peers is called an adjacency. This slide shows the initial interchange between two peers that are forming an adjacency. Any new adjacency goes through different states: Init, 2-way, ExStart, Exchange, Loading, and Full. The Full state indicates that the adjacency has successfully formed, and both routers have identical copies of the LSDB.

Adjacency Requirements

- Requirements for forming an adjacency:
  - Primary IP addresses of peers are in the same subnet with the same mask
  - Interfaces of peers are the same type and in the same OSPF area
  - Hello and dead intervals of peers match
  - Each peer has an unique router ID
  - OSPF IP MTUs match
  - OSPF authentication, if enabled, is successful

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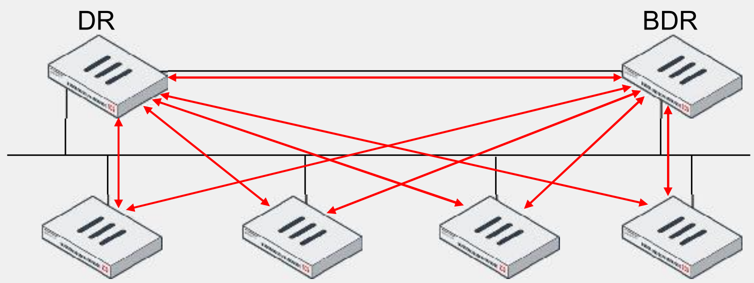
This slide lists the requirements for two peers to form an OSPF adjacency. If any of the requirements are not met, the adjacency fails and will not reach the Full state.

Designated Router

- In multiaccess networks, one designated router (DR) and one backup DR (BDR) are elected
  - Router priority: highest priority wins, 0 = never become a DR
  - Router ID: highest ID wins
- If the DR fails, the BDR takes the DR role
- Full adjacencies are only formed to the DR and BDR to reduce resource utilization



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In any multiaccess network there is one DR and one BDR. The OSPF network elects the router with the highest priority as the DR. If two or more routers are tied with the highest priority, the network elects the router with the highest OSPF ID.

The BDR monitors the DR status. If the DR fails, the BDR takes the DR role.

Other routers form adjacencies only with the DR and BDR. The DR forwards the link state information from one router to another. This simplifies the number of adjacencies required in multiaccess networks.

OSPF Destination Addresses

- On broadcast networks, OSPF uses two multicast destination addresses
  - `224.0.0.5` AllSPFRouters
    - Hello packets
    - Either the DR or BDR sends LSA updates and acknowledgements
  - `224.0.0.6` AllDRouters
    - All other routers send LSA updates and acknowledgements
- On point-to-point networks
  - `224.0.0.5` AllSPFRouters
    - Routers send all packets to this address
- Some OSPF packets may be unicast
  - Database description (DD) packets exchanged
  - LSA retransmissions

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This slide shows the multicast addresses that OSPF uses in broadcast multiaccess, and point-to-point networks. Keep in mind that OSPF also uses unicast addresses for LSA retransmissions and database description packets.

LSA Types

- There are 11 LSA types. These are the five most common types:
  - Type 1: Router link advertisement
    - Describes the links of a router. Confined within an area.
  - Type 2: Network link advertisement
    - Describes all the routers (if more than one) in a multiaccess network. Confined within an area.
  - Type 3: Summary link advertisement
    - Describes summarized networks within an area. Generated by an ABR.
  - Type 4: AS summary link advertisement
    - Describes the path to an ASBR router
  - Type 5: AS external link advertisement
    - Describes external destinations originated on an ASBR. Generated by the ASBR.

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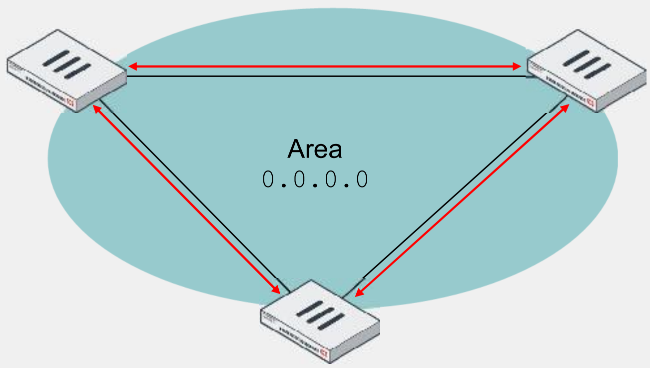
There are 11 LSA types. This lesson covers the following five most commonly used types:

- Type 1 describes all the links connected to a router.
- Type 2 describes all the routers (if more than one) in a multiaccess network.
- Type 3 describes the networks within an area (only generated by an ABR).
- Type 4 describes the path to reach an ASBR.
- Type 5 describes the external destinations originated by an ASBR.

You will see examples of each of these five types in the next slides.

Router Link Advertisements (Type 1)

- Advertised by every OSPF router in an area
- Describe router network connections within that area



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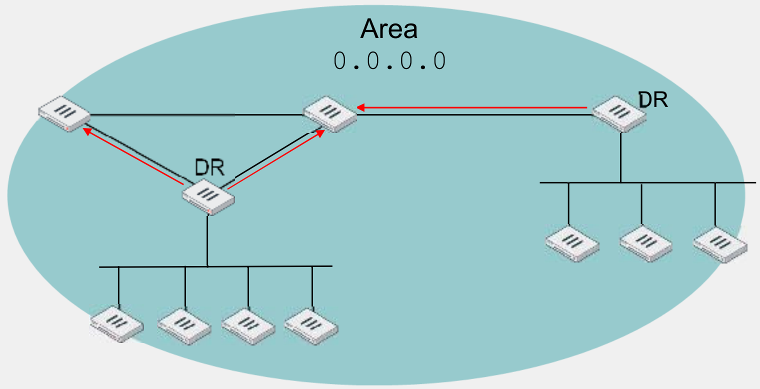
Type 1 describes the networks connected to a router. They are advertised by all the routers in an area. Type 1 LSAs are not advertised outside the area where they originate.

Network Link Advertisements (Type 2)

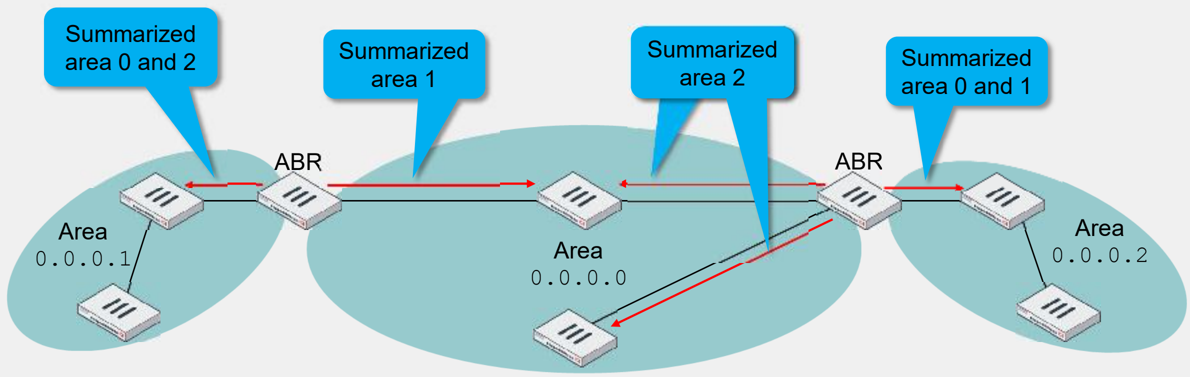
- Advertised by every DR
- Contains the list of routers connected to a multiaccess network



Only DRs advertise Type 2 LSAs. In the example shown on this slide, the area has two multiaccess networks, each of them with one DR. The two DRs advertise type 2 LSAs, which contain information about the other routers connected to their multiaccess networks.

Summary Link Advertisements (Type 3)

- Generated only by ABRs
- Can be used to summarize networks
  - One LSA can represent a range of networks



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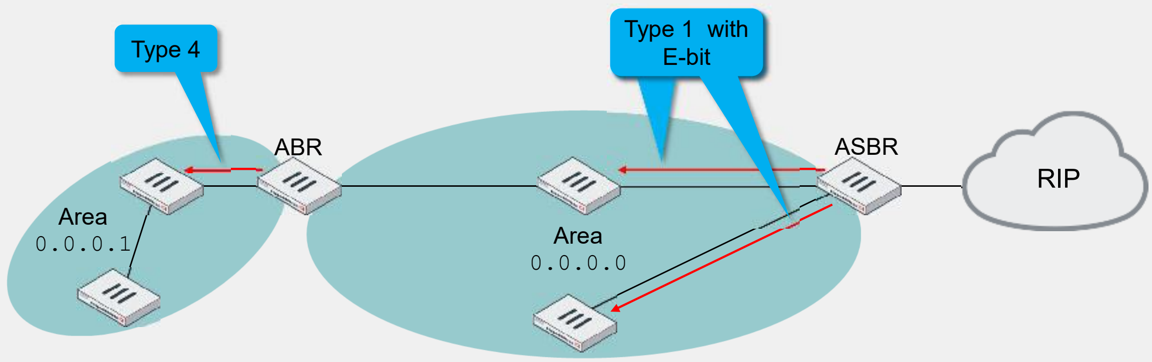
Type 3 LSAs contain summarized link state information. They are advertised only by ABRs. In the example shown on this slide, the ABR on the left sends type 3 LSAs to area 1. They contain link state information for the summarized subnets in areas 0 and 2. This same ABR also sends type 3 LSAs to the backbone area, with a summary of the subnets in area 1.

Something similar happens with the ABR shown on the right side of the diagram. It sends type 3 LSAs to area 2. They contain link state information for the summarized subnets in areas 0 and 1. This same ABR also sends type 3 LSAs to the backbone area, with a summary of the subnets in area 2.

AS Summary Link Advertisements (Type 4)

- An ASBR announces itself by setting the E-bit in its router link advertisements
- The ABRs in the same area generate a type 4 advertisement to the other areas



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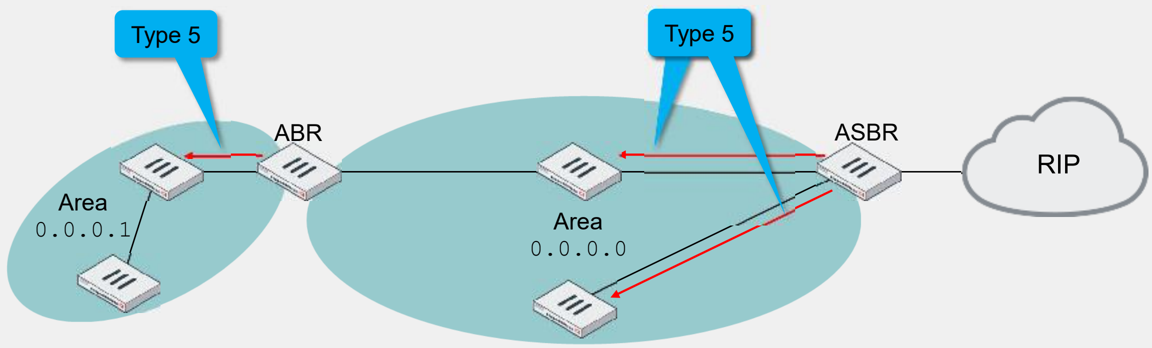
An ASBR advertises itself by sending type 1 LSAs. These LSAs have the E-bit on in the OSPF header. Like any other type 1, the LSAs with the E-bit are confined to the area where they originate. However, ABRs in the same area send a type 4 LSA to the other areas with information about how to reach the ASBR. In the example shown on this slide, an ASBR that is redistributing RIP routes into OSPF announces itself by sending type 1 LSAs to the backbone area. The ABR receives that LSA and sends a type 4 LSA to area 1.

AS External Link Advertisements (Type 5)

- OSPF views non-OSPF networks as external
- OSPF considers the following to be external:
  - A directly connected interface not running OSPF
  - A static route
  - A route derived from a different routing protocol



The last type of LSA covered in this lesson is type 5. Type 5 LSAs are sent only by the ASBRs and are not confined to one area. They reach all the standard areas. They contain link state information for routes redistributed to OSPF (also called external routes).

Note that all the area examples in this lesson are standard areas. There are also stub and not-so-stubby areas (NSSA), which this lesson does not cover. Type 5 LSAs are not advertised to stub or NSSAs.

External Route Metrics Types

- Type 1
  - Considered *close* to the AS
  - Metric is based on the sum of the internal and external costs
- Type 2
  - Considered *far away* from the AS
  - Metric is based on the external cost only
- OSPF router prefers a type 1 external router over a type 2

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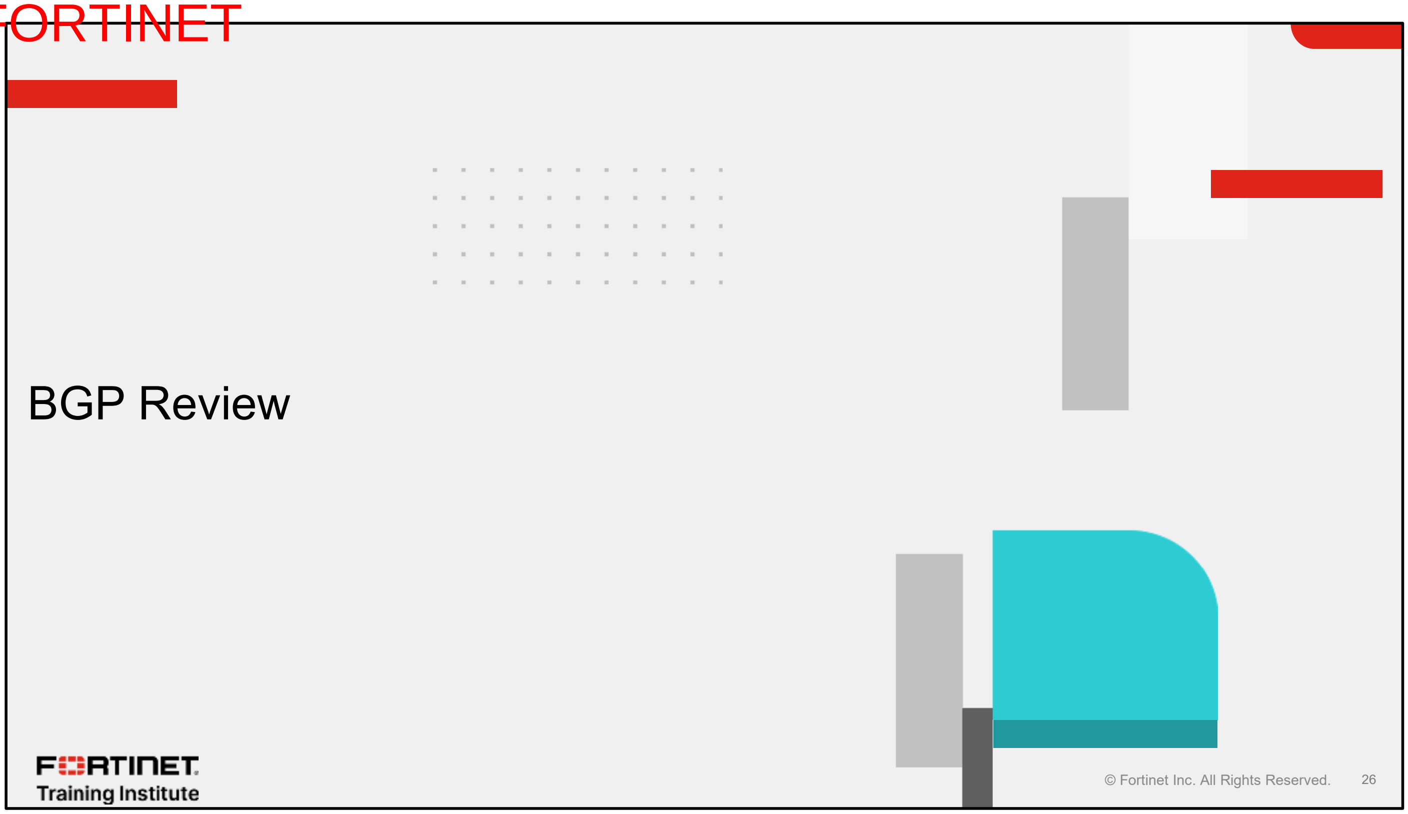
Each external route is assigned a metric. There are two types of external-route metrics. A type 1 metric is the sum of the external cost plus the internal cost to reach the ASBR. A type 2 metric is only the external cost (the internal cost is not considered). If there are two external routes to the same destination, one type 1 and one type 2, an OSPF router selects the type 1 route over the type 2 route.



In this section, you will review BGP concepts and how to configure BGP on FortiGate.

AS

- Set of routers and networks under the same, consistent administration
- An AS administrator is free to choose any internal routing architecture:
  - OSPF, RIP, and so on
- Each AS is identified by a unique number

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An autonomous system (AS) is a set of routers and networks under the same administration. Each AS is identified by a unique number, and usually runs an interior gateway protocol, such as OSPF or RIP.

BGP

- You can configure BGP to operate as EBGP or IBGP:
  - EBGP advertises routing updates across multiple autonomous systems
  - IBGP advertises routing updates within the same AS
- BGP is typically used when the network requires:
  - Large numbers of routes
  - Strict control over which routes are announced or accepted
- BGP4 is the dominant EGP today
  - Example: the internet

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BGP can serve one of two purposes: external BGP (EBGP) or internal BGP (IBGP).

An exterior gateway protocol (EGP) exchanges routing information between autonomous systems. BGP4, which runs in the internet, is the dominant EGP protocol today. EBGP is typically used when strict control is required over a large number of routes.

Two EBGP routers exchange AS path information for destination prefixes or subnets. When two routers start an EBGP communication, the whole BGP routing table is exchanged. After that, only network updates are sent.

BGP Components

- Speaker or peer:
  - Router that transmits and receives BGP messages, and acts on those messages
- Session
  - Connectivity between two peers

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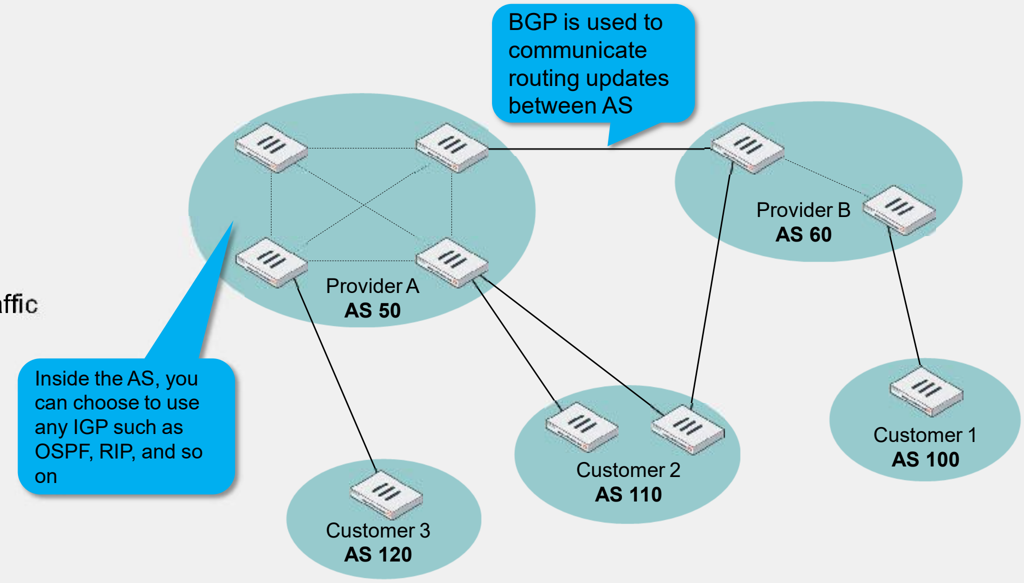
A BGP speaker or peer is a router that sends and receives BGP routing information. The connection between two BGP peers is called a BGP session.

AS Types

- Stub AS
  - Single exit point
  - Local traffic
- Multihomed AS
  - Multiple exit points
  - Local traffic
- Transit AS
  - Local and transit traffic

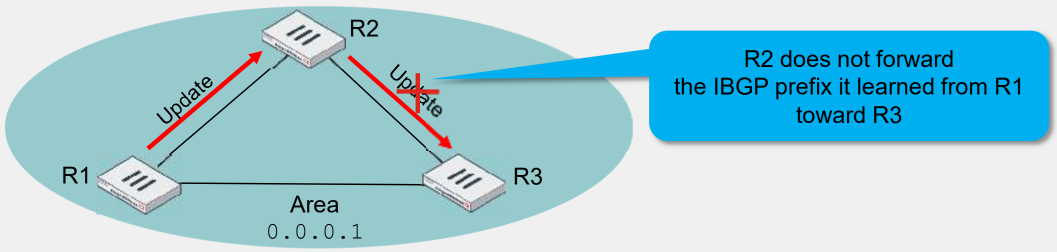


There are three types of autonomous systems:

- A stub AS handles and routes local traffic only and has only one connection to another AS. One example is a company that is running BGP, and has its own AS number and one ISP connection.
- Multihomed AS also handles and routes local traffic only, but it has multiple connections to different autonomous systems. One example is a company that is running BGP, and has its own AS number and multiple ISP connections.
- Transit AS handles and routes local traffic as well as traffic that originates and terminates in different autonomous systems (transit traffic). An ISP is an example of a transit AS.

Split Horizon

- According to the IBGP split horizon rule, prefixes are not advertised between iBGP neighbors



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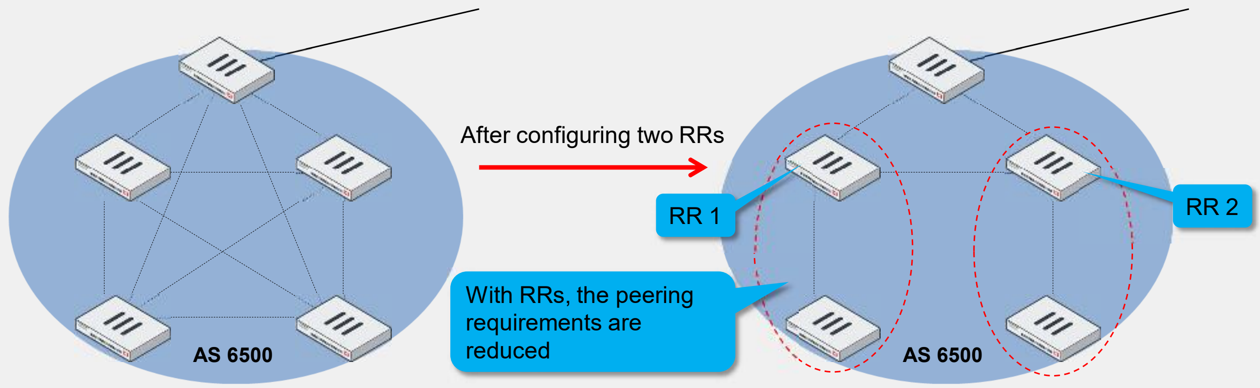
By default, IBGP does not advertise prefixes between iBGP neighbors. This IBGP split horizon rule prevents routing loops. Within an AS, you must then establish full mesh IBGP peerings to learn all the prefixes within an AS.

Route Reflectors

- Running IBGP usually requires full mesh peering
- Route reflectors (RRs) simplify the network and the configuration by:
  - Reducing the number of IBGP peers
  - Forwarding the routes learned from one peer to the other peers

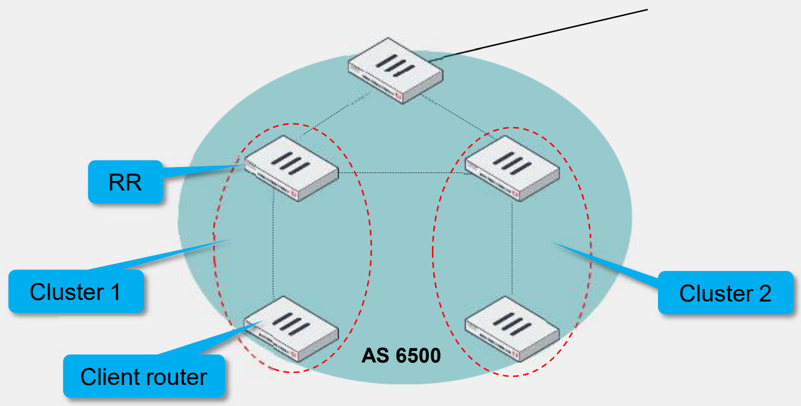


When running IBGP, you usually need to configure full mesh peering between all the routers. In large networks, full mesh peering between routers can be difficult to administer and is not scalable.

RRs help to reduce the number of IBGP sessions inside an AS. An RR forwards the routes learned from one peer to the other peers. If you configure RRs, you don't need to create a full mesh IBGP network. RRs pass the routing updates to other RRs and border routers within the AS.

Route Reflectors (Contd)

- An RR and its clients form a cluster
- All clients in a cluster communicate with the RR only for routing updates
- The RR communicates all routing updates to peers external to the cluster



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In a BGP RR configuration, the AS is divided into different clusters that each include an RR and clients. The client routers communicate route updates only to the RR in the cluster. The RR communicates with other RRs and border routers. FortiGate can be configured as either an RR or a client.

RIBs

- Routes are stored in routing information bases (RIBs)
  - RIB-in
    - Routing information learned from inbound update messages
    - Contains unprocessed routing information advertised to the local BGP speaker by its peers
  - Local RIB
    - Routing information that the BGP speaker selects after applying its local policies to the RIB-in
  - RIB-out
    - Routing information that the local BGP speaker selects to advertise to its peers

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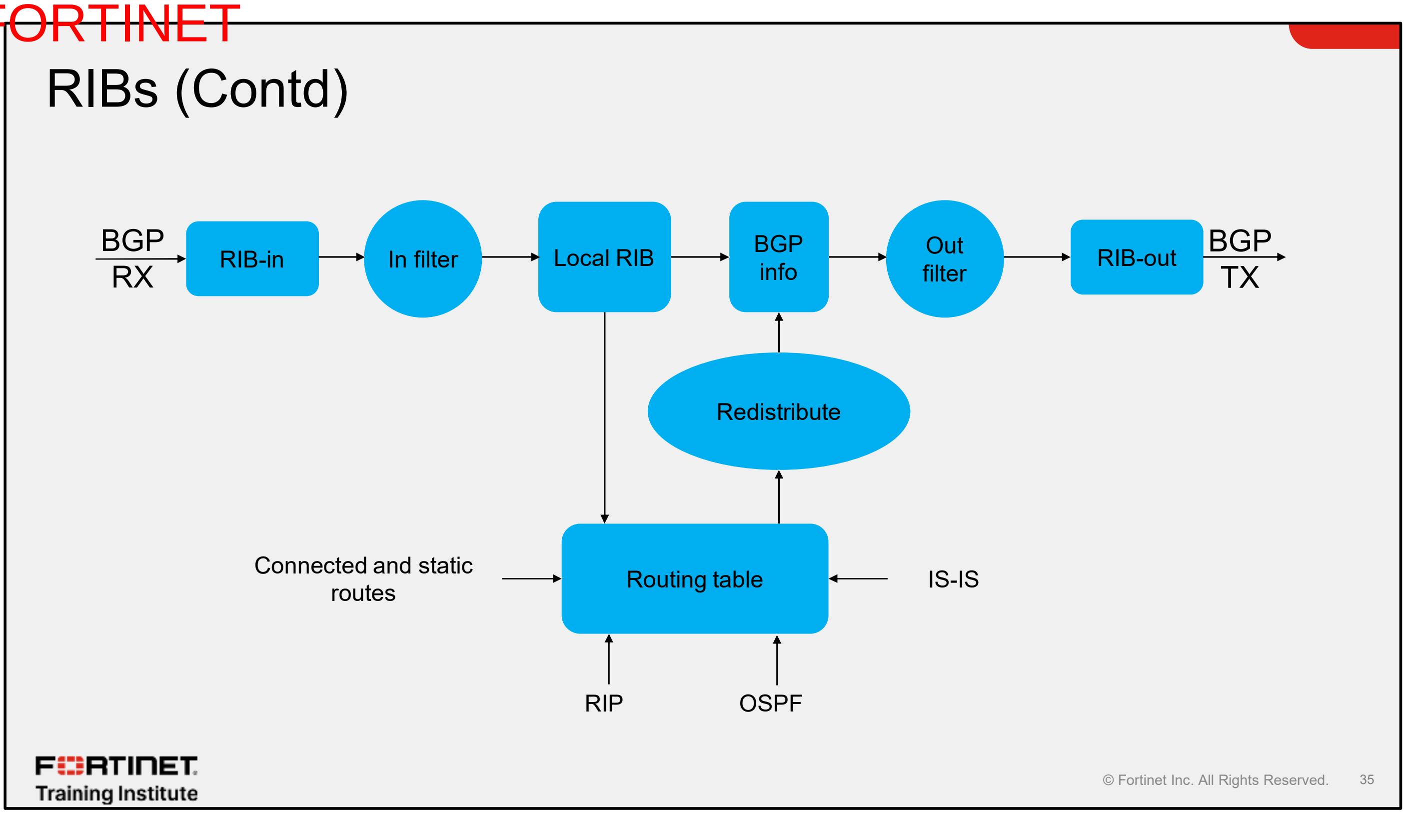
A BGP router stores routing information in three logical tables. The RIB-in table contains all routing information received from other BGP routers before any filtering. The local RIB table contains that same information after the filtering. The RIB-out table contains the BGP routing information selected to advertise to other BGP routers.



This slide shows a flowchart that summarizes the BGP process. The BGP router stores the BGP routes it receives from other routers in the RIB-in table. The BGP router applies a filter, and the resulting routes are stored in the local RIB table. Then, the BGP router adds routes that were redistributed from the routing table, and applies another filter (outbound). The BGP router advertises the resulting routes.

BGP Attributes

- BGP processes routes based on path information
- A path is a route to a destination
  - Paths are described by a set of attributes, including an AS list
  - Attributes help routers select the route to each destination

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BGP routes traffic based on AS paths. Each AS path includes attributes, which BGP uses to select the best route to each destination. One of the attributes is the AS list, which contains the autonomous systems through which the traffic must pass to reach the destination.

BGP Attributes (Contd)

- BGP path attribute categories
  - Well-known mandatory
    - Attributes are mandatory
  - Well-known discretionary
    - Attributes may or may not be included
  - Optional transitive
    - Attributes may or may not be accepted and can be passed outside of the local autonomous system
  - Optional non-transitive
    - Attributes may or may not be accepted and can't be passed outside of the local autonomous system

Supported Attributes	
ORIGIN	Well-known mandatory
AS_PATH	Well-known mandatory
NEXT_HOP	Well-known mandatory
MULTI_EXIT_DISC	Optional non-transitive
LOCAL_PREF	Well-known discretionary
ATOMIC_AGGREGATE	Well-known discretionary
AGGREGATOR	Optional transitive
COMMUNITY	Optional transitive

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There are four types of BGP attributes:

- Well-known mandatory
- Well-known discretionary
- Optional transitive, which can be passed from one AS to another
- Optional non-transitive, which can't be passed from one AS to another

This slide shows a list of the BGP attributes and their attribute types that FortiGate supports.

Route Selection

- Route selection tie breakers:
  1. Highest weight
  2. Highest local preference
  3. Prefer the path that originated locally
  4. Shortest AS path
  5. Lowest origin type
  6. Lowest multi-exit discriminator (MED)
  7. Lowest IGP metric to the BGP next hop
  8. Prefer external paths (EBGP) over internal paths (IBGP)
  9. If ECMP is enabled, insert up to 10 routes in the routing table
  10. Lowest router ID

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FortiGate uses some of the BGP attributes during the routing selection process. If all the attributes for multiple routes to the same destination match, and if ECMP is enabled, FortiGate shares the traffic among up to 10 BGP routes. If you don't enable ECMP, FortiGate uses the route that goes to the router with the lowest BGP router ID.

FortiGate BGP Implementation

- Scaling capabilities
  - No hard limits, system memory is the limitation
  - Number of neighbors, routes, and policies have an impact on the scaling capabilities
- By default, BGP doesn't originate any prefix
  - Redistribution or policies are required
- By default, all routes received are accepted

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There are three important things to consider when you implement BGP on FortiGate.

First, there are no hardcoded limits. Limitations on the number of neighbors, routes, and policies depend exclusively on the available system memory.

Second, by default, FortiGate doesn't originate any prefix. You must enable redistribution, or manually indicate the prefixes that FortiGate originates.

Third, by default, FortiGate accepts all the prefixes it receives. Optionally, you can filter out or modify some prefixes.

Protocol Redistribution

- A non-BGP route can be redistributed into BGP
- Optional route maps can filter out the redistribution of specific routes

```
# config router bgp
    config redistribute "static"
    set status enable
    end
  end
```

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By default, FortiGate BGP doesn't advertise prefixes. You can use the redistribution command to configure FortiGate to advertise prefixes. You can redistribute connected and static routes, and routes learned from other routing protocols, into BGP. Optionally, you can add route maps to filter the prefixes or modify some of their BGP attributes.

Network Command

- An alternative to redistribution of protocols into BGP:

```
# config router bgp
    config network
      edit <id>
        set prefix <prefix>
  ...
```

- By default, the prefix is advertised only when it matches the destination subnet of an active route in the routing table
- To always advertise the prefix (regardless of the active routes):

```
# config router bgp
        set network-import-check disable
  end
```

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You can also use the `network` command to configure FortiGate BGP to advertise prefixes. However, an exact match of the prefix in the `network` command must be active in the routing table. If the routing table doesn't contain an active route with a destination subnet that matches the prefix, FortiGate doesn't advertise the prefix. You can change this behavior by disabling the `network-import-check` setting. After you disable the setting, FortiGate advertises all prefixes in the BGP network table, regardless of the active routes present in the routing table.

Prefix Lists

- Prefix lists can be used to filter out the subnets being advertised to and being received from each neighbor

```
config router prefix-list
   edit filter-subnets
      config rule
         edit 1
           set prefix 10.1.0.0/16
           set action deny
         next
         edit 2
           set prefix 10.0.0.0/8
           set action permit
         next
      end
   end
end
```

```
config router bgp
  config neighbor
    edit 10.3.1.254
      set prefix-list-in filter-subnets
    next
  end
end
```

By default, traffic not matching the prefix list is denied

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By default, the subnets under the `config network` command, and the subnets redistributed from other routing protocols, are advertised to all the neighbors.

With a prefix list, you can be more selective about which prefixes to advertise to each neighbor. Additionally, prefix lists allow you to select which prefixes you want to use from each neighbor. This example shows a prefix list that allows the prefix `10.0.0.0/8`, but blocks the prefix `10.1.0.0/16`. By default, all traffic that does not match a prefix list is denied. The prefix list is applied in the incoming direction from the neighbor `10.3.1.254`. The local FortiGate applies this filter for all prefix advertisements coming from `10.3.1.254`.

When applying a prefix list, all the prefixes that don’t match an entry in the list are denied by default.

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FORTINET CERTIFIED SOLUTION SPECIALIST
Network Security

Enterprise Firewall Administrator

Use Case Solutions

FortiOS 7.6

Last Modified: 28 January 2026

Objectives

- Exercise 1
  - Implement segmentation on the HQ-PC-3 HR network
  - Add dynamic routing
  - Allow the HR segment to access the internet and HQ-Web-1
- Exercise 2
  - Implement a VPN with HQ-NGFW as a hub, and BR2-FGT-1 and BR3-FGT-1 as spokes
  - Implement ADVPN on the hub and spokes
  - Allow traffic between the hub and spokes
  - Implement IBGP on the hub and spokes
- Exercise 3
  - Implement the Security Fabric within the LAN FortiGate devices
  - Implement the automatic configuration backup of the devices in the Security Fabric environment

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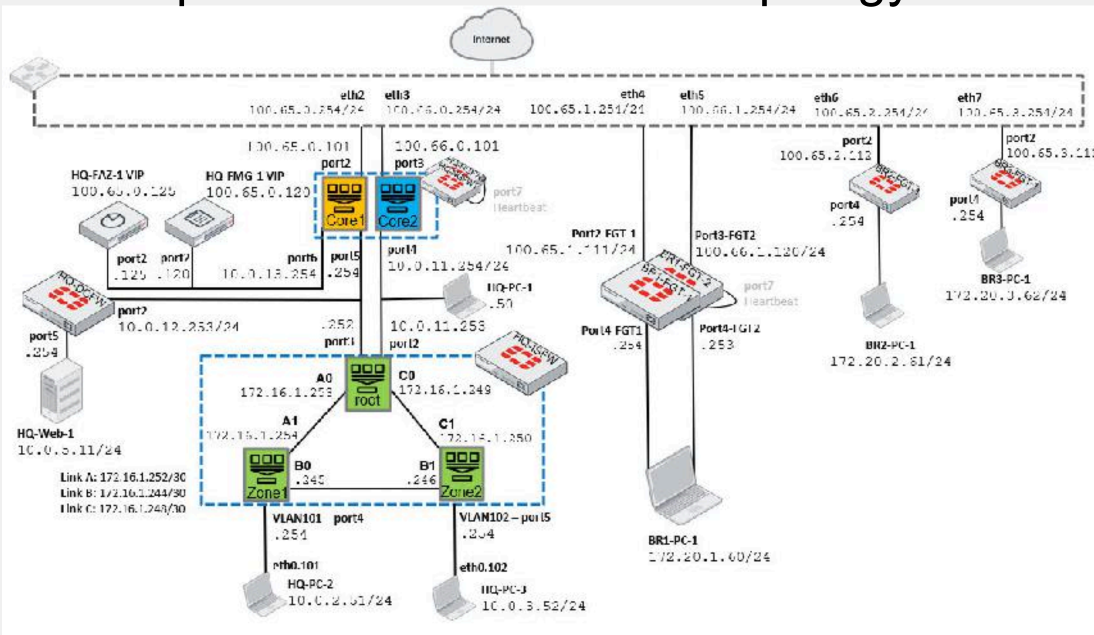
 2

Lab—Enterprise Firewall Network Topology



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This slide shows the enterprise network topology used in the labs.

Exercise 1

Objectives

- Implement segmentation on HQ-PC-3 HR network
- Add dynamic routing
- Allow the HR segment to access the internet and HQ-Web-1

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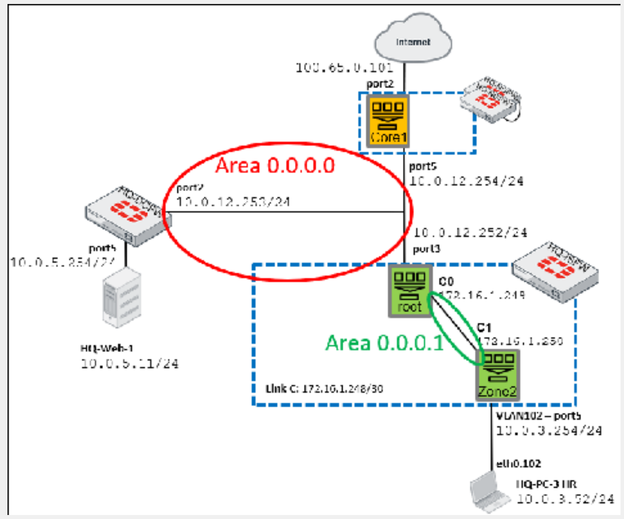
Exercise 1

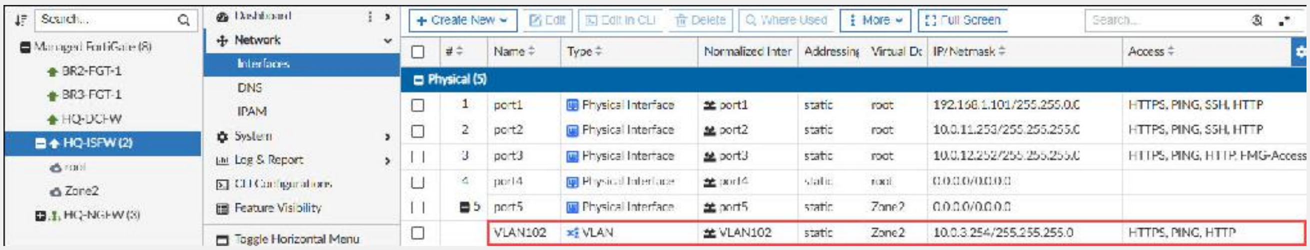
- To implement segmentation in the HR network
  - On HQ-ISFW, create the VLAN102 interface with the **PING** option
  - Create the inter-VDOM link between the root VDOM and Zone2 VDOM
- To add dynamic routing
  - On HQ-ISFW, configure the 0.0.0.1 OSPF area between the root VDOM and new Zone2 VDOM
  - Allow the OSPF protocol to inject the 10.0.3.0/24 subnet into the backbone area
- To allow the HR segment to access HQ-Web-1 and internet
  - Allow only ping from the HR segment to HQ-Web-1
  - Allow only ping and web traffic from the HR segment to the internet



 5

To Implement Segmentation in the HR Network

- On the FortiManager GUI, click **EFW_Use_Case** > **Device Manager** > **HQ-ISFW**
- In **Network** > **Interfaces** > **Create New** interface: **VLAN102** on port5, VLANID 102 linked to the **Zone2** VDOM with IP address 10.0.3.254/24 (**PING** and **HTTPS**)



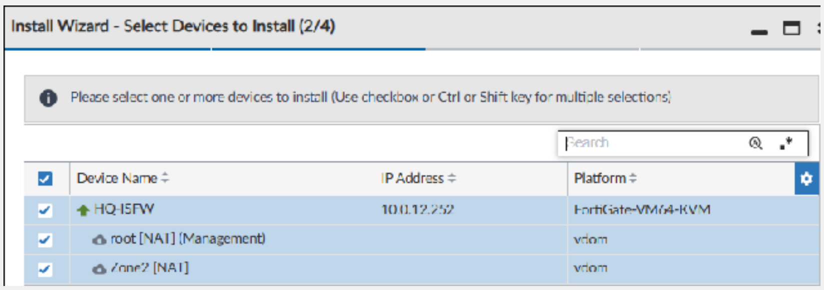
To Implement Segmentation in the HR Network (Contd)

- Create new **VDOM Link** named **C**, for the **root** VDOM, IP address 172.16.1.249/30 (**PING**) and the **Zone2** VDOM, IP address 172.16.1.250/30 (**PING**)

VDOM Link (2)

☐	13	C0	VDOM Link	C0	static	root	172.16.1.249/255.255.255.252	PING
☐	14	C1	VDOM Link	C1	static	Zone2	172.16.1.250/255.255.255.252	PING

- **Install Wizard** > **Install Device Settings (only)**



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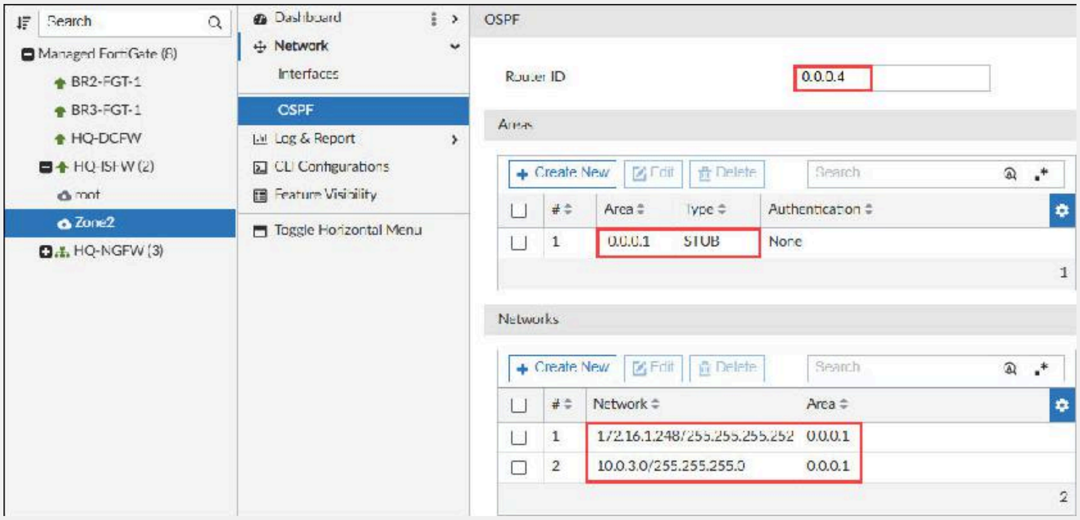
 7

To Add Dynamic Routing

- In **HQ-ISFW** > **Zone2** > **Network** > **OSPF**, configure **Router ID** `0.0.0.4`, **Area** `0.0.0.1` (type **STUB**), **Network** `172.16.1.248/30` and `10.0.3.0/24`



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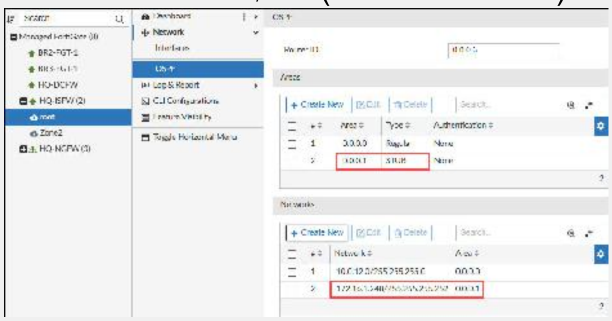
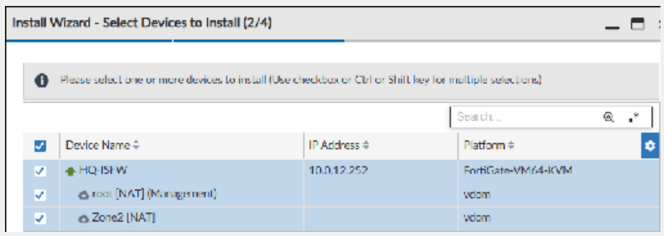
To Add Dynamic Routing (Contd)

- In **HQ-ISFW** > **root** > **Network** > **OSPF**, configure **Router ID** `0.0.0.3`, **Area** `0.0.0.1` (type **STUB**), **Network** `172.16.1.248/30` (for **Area** 0.0.0.1)



- Click **Install Wizard**



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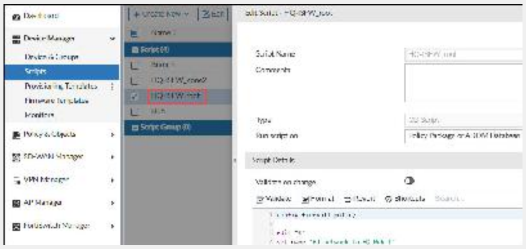
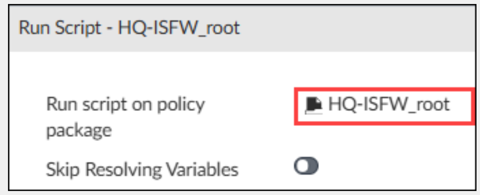
 9

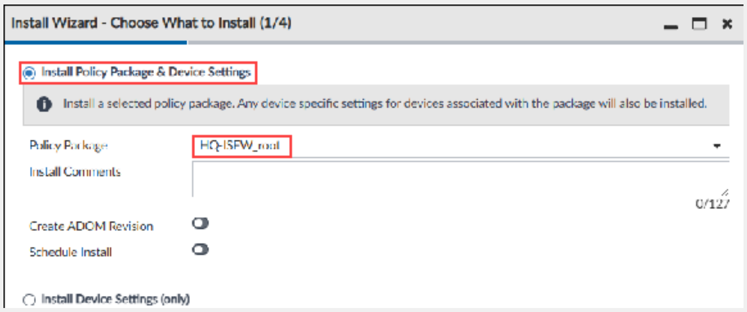
To Allow the HR Segment to Access HQ-Web-1 and Internet

- In **Device Manager** > **Scripts** > **HQ-ISFW_root** > run script on **HQ-ISFW_root**





- In **Policy & Objects** > **Policy Packages** > **HQ-ISFW_root** > **Firewall Policy** > click **Install Wizard**



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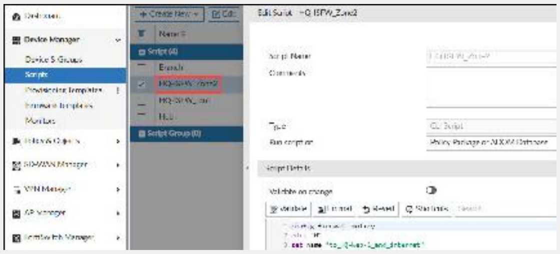
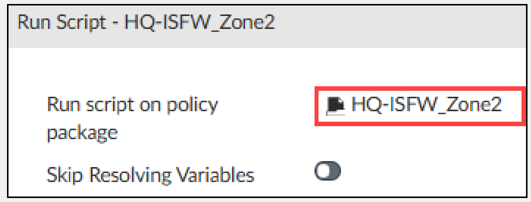
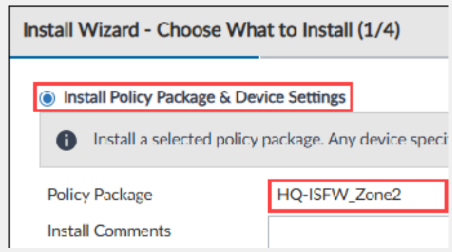
To Allow the HR Segment to Access HQ-Web-1 and Internet (Contd)

- In **Device Manager** > **Scripts** > **HQ-ISFW_Zone2** > run script on **HQ-ISFW_Zone2**





- In **Policy & Objects** > **Policy Packages** > **ISFW_Zone2** > **Installation Targets** > select **HQ-ISFW-Zone2**



- In **Policy & Objects** > **Policy Packages** > **HQ-ISFW_Zone2** > **Firewall Policy** > click **Install Wizard**

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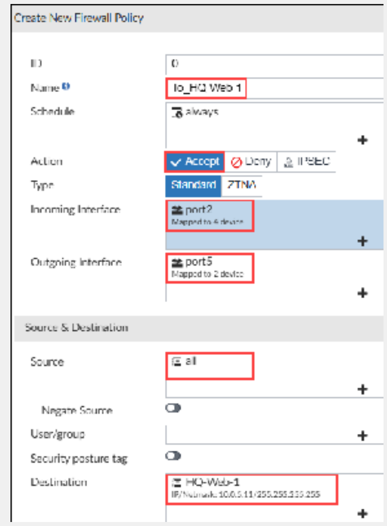
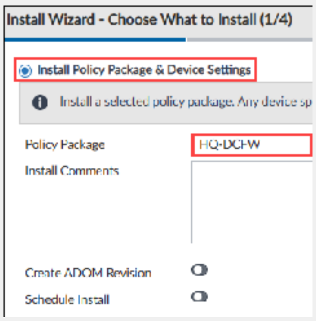
To Allow the HR Segment to Access HQ-Web-1 and Internet (Contd)

- In **Policy & Objects** > **Policy Packages** > **HQ-DCFW** > **Firewall Policy** > Click **Create New**



- In **Policy & Objects** > **Policy Packages** > **HQ-DCFW** > **Firewall Policy** > click **Install Wizard**



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To Test the Configuration

- To test basic connectivity
  - Ping `10.0.3.254` (VLAN102 IP address) from HQ-PC-3

```
Administrator@ubuntu-2204-desktop:~$ ping 10.0.3.254
PING 10.0.3.254 (10.0.3.254) 56(84) bytes of data.
64 bytes from 10.0.3.254: icmp_seq=1 ttl=255 time=1.83 ms
64 bytes from 10.0.3.254: icmp_seq=2 ttl=255 time=0.612 ms
```

- To verify dynamic routing
  - On HQ-ISFW root, get the output from `get router info ospf status`
  - On HQ-DCFW, get the output from `get router info routing-table all`

```
HQ-ISFW (root) # get router info ospf status
 Routing Process "ospf 0" with ID 0.0.0.3
 Process uptime is 1 hour 12 minutes
 Process bound to VRF default
 Conforms to RFC2328, and RFC1583Compatibility flag is disabled
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 Do not support Restarting
 This router is an ABR, ABR Type is Standard (RFC2328)
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Refresh timer 10 secs
 Number of incomming current DD exchange neighbors 0/5
 Number of outgoing current DD exchange neighbors 0/5
 Number of external LSA 0. Checksum 0x000000
 Number of opaque AS LSA 0. Checksum 0x000000
 Number of non-default external LSA 0
 External LSA database is unlimited.
 Number of LSA originated 7
 Number of LSA received 15
 Number of areas attached to this router: 2
    Area 0.0.0.0 (BACKBONE)
        Number of interfaces in this area is 1(1)
        Number of fully adjacent neighbors in this area is 2
        Area has no authentication
        SPF algorithm last executed 00:23:32.830 ago
        SPF algorithm executed 3 times
        Number of LSA 6. Checksum 0x01deba
    Area 0.0.0.1 (Stub)
        Number of interfaces in this area is 1(1)
        Number of fully adjacent neighbors in this area is 1
        Number of fully adjacent virtual neighbors through this area is 0
```

- To test the HR segmentation
  - Ping `10.0.5.11` (HQ-Web-1) from HQ-PC-3
    - Ping should be successful
  - On HQ-PC-3, browse internet
    - Should be able to browse
  - Ping `10.0.3.52` (HQ-PC-3 IP address) from HQ-DCFW
    - Ping should not be successful

```
HQ-DCFW # exec ping 10.0.3.52
PING 10.0.3.52 (10.0.3.52): 56 data bytes
^C
--- 10.0.3.52 ping statistics ---
5 packets transmitted, 0 packets received, 100% packet loss
```

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Exercise 2

Objectives

- Implement a VPN with HQ-NGFW as a hub, and BR2-FGT-1 and BR3-FGT-1 as spokes
- Implement ADVPN on the hub and spokes
- Allow traffic between the hub and spokes
- Implement IBGP on the hub and spokes

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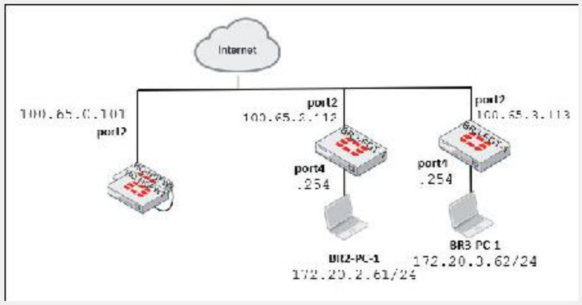
Exercise 2



- To implement a VPN with NGFW as a hub, and BR2-FGT-1 and BR3-FGT-1 as spokes
  - Create a VPN tunnel as a hub on the Core1 VDOM
  - Create the corresponding VPN tunnels as spokes on BR2-FGT-1 and BR3-FGT-2
- To Implement ADVPN on the hub and spokes
  - Configure the tunnel interfaces with the overlay local IP addresses and remote subnet
  - Enable **auto-discovery-sender** on the hub
  - Enable **auto-discovery-receiver** on the spokes

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Exercise 2 (Contd)

- To allow traffic on the hub
  - Allow traffic from the spokes to the Core1 VDOM
  - Allow traffic from the Core1 VDOM to the spokes
  - Allow traffic between the spokes
- To allow traffic on the spokes
  - Allow traffic from the spokes to the hub
  - Allow traffic from the hub to the spokes
- To implement IBGP on the hub and spokes
  - Use the same BGP autonomous system (AS)
  - Configure the hub as a BGP route reflector
  - Configure the hub with neighbor groups and ranges
  - Configure the spokes with their corresponding network prefixes

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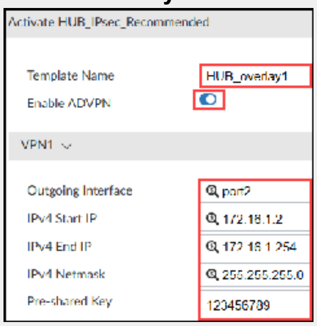
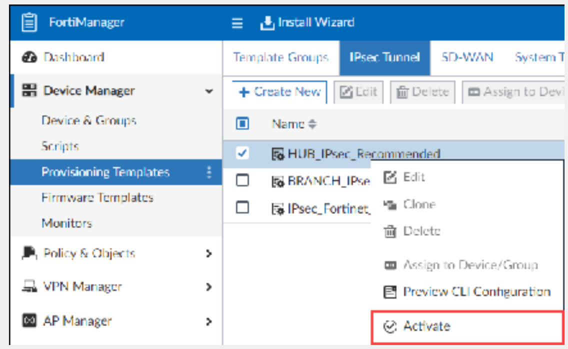
To Implement a VPN With HQ-NGFW-1 as a Hub, and BR2-FGT-1 and BR3-FGT-1 as Spokes

- On the FortiManager GUI, **EFW_Use_Case** > **Device Manager** > **Provisioning Templates** > **IPsec Tunnel** > right-click **HUB_IPsec_Recommended**, and then select **Activate** with:
  - **Template Name**: `HUB_overlay1`
  - **Enable ADVPN**: Enabled
  - **Outgoing Interface**: `port2`
  - **IPv4 Start IP**: `172.16.1.2`
  - **IPv4 End IP**: `172.16.1.254`
  - **IPv4 Netmask**: `255.255.255.0`
  - **Pre-shared Key**: `123456789`





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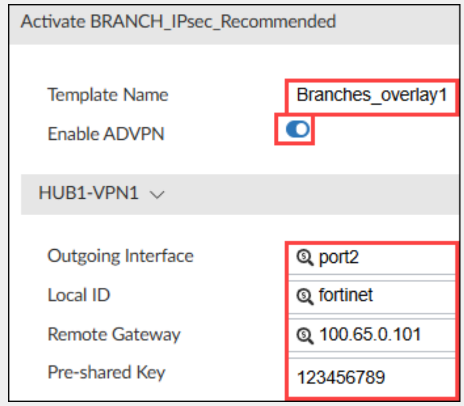
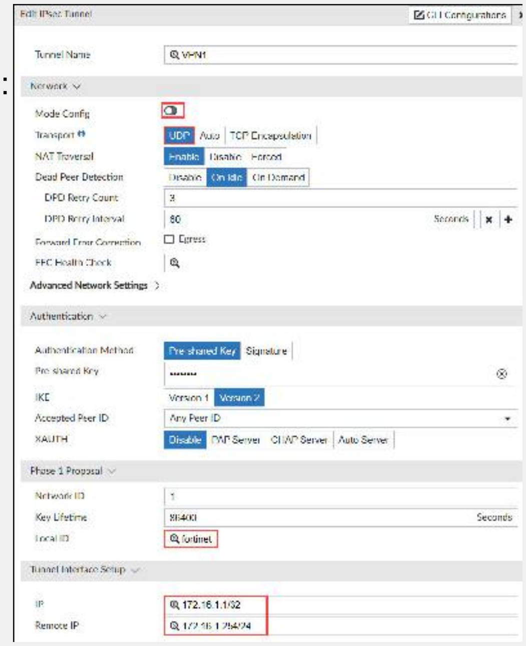
To Implement a VPN With HQ-NGFW-1 as a Hub, and BR2-FGT-1 and BR3-FGT-1 as Spokes (Contd)

- On the FortiManager GUI, **Device Manager** > **Provisioning Templates** > **IPsec Tunnel** > right-click **BRANCH_IPsec_Recommended**, and then select **Activate** with:
  - **Template Name**: `Branches_overlay1`
  - **Enable ADVPN**: Enabled
  - **Outgoing Interface**: `port2`
  - **Local ID**: `fortinet`
  - **Remote Gateway**: `100.65.0.101`
  - **Pre-shared Key**: `123456789`



To Implement ADVPN on the Hub and Spokes

- Edit **HUB_overlay1** > edit **VPN1**, and then configure:
  - **Mode Config**: Disable
  - **Transport**: `UDP`
  - **Local ID**: `fortinet`
  - In **Tunnel Interface Setup** section:
    - **IP**: `172.16.1.1/32`
    - **Remote IP**: `172.16.1.254/24`



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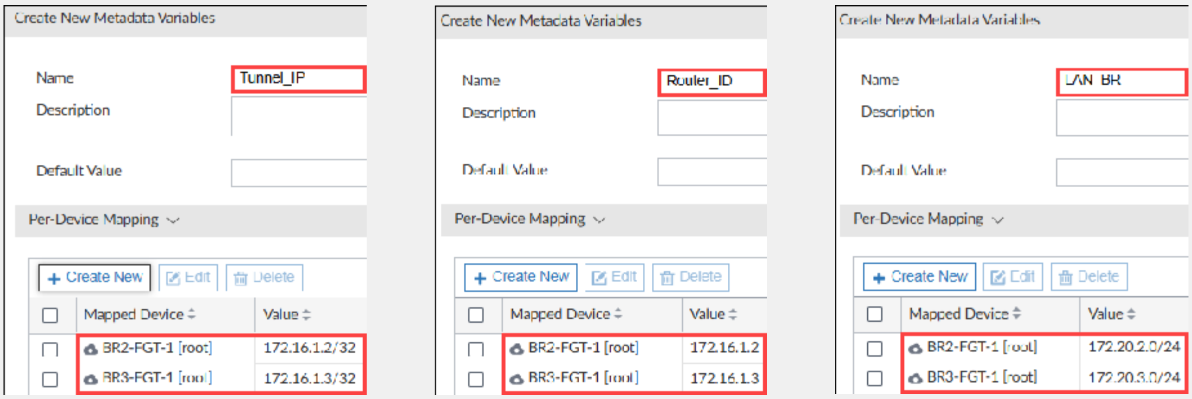
To Implement ADVPN on the Hub and Spokes (Contd)

- For the spokes, create three metadata variables (in **Policy & Objects** > **Advanced**):
  - **Name**: `Tunnel_IP` with `172.16.1.2/32` for BR2-FGT-1 and `172.16.1.3/32` for BR3-FGT-1
  - **Name**: `Router_ID` with `172.16.1.2` for BR2-FGT-1 and `172.16.1.3` for BR3-FGT-1
  - **Name**: `LAN_BR` with `172.20.2.0/24` for BR2-FGT-1 and `172.20.3.0/24` for BR3-FGT-1



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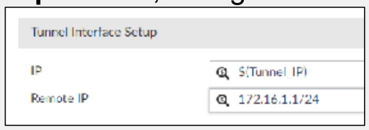
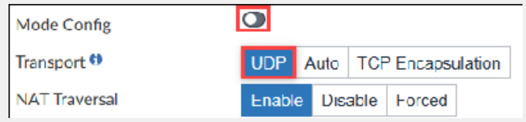
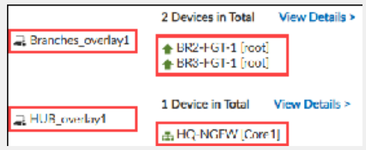
To Implement ADVPN on the Hub and Spokes (Contd)

- Edit **Branches_overlay1** > edit **HUB1-VPN1**, and then, disable **Mode Config**, set **Transport** to **UDP** in **Tunnel Interface Setup** section, configure:
  - **IP**: `$(Tunnel_IP)`
  - **Remote IP**: `172.16.1.1/24`

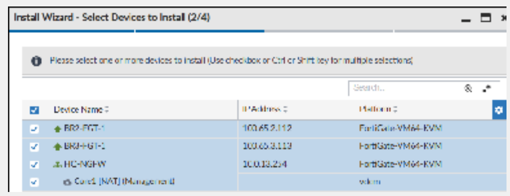




- Assign **Branches_overlay1** to **BR2-FGT-1 [root]** and **BR3-FGT-2 [root]** and **HUB_overlay1** to **HQ-NGFW [Core1]**



- **Install Wizard** > **Install Device Settings (only)**



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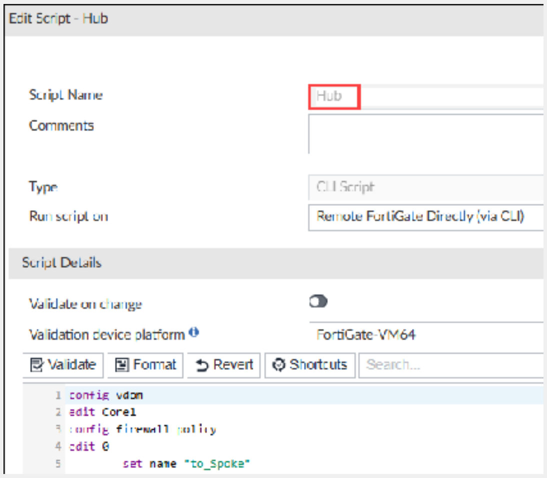
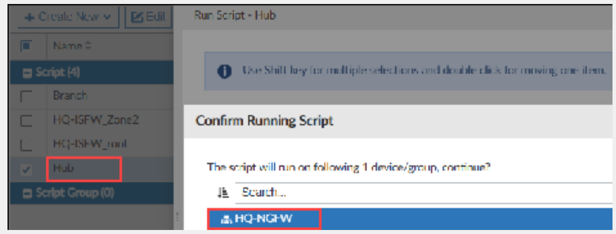
To Allow Traffic on the Hub

- Use a script (but other ways are possible)
- On the FortiManager GUI, **Device Manager** > **Scripts** > **Hub**



- Run the **Hub** script on **HQ-NGFW**



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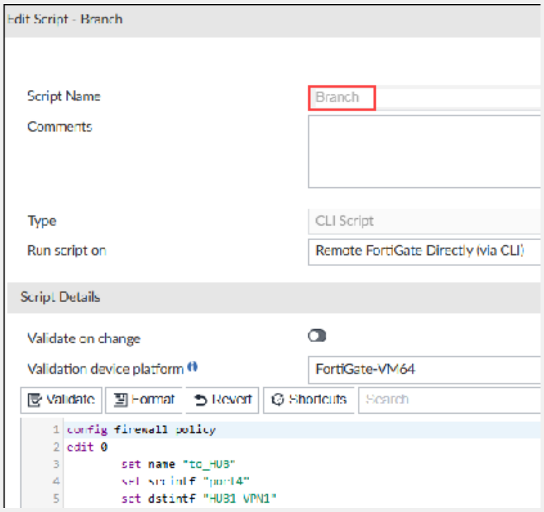
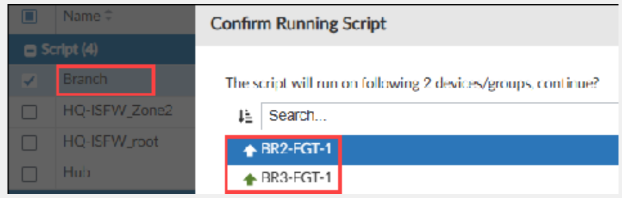
To Allow Traffic on the Spokes

- **Device Manager** > **Scripts** > **Branch**



- Run the **Branch** script on **BR2-FGT-1** and **BR3-FGT-1**



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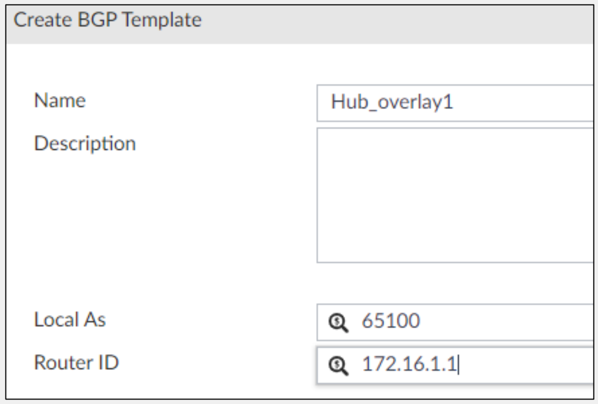
To Implement IBGP on the Hub and Spokes

- On the FortiManager GUI, **Device Manager** > **Provisioning Templates** > **Feature Visibility** > enable **BGP**
- In **Device Manager** > **Provisioning Templates** > **BGP** > **Create New** BGP template with:
  - **Name**: `Hub_overlay1`
  - **Local As**: `65100`
  - **Router ID**: `172.16.1.1`



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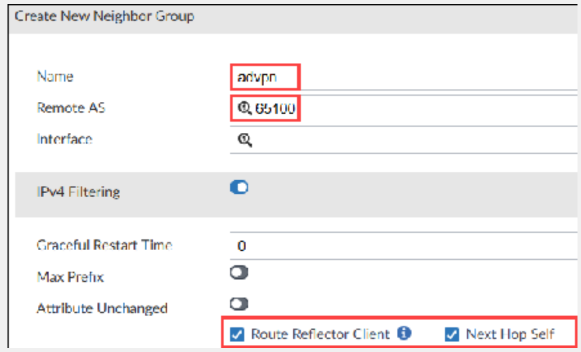
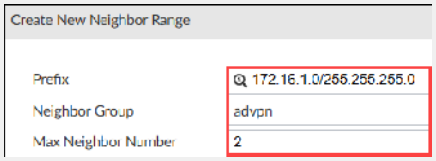
To Implement IBGP on the Hub and Spokes (Contd)

- In **Neighbor Group** section, **Create New** with:
  - **Name**: `advpn`
  - **Remote AS**: `65100`
  - **Route Reflector Client**: enabled
  - **Next Hop Self**: enabled
- In **Neighbor Range** section, **Create New** with:
  - **Prefix**: `172.16.1.0/255.255.255.0`
  - **Neighbor Group**: `advpn`
  - **Max Neighbor Number**: `2`





- **Networks**: `10.0.12.0/24`

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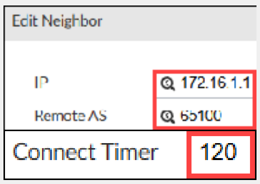
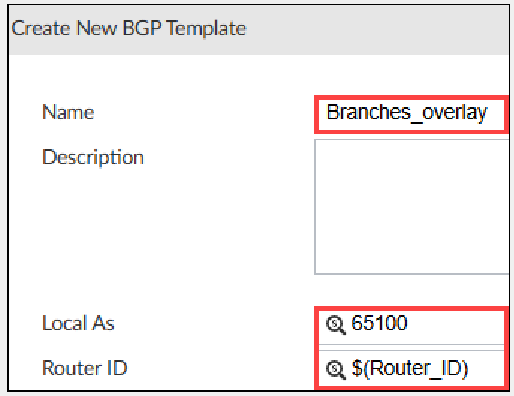
To Implement IBGP on the Hub and Spokes (Contd)

- In **Device Manager** > **Provisioning Templates** > **BGP** > **Create New** BGP template with:
  - **Name**: `Branches_overlay`
  - **Local As**: `65100`
  - **Router ID**: `$(Router_ID)`
  - In **Neighbors** section, **Create New** with:
    - **IP**: `172.16.1.1`
    - **Remote AS**: `65100`
    - **Connect Timer**: `120`



  - **Networks**: `$(LAN_BR)`



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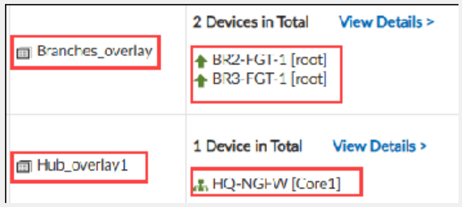
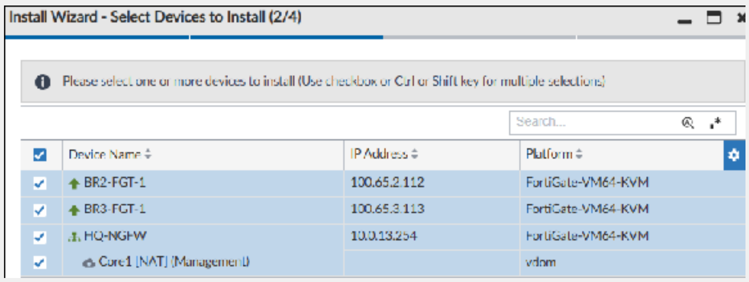
To Implement IBGP on the Hub and Spokes (Contd)

- Assign **Branches_overlay** to **BR2-FGT-1 [root]** and **BR3-FGT-2 [root]** and **Hub_overlay1** to **HQ-NGFW [Core1]**



- **Install Wizard** > **Install Device Settings (only)**



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To Test the Configuration

- In BR2-FGT-1 console

```
BR2-FGT-1 # diagnose vpn tunnel list
list all ipsec tunnel in vd 0
------------------------------------------------------
name=HUB1-VPN1 ver=2 serial=1 100.65.2.112:0->100.65.0.101:0 nexthop=100.65.2.254 tun_id=100.65.0.101 tun_id6=
bound_if=4 real_if=4 lgwy=static/1 tun=intf mode=auto/1 encap=none/568 options[0238]=npu create_dev frag-rfc

proxyid_num=1 child_num=0 refcnt=4 ilast=38 olast=38 ad=r/2
stat: rxp=12 txp=13 rxb=884 txb=1970
dpd: mode=on-demand on=1 status=ok idle=20000ms retry=3 count=0 seqno=0
natt: mode=none draft=0 interval=0 remote_port=0
fec: egress=0 ingress=0
proxyid=HUB1-VPN1 proto=0 sa=1 ref=3 serial=1 auto-negotiate adr
  src: 0:0.0.0.0-255.255.255.255:0
  dst: 0:0.0.0.0-255.255.255.255:0
  SA:  ref=3 options=3a203 type=00 soft=0 mtu=1438 expire=42304/0B replaywin=2048
       seqno=e esn=0 replaywin_lastseq=0000000d qat=0 rekey=0 hash_search_len=1
  life: type=01 bytes=0/0 timeout=42899/43200
  dec: spi=2b097774 esp=aes key=32 d7baac080f775cc4b44f1490852bcfdf99db500b1f308a48ecfd86cb1df1fde8
       ah=sha256 key=32 f652c53f72cd36ba0ea095804c41ab76f30714c3493da3125ade9ff0bbcc577b
  enc: spi=a1e5480a esp=aes key=32 b5e0462bbed2a766aa802c5f8dfb30d6c55bfcc9c0e7ff3b8f3b85e4bc96a05f
       ah=sha256 key=32 0e17867178b27c94b8ba96139a65cde41e026a37b58cd70c5020669a41b25409
  dec:pkts/bytes=12/884, enc:pkts/bytes=13/1788
  npu_flag=00 npu_rgwy=100.65.0.101 npu_lgwy=100.65.2.112 npu_selid=0 dec_npuid=0 enc_npuid=0
```

- In Core1 console

```
HQ-NGFW-1 (Core1) # diagnose vpn tunnel list
list all ipsec tunnel in vd 2
------------------------------------------------------
name=VPN1_0 ver=2 serial=5 100.65.0.101:0->100.65.3.113:0 nexthop=100.65.0.254 tun_id=172.16.1.2 tun_id6=
bound_if=4 real_if=4 lgwy=static/1 tun=intf mode=dial_inst/3 encap=none/74408 options[122a8]=npu rgwy-chg

parent=VPN1 index=0
proxyid_num=1 child_num=0 refcnt=5 ilast=27 olast=27 ad=s/1
stat: rxp=19 txp=18 rxb=1243 txb=2772
dpd: mode=on-idle on=1 status=ok idle=60000ms retry=3 count=0 seqno=7
natt: mode=none draft=0 interval=0 remote_port=0
fec: egress=0 ingress=0
proxyid=VPN1 proto=0 sa=1 ref=3 serial=1 ads
  src: 0:0.0.0.0-255.255.255.255:0
  dst: 0:0.0.0.0-255.255.255.255:0
  SA:  ref=3 options=20a02 type=00 soft=0 mtu=1438 expire=42446/0B replaywin=2048
       seqno=13 esn=0 replaywin_lastseq=00000014 qat=0 rekey=0 hash_search_len=1
  life: type=01 bytes=0/0 timeout=43187/43200
  dec: spi=a1e5480b esp=aes key=32 5bd0928dfb028355e71929573a566cdaf408f2d5d7c983a2687d0027f42285a5
       ah=sha256 key=32 48aa6fab697eaf0f723ce14ee2200db9a2bd1c6d8af4e9483a70da7db0a2ef3c
  enc: spi=27691f93 esp=aes key=32 98930232b06051ab82770a2a589e2249439bde21efe159715efe34028bfbf63e
       ah=sha256 key=32 766662e6154b3f8cd889a0cef0bf21407468ef61c2daf5c446df7d192d17455b
  dec:pkts/bytes=19/1243, enc:pkts/bytes=18/2520
  npu_flag=00 npu_rgwy=100.65.3.113 npu_lgwy=100.65.0.101 npu_selid=3 dec_npuid=0 enc_npuid=0
------------------------------------------------------
name=VPN1_1 ver=2 serial=4 100.65.0.101:0->100.65.2.112:0 nexthop=100.65.0.254 tun_id=172.16.1.3 tun_id6=
bound_if=4 real_if=4 lgwy=static/1 tun=intf mode=dial_inst/3 encap=none/74408 options[122a8]=npu rgwy-chg

parent=VPN1 index=1
proxyid_num=1 child_num=0 refcnt=5 ilast=41 olast=41 ad=s/1
stat: rxp=19 txp=18 rxb=1243 txb=2772
dpd: mode=on-idle on=1 status=ok idle=60000ms retry=3 count=0 seqno=6
natt: mode=none draft=0 interval=0 remote_port=0
fec: egress=0 ingress=0
proxyid=VPN1 proto=0 sa=1 ref=3 serial=1 ads
  src: 0:0.0.0.0-255.255.255.255:0
  dst: 0:0.0.0.0-255.255.255.255:0
  SA:  ref=3 options=20a02 type=00 soft=0 mtu=1438 expire=42436/0B replaywin=2048
       seqno=13 esn=0 replaywin_lastseq=00000014 qat=0 rekey=0 hash_search_len=1
  life: type=01 bytes=0/0 timeout=43190/43200
  dec: spi=a1e5480a esp=aes key=32 b5e0462bbed2a766aa802c5f8dfb30d6c55bfcc9c0e7ff3b8f3b85e4bc96a05f
       ah=sha256 key=32 0e17867178b27c94b8ba96139a65cde41e026a37b58cd70c5020669a41b25409
  enc: spi=2b097774 esp=aes key=32 d7baac080f775cc4b44f1490852bcfdf99db500b1f308a48ecfd86cb1df1fde8
       ah=sha256 key=32 f652c53f72cd36ba0ea095804c41ab76f30714c3493da3125ade9ff0bbcc577b
  dec:pkts/bytes=19/1243, enc:pkts/bytes=18/2520
  npu_flag=00 npu_rgwy=100.65.2.112 npu_lgwy=100.65.0.101 npu_selid=2 dec_npuid=0 enc_npuid=0
------------------------------------------------------
name=VPN1 ver=2 serial=1 100.65.0.101:0->0.0.0.0:0 nexthop=100.65.0.254 tun_id=10.0.0.1 tun_id6=::10.0.0.
bound_if=4 real_if=0 lgwy=static/1 tun=intf mode=dialup/2 encap=none/552 options[0228]=npu frag-rfc  role

proxyid_num=0 child_num=2 refcnt=4 ilast=42986635 olast=42986635 ad=/0
stat: rxp=38 txp=36 rxb=2486 txb=5544
dpd: mode=on-idle on=0 status=ok idle=60000ms retry=3 count=0 seqno=0
natt: mode=none draft=0 interval=0 remote_port=0
fec: egress=0 ingress=0
```

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To Verify the IBGP and ADVPN Implementation

- on Core1 GUI

Network	Gateway IP	Interfaces	Distance	Type
0.0.0.0/0	100.65.0.254	port2	10	Static
10.0.3.0/24	10.0.12.252	port5	110	OSPF (Inter area)
10.0.5.0/24	10.0.12.253	port5	110	OSPF
10.0.12.0/24	0.0.0.0	port5	0	Connected
10.0.13.0/24	0.0.0.0	port6	0	Connected
100.65.0.0/24	0.0.0.0	port2	0	Connected
172.16.1.0/24		VPN1	0	Connected
172.16.1.1/32		VPN1	0	Connected
172.16.1.248/30	10.0.12.252	port5	110	OSPF (Inter area)
172.20.2.0/24		VPN1	200	BGP
172.20.3.0/24		VPN1	200	BGP

- On BR2-FGT-1 console, ping 172.20.3.254 with source as 172.20.2.254

```
BR2-FGT-1 # execute ping-options source 172.20.2.254

BR2-FGT-1 # execute ping 172.20.3.254
PING 172.20.3.254 (172.20.3.254): 56 data bytes
64 bytes from 172.20.3.254: icmp_seq=0 ttl=254 time=5.2 ms
64 bytes from 172.20.3.254: icmp_seq=1 ttl=254 time=2.2 ms
^C
--- 172.20.3.254 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 2.2/3.7/5.2 ms
```

- On BR3-FGT-1 console, confirm the routing table

```
BR3-FGT-1 # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4
       * - candidate default

Routing table for VRF=0
S*      0.0.0.0/0 [10/0] via 100.65.3.254, port2, [1/0]
B       10.0.12.0/24 [200/0] via 172.16.1.1 (recursive via HUB1-VPN1 tunnel 100.65.0.101), 00:03:25, [1/0]
C       100.65.3.0/24 is directly connected, port2
S       172.16.1.0/24 [5/0] via HUB1-VPN1 tunnel 100.65.0.101, [1/0]
S       172.16.1.1/32 [15/0] via HUB1-VPN1 tunnel 100.65.0.101, [1/0]
C       172.16.1.2/32 is directly connected, HUB1-VPN1_0
C       172.16.1.3/32 is directly connected, HUB1-VPN1
                      is directly connected, HUB1-VPN1_0
B       172.20.2.0/24 [200/0] via 172.16.1.2 (recursive is directly connected, HUB1-VPN1_0), 00:02:56, [1/0]
C       172.20.3.0/24 is directly connected, port4
C       192.168.0.0/16 is directly connected, port1
```

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Exercise 3

Objectives

- Implement the Security Fabric within the LAN FortiGate devices
- Implement the automatic configuration backup of the devices in the Security Fabric environment

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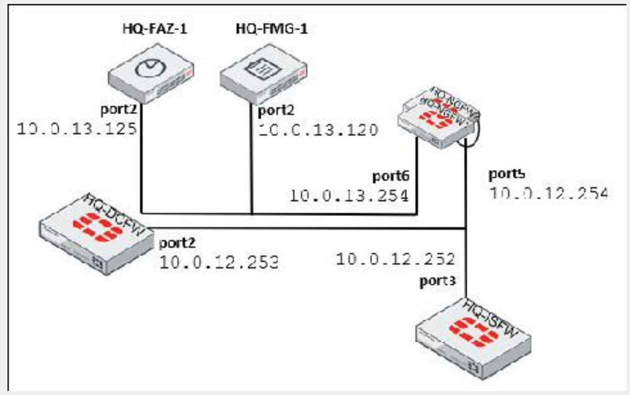
Exercise 3

- To configure the Security Fabric
  - Configure HQ-NGFW to be the root FortiGate
  - Configure HQ-DCFW and HQ-ISFW to join the Security Fabric environment
- To achieve automatic configuration backup
  - Select the automation trigger based on configuration changes
  - Configure the automation action configuration backup
  - Create the stitch based on the automation trigger and action that you configured



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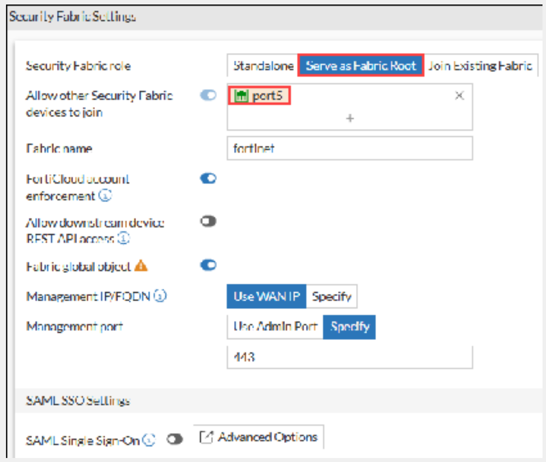
To Configure the Security Fabric

- In **HQ-NGFW-1[Global]** > **Security Fabric** > **Fabric Connectors** > **Security Fabric Setup** configure:
  - **Security Fabric role**: **Serve as Fabric Root**
  - **Allow other Security Fabric devices to join**: **port5**
  - **Fabric name**: `fortinet`



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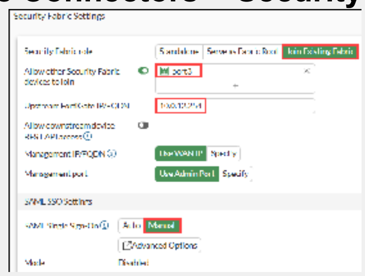
To Configure the Security Fabric (Contd)

- In **HQ-ISFW** and **HQ-DCFW** > **Security Fabric** > **Fabric Connectors** > **Security Fabric Setup** configure:
  - **Security Fabric role**: **Join Existing Fabric**
  - **Allow other Security Fabric devices to join**:
    - **port3** for **HQ-ISFW** and **port2** for **HQ-DCFW**
  - **Upstream FortiGate IP/FQDN**: `10.0.12.254`
  - **SAML Single Sign-On**: **Manual**



- In **HQ-NGFW-1 [Global]** > **System** > **Firmware & Registration** > authorize **HQ-ISFW** and **HQ-DCFW**

Device	Status	Registration Status
HQ-NGFW-1	Online	Registered
HQ-ISFW	Online	Registered
HQ-DCFW	Online	Registered

Caution: On the **HQ-NGFW-1** GUI, you can refresh the page to see HQ-ISFW and HQ-DCFW registration requests

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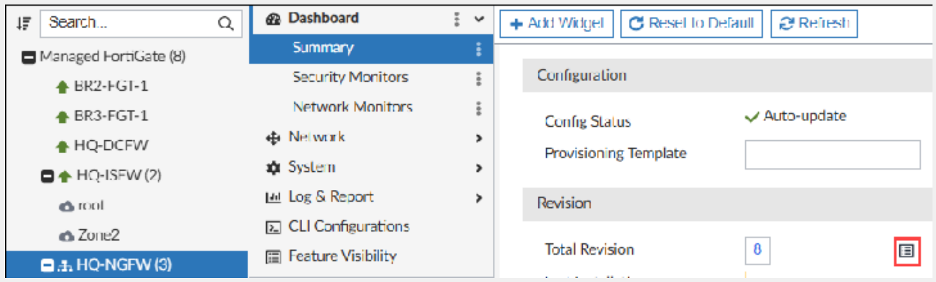
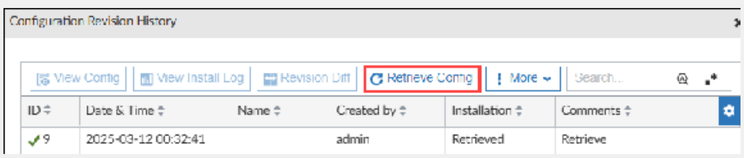
To Achieve Automatic Configuration Backup

- On the FortiManager GUI, **EFW_Use_Case** > **Device Manager** > **Device & Groups** > **Managed FortiGate** > click **HQ-NGFW**
- In **Summary** > **Configuration and Installation** > **Revision** > **Total Revision** > click the revision history icon



- **Retrieve Config**



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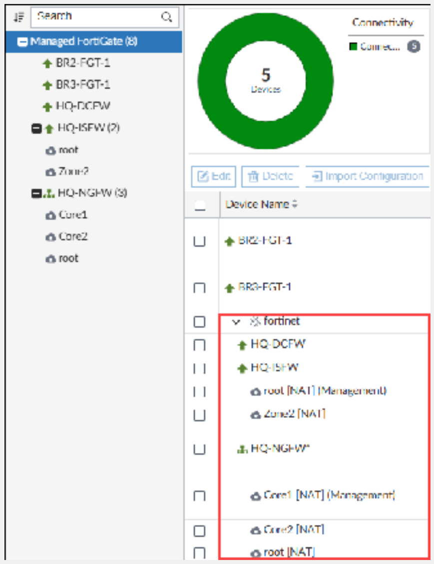
To Achieve Automatic Configuration Backup (Contd)

- You can view the Security Fabric environment on the FortiManager GUI:



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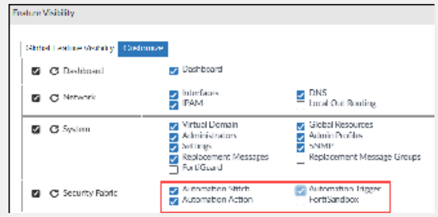
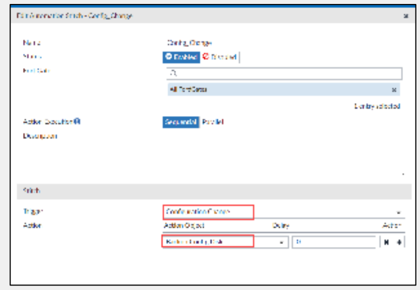
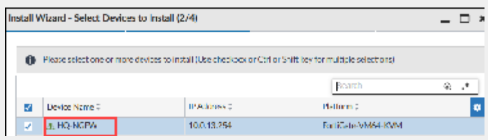
To Achieve Automatic Configuration Backup (Contd)

- On the FortiManager GUI, **Device Manager** > **Managed FortiGate** > **HQ-NGFW** > **Feature Visibility** > **Customize** > select **Automation Stitch**, **Automation Trigger**, and **Automation Action** checkboxes



- In **HQ-NGFW** > **Security Fabric** > **Automation Stitch** > **Create New** stitch with:
  - **Name**: `Config_Change`
  - **Trigger**: **Configuration Change**
  - **Action**: **Backup Config Disk**
- Click **Install Wizard**





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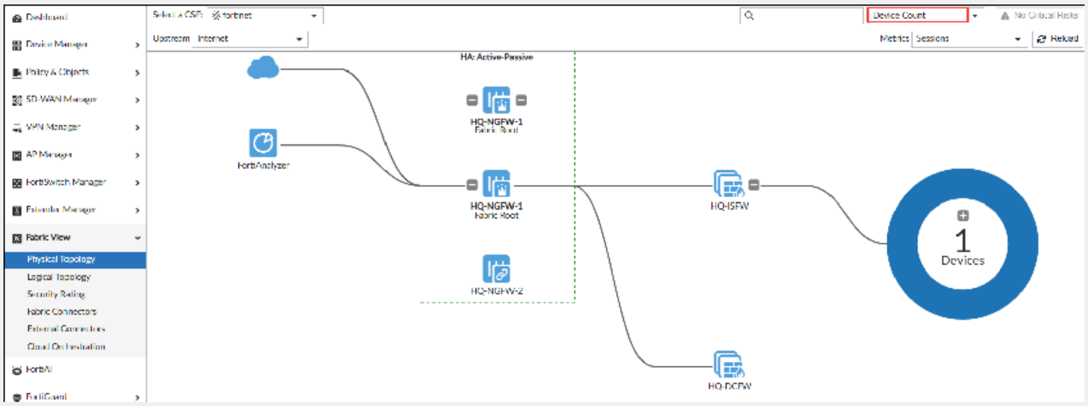
To Test the Configuration

- To verify the Security Fabric configuration:
  - On the FortiManager GUI, **EFW_Use_Case** > **Fabric View** > **Physical Topology**

Caution: Select **Device Count** to see all the devices correctly
You may need to log out and log in again to see **Physical Topology** in the menu



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To Test the Configuration (Contd)

- To test the automatic configuration backup:
  - In **HQ-DCFW** SSH, enter:

```
config system interface
    edit port2
    set description modification
    next
end
exit
```

  - On the **HQ-DCFW** GUI > **admin** > **Configuration** > **Revisions** > expand **7.6.3 build 3510**

Config ID	Username	Date	Comments
7.6.3 build 3510 (3)			
4	daemon_admin	2025/12/29 06:51:13	Autod backup config by stitch: Config_Change

  - On the **HQ-DCFW** GUI > **Log & Report** > **System Events** > **Logs**, you must see:

Date/Time	Level	User	Message	Log Description
2025/12/29 06:51:13	Notice		stitch:Config_Change is triggered.	Automation stitch triggered
2025/12/29 06:51:12	Alert Notification	admin	Configuration is changed in the admin ses...	Configuration changed
2025/12/29 06:51:12	Information	admin	Administrator admin logged out from ssh(...	Admin logout successful
2025/12/29 06:51:09	Information	admin	Edit system.interface port2	Object attribute configured
2025/12/29 06:50:49	Information	admin	Administrator admin logged in successfull...	Admin login successful

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